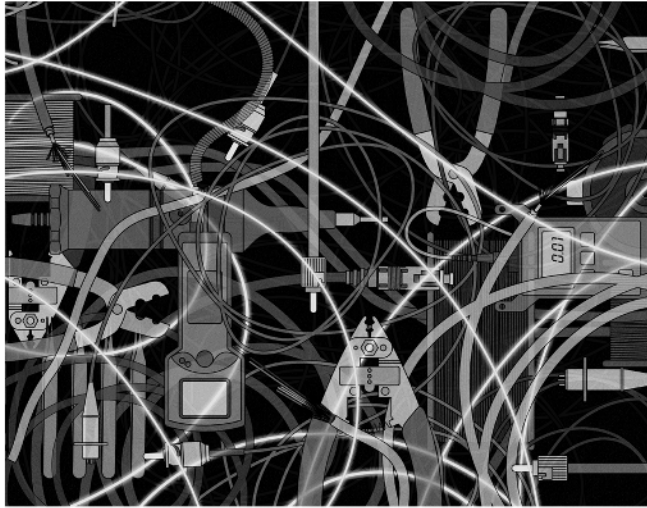


Fiber Optics Installer and Technician Guide

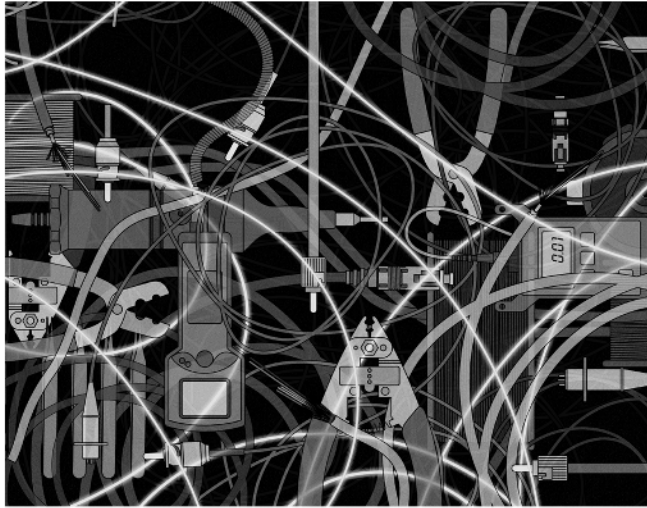
Bill Woodward
Emile B. Husson

SYBEX®

Fiber Optics Installer and Technician Guide



Fiber Optics Installer and Technician Guide



Bill Woodward
Emile B. Husson

San Francisco • London



Publisher: Neil Edde
Acquisitions and Developmental Editor: Maureen Adams
Production Editor: Mae Lum
Technical Editor: Charles Husson
Copyeditor: Suzanne Goraj
Compositor: Jeffrey Wilson, Happenstance Type-O-Rama
Graphic Illustrator: Jeffrey Wilson, Happenstance Type-O-Rama
CD Coordinator: Dan Mummert
CD Technician: Kevin Ly
Proofreaders: Jim Brook, Ian Golder, Rachel Gunn, Jennifer Larsen, Nancy Riddiough
Indexer: Nancy Guenther
Book Designers: Bill Gibson, Judy Fung
Cover Designer and Illustrator: Richard Miller, Calyx Design

Copyright © 2005 SYBEX Inc., 1151 Marina Village Parkway, Alameda, CA 94501. World rights reserved. No part of this publication may be stored in a retrieval system, transmitted, or reproduced in any way, including but not limited to photocopy, photograph, magnetic, or other record, without the prior agreement and written permission of the publisher.

Library of Congress Card Number: 2004115078

ISBN: 0-7821-4390-3

SYBEX and the SYBEX logo are either registered trademarks or trademarks of SYBEX Inc. in the United States and/or other countries.

The CD interface was created using Macromedia Director, COPYRIGHT 1994, 1997–1999 Macromedia Inc. For more information on Macromedia and Macromedia Director, visit <http://www.macromedia.com>.

SYBEX is an independent entity from the Electronic Technicians Association (ETA) and is not affiliated with the ETA in any way. This publication may be used in assisting students to prepare for the FOI and FOT exams, but neither SYBEX nor the ETA warrants that use of this publication will ensure passing the relevant exams.

TRADEMARKS: SYBEX has attempted throughout this book to distinguish proprietary trademarks from descriptive terms by following the capitalization style used by the manufacturer.

The author and publisher have made their best efforts to prepare this book, and the content is based upon final release software whenever possible. Portions of the manuscript may be based upon pre-release versions supplied by software manufacturer(s). The author and the publisher make no representation or warranties of any kind with regard to the completeness or accuracy of the contents herein and accept no liability of any kind including but not limited to performance, merchantability, fitness for any particular purpose, or any losses or damages of any kind caused or alleged to be caused directly or indirectly from this book.

Manufactured in the United States of America

10 9 8 7 6 5 4 3 2 1



To Our Valued Readers:

Thank you for looking to Sybex for your Fiber Optics Installer or Fiber Optics Technician exam prep needs. We at Sybex are proud of our reputation for providing certification candidates with the practical knowledge and skills needed to succeed in the highly competitive marketplace. Certification candidates have come to rely on Sybex for accurate and accessible instruction on today's crucial technologies. Just as the Electronic Technicians Association is committed to establishing measurable standards for certifying individuals working in the demanding field of fiber optics installation and support, Sybex is committed to providing those individuals with the skills needed to meet those standards.

The authors and editors have worked hard to ensure that the *Fiber Optics Installer and Technician Guide* that you hold in your hands is comprehensive, in-depth, and pedagogically sound. We're confident that this book will exceed the demanding standards of the certification marketplace and help you, the FOI and FOT candidate, succeed in your endeavors.

As always, your feedback is important to us. If you believe you've identified an error in the book, please send a detailed e-mail to support@sybex.com. And if you have general comments or suggestions, feel free to drop me a line directly at nedde@sybex.com. At Sybex, we're continually striving to meet the needs of individuals preparing for certification exams.

Good luck in pursuit of your Fiber Optics Installer or Fiber Optics Technician certification!

Neil Edde
Publisher—Certification
Sybex, Inc.

Software License Agreement: Terms and Conditions

The media and/or any online materials accompanying this book that are available now or in the future contain programs and/or text files (the “Software”) to be used in connection with the book. SYBEX hereby grants to you a license to use the Software, subject to the terms that follow. Your purchase, acceptance, or use of the Software will constitute your acceptance of such terms. The Software compilation is the property of SYBEX unless otherwise indicated and is protected by copyright to SYBEX or other copyright owner(s) as indicated in the media files (the “Owner(s)”). You are hereby granted a single-user license to use the Software for your personal, noncommercial use only. You may not reproduce, sell, distribute, publish, circulate, or commercially exploit the Software, or any portion thereof, without the written consent of SYBEX and the specific copyright owner(s) of any component software included on this media.

In the event that the Software or components include specific license requirements or end-user agreements, statements of condition, disclaimers, limitations or warranties (“End-User License”), those End-User Licenses supersede the terms and conditions herein as to that particular Software component. Your purchase, acceptance, or use of the Software will constitute your acceptance of such End-User Licenses.

By purchase, use or acceptance of the Software you further agree to comply with all export laws and regulations of the United States as such laws and regulations may exist from time to time.

Software Support

Components of the supplemental Software and any offers associated with them may be supported by the specific Owner(s) of that material, but they are not supported by SYBEX. Information regarding any available support may be obtained from the Owner(s) using the information provided in the appropriate read.me files or listed elsewhere on the media.

Should the manufacturer(s) or other Owner(s) cease to offer support or decline to honor any offer, SYBEX bears no responsibility. This notice concerning support for the Software is provided for your information only. SYBEX is not the agent or principal of the Owner(s), and SYBEX is in no way responsible for providing any support for the Software, nor is it liable or responsible for any support provided, or not provided, by the Owner(s).

Warranty

SYBEX warrants the enclosed media to be free of physical defects for a period of ninety (90) days after purchase. The Software is not available from SYBEX in any other form or media than that enclosed herein or posted to www.sybex.com. If you discover a defect in the

media during this warranty period, you may obtain a replacement of identical format at no charge by sending the defective media, postage prepaid, with proof of purchase to:

SYBEX Inc.
Product Support Department
1151 Marina Village Parkway
Alameda, CA 94501
Web: <http://www.sybex.com>

After the 90-day period, you can obtain replacement media of identical format by sending us the defective disk, proof of purchase, and a check or money order for \$10, payable to SYBEX.

Disclaimer

SYBEX makes no warranty or representation, either expressed or implied, with respect to the Software or its contents, quality, performance, merchantability, or fitness for a particular purpose. In no event will SYBEX, its distributors, or dealers be liable to you or any other party for direct, indirect, special, incidental, consequential, or other damages arising out of the use of or inability to use the Software or its contents even if advised of the possibility of such damage. In the event that the Software includes an online update feature, SYBEX further disclaims any obligation to provide this feature for any specific duration other than the initial posting.

The exclusion of implied warranties is not permitted by some states. Therefore, the above exclusion may not apply to you. This warranty provides you with specific legal rights; there may be other rights that you may have that vary from state to state. The pricing of the book with the Software by SYBEX reflects the allocation of risk and limitations on liability contained in this agreement of Terms and Conditions.

Shareware Distribution

This Software may contain various programs that are distributed as shareware. Copyright laws apply to both shareware and ordinary commercial software, and the copyright Owner(s) retains all rights. If you try a shareware program and continue using it, you are expected to register it. Individual programs differ on details of trial periods, registration, and payment. Please observe the requirements stated in appropriate files.

Copy Protection

The Software in whole or in part may or may not be copy-protected or encrypted. However, in all cases, reselling or redistributing these files without authorization is expressly forbidden except as specifically provided for by the Owner(s) therein.

To my grandparents, parents, aunts, and uncles for sharing their knowledge and providing encouragement. And to my son Mike for all the encouragement, my son Brandon for taking care of the household and his sister when I couldn't, and to my daughter, Kathryn, for being patient over the last seven months.

—Bill Woodward

For my parents, who always knew I'd get here. For Diane. Yes, I will now get back to working on the house. Thank you for your patience.

—Emile B. Husson

Foreword

This text is intended for students in fiber optics installation, design, and maintenance courses. The 16 chapters encompass the latest techniques, skills, and knowledge required of the technologists who are now rewiring the business and residential worlds with high-speed broadband optical fiber. While only months ago, some telecommunications industry observers were predicting that copper and fiber were soon to be replaced in the main by wireless technologies, that has proven not to be the case. Instead, the major telephone and communications companies have set in motion some of the industry's largest and most expensive construction projects by initiating new fiber networks. The cable, telephone, and Internet technology companies have expanded their systems worldwide and have driven fiber cabling from the trunk lines to the curb, to the premises, and into the home. Local and wide area networks are heavily fibered. Ships, aircraft, and automobiles now include fiber transmission media.

The Electronics Technicians Association International began the FOI certification program in 1996. Nearly 20,000 workers now hold the Certified Fiber Optics Installer (CFOI) or Technician (CFOT) credential. It is a rare day when one hears of a certified fiber professional who does not hold a well-paying job. Telecommunications companies are hiring workers with fiber skills and knowledge and are training existing employees to handle the growing projected future needs.

During the last decade, the training schools have used one or more of the existing study textbooks in their courses. Suppliers, training institutions, and technical publishers have produced several fine books that have been critical in helping students understand the principles and skills needed to safely and correctly install cable infrastructure. This book is an outgrowth of previous efforts to produce a comprehensive study guide that includes virtually everything needed to become a fiber professional.

The primary author, William Woodward, P.E., CFOT, has taught fiber courses at commercial training schools as well as in industrial settings. Not only does he have a background in copper, coax, and fiber cabling, but his life's work has been in electronics communications. This includes military and civilian research, development, and quality control experiences. He has served as the Cabling Division Committee Chairman for ETA-I for three years and has been a major part of the certification examination development teams in the Fiber, Copper, Telecommunications, FDR Line Sweeping, and Wireless Communications areas. Few others have the extensive background directly related to fiber, as well as related technologies, that Mr. Woodward has.

Both students and cabling instructors will find this guide invaluable. It not only covers the theoretical, but digs into the practical hands-on practices needed by fiber installers and technicians. It has the most extensive chapter ever written on the functions and usage of all the test equipment now being used by fiber technicians. It is heavy on standards recognition and is an excellent reference manual for cabling professionals. Yes, it is a lengthy textbook, but once you start your studies, you will quickly discover that the easy-to-understand style make it fun, rather than a chore, to learn all about fiber cabling.

Lastly, this text prepares you to pass the ETA CFOI and CFOT certification exams. As you reach the end of the book, the practice exams, and perhaps the end of your classroom training, you will know that you are ready to become a Certified Fiber Optics professional.

—Dick Glass, CETsr

President, Electronics Technicians Association, International

President, NCEE, National Coalition for Electronics Education

Acknowledgments

Writing a book is a team effort that takes a dedicated group of professionals. This is my first book and I am very fortunate to have been able to work with a team of talented and dedicated individuals. The talented staff at Sybex, my coworkers at the ECPI College of Technology and at WR Systems, and my friends and mentors have made this possible.

First, I'd like to especially thank Sybex for giving me the opportunity to write this book. I can't express how grateful I am that you took a chance on a new venue and on me. Special thanks to Maureen Adams for the outstanding job you did in guiding me through this project from start to finish and keeping the team focused. Thanks to Mae Lum for putting everything together and for being so patient as the project came to a close. Thanks to Suzanne Goraj for the great job you did in editing the text. My grammar has improved tremendously, thanks to you. Thanks also to Margaret Rowlands for creating the eye-catching book cover. And thanks to Charlie Husson for the outstanding job with the technical edits. You are an exceptional engineer and a great mentor.

Thank you, Lori Skowronski, for your significant contribution to Chapter 12. You spent many hours away from your family to write this chapter and to keep the book on track while I was recovering from surgery. You are a great friend.

I'd like to thank Karl Kuhn for his outstanding job on providing the many illustrations, and Teresa Jones for her outstanding job on many of the tables. Thanks to John Jeffcoat and Chuck Casbeer from the ECPI College of Technology for providing the test equipment and the industry standards required for this effort, not to mention all of Chuck's help in reviewing the text and providing feedback. Special thanks to Marcus Friedman from the ECPI College of Technology for believing in me and giving many opportunities in my career in fiber optics.

Many companies provided technical information, equipment, and photographs. Special thanks to Mark Roehm and Mark Joseph from Stran Technologies, Scott Kale from Norfolk Wire, Christine Pons from OptiConcepts, Bill Troemel from Aerotech, and Dave Edwards from WR Systems.

I really need to thank my coauthor Emile Husson. Emile did a fantastic job and was an inspiration to work with. He spent many sleepless nights putting this manuscript together. His many talents and professionalism are greatly appreciated.

For many years, Dick Glass has been a friend, mentor, and coworker. Dick has spent many hours guiding me through this project and my career. I feel very blessed to have met Dick and greatly appreciate his guidance over the years and assistance with this project.

Thanks to the host of people behind-the-scenes that I did not mention for all your efforts to make this book the best that it can be.

Last but not least, thank you to my children, Mike, Brandon, and Kathryn; the love of my life, Susan; and her sons, Eric and Nathan, for your patience, inspiration, encouragement, and prayers. I am the luckiest man alive to have all of you in my life.

—Bill Woodward

Contents at a Glance

<i>Introduction</i>	<i>xix</i>
<i>Assessment Test</i>	<i>xxxi</i>
Chapter 1	History of Fiber Optics 1
Chapter 2	Principles of Fiber Optic Transmission 13
Chapter 3	Basic Principles of Light 41
Chapter 4	Optical Fiber Construction and Theory 61
Chapter 5	Optical Fiber Characteristics 85
Chapter 6	Safety 111
Chapter 7	Fiber Optic Cables 129
Chapter 8	Splicing 163
Chapter 9	Connectors 183
Chapter 10	Fiber Optic Light Sources 219
Chapter 11	Fiber Optic Detectors and Receivers 251
Chapter 12	Passive Components and Multiplexers 271
Chapter 13	Cable Installation and Hardware 299
Chapter 14	Fiber Optic System Design Considerations 321
Chapter 15	Test Equipment and Link/Cable Testing 355
Chapter 16	Link/Cable Troubleshooting 401
Glossary	427
<i>Index</i>	<i>443</i>

Contents

<i>Introduction</i>	<i>xix</i>
<i>Assessment Test</i>	<i>xxxi</i>
Chapter 1	History of Fiber Optics 1
Evolution of Light in Communication	2
Early Forms of Light Communication	2
The Quest for Data Transmission	3
Evolution of Optical Fiber Manufacturing Technology	4
Controlling the Course of Light	4
Extending Fiber's Reach	6
Evolution of Optical Fiber Integration and Application	7
Summary	8
Exam Essentials	8
Review Questions	10
Answers to Review Questions	11
Chapter 2	Principles of Fiber Optic Transmission 13
The Fiber Optic Link	14
Transmitter	15
Receiver	15
Optical Fibers	15
Connectors	17
Amplitude Modulation	17
Analog Transmission	18
Digital Data Transmission	19
Analog Data Transmission vs. Digital Data Transmission	20
Analog to Digital (A/D) Conversion	21
Sample Rate	21
Quantizing Error	22
Digital to Analog (D/A) Conversion	23
Pulse Code Modulation (PCM)	25
Multiplexing	26
Decibels (dB)	26
The Rules of Thumb	31
Absolute Power Gains and Losses	32
Summary	34
Exam Essentials	34
Review Questions	36
Answers to Review Questions	39

Chapter 3	Basic Principles of Light	41
	Light as Electromagnetic Energy	42
	The Electromagnetic Spectrum	45
	Refraction	47
	What Causes Refraction?	48
	Total Internal Reflection	51
	Fresnel Reflections	54
	Summary	55
	Exam Essentials	56
	Review Questions	57
	Answers to Review Questions	60
Chapter 4	Optical Fiber Construction and Theory	61
	Optical Fiber Components	62
	Core	63
	Cladding	63
	Coating	63
	Standards	64
	Materials	64
	Tensile Strength	67
	Manufacturing Optical Fiber	68
	Modified Chemical Vapor Deposition (MCVD)	69
	Outside Vapor Deposition (OVD)	70
	Vapor Axial Deposition (VAD)	70
	Plasma Chemical Vapor Deposition (PCVD)	70
	Modes	71
	Refractive Index Profiles	73
	Dispersion-Shifted Fiber	76
	Summary	78
	Exam Essentials	78
	Review Questions	79
	Answers to Review Questions	82
Chapter 5	Optical Fiber Characteristics	85
	It All Adds Up	86
	Dispersion	87
	Modal Dispersion	88
	Material Dispersion	89
	Waveguide Dispersion	89
	Chromatic Dispersion	90
	Polarization-Mode Dispersion	93
	How Dispersion Affects Bandwidth	94
	Attenuation	94
	Absorption	95

	Scattering	96
	Total Attenuation	97
	Numerical Aperture	98
	Bending Losses	100
	Microbends	100
	Macrobends	100
	Equilibrium Mode Distribution	101
	Fiber Specifications	102
	Summary	103
	Exam Essentials	103
	Review Questions	104
	Answers to Review Questions	108
Chapter 6	Safety	111
	Basic Safety	112
	Engineering Controls	112
	Personal Protective Equipment (PPE)	113
	Good Work Habits	113
	Light Sources	114
	Laser Service Groups	114
	Laser Safety	115
	Handling Fiber	117
	Chemicals	118
	Isopropyl Alcohol	119
	Solvents	119
	Anaerobic Epoxy	120
	Site Safety	120
	Electrical	120
	Ladders	121
	Trenches	122
	Emergencies	122
	Injury	122
	Chemical Exposure	122
	Fire	123
	Summary	123
	Exam Essentials	124
	Review Questions	125
	Answer to Review Questions	127
Chapter 7	Fiber Optic Cables	129
	Basic Cable	130
	Cable Components	131
	Buffer	131
	Strength Members	134
	Jacket	136

	Cable Types	137
	Cordage	138
	Distribution Cable	139
	Breakout Cable	139
	Armored Cable	140
	Messenger Cable	140
	Ribbon Cable	141
	Submarine Cable	143
	Hybrid Cable	145
	Composite Cable	145
	Cable Duty Specifications	145
	Cable Termination Methods	146
	Fanout Kit	146
	Breakout Kit	147
	Blown-in Fiber	148
	NEC Standards for Optical Fiber	149
	NEC-Listed Cable Types	149
	NEC-Listed Raceways	151
	Cable Markings and Codes	152
	External Markings	152
	Color Codes	152
	Bend Radius Specifications	155
	Summary	156
	Exam Essentials	156
	Review Questions	157
	Answers to Review Questions	161
Chapter 8	Splicing	163
	Putting It Together	164
	Intrinsic Factors	164
	Extrinsic Factors	167
	Splicing Equipment	170
	Mechanical Splicers	170
	Fusion Splicers	171
	Splicing Procedures	173
	Mechanical Splicing Procedure	173
	Fusion Splicing Procedure	174
	Splice Requirements	176
	Summary	178
	Exam Essentials	178
	Review Questions	179
	Answers to Review Questions	181

Chapter 9	Connectors	183
	The Fiber Optic Connector	184
	Connector Performance	187
	Roughness	187
	Geometry	187
	Connector Types	188
	Single-Fiber Connectors	189
	Multiple-Fiber Connectors	192
	Connector Termination	197
	Epoxy	197
	Tools	199
	Assembling the Connector	202
	Endface Examination	208
	Connector Performance	212
	Summary	212
	Exam Essentials	213
	Review Questions	214
	Answers to Review Questions	217
Chapter 10	Fiber Optic Light Sources	219
	Semiconductor Light Sources	220
	LED Sources	220
	Laser Sources	222
	Light Source Performance Characteristics	223
	Output Pattern	223
	Source Wavelengths	226
	Source Spectral Output	227
	Source Output Power	229
	Source Modulation Speed	230
	Transmitter Performance Characteristics	231
	LED Transmitter Performance Characteristics	231
	Laser Transmitter Performance Characteristics	235
	Light Source Safety	240
	Classifications	241
	Safety Handling Precautions	242
	Summary	242
	Exam Essentials	242
	Review Questions	244
	Answers to Review Questions	248
Chapter 11	Fiber Optic Detectors and Receivers	251
	Photodiode Fundamentals	252
	PIN Photodiode	253

Avalanche Photodiode	254
Responsivity	254
Quantum Efficiency	255
Switching Speed	256
Fiber Optic Receiver	256
Receptacle	256
Optical Subassembly	256
Electrical Subassembly	258
Receiver Performance Characteristics	258
Dynamic Range	259
Operating Wavelength	259
LED Receiver Performance Characteristics	259
Laser Receiver Performance Characteristics	262
Summary	265
Exam Essentials	266
Review Questions	267
Answers to Review Questions	269
 Chapter 12	 Passive Components and Multiplexers
	271
Couplers	272
The Tee Coupler	273
The Star Coupler	276
Optical Switches	278
Optomechanical	279
Thermo-Optic	280
Electro-Optic	280
Optical Attenuators	280
Gap-Loss Principle	281
Absorptive Principle	282
Reflective Principle	283
Fixed Attenuators	283
Stepwise Variable Attenuators	284
Continuously Variable Attenuators	284
Optical Isolator	284
Polarized	284
Magnetic	286
Wavelength Division Multiplexing	286
Optical Amplifier	291
Optical Filter	293
Summary	294
Exam Essentials	295
Review Questions	296
Answers to Review Questions	298

Chapter 13	Cable Installation and Hardware	299
	Installation Specifications	300
	Minimum Bend Radius	300
	Maximum Tensile Rating	301
	Installation Hardware	302
	Pulling Eye	303
	Pullbox	303
	Splice Enclosures	304
	Patch Panels	305
	Installation Methods	306
	Tray and Duct	306
	Conduit	308
	Direct Burial	309
	Aerial	309
	Blown Fiber	310
	Electrical Safety	310
	Hardware Management	312
	Cleanliness	312
	Organization	312
	Labeling	313
	Documentation	313
	Labeling Requirements	313
	Summary	314
	Exam Essentials	315
	Review Questions	316
	Answers to Review Questions	319
Chapter 14	Fiber Optic System Design Considerations	321
	Basic Fiber Optic System Design Considerations	322
	The Advantages of Optical Fiber over Copper	323
	Bandwidth	323
	Attenuation	325
	Electromagnetic Immunity	328
	Size and Weight	329
	Security	331
	Safety	331
	Link Performance Analysis	332
	Cable Transmission Performance	333
	Splice and Connector Performance	333
	Power Budget	335
	Summary	346
	Exam Essentials	346
	Review Questions	349
	Answers to Review Questions	353

Chapter 15	Test Equipment and Link/Cable Testing	355
	Continuity Tester	356
	Visible Fault Locator	359
	Fiber Identifier	361
	Optical Return Loss Test Set	364
	Light Source and Optical Power Meter	365
	Multimode	365
	Single-Mode	367
	Patch Cord	368
	Test Jumper	368
	Mode Filter	369
	TIA/EIA-526-14A Optical Loss Measurement	372
	Method A	373
	Method B	374
	Method C	375
	Patch Cord Optical Power Loss Measurement	376
	OTDR	376
	OTDR Theory	377
	OTDR Display	381
	OTDR Setup	383
	Cable Plant Test Setup	386
	Testing and Trace Analysis	388
	Documentation	396
	Summary	396
	Exam Essentials	396
	Review Questions	398
	Answers to Review Questions	400
Chapter 16	Link/Cable Troubleshooting	401
	Connector Inspection	402
	Connector Endface Evaluation	403
	Continuity Tester Fault Location Techniques	407
	Visible Fault Locator	411
	Fiber Identifier	414
	OTDR Fault Location Techniques	416
	Restoration Practices	420
	Summary	422
	Exam Essentials	422
	Review Questions	424
	Answers to Review Questions	426
	Glossary	427
	<i>Index</i>	443

Introduction

This book focuses on building a solid foundation in fiber optic theory. In addition, it describes in great detail fiber optic cable technology, connectorization, splicing, and passive devices. It examines the electronic technology built into fiber optic receivers, transmitters, and test equipment. It also incorporates many of the current industry standards pertaining to optical fiber, connector, splice, and network performance.

This book is an excellent reference for anyone currently working in fiber optics or for the person who just wants to start learning about fiber optics. The book covers in detail all of the competencies of the Electronics Technicians Association International (ETA) fiber optic installer (FOI) and fiber optic technician (FOT) certification.

ETA's FOI and FOT Programs

The ETA's FOI and FOT programs are the most comprehensive in the industry. Each program requires the student to attend an ETA-approved training school. Each student must achieve a score of 75% or greater on the written exam and satisfactorily complete all the hands-on requirements. Persons interested in obtaining ETA FOI or FOT certification can visit the ETA's website at www.eta-i.org and get the most up-to-date information on the program and a list of approved training schools.

The ETA FOI certification requires no prerequisite. The FOI program is designed for anyone who is interested in learning how to become a fiber optic installer. The FOI certification is recommended as a prerequisite for the FOT certification. The FOT certification is recommended for anyone who wants to learn how to test a fiber optic link to the current industry standards and how to troubleshoot. Fiber optic certification demonstrates to your employer that you have the knowledge and hands-on skills required to install, test, and troubleshoot fiber optic links and systems. With the push to bring fiber optics to every home, these skills are highly sought after.

What Does This Book Cover?

We've put this book together to provide you with a solid foundation in fiber optic technologies and practices. The book is loaded with valuable information, including the following elements:

Assessment test Directly following this introduction is an assessment test that you should take. It is designed to help you determine how much you already know. Each question is tied to a topic discussed in the book. Using the results of the assessment test, you can figure out the areas where you need to focus your study. Of course, we do recommend that you read the entire book.

Objective-by-objective coverage of the topics you need to know Each chapter lists the exam objectives covered in that chapter, followed by detailed discussion of each objective. Each objective meets or exceeds an ETA FOI or FOT competency.

Chapter exercises In each chapter, you'll find exercises designed to give you the important hands-on experience that is critical for your exam preparation. The exercises support the topics of the chapter, and they walk you through the steps necessary to perform a particular function.

Real World Scenarios Because reading a book isn't enough for you to learn how to apply these topics in your everyday duties, we have provided Real World Scenarios in special sidebars. These explain when and why a particular solution would make sense, in a working environment that you'd actually encounter.

Exam Essentials To highlight what you learn, you'll find a list of Exam Essentials at the end of each chapter. The Exam Essentials section briefly highlights the topics that need your particular attention as you prepare for the FOI or FOT exam.

Review questions, complete with detailed explanations Each chapter is followed by a set of review questions that test what you learned in the chapter. The questions are written with the exam in mind, meaning that they are designed to have the same look and feel as what you'll see on the exam.

Glossary Throughout each chapter, you will be introduced to important terms and concepts that you will need to know for the FOI or FOT exam. These terms appear in *italics* within the chapters. At the end of the book, a detailed glossary gives definitions for these terms, as well as other general terms you should know.

How Do You Use This Book?

This book provides a solid foundation for the serious effort of preparing for the ETA FOI or FOT certification exam. To best benefit from this book, you might want to use the following study method:

1. Take the assessment test to identify your weak areas.
2. Study each chapter carefully. Do your best to fully understand the information.
3. Read over the Real World Scenarios to improve your understanding of how to use what you learn in the book.
4. Study the Exam Essentials to make sure that you are familiar with the areas you need to focus on.
5. Answer the review questions at the end of each chapter. If you prefer to answer the questions in a timed and graded format, install the test engine from the book's companion CD and answer the chapter questions there instead of from the book.
6. Take note of the questions you did not understand, and study the corresponding sections of the book again.
7. Go back over the Exam Essentials.
8. Go through this book's other training resources, which are included on the book's accompanying CD. These include electronic flashcards, the electronic version of the assessment test and chapter review questions (try taking them by objective), and two bonus exams.

To learn all the material required to pass the exam, you will need to study regularly and with discipline before and while attending an ETA-approved training course. Try to set aside the

same time every day to study, and select a comfortable and quiet place in which to do it. Do not wait until the break before the exam to start studying. Remember: if you have any questions about the material you are learning, ask your instructor.

What's on the CD?

This book's companion CD includes numerous simulations, bonus exams, and flashcards to help you study for the exam. We have also included the complete contents of the book in electronic form. The CD's resources are described here:

The Sybex test engine preparation software These are a collection of multiple-choice questions that will help you prepare for your FOI and FOT exams. You'll find the following:

- Two bonus exams designed to simulate the actual live exam.
- All the chapter review questions from the book. You can review questions by chapter or by objective, or you can take a random test.
- The assessment test.

Electronic flashcards for PCs and Palm devices The "flashcard" style of question is an effective way to quickly and efficiently test your understanding of the fundamental concepts covered in the exam. The Sybex flashcards set consists of 150 questions presented in a special engine that can run either on your PC or on your hand-held device.

***Fiber Optics Installer and Technician Guide* in PDF** Many people like the convenience of being able to carry their book on a CD. They also like being able to search the text via computer to find specific information quickly and easily. For these reasons, the entire contents of this book are supplied on the companion CD in PDF. We've also included Adobe Acrobat Reader, which provides the interface for the PDF contents as well as the search capabilities.

ETA-Approved Certified Fiber Optics Installer Training Schools

These training schools are listed in ZIP code order.

Telecommunications Training Academy of New England
 32 Boulevard Road
 Wellesley, MA 02481
 617-784-1844
 Barry McLaughlin, RCDD: barry@barrymclaughlin.com
www.ttane.com

Briarcliffe College
 1055 Stewart Avenue
 Bethpage, NY 11714
 516-918-3700
 Nancy Klein: nklein@bcl.edu

New Horizons Computer Learning Center of Long Island
6080 Jericho Turnpike
Commack, NY 11725
631-499-7929, ext. 127
Stuart Tenzer: stuart@nhli.com
www.nhli.com

Computer Education Services Corp.
920 Albany Shaker Road
Latham, NY 12110
860-243-1000, ext. 191
Ralph Fraley: rfraley@computeredservices.com
860-243-1000, ext. 174
Holly Banak: hbanak@computeredservices.com

Pittsburgh Job Corps Center
341 Third Street
Pitcairn, PA 15140
412-401-0846
Edward Parady, CET: eeepar@aol.com

TBK Technologies
RD#1, Box 546
Adrian, PA 16210
412-600-8185
Robert Keys, FOI: rkeys@teksystems.com

Philadelphia Wireless Technical Institute
1533 Pine Street
Philadelphia, PA 19102
215-928-9960
Richard Agard, FOI: ragard@aol.com

Quality Telecommunications Services, Inc.
5410 Indianhead Highway
Oxon Hill, MD 20745-2021
301-686-0500
Bennie Davis: info@hqtsi.com

Howard Community College
 10901 Little Patuxent Parkway
 Columbia, MD 21044
 410-772-4123 (Dave Rader)
 410-772-4856 (Admissions)
 Dave Rader: drader@howardcc.edu

Honeywell Technology Solutions, Inc.
 7000 Columbia Gateway Drive
 P.O. Box 5555
 Columbia, MD 21046
 410-964-7274
 Jeffry Miller, FOI

IES Training Facility
 220 8th Avenue N.W.
 Glen Burnie, MD 21061
 410-760-2990
 Craig Jones: cjones@iestraining.com

Northern Virginia Community College
 7630 Little River Turnpike, Suite 600
 Annandale, VA 22003
 703-323-3102
 Rickie Harris: riharris@nvcc.edu

Priest Electronics, Inc.
 1525 Technology Drive
 Chesapeake, VA 23320
 800-777-3532
 John Hogan: Haggard23434@yahoo.com
 Ted Green, FOI: ted@priestelectronics.com

Advanced Technology Center
 1800 College Crescent
 Virginia Beach, VA 23453
 757-468-8960
 Robert Stover, FOI: rstover@vbcps.k12.va.us
 www.vbatc.com

ECPI
5555 Greenwich Road
Virginia Beach, VA 23462
757-858-6000
Chuck Casbeer, FOI: ccasbeer@ecpi.edu
Bill Woodward, FOI: wwoodwar@wrsystems.com

KITCO Fiber Optics
5269 Cleveland Street, Suite 109
Virginia Beach, VA 23462
888-548-2636
Dan Morris: dmorris@kitcofo.com

WR Systems
2500 Alameda Avenue, Suite 214
Norfolk, VA 23513
757-858-6000, ext. 606
William Woodward, FOI: wwoodwar@wrsystems.com

Yeager Career Center
10 Marland Avenue
Hamlin, WV 25523
304-824-5449
Gregory A Gosnay: ggosnay@access.k12.wv.us

Calhoun Community College
6250 U.S. 31 N.
Tanner, AL 35671
256-306-2972
Sherman Banks: smb@calhoun.edu

Communications Apprenticeship & Training
1400 E. Schaaf Road
Cleveland, OH 44131
216-635-1313
Richard Bowers: rickcatcwa@hotmail.com

Midwest Telecom Training, FiberCamp
2518 Waller Drive
Washington, IN 47501
812-254-3488
Kent Norris: kent@fibercamp.com

Diversified Wiring and Cable, Inc.
 6250 Fifteen Mile Road
 Sterling Heights, MI 48312
 586-264-6500, ext. 245
 Al Jankowski, FOI: a.jankowski@dw-c.com

Breakthru Training Solutions
 8608 N. Richmond Avenue, 1st Floor
 Kansas City, MO 64157
 816-584-8177
 Christopher Kehoe: ckehoe@btstraining.com
 www.BTStraining.com

Central Community College
 3134 W. Highway, Suite 34
 Grand Island, NE 68802-4903
 308-398-7490
 Tim Ziller: tziller@cccneb.edu

Louisiana Technical College: Slidell Campus
 1000 Canulette Road
 Slidell, LA 70458
 985-646-6430, ext. 128
 William L. Little, FOI: wlittle@theltc.net

Elayn Hunt Correctional Center
 Education Department
 P.O. Box 174
 St. Gabriel, LA 70776-0174
 225-319-4266
 Madeline McCaleb: eeducation@corrections.state.la.us

Texas State Technical College
 3801 Campus Drive
 Waco, TX 76705
 Sandra Herinckx, FOI: sherinckx@tstc.edu

Cricket Institute of Technology
 3727 Pinemont Drive
 Houston, TX 77018
 713-682-7352
 Michael Brittain, FOI: Michael@cricketfiber.com

The Institute of Robotics
957 NASA Road 1, Suite 261
Houston, TX 77058
281-535-3030
Scarlet Black: rov@irov.com

Montgomery College
102 Longview Drive
Conroe, TX 77301
936-271-6033
David Boden, FOI: boden@nhmccd.edu
www.mc.nhmccd.edu

Texas A&M Riverside Campus
Telecom Training Division
301 Tarrow, Suite 119
College Station, TX 77843-8000
800-645-0686
Joe Smith, FOI: joesmith@tamu.edu

Rocky Mountain Technical Institute
6229 S. Krameria
Greenwood Village, CO 80111
720-200-0784
Tom Janca, CETsr, FOI: trjanca@lucent.com

Casper College
125 College Drive
Casper, WY 82601
307-268-2521
David Arndt, FOI: darndt@caspercollege.edu

FNT Fiber Network Training
3908 E. Broadway, Suite 100
Phoenix, AZ 85040
866-818-8050
Jeffrey Dominique: jeff@f-n-t.com
www.f-n-t.com

Southern Arizona Institute for Advanced Technology
3000 East Valencia, Suite 190
Tucson, AZ 85706
520-573-7399 ext. 109
Kimberly Nichols: knichols@saiat.org
www.saiat.org

Integrated Training Center
4801 Hardware Avenue N.E.
Albuquerque, NM 87109
877-883-4130
Melody Dudley: Melodyd@itc4u.com
www.itc4u.com

JM Fiber Optics, Inc.
6251 Schaefer Avenue, Suite D
Chino, CA 91710-9065
909-628-3445
Kenneth Rivera: krivera@jmfiberoptics.com
www.jmfiberoptics.com

Advanced Training Associates
1900 Joe Crosson Drive, Suite C
El Cajon, CA 92020-1236
619-596-2766
Jose Villaman: tony@advancedtraining.edu

Cable Links Consulting/West Hills College
5100 N. 6th Street, Suite 174
Fresno, CA 93710
877-995-2555
559-225-2555
Sandy Slumberger: slummyclc02@sc.com

Technical Training Seminars
P.O. Box 596
Concord, CA 94522
510-331-1124
Joseph I. Pappaly, FOI: molu@attbi.com

Aviation and Electronic Schools of America
P.O. Box 1810
201 South Railroad Street
Colfax, CA 95713
800-345-8466
Evan Neilsen: eneilsen@aesa.com

CORADI Telecom Training Center
184 Lizama Street
Barrigada, Guam 96913
671-734-6897
Al Alicto, FOI: coradia@netpci.com

Guam Community College
P.O. Box 23069
Barrigada, Guam 96921
671-735-5610
John Limtiaco, FOI: jlimtiaco@guamcc.net

The Light Brigade
7691 S. 180th Street
Kent, WA 98032
800-451-7128
Larry Johnson: larry@lightbrigade.com
www.lightbrigade.com

Renton Technical College
3701 N.E. 10th Street
Renton, WA 98056
425-235-2352
John Cambroto: jcambroto@rtc.ctc.edu

Vector Technology Institute
35a Eastwood Park Road
Kingston, Jamaica KGN10
876-929-3434
Rohan Morris: rohmor@cwjamaica.com
www.vti-institute.com

Approved Military Schools

These training schools are listed in ZIP code order.

Fleet Training Center Norfolk
 9459 Bainbridge
 CCMM/N752/Fiber Optics
 Norfolk, VA 23511
 757-444-1262 ext. 3041
 Anthony Corey, FOI: ET2-anthony.m.corey@cnet.navy.mil

Sheppard Air Force Base
 364th TRS (Fiber Optics)
 Building 1950
 Wichita Falls, TX 76311
 940-676-5541
 Ronald Cook: Ronald.Cook@Sheppard.AF.Mil
 MSgt. Wayne Siverling: Wayne.Siverling@Sheppard.AF.Mil

Goodfellow AFB Air Education and Training Command
 316th TRS/DOBB
 17th Training Wing
 156 Marauder Street
 Goodfellow AFB, TX 76908-5000
 325-654-4535
 James Beam, FOI: james.beam@goodfellow.af.mil

Fleet Training Center San Diego
 3975 Norman Scott Road, Suite 1
 Code N7623/Fiber Optics
 San Diego, CA 92136-5588
 619-556-7059

Marine Corps/Communications-Electronics
 Marine Corps Air Ground
 Combat Center
 Box 788251
 29 Palms, CA 92278-8251
 760-830-5028
 760-830-6831
 John A. Walters: waltersja@29palms.usmc.mil

United States Coast Guard
TRACENPET
599 Tomales Road
Petaluma, CA 94952
TT1 Brian K. Bonner: Bbonner@d11.uscg.mil

Contacts and Resources

To find out more about the ETA's Fiber Optic Installer and Fiber Optic Technician certification program and approved training institutes, visit them on the Web at www.eta-i.org.

About the Authors and Technical Editor

Bill Woodward is a senior electrical engineer with WR Systems, an engineering services company. While at WR Systems, Bill introduced the industry's first fiber optic polishing cloth, which is marketed by WR Systems under the name Innovative Fiber Technologies. Bill has been teaching fiber optics and other technical courses for the ECPI College of Technology since 1992.

Bill is licensed in the Commonwealth of Virginia as a professional electrical engineer. He is currently serving his second term as chairman of the Electronics Technicians Association International (ETA). In addition, he is the chairman of the ETA's fiber optic committee that is responsible for the fiber optic installer, technician, and designer certifications. Bill is working with the Society of Automotive Engineers (SAE) on ARP5602, a guideline for aerospace platform fiber optic training and awareness education. Bill lives with his son Brandon and daughter Kathryn in Virginia Beach, Virginia.

Emile B. Husson is a full-time electronic media consultant in technical training covering areas including electronic control systems, software, safety, and inertial navigation. He has won industry awards as a producer of technical training videos and interactive courseware and has specialized in electronic training media content development since 1986. Emile and his wife, Diane, make their home in Virginia.

Charles Husson is a retired electrical engineer from NASA. Charlie was at the forefront of many of the developments in optical fiber and the semiconductor technology used in fiber optic light sources and detectors. Charlie is still sharing his knowledge with his students at the ECPI College of Technology and his coworkers at WR Systems.

Assessment Test

1. In which decade did the loss for 1 km of optical fiber fall below 20 dB?
 - A. 1960
 - B. 1970
 - C. 1980
 - D. 1990
2. The component in a fiber optic system that converts light energy into electrical energy is the _____.
 - A. Transmitter
 - B. Receiver
 - C. Optical fiber
 - D. Coupler
3. A 50% reduction in signal strength is a loss of _____.
 - A. 3 dB
 - B. 3 dBm
 - C. 7 dB
 - D. 7 dBm
4. The velocity of light traveling through a medium with a refractive index of _____ has a velocity of approximately 225,000 km/s.
 - A. 1.15
 - B. 1.25
 - C. 1.33
 - D. 1.50
5. The _____ is the layer of glass that surrounds the core of an optical fiber.
 - A. Buffer
 - B. Jacket
 - C. Coating
 - D. Cladding
6. A path for light through an optical fiber is called a _____.
 - A. Mode
 - B. Route
 - C. Highway
 - D. Multimode

7. Modal _____ in an optical fiber is when light rays take different paths through the optical fiber.
- A. Mixing
 - B. Attenuation
 - C. Gain
 - D. Dispersion
8. The laser light sources used in fiber optic communication have a longer _____ than visible light.
- A. Frequency
 - B. Wavelength
 - C. Amplitude
 - D. Pulse
9. Article _____ of the National Electric Code covers optical fiber cables and raceways.
- A. 250
 - B. 660
 - C. 770
 - D. 810
10. When the NA of the transmitting optical fiber is _____ than the receiving optical fiber, a _____ occurs.
- A. Less, gain
 - B. Greater, loss
 - C. Less, loss
 - D. Greater, gain
11. The connector with a PC finish has a(n) _____ endface geometry.
- A. Flat
 - B. Rough
 - C. Angled
 - D. Rounded
12. The polishing puck is designed to keep a connector ferrule with a UPC finish _____ to the polishing surface.
- A. Adjacent
 - B. Parallel
 - C. Perpendicular
 - D. Below

13. Per TIA/EIA-568-B, a multimode optical fiber has a _____ bandwidth at _____ than at _____.
- A. Lower, 1300 nm, 850 nm
 - B. Higher, 850 nm, 1300 nm
 - C. Lower, 1310 nm, 850 nm
 - D. Higher, 1300 nm, 850 nm
14. _____ is typically used to define the error generation of a digital fiber optic receiver.
- A. BER
 - B. BRE
 - C. RER
 - D. FORER
15. A tree coupler has _____ port(s) and _____ port(s).
- A. Many input, one output
 - B. One input, many output
 - C. Many input, many output
 - D. One input, one output
16. Fiber optic cable should always be pulled by the _____.
- A. Jacket
 - B. Buffer
 - C. Coating
 - D. Strength member
17. Optical fiber is immune to the effects of _____ because it's a dielectric.
- A. EMI
 - B. EIM
 - C. CAT
 - D. MMF
18. The VFL typically uses a _____ 1 mW laser to illuminate breaks in an optical fiber.
- A. 400 nm
 - B. 450 nm
 - C. 650 nm
 - D. 850 nm

19. A _____ splice does not produce a back reflection on the OTDR trace.
- A. Fusion
 - B. Epoxy
 - C. Mechanical
 - D. Cured
20. Defects in the outer _____ of a _____ connector are typically acceptable.
- A. Core, multimode
 - B. Cladding, multimode
 - C. Core, single-mode
 - D. Cladding, single-mode

Answers to Assessment Test

1. B. In 1970, the first optical fiber with a loss of less than 20 dB was produced. See Chapter 1 for more information.
2. B. The fiber optic receiver converts light energy from the optical fiber into electrical energy. See Chapter 2 for more information.
3. A. Reducing signal strength by one half is a 3 dB loss. See Chapter 2 for more information.
4. C. Light travels through a medium at approximately 225,000 km/s when the refractive index of the material is 1.33. See Chapter 3 for more information.
5. D. A glass optical fiber is made up of the core, cladding, and coating. The cladding surrounds the core and the coating surrounds the cladding. See Chapter 4 for more information.
6. A. In fiber optics, “mode” describes the propagation of light through an optical fiber. See Chapter 4 for more information.
7. D. Light rays taking different paths through an optical fiber arrive at the end of the optical fiber at different times because of modal dispersion. See Chapter 5 for more information.
8. B. The infrared laser light sources used in fiber optic communication have a longer wavelength than visible light. See Chapter 3 for more information.
9. C. Article 770 of the National Electric Code defines requirements for optical fiber and raceways. See Chapter 7 for more information.
10. B. A loss from a NA mismatch occurs when the transmitting optical fiber has a greater NA than the receiving optical fiber. See Chapter 8 for more information.
11. D. The endface geometry of a connector ferrule with a PC finish is rounded. See Chapter 9 for more information.
12. C. The polishing puck is designed to keep the ferrule of a UPC connector perpendicular with the polishing surface. See Chapter 9 for more information.
13. D. TIA/EIA-568-B defines the bandwidth of multimode optical fiber at only 850 nm and 1300 nm. Multimode optical fiber has the greatest bandwidth at 1300 nm. See Chapter 10 for more information.
14. A. The error generation of a digital fiber optic receiver is typically described by the receiver’s bit error rate, or BER. See Chapter 11 for more information.
15. B. A tree coupler has only one input port and three or more output ports. See Chapter 12 for more information.
16. D. When installing fiber optic cable, the strength member should always be used to pull it. See Chapter 13 for more information.

- 17. A. Because optical fiber is a dielectric, it is immune to the effects of electromagnetic interference, or EMI. See Chapter 14 for more information.
- 18. C. The VFL typically uses a red laser with an operating wavelength around 650 nm. See Chapter 15 for more information.
- 19. A. Unlike the mechanical splice, the fusion splice does not produce a back reflection on the OTDR trace. See Chapter 15 for more information.
- 20. B. As a general rule of thumb, defects in the outer cladding of a multimode connector are typically acceptable. See Chapter 16 for more information.



Chapter

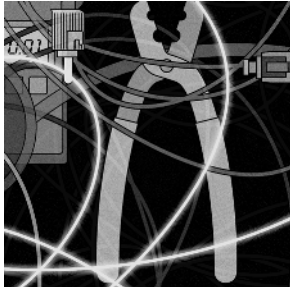
1

History of Fiber Optics

OBJECTIVES COVERED IN THIS CHAPTER:

✓ History of Fiber Optics

- Trace the evolution of light in communication.
- Understand the evolution of optical fiber manufacturing technology.
- Track the evolution of optical fiber integration and application.



Like many technological achievements, fiber optic communications grew out of a succession of quests, some of them apparently unrelated. It is important to study the history of fiber optics to understand that the technology as it exists today is relatively new and still evolving.

This chapter discusses the major accomplishments that led to the creation of optical-quality fibers and their use in high speed communications and data transfer, as well as their integration into existing communications networks.

Evolution of Light in Communication

Hundreds of millions of years ago, the first bioluminescent creatures began attracting mates and luring food by starting and stopping chemical reactions in specialized cells. Over time, these animals began to develop distinctive binary, or on-off, patterns to distinguish one another and communicate intentions quickly and accurately. Some of them have evolved complex systems of flashing lights and colors to carry as much information as possible in a single glance. These creatures were the first to communicate with light, a feat instinctive to them but tantalizing and elusive to modern civilization until recently.

Early Forms of Light Communication

Some of the first human efforts to communicate with light consisted of signal fires lit on hilltops or towers to warn of advancing armies, and lighthouses that marked dangerous coasts for ancient ships and gave them reference points in their journeys. To the creators of these signals, light's tremendous speed (approximately 300,000 kilometers per second) made its travel over great distances seem instantaneous.

An early advance in these primitive signals was the introduction of relay systems to extend their range. In some cases, towers were spread out over hundreds of kilometers, each one in the line of sight of the next. With this system, a beacon could be relayed in the time it took each tower guard to light a fire—a matter of minutes—while the fastest transportation might have taken days. Because each tower only needed in its line of sight the sending and receiving towers, the light, which normally travels in a straight line, could be guided around obstacles such as mountains as well as over the horizon. As early as the fourth century AD, Empress Helene, the mother of Constantine, was believed to have sent a signal from Jerusalem to Constantinople in a single day using a relay system.



The principle behind signal relay towers is still used today in the form of repeaters, which amplify signals attenuated by travel over long distances through optical fibers.

Early signal towers and lighthouses, for all their usefulness, were still able to convey only very simple messages. Generally, no light meant one state, while a light signaled a change in that state. The next advance needed was the ability to send more detailed information with the light. A simple but notable example is the signal that prompted Paul Revere's ride at the start of the American Revolution. By prearranged code, one light hung in the tower of Boston's Old North Church signaled a British attack by land, while two lights meant an invasion by sea. The two lamps that shone in the tower not only conveyed a change in state, but also provided a critical detail about that change.

The Quest for Data Transmission

Until the 1800s, light had proven to be a speedy way to transmit simple information across great distances, but until new technologies were available, its uses were limited. It took a series of seemingly unrelated discoveries and inventions to harness the properties of light through optical fibers.

The first of these discoveries was made by Willebrord Snell, a Dutch mathematician who in 1621 wrote the formula for the principle of *refraction*, or the bending of light as it passes from one medium into another. The phenomenon is easily observed by placing a stick into a glass of water. When viewed from above, the stick appears to bend because light travels more slowly through the water, which is optically denser than the air. Snell's formula, which was only published 70 years after his death, stated that every transparent substance had a particular *index of refraction*, and the amount that the light would bend was based on the relative refractive indices of the two materials through which the light was passing. Air has a refractive index of 1, for example, while water has a refractive index of 1.33.

The next breakthrough came from Daniel Colladon, a Swiss physicist, and Jacques Babinet, a French physicist. In 1840, Colladon and Babinet demonstrated that bright light could be guided through jets of water through the principle of *total internal reflection (TIR)*. In their demonstration, an electric arc light was shone through a container of water. Near the bottom of the container was a hole through which the water could escape. As the water poured out of the hole, the light shining into the container followed the stream of water through its arc. Their use of this discovery, however, was limited to illuminating decorative fountains and special effects in operas. It took John Tyndall, a natural philosopher and physicist from Ireland, to bring the phenomenon to greater attention. In 1854, Tyndall performed the demonstration before the British Royal Society and made it part of his published works in 1871, casting a shadow over the contribution of Colladon and Babinet. Tyndall is now widely credited with discovering TIR, although Colladon and Babinet had demonstrated it 14 years previously.

Total internal reflection takes place when light passing through a medium with a higher index of refraction (the water in the experiment) hits a boundary layer with a medium that has

a lower index of refraction (the air). When this takes place, the boundary layer becomes reflective, and the light bounces off of the boundary layer, remaining contained within the medium.

Shortly after Tyndall, Colladon, and Babinet laid the groundwork for routing light through a curved medium, another experiment took place that showed how light could be used to carry higher volumes of data.

In 1880, Alexander Graham Bell demonstrated his photophone, one of the first true attempts to carry complex signals with light. It was also the first device to transmit signals wirelessly. The photophone gathered sunlight onto a mirror attached to a mouthpiece that vibrated when a user spoke into it. The vibrating mirror reflected the light onto a receiver coated with selenium, which produced a modulated electrical signal that varied with the light coming from the sending device. The electrical signal went to headphones where the original voice input was reproduced.

Bell's invention suffered from the fact that outside influences such as dust or stray light confused the signals, and clouds or other obstructions to light rendered the device inoperable. Although Bell had succeeded in transmitting a *modulated* light signal nearly 200 meters, the photophone's limitations had already fated it to be eclipsed by Bell's earlier invention, the telephone. Until the light could be modulated and guided as well as electricity could, inventions such as the photophone would continue to enjoy only novelty status.

Evolution of Optical Fiber Manufacturing Technology

John Tyndall's experiment in total internal reflection had led to attempts to guide light with more control than could be achieved in a stream of water. One such effort by William Wheeler in 1880, the same year that Bell's photophone made its debut, used pipes with a reflective coating inside that guided light from a central arc light throughout a house. As with other efforts of the time, there was no attempt to send meaningful information through these conduits—merely to guide light for novelty or decorative purposes. The first determined efforts to use guided light to carry information came out of the medical industry.

Controlling the Course of Light

Doctors and researchers had long tried to create a device that would allow them to see inside the body with minimal intrusion. They had begun experimenting with bent glass and quartz rods, bringing them tantalizingly close to their goal. These tools could transmit light into the body, but they were extremely uncomfortable and sometimes dangerous for the patient, and there was no way yet to carry an image from the inside of the body out to doctors. What they needed was a flexible medium that could carry whole images for about half a meter.

One such material was in fact pioneered for quite a different purpose. Charles Vernon Boys was a British physics teacher who needed extremely sensitive instruments for his continuing research in heat and gravity. In 1887, to provide the materials he needed, he began drawing fine fibers out of molten silica. Using an improvised miniature crossbow, he shot a needle that

dragged the molten material out of a heat source at high speed. The resulting fiber—more like quartz in its crystalline structure than glass—was finer than any that had been made to date, and was also remarkably even in its thickness. Even though glass fibers had already been available for decades before this, Boys' ultra-fine fibers were the first to be designed for scientific purposes, and were also the strongest and smallest that had been made to date. He did not, however, pursue research into the optical qualities of his fibers.

Over the next four decades, attempts to use total internal reflection in the medical industry yielded some novel products, including glass rods designed by Viennese researchers Roth and Ruess to illuminate internal organs in 1888, and an illuminated dental probe patented in 1898 by David Smith. A truly flexible system for illuminating or conveying images of the inside of the body remained elusive, however.

The next step forward in the optical use of fibers occurred in 1926. In that year, Clarence Weston Hansell, an electrical engineer doing research related to the development of television at RCA, filed a patent for a device that would use parallel quartz fibers to transmit a lighted image over a short distance. The device remained in the conceptual stage, however, until a German medical student, Heinrich Lamm, developed the idea independently in an attempt to form a flexible gastroscope. In 1930, Lamm bundled commercially produced fibers and managed to transmit a rough image through a short stretch of the first fiber optic cable. The process had several problems, however, including the fact that the fiber ends were not arranged exactly, and they were not properly cut and polished. Another issue was to prove more daunting. The image quality suffered from the fact that the quartz fibers were bundled against each other. This meant that the individual fibers were no longer surrounded by a medium with a lower index of refraction. Much of the light from the image was lost to *crosstalk* created when the light passed across fibers.

The poor focus and resolution of Lamm's experimental image meant that a great deal more work would be needed, but Lamm was confident enough to write a paper on the experiment. The rise of the Nazis, however, forced Lamm, a Jew, to leave Germany and abandon his research. The dream of Hansell and Lamm languished until a way could be found to solve the problems that came with the materials available at the time.

Also in 1930, the chemical company DuPont invented a clear plastic material that it branded Lucite. This new material quickly replaced glass as the medium of choice for lighted medical probes. The ease of shaping Lucite pushed aside experiments with bundles of glass fiber, along with the efforts to solve the problems inherent in Lamm's probe.

The problems surfaced again twenty years later, when the Dutch government began looking for better periscopes for its submarines. They turned to Abraham van Heel, who was at the time the president of the International Commission of Optics and a professor of physics at the Technical University of Delft, the Netherlands. Van Heel and his assistant, William Brouwer, revived the idea of using fiber bundles as an image-transmission medium. Fiber bundles, Brouwer pointed out, had the added advantage of being able to scramble and then unscramble an image—an attractive feature to Dutch security officials.

When van Heel attempted to build his image carrier, however, he rediscovered the problem that Lamm had faced. The refractive index of adjacent fibers reduced a fiber's ability to achieve TIR, and the system lost a great deal of light over a short distance. At one point, van Heel even tried coating the fibers with silver to improve their reflectivity, but the effort provided little benefit.

At his government's suggestion, van Heel approached Brian O'Brien, president of the Optical Society of America, in 1951. O'Brien suggested a procedure that is still the basis for fiber optics today: surrounding, or "cladding," the fiber with a layer of material with a lower refractive index.

Following O'Brien's suggestion, van Heel ran the fibers through a liquid plastic that coated them, and in April 1952, he succeeded in transmitting an image through a 400-fiber bundle over a distance of half a meter.

Van Heel's innovation, along with research performed by Narinder Kapany, who also coined the term fiber optics, and Harold Hopkins, helped make the 1950s the pivotal decade in the development of modern fiber optics.

Working in England, Kapany and Hopkins developed a method for ensuring that the fibers at each end of a cable were in precise alignment. They wound a single fine strand several thousand times in a figure-eight pattern and sealed a section in clear epoxy to bind the fibers together throughout the bundle. They then sawed the sealed portion in half, leaving the fiber ends bonded in exact alignment. The image transmitted with this arrangement was clearly an improvement, but the brightness degraded quickly since the fibers were unclad.

Extending Fiber's Reach

In January 1954, the British journal *Nature* chanced to publish papers on the findings of van Heel as well as Kapany and Hopkins in the same issue. Although their placement in the journal was apparently coincidental, the two advancements were precisely the right combination of ideas for Professor Basil Hirschowitz, a gastrosurgeon from South Africa who was working on a fellowship at the University of Michigan. Hirschowitz assembled a team to study the uses of these new findings as a way to finally build a flexible endoscope for peering inside the body. Assisting Hirschowitz were physicist C. Wilbur Peters and a young graduate student named Lawrence Curtiss.

Curtiss studied the work of Kapany and Hopkins and used their winding method to create a workable fiber bundle, but his first attempt at cladding used van Heel's suggestion of cladding glass fibers with plastic. The results were disappointing.

In 1956, Curtiss began working with a new type of glass from Corning, one with a lower refractive index (RI) than the glass he was using in his fibers. He placed a tube made of the new glass around a core made from the higher RI glass and melted the two together. The clad-glass fiber that he drew from this combination was a success. On December 8, 1956, Curtiss made a fiber with light-carrying ability far superior to that of any fiber before it. Even when he was 12 meters away from the glass furnace, he could see the glow of the fire inside the fiber that was being drawn from it. By early 1957, Hirschowitz and Curtiss had created a working endoscope, complete with lighting and optics. This event marked the first practical use of optical fibers to transmit complex information.

Curtiss' fibers were well suited for medical applications, but their ability to carry light was limited. Suffering a signal loss of one *decibel* per meter, the fibers were still not useful for long-distance communications. Many thought that glass was inherently unusable for communications, and research in this area remained at a minimal level for nearly a decade.

In the meantime, the electronic communications industry had been experimenting with methods of improving bandwidth for the higher volumes of traffic they expected to carry. The obvious choice for increasing the amount of information a signal could carry was to increase the frequency, and throughout the 1950s, researchers had pushed frequencies into the tens of *gigahertz*, which produced *wavelengths* of only a few millimeters. Frequencies in this range—just below the lowest infrared frequencies—required hollow pipes to be used as *waveguides*, because the signals were easily disturbed by atmospheric conditions such as fog or dust.

With the invention of the laser in 1960, the potential for increasing communication bandwidths literally increased exponentially. Wavelengths had been slashed from the millimeter range to the micrometer range, and true optical communications seemed within reach. The problems of atmospheric transmission remained, however, and waveguides used for lower frequencies were proving inadequate for optical wavelengths unless they were perfectly straight. Optical fibers, too, were all but ruled out as a transmission medium because at losses of 1000 decibels per kilometer, their *attenuation* was still too great.

One researcher did not give up on fiber, however. Charles K. Kao, working at Standard Telecommunications Laboratories, began studying the problems encountered in optical fibers. His conclusions revived interest in the medium after he announced in 1966 that signal losses in glass fibers were not caused by inherent deficiencies of the material, but by flaws in the manufacturing process. Kao proposed that improved manufacturing processes could lower attenuation to levels of 20 decibels per kilometer or better, while providing the ability to carry up to 200,000 telephone channels in a single fiber.

Kao's pronouncement sparked a race to find the lower limit of signal loss in optical fibers. In 1970, Corning used pure silica to create a fiber with a loss that achieved Kao's target of only 20 decibels per kilometer. That was just the beginning. Six years later, the threshold had dropped to just half a decibel per kilometer, and in 1979 the new low was 0.2 decibel per kilometer. Optical fiber had passed well into the realm of practicality for communications and could begin showing its promise as a superior medium to copper.

Evolution of Optical Fiber Integration and Application

Once signal losses in fiber dropped below Kao's projected figure of 20 decibels per kilometer, communications companies began looking seriously at fiber optics as a new transmission medium. The technology required for this fledgling medium was still expensive, however, and fiber optic communications systems remained in closed-circuit, experimental stages until 1973. In that year, the U.S. Navy installed a fiber optic telephone link aboard the USS Little Rock. Fiber optics had left the lab and started working. Further military tests showed fiber's advantages over copper in weight and information-carrying capacity.

The first full-scale commercial application of fiber optic communication systems occurred in 1977, when both AT&T and GTE began using fiber optic telephone systems for commercial customers. During this period, the U.S. government breakup of the Bell Telephone system

monopoly began a boom time for smaller companies seeking to market long-distance service. A number of companies had positioned themselves to build microwave towers throughout the country to create high-speed long-distance networks. With the rise of fiber optic technology, however, the towers were obsolete before they had even been built. Plans for the towers were scrapped in the early 1980s in favor of fiber optic links between major cities. These links were then connected to local telephone companies that leased their capacity from the operators. The result was a bandwidth feeding frenzy. The fiber optic links had such high capacities that extra bandwidth was leased to other local and long-distance carriers, which often undercut the owners of the lines, driving some out of business. The survivors, such as Sprint and MCI, have become major players in today's telecommunications industry.

Following the success of fiber optics in the telecommunications industry, other sectors began taking advantage of the medium. During the 1990s, fiber optic networks began to dominate in the fields of industrial controls, computers, and information systems. Improvements in lasers and fiber manufacturing continued to drive data rates higher and bring down operating costs.

Today, fiber optics have become commonplace in many areas as the technology continues to improve. Until recently, the transition to fiber optics was cost effective only for business and industry; equipment upgrades made it too expensive for telephone and cable companies to run fiber to every home. Manufacturing improvements have reduced costs, however, so that running fiber to the home is now an affordable alternative for telephone and cable companies. With recent FCC approval, telephone and cable companies are preparing to bring fiber to over 100 million homes over the next 10 years.

Summary

This chapter discussed the history of fiber optics in communications, beginning with the first use of light to carry messages. It covered early experiments in the control of light for carrying sound, along with the problems faced by early experimenters.

This chapter also covered the discovery of principles that are essential to fiber optics and the ways in which they were adapted to control the path of light. It described the invention of processes used to improve the ability of glass fibers to carry light, as well as the refinements that made efficient, long-distance light transmission possible.

This chapter described the growth of fiber optics as an experimental data transmission medium, then as a new technology for the telecommunications industry. It also described advances that will make fiber optics even more widespread as a voice and data carrier.

Exam Essentials

Understand the evolution of light in communication Make sure that you understand the qualities of light that make it a desirable form of communication, the limitations of early communication with light, and some of the ways in which those limitations were overcome. Also be sure you understand the principles behind early experiments with materials used to modulate and guide light.

Understand the evolution of optical fiber manufacturing technology. Make sure that you understand the problems encountered by researchers looking for more efficient ways to guide light over long distances and the breakthroughs that made modern optical fibers practical.

Understand the evolution of optical fiber integration and application. Make sure that you understand how fiber optics made the transition from experiments in guided light to a wide-spread communication medium. Pay attention to events that first used fiber optics in “real world” applications and proved that the technology could be used practically.

Review Questions

1. What was one of the earliest advantages that light held over other forms of communication?
 - A. Ability to communicate complex ideas
 - B. Ability to carry messages quickly
 - C. Ability to carry messages privately
 - D. Ability to carry coded messages
2. The experiments in which light could be seen in water draining from a vessel proved that:
 - A. Light could travel through water.
 - B. Light was refracted in some materials.
 - C. Light could be guided through some materials.
 - D. Light could travel in a straight line.
3. Some of the earliest attempts to carry light through glass fibers took place because researchers were trying to:
 - A. View images from inside the human body.
 - B. Send messages from one building to another.
 - C. Compete with the telephone.
 - D. Collect numerous light sources in one area.
4. One of the first goals in creating a fiber that could be used for communications was to:
 - A. Make the fibers smaller.
 - B. Make the fibers longer.
 - C. Reduce signal loss to 40 decibels per kilometer.
 - D. Reduce signal loss to 20 decibels per kilometer.
5. The decade that saw the greatest advances in the use of fiber optics for data and communications was:
 - A. The 1880s
 - B. The 1920s
 - C. The 1960s
 - D. The 1970s

Answers to Review Questions

1. B. Light travels at approximately 300,000 kilometers per second in air, and it allowed simple messages to be sent quickly over great distances through the use of signal fires.
2. C. The experiments performed by Colladon, Babinet, and Tyndall proved that light could be guided through water if the air around it had a lower index of refraction. This demonstrated the principle of total internal reflection (TIR).
3. A. Most of the early experiments done with glass fibers involved bundling them to try to carry an image from one end to the other.
4. D. In 1966, Charles K. Kao announced that signal losses in glass fibers could be reduced to 20 decibels per kilometer or less through improved manufacturing processes. This announcement was the beginning of a quest for lower and lower signal loss through optical fiber.
5. D. Beginning with U.S. Navy trials in 1973, the 1970s saw fiber optics move out of the laboratory and into the mainstream of high-speed, high-tech communications. Once the technology was proven, communications companies began laying fiber networks throughout the country, spurring further innovation in the medium.



Chapter

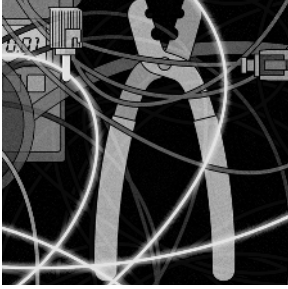
2

Principles of Fiber Optic Transmission

OBJECTIVES COVERED IN THIS CHAPTER:

✓ Principles of Fiber Optic Transmission

- Describe the basic parts of a fiber optic link.
- Describe the basic operation of a fiber optic transmitter.
- Describe the basic operation of a fiber optic receiver.
- Explain amplitude modulation (AM) as it applies to a fiber optic transmitter.
- Explain analog data transmission as it applies to a fiber optic communication system.
- Explain the basic components of a digital transmission.
- Explain digital data transmission as it applies to a fiber optic transmitter.
- Graphically explain how an analog to digital conversion (A/D) is accomplished.
- Graphically explain how a digital to analog conversion (D/A) is accomplished.
- Explain pulse coded modulation (PCM).
- Explain the fundamentals of multiplexing signals.
- Demonstrate how to express gain or loss using dB.
- Demonstrate how to express optical power in dBm.



Like Bell's photophone, the purpose of fiber optics is to convert a signal to light, move the light over distance, and then reconstruct the original signal from the light. The equipment used to do this job has to overcome all of the same problems that Bell encountered, while carrying more data over a much greater distance.

In this chapter, you will learn about the basic components that transmit, receive, and carry the optical signal. You will also learn some of the methods used to convert signals to light, and light back to the original signals, as well as how the light is carried over the distances required.

The Fiber Optic Link

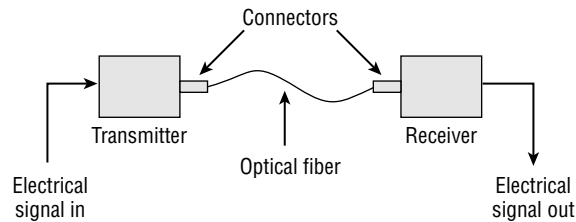
A link is a signal pathway between two points using some kind of generic cable. The pathway includes a means to send the signal into the cable and a way to receive it at the other end in a useful way.

Any time we send a signal from one point to another over a wire, we are using a link. A simple intercom, for example, consists of the sending station (which converts voice into electrical signals), the wire over which the signals are transmitted, and the receiving station (which converts the electrical signal back into voice).

Links are often described in terms of their ability to send and receive signals as part of a communication system. When described in these terms, they are broken down into *simplex* and *duplex*. Simplex means that the link can only send at one end and receive at the other end. In other words, the signal goes only one way. Duplex means that the link has a transmitter and a receiver at each end. A *half-duplex* system allows signals to go only one way at a time, similar to an intercom system. A *full-duplex* system allows users to send and receive at the same time. A telephone is a common example of a full-duplex system.

A fiber optic link, shown in Figure 2.1, is like any other link, except that it uses optical fiber instead of wire. A fiber optic link comprises four basic components:

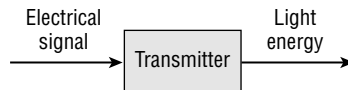
- Transmitter to convert a signal into light and send the light
- Receiver to capture the light and convert it back to a signal
- The optical fiber that carries the light
- The connectors that link the cable to the transmitter and receiver

FIGURE 2.1 The fiber optic link

Now let's look at each component in a little more detail.

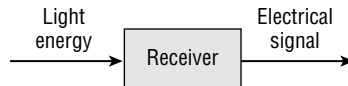
Transmitter

The transmitter, shown in Figure 2.2, converts an electrical signal into light energy to be carried through the fiber optic link. The signal could be generated by a computer, a voice over a telephone, or data from an industrial sensor.

FIGURE 2.2 The fiber optic transmitter

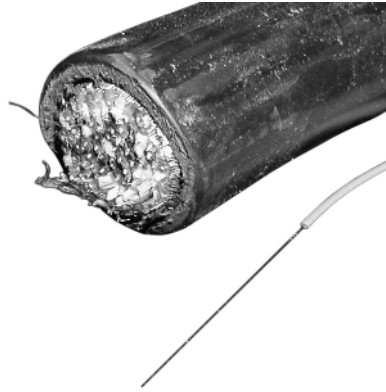
Receiver

The receiver is an electronic device that collects light energy and converts it into electrical energy, which can then be converted into its original form, as shown in Figure 2.3. The receiver typically consists of a photo detector to convert the received light into electricity, and circuitry to amplify and process the signal.

FIGURE 2.3 The fiber optic receiver

Optical Fibers

Optical fibers carry light energy from the transmitter to the receiver. An optical fiber may be made of glass or plastic, depending on the requirements of the job that it will perform. The advantage of optical fiber over transmission through air is that the fiber can carry light around corners and over great distances.

FIGURE 2.4 Comparison of fiber and copper cables**FIGURE 2.5** Fiber optic connectors

Many fibers used in a fiber optic link have a *core* between 8 and 100 microns (millionths of a meter) in diameter. For comparison, a typical human hair is about 100 microns in diameter. The *cladding* which surrounds the fiber may be as much as 140 microns in diameter.

The optical fiber's coating protects the cladding from abrasion. Even with the thickness of the coating, however, optical fiber cabling is much smaller and lighter than copper cabling, as shown in Figure 2.4, and can carry many times the information.

Connectors

The connector is attached to the optical fiber and allows it to be mated to the transmitter or receiver to provide solid contact. The connector must align the fiber end precisely with the light source or receiver to prevent signal loss.

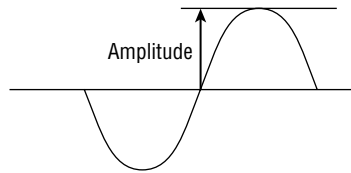
The connector, shown in Figure 2.5, could be considered the element that makes it possible for us to use fiber optics, because it allows large hands to handle the small, fragile fibers.

Now that you've seen the components required for a fiber optic link, let's look at some of the methods that make it possible to transmit data with light.

Amplitude Modulation

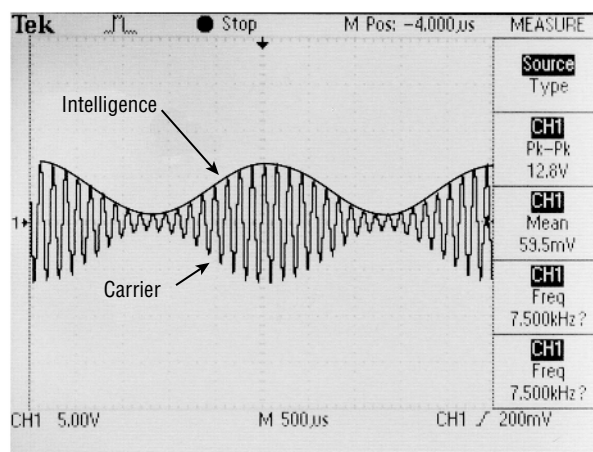
One method used for converting electrical signals into light signals for transmission is *amplitude modulation (AM)*. Amplitude refers to the strength of a signal, represented by a waveform as shown in Figure 2.6. In amplitude modulation, electrical energy with continuously varying voltage is converted into light with continuously varying brightness.

FIGURE 2.6 Amplitude on a waveform



Amplitude modulation requires two components: a carrier and a signal that is imposed on the carrier—also known as the *intelligence*—to change it in some way. When we speak, we impose the intelligence created by the vibration of our vocal cords on air, which is the carrier. Similarly, Bell's photophone used sound to vibrate a mirror, which *modulated* the light reflected from it. At the receiving end, a similar arrangement worked in reverse to *demodulate* the light, retrieving the intelligence from it and creating the sound again.

To modulate the amplitude of the light in a fiber optic transmitter, the intelligence is sent through a circuit that changes it to a continuously varying voltage. As the intelligence changes, the voltage controlling the light rises and falls, varying the light's intensity to match the intelligence. Figure 2.7 shows the basic process of amplitude modulation.

FIGURE 2.7 Amplitude modulation

The intelligence imposed on the light changes the amplitude of the light, but not its wavelength. Amplitude modulation suffers from two problems that can affect the quality of the signal: attenuation and noise.

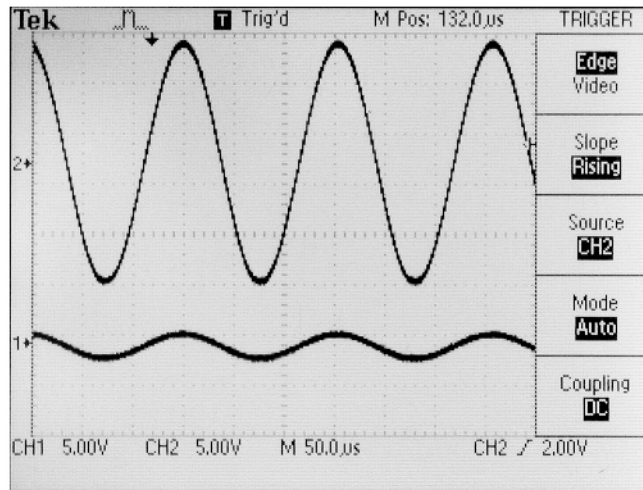
Attenuation is the loss of optical power as the signal passes through the fiber. Attenuation occurs as light is absorbed or scattered by impurities in the fiber, or as light energy is absorbed by the cladding and the coating.

As an amplitude-modulated signal is attenuated, its average power decreases and small differences in amplitude become even smaller, or disappear entirely, as shown in Figure 2.8. When the light energy is converted back to electrical energy, these small differences are lost, and cannot be reconstructed.

Noise is the introduction of unwanted energy into a signal. An example is static on an AM radio, especially when passing near high voltage power lines. The unwanted energy changes the amplitude of the signal, sometimes rendering it unusable if the noise is powerful enough in comparison to the original signal.

Analog Transmission

Amplitude modulation is a form of *analog* transmission. An analog signal is one that varies continuously through time in response to an input. In addition, the response is infinitely variable within the specified range. In other words, a smooth change in the input will produce a smooth change in the signal.

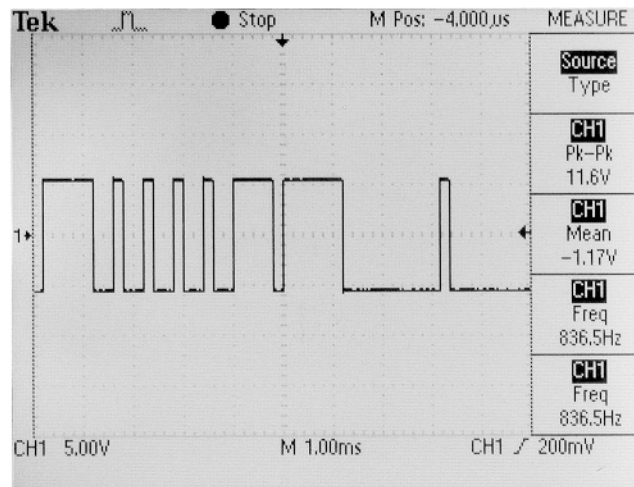
FIGURE 2.8 Attenuation of an AM signal

A common example of an analog system is an electrical temperature sensor such as a thermocouple, which generates a small voltage that changes with the temperature difference registered between its hottest and coldest points. As the temperature difference that the thermocouple is sensing rises, the voltage increases. Because the relationship between temperature and voltage from the device is predictable, the thermocouple's output can be translated into a temperature reading. A reading of 3 millivolts (mV), for example, could indicate a temperature difference of 140° F.

When amplitude modulation is used with fiber optics, the amplitude of the optical transmission changes smoothly in relation to the strength of the incoming signal. Because of their infinitely variable response within a given range, analog signals are commonly used in Community Access Television (CATV) to carry sound, video, and sensor data, which originate as analog components.

Digital Data Transmission

In spite of the problems caused by noise, analog signals are still used in fiber optic communications. If information is to be stored, carried, or manipulated by computers, however, it must be in a *digital* form—that is, represented by a series of on-off or high-low voltage readings. Figure 2.9 shows a digital waveform. The voltage readings are often represented as ones and zeros, with the high or on state being a one, and the low or off state being a zero. Because only two states—or digits—are used, the numbering system is referred to as *binary*.

FIGURE 2.9 Digital waveform

Recall that early signal fires, a form of digital communication, could only announce a change in state by being lit, but could not communicate complex information. To make digital information more detailed, *binary digits*, or *bits*, are combined into eight-place sequences called *bytes*. A byte can be used to represent a single number in the same way that a voltage reading would be used in an analog transmission. For example, the above temperature reading of 140° F might be transmitted digitally as 10001100, the binary equivalent of 140.

Analog Data Transmission vs. Digital Data Transmission

One of the main reasons that digital transmission is considered superior to analog transmissions is the fact that a digital signal is not affected by noise or attenuation the way an analog signal is.

Digital information can be stored and transmitted accurately because noise that would interfere with the analog reading do not affect digital data. Each voltage in the sequence is either high or low, and voltages that do not match either the high or low levels do not change the meaning of the digital sequence.

The difference between the two is like the difference between a tape recording of a musical performance and a CD of the same performance. The analog recording may carry the same detail, but it would also contain a certain amount of hiss caused by electrical noise. The CD would be free of hiss, because the stray voltages do not register as either high or low signals.

More and more, digital transmissions are replacing analog transmissions, even in radio and television transmission. Many radio stations now broadcast digital signals to receivers. In addition

to carrying the regular programming as digital data, the broadcast can also carry digital data for display on the receiver, such as program details, announcers' names, and song titles.

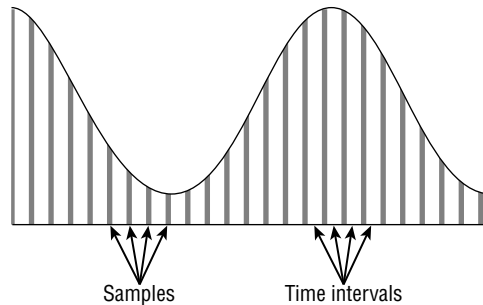
In fiber optic transmission, digital signals make it possible to carry many thousands of conversations over a single fiber through the use of multiplexing, which will be explained later in this chapter.

Analog to Digital (A/D) Conversion

To transmit an analog signal such as a voice through a digital system, it is necessary to *digitize*, or *encode*, it. This is also known as analog to digital, or A/D, conversion.

In A/D conversion, the smooth, continuously variable analog signal is translated into a digital signal that carries the same information. To do this, the signal's voltage is "sampled" at regular intervals and converted into binary numbers that represent the voltage at each interval. In Figure 2.10, for example, each vertical line represents a sampling of the analog signal at a given time.

FIGURE 2.10 Sampling an analog signal

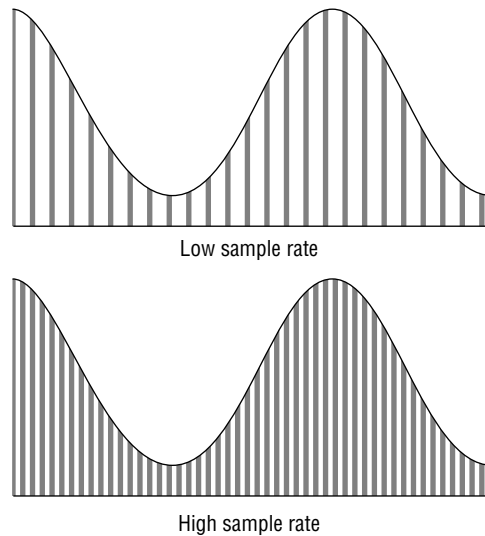


As with frequency measurements, the sample rate is measured in terms of cycles, or hertz, so a rate of one sample per second would be designated 1 Hz. A rate of 1000 samples per second would be 1 kilohertz, or 1 kHz.

There are two factors that affect the quality of the digital sample: sample rate and quantizing error.

Sample Rate

When an analog signal is digitized, any information between the samples is lost, so instead of a smooth transition over time, the digital information jumps from one voltage to the next in the signal. To smooth out the transitions and retain more of the information from the original analog signal, more samples must be taken over time. The higher the sampling rate, the more accurately the original analog signal can be digitized. Typically, audio signals for CDs and other digital music are sampled at 44.1 kHz or 48 kHz.

FIGURE 2.11 Low sample rate vs. high sample rate

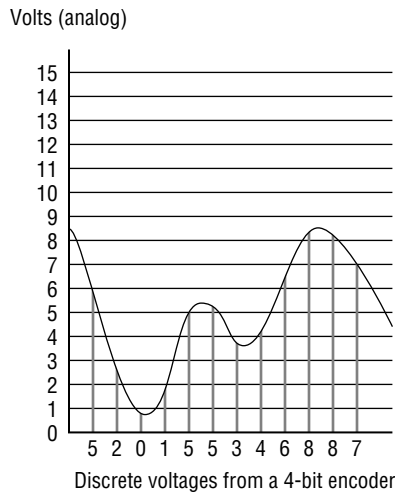
Quantizing Error

The second factor affecting digital signal quality is called quantizing error. Quantizing error is caused by the inability of a binary number to capture the exact voltage of a digital sample.

Because an analog signal is infinitely variable, the sample's voltage could be any number within a specified range. If the binary number used to represent the voltage does not have enough bits, it cannot represent the voltage accurately.

In Figure 2.12, for example, a 4-bit number can represent 16 voltage levels—from 0 to 15. Subtract the 0 voltage level and you have 15 increments available for a 4-bit number. Therefore, on a scale from 0 V to +15 V, each binary number represents a change of 1 whole volt. If a sample returned a reading of +1.5 V, the binary number would still read 0001, or +1 V. You can calculate the maximum error by dividing the voltage range by the number of increments. In this case, $15 \div 15 = 1$, so you have a maximum error of 1 V, and the average error is one-half of that, or 0.5 V.

Increase the number of bits to eight, however, and you have 255 increments plus zero. The voltage between increments, and the maximum error, is now $15 \div 255 = 58.82 \times 10^{-3}$ or 58.82 mV. Now, a reading of +1.5 V is 0001 1001, or 25 steps from 0 instead of just one. Multiplying the number of increments by the voltage between them gives us $25 \times 58.82 \text{ mV} = 1.4705 \text{ V}$. This result is much closer to the analog reading of +1.5 V.

FIGURE 2.12 Sampling with a 4-bit number

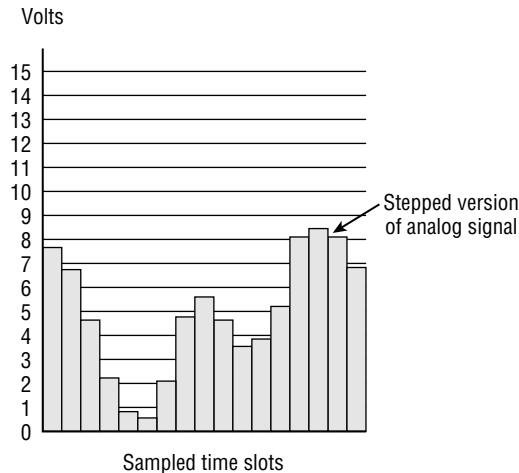
As with the sample rate, the more bits used in encoding, the more accurate each sample can be. CD-quality audio signals are usually encoded at 16 bits, which means that there are 65,535 increments available, plus zero.

Digital to Analog (D/A) Conversion

When digital information is used to control analog devices such as temperature controls, or when analog information has been converted to digital data for transmission and must be converted back to analog data, digital to analog (D/A) conversion is used.

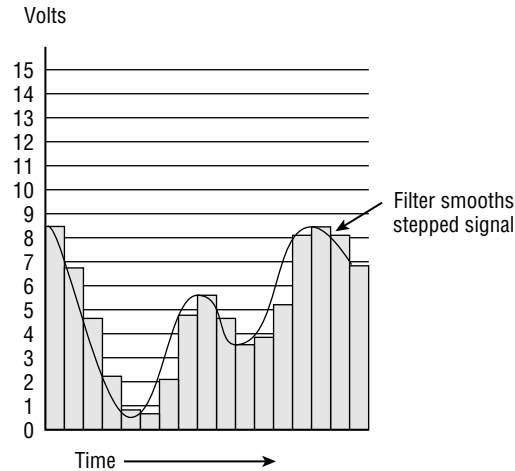
When digital data is converted to analog, two processes take place. First, a digital-to-analog converter converts each sequential binary sample to a proportional voltage. From our previous example of a 0 V to +15 V range represented by an 8-bit number, the binary sample 0001 1001 would be converted to +1.4705 V. If the same binary sample were applied to a different voltage range, the result would be proportional to that range. The D/A converter puts out a stepped version of the analog signal, as shown in Figure 2.13. When reconstructing an encoded analog signal, the higher the sampling rate and the greater the number of bits in each sample, the more accurate the analog reconstruction can be.

FIGURE 2.13 Converting analog to digital



Next, the steps between each digital sample must be smoothed out to provide a transition from one voltage to another. No matter how many samples are used, the digital output will always produce a signal that jumps from one voltage to another, and then holds each voltage for the amount of time between samples. When a smooth analog signal is required, D/A converters have circuits that filter the stairstep voltage into a smooth waveform, as shown in Figure 2.14.

FIGURE 2.14 Filters convert stairstep voltage to a smooth waveform.

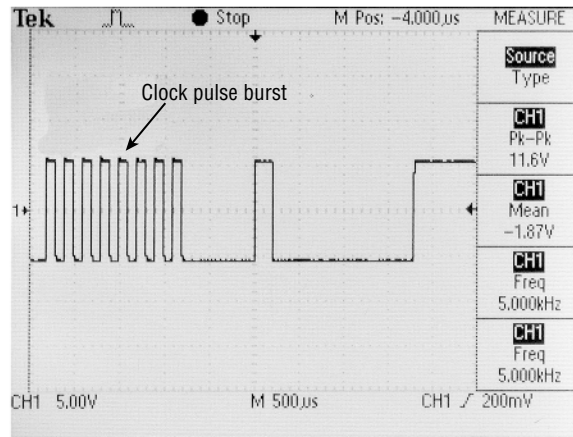


Pulse Code Modulation (PCM)

When an analog signal has been digitally encoded, transmitted, and reconstructed at the receiving end as an analog signal, it has undergone a process known as *pulse code modulation (PCM)*. Pulse code modulation is the most common method of digitizing data for transmission.

Data transmission using PCM is *serial*, which means that the binary words are sent one after another in the order they were generated. The circuitry that converts the data also sends a timing, or *clock*, signal so the receiver can synchronize itself with the data that is being transmitted and reconstruct it accurately. Figure 2.15 shows a typical PCM sequence with a clock pulse burst.

FIGURE 2.15 In this PCM transmission, the binary numbers are sent along with a clock signal.



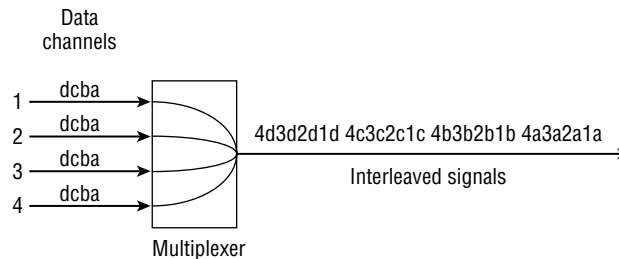
In order for pulse code modulation to be effective, an analog signal must be sampled at a rate that is at least twice the highest expected frequency. This number is referred to as the Nyquist Minimum. In practice, though, the sampling rate is usually closer to three times the highest expected frequency or more. This formula ensures that sampling will capture some portion of even the highest frequencies. For example, in a telephone conversation, the highest frequency encountered is about 4 kHz. That means sampling must take place at the *Nyquist Minimum* of 8 kHz to maintain a basic signal quality.

Multiplexing

Most fiber optic data transmission systems can send data at speeds that far exceed the requirements of single stream of information. To take advantage of this fact, multiplexing can be used to carry several information channels, such as telephone conversations, nearly simultaneously. *Multiplexing* is the process of transmitting many channels of information over one link or circuit.

Figure 2.16 shows a simple multiplexing process. A multiplexer first divides each channel into several parts. In a process known as interleaving, the multiplexer sends the first part of each channel, then the second part of each channel, continuing the process until all of the transmissions are completed. At the receiving end, a demultiplexer separates the transmissions into their individual channels and reassembles them in their proper order.

FIGURE 2.16 Multiplexing allows thousands of conversations to be carried in a single fiber.



Decibels (dB)

As light travels away from its source through a medium, it loses energy. Energy loss can be caused by absorption or scattering of light by impurities in the fiber, or by light passing through the core and cladding and being absorbed in the coating.

It is important to measure the amount of light energy lost in the fiber optic link so that receiving equipment can process it properly for conversion or retransmission farther down the line. In addition, an understanding of light loss can lead to the development of improved equipment that will further reduce the loss.

One of the more common terms used when discussing the quality of a signal in fiber optics is the decibel (dB). The decibel was originally used to measure the strength of sounds as perceived by the human ear. Its name means “one-tenth of a bel.”



A *bel* is a sound measurement named for telephone inventor Alexander Graham Bell.

In fiber optics, decibels are most commonly used to measure signal loss through the system after the light has left the transmitter. Recall that in the 1960s, a signal loss of 20 decibels per kilometer was considered the goal for making fiber optics practical for communications. A 20-decibel loss means that of the original power put into the signal, only 1 percent remains.

To calculate the decibel value of a gain or loss in signal power, use the following equation:

$$\text{dB} = 10\text{Log}_{10}(P_{\text{out}} \div P_{\text{in}})$$

If you know the decibel value and want to calculate the gain or loss, you will have to rearrange the equation as shown here:

$$(P_{\text{out}} \div P_{\text{in}}) = \text{antilog}(\text{dB} \div 10)$$

EXERCISE 2.1

Calculate the power loss in decibels of a signal that starts at 3 milliwatts and arrives at 1.8 milliwatts.

1. Use the dB equation $\text{dB} = 10\text{Log}_{10}(P_{\text{out}} \div P_{\text{in}})$.
2. Plug in the P_{in} and P_{out} values: $\text{dB} = 10\text{Log}_{10}(0.0018 \div 0.003) = 10\text{Log}_{10}(0.6)$.
3. $10\text{Log}_{10}(0.6) = 10 \times -0.2218 = -2.218$.
4. The power loss in dB is -2.218 .

EXERCISE 2.2

Calculate the remaining power after a 3.5 milliwatt signal is subjected to a loss of 0.9 decibels.

1. Use the dB equation $(P_{\text{out}} \div P_{\text{in}}) = \text{antilog}(\text{dB} \div 10)$.
2. Plug in the P_{in} and dB values: $(P_{\text{out}} \div 0.0035) = \text{antilog}(-0.9 \div 10) = \text{antilog } -0.09 = 0.813$.
3. $P_{\text{out}} = 0.0035 \times 0.813 = 0.0028$.
4. The remaining power is 2.8 milliwatts.

Signals may be decreased, or attenuated, just about anywhere in the system. In addition to the optical fiber itself, connectors, splices, and other equipment can cause considerable loss in the signal.

When measuring signal gain or loss, decibels are calculated relative to the original power, rather than as an absolute number. For example, a loss of 0.1 decibel means that the signal has 97.7 percent of its power remaining, and a loss of 3 decibels means that only 50 percent of the original power remains. This relationship is logarithmic, rather than linear, meaning that with each 10 decibels of loss, the power is 10 percent of what it was, and with each 10 decibels of gain, the power is 10 times what it was, as shown in Tables 2.1 and 2.2, respectively.

TABLE 2.1 Decibel Losses Expressed in Percentages

Loss in dB	% Power Lost	% Power Remaining
0.1	2.3%	97.7%
0.2	4.5%	95.5%
0.3	6.7%	93.3%
0.4	8.8%	91.2%
0.5	10.9%	89.1%
0.6	12.9%	87.1%
0.75	15.9%	84.1%
0.8	16.8%	83.2%
0.9	18.7%	81.3%
1	21%	79%
3	50%	50%
6	75%	25%
7	80%	20%
9	87%	13%
10	90%	10%
13	95.0%	5%
16	97%	3%

TABLE 2.1 Decibel Losses Expressed in Percentages *(continued)*

Loss in dB	% Power Lost	% Power Remaining
17	98.0%	2.0%
19	98.7%	1.3%
20	99.00%	1.00%
23	99.50%	0.50%
30	99.9%	0.1%
33	99.95%	0.05%
40	99.99%	0.01%
50	99.999%	0.001%
60	99.9999%	0.0001%
70	99.99999%	0.00001%

TABLE 2.2 Decibel Gains Expressed in Percentages

Gain in dB	% Power Increase	% Total Power
0.1	2.3%	102.3%
0.2	4.7%	104.7%
0.3	7.2%	107.2%
0.4	9.6%	109.6%
0.5	12.2%	112.2%
0.6	14.8%	114.8%
0.75	18.9%	118.9%
0.8	20.2%	120.2%
0.9	23.0%	123.0%

TABLE 2.2 Decibel Gains Expressed in Percentages *(continued)*

Gain in dB	% Power Increase	% Total Power
1	26%	126%
3	100%	200%
6	298%	398%
7	401%	501%
9	694%	794%
10	900%	1,000%
13	1,895%	1,995%
16	3,881%	3,981%
17	4,912%	5,012%
19	7,843%	7,943%
20	9,900%	10,000%
23	19,853%	19,953%
30	99,900%	100,000%
33	199,426%	199,526%
40	999,900%	1,000,000%
50	9,999,900%	10,000,000%
60	99,999,900%	100,000,000%
70	999,999,900%	1,000,000,000%

One of the advantages of using decibels when calculating gain and loss is their relative ease of use compared to using percentages. When measuring loss through different components, you can algebraically add the decibels from each and arrive at a total signal loss for the system. If

you were to use percentages, you would have to calculate the remaining power after the signal passed through each component, then take another percentage off for the next component, and so on, until the loss through the entire system had been calculated.

The Rules of Thumb

When calculating gains and losses in a system, it is useful to remember the three rules of thumb shown in Table 2.3, which make it easier to perform some common decibel calculations. These rules help you calculate how much power you have after the indicated decibel gain or loss.

TABLE 2.3 Rules of Thumb

Decibel	Loss	Gain
3 dB	$\frac{1}{2}$	$\times 2$
7 dB	$\frac{4}{5}$	$\times 5$
10 dB	$\frac{9}{10}$	$\times 10$

For example, a loss of 3 decibels means that you have about $\frac{1}{2}$ of the original power. A gain of 3 decibels means that you have about twice the original power. These rules are meant for rough calculations only, because a 3-decibel loss actually leaves 50.1187 percent of the original power, and a 7-decibel loss leaves 19.953 percent of the original power.

Because decibels can be algebraically added and subtracted, you can use combinations of the decibel values to determine total gains or losses.

EXERCISE 2.3

Use the rules of thumb to calculate the power remaining after 3 watts of power suffers a loss of 27 decibels.

1. Determine whether or not you can use the rules of thumb in any combination to achieve the decibels expressed: $10 + 10 + 7 = 27$.
2. Apply the first 10-decibel loss: $3 \div 10 = 0.3$.
3. Apply the second 10-decibel loss to the result: $0.3 \div 10 = 0.03$.
4. Apply the 7-decibel loss to the result: $0.03 \div 5 = 0.006$.
5. The remaining power is 6 milliwatts.

Absolute Power Gains and Losses

Taken by itself, the decibel is only a relative number, and it only has meaning when it is referenced to a known input or output power. In many cases, however, it is important to have an absolute value to use for comparison or for equipment specifications. When such a value is required, the decibel referenced to 1 milliwatt (dBm) is used as a standard.

As an example, a value of -10 dBm means that 1 mW of power has been attenuated by 10 dB, so only 10 percent of its power, or 100 microwatts, remains. Table 2.4 shows how dBm values convert to power measurements.

TABLE 2.4 Converting dBm to Power

Power	Gain or Loss
100 mW	20 dBm
20 mW	13 dBm
10 mW	10 dBm
8 mW	9 dBm
5 mW	7 dBm
4 mW	6 dBm
2 mW	3 dBm
1 mW	0 dBm
500 μ W	-3 dBm
250 μ W	-6 dBm
200 μ W	-7 dBm
125 μ W	-9 dBm
100 μ W	-10 dBm
50 μ W	-13 dBm
25 μ W	-16 dBm

TABLE 2.4 Converting dBm to Power (*continued*)

Power	Gain or Loss
20 μ W	−17 dBm
12.5 μ W	−19 dBm
10 μ W	−20 dBm
5 μ W	−23 dBm
1 μ W	−30 dBm
500 nW	−33 dBm

The formula used to convert dBm to a power measurement is:

$$\text{dBm} = 10\text{Log}_{10}(P \div 0.001)\text{W}$$

This equation is similar to the equation for finding a decibel value from an input power and an output power, except that in this case the input power is fixed at 1 mW.

EXERCISE 2.4

Calculate the dBm of an amplifier with an output of 1.5 watts.

1. Use the dBm formula: $\text{dBm} = 10\text{Log}_{10}(1.5 \div 0.001) = 10\text{Log}_{10}(1.5 \div 0.001) = 10\text{Log}_{10}(1500) = 10 \times 3.176 = 31.76$.
2. The output is 31.76 dBm.



Real World Scenario

Getting Lost in Losses

If you are measuring the loss over a link from a source rated at −20 dBm to −26 dBm at the end of the link, make sure you express the loss correctly. If you say that the loss is 6 dBm, you are actually saying that you have lost 4 milliwatts, an absolute value that has nothing to do with the actual loss of 6 decibels, or 7.5 microwatts (based on the starting value of 10 microwatts). So you have 2.5 microwatts remaining at the end of the link.

Remember that dBm is a measurement based on 1 milliwatt, and power is measured based on that value. On the other hand, dB is a relative value applied to whatever the original power measurement is.

Summary

This chapter discussed some of the basic principles and components that are necessary for fiber optic communications, including the physical means of transmission and the methods for turning electrical signals carrying information into light.

We specifically discussed amplitude modulation, which is a form of analog transmission, and the binary coding that is the basis of digital transmission. In addition, we described how a signal is converted from analog to digital and from digital to analog, along with some of the problems that can be encountered in the process.

We also discussed how the information capacity of a fiber optic link can be used to carry more than one channel of information over a single link.

Finally, we described methods for measuring changes in the intensity of light using decibels and decibels referenced to 1 milliwatt, measurements that are important to creating fiber optic networks that can operate over long distances.

Exam Essentials

Know the components of the fiber optic link. Make sure you understand the four components that make up any fiber optic link. Also, you should know the function of each within the link.

Understand amplitude modulation. Be aware of how amplitude modulation is used to carry information with a fixed frequency of light. Understand the elements of an amplitude-modulated signal and the problems that can affect signal quality.

Understand analog data transmission. Know what an analog signal is, and how its energy output responds to an input. Understand the uses of analog transmission with fiber optics.

Understand digital data transmission. Make sure you know what a digital signal is and how its waveform appears. Understand how binary signals can be used to carry complex information.

Know the difference between analog data transmission and digital data transmission. Make sure you understand the differences between analog and digital data transmission, especially the advantages and disadvantages of each. Know the capabilities of each type of transmission in terms of signal quality and ability to recapture data from transmissions over long distances.

Understand analog to digital (A/D) conversion. Be able to describe the process used to convert an analog signal into a digital signal. Understand the importance of concepts including:

- Sampling
- Sample rate
- Quantizing error

Understand digital to analog (D/A) conversion. Be able to describe how a digital signal is converted into an analog signal. Know the importance of filtering circuits in creating an analog signal from a digital one.

Be able to describe pulse code modulation (PCM). Make sure you know how pulse code modulation allows digital data to be carried through the fiber optic link. Understand the nature of a serial transmission and the use of a clock signal.

Know about multiplexing. Understand how digital data transmission makes multiplexing possible, and how several channels of information can be carried through a single fiber using multiplexing.

Understand the importance of decibels (dB). Know why it is important to measure energy loss through a fiber optic link. Make sure you understand how decibels relate to the percentage of power gained or lost from the original signal.

Know the difference between absolute power gains and losses. Understand how decibels referenced to a standard power level can be used to create an absolute value of gain or loss in light signals. Understand how decibels referenced to a standard translate into power measurements.

Review Questions

1. Which of the following groups describes the components in a fiber optic link?
 - A. Transmitter, receiver, optical fiber, decoder
 - B. Transmitter, receiver, optical fiber, connectors
 - C. Transmitter, receiver, optical fiber
 - D. Transmitter, receiver, optical fiber, cabling
2. In a fiber optic link, which component has an input of electrical energy and an output of light?
 - A. Connector
 - B. Optical fiber
 - C. Receiver
 - D. Transmitter
3. In amplitude modulation, the signal that is imposed on the carrier is also known as the _____.
 - A. Intelligence
 - B. Imposition
 - C. Inductance
 - D. Illumination
4. If an optical signal is attenuated, you may notice that it has turned _____.
 - A. More blue
 - B. More red
 - C. Brighter
 - D. Dimmer
5. How many different readings between zero and +10 volts are possible on an analog voltmeter?
 - A. 10
 - B. 100
 - C. 1000
 - D. Infinite
6. A digital transmission is made up entirely of _____.
 - A. Light
 - B. Electricity
 - C. Ones and zeros
 - D. Waveforms

7. If noise enters a digital transmission, what is the most likely effect?
 - A. It will introduce a number between zero and one.
 - B. It will likely have no effect.
 - C. It will introduce a number above one.
 - D. It will introduce a number below zero.
8. If an analog signal has been “sampled,” what has happened to it?
 - A. It has been attenuated.
 - B. It has been measured at regular intervals and its voltages have been converted to binary numbers.
 - C. It has been measured at regular intervals and its voltages have been converted into light.
 - D. A portion of it has undergone an accuracy test.
9. A high sample rate is used to _____.
 - A. Convert data more rapidly.
 - B. Keep information moving quickly.
 - C. Retain more accurate information from the original analog signal.
 - D. Record voltages above the specified range.
10. Quantizing errors can be reduced by _____.
 - A. Decreasing the sample rate
 - B. Decreasing the number of bits in a sample
 - C. Increasing the sample rate
 - D. Increasing the number of bits in a sample
11. When a digital signal is first converted to analog, the result is _____.
 - A. A smooth waveform
 - B. A stepped version of an analog waveform
 - C. A combination of analog and digital waves
 - D. Extremely accurate
12. When you are told that a signal uses pulse code modulation, you know that the signal is _____.
 - A. Digital
 - B. Analog
 - C. Attenuated
 - D. Noisy

13. Multiplexing takes advantage of digital signals to _____.
A. Carry a separate channel on each fiber optic cable.
B. Carry analog and digital signals on the same fiber.
C. Carry multiple channels on the same fiber.
D. Carry signals faster than light.
14. If a signal has lost 30 decibels, how much of the original signal is left?
A. 30 percent
B. 0.3 percent
C. 0.01 percent
D. 0.1 percent
15. A zero dBm loss means that the power remaining is _____.
A. 1 watt
B. 0.1 watt
C. 1 milliwatt
D. 0.1 milliwatt

Answers to Review Questions

1. B. The fiber optic link requires a transmitter to send the signal, a receiver to capture the signal, optical fiber to carry the signal, and connectors to provide an interface for the optical fiber and the equipment.
2. D. In order for a signal to be sent through optical fiber, it must first be converted from electrical energy into light by the transmitter.
3. A. When the intelligence is imposed on the carrier, the carrier is changed in a way that allows it to transmit information contained in the intelligence.
4. D. When light has attenuated, it means that it has lost energy, and as a result is not as bright.
5. D. One of the characteristics of analog signals is that they are infinitely variable within any specified range.
6. C. A digital transmission is defined by voltages or light energy being in either a high or low state, which is represented by a one or a zero, respectively.
7. B. Digital transmissions are less likely to be affected by noise, because the only signals recognized by a digital system are zero and one.
8. B. Sampling is the process used to convert analog signals to digital signals by regularly measuring their voltages and converting them to binary numbers.
9. C. A high sample rate reduces the time between samples, enabling more samples to be taken in a given time and producing a more detailed digital version of the waveform.
10. D. Quantizing error occurs because a sample can never have enough bits to represent all of the possible voltages in a specified range. Increasing the number of bits enables more binary numbers to be used in that range.
11. B. When voltages are created from a digital signal, each voltage matches the value and length of the sample, resulting in a stepped waveform.
12. A. Pulse code modulation is the transmission of binary words in the order they were generated.
13. C. Multiplexing takes advantage of the high transmission capacities available in optical fibers to carry multiple channels almost simultaneously on a single fiber.
14. D. The decibel scale is logarithmic, so each 10-decibel loss divides the percentage of remaining power by another factor of 10.
15. C. The abbreviation dBm means that gain or loss is referenced to a standard 1 milliwatt.



Chapter

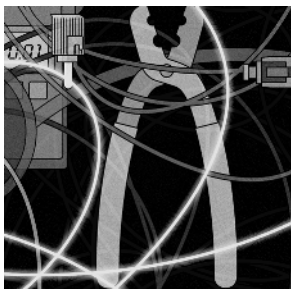
3

Basic Principles of Light

OBJECTIVES COVERED IN THIS CHAPTER:

✓ Basic Principles of Light

- Describe light as electromagnetic energy.
- Convert various wavelengths to corresponding frequencies.
- Describe light as particles and waves.
- Describe the electromagnetic spectrum and locate light frequencies within the spectrum in relation to radio and microwave communication frequencies.
- Describe the refraction of light.
- Explain how the index of refraction is used to express the speed of light through a transparent medium.
- Explain reflection to include angle of incidence, critical angle, and angle of refraction.
- Explain Snell's law and how it is used to calculate the critical angle of incidence.
- Explain Fresnel reflections and how they can impact the performance of a fiber optic communication system.



Even the simplest fiber optic link is a triumph of innovative design, manufacturing precision, and technical creativity. One of the factors contributing to the high data rates and long transmission distances of fiber optics is knowledge of the principles of light. While the optical fiber is the transmission medium, light is the carrier on which the signals are imposed. Its small wavelengths and high transmission velocity make possible bandwidths that would be unimaginable using other forms of transmission. In order to get the best performance from every part of the fiber optic link, it is important to understand the characteristics of light and the factors within the fiber optic link that affect it.

This chapter describes the basic characteristics of light, especially the type of light used in fiber optic communications. It also discusses the principles and materials that make fiber optic communications possible, along with some of the problems that must be overcome.

Light as Electromagnetic Energy

Whether it comes from the sun, an electric bulb, or a laser, all light is a form of electromagnetic energy. Electromagnetic energy is emitted by any object that has a temperature above absolute zero (-273°C), which means that the atoms in the object are in motion. The electrons orbiting the atoms pick up energy from the motion, and the energy causes them to move to higher orbits, or energy levels. As they drop back to their original energy levels, they release the energy again. The energy takes two forms: an electrical field and a magnetic field, formed at right angles to each other and at right angles to their path of travel, as shown in Figure 3.1.

FIGURE 3.1 The three-dimensional nature of electromagnetic energy

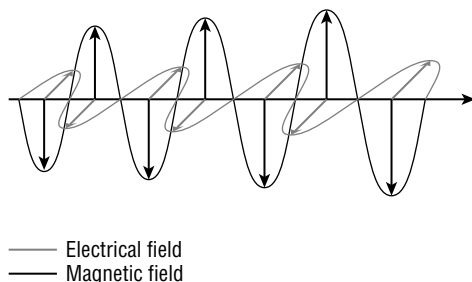
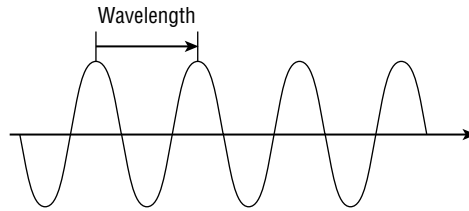


FIGURE 3.2 Electromagnetic energy is often shown as a sine wave.

The combination of these electrical and magnetic fields is an electromagnetic wave, which travels through open space or air at approximately 300,000 km/s (kilometers per second). Although the wave exists three-dimensionally, it is often represented as a two-dimensional sine wave, as shown in Figure 3.2. It is important to understand the three-dimensional nature of electromagnetic waves because different types of fiber optic transmission take advantage of it, as will be discussed in later chapters.

One of the most important characteristics of electromagnetic energy is its wavelength. Wavelength is the distance between corresponding points on two consecutive waves, and it defines the different ways in which radiation behaves at different wavelengths. Depending on its wavelength, electromagnetic energy can occur as radio waves, light waves, X-rays, and more.

Wavelength is important because it defines the electromagnetic energy's frequency, which is the number of waves that pass a given point in one second. Frequency is measured in cycles per second, or hertz (Hz).

A way to express the relationship between wavelength and frequency is that wavelength equals the velocity of the wave (v) divided by its frequency (f), or

$$\lambda = v \div f$$

Note that v is usually the speed of light in vacuum or open air, approximately 300,000 km/s.

Likewise, you can calculate the frequency using the wavelength with the equation:

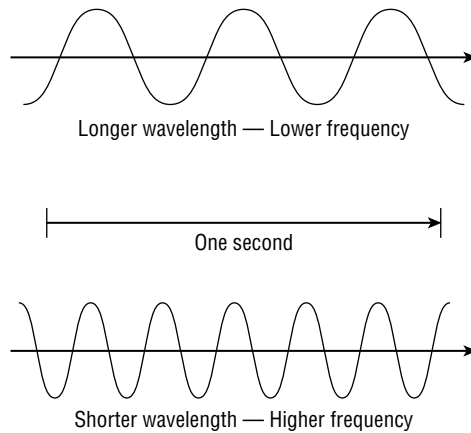
$$f = v \div \lambda$$

For example, one of the infrared light frequencies used in fiber optics has a wavelength of 850 nanometers (billionths of a meter), which translates to a frequency of 352.9 terahertz (THz), or 352.9 trillion hertz.

Note in Figure 3.3 that as the wavelength (λ) becomes smaller, more waves will occur in one second, which means that the frequency will increase as wavelength decreases.

To illustrate this, an AM radio wavelength of 790 kilohertz (kHz), or 790,000 Hz, has a wavelength of 379 meters. If we choose a higher frequency, such as 850 kHz, the wavelength goes down to 353 meters.

FIGURE 3.3 As wavelength decreases, frequency increases.



Some Useful Terms

Many of the terms used to express electromagnetic wavelengths and frequencies describe large multiples of cycles or small fractions of meters. To understand these terms, it is helpful to know the prefixes used with them.

Each prefix expresses a multiple of 10 or one divided by a multiple of 10 (as in $1/1000$) applied to the measurement unit, such as meters or cycles. Some terms may already be familiar, such as *kilometer*, for a thousand meters, or *centimeter*, for a hundredth of a meter.

Below is a list of prefixes, along with their translations and mathematical equivalents, in descending order of magnitude.

Prefix	Meaning	Magnitude
tera-	Trillion	10^{12}
giga-	Billion	10^9
mega-	Million	10^6
kilo-	Thousand	10^3
centi-	Hundredth	10^{-2}
mili-	Thousandth	10^{-3}
micro-	Millionth	10^{-6}
nano-	Billionth	10^{-9}
pico-	Trillionth	10^{-12}

Electromagnetic radiation, like most radiated energy, has characteristics of both waves and particles. As a wave, it propagates through a medium and transfers energy without permanently displacing the medium. However, light particles, called *photons*, can be emitted by light-emitting diodes (LEDs) and collected by photodetectors.

A photon, which is emitted from an electron as it changes energy levels, is the basic unit, or *quantum*, of electromagnetic energy. The amount of energy in each photon, however, depends on the electromagnetic energy's frequency: the higher the frequency, the more energy in the particle.

To express the amount of energy in a photon, we use the equation

$$E = hf$$

where E is the energy expressed in watts, h is Planck's constant, or 6.626×10^{-34} joule-seconds, and f is the frequency of the electromagnetic energy.

So, to find the energy of a photon of infrared light at 352.9 THz:

$$(3.529 \times 10^{14})(6.625 \times 10^{-34}) = 2.338 \times 10^{-19} \text{ w}$$



Much of what we take for granted in the field of fiber optics comes from work done by pioneers in the field of physics in the late 1800s and early 1900s. The *joule* is a unit of energy named for James Prescott Joule, who studied the relationship between heat and mechanical work. One joule is equal to the amount of work required to produce 1 watt in 1 second. Planck's constant was defined in 1899 by Nobel Laureate Max Planck, who is regarded as the founder of quantum theory.

The Electromagnetic Spectrum

In 1964, scientists at a Bell Laboratories facility in New Jersey accidentally discovered electromagnetic radiation associated with the very beginnings of our universe. The radiation, predicted by certain cosmological theories, was emitted by hydrogen atoms at a temperature just 3° C above absolute zero, and is practical evidence that anything with a temperature above absolute zero puts out electromagnetic energy.

The wavelengths emitted by the cosmic hydrogen atoms, about 2.5 mm to 0.5 mm, are in the *microwave* range of the electromagnetic spectrum, just above radio waves in frequency, and just below infrared light.

As with sound, some characteristics of electromagnetic energy are dependent on wavelength. Longer wavelengths require less energy for propagation than shorter wavelengths of the same amplitude, making them useful for long distance communication. By the same rule, particles of higher frequency electromagnetic radiation have more energy than particles from lower frequency emissions of the same amplitude. This fact is important when determining which wavelengths to use in fiber optic transmissions.

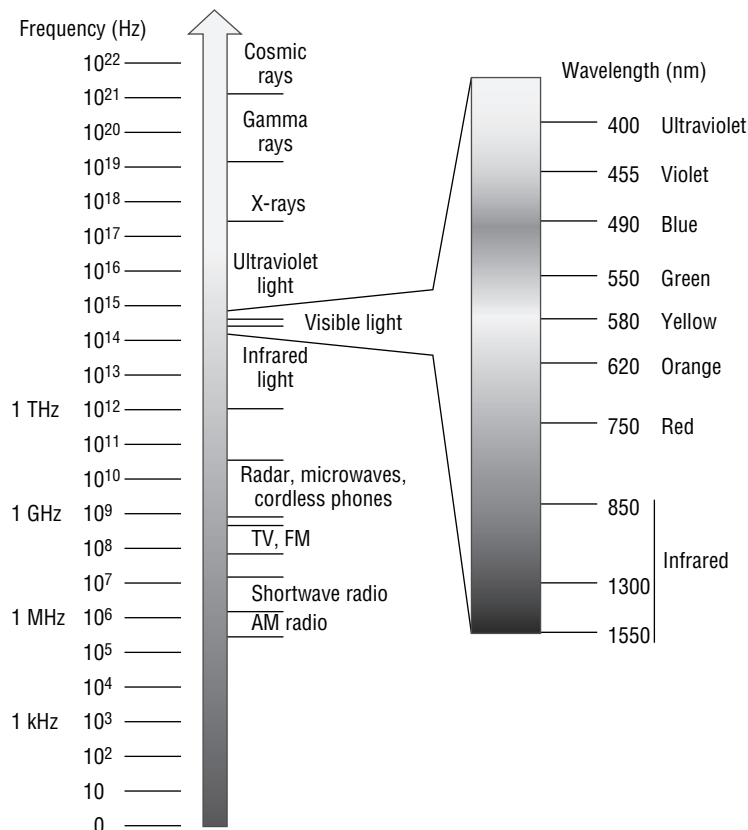
These transmissions consist of turning the carrier, or the light, on and off at high switching rates. At the higher rates desired for communications, higher frequencies of light are needed if an extremely short “on” cycle is to have enough energy to be detected.

Remember also that higher frequencies can carry more data, because more waves per second allow more bits per second to be carried. This fact extends the principle that had been applied first to radio and then to television, which relied on ever higher frequencies to carry greater amounts of information.

What we commonly call light is just one small part of the electromagnetic spectrum, shown in Figure 3.4. Visible light is an even smaller component, bordered by infrared, with longer wavelengths, and ultraviolet, with shorter wavelengths.

The wavelengths most commonly used for fiber optics are in the infrared range, at *windows* of 850 nanometers (nm), 1300 nm, and 1550 nm. The spectrum range of these wavelengths provides an important combination of characteristics: it is high enough to make high data rates possible, but low enough to require relatively low power for transmission over long distances.

FIGURE 3.4 The electromagnetic spectrum



The specific wavelength windows have been selected because they provide the best possible characteristics for transmission. Even within the range between 850 nm and 1550 nm, certain regions have high losses due to materials in the fiber, such as stray water molecules, absorbing light at a wavelength of 1380 nm. Other wavelengths, such as 1550 nm, are favored because they have a low loss, allowing longer transmission distances. On the other hand, wavelengths near 1300 nm suffer less from *dispersion*, which will be discussed at length in a later chapter.

Without specially manufactured optical fibers, system designers must often choose the characteristics that are most important to their needs and create a system accordingly.



Windows were created as a way of standardizing useful wavelengths in fiber optics, making it easier to build interoperable equipment.

Refraction

Refraction, or the bending of light as it passes from one material into another, is a key component in fiber optic transmissions. The principles that cause an object in water to look like it is bent, as seen in Figure 3.5, are the same principles that keep light contained within the core of an optical fiber even through curves.

FIGURE 3.5 Refraction of light through water

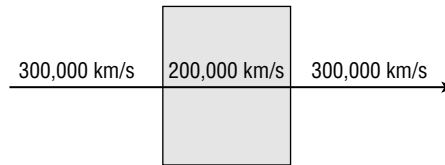


What Causes Refraction?

Refraction occurs when light waves change speed as they cross the boundary between two materials with different optical densities. We commonly think of the speed of light as constant at about 300,000 km/s (299,792.458 km/s, to be precise), but that figure is actually the theoretical top speed of light, which only applies to its speed through the vacuum of space. In reality, light travels at lower velocities through various materials or *media* such as the earth's atmosphere, glass, plastic, and water. A medium's optical density, which is different from its physical density, determines how quickly light passes through it.

It is important to note that even though light slows down when passing from a less dense medium to a denser one, it speeds up again when it passes into a less dense medium (see Figure 3.6).

FIGURE 3.6 Light speeds up when it passes into a less dense medium.



Even though light has less velocity in a medium than it does in a vacuum, its frequency stays the same.

When light changes velocity as it crosses the boundary between two media at an angle, it also bends, or refracts. The amount of refraction is based on the relative difference in the light's velocity through each of the materials. The greater the difference, the greater the refraction.

In Figure 3.5, shown earlier, the portion of the oar in the water looks bent because the light rays reflected from it are bent as they pass through the boundary between the water and the air, as seen in Figure 3.7. Notice also that the portion of the oar that is underwater is distorted because the light rays reflected from it have been bent.

To explain refraction, we have to look at the wave nature of light. Recall that light consists of two perpendicular waves, and that the light travels in a path that forms right angles with both waves.

As each wave meets the boundary layer with a medium of a higher optical density, the portion of the wavefront that hits the boundary first will slow down, while the rest of the wavefront maintains its original velocity, as shown in Figure 3.8. Each portion of the wave that crosses the boundary continues in its original direction, but the rest of the wave, which is moving at higher velocity, has traveled farther before reaching the boundary layer. As a result, the portion of the wavefront that crosses the boundary layer last has traveled a greater distance in the same time than the portion that crossed over first, changing its orientation relative to the boundary layer. Once the entire wave has crossed the boundary, its path, still perpendicular to the wavefront, has changed.

FIGURE 3.7 Light rays from the oar are refracted as they pass into the air.

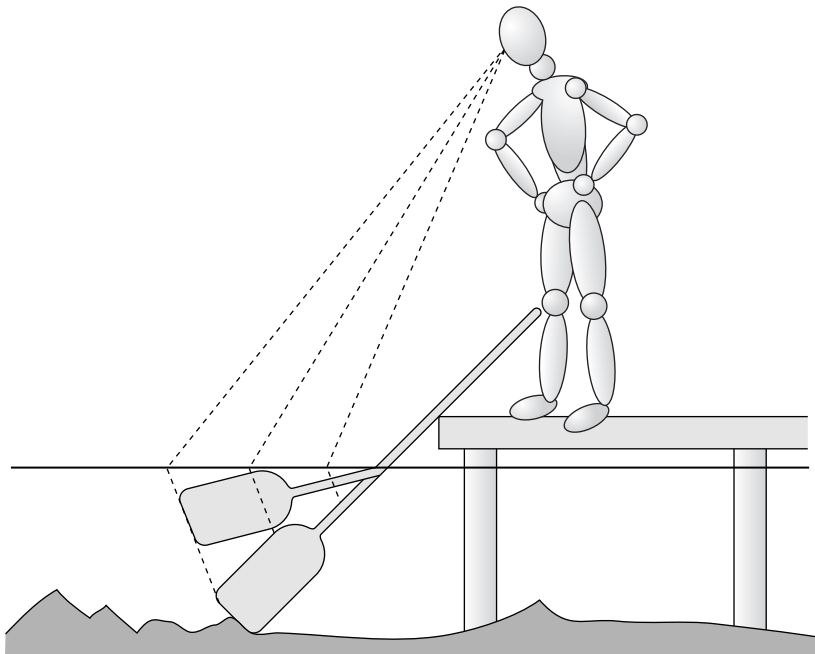
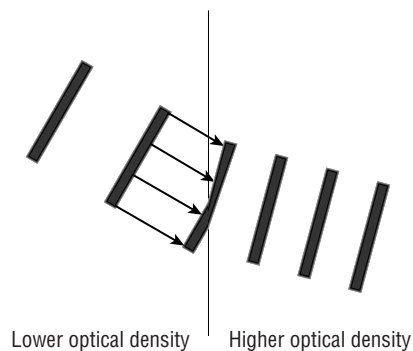


FIGURE 3.8 Arrows indicate distance traveled by the wavefront over the same period.

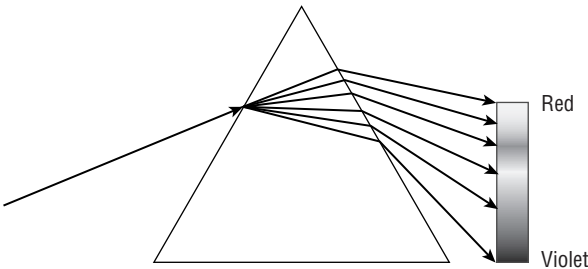


The velocity of light through different media also depends on the light's wavelength. Shorter wavelengths will travel more slowly through a medium than longer wavelengths. Because they travel more slowly, the shorter wavelengths will also be refracted more than the longer ones.

One of the most common ways to observe this is by seeing how white light is refracted through a prism and broken up into its component wavelengths, as shown in Figure 3.9. Notice

that violet, which has the shortest visible wavelength, is refracted more than red, which has the longest visible wavelength.

FIGURE 3.9 Refraction of white light into component colors



Because we know the velocity of light through certain materials, we can assign each a relative value, or *refractive index (RI)*, based on the velocity of light through a vacuum, which has an RI of 1.

TABLE 3.1 Refractive Indices of Different Materials

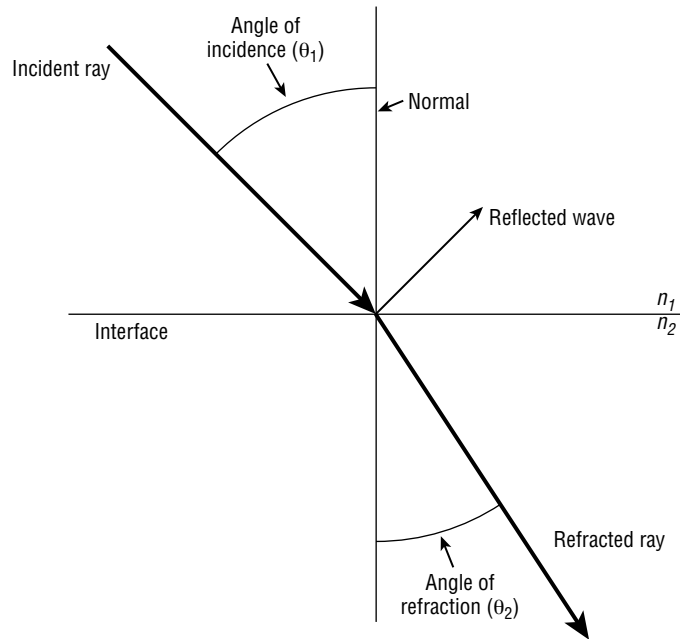
Material	C (Speed of Light)	Refractive Index
Vacuum	300,000 km/s	1.0000
Air	300,000 km/s	1.0003
Water	225,056	1.333
Ethyl alcohol	220,588	1.36
Optical fiber cladding (typ.)	205,479	1.46
Optical fiber core (typ.)	204,082	1.47
Glass	200,000	1.5

We can calculate the refractive index with the equation

$$n = c/v$$

where n is the refractive index of the material, c is the velocity of light through vacuum, and v is the light's velocity through the material. So if light passes through a theoretical material at 210,000 km/s, the material's RI is

$$n = 300,000 \div 210,000 = 1.43$$

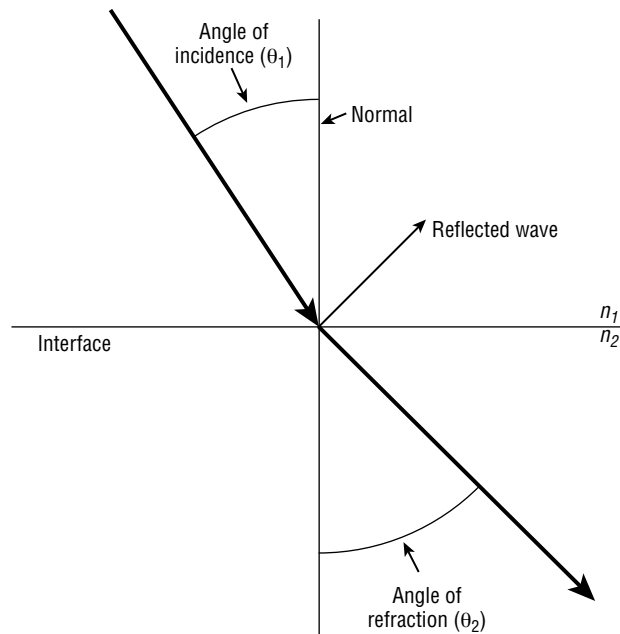
FIGURE 3.10 Model used to calculate refraction

To calculate the amount of refraction that will take place when light passes from one material to another, we'll need a model and some basic terms. The model, shown in Figure 3.10, illustrates light passing from a medium with a lower RI (n_1) to a medium with a higher RI (n_2). The boundary layer, or *interface*, is represented by a horizontal line. A path perpendicular to the interface is known as *normal*. Normal is the only condition in which the light will not refract, because all portions of the wavefront cross the boundary at the same time.

The *angle of incidence* represents the angle between the incident ray and normal. The *angle of refraction* represents the angle between the refracted ray and normal. Note that a small amount of light is also reflected from the interface at an angle equal to the angle of incidence.

Total Internal Reflection

As shown earlier in Figure 3.10, light passing from a lower RI to a higher RI refracts toward normal. When light passes from a higher RI to a lower RI, as shown in Figure 3.11, the light refracts away from normal.

FIGURE 3.11 Light passing from higher RI to lower RI refracts away from normal.

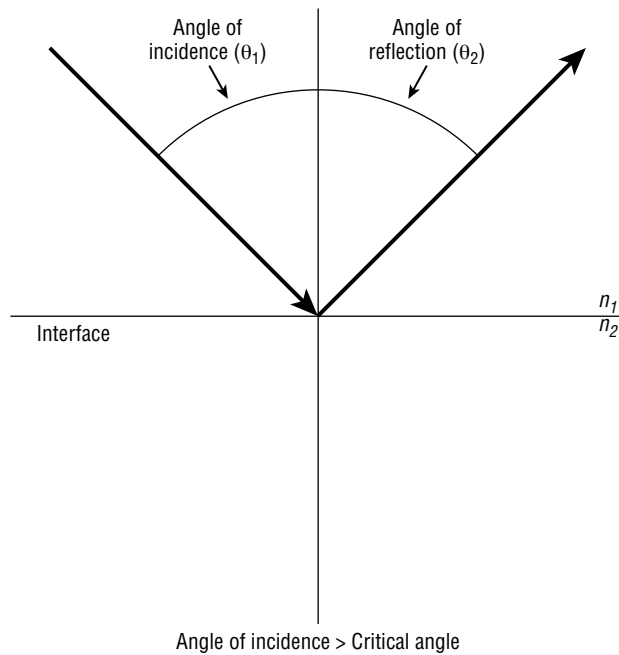
We can calculate the amount of refraction using Snell's law, which shows the relationship between incident radiation and refracted rays:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

where n_1 and n_2 are the RI values of each material, θ_1 is the angle of incidence, and θ_2 is the angle of refraction.

Recall that the phenomenon that makes fiber optic transmission possible, *total internal reflection*, is caused by the same principles that cause refraction. In the case of TIR, the light is passing from a higher RI to a lower RI at an angle that causes all of the light to be reflected. What has happened is that the angle of refraction has exceeded 90° from normal.

The incident angle required to produce a refracted angle of 90° is called the *critical angle*. As the incident ray moves from normal toward the critical angle, less and less of the incident ray's energy is carried into the refracted ray. At the critical angle, all of the incident ray's energy is refracted along the interface. As the incident angle exceeds 90° , the light is reflected, as shown in Figure 3.12.

FIGURE 3.12 Reflection of light exceeding critical angle

To find the critical angle of two materials, we can use a modified version of Snell's equation:

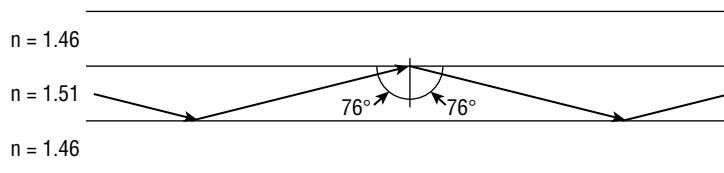
$$\theta_c = \arcsin(n_2 \div n_1)$$

where θ_c produces a refracted angle of 90° from normal.

So if we want to know the critical angle of an optical fiber having a core RI of $n_1 = 1.51$ and a cladding RI of $n_2 = 1.46$:

$$\theta_c = \arcsin(1.46 \div 1.51) = 75.211^\circ$$

So, if light is passing through the core at an angle greater than 75.211° , it will be reflected from the interface with the cladding. As long as the interface remains parallel, the light will continue to reflect at the same angle, as shown in Figure 3.13.

FIGURE 3.13 Total internal reflection

EXERCISE 3.1

Calculate the critical angle of an optical fiber with a core RI of 1.48 and a cladding RI of 1.46.

1. Divide the cladding RI by the core RI:

$$1.46 \div 1.48 = 0.986$$

2. Next, find $\arcsin 0.986$:

$$\text{Arcsin} 0.986 = 80.401$$

The critical angle for this combination of core and cladding is 80.401°.

Fresnel Reflections

Recall that when light passes from one medium into another, most of it refracts according to Snell's law, but a small part of it, known as *Fresnel reflection*, is reflected at an angle equal to the angle of incidence. The greater the difference in RI between the two materials, the more light will be reflected. You experience Fresnel reflection whenever you look through a window and see a faint reflection of yourself in the glass.

The reflected light must be considered a loss when dealing with some areas of fiber optics. One such area is connections, where a small air gap exists between fiber ends. At this point, light passing from the air, with an RI of 1.0003, to the fiber, with an RI of about 1.5, could reflect enough to create a noticeable loss. The same loss occurs as the light passes from fiber to air.

Augustin Fresnel determined how to calculate the amount of light lost through Fresnel reflection when light passes from one medium into another with the following equation:

$$\rho = ((n_1 - n_2) \div (n_1 + n_2))^2$$

where ρ is the amount of light reflected and n is the RI of the medium.

If we calculate the reflection value for the interface between air, with an RI of 1 and glass with an RI of 1.5,

$$\rho = ((1.5 - 1) \div (1.5 + 1))^2 = 0.2^2 = 0.04$$

To calculate the loss in decibels, we use the equation

$$\text{dB} = 10\text{Log}_{10} (1 - \rho)$$

plugging in the Fresnel reflection value from above,

$$\text{dB} = 10\text{Log}_{10} 0.96 = 10 \times -0.018 = -0.18$$

which gives us a loss of 0.18 dB.

EXERCISE 3.2

Calculate the dB loss due to Fresnel reflection of a light from the air entering a fiber core with an RI of 1.48.

1. Determine the value of ρ , the amount of light reflected.

$$\rho = ((1.48 - 1) \div (1.48 + 1))^2 = 0.0375$$

2. Next, determine the dB loss based on the value of ρ .

$$\text{dB} = 10\text{Log}_{10}0.9625 = -0.166$$

**Real World Scenario****Making the Connection**

Even in the best of worlds, connectors and other junctions between fibers cannot always match up perfectly, and a small air gap could create Fresnel reflections. If enough of these connections occur in a line, the losses due to Fresnel reflection could be significant, especially if the fiber is running over long distances.

To reduce Fresnel reflections in connections, installers rely on *index matching gel*, a substance that fills the air space within a connection with a material matching the RI of the fiber core.

Summary

This chapter covered the properties of light and their effects on fiber optic transmission. We explored the nature of light as an electromagnetic wave and as a moving particle, including its generation and transmission.

We also described the characteristics of light, including speed, frequency, and wavelength, that influence its behavior as it passes through different materials. In addition, we discussed the principles and equations that can be used to predict light's refraction, reflection, and loss through optical fibers.

Exam Essentials

Understand light as electromagnetic energy. Make sure you understand the nature of light as a form of electromagnetic energy. Also, be able to describe light as a three-dimensional wave with motion, and its characteristics as both a wave and a particle. Understand the relationship between frequency and wavelength.

Understand the electromagnetic spectrum. Make sure you understand the makeup of the electromagnetic spectrum. Be able to associate wavelength and frequency ranges with different types of electromagnetic energy. Understand the position of light, including infrared and ultraviolet, on the spectrum.

Understand refraction. Make sure you understand what refraction is and what causes it. Also, be able to explain how the refractive index is used to express the speed of light through a transparent medium. Be able to explain reflection in terms of angle of incidence, critical angle, and angle of refraction.

Know what Snell's law is. Be able to explain Snell's law as it applies to fiber optic systems. Be able to use Snell's law to determine refraction and critical angle.

Understand Fresnel reflection. Be able to describe Fresnel reflection and the effect it has on fiber optic systems. Be able to calculate loss in decibels due to Fresnel reflection.

Review Questions

1. What relation do the electrical and magnetic waves and the direction of travel have in light?
 - A. They are parallel to one another.
 - B. They are at right angles to one another.
 - C. They are multiples of one another.
 - D. They are unrelated.
2. If an electromagnetic wave has a frequency of 800 kHz, what is its wavelength?
 - A. 375 m
 - B. 3.75 m
 - C. 800 m
 - D. 8 km
3. One of light's characteristics is that it can travel through a medium without permanently displacing it, and it travels in the form of "photons." Because of this, we must consider light to:
 - A. Be neither a wave nor a particle.
 - B. Be primarily a wave.
 - C. Have characteristics of both a wave and a particle.
 - D. Be primarily a particle.
4. At what point does an object no longer put out electromagnetic energy?
 - A. When it becomes a black body
 - B. When its temperature is absolute zero (-273°C)
 - C. When it drops below microwave levels
 - D. When it begins absorbing electromagnetic energy
5. Which of the following forms of electromagnetic energy has a lower frequency than the other three?
 - A. Infrared light
 - B. Visible light
 - C. Gamma rays
 - D. Microwaves
6. Which of the following wavelengths would you choose for the best combination of high data rate and low power requirement?
 - A. 1300 nm
 - B. 620 nm
 - C. 12 m
 - D. 375 m

7. Refraction is caused by
 - A. Light changing color as it passes from one medium into another
 - B. Light changing speed as it passes from one medium into another
 - C. Light changing frequency as it passes from one medium into another
 - D. Light changing direction within a single medium
8. If light is passing through a medium with a refractive index of 1.27, what will the velocity of the light through the medium be?
 - A. 381,000 km/s
 - B. 300,000 km/s
 - C. 299,998 km/s
 - D. 236,220 km/s
9. If a beam of light passing from medium A to medium B refracts toward normal, what do we know about the two media?
 - A. Medium A is thicker than medium B.
 - B. Medium A has a higher RI than medium B.
 - C. Medium A has a lower RI than medium B.
 - D. Medium A has an RI of 1.
10. If we want to create total internal reflection, we have to pass light from
 - A. A lower RI to a higher RI
 - B. A higher RI to a lower RI
 - C. An RI of 1 to a higher RI
 - D. A higher RI to an RI of 1
11. The critical angle is the angle of incidence required for two materials to produce a refracted angle of ____.
 - A. 180°
 - B. 0°
 - C. 45°
 - D. 90°
12. Calculate the critical angle of two materials having RIs of 1.48 and 1.52.
 - A. 82.838°
 - B. 79.217°
 - C. 76.826°
 - D. 75.913°

13. If the critical angle for two media is 78.895° , what is the nearest angle of incidence that will produce total internal reflection?
- A. 79°
 - B. 82°
 - C. 78°
 - D. 75°
14. Fresnel reflection occurs when
- A. The light is at the critical angle.
 - B. A part of the incident ray is reflected from the interface.
 - C. A part of the refracted ray is reflected from the interface.
 - D. The reflected light changes speed.
15. Which part of a fiber optic system can suffer due to Fresnel reflection?
- A. Transmitter
 - B. Receiver
 - C. Connector
 - D. Optical fiber

Answers to Review Questions

1. B. The waves form right angles to each other and travel in a path that forms a right angle to both waves.
2. A. By dividing the speed of the wave, 300,000 km/s, by the frequency, 800 kHz, we arrive at the wavelength.
3. C. Because light travels as a wave, and because in its emission and detection it behaves as a particle, we treat light as having characteristics of both.
4. B. When an object reaches absolute zero, all motion in its atoms stops. Because it is the motion of the atoms that gives electrons the energy to release electromagnetic waves, the generation of electromagnetic energy requires the temperature to be above absolute zero.
5. D. On the electromagnetic spectrum, the choices would run, in descending frequency: gamma rays, visible light, infrared light, microwaves.
6. A. Considering that longer wavelengths require less power for their amplitude to carry long distances, and that smaller wavelengths can carry higher data rates, it is important to find a compromise. The 1300 nm wave corresponds to one of the infrared windows used in most fiber optics transmissions, and represents an efficient combination of power use and data rate.
7. B. When light passes the boundary between two media, the change in its speed can cause the light to bend, or refract, at a predictable angle.
8. D. To find the speed of light in a material with a known refractive index, we divide the speed of light in material with RI of 1, or 300,000 km/s, by the medium's RI—in this case, 1.27.
9. C. If the light refracts toward normal, we know that the light has slowed down upon entering medium B, so we know that medium A has a lower RI than medium B. We cannot know what the RI of medium A is unless we know the angle of incidence, the angle of refraction, and the RI of medium B.
10. B. Total internal reflection takes place when light passing through a medium with a higher RI hits the boundary layer of a medium with a lower RI at a shallow angle, causing it to reflect from the boundary layer.
11. D. The critical angle is the angle of incidence that produces a refracted ray that goes into the boundary layer, which is 90° to normal.
12. C. The critical angle can be found by dividing the lower RI by the higher RI, then finding the arc-sin of the result.
13. A. As long as the angle of incidence is greater than the critical angle, all of the light will be reflected.
14. B. Fresnel reflection occurs whenever refraction takes place, resulting in the loss of a small portion of the light.
15. C. Fresnel reflection can occur in the small air gap at connectors, causing a loss of light energy.



Chapter

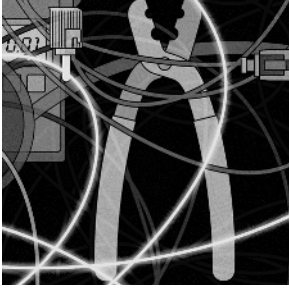
4

Optical Fiber Construction and Theory

OBJECTIVES COVERED IN THIS CHAPTER:

✓ Optical Fiber Construction and Theory

- Describe the basic parts of an optical fiber.
- List the different materials that can be used to construct an optical fiber.
- Describe the tensile strength of an optical fiber.
- Describe optical fiber manufacturing techniques.
- Describe total internal reflection.
- Describe mode in an optical fiber.
- Describe the three refractive index profiles commonly found in optical fiber.
- Explain the propagation of light through a multimode step index optical fiber.
- Explain the propagation of light through a multimode graded index optical fiber.
- Explain the propagation of light through a single-mode optical fiber.
- Describe dispersion shifted single-mode optical fiber.
- Describe the TIA/EIA-568-B.3 recognized multimode optical fibers.
- Describe the TIA/EIA-568-B.3 recognized single-mode optical fibers.
- Describe commercially available PCS and HCS optical fiber.
- Describe commercially available plastic optical fiber.



Optical fibers are called on to operate in a wide variety of conditions. Some of them must carry high volumes of data over many kilometers, while others carry smaller amounts of data inside an office building or aboard a ship. The type of job an optical fiber will do determines the type of fiber you'll choose to run. It is important to understand the types of fibers that are available and the ways in which they are built so that you can select and use them properly.

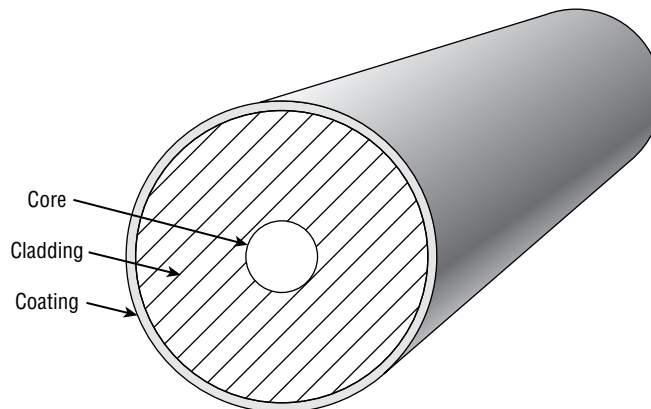
This chapter describes the construction of optical fibers and the components that make them up. We will discuss some of the important factors that must be considered in the manufacture and use of optical fibers, as well as the designs used to optimize them for different types of data transmission. Finally, the chapter will introduce you to some of the commonly used commercial optical fibers and describe their features.

Optical Fiber Components

Today's standard optical fiber is the product of precision manufacturing techniques and exacting standards. Make no mistake: even though it is found in almost any data or communications link, optical fiber is a finely tuned instrument requiring care in its production, handling, and installation.

As shown in Figure 4.1, a typical optical fiber comprises three main components: the core, which carries the light; the cladding, which surrounds the core with a lower refractive index and contains the light; and the coating, which protects the fragile fiber within.

FIGURE 4.1 Optical fiber components include the core, cladding, and coating.



Let's look at these components individually.

Core

The core, which carries the light, is the smallest and most fragile part of the optical fiber. The optical fiber core is usually made of glass, although some are made of plastic. The glass used in the core is extremely pure silicon dioxide (SiO_2), a material so clear that you could look through five miles of it as though you were looking through a household window.

In the manufacturing process, the glass used in the core has impurities such as germanium or phosphorous added to raise the refractive index under controlled conditions.

Optical fiber cores are manufactured in different diameters for different applications. Typical glass cores range from as small as $3.7\text{ }\mu$ up to $200\text{ }\mu$. Core sizes commonly used in telecommunications are $9\text{ }\mu$, $50\text{ }\mu$, and $62.5\text{ }\mu$. Plastic optical fiber cores can be much larger than glass. A popular plastic core size is about $1000\text{ }\mu$.

Cladding

Surrounding and protecting the core, and providing the lower refractive index to make the optical fiber work, is the cladding. When glass cladding is used, the cladding and the core are manufactured together from the same silicon dioxide-based material in a permanently fused state. The manufacturing process adds different amounts of impurities to the core and the cladding to maintain a difference in refractive indices between them of about 1 percent. Typically, the core will have a refractive index of 1.48, while the cladding will have a refractive index of 1.46.

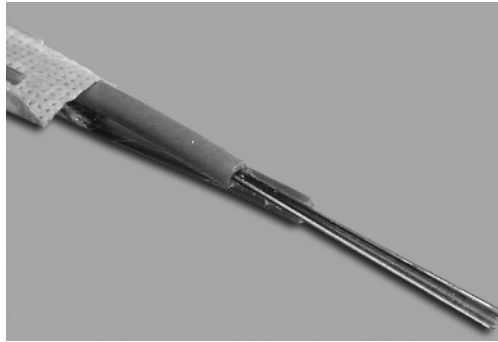
Like the core, cladding is manufactured in standard diameters. The two most commonly used diameters are $125\text{ }\mu$ and $140\text{ }\mu$. The $125\text{ }\mu$ cladding typically supports core sizes of $9\text{ }\mu$, $50\text{ }\mu$, $62.5\text{ }\mu$, and $85\text{ }\mu$. The $140\text{ }\mu$ cladding typically has a $100\text{ }\mu$ core.

Coating

The coating is the true protective layer of the optical fiber. Generally made of plastic or acrylate, the coating absorbs the shocks, nicks, scrapes, and even moisture that could damage the cladding. Without the coating, the optical fiber is very fragile. A single microscopic nick in the cladding could cause the optical fiber to break when it's bent. Coating is essential for all-glass fibers, and they are not sold without it.

The coating is solely protective. It does not contribute to the light-carrying ability of the optical fiber in any way. The outside diameter of the coating is typically either $250\text{ }\mu$ or $500\text{ }\mu$. The coating is typically colorless. In some applications, however, the coating is colored, as shown in Figure 4.2, so that individual optical fibers in a group of optical fibers can be identified.

FIGURE 4.2 The fiber coating can be color-coded to help identify it.



Standards

While many combinations of core and cladding sizes are possible, standards are necessary to ensure that connectors and equipment can be matched properly. This is especially important when dealing with components as small as those used in fiber optics, where even slight misalignments can render the entire system useless.

The Telecommunications Industry Association (TIA) and the Electronic Industries Alliance (EIA) recognize certain standard core/cladding size combinations in their TIA/EIA-568-B.3 standard.

Optical fibers are commonly described by a pair of numbers representing the core diameter and cladding diameter in microns. For example, a fiber with a core diameter of $50\ \mu$ and a cladding diameter of $125\ \mu$ is classified as a 50/125. This size is referred to as the European Standard, while the 62.5/125 is referred to as the North American Standard.

Section 4.3 of TIA/EIA-568-B.3 states that the optical fiber cable construction shall consist of 50/125 μ or 62.5/125 μ multimode optical fiber or single-mode optical fibers, or a combination of these media. TIA/EIA-568-B.3 does not define core or cladding diameters for single-mode optical fibers. However, virtually all single-mode fiber has a cladding diameter of $125\ \mu$. Figure 4.3 shows the comparative sizes of various fiber cores and cladding.

Materials

Optical fibers are commonly made with a glass core and glass cladding, but other materials may be used if the fiber's performance must be balanced with the cost of installing the fiber, fitting it with connectors, and ensuring that it is properly protected from damage. In many cases, fibers must run only a short distance, and the benefits of high-quality all-glass fibers become less important than simply saving money. There are also circumstances in which the fibers are exposed to harsh conditions, such as vibration, extreme temperature, repeated handling, or constant movement. Different fiber classifications have evolved to suit different conditions, cost factors, and performance requirements. The major fiber classifications by material are:

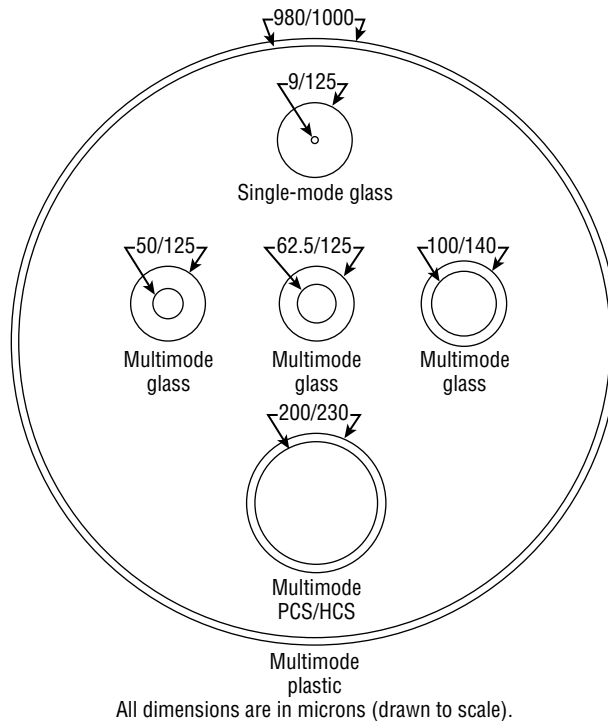
Glass fibers These have a glass core and glass cladding. They are used when high data rates, long transmission distances, or a combination of both are required. As you will see in later chapters, connectors for glass fibers must be built with a high degree of precision and can take time to apply correctly. Glass fibers are the most fragile of the various types available, and as a result they must be installed in environments where they will not be subjected to a great deal of abuse, or they must be protected by special cables or enclosures to ensure that they are not damaged.

Glass fibers are commonly found in long-distance data and voice lines, high-quality video and voice transmission systems, and interbuilding and interoffice networking applications.

Plastic-clad silica (PCS) These fibers have a glass core and plastic cladding. The core is larger than all-glass fiber, but the cladding is much thinner in relation to the core. A typical PCS fiber specification would be 125/150. The plastic cladding also serves as a protective layer, so the coating normally found on all-glass fiber is not included on PCS fibers. PCS fiber is designed to provide a compromise between the high performance signal-carrying ability of glass fibers and the durability, low cost, and relatively easy handling of plastic fibers.

These fibers are used in areas where a moderate degree of ruggedness, low cost, and high performance over short distances are important. Examples include industrial controls, automotive systems, and medical monitoring systems. PCS fibers are also commonly used in consumer electronics and home networking.

FIGURE 4.3 Comparison of fiber core and cladding diameters



Hard-clad silica (HCS) These fibers are similar to PCS fiber but they have a glass core with cladding made of a hard polymer or other material, typically stronger than other cladding materials. Hard-clad silica fiber is commonly used in locations where ruggedness is a prime consideration, such as manufacturing, aircraft, and other locations where shock and vibration would render standard glass fibers unreliable.

Plastic fiber These fibers have a plastic core and plastic cladding. They are selected for their low cost, ruggedness, and ease of use, and are installed where high bandwidth and long transmission distances are not required. While plastic fibers are unsuited for long-distance, high-performance transmissions, they can still carry signals with useful data rates over distances of less than 100 m. Plastic fibers are much thicker than glass fibers, often reaching a diameter of 1.0 mm with the cladding. Plastic fiber is used with visible light in the 650 nm range. Some typical locations for plastic fiber include home entertainment systems, telecommunications switching systems, and manufacturing control systems. They may also be used in links between computers and peripherals and in medical equipment.



Real World Scenario

Don't Rule Out Plastic

Superior network performance is a never-ending quest for higher data rates, longer transmission distances, and—let's face it—bragging rights over the sheer quality of the components used. As you look for the best of the best, however, don't forget to take a serious look at plastic optical fiber. No, it doesn't have the prestige of all-glass fiber, nor will it ever carry HDTV across the country. What it does, though, is provide a serious alternative to high-cost, hard-to-handle, fragile glass fibers.

Plastic fibers, with their large core diameters and high attenuation, have often been associated with non-communication applications, such as fiber optics demonstrations, carrying light for illumination to hard-to-reach places, and even in the toy industry as working headlights for model trains and cars. With a work record like that, it's hard to shake a lightweight image and be considered seriously for any kind of communications work.

Consider the following, though. While plastic fiber's high attenuation may rule it out for long-distance use, it is still a strong candidate for use in short-run applications. Additionally, plastic fiber has a number of benefits that render it far superior to glass in some circumstances:

Safety As you will see later, glass fiber requires a number of safety precautions if you will be working with it. The tiny shards of glass that may be removed or that may break off accidentally can become dangerous, nearly invisible splinters. This can be especially important as fiber is used more and more in household applications. Plastic fibers require no such precautions, because their size and their flexibility make them less likely than glass to penetrate the skin, eyes, or mucous membranes.

Handling Glass fibers are more brittle and fragile than their plastic counterparts, and they are extremely small compared to plastic. These facts make them difficult to handle, even for experienced installers. Plastic fibers are tougher, more flexible, and just plain larger than glass fibers, so they can be handled more easily, making installations go more quickly and reducing the risk of damage to the fiber.

Connectors In later chapters, you will learn about the precision process necessary to make a connector for a glass fiber. For now, all you have to know is that it is time-consuming and unforgiving compared to making connectors for plastic fiber. Plastic's large core diameter leaves a great deal of leeway in case the connector assembly is not perfect.

Testing Because glass fibers carry light at wavelengths in the infrared ranges, it is difficult to determine if a fiber is energized without specific test equipment. This can slow down working time and actually endanger your eyes, because the lasers used to drive the infrared light can cause damage without your ever seeing the light. Plastic fibers use light-emitting diodes (LEDs) in the visible light range of about 650 nm, so you can tell immediately if a signal is running through the fiber. In addition, even if you look at the fiber end with the light in it, the LED's power is not great enough to hurt your eyes.

Once you have assessed the job requirements, determine whether plastic fiber will fill the bill in terms of performance. If it will, congratulations—you will be able to work with a convenient type of fiber that is safer, easier to handle, and easier to connect than glass.

Tensile Strength

In addition to the coating, optical fibers may have several layers of shielding, armor, and other materials designed to protect the fiber and keep it together in cables. Like copper, though, optical fiber is still subject to hazards caused by handling, installation, careless digging, and bad weather.

One characteristic of optical fiber that deserves special attention is its tensile strength, or the amount of stretching it can handle.

Tensile strength is important for several reasons. It affects the way fiber must be handled during installation, the amount of curvature it can take, and the way it will perform throughout its lifespan.

To understand optical fiber's tensile strength, let's look at a standard piece of plate glass. To cut the glass, all you have to do is scribe a sharp line through the surface layer. Once the strength of that layer is compromised, the glass snaps easily along the scribe, even if it follows a curve.

Optical fiber follows the same rule. The outer layer of the cladding provides much of the fiber's tensile strength, which is often measured in thousands of pounds per square inch (kpsi). A typical breaking strength for an optical fiber is 100 kpsi. That means that a typical fiber with a cladding thickness of 125 μ can withstand a pull of about 1.9 pounds.

Once the outer layer is scratched or cracked, however, the tensile strength is gone at that location. Like a scribed line, a scratch or crack compromises the integrity of the glass and allows the fiber to break more easily under stress. Scratching or cracking can occur due to mishandling, sharp blows, or bending beyond the fiber's *minimum bend radius*, especially if the bending takes place while the fiber is under tension.

Manufacturing Optical Fiber

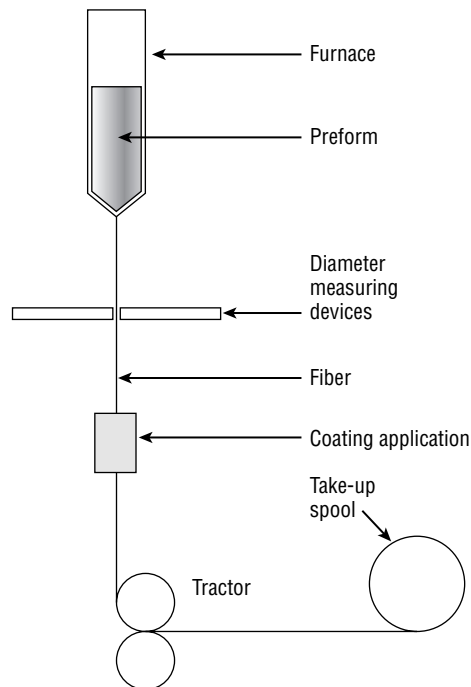
Optical fiber is manufactured to very high standards, because the core diameter and refractive indices of the core and cladding must remain consistent over stretches of up to 80 km.

There are four different methods commonly used to make optical fiber:

- Modified chemical vapor deposition (MCVD)
- Outside vapor deposition (OVD)
- Vapor axial deposition (VAD)
- Plasma chemical vapor deposition (PCVD)

Each method uses variations on a process to create a *preform*, a short, thick glass rod that has a similar cross section to the final fiber. As shown in the block diagram in Figure 4.4, the preform is heated to 2500° C in a drawing tower until it begins to melt and a blob falls from the end, drawing a small thread of glass after it. The thread contains the core and cladding of the optical fiber, their relative thicknesses preset by the preform's construction.

FIGURE 4.4 Drawing the preform into fiber



This thread is taken up by a pulling machine, or *tractor*, at a constant rate to maintain a consistent thickness. A thinner, longer fiber can be created by speeding up the draw rate. The entire process is closely monitored by laser measuring devices to ensure that the thickness remains consistent over the entire length of the fiber, which can be anywhere from 10 to 25 km, depending on the thickness.

The fiber is then drawn through another process, which deposits the coating and makes final measurements before it is taken up on a spool.

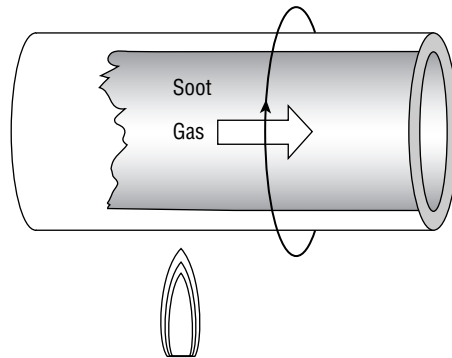
Each of the preceding methods is best suited for different types of fiber, depending on the type of signals it will carry, the distance it will cover, and other factors, which will be discussed later in this chapter.

Let's look at the differences in the manufacturing methods.

Modified Chemical Vapor Deposition (MCVD)

Fibers manufactured using MCVD begin with a hollow glass tube about 1 m long and about 2.5 cm in diameter, as shown in Figure 4.5.

FIGURE 4.5 Modified chemical vapor deposition



The cladding is created first by placing the hollow tube, or *bait*, on a lathe and spinning it rapidly. As the bait is spinning, it is heated by an oxygen/hydrogen torch passing lengthwise underneath as a gaseous mixture of vaporized silicon dioxide is introduced into the tube. The gas, mixed with carefully controlled impurities or *dopants*, forms a soot, which fuses to the inside of bait in successive layers as the heat passes beneath it. The dopants are used to increase the refractive index of the fused material.

Every time the flame passes beneath the tube to heat it, another thin layer of soot adheres to the inside, building a thicker and thicker layer of glass.

The core is formed next by changing the gas/dopant mixture to create a vapor with a higher refractive index. This, too, is introduced into the spinning, heated tube, inside the first layers that were laid down. When the deposited material has reached the desired thickness, the tube is heated to an even higher temperature to consolidate the soot into glass without melting it, a

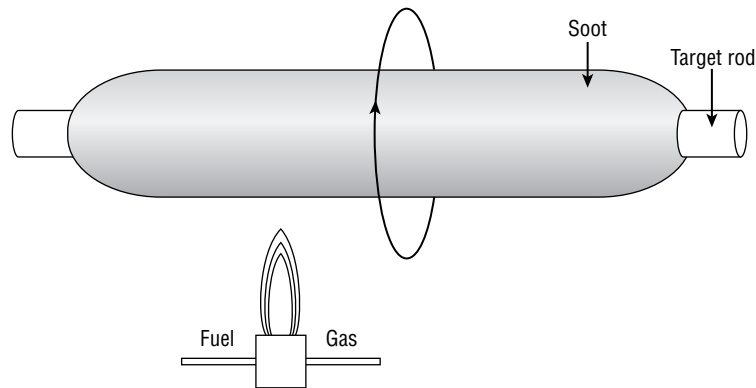
process known as *sintering*. A drying gas is passed through the core to remove water contamination, and the tube is compressed into a preform and taken to the drawing tower.

Outside Vapor Deposition (OVD)

The OVD method is similar to MCVD, except that the fiber preform is built from the inside out.

In the OVD method, a glass target rod serves as the bait, as shown in Figure 4.6. The rod is placed on the lathe and spun as heat is applied to it. The core is laid down first by introducing a gas mixture between the heat and the rod. When the core layer is thick enough, the mixture is changed and the cladding is laid down.

FIGURE 4.6 Outside vapor deposition



After the layers have been deposited, the rod is removed and the layers are dried and sintered, and then collapsed into the preform.

Vapor Axial Deposition (VAD)

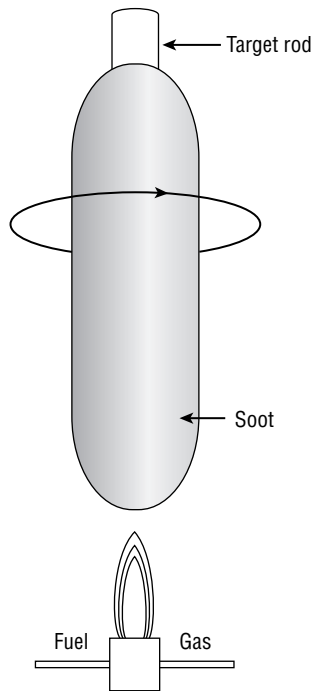
As with MCVD and OVD, the VAD process uses a heated glass bait to collect a soot of silicon dioxide and dopant. With VAD, however, the glass rod is suspended vertically and the heat source is at the lower end, as shown in Figure 4.7.

The gas is introduced between the end of the rod and the heat source, and the soot builds up in a radial pattern. This method gives the manufacturer a great deal of control over not only the refractive index of the various layers, but also the pattern in which they are laid down.

As with OVD, the rod is then removed, and the layers are dried, sintered, and collapsed into the preform before being taken to the drawing tower.

Plasma Chemical Vapor Deposition (PCVD)

The PCVD is similar to MCVD, but allows a finer layer of material to be deposited.

FIGURE 4.7 Vapor axial deposition

In PCVD, the gas particles are heated with microwaves, causing a plasma to form inside the tube. The plasma contains electrons with extremely high energy levels, approximating those found in a gas heated to 60,000° C, even though the temperature inside the tube is no higher than it is with MCVD—about 1200° C. As the high-energy electrons meet the soot forming inside the tube, they transfer their energy to it in the form of heat, causing it to fuse into the final glass form on the tube walls, making the step of consolidation unnecessary.

As with the other methods, the tube can then be compressed and taken to the drawing tower.

Modes

One of the most important characteristics used to distinguish types of fiber is the number of potential paths light can take through it. It may seem that light would go straight through the fiber core, following all of its curves, until it comes out the other end. However, the light itself is a complex combination of electrical and magnetic waves, and their wavelengths can be many times smaller than the core of the fiber. Like a rubber ball shot through a sewer pipe, the light actually has many potential paths, or *modes* it can follow, depending upon the size of the core and the light's angle of entry.

In order to understand how modes operate in optical fibers, let's take another look at the basis for fiber optic transmission, total internal reflection.

Recall that TIR depends on the principle that light passing through a medium of a given refractive index will be reflected off the interface with a medium of a lower refractive index if it hits the interface at or above the critical angle, which is determined by the difference between the refractive indices of the two media.

To satisfy this requirement, optical fiber is manufactured with a core having an RI slightly higher than that of the cladding surrounding it. As a result, light entering the end of the fiber will be reflected off the interface with the cladding, and guided through the fiber. By definition, each reflection will be at the same angle as the angle of incidence.

The number of modes possible in a length of fiber depends on the diameter of the core, the wavelength of the light, and the core's numerical aperture. You can find the numerical aperture using the formula

$$NA = \sqrt{n_{\text{core}}^2 - n_{\text{cladding}}^2}$$

For example, if the core has an RI of 1.48 and the cladding has an RI of 1.46, we can calculate:

$$NA = \sqrt{1.48^2 - 1.46^2} = 0.242$$

Note that the numerical aperture has no dimension. It is intended as a relative indication of the light-gathering capacity of the fiber core.

To find the number of modes, we can use the equation

$$M = (2 \times \pi \times D \times NA/\lambda)^2/2$$

where D is the diameter of the core and λ is the wavelength of the light. For example, if we use our previously derived value for the numerical aperture with a core diameter of 50 μ and a light wavelength of 1300 nm:

$$M = (100 \times 10^{-6} \times 0.242 \times \pi/1300 \times 10^{-9})^2/2 = 1710.07$$

Because light cannot have part of a mode, we must round down to the nearest whole number if we come up with a decimal, so the answer is 1710 modes, or potential paths for the light to follow.

EXERCISE 4.1

Calculate the number of modes in an optical fiber with the following specifications:

Core diameter = 62.5 μ

Core RI = 1.50

Cladding RI = 1.46

Wavelength of light used = 1300 nm

1. First, determine the numerical aperture of the fiber's core.

$$NA = \sqrt{1.50^2 - 1.48^2} = 0.244$$

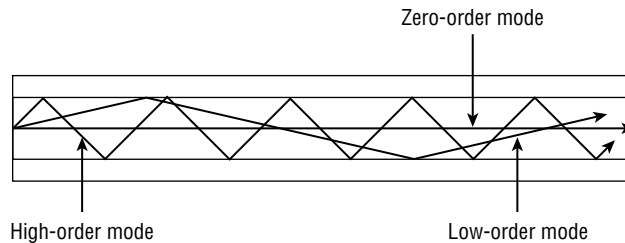
2. Next, use the equation $M = (2D \times \pi \times NA/\lambda)^2/2$ to find the number of modes.

$$M = (125 \times 10^{-6} \times 0.244 \times \pi/1300 \times 10^{-9})^2/2 = 2716.33 \text{ modes}$$

Refractive Index Profiles

What happens when the light rays entering a fiber take slightly different paths, some entering at sharper angles, some at shallower angles? As shown in Figure 4.8, the light can follow modes ranging from a straight line through the fiber (*zero-order mode*) to a low number of reflections (*low-order mode*) to a high number of reflections (*high-order mode*). Depending on a fiber's construction, it can allow only one mode, or more than 100,000 modes. The difference is in the fiber's *refractive index profile*.

FIGURE 4.8 The three types of light modes



The refractive index profile defines the relationship between the refractive index of the core and that of the cladding. There are two criteria used to define the relationship: the mode, and the index, which describes the interface between the core and the cladding.

So far, we have described a simple relationship in which the core has one RI and the cladding another, and there is a single interface between the two. This is known as *step-index* fiber. If the fiber has a large enough diameter, there is room for the light to take a number of different possible paths, or modes, by way of reflection. This is referred to as *multimode* fiber.

As shown in Figure 4.9, the three main refractive index profiles are:

- Multimode step-index fiber
- Multimode graded-index fiber
- Single-mode step-index fiber

Let's look at these individually.

Multimode Step-Index Fiber

Also known simply as step-index fiber, multimode step-index fiber, as already discussed, has a core with a single RI and a cladding with another, slightly lower RI. These are separated by a single interface, which reflects light that hits it at any angle greater than its critical angle. In addition, the diameter of the fiber core is large enough ($62.5\ \mu$ to $100\ \mu$) to allow a ray of light to take many possible paths.

The result is that even though all of the rays pass through the same length of fiber, the reflections create a longer path for the light to follow. The more reflections, the longer the path. You would get the same result if you and a friend walked down the same road at the same rate of speed, with you walking down the middle of the road and your friend zig-zagging from one side of the road to the other. Even if your friend started out first, you would eventually pass him,

because he is taking a longer path. As a result, you would arrive at the end of the path before your friend, even though you started out together.

The same effect occurs as light is reflected through the fiber. Even though light rays enter the fiber in the order 1,2,3, they may arrive in the order 3,1,2, or even overlap each other as they arrive, causing the information they carry to become muddled and useless, as demonstrated in Figure 4.10. This effect is called *modal dispersion*, and is an important factor limiting bandwidth in optical fiber.

FIGURE 4.9 Refractive index profiles

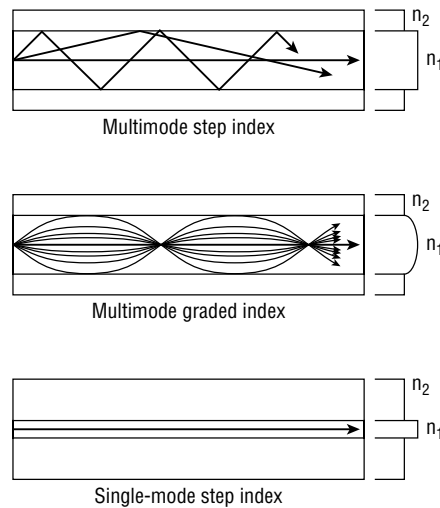
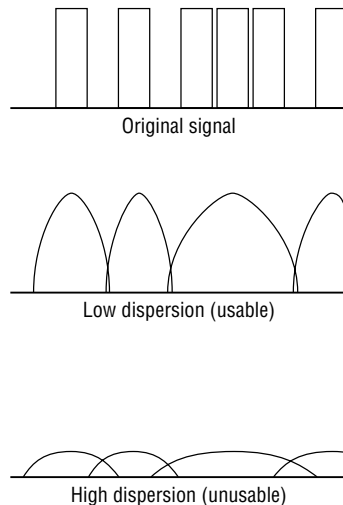


FIGURE 4.10 Dispersion can render data unusable if it becomes too great.



Typically, modal dispersion in step-index fiber offsets light rays by about 15 to 30 ns/km, depending on the diameter of the fiber and the wavelength of the light. In other words, if one ray takes the most direct path through the fiber, and another ray takes the longest possible path by reflecting off of the fiber walls, the ray taking the longest path will follow the ray taking the shortest path by anywhere from 15 to 30 nanoseconds, or 15 to 30 billionths of a second, for every kilometer of fiber length. That means that for every kilometer of fiber length, each bit would have to be separated from the one before it by at least 30 ns, or roughly 6 m, to ensure that it would not overlap another bit. In this example, it results in a data rate limited to 33.33 MHz for a 1 km fiber, 16.67 MHz for a 2 km fiber, 11.11 MHz for a 3 km fiber, and so on, far below the capabilities of transmitters, receivers, and processors, and completely unsuited for long-distance fiber optics transmission.

Two methods used to reduce modal dispersion are multimode graded-index fiber and single-mode step-index fiber.

Multimode Graded-Index Fiber

Multimode graded-index fiber tackles the problem of modal dispersion by increasing the speed of the high-order mode light rays, allowing them to keep up with the low-order mode rays.

The fiber accomplishes this by using the very principles upon which fiber optics is based: the laws of refraction.

Remember that when light passes from a higher RI medium to a lower RI medium, it gains velocity. A *graded-index* fiber core actually consists of many concentric glass layers with refractive indices that decrease with the distance from the center. Viewed on end, it resembles the rings of a tree.

Any light that passes straight through the fiber travels at a constant speed. If a light ray enters at an angle, however, it passes through the graded layers. As it does, two things happen:

First, the light is refracted away from normal, because the refractive index has decreased. Second, the light propagation velocity increases. This happens in every new layer the light traverses, until it has reached the cladding or has been bent to the critical angle for one of the layers. At this point, the light is reflected and begins its new direction, this time refracting toward normal and slowing down until it reaches the highest RI, at the center of the optical fiber core. It then passes through the center and begins the cycle again on the opposite side of the fiber core. The path resembles a segmented sine wave.

Note that as the light follows a longer path, its average velocity increases, offsetting the extra distance it must travel. The time it takes to move from one end of the fiber to the other is now much closer to that of light following a straight path through the fiber. In fact, modal dispersion in graded-index fiber can be reduced to as little as 1 ns/km.

Single-Mode Step-Index Fiber

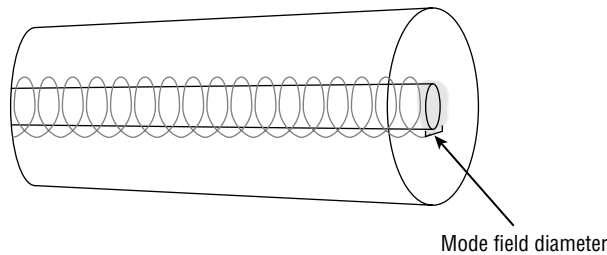
Single-mode step-index fiber uses another approach to reduce modal dispersion. It doesn't give the light enough space to follow anything but a single path through the fiber.

Single-mode fiber uses a core so small that light can only travel in one mode. The diameter of the single-mode fiber is typically only about 5 to 10 μ , but its performance also depends on the wavelength of the light.

For example, if a fiber is designed to carry a wavelength of 1300 nm in only one mode, there is room for light at 850 nm to travel in several modes, so the fiber has become a multimode fiber at the smaller wavelength. The smallest wavelength at which the fiber carries only one mode is called the *cutoff* wavelength. The cutoff wavelength for a fiber designed to carry 1300 nm in single mode is about 1260 nm.

Even in single-mode fiber, light does not follow a straight path. Because of the way light propagates, it actually follows a corkscrew-like path through the fiber core and propagates in a portion of the cladding, as shown in Figure 4.11. As a result, manufacturers must make single-mode fiber with cladding that will carry the light with less attenuation.

FIGURE 4.11 Light propagation in single-mode fiber



The unique light propagation characteristics of single-mode fiber also make it necessary to take the light propagated in the cladding into account when matching fiber sizes for connections, so the light in the cladding is not lost. In addition to knowing the core and cladding sizes, you must also know the *mode field diameter* of a step-index fiber. As you can see in Figure 4.11, the mode field diameter is the real estate used by the light within the core and the cladding as it propagates.

Dispersion-Shifted Fiber

With single-mode fiber used to overcome modal dispersion, it is possible to transmit data at much higher rates, around 10 Gb/s. In this range, however, other dispersion problems arise. One of these is *chromatic dispersion*.

Chromatic dispersion occurs because the light that is propagated through the fiber, even from a laser, is not confined to a single wavelength. Instead, it is made up of several wavelengths traveling down the same path. The range of these wavelengths, which can run from 0.1 nm to 5 nm, is called the *spectral width*.

As we've already learned, different wavelengths travel through a given material at different speeds. If the difference in their arrival times is great enough, the signal could again be muddled and unusable, just as with modal dispersion.

But at the wavelengths used for fiber optic communications, an interesting thing happens. In the 850 nm range, the longer wavelengths travel faster than the shorter wavelengths. In the 1550 nm range, however, the shorter wavelengths travel faster than the longer ones.

Somewhere between the two, there must be a point where the speeds cross—in other words, where all of the wavelengths within a certain range travel at the same speed through the fiber. This *zero dispersion point* occurs around 1300 nm.

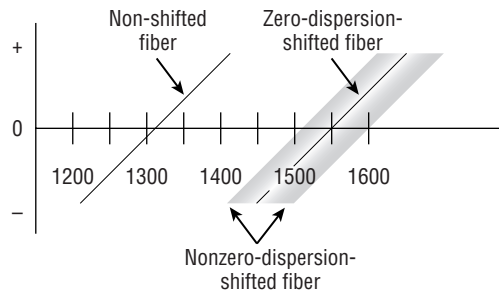
It would make sense, then, to transmit fiber optic signals in this range, right?

Unfortunately, the 1300 nm range suffers more attenuation through glass fibers than does the 1550 nm range, limiting the distance the signal can travel before it has to be boosted. On the other hand, chromatic dispersion at the 1550 nm range is about five times greater than it is at 1300 nm, which limits the rate at which the data can be transmitted.

It would seem that fiber optic system designers would have to choose between the two, depending on whether they want the optimum data rate or the longest transmission distance.

One compromise between the two choices is available with *dispersion-shifted fiber (DSF)*. Dispersion-shifted fiber, sometimes called *zero-dispersion-shifted fiber*, is made so that the zero dispersion point is shifted to 1550 nm, where the attenuation is lowest, as shown in Figure 4.12. Using dispersion-shifted fiber, high data rates can be transmitted over long distances.

FIGURE 4.12 Zero dispersion points of nonshifted, dispersion-shifted, and nonzero-dispersion-shifted fibers



A variation of the dispersion-shifted fiber, *nonzero-dispersion-shifted fiber (NZ-DSF)*, is used when multiple frequencies are being used to send more than one channel through the fiber, a process known as *wavelength division multiplexing (WDM)*. When WDM takes place at the zero dispersion point, an effect known as *four-wave mixing* occurs, in which multiple wavelengths combine to form new wavelengths. These new wavelengths can interfere with the signal carrying wavelengths, reducing their power and introducing noise into the system.

To reduce four-wave mixing, NZ-DSF is used to move the dispersion point just off the 1550 nm mark. The slight amount of dispersion introduced minimizes four-wave mixing while preserving most of the benefits of dispersion-shifted fiber.



Real World Scenario

Making the Connection

You're planning to run an optical fiber network in a small office. None of the cable runs will be more than 100 feet apart, but the office needs to be able to use data rates of up to 3 Gb/s. Which type of optical fiber is best suited to the job? Consider the characteristics of each type of fiber:

Multimode step-index fiber This type is low-cost, and lower data rates might be fine for the short runs. A high data rate, however, could cause problems due to modal dispersion. These fibers are best suited for use within an office or between a computer and a peripheral device, where short transmission distances keep the effects of dispersion to a minimum.

Single-mode fiber This type is ideal for handling high data rates over long distances, but the cost of using fiber suited for runs in the tens of kilometers may not be justified for a series of short runs. This type of fiber might be ideal for sending video signals throughout a campus or even across town, but for running between walls, it's overkill.

Multimode graded-index fiber This type represents the best compromise in this situation. It corrects the dispersion while costing less than the higher-quality single-mode fiber.

Summary

This chapter covered the structure and components of optical fiber. We discussed the different configurations and materials used in optical fibers, and the benefits and drawbacks of each.

We also discussed how optical fiber is manufactured, paying attention to different manufacturing techniques leading to the final production process.

Finally, we looked at the principles behind the construction of different types of optical fiber, and the importance of selecting the correct grade of fiber for the requirements that it must meet.

Exam Essentials

Describe optical fiber components. Be able to describe the basic parts of an optical fiber and list the materials that go into its construction. Understand the purpose of each component.

Understand manufacturing optical fiber. Understand the different processes used in the first stages of manufacturing optical fiber, as well as the final process of drawing the fiber. Know the basic terms used in the manufacturing process.

Understand modes. Understand what modes are and how they affect the transmission of data through optical fibers. Be able to describe the propagation of light through different types of optical fibers. Finally, be able to describe the methods used to overcome dispersion in optical fibers.

Review Questions

1. Listed from innermost to outermost, the components of an optical fiber are:
 - A. Cladding, core, coating
 - B. Core, cladding, coating
 - C. Cladding, coating, core
 - D. Core, coating, cladding
2. A fiber's tensile strength refers to:
 - A. How much stretching it can handle
 - B. How much light it can carry
 - C. How much weight it can support
 - D. How much heat it can resist
3. The preform used in manufacturing optical fiber is:
 - A. A mold used to shape the fiber
 - B. A process of heating the fiber to shape it
 - C. The raw materials that go into the fiber
 - D. A shorter, thicker version of the fiber
4. In most manufacturing processes, the actual fiber starts out as:
 - A. Sand
 - B. Soot
 - C. Cloth
 - D. Powder
5. The bait used in modified chemical vapor deposition is in the form of:
 - A. A glass rod
 - B. A glass tube
 - C. A glass beaker
 - D. Glass balls
6. The process that uses a glass rod suspended at one end is called:
 - A. Chemical vapor deposition
 - B. Outside vapor deposition
 - C. Vapor axial deposition
 - D. Plasma chemical vapor deposition

7. Modes are:
 - A. Potential paths for light
 - B. Light wavelengths
 - C. Potential paths for optical fiber
 - D. Potential uses for optical fiber
8. Which of these numbers could describe the number of modes in a fiber?
 - A. 27.3
 - B. 425.1
 - C. 675
 - D. 218.75
9. A refractive index profile expresses:
 - A. The RI of the core
 - B. The RI of the cladding
 - C. The sum of the core RI and the cladding RI
 - D. The relationship between the core RI and the cladding RI
10. Modal dispersion is caused when:
 - A. Light escapes into the cladding from the core.
 - B. Light rays follow different paths along the fiber core.
 - C. A fiber is split into several parts.
 - D. Light encounters a break in the fiber.
11. Graded-index fiber overcomes modal dispersion by:
 - A. Eliminating the rays that do not travel a straight path
 - B. Increasing the amplitude of the rays that travel the farthest
 - C. Increasing the velocity of the rays that travel the farthest
 - D. Eliminating the weakest rays
12. Single-mode fibers overcome modal dispersion by:
 - A. Routing each mode to a different receiver
 - B. Assigning each mode a different wavelength
 - C. Eliminating one mode
 - D. Only allowing one mode to pass

- 13.** A single-mode fiber can become a multimode fiber if:
- A.** The wavelength is short enough.
 - B.** The light is bright enough.
 - C.** The data rate is high enough.
 - D.** The fiber is long enough.
- 14.** In a single-mode fiber, the mode field diameter measures:
- A.** The diameter of the core
 - B.** The diameter of the cladding
 - C.** The diameter of the light
 - D.** The diameter of the entire fiber
- 15.** The purpose of dispersion-shifted fiber is to:
- A.** Move dispersion into the cladding.
 - B.** Eliminate dispersion at all wavelengths.
 - C.** Use dispersion for encryption.
 - D.** Combine high bandwidth and low attenuation.

Answers to Review Questions

1. B. The core, at the center of the fiber, carries the light. The cladding provides a medium of lower RI to create total internal reflection, and the coating protects the fiber.
2. A. Because optical fibers must undergo a great deal of handling when they are being installed, it is important for them to have enough tensile strength to survive the bending and pulling that takes place.
3. D. The preform is a short, thick piece of glass with the same cross-section as the fiber. It is heated and drawn into the thin fiber in a drawing tower.
4. B. The material that will become the core and cladding of the glass fiber is originally laid down as a soot produced by burning a gas with the materials in it. The soot is then heated and compressed into the high-quality glass fiber.
5. B. In MCVD, gas is passed through a heated glass tube, where the soot from the gas collects to become the layers of the fiber.
6. C. In vapor axial deposition, a glass rod is suspended over a flame where gases are burned. The soot from the gases collects at one end of the rod to build up the layers of the preform.
7. A. When the diameter of the fiber core is large enough, the light used in fiber optic transmissions may have several potential paths they can follow. These paths, or modes, can be derived mathematically based on the fiber characteristics.
8. C. Because light can only follow a whole path, and not a part of one, modes are expressed in whole numbers. If the calculation for modes results in a fraction, the result is rounded down to the nearest whole number.
9. D. The refractive index profile describes the way in which the optical fiber handles modes and the interface between the core and the cladding.
10. B. Modal dispersion is caused when light rays follow different paths in the fiber and overlap each other or arrive in a different order from how they started out, making the transmission difficult or impossible to use.
11. C. Graded-index fibers gradually decrease the RI of the core as it moves out from the center, causing the rays that bounce the most to move the fastest. This reduces many of the effects of modal dispersion.
12. D. Single-mode fiber has a core so small that only one mode can pass through it. This eliminates all other modes and allows much higher data rates than multimode fibers.
13. A. A single-mode fiber remains single mode as long as the light entering it has a wavelength greater than the cutoff, or the minimal wavelength for a fiber to be single mode. If the wavelength is too short, there may be room in the core for multiple modes to propagate.

14. C. In a single-mode fiber, the light propagates through part of the cladding as well as the core. To make sure all of the light is collected, fibers are matched based on the mode field diameter, or the area of the fiber in which the light propagates.
15. D. In nonshifted fiber, the wavelength that provides the lowest dispersion is different from the wavelength that provides the lowest attenuation. Dispersion-shifted fiber places the lowest dispersion and the lowest attenuation at the same wavelength, providing the best of both worlds.



Chapter

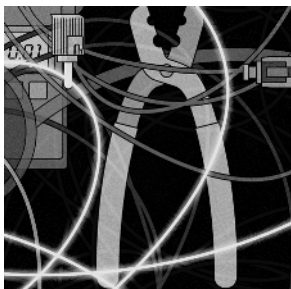
5

Optical Fiber Characteristics

OBJECTIVES COVERED IN THIS CHAPTER:

✓ Optical Fiber Characteristics

- Describe dispersion in an optical fiber.
- Describe modal dispersion and its effects on the bandwidth of an optical fiber.
- Describe material dispersion and its effects on the bandwidth of an optical fiber.
- Explain chromatic dispersion in an optical fiber.
- Explain waveguide dispersion in a single-mode optical fiber.
- Explain polarization mode dispersion in a single-mode optical fiber.
- Describe the effects dispersion has on bandwidth.
- Describe the causes of attenuation in an optical fiber.
- Describe attenuation versus wavelength in a multimode optical fiber.
- Describe attenuation versus wavelength in a single-mode optical fiber.
- Describe the numerical aperture of an optical fiber.
- Describe how the number of modes in an optical fiber are defined by core diameter and wavelength.
- Explain equilibrium mode distribution in an optical fiber.
- Describe microbends in an optical fiber.
- Describe macrobends in an optical fiber.
- List the bandwidth and attenuation characteristics of TIA/EIA-568-B.3 recognized multimode optical fibers.
- List the bandwidth and attenuation characteristics of TIA/EIA-568-B.3 recognized single-mode optical fibers.
- List the bandwidth and attenuation characteristics of PCS and HCS optical fibers.



In addition to factors such as construction, materials, and size as discussed in the previous chapter, optical fibers have certain performance or operational characteristics that define them. These characteristics may describe limitations or features of the fiber with regard to its light-carrying ability under various conditions, and are generally affected by its physical properties.

In this chapter, we describe the characteristics of optical fiber that affect the way it is selected, handled, installed, and used. We'll cover in detail how these characteristics change a fiber's ability to carry light, as well as the methods used to take advantage of some characteristics while minimizing the effects of others.

It All Adds Up

If you've ever looked at your bank balance and wondered where it all went, you have some idea of why it's important to understand fiber characteristics in great detail.

Chances are, when you took another look at your account, you remembered that some of it went to necessities, some to entertainment, some to service charges, and so on. It didn't all go away at once: it was spent little by little, in many different places. Any single expenditure may have gone almost unnoticed, but over time all of those expenses can add up to a significant amount.

This is typically the fate of light as it passes through an optical fiber. By the time it reaches the other end, it is diminished in several ways—most notably in its power and its ability to carry a signal. In spite of the purity of the materials that go into fibers, they are still not perfect—in part by design, and in part because there are many factors that simply cannot be overcome with current technology.

Going back to your bank account, if you know where the money goes, you can take steps to make it last longer or make sure that when it is spent, you get more use out of it.

The same goes for our light beam. If you understand the characteristics in an optical fiber that take away from its ability to carry a signal at high speed and over long distances, it is possible to overcome them, or use them to advantage. Remember, though, that propagation of light within a fiber is a complex mix of influences, and each of the characteristics that we'll be discussing affects different portions of this mix.

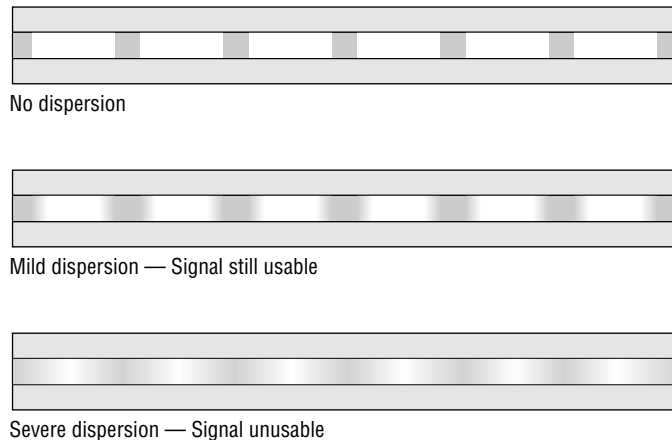
Some aspects of these characteristics have been covered in previous chapters to describe why fiber is constructed in certain ways. Now we are going to look at them more closely.

Dispersion

In general, dispersion is the spreading of light as it travels away from its source. The light spreads because different components of it travel at slightly different velocities, depending on the conditions in the medium through which it is traveling and the wavelengths that make up the light. There are different kinds of dispersion, however, and the kind that is taking place depends on several factors in the fiber and in the light itself.

The greatest effect of dispersion is that as the light spreads, it can degrade or destroy the distinct pulses of the digital signals in the light by making them overlap each other, as shown in Figure 5.1, blurring and blending them to the point that they are unusable. The effect grows more pronounced as the distance the light travels increases.

FIGURE 5.1 The effects of dispersion on a signal



The effect is similar to looking into a hallway through a frosted glass window. If people are moving through the hallway close together, the glass spreads their images so much that they merge with one another and look like a single mass rather than individuals. If they spread out far enough from each other, however, you can see each person moving past the window. The images are still spread out, but the space between each person is great enough to see.

To prevent signal loss due to dispersion, it is necessary to keep the pulses far enough apart to ensure that they do not overlap. This limits the signals to a *bit rate* that is low enough to be only minimally affected.

Restricting the bit rate places a limit on the fiber's *bandwidth*, or the amount of information it can carry.

The types of dispersion that affect optical fiber are:

- Modal dispersion
- Material dispersion

- Waveguide dispersion
- Chromatic dispersion
- Polarization-mode dispersion

Let's look at these types of dispersion more closely.

Modal Dispersion

We mentioned modal dispersion in the last chapter to explain why fibers are classified as multi-mode and single-mode. It will help to review some of the important points.

Modal dispersion results from light taking different paths, or modes, as it passes through the fiber. The number of modes the light can take is determined by the diameter of the fiber core, the refractive indices of the fiber core and cladding, and the wavelength of the light.

A mode can be a straight line through the fiber, or it can follow an angular path, resulting in reflections every time the light meets the interface between the core and the cladding. The more reflections, the longer the path through the fiber, and the longer the light takes to pass through it.

Depending on the mode, some parts of the light will pass through the fiber more quickly than others. The difference in travel time can cause parts of the light pulses to overlap each other, or in extreme cases to arrive in a different order from the order they were transmitted. The signal is then no longer usable.

Methods for overcoming modal dispersion include:

Lower bit rate Lowering the bit rate increases the gap between bits in the signal. While dispersion will still affect them, they will not overlap one another, and will still be usable. The drawback of this method is a reduction in bandwidth, reducing the fiber's ability to carry data.

Graded index fiber Graded index fiber gradually reduces the refractive index of the fiber core from the center toward the cladding, allowing the light that follows a more angled path to speed up as it leaves the center and causing it to slow down again as it reaches the center. This effect reduces the difference in travel time between modes and allows wider bandwidths. Graded index fiber is a moderately priced solution that allows wider bandwidths than multimode step index fiber. In addition, the gradual change of indices as the light heads for the cladding causes the light to curve back into the core of the fiber before it has a chance to approach the cladding at a penetrating angle and be lost or reflected with a destructive time delay.

Single-mode fiber Single-mode fiber has a core that is narrow enough for only one mode to propagate, eliminating the problems caused by multiple modes. This type of fiber requires more expensive connectors and equipment because of the small core size and is typically used when very wide bandwidth requirements justify the cost.

Remember that modal dispersion is measured in nanoseconds per kilometer (ns/km). For example, if a fiber has a modal dispersion of 15 ns/km, the beam that takes the longest path will fall behind the beam that takes the shortest path by 15 ns for every kilometer of fiber length. This figure is used to determine the maximum bit rate possible for a length of fiber.

Material Dispersion

Material dispersion is the result of different wavelengths of light traveling at different velocities in the fiber. Even the light from a laser is made up of several wavelengths within a narrow range called the *spectral width*, which varies depending on the source. A light-emitting diode (LED) has a spectral width of 20 nm to 170 nm, while a laser diode's spectral width is much smaller, between 1 nm and 3 nm.

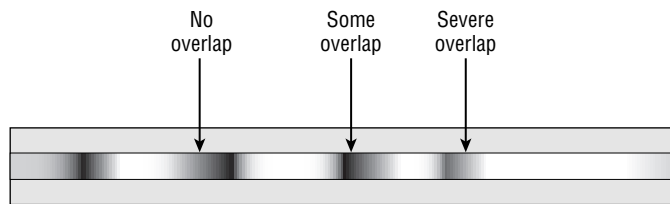
Recall the formula for determining the refractive index of a material,

$$n = c/v$$

where n is the refractive index, c is the speed of light in a vacuum, and v is the speed of the wavelength of light through the material. In this equation, n changes with the wavelength of the light passing through the material. Remember that this is the cause of white light breaking into its component colors in a prism.

When the different wavelengths travel at different velocities, the slower wavelengths begin to lag behind as the light travels down the fiber core, causing the light to spread as shown in Figure 5.2. If the light must travel a great distance, the lag in the slower wavelengths can cause them to overlap the faster wavelengths of the bits following them. As with modal dispersion, these overlaps can degrade and ultimately destroy the signal.

FIGURE 5.2 Material dispersion in fiber causes some wavelengths to travel more slowly than others.

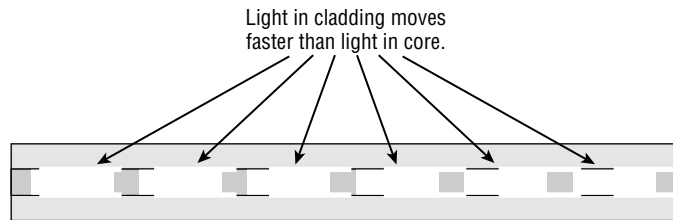


Because the wavelengths used in fiber optic transmissions have a narrow spectral width, material dispersion takes place on a much smaller scale than modal dispersion. Its effects in a fiber are measured in picoseconds per nanometer of spectral width per kilometer (ps/nm/km), and are insignificant in a multimode fiber when compared to the effects of modal dispersion.

Material dispersion only becomes a problem when modal dispersion is overcome with single-mode fiber and higher data rates are used over long distances.

Waveguide Dispersion

Waveguide dispersion occurs in single-mode fiber as the light passes through not only the core, but also part of the cladding, as shown in Figure 5.3. Because, by design, the core has a higher refractive index than the cladding, the light will be traveling more slowly through the core than through the cladding.

FIGURE 5.3 Waveguide dispersion in optical fiber

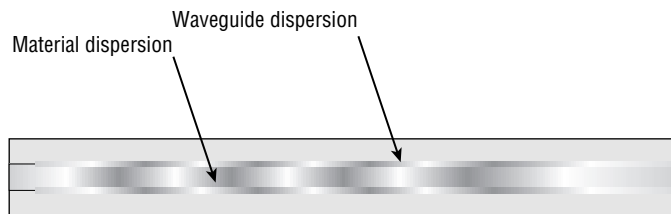
While the difference in refractive indices of single-mode fiber core and cladding are minuscule, they can still become a factor over great distances. In addition, waveguide dispersion can combine with material dispersion to create another problem for single-mode fiber: chromatic dispersion.

Various tweaks in the design of single-mode fiber can be used to overcome waveguide dispersion, and manufacturers are constantly refining their processes to reduce its effects.

Chromatic Dispersion

Chromatic dispersion occurs in single-mode fiber, and results from the combination of effects from material dispersion and waveguide dispersion.

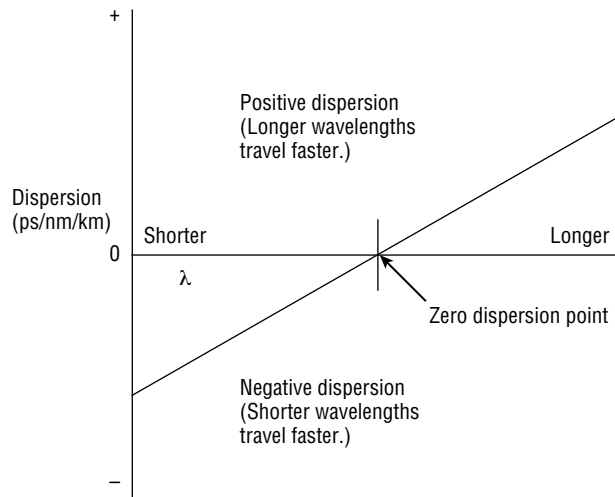
When chromatic dispersion occurs, the effects of material dispersion, as shown in Figure 5.4, compound the effects of waveguide dispersion. At lower data rates and in multimode fiber, the effects of chromatic dispersion are so small as to be unnoticed, especially when buried under modal dispersion. It is mostly a problem in single-mode fiber carrying bit rates up to 10 Gbps over long distances, where the detrimental effects build up.

FIGURE 5.4 Waveguide dispersion and material dispersion combine to create chromatic dispersion.

One way to reduce chromatic dispersion is by taking advantage of the fact that the relationship between wavelength, refractive index, and velocity is not linear. In the infrared range of most fiber optic transmissions, the light's velocity through the medium drops as the wavelength increases until it reaches the range between 1300 nm and 1550 nm. At wavelengths greater than 1550 nm, the longer wavelengths have a higher velocity. Somewhere in the 1300 nm to 1550 nm

range there is a crossover where, depending on the specific composition of the fiber, the refractive index is the same for the wavelengths within the narrow spectral width of the light being transmitted. In other words, as shown in Figure 5.5, as the wavelength approaches this range, dispersion drops to zero. This *zero-dispersion point* normally occurs at 1300 nm in a standard single-mode fiber. Unfortunately, other characteristics of the fiber attenuate the signal at this wavelength, making it unusable for long-distance runs.

FIGURE 5.5 Dispersion profile of a typical optical fiber



There are two ways to reduce chromatic dispersion in fiber while maintaining the energy of the signal: dispersion-shifted fiber and reduced spectral width.

Dispersion-Shifted Fiber

As we discussed in the last chapter, dispersion-shifted fiber is a specially formulated single-mode fiber that shifts the zero-dispersion point to 1550 nm, where the signal can travel a greater distance through the fiber without significant attenuation. The refractive index profile of dispersion-shifted fiber is shown in Figure 5.6. The peak in the center of the profile reveals an inner core that has its highest refractive index at the center. The refractive index gradually decreases toward a thin inner layer of cladding. The smaller peaks represent a ring of silica with a higher refractive index surrounding the inner cladding, slowing the light that would normally increase its velocity in the cladding. This effect reduces waveguide dispersion, which in turn reduces chromatic dispersion. The use of dispersion-shifted fiber allows much higher bit rates to travel over greater distances, and as a result, it may also be used to carry several different channels at different wavelengths. This can lead to *four-wave mixing*, which we will discuss shortly.



Real World Scenario

Mixing It Up

It seems that nobody can leave a good thing alone. Once engineers overcame the problem of chromatic dispersion in single-mode fiber using dispersion shifting, they decided to squeeze all the use they could out of it by piling on different wavelengths to create multiple transmission channels. The idea behind this is that different wavelengths can actually occupy the same space but remain distinct from one another until they are sorted out at the other end of the fiber link.

It makes very good sense, but then another problem cropped up. The wavelengths used in the multiple channels must stay near the zero-dispersion range of 1550 nm, so you end up with individual channels only 2 nm apart, typically at 1546, 1548, 1550, and 1552 nm, for example. It's difficult for anything to be only 2 nm apart and not interact, so interact they do. In fact, they create new wavelengths that can interfere with the wavelengths that are part of the transmission.

The problem gets exponentially worse with the number of wavelengths being transmitted. The formula for predicting the number of new waves created is:

$$\text{FWM} = (n^2 (n - 1))/2$$

where FWM is the number of waves created and n is the number of wavelengths being transmitted through the fiber. So if two wavelengths are being used, an extra two wavelengths will appear. That's not too bad in itself. But if you are transmitting four original wavelengths:

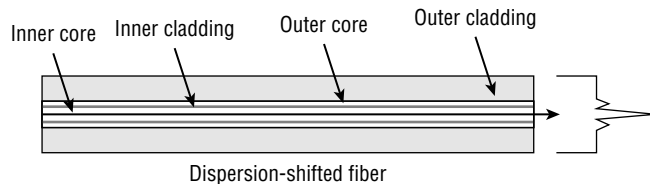
$$\text{FWM} = (16 \times 3)/2 = 24$$

And eight wavelengths will produce:

$$\text{FWM} = (64 \times 7)/2 = 224$$

The solution to four-wave mixing actually involves creating just enough dispersion in the fiber to render the newly created wavelengths harmless to the signals while leaving the original signal clear enough to use. The fiber created for this purpose is *nonzero-dispersion-shifted fiber (NZ-DSF)*. Nonzero-dispersion shifting moves the zero-dispersion point slightly away from the wavelength used for the transmission, usually about 10 nm above or below the transmission wavelength, so there is sufficient dispersion to keep the effects of four-wave mixing to minimum.

FIGURE 5.6 Refractive index profile of dispersion-shifted fiber



Reduced Spectral Width

Because material dispersion is caused by an overabundance of wavelengths in the optical signal, the simplest solution is to reduce the number of wavelengths by reducing the spectral width. Recall that dispersion is expressed as picoseconds per nanometer of spectral width per kilometer of fiber, so any reduction in spectral width will have a significant effect on the amount of material dispersion.

Polarization-Mode Dispersion

Polarization-mode dispersion (PMD) is masked by other forms of dispersion until the bit rate exceeds 2.5 Gbps. In order to understand PMD, we must look at an information pulse more closely.

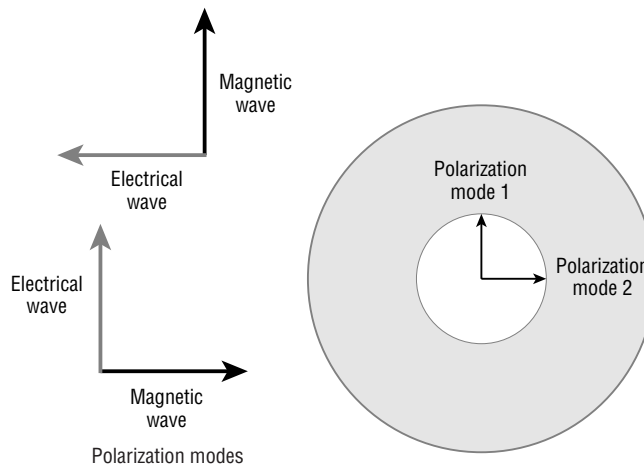
Recall that light is an electromagnetic wave, consisting of an electrical and a magnetic wave traveling at right angles to one another. The orientation of the two waves along the path of propagation determines the light's *polarization mode*, or *polarity*.

As shown in Figure 5.7, it is possible to have different polarities of light traveling through the fiber in a signal, occupying different parts of the fiber as they pass through it. Because no fiber is perfect, there will be obstacles in one part of the fiber that are not present in another. As a result, the light having one polarity may pass an area without interference, while another polarity may pass through a defective region, slowing it down.

Polarization-mode dispersion is not so much a function of the fiber's overall characteristics as it is a result of irregularities, damage, or environmental conditions such as temperature. Small areas of damage called microbends can cause PMD, as can fiber that is not perfectly round or concentric.

Because PMD is caused by specific conditions within the fiber, and not the fiber's overall characteristics, it is difficult to assign a consistent PMD value to a length of fiber. The exact amount of PMD changes with external conditions, the physical condition of the fiber, and the polarization state of the light passing through it at any given moment. For this reason, PMD is measured in terms of the total difference in the travel time between the two polarization states, referred to as the *differential group delay (DGD)* and measured in picoseconds. The amount of PMD itself may be measured in $\text{ps/km}^{1/2}$.

FIGURE 5.7 Polarized light shown in a cross-section of optical fiber



How Dispersion Affects Bandwidth

While different types of dispersion have different causes and are measured at different rates (ns/km, ps/nm/km), all of them have one effect in common: they place a limit on the bandwidth of optical fibers. As we saw in the previous chapter, in fact, modal dispersion can cause the bandwidth of a fiber to narrow dramatically as the distance increases.

It seems that no matter what kind of dispersion a fiber is manufactured to overcome, it is always going to be subject to another kind, at a higher bandwidth. For this reason, one way of grading fibers is by the limit that dispersion places on its ability to carry a signal. Depending on the type of fiber, this limit can be expressed in one of two ways. In multimode fiber, where modal dispersion is the overwhelming factor, it is expressed in megahertz of bandwidth per kilometer of fiber (MHz/km), and in single-mode fiber it is expressed in picoseconds per nanometer of source spectral width per kilometer of fiber (ps/nm/km).

The MHz/km figure expresses how much bandwidth the fiber can carry per kilometer of its length. The fiber's designation must always be greater than or equal to the product of the source bandwidth and the length of the cable.

EXERCISE 5.1

Calculate the usable bandwidth of a 500 MHz/km fiber running 3.2 km.

Remember that $\text{bandwidth} \times \text{length} \leq \text{fiber designation}$. To find the bandwidth that you can use with the fiber, we have to turn the equation around to read:

$$\text{Fiber designation/length} \geq \text{bandwidth}$$

Now let's plug in the values:

$$500/3.2 = 156.25$$

So the widest bandwidth that can be used with this fiber is 156.25 MHz.

When working with single-mode fiber, where chromatic dispersion is a limiting factor, the more precise designation of ps/nm/km is necessary as a gauge of a fiber's ability to carry a signal. Instead of specifying the bandwidth that the fiber can carry, the classification describes the dispersion that takes place within the fiber. This is done because the bandwidth is no longer simply a factor of distance. It can also be changed by narrowing the spectral width of the source. As we've already seen, a small change in spectral width can significantly affect the fiber's bandwidth.

Attenuation

Attenuation in a fiber optic signal is the loss of optical power as the signal travels through the fiber. Attenuation is caused by the fact that no manufacturing process can produce a perfectly pure fiber. Either by accident or by design, the fiber will always have some characteristic that attenuates the signal passing through it.

The wavelength of the light passing through the fiber also affects attenuation. In general, attenuation decreases as wavelength increases, but there are certain wavelengths that are more easily absorbed in plastic and silica fibers than others. One of the reasons for establishing standard operating wavelengths of 850 nm, 1300 nm, and 1550 nm in silica fiber and in the visible range of 650 nm for plastic is because the wavelengths in between are considered high-loss regions. Specifically, these wavelengths are in the ranges of 730, 950, 1250, and 1380 nm.

Attenuation must be measured differently from dispersion. As with dispersion, the effect on a signal due to attenuation increases with the length of the fiber. If the fiber's makeup is consistent, the amount of attenuation can be predicted and accounted for to some extent by adjusting the power of the source or by adding *repeaters*, which collect a weak signal and amplify it.



Attenuation provides a good example of the superiority of fiber over copper for carrying signals. When an electrical signal is carried through copper wire, attenuation increases with the data rate of the signal, requiring an increase in transmission power or, more often, the use of repeaters. Attenuation per unit length in an optical signal for a fiber of a given type is constant no matter what the data rate, so repeaters can be farther apart, requiring fewer of them.

Attenuation behaves differently from dispersion, however, in the way that its effects accumulate. As we have seen, dispersion is determined by factors within the fiber and the signal's wavelength and spectral width. None of these factors changes as the signal passes through the fiber, so the amount of change caused by dispersion can be calculated fairly simply.

Attenuation, however, is referenced to the signal's reduction in power, and any calculations must take into account the fact that as power is reduced, attenuation will affect only the power that remains, thus altering the equation over the length of the fiber.

For example, if attenuation reduces power by 1% over the distance of 1 km, then only 99% of the original power will be left at the end of 1 km. At the end of another kilometer, the *remaining* power is reduced by 1%, and so on. This produces a more complex equation for determining attenuation, but it can still be done.

Remember from Chapter 2, "Principles of Fiber Optic Transmission," that attenuation is measured in *decibels* (dB). Decibels help us account for the constant loss of power when we are measuring attenuation in a fiber. In silica fiber, typical attenuation is about 2.1 dB/km at 1500 nm, and attenuation in plastic fiber can be over 300 dB/km at 650 nm.

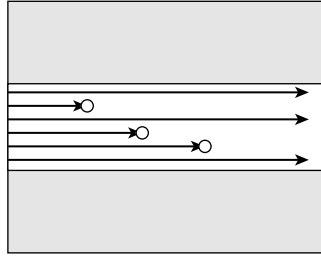
While decibels are useful in measuring *total attenuation*, we can also divide attenuation into two types: *absorption* and *scattering*.

Absorption

All materials, even the clearest glass, absorb some light. The amount of absorption depends on the type of material and the wavelength of the light passing through it. You can see absorption easily in sunglasses. Even on the brightest days, only a fraction of the light energy passes through the tinted lenses. The wavelengths that do not pass through are mostly absorbed by impurities that have been placed in, or coated on, the lens material.

In an optical fiber, absorption occurs when impurities such as water or ions of materials such as copper or chromium absorb certain wavelengths, as shown in Figure 5.8. By keeping these impurities as low as possible, manufacturers can produce fibers with a minimum of attenuation.

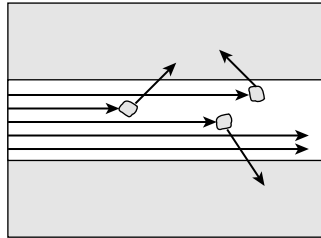
FIGURE 5.8 Absorption in optical fiber



Scattering

Scattering is caused by atomic structures and particles in the fiber redirecting light that hits them, as shown in Figure 5.9. The process is called *Rayleigh scattering*, for Lord Rayleigh, a British physicist who first described the phenomenon in the late nineteenth century.

FIGURE 5.9 Scattering in optical fiber



Rayleigh scattering is also the answer to the age-old question “Why is the sky blue?” The blue that we see is actually the more prevalent blue wavelengths of light from the sun being scattered by particles in the atmosphere. As the sun moves toward the horizon and the light must pass through more of the atmosphere, the scattering increases to the point where the blue light is almost completely attenuated, leaving the red wavelengths, which are less affected by the scattering for reasons that we’ll see shortly.

Rayleigh scattering depends on the relationship between wavelength and the size of the structures in the fiber. Scattering increases as the wavelength of the light approaches the size of the structures, which means that as the wavelength decreases, it is more likely to be scattered. This is one of the main reasons that infrared wavelengths are used in fiber optics. Their relatively long wavelengths are less subject to scattering than visible wavelengths. It also explains why the

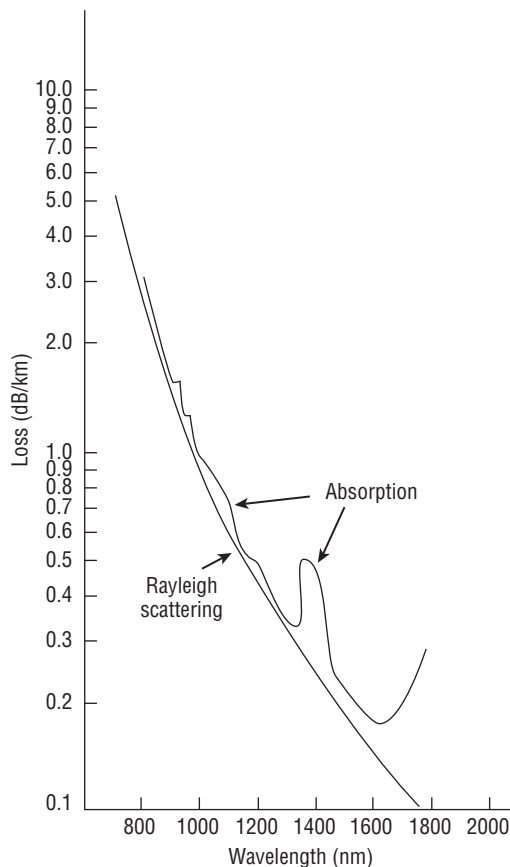
sun turns red on the horizon. The shorter blue wavelengths are more likely to be scattered by the similarly sized particles in the atmosphere than are the red wavelengths.

Total Attenuation

Total attenuation is the combination of the effects of absorption and scattering in a fiber. Figure 5.10 shows a typical attenuation curve for an optical fiber with the effects of absorption and scattering combined. Note that the general curve is caused by the effects of scattering, while the irregularities in the plot are caused by specific impurities, such as hydroxyl molecules, absorbing light in those wavelengths.

Note also the windows at the 850, 1300, and 1550 nm ranges. Remember that while the 1300 nm range is better in terms of dispersion, it still has a higher attenuation than the 1550 nm range, which is the reason for dispersion-shifted fiber.

FIGURE 5.10 An optical fiber's attenuation curve



Numerical Aperture

As we saw in Chapter 4, “Optical Fiber Construction and Theory,” many of a fiber’s characteristics are determined by the relationship between the core and the cladding. The *numerical aperture (NA)* expresses the light-gathering ability of the fiber. The NA is a dimensionless number, meaning that it is to be used as a variable in determining other characteristics of the fiber, or as a means of comparing two fibers.

As discussed earlier, the numerical aperture is determined by the refractive indices of the core and the cladding:

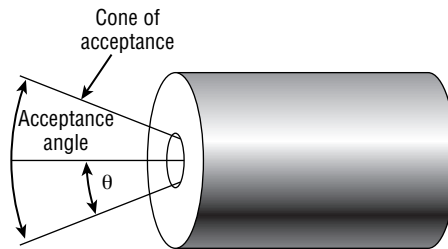
$$NA = \sqrt{(n_1)^2 - (n_2)^2}$$

where n_1 is the refractive index of the core and n_2 is the refractive index of the cladding.

Recall that in order for light to be contained within a multimode fiber it must stay above the *critical angle*, or the angle at which it reflects off of the boundary between the core and the cladding, rather than penetrating the boundary and refracting through the cladding.

In order to maintain the critical angle, light must enter within a specified range called the *cone of acceptance* also known as the *acceptance angle*. As shown in Figure 5.11, this region is defined by a cone extending outside the fiber core. Light entering the core from outside of the cone will either miss the core or enter at an angle that will allow it to pass through the boundary with the cladding and be lost.

FIGURE 5.11 The cone of acceptance



The cone of acceptance is determined using the numerical aperture:

$$NA = \sin\theta$$

where θ is $1/2$ of the angle measuring the cone of acceptance. Another useful term is the *maximum coupling angle*.

The acceptance angle is also used to determine how light emerges from a fiber. The light that comes out of a fiber end is the light that has not been absorbed or lost in the cladding, so with the exception of a small percentage of light that has propagated in the boundary between the

EXERCISE 5.2

Determine the acceptance angle of a fiber with a core RI of 1.47 and a cladding RI of 1.48.

First, we must use the RI information to determine the NA of the fiber:

$$NA = \sqrt{[(n_1)^2 - (n_2)^2]} = \sqrt{(2.19 - 2.16)} = \sqrt{0.03} = 0.1732$$

Next, we can use the NA to determine the acceptance angle. Remember that the acceptance angle is twice the value of θ .

$$NA = \sin\theta$$

So

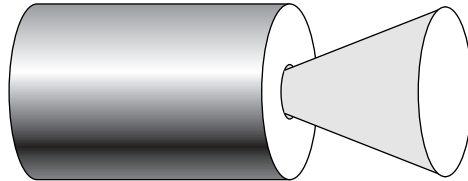
$$\theta = \arcsin NA = \arcsin 0.1732 = 9.9739$$

The acceptance angle is 2×9.9739 or 19.9478° .

(Note: The exact answer will depend on how much you round off your results during the calculations.)

core and the cladding, what emerges is coming out at an angle equal to or greater than the critical angle, as shown in Figure 5.12.

FIGURE 5.12 Light emerges from the fiber in the cone of acceptance.



Although NA can be calculated for a single-mode fiber, it is not a useful exercise since light does not propagate in a single-mode fiber through total internal reflection.

This is a good time to review another use for the NA: determining the number of modes possible in a multimode fiber core.

Remember that in a step-index multimode fiber, the number of *modes*, or potential paths for a ray of light, is determined by the wavelength of the light, the diameter of the fiber core, and the numerical aperture. This relationship is expressed with the equation:

$$M = (D \times NA \times \pi/\lambda)^2/2$$

where D is the diameter of the fiber and λ is the wavelength of the light.

Bending Losses

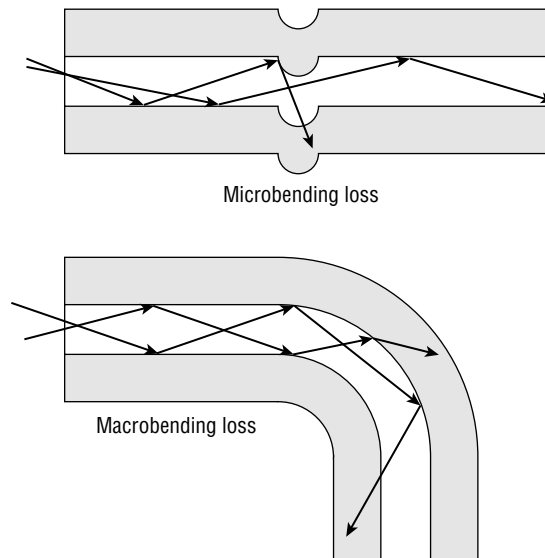
In addition to characteristics within the fiber material, the actual condition of the fiber can lead to losses as well. Because of bending, high-order mode light rays can be lost in the cladding as the angle of the boundary layer changes in relation to the light.

The types of bending we will look at are *microbends* and *macrobends*.

Microbends

Microbends are small distortions of the boundary layer between the core and cladding caused by crushing or pressure. Microbends change the angle of incidence within the fiber, as shown in Figure 5.13. Changing the angle of incidence forces high-order light rays to reflect at angles that prevent further reflection, causing them to be lost in the cladding and absorbed.

FIGURE 5.13 Losses caused by microbending and macrobending



Macrobends

Macrobends occur when the fiber is bent around a radius that can be measured in centimeters. As shown in Figure 5.13, these tight radii change the angle of incidence within the fiber, causing some of the light rays to reflect outside of the fiber and, as with microbending, be lost in the cladding and absorbed.

Equilibrium Mode Distribution

Equilibrium mode distribution (EMD) is a condition in which the light traveling through the fiber populates the available modes in an orderly way.

When light energy first enters a fiber, it does not automatically fill every available mode with an equal amount of energy. Some modes will carry a high amount of light energy, while others may carry nothing at all. Over distance, this condition balances out as light transfers between modes because of imperfections or bends in the fiber. Until EMD is achieved, however, light may be traveling in inefficient modes that will eventually lose energy. These modes may include paths that carry the light through the cladding, or high-order modes that produce a high number of reflections through the fiber core.

Let's back up to where the light enters the fiber and see how EMD is reached.

A light source rarely, if ever, perfectly matches the numerical aperture of the fiber. Typically, the beam will either be larger or smaller than the NA. If it is larger, as is the case with most LEDs, the fiber will be overfilled. This means that the light energy will occupy most of the modes, including those that are less efficient and are doomed to loss somewhere down the length of the fiber.

If the source has a smaller beam, as with most lasers, the fiber will be underfilled. The light energy will occupy only a few of the available modes until twists, turns, and imperfections in the fiber distribute some of the light energy into higher-order modes.

The effect is similar to traffic entering a highway. If eight lanes of cars are trying to get onto a six-lane highway, you'll see cars jockeying for position in the available lanes, and possibly even a few driving on the shoulder until they are attenuated by the highway patrol. This condition will persist until the drivers settle down into their new pattern. On the other hand, if a single lane of cars enters the same six-lane highway, they will first occupy only a single lane, but eventually spread out to a more evenly distributed pattern.

Even when EMD is achieved, the effect is short-lived due to minute changes in the fiber characteristics and the effects of connectors. It does not take much to throw the modes out of equilibrium.

The distance required to achieve EMD depends on the fiber material. Light passing through plastic fibers reaches EMD in a few meters, but light passing through glass fibers may reach equilibrium only after several kilometers.

It is important to understand EMD because it affects the measurement of light energy in a fiber. In an overfilled fiber, light that has not yet reached EMD still has much of the energy traveling in the inefficient or high-order modes, and this energy can be measured over a short distance. Over a longer distance, however, once EMD has been reached, the energy in the less efficient modes has been lost, and the energy reading may drop significantly.

Once EMD has been reached, however, energy loss drops off, because the light is traveling in more efficient modes and is less likely to be lost. For this reason, loss before EMD is proportional to the length of the fiber, and loss after EMD has been reached is proportional to the square root of the fiber length. In other words, if the energy loss before EMD is 0.2 dB/km, the loss over 2 km would be 0.4 dB. After reaching EMD, however, the loss would be $0.2 \times \sqrt{2}$, or 0.283 dB over 2 km.

Fiber Specifications

In order for optical fibers to work properly within a system or network, they must meet certain minimum standards established in TIA/EIA-568-B.3 for size, attenuation, and bandwidth. These standards ensure compatibility with other components within the system, even if the manufacturers vary.

Table 5.1 describes the attenuation and bandwidth characteristics for fibers recognized in TIA/EIA-568-B.3.

TABLE 5.1 Characteristics of TIA/EIA-Recognized Optical Fibers

Optical Fiber Cable Type	Wavelength (nm)	Maximum Attenuation (dB/km)	Maximum Information Transmission Capacity for Overfilled Launch (MHz/km)
50/125 μ multimode	850	3.5	500
	1300	1.5	500
62.5/125 μ multimode	850	3.5	160
	1300	1.5	500
Single-mode inside plant cable	1310	1.0	N/A
	1550	1.0	N/A
Single-mode outside plant cable	1310	0.5	N/A
	1550	0.5	N/A

Note: Bandwidth and numerical aperture are not typically specified in single-mode fiber.

Table 5.2 describes the attenuation and bandwidth characteristics for plastic fibers and HCS (hard-clad silica) and PCS (plastic-clad silica) fibers.

TABLE 5.2 Characteristics of HCS/PCS and Plastic Fibers

Optical Fiber Cable Type	Wavelength (nm)	Maximum Attenuation	Maximum Information Transmission Capacity for Overfilled Launch (MHz/km)
200/230 μ HCS/PCS	650	10 dB/km	17
	850	8 dB/km	20
1 mm plastic fiber	650	19 dB/m	*

*Plastic fiber is a low-bandwidth, high-attenuation fiber used for short-run applications.

Summary

This chapter covered characteristics of optical fiber relating to their ability to carry light and optical signals. It described conditions in the fiber that limit both bit rate and signal strength, as well as specific methods used to overcome some of the conditions.

The chapter described fiber characteristics that cause different types of dispersion, which affects signal quality, and different types of attenuation, which affects signal strength. It also described standards applied to different types of fiber with regard to the characteristics discussed.

Exam Essentials

Understand dispersion in optical fiber. Make sure you understand the different types of dispersion that occur in optical fiber, the type of fiber in which each occurs, and the causes. Be able to explain the effects of dispersion and the methods used to overcome each type. Be able to describe the methods of measuring dispersion.

Understand attenuation in optical fiber. Make sure you understand the different causes of attenuation, the methods used to measure it, and its effects on an optical signal. Be able to describe the methods used to overcome attenuation.

Be familiar with modes in optical fiber. Be able to explain modes in optical fiber and how they affect an optical signal. Be familiar with the methods used to overcome problems related to modes. Understand the types of fibers as they relate to modes.

Review Questions

1. What is dispersion?
 - A. The weakening of light
 - B. The spreading of light
 - C. The coloring of light
 - D. The diverting of light
2. What does dispersion affect most in a fiber optic signal?
 - A. Bandwidth
 - B. Signal strength
 - C. Wavelength
 - D. Color
3. What causes modal dispersion?
 - A. Light rays change wavelength as it passes through the fiber.
 - B. Light rays rotate as they pass through the fiber.
 - C. Light rays take different possible paths through the fiber.
 - D. Light rays become weaker as they travel through the fiber.
4. Material dispersion is partly dependent on:
 - A. The power of the source
 - B. The bit rate of the source
 - C. The manufacturer of the source
 - D. The spectral width of the source
5. Where does waveguide dispersion take place?
 - A. Only in the core
 - B. Only in the cladding
 - C. In the core and the cladding
 - D. In fiber without cladding
6. Chromatic dispersion is:
 - A. Another name for modal dispersion
 - B. Another name for material dispersion
 - C. Another name for waveguide dispersion
 - D. A combination of waveguide dispersion and material dispersion

7. How does dispersion-shifted fiber reduce chromatic dispersion?
 - A. It shifts the point of no dispersion to the wavelength of the light that travels through the fiber best.
 - B. It adds enough modal dispersion to counteract the chromatic dispersion.
 - C. It shifts the wavelength of the light passing through it.
 - D. It shifts the light from the core to the cladding.
8. Another solution for chromatic dispersion is to:
 - A. Reduce the power of the light.
 - B. Reduce the wavelength of the light.
 - C. Reduce the spectral width of the light.
 - D. Reduce the length of the fiber.
9. Four-wave mixing takes place when:
 - A. Four different wavelengths combine into one.
 - B. Four different wavelengths combine into two.
 - C. One wavelength is split into four.
 - D. Two wavelengths react with each other to produce two new wavelengths.
10. Polarization-mode dispersion takes place when:
 - A. Light travels in opposite directions in the fiber.
 - B. Two polarities of light are affected differently by conditions in the fiber.
 - C. Light passes near a magnet, splitting it into north and south polarities.
 - D. Light passes through an electrical field, splitting it into positive and negative components.
11. Differential group delay refers to:
 - A. The distance between pulses in a signal
 - B. The time taken to travel the length of the fiber
 - C. The difference in travel time between light polarization states
 - D. The difference between the longest and shortest wavelength in the light
12. For a fiber of a given bandwidth, as the length of the fiber increases, the bandwidth:
 - A. Increases
 - B. Decreases
 - C. Does not change
 - D. Is independent of the length

13. What is attenuation?
- A. The amount of light that the fiber can collect
 - B. The increase in power of a signal
 - C. The decrease in power of a signal
 - D. The change in wavelength of a signal
14. Windows are wavelength regions in fiber where attenuation:
- A. Is lowest
 - B. Is highest
 - C. Drops to zero
 - D. Allows light to escape the fiber
15. Attenuation is measured in:
- A. MHz/km
 - B. ps/km/nm
 - C. dB/km
 - D. GHz
16. The numerical aperture of the fiber expresses its:
- A. Core diameter
 - B. Cladding diameter
 - C. Light gathering ability
 - D. Bandwidth
17. In order for light to enter the fiber at the proper angle, it must be within the:
- A. Core diameter
 - B. Cladding diameter
 - C. Spectral width
 - D. Cone of acceptance
18. Microbends and macrobends can attenuate a signal by:
- A. Changing the angle of incidence
 - B. Changing the cone of acceptance
 - C. Changing the numerical aperture
 - D. Changing the refractive index

19. Macrobends are caused by:
- A. Twisting the fiber
 - B. Damage to the fiber
 - C. Curving the fiber
 - D. Pressurizing the fiber
20. Equilibrium mode distribution is achieved when:
- A. All available modes are carrying energy.
 - B. Light populates the available modes in an orderly way.
 - C. A fiber is overfilled.
 - D. A fiber is underfilled.

Answers to Review Questions

1. B. Dispersion is the spreading of light for various reasons as it moves away from its source.
2. A. Dispersion affects bandwidth, since signal pulses must be kept apart to keep from overlapping as they spread.
3. C. Modes are possible paths that light can travel through the fiber, and modal dispersion is caused when some light takes paths that are subject to more reflection off of the boundary between the core and the cladding, causing its path to be longer.
4. D. The spectral width of the source is the range of wavelengths being sent through the fiber. Because material dispersion causes different wavelengths to travel at different speeds, a large spectral width can increase the amount of dispersion taking place.
5. C. Waveguide dispersion takes place as the light passing through a single-mode fiber passes through part of the cladding. Because the cladding has a lower refractive index, the light speeds up, arriving ahead of the light in the core.
6. D. Chromatic dispersion takes place when the combination of waveguide dispersion and material dispersion causes signals in a single-mode fiber to overlap.
7. A. Dispersion-shifted fiber shifts the “zero-dispersion point,” or the point where chromatic dispersion drops to zero, to the wavelength that allows light to travel through the fiber with the least energy loss.
8. C. Reducing the spectral width of the light cuts down the difference between the slowest and the fastest wavelengths in the fiber, thus reducing the material dispersion component of chromatic dispersion.
9. D. Four-wave mixing takes place in multiple-channel transmissions when wavelengths that are close to one another interact, creating new wavelengths that act as noise and interfere with the signals being transmitted.
10. B. Light waves travel in different orientations, or polarities. When two different polarities travel through a fiber, they can be affected differently by conditions within the parts of the fiber in which they are traveling, causing one polarity to travel more slowly than the other.
11. C. Differential group delay, measured in picoseconds, is an indicator of the amount of polarization-mode dispersion in the fiber.
12. B. Because dispersion increases with the length of the fiber, the usable bandwidth of the fiber decreases as the signal pulses must be kept farther apart to avoid overlapping.
13. C. Attenuation is the loss of power in a signal as it travels through the fiber.
14. A. Windows are spectral regions determined by the composition of the fiber where light suffers the lowest attenuation. Standard windows for fiber optic signals are at 850 nm, 1300 nm, and 1550 nm.

- 15. C. Because attenuation takes a percentage of power and then takes the same percentage of the remaining power, decibels (dB) are used to express the constantly changing relationship between the signal level and the amount of loss.
- 16. C. The numerical aperture is a dimensionless number used to compare light-gathering ability with other fibers.
- 17. D. The cone of acceptance is the area defined by the numerical aperture. Light entering within the cone of acceptance will strike the core/cladding interface at the proper angle to reflect off of it and travel through the fiber.
- 18. A. Microbends and macrobends change the angle of incidence of the core/cladding interface, causing light rays to reflect at angles that send them into the cladding where they are absorbed.
- 19. C. Macrobends occur when the fiber is bent in a sharp radius.
- 20. B. Equilibrium mode distribution takes place when the modes carrying light have stabilized and no energy is jumping between modes.



Chapter

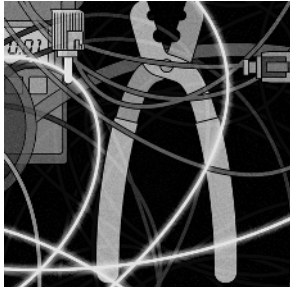
6

Safety

OBJECTIVES COVERED IN THIS CHAPTER:

✓ Safety

- Explain how to safely handle and dispose of fiber optic cable.
- List the safety classifications of fiber optic light sources.
- Explain the potential chemical hazards in the fiber optic environment and the purpose of the material safety data sheet (MSDS).
- Explain potential electrical hazards in fiber optic installation environment.
- Describe typical work place hazards in the fiber optic environment.



Whether you work as a technician or an installer, your work with fiber optics can expose you to several workplace hazards that are defined and regulated by the Occupational Safety and Health Administration (OSHA). OSHA has published numerous regulations on workplace hazards ranging from laser light sources to ladders, and employers are required to be familiar with these regulations and follow them to keep the workplace safe.

You are also responsible for your own safety, however, as well as for the safety of your co-workers. It is up to you to know and incorporate safe work practices in everything you do.

This chapter describes the types of hazards that you will encounter as you work with fiber optics. Some of the hazards are unique to fiber optics work, but others are more common. This chapter discusses the dangers that these hazards create, and inform you of different methods of working safely around them.

Basic Safety

Whenever you work in a hazardous environment, such as a construction site, a lab, or a production facility, you must always be aware of the potential dangers you face. Your workplace is required by law to provide you with equipment and facilities that meet standards set by OSHA, but you will have to be an active participant in your own safety.

There are three lines of defense that you can use to help you get through the day safely: engineering controls, personal protective equipment (PPE), and good work habits.

Engineering Controls

Engineering controls are the mechanisms that your facility has established to make a hazardous situation safer. They may include ventilation in the form of exhaust fans or hoods, special cabinets for storing flammables, or workstations that minimize the hazards of specialized work, such as cutting optical fibers.

Do not ignore or try to get around the engineering controls set up in your workplace. By doing so, you only endanger yourself and others. Make sure that fans and ventilation systems are working properly. If they are not, report any problems to your facility immediately.

Do not try to alter or modify the engineering controls unless the modifications have been approved by your safety officer. Improper modifications could reduce the effectiveness of the controls and create a greater hazard.

Personal Protective Equipment (PPE)

Personal protective equipment consists of anything that you would wear to protect yourself from materials or situations. They can include protective gloves and eyewear for cutting and grinding operations, respirators for working with chemicals that put out harmful vapors, and specialized goggles for working with lasers.

Your PPE protects you not only from short-term accidents, such as cuts or flying shards of glass, but also from damage that can build up over time. Such damage may include dust from construction operations such as drywall sanding that can build up over time in your lungs and cause diseases such as silicosis, or exposure to chemicals such as solvents that can have harmful long-term or chronic effects as well as harmful short-term effects.

Whenever you use PPE, inspect it carefully to ensure that it is in good condition. Look for cuts, tears, or other signs of damage in protective outerwear such as gloves or aprons. Inspect eyewear for cracks or pitting. If you use goggles designed to protect you from certain light wavelengths, make sure they are clean and free from scratches that could reduce their effectiveness.

If you wear contact lenses, be sure your facility allows them in your work area. If you work with adhesives or solvents, you should avoid wearing them anyway, because splashed chemicals could be trapped in the lens and be more difficult to wash out. You may be able to obtain safety goggles with prescription lenses if you have to use them on a regular basis.

If you work with a respirator, test it every time you put it on. Cover the canisters with your hands and try to inhale, then cover the exhaust port and try to exhale. The respirator should form a good seal with your face, and no air should leak through the canisters or exhaust.

Some construction areas may require hardhats. Do not take these warnings lightly. Even a small hand tool dropped from a few feet can injure or kill you if you are not protected. Hardhats are designed to absorb the shock from falling objects so your head doesn't have to. To make sure the hard hat fits properly, adjust the inner band so that it fits snugly against your forehead and does not allow the hat to move around on your head. Make sure there is enough room between the suspension and the hard hat shell to absorb any blows.

Good Work Habits

Good work habits are in some ways the simplest and most effective means to working safely. Good work habits can help you prevent accidents and spot potential problems in time to correct them.

Here are some general rules for working safely:

- Keep a clean workspace. Clean up at the end of your work day and store tools properly. A “rat’s nest” can hide problems and add to confusion.
- Observe your surroundings. Look up from what you are doing once in a while to make sure everything around you is the way it should be.
- Use tools for the job they were designed to perform. Misuse of tools is one of the most common causes of accidents in the workplace.

- Do not eat or drink in the work area. In addition to accidentally drinking from the wrong bottle, you could accidentally ingest glass fiber or other dangerous materials that might get mixed in with your food.
- Report problems or injuries immediately. Let your facility supervisors know about hazards so they can correct them as soon as possible.
- Know how to reach emergency personnel. Have emergency numbers posted by the nearest telephone so you don't waste time fumbling through a directory in an emergency.

Let's look at some of the hazards directly related to your work with fiber optics.

Light Sources

Even though most lasers and LEDs used in fiber optics operate in the near-infrared and infrared (IR) wavelengths and are invisible to the eye, they can still cause damage if they are delivered at high intensity, or if the exposure is long enough. The possibility of damage is even greater because you cannot see the beam, and in many cases the damage is done before you know it.

A laser can be especially dangerous because it can concentrate a great amount of power into a small beam of coherent light. Many lasers used in fiber optics operate below dangerous levels, but some, such as those used for transmission over long distances, put out enough power to cause damage in a very short time.

Injuries from the infrared wavelengths used in fiber optic system lasers include cataracts, as well as corneal and retinal burns. Other light sources include test equipment used to ensure fiber quality or look for breaks. These should only be used with the proper eye protection or measuring devices to prevent eye damage.

Laser Service Groups

To distinguish the hazard levels posed by different types of lasers, OSHA and the American National Standards Institute (ANSI) have divided lasers into four categories. These categories are called "Hazard Classes" by OSHA and "Service Groups" by ANSI. We'll use the ANSI categories for our discussion.

This list of Service Groups shows you the conditions under which different types of lasers can cause damage or injury.

SG1 These lasers typically operate at 400 nW and do not emit laser radiation at known hazard levels. According to OSHA, users of these lasers do not have to wear protective gear while using them, but may have to do so while performing service on them.

SG2 These lasers typically operate only in visible wavelengths and above power levels below 1 mW. These lasers can cause damage if you look directly at them for more than one-quarter of a second, but have limited requirements for protection. It is assumed that the normal aversion to pain will cause anyone looking at the bright light to turn away or close their eyes before any damage can take place.

SG3a These lasers are commonly used with fiber optics communications systems and operate in the range between 1.0 and 5.0 mW. Short-term exposure will not harm the naked eye, but if you look at the IR beam through a microscope or other magnifier, you could cause damage to your retina and not know it immediately, because there are no nerve endings in that part of your eye.

SG3b These communication lasers operate in the 5–500 mW range (up to +27 dBm) and can cause damage to the naked eye, even if the beam is reflected.

SG4 These lasers used in fiber optic systems operate above 500mW, or +27 dBm, and can burn almost any living tissue they contact. They also pose a fire hazard if directed at flammable materials.

Laser Safety

Because most lasers used in fiber optic systems emit IR radiation, you cannot see the beam, no matter how powerful it is. As a result, you will not be able to tell if the system is powered, especially if you are working on a piece of fiber far from the transmitter.

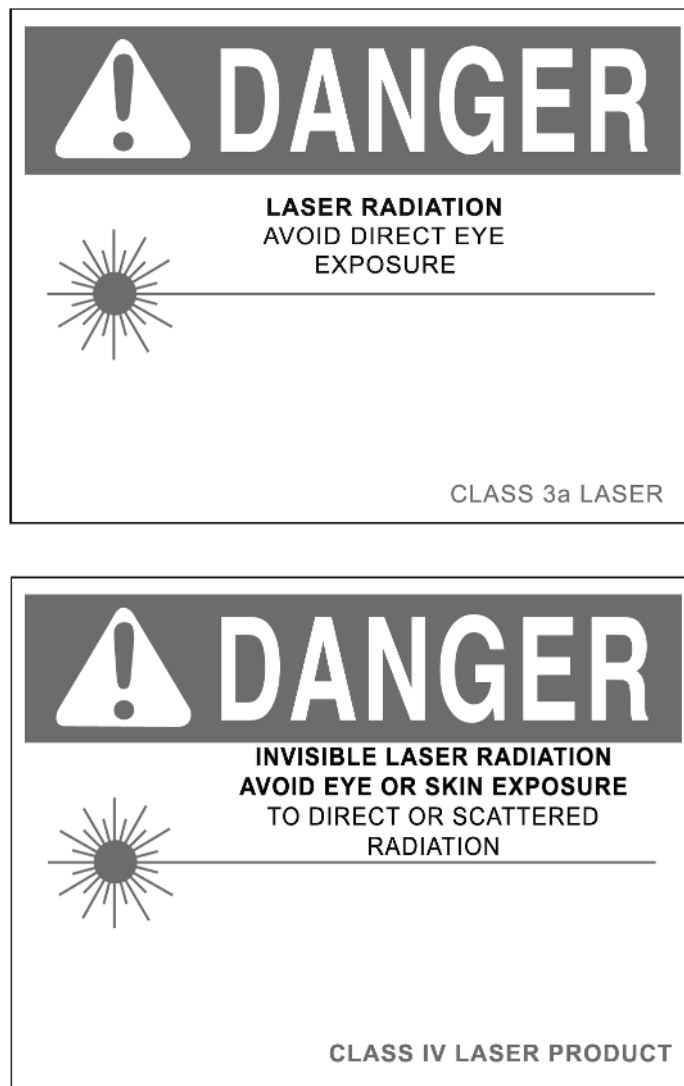
You should treat an optical fiber coupled to a laser with the same caution that you would treat electrical cables connected to a breaker panel. Do not assume that the system is turned off, especially if you have to use a microscope or fiberscope to look at the fiber end. Do not take anyone else's word that the system is off, or that the fiber is uncoupled from the laser. You will have to endure the results for the rest of your life.

Unless you can be sure that the fiber is not coupled to a laser, do not look at the fiber end without some kind of protection. Use filters and protective eyewear that block out the specific wavelengths used by the lasers.

Hazardous laser areas should be clearly identified with warning placards and signs stating that access is limited to personnel with proper safety gear and authorized access, as shown in Figure 6.1. Do not ignore these signs or think that they don't apply to you. They are there for your protection and for the safety of those working inside the restricted areas.

If the lab has a separate door and a hazardous laser is operating inside, the door should have interlocks to kill the laser before the door is opened. Some of these doors may have separate combination locks to prevent unauthorized entry.

FIGURE 6.1 Warning placards for lasers



Handling Fiber

In spite of optical fiber's flexibility, remember that it is glass. In short pieces, it is stiff enough to pierce your skin or eyes and cause discomfort, pain, or damage. If the pieces become airborne, you may even accidentally inhale or swallow them, risking damage to your throat or respiratory system. You can protect yourself with correct procedures and the right PPE.

When you cut, cleave, scribe, or accidentally break optical fibers, the ends can get lost easily, either by becoming airborne or by rolling along a surface. These ends can have extremely sharp edges, and if they are mishandled, they can lodge in your skin or eyes. If they are not removed immediately, the pieces can work themselves in further, increasing the risk of damage or infection. As shown in Figure 6.2, always work over a nonreflective black surface, which makes it easier to keep track of cut fiber ends. Also, keep a separate labeled container with a lid nearby for cut fiber ends.

FIGURE 6.2 A nonreflective black surface, a fiber waste container, and safety glasses can help prevent injury from fiber ends.



To prevent injury to your hands, always handle cut pieces of fiber with tweezers. To prevent eye injury, always wear proper eye protection. It takes only one piece of glass to damage your vision permanently. If you have been handling fiber, do not rub your eyes or put your hands near them until you have washed your hands. If you do get a piece of fiber in your skin, remove it immediately with a pair of tweezers or seek medical attention.

You may not always have the convenience of a laboratory or workshop environment for your fiber work. Work areas for splicing, building connectors, or other tasks may include basements, crawlspaces, underground vaults, an attic, or the back of a van. Don't take shortcuts just because you don't have the luxury of a full workshop at your disposal. Make sure you have an appropriate work surface and the proper tools and safety equipment before you start working.



Real World Scenario

Removing Fiber from Your Diet

Whenever you are scribing or cleaving fiber, you are going to end up with ends. No matter how short or long they are, rest assured that they will be hard to spot when it's time to clean up. Your first line of defense against these hard-to-spot fibers is immediate response. In other words, as soon as you cleave, pick up and dispose of the end.

If this discipline doesn't work for you, or if you find yourself with stray ends anyway, you'll still have to round them up and dispose of them properly. There are some low-tech and high-tech strategies to help you, as long as you can prepare them beforehand.

First, to prevent fiber ends from traveling beyond your work area, create a barrier around your work surface. This will keep bouncers and fliers confined to the work surface, where you can find them more easily.

If you want to make sure you've collected all of your strays, or if you know that you created more ends than you have collected, use a bright light source, such as an LED flashlight or even better, a pocket laser. To find the fibers, place the light at the level of the work surface and rotate along the surface. The fiber ends will pick up the light and Fresnel reflection will cause it to shine brightly.

As a last, low-tech measure, double up a piece of adhesive tape and pat it all over the work surface. This will pick up any minute silica particles that you could not spot or retrieve using other methods.

Chemicals

In your work with fiber optics, you will use several different types of chemicals, including 90-percent isopropyl alcohol for cleaning components, solvents for removing adhesives and other materials, and anaerobic epoxy for making connectors.

Each of these chemicals poses a number of different hazards and should be handled carefully. Each chemical that you use is accompanied by a Material Safety Data Sheet (MSDS), which provides important information on the chemical's properties, characteristics such as appearance and odor, and common uses. The MSDS also gives you information on specific hazards posed by each chemical and ways to protect yourself through specific handling procedures, protective clothing and equipment, and engineering controls such as ventilation. Finally, the MSDS describes emergency procedures including first aid for exposure to the chemical, methods for fighting fires in the case of flammable chemicals, and cleanup procedures for spills.

Even if you think you're familiar with the chemicals you handle, take time to read the MSDS. The information you gain could help prevent an accident or save valuable time in an emergency.

Let's look at some of the most important hazards associated with the chemicals you'll be handling.

Isopropyl Alcohol

Even though alcohol is commonly used in the home and the lab, its hazards should not be ignored or taken lightly. Alcohol vapors escape into the air easily, and can cause damage to your liver and kidneys if they are inhaled. The vapors are also highly flammable and can ignite if exposed to a spark or flame in high enough concentrations.

Alcohol can also cause irritation to your eyes, skin, and mucous membranes (nose and mouth) if it comes in direct contact with them.

Always use alcohol in a proper dispenser, shown in Figure 6.3. Store and transfer it carefully to avoid spills and excess evaporation. If you do spill any alcohol, clean it up with a dry cloth and dispose of the cloth in a container designed for flammable waste materials. As with all flammables, do not use alcohol in areas where sparks, open flames, or other heat sources will be present.

FIGURE 6.3 An alcohol dispenser helps reduce the risk of spills and fire.



It's important to remember that alcohol flames are almost invisible, and spills could lead to a mad scramble as you try to dodge flames you cannot see. Alcohol fires can be extinguished with water or a Class A fire extinguisher.

Solvents

Many solvents have similar properties that require that you handle them with great care. Like alcohol, solvents are very volatile and sometimes flammable. Their primary danger, however, is their hazard to your health.

One of the health hazards posed by solvents comes from the fact that they can cause excessive drying in your skin and mucous membranes, and the resulting cracking of the surface layers can

leave you open to infection. The hazard is more serious if you inhale the solvent vapors, as they can damage your lungs and respiratory system.

Solvents can also cause organ damage if inhaled or ingested. The molecules that make up most solvents can take the place of oxygen in the bloodstream and find their way to the brain and other organs. As the organs are starved of oxygen, they can become permanently damaged. One of the first signs that this kind of damage is taking place is dizziness or a reaction similar to intoxication. If you feel these symptoms, get to fresh air immediately.

Solvents may come in glass or plastic containers. Make sure that the container you are using is properly marked for the solvent it contains. Do not leave solvent in an unmarked container.

If you are carrying solvent in a glass bottle, use a rubber cradle to carry the bottle. The cradle protects the bottle from breaking if it falls.

Never leave a solvent container open. Keep the top off just long enough to transfer the amount necessary for the job, and replace it firmly to prevent the vapors from escaping.

Anaerobic Epoxy

The two-part epoxy used for making connectors is typically used in small quantities and does not present any immediate health hazards. If you are working in an enclosed space, however, such as the back of a van or an access area, vapors from the adhesive portion can irritate your eyes, nose, and throat. If you feel any of these symptoms, get to an open space immediately. The adhesive can also irritate your skin or eyes on contact. If you get any of the adhesive on you, wash the area immediately. If any material splashes in your eyes, flush them for 15 minutes at an eyewash station or sink.

Use caution when working with the primer portion of these adhesives. With a flash point of -18°C , it is highly flammable. Do not use anaerobic epoxy where there is an open spark or flame, or where heating components or elements are being used.

As with solvents, do not leave epoxy containers open any longer than necessary to dispense the amount you are using.

Site Safety

Many of the locations for fiber optic components may be in areas that require special safety precautions. These may include construction sites, enclosed areas, locations near high-voltage power lines, or areas requiring access by ladder or scaffold.

Always follow the on-site safety requirements and observe all warning signs. Here are some general safety rules to help you.

Electrical

When fiber optic systems run through the same area as electrical wiring or cabinets, use extreme caution with tools and ladders. One wrong move can send enough voltage through your body

to kill you. Remember that electrical fields can exist beyond a cable's insulation if high voltages are present, so use wooden ladders to reduce the possibility of exposure to induced voltages. Use care with cutters and other tools to avoid accidental contact with electrical wires, and report any hazardous conditions that may exist.

Remember that high voltage causes most of its damage by making muscles seize up, including the heart and lungs. The greatest chance of damage comes when voltage passes through your heart to get the ground, such as when you touch a wire with your hand and your opposite leg provides the path to ground, or when the current passes from one hand to the other.

If you accidentally grab a live wire, the voltage may keep your hand clenched, making it nearly impossible to release the wire. If you see someone who is in this situation, do not try to pull them away with your hands. You may be caught up in the circuit as well. Instead, use a non-conducting stick, such as a wooden or fiberglass broom handle, to knock the victim away from the voltage source.

If the victim is not breathing, artificial respiration may be necessary to get the heart and lungs operating again.

Ladders

You may often find that you need a ladder to reach a work area. When choosing a ladder, make sure you select one that matches your requirements. Self-supporting ladders, such as stepladders, should be used only if the work area is not near a vertical support such as a wall, and the floor beneath the work area is even and firm.

Non-self-supporting ladders, such as extension ladders, are useful when there is a firm vertical support near the work area and there is a stable, nonslip surface on which to rest the ladder. When setting up a non-self-supporting ladder, it is important to place it at an angle of $75\frac{1}{2}^{\circ}$ to support your weight and be stable. A good rule for finding the proper angle is to divide the working height of the ladder (the length of the ladder from the feet to the top support) by four, and place the feet of the ladder that distance from the wall. For example, if the ladder is twelve feet tall, the bottom should be three feet from the wall.

Be sure that the ladder you select can carry your weight along with the weight of any tools and equipment you are carrying. Read the labels and warnings on the ladder you select to make sure that it is the right one for the work you are performing.

If you are working near live electrical systems, be sure to use a nonconducting ladder of wood or fiberglass. If you are working near high heat, select an aluminum ladder to avoid scorching or melting your only means of support.

In the work area, place the ladder so that you can work comfortably without having to reach too high or too far to the side. Overreaching can cause you to lose your balance and fall, or place too much weight above or to the side of the ladder, causing the entire ladder to come down. If your work area extends beyond a comfortable reach, climb down the ladder and move it. Do not try to "walk" the ladder to a new work area.

Inspect ladders before using them. Make sure the rungs and rails are in good shape, and are not split, broken, or bent. Make sure all fittings and fasteners are secure, and that all locking mechanisms are working properly.

When carrying a ladder to the worksite, reach through the rail and balance it on your shoulder. Be aware of obstacles and corners as you carry it, and make sure others are aware that you have an awkward load, especially if you are walking through hallways or other limited visibility areas.

Trenches

If you are working on a fiber system in a trench, be sure the trench is properly dug and shored before entering it. Never work in a trench without someone else around, in case of a collapse.

If you have never worked in a trench before, learn the proper way to enter and exit a trench. Always use a ladder. Never jump into a trench or try to climb down the sides. You could trigger a collapse.

If you witness a collapse and others are trapped but not in immediate danger, do not try to dig them out yourself. You risk making the problem worse. Get help immediately. Special training is required to recover victims from a trench collapse, so leave it to the experts.

Emergencies

It takes only one slip-up to create an emergency. It could come from a moment of carelessness, an attempt at taking a shortcut, or ignorance of the proper procedures. Whatever the cause, the first response is always the same. Remain calm. Panic can cause even more damage and complicate matters beyond repair.

The best way to handle emergencies is to accept the fact that they will occur, and be ready for them. Make sure you know what can go wrong with the materials and chemicals you handle, and what you can do to minimize the damage.

Injury

Injuries can be caused by misuse of tools, fibers penetrating your skin or eyes, burns, falls, or any number of other mishaps. Make sure that you and your co-workers know the location of first-aid kits in your work area. Also, make sure you know how to reach emergency personnel. If you are on a new job site, make it a priority to familiarize yourself with emergency procedures and contact information.

Chemical Exposure

Accidental chemical exposure can result in anything from temporary discomfort to permanent injury or death. The first few seconds of an emergency involving chemical exposure can be critical in the victim's recovery.

If the chemical is splashed on the skin or in the eyes, flush the affected area with clean water immediately, using a shower or eyewash station if they are available. Continue flushing the area for at least 10 minutes. This washes the chemical away, but also dilutes its effects if it has been absorbed by the skin or eyes.

In case of inhalation, remove the victim to fresh air immediately and call for medical attention. If a chemical has been accidentally swallowed, induce vomiting unless the chemical is corrosive and could damage the esophagus and throat as it comes back up. Use a neutralizing liquid such as milk to dilute corrosive chemicals if they have been swallowed, and seek medical attention immediately.

Fire

If a fire breaks out in your work area, it may be small enough for you to handle alone. If it is small, you can smother it with a damp cloth. If it is larger, but contained in a trash can or other enclosure, use the appropriate fire extinguisher for the material that is burning.

To use a fire extinguisher properly, remember the acronym PASS as you use the following procedure:

Pull Pull the pin from the fire extinguisher trigger.

Aim Aim the extinguisher at the base of the fire.

Squeeze Squeeze the handle firmly to activate the extinguisher.

Sweep Sweep the extinguisher discharge at the base of the fire until the flames are out.

Do not give the fire a chance to trap you. If you think that the extinguisher will not put out the fire completely and there is a risk that your exit will be cut off, leave immediately and call for help. You can do more good with a phone call than you can in a failed attempt at being a hero.

Summary

This chapter covered basic safety guidelines for your work with fiber optic systems. It described hazards specifically associated with fiber optics work, as well as general safety concerns.

We described the different types of lasers that you may encounter and the ways in which they are classified. We also discussed methods for working safely with optical fiber in the lab and in various work situations.

Exam Essentials

Understand basic safety. Make sure you understand the basic rules for working safely in most situations. Be sure you know the importance of using engineering controls and personal protective equipment properly. Be able to describe good work practices to help keep you safe.

Be familiar with light sources. Make sure you understand the dangers of working with lasers and other high-energy light sources. Be able to describe the Service Groups of different types of lasers and what they mean. Be sure you know how to protect yourself from the hazards of laser light.

Know the hazards of chemicals. Be familiar with the hazards of the chemicals you will use in working with fiber optics. Be able to describe their effects and the ways in which you can protect yourself. Also, make sure you understand what a Material Safety Data Sheet is and why it is important to your safety.

Understand site safety. Be able to describe the hazards of different sites where you may work. Make sure you can discuss how to work safely under different conditions.

Know emergency procedures. Be able to describe emergency procedures and the proper responses for different types of emergencies. Understand the importance of being prepared for emergencies.

Review Questions

1. What government agency writes and enforces safety rules for the workplace?
 - A. ANSI
 - B. OSHA
 - C. NASA
 - D. FBI
2. How often should you inspect your personal protective equipment?
 - A. Annually
 - B. Monthly
 - C. Weekly
 - D. Whenever you use it
3. Because infrared lasers cannot be seen, they _____.
 - A. Cannot hurt your eyes
 - B. Are not useful in fiber optics
 - C. Are especially dangerous to your eyes
 - D. Require special instruments
4. Lasers in Service Group 1 are considered _____.
 - A. The most hazardous
 - B. The least hazardous
 - C. Hazardous only under certain conditions
 - D. Hazardous only to living tissue
5. Service Group 3a lasers operate at power levels of _____.
 - A. 400 nW
 - B. Below 1 mW
 - C. 1–5 mW
 - D. Above 5 mW
6. Service Group 4 lasers are different from the other service groups because they _____.
 - A. Operate in the infrared range
 - B. Can damage the eyes
 - C. Can start fires
 - D. Whistle

7. Isopropyl alcohol can be hazardous because it is _____.
 - A. Corrosive
 - B. Flammable
 - C. A skin irritant
 - D. Poisonous to inhale
8. One of the symptoms of exposure to solvent vapors is _____.
 - A. The appearance of intoxication
 - B. Yellowing skin
 - C. Irritability
 - D. Hunger
9. The adhesive portion of anaerobic epoxy is _____.
 - A. Flammable
 - B. Noxious
 - C. Explosive
 - D. A skin irritant
10. When setting up a non-self-supporting ladder, it should be at an angle of _____.
 - A. 60°
 - B. $65\frac{1}{2}^{\circ}$
 - C. 70°
 - D. $75\frac{1}{2}^{\circ}$

Answer to Review Questions

1. B. The Occupational Safety and Health Administration is responsible for creating and enforcing workplace safety regulations.
2. D. You should inspect your PPE every time you use it to make sure it is not damaged or defective.
3. C. Because infrared lasers are invisible, you cannot know when they are on, even if you are looking directly into a fiber. You should always make sure the fiber is disconnected from the light source before looking into it with your bare eyes or optical instruments.
4. B. Service Group 1 lasers do not emit hazardous radiation at any known levels, so protective equipment is not required when using them.
5. C. Service Group 3a lasers operate at power level between 1 and 5 mW, and are considered safe for the naked eye for short exposures, but not through optical instruments.
6. C. Service Group 4 lasers are powerful enough to burn any living tissue and can even start fires if aimed at flammable objects.
7. B. Isopropyl alcohol is extremely flammable and should only be used in a proper dispenser to avoid vapors escaping into the air.
8. A. Because solvent molecules can displace oxygen molecules in the blood, they can starve the brain of oxygen and give the appearance of intoxication.
9. D. The adhesive portion of anaerobic epoxy can irritate your eyes or skin if it comes in contact with them.
10. D. An angle of $75\frac{1}{2}^\circ$ provides the proper combination of strength and stability for the ladder.



Chapter

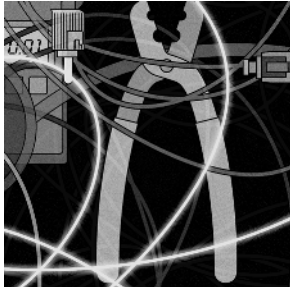
7

Fiber Optic Cables

OBJECTIVES COVERED IN THIS CHAPTER:

✓ Fiber Optic Cables

- Draw a cross section of a fiber optic cable and explain the purpose of each segment.
- Explain why and where loose buffer fiber optic cable is used.
- Describe tight buffer fiber optic cable.
- Describe common strength members found in fiber optic cables.
- Describe common jacket materials found in fiber optic cables.
- Describe simplex and duplex cordage and explain the difference between cordage and cable.
- Describe distribution cable.
- Describe breakout cable.
- Describe armored cable.
- Describe messenger cable.
- Describe ribbon cable.
- Describe submarine cable.
- Describe hybrid cable.
- Describe composite cable.
- Explain fiber optic cable duty specifications.
- Explain how and when a fan-out kit is used.
- Explain how and when a breakout kit is used.
- Describe the National Electric Code (NEC) optical fiber cable types.
- Describe the NEC listing requirements for optical fiber cables.
- Describe the NEC listing requirements for optical fiber raceways.
- Describe the TIA/EIA-598-B color code and cable markings.
- List the TIA/EIA-568-B.3 bend radius specifications.



So far we have studied the principles and characteristics of individual optical fibers. While they are certainly adequate for the job of carrying signals from one place to another, they are neither large enough nor strong enough to withstand the rigors of handling, transportation, and installation. In addition, some installations require multiple fibers for sending and receiving or for routing to a number of sites.

For a fiber to be suitable for everyday use, it must be incorporated into cables that provide standardized fiber groupings, protection for the fibers, and suitable size for handling.

In this chapter, we will describe standard fiber optic cables used in most installations. We will detail different types of fiber optic cables and the uses for which they were designed. We will also describe some of the basic techniques used for handling and installing fiber optic cable.

Basic Cable

You may already be familiar with cables used for electrical wiring. These cables typically consist of two or more wires bundled together and held by a protective outer covering. In addition to holding the wires in place, cables also protect the wires from damage and insulate them from electrical interference.

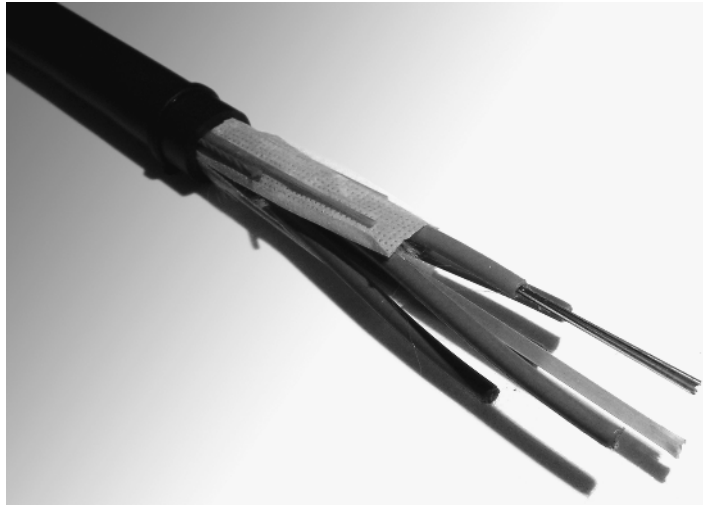
Some of the largest cables, used for telephone transmissions, can be several inches in diameter and carry a hundred pairs of wires, with each pair being used for a single telephone conversation.

Fiber optic cables, on the other hand, are usually much smaller, because a single fiber can carry several thousand transmissions. With the addition of a second fiber, two-way (duplex) transmissions are possible.

Other installations may require even more fibers in a single cable for networking or future expansion. Figure 7.1 shows a typical optical fiber cable with multiple fibers running through it.

Because optical fibers are used in many different configurations and circumstances, manufacturers have created a wide variety of cables to meet specific needs. The type of signal being carried and the number of fibers needed are just two of the many considerations when selecting the right cable for an application. Other factors include:

- Tensile strength
- Temperature resistance
- Ruggedness
- Environmental extremes
- Appearance
- Durability
- Flexibility

FIGURE 7.1 Typical fiber optic cable

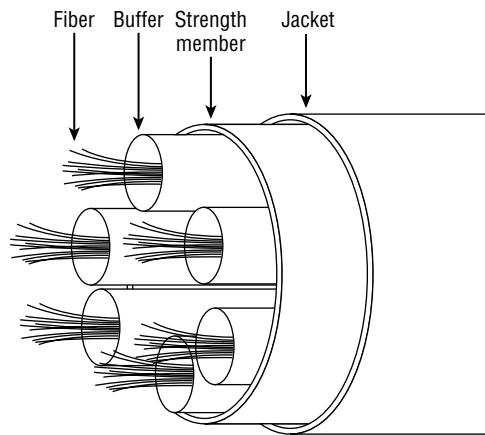
The exact combination of these factors will vary depending on the cable installation. A cable installed inside an office building, for example, will be subject to less extreme temperatures than one installed outdoors or in an open warehouse. Cabling installed in a manufacturing facility for instrumentation may be exposed to abrasive dusts, corrosive chemicals, or hotwork, such as welding, requiring special protection. Some cables may be buried underground, where they are exposed to burrowing or chewing animals, while others may be suspended between poles, subject to their own weight plus the weight of animals and birds who think the cables were put there for them.

Cable Components

Whether a cable contains a single optical fiber or several, it has a basic structure in common with other cables. As shown in Figure 7.2, a typical fiber optic cable consists of the optical fiber (made up of the core, cladding, and coating), plus a buffer, strength members, and an outer protective jacket. Let's look at these components individually.

Buffer

While the fiber's coating is the first non-optical protective layer surrounding the fiber, the buffer, which in turn surrounds the coating, provides a greater measure of protection as well as some tensile strength, which is useful when pulling the cable to install it or when it must hang between two suspension points. The buffer is also the first layer used to define the type of cable construction. Depending on the application, manufacturers can provide *loose-buffered* or *tight-buffered* cables.

FIGURE 7.2 Fiber optic cable components

Loose Tube Buffering

Loose tube buffering consists of a buffer layer that has an inner diameter much larger than the diameter of the fiber, as shown in Figure 7.3. The primary purpose of loose tube buffering is to allow the fiber more room to move independently of the buffer and the rest of the cable—an important factor if the cable will be subjected to temperature extremes that cause expansion or contraction, changes in tension, or excessive bending. Figure 7.4 shows how the fiber inside the loose tube buffer is isolated from movement of the buffer and the rest of the cable.

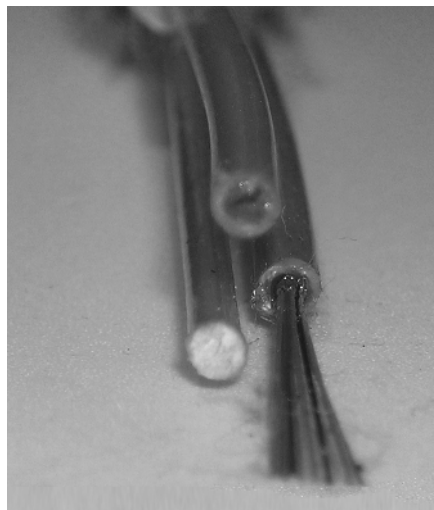
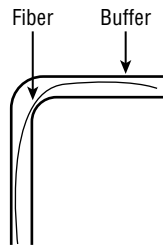
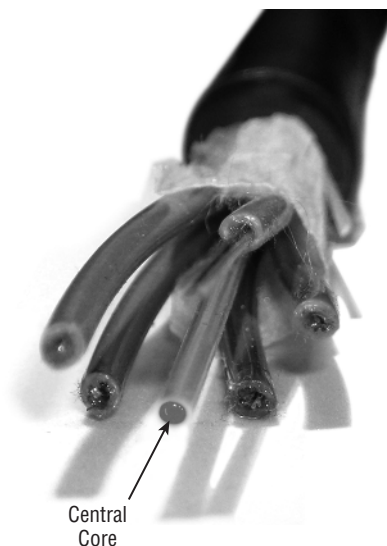
FIGURE 7.3 Loose tube buffered cable has a buffer diameter greater than the fiber diameter.

FIGURE 7.4 Loose tube buffering protects the fiber from cable movement.

Because the fiber is not connected to the buffer, it will usually follow a gently meandering path through the buffer, giving it some extra length. This is useful when the cable is stretched—as it inevitably will be—since a small amount of stretching will only straighten the fiber out without putting any damaging tension on it.

Loose tube buffered cable may be single fiber or multifiber (not to be confused with single-mode and multimode fiber), meaning that it may have one or many fibers running through it. In addition, a cable may contain a number of loose tube buffers grouped together, as shown in Figure 7.5. In such cables, loose tube buffers are grouped around a central core that provides added strength.

Loose tube buffered cable that is made for outdoor use may be filled with a gel that prevents water from getting in the cable. The gel also helps to cushion the fiber against any damage from shock or pressure and insulates it against rapid temperature changes. In catalogs, these cables are referred to as *loose tube, gel-filled* or LTGF cables.

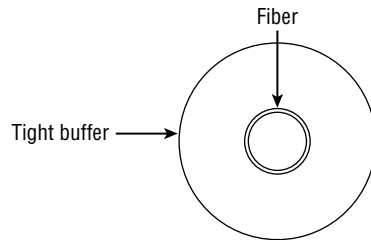
FIGURE 7.5 Loose tube buffers in a cable

Tight Buffering

Tight-buffered cable is used in more controlled environments where the cable is not subjected to changes in position, temperature, tension, or moisture. In short, tight-buffered cable is mostly used indoors.

As shown in Figure 7.6, tight-buffered cable begins with a 250 μ optical fiber. The plastic buffer itself is 900 μ in diameter and is applied directly to the outer coating layer of the fiber. In this way, it resembles a conventional insulated copper wire. The buffer may have additional strength members running around it for greater resistance to stretching. Tight-buffered cable is more flexible than loose tube buffered cable. With added protection against outside forces, tight-buffered cable can be used outdoors or in temporary setups where greater flexibility is required.

FIGURE 7.6 Tight-buffered cable uses a buffer attached to the fiber coating.



One of the benefits of tight-buffered cable is that the buffered fiber can be run outside of the larger cable for short distances, making it easier to attach connectors. The standard thickness of the buffer allows many different types of conventional connectors to be applied. When several tight-buffered cables are run together, as shown in Figure 7.7, they can then be broken out and prepared with connectors. As with loose tube buffered cables, tight-buffered cables usually have several fibers grouped around a central strength member. These cables may in turn be grouped into larger cables with similar construction.

The tight buffer provides added tensile strength to the fiber, but does not isolate it from stretching and bending the way loose buffering does. Since the plastic, which is attached directly to the fiber, expands and contracts with temperature at a different rate than the fiber itself, the different expansion rates can cause loss-inducing microbends. For this reason, tight-buffered cable is most often used in areas where the cable is fixed in place and maintained in controlled conditions. If any microbending takes place, however, the cable runs indoors are typically short enough to minimize attenuation problems.

Strength Members

We've already seen a little bit of how strength members help increase a cable's tensile strength. The primary importance of these members is to allow the cables to survive installation, which requires a great deal of pulling through conduits or cutouts. Strength members must also protect the fiber against stretching and excessive bending. In addition to running through the middle of a cable,

strength members can also run between the buffer and the outer jacket. The exact combination of strength members and materials depends on the stresses that the cable must withstand.

Strength members are the parts of the cable that are designed to improve its tensile strength. Depending on where they occur in the cable and the type of cable in which they are being used, strength members may be made of many different materials. Some of the most common are:

- Aramid yarns, usually Kevlar
- Fiberglass/epoxy rods
- Steel members

Aramid yarns, as shown in Figure 7.8, are useful when the entire cable must be flexible. These fine, yellowish or gold yarns are the same material used in high-performance sails, bulletproof vests and fire protection gear. They have the advantages of being light, flexible, fire-resistant and quite strong. Aramid yarns are typically used in cable subgroups if they will be bundled into larger cables, but they may also be used as central members.

FIGURE 7.7 Tight-buffered cables in multiple units

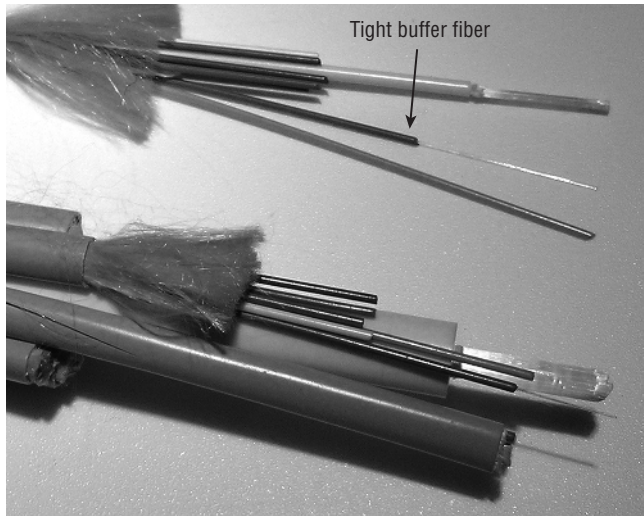
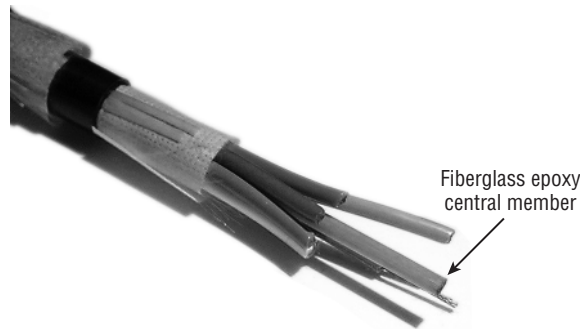


FIGURE 7.8 Aramid (Kevlar) yarns used as strength members



Larger cables and those that will see heavier use will usually have a central member of either fiberglass and epoxy or steel added, as shown in Figure 7.9. This central member is more resistant to stretching than aramid fibers, but it is not as flexible.

FIGURE 7.9 A fiberglass epoxy central member adds strength when a cable will be subjected to strong pulling forces.



Although steel is preferred over fiberglass for its greater tensile strength, many applications require a *dielectric*, or nonconductive cable. For example, steel running through a cable outdoors makes an excellent lightning rod. Fiberglass is nonconductive while satisfying most strength requirements for cable.

Jacket

The jacket is the fiber's outer protective layer. This is the part that must protect the fiber from the worst of the world outside, including sunlight, ice, animals, equipment accidents, and ham-handed installers. The jacket must also protect the fiber from abrasion, oil, corrosives, solvents, and other chemicals that would attack and destroy it.

Jacket materials vary with the amount of protection they must provide. Typical jacket materials include:

Polyvinyl chloride (PVC) PVC is used primarily for indoor cable runs. It is fire-retardant and flexible, and is available in different grades to meet different conditions. PVC is not highly water resistant, and it does not stand up well to solvents. In addition, it loses much of its flexibility at low temperatures.

Polyethylene This material is only used outdoors because it does not meet fire safety standards. It is a standard protective covering, shielding the fiber from water and abrasion while remaining stable over a range of temperatures.

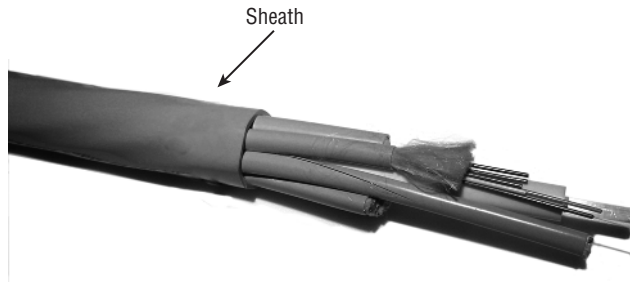
Polyvinyl difluoride (PVDF) This cable is chosen for its low smoke and fire retardant properties for use in cables that run through airways or *plenums* in a building. PVDF cables are not as flexible as other types, so their fire safety properties are their primary draw.

Low-smoke, no halogen (LSNH) Another jacket material chosen for its fire safety ratings, LSNH produces little smoke under fire conditions and none of the halogen compounds that are poisonous to breathe. These fibers are chosen for use in enclosed spaces and where smoke from other jacket materials would damage electronic equipment.

Steel Steel jackets are used when the cable is at risk from crushing, impact, or encounters with chewing animals such as rodents. Steel jackets may be used inside or outside, but are obviously used only when necessary due to their cost and the difficulty of running them compared to the plastic-jacketed cables.

Some cables contain multiple layers of jacketing and strength members, as shown in Figure 7.10. In such cables, the outermost jacket is called a *sheath*, while the inner protective layers are still called jackets.

FIGURE 7.10 The sheath is the outer layer of a cable with multiple jacket layers.



The *ripcord* is a piece of strong thread running through the jacket. When the ripcord is pulled, it splits the jacket easily to allow the fibers to be separated for connectorization. The ripcord reduces the risk incurred when cutters or knives are used to split the jacket.

Cable Types

As uses for fiber have become more varied, manufacturers have begun producing cables to meet specific needs. Cable configurations vary based on the type of use, the location, and future expansion needs, and it is likely that more will be created as future applications emerge.

Bear in mind that different cable arrangements are variations on a theme. Different combinations of buffer type, strength members, and jackets can be used to create cables to meet the needs of a wide variety of industries and users.

Let's look at some of the commonly available optical fiber cables.

Cordage

The simplest types of cables are actually called *cordage*, and are used in connections to equipment and patch panels and are typically made into patch cords. The major difference between cordage and cables is that cordage only has one fiber/buffer combination in a jacket, whereas cables may have multiple fibers inside a single jacket, or multiple cords inside an outer jacket or sheath.

The two common types of cordage are *simplex* and *duplex*.

Simplex Cordage

Simplex cordage, shown in Figure 7.11, consists of a single fiber with a tight buffer, an aramid yarn strength member, and a PVC jacket. Simplex cordage with plastic fiber uses no strength members, but has a thicker jacket.

FIGURE 7.11 Simplex cordage



Simplex cordage gets its name from the fact that, because it is a single fiber, it can be used only for one-way, or simplex, transmission, although bidirectional communications have been accomplished using a single fiber.

Duplex Cordage

Duplex cordage, also known as *zipcord*, is similar in appearance to household electrical cords, as you can see in Figure 7.12. Duplex cordage is a convenient way to combine two simplex cords to achieve duplex, or two-way, transmissions without individual cords getting tangled or switched around accidentally.

FIGURE 7.12 Duplex cordage



Simplex and duplex cordage are meant for short-distance connections between equipment and for temporary connections only, and should not be used as permanent cabling for network or long-distance transmission.

Distribution Cable

When it is necessary to run a large number of fibers through a building, distribution cable is often used. Distribution cable consists of multiple tight-buffered fibers bundled in a jacket with a strength member. Typically, these cables may also form subcables within a larger distribution cable, as shown in Figure 7.13.

FIGURE 7.13 Distribution cable with subcables



Distribution cables usually end up at patch panels or communication closets, where they are hooked into devices that communicate with separate offices or locations. These fibers are not meant to run outside of office walls or be handled beyond the initial installation, because they do not have individual jackets.

Distribution cables often carry up to 144 individual fibers, many of which may not be used immediately but should be considered for future expansion.

Breakout Cable

Breakout cables are used to carry fibers that will have individual connectors attached, rather than being connected to a patch panel.

Breakout cables consist of two or more simplex cables bundled around a central strength member and covered with an outer jacket, as shown in Figure 7.14. Like distribution cable, breakout cables may be run through a building's walls, but the individual simplex cords can then be broken out and handled individually.

FIGURE 7.14 Breakout cable contains simplex cords bundled around a central strength member.

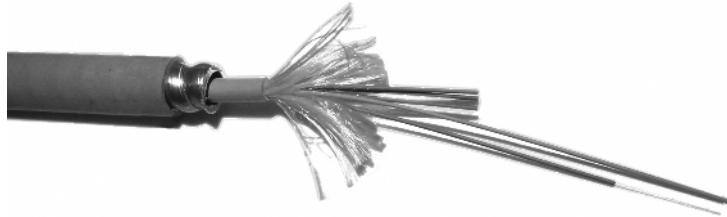


As is the case with distribution cable, breakout cables may end up in communication closets, but in the case of breakout cables, users can manually change connections. Breakout cables may also be used to connect directly to equipment.

Armored Cable

Armored cable, shown in Figure 7.15, addresses the special needs of outdoor cable that will be exposed to potential damage from equipment, rodents, and other especially harsh attacks.

FIGURE 7.15 Armored cable protects the fiber from harsh conditions and gnawing animals.



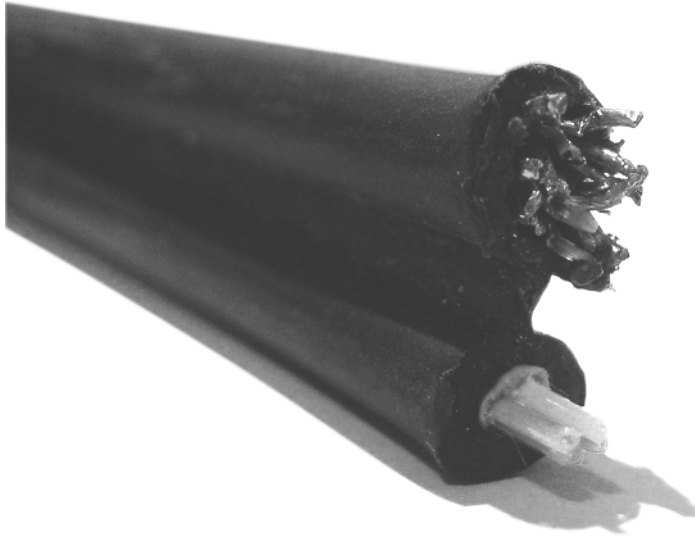
Armored cable consists of a cable surrounded by a steel or aluminum jacket which is then covered with a polyethylene jacket to protect it from moisture and abrasion. It may be run aerially, installed in ducts, or placed in underground enclosures with special protection from dirt and clay intrusion.

Messenger Cable

When a fiber optic cable must be suspended between two poles or other structures, the strength members alone are not enough to support the weight of the cable. Installers must use a *messenger cable*, which incorporates a steel or dielectric line known as a messenger to take the weight of the cable. The cable carrying the fiber is attached to the messenger by a thin web and hangs below it, as shown in Figure 7.16.

Also called *figure 8* cable for the appearance of its cross section, messenger cable greatly speeds up installation of aerial cable by eliminating the need to lash a cable to a pre-run messenger line.

FIGURE 7.16 Messenger cable is used for aerial installations.



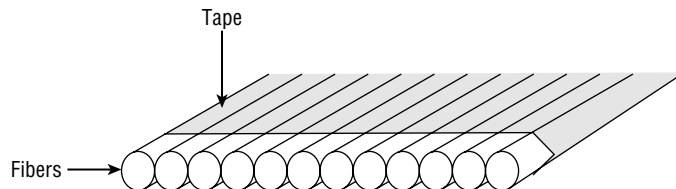
In applications that will run near power lines, the dielectric messenger is often used to minimize the risk of energizing the cable through induced current, which is created when the electrical field from a high voltage alternating current line expands and contracts over a nearby conductor. If a conductive cable is close enough to the alternating current, the induced current may be strong enough to injure someone working near the cable.

It's a good practice, in fact, to use dielectric strength members wherever tension considerations permit, as this will help avoid any potential conductivity problems in the cable.

Ribbon Cable

Ribbon cable is a convenient solution to some space and handling problems. The cable contains fiber ribbons, which are actually fibers bonded side by side by Mylar tape in a single tight jacket such as the one shown in Figure 7.17, similar to a miniature version of wire ribbons used in computer wiring.

FIGURE 7.17 Ribbon cables consist of parallel fibers bonded together.



Because the fibers are bonded together so closely, fiber ribbons take up much less space than individually jacketed ones. As a result, the ribbons allow more fibers to be packed into loose tube cables, saving installation time and greatly reducing the required space for installation.

As shown in Figure 7.18, ribbon cables come in two basic arrangements. In the *central tube ribbon cable*, fiber ribbons are stacked on top of one another inside a loose tube buffer. This type of arrangement can hold several hundred fibers in close quarters. The buffer, strength members, and cable jacket carry any strain while the fiber ribbons float inside the buffer tube.

FIGURE 7.18 The central tube ribbon cable (bottom) and the jacketed ribbon cable (top)



The *jacketed ribbon cable* looks like a regular tight buffer cable, but it is elongated to contain a fiber ribbon.

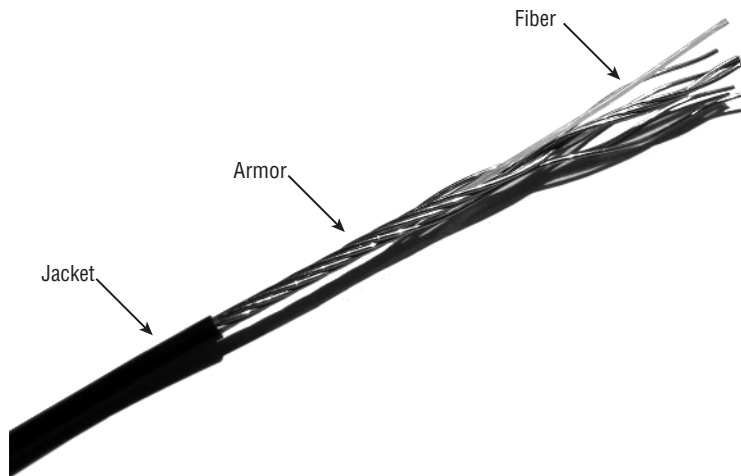
While ribbon fiber provides definite savings in conduit and cabinet space and splicing and connectorizing time, it does require special equipment and training to take advantage of those benefits. Connectors, strippers, and cleavers must all be tailored to the ribbon fiber dimensions, and installers must know how to use the equipment properly. For these reasons, ribbon fiber may not be the best solution in all situations. It should be considered, however, for high-volume installations where the initial equipment and training investment is offset by the time and space savings it provides.

Submarine Cable

Submarine cable is specially designed for carrying fiber underwater. Not all submarine cable is the same, however. Depending on the distance it will span and the type of service it will provide, submarine cable can take many different forms.

Submarine cable may be laid in trenches under the bottom of waterways where shipping or fishing activities threaten to snag or damage the cable, or they may be laid directly on the bottom of less-traveled waterways, or on the deep ocean floor where such activities do not penetrate. Figure 7.19 shows submarine cable with armor.

FIGURE 7.19 Submarine cable with armor



Short Runs

Short runs are a relative term when we are discussing submarine cables. Generally, if it is easier to go around a body of water without expending much more cable, that's the way the cable will go. Bodies of water that require short runs of cable may be up to a hundred kilometers across.

Short runs of submarine cable may be used to cross rivers, bays and harbors. In addition, they may be used to cross large lakes such as the Great Lakes, or even the water between islands such as those in Hawaii or Japan. They may also be used to achieve shorter cable runs between towns or cities on a coastline than could be done running the cable over land.

Short runs are usually part of a larger communication network within a country and the water is simply seen as another obstacle to cross. Cables for this use are typically beefed-up versions of standard cables used on land, with extra protection to keep water from getting to the fibers.

Long Runs

Cables used for long runs cover distances of up to 1000 km, often linking islands that cover a large expanse of water to each other and to the mainland, or crossing larger bodies of water that separate countries from one another. The runs will often include repeaters for older cables or optical amplifiers for newer ones along with the copper cable to power them. Typical areas for these runs include the Caribbean, South Pacific, and Mediterranean.

Cables used in long runs are usually connecting countries with one another, and therefore are likely to carry high volumes of traffic. In addition, they may have to serve several purposes, so their capacity will be higher than the shorter-run cables, and extra capacity may be provided for future growth.

Intercontinental

Intercontinental cables are designed to run thousands of kilometers across the ocean floor and handle the tremendous traffic of entire continents. These cables are essential to global commerce and security, and because they are difficult to reach once they have been laid, they are designed for long life and survivability with several layers of metal, plastic and fiber protection.



Real World Scenario

More Underwater Hazards

While fishing operations, anchoring, and other hazards pose a threat to underwater cables of all types, some fiber optic cables running in the Grand Canary Islands encountered a special danger soon after they were laid.

The cables were part of a fiber optic test system run by AT&T in 1985. Only three weeks after the cables were laid about $\frac{3}{4}$ of a mile below the surface, the cable stopped operating and was out of commission for a week. Technicians first thought that the cable had separated because of abrasion due to scraping across the ocean floor, but when the cable was examined more closely, it was found to have small shark's teeth embedded in it. The shark attacks were repeated twice over the next several months.

Why the repeated shark bites on fiber optic cable? Sharks are known to be drawn to prey by their electromagnetic emissions, and attacks on standard coaxial cable would have been much more likely if the cable were not shielded against electromagnetic interference. But the fiber did not carry any electricity, only light, so the sharks should have ignored it.

The answer lay in the fact that light does have an electromagnetic signature, but because the optical fiber is not subject to electromagnetic interference, no shielding was applied to the cable. The small sharks were drawn to the weak electrical fields put out by the light traveling through the fiber, and in the dark water attacked it as if it were a meal.

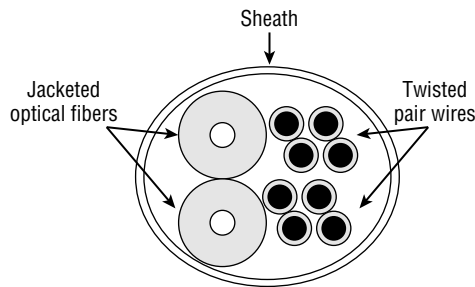
Hybrid Cable

Hybrid cable, as applied to fiber optics, combines multimode and single-mode fibers in one cable. Hybrid cable should not be confused with composite cable, although the terms have been used interchangeably in the past.

Composite Cable

Composite cable, as defined by the National Electrical Code (NEC), is designed to carry optical fiber and current-carrying electrical conductors in the same run. As shown in Figure 7.20, composite cable consists of optical fibers along with twisted-pair wiring typical of telephone wiring. This arrangement is convenient for networks that carry fiber optic data and conventional telephone wiring to the same user. Other uses for composite cable include providing installers with a way to communicate during fiber installation and providing electrical power to remote equipment along the fiber's route, such as repeaters.

FIGURE 7.20 Composite cable carries fiber and wiring in the same run.



Cable Duty Specifications

The various combinations of strength members, jacket materials, and fiber arrangements are determined by the specific requirements of an installation. Among the factors considered are the amount of handling a cable will take, the amount of stress the cable must endure in normal use, and the locations where it will run.

Cable duty specifications are divided into four basic types:

Light-duty cables These are designed for basic protection of the fiber within and minimal handling. A good example of a light-duty fiber cable is a simplex cord, which consists of a tight-buffered fiber with a jacket. It is not meant for excessive pulling or abuse.

Heavy-duty cables These are designed for more and rougher handling, with extra layers of strength members and jacketing around the fiber. They are made for harder pulling during installation, and to protect the fiber within from damage in exposed or extreme environments.

Plenum cable This is meant to go in the plenum between walls, floors, and drop ceilings. Because fires find their way into these spaces easily, the jacket on the cable must be made of a material that will not give off noxious fumes when it gets hot. The low-smoke, no halogen (LSNH) jacket materials, as described earlier in this chapter, are chosen for their ability to slow the spread of fire and reduce the amount of poisonous smoke created if they do start to burn. National Electrical Code article 770 requires that cables running through plenums must either have this type of jacketing or be enclosed in a fireproof conduit.

Riser cable Like plenum cables, these are built with fire safety in mind. Riser cables run in the vertical spaces spanning one or more floors, and have the potential to carry fire from one floor to the next if they ignite. For this reason, riser cables must pass the Underwriters Laboratories (UL) 1666 “Standard Test for Flame Propagation Height of Electrical and Optical-Fiber Cables Installed Vertically in Shafts,” as specified in NEC article 770.

Cable Termination Methods

Some fibers, such as those found in simplex and duplex cords and breakout cable, are already set up to receive connectors and can be handled easily. Others, including loose tube buffered cable, must be prepared for connectors and handling with special kits.

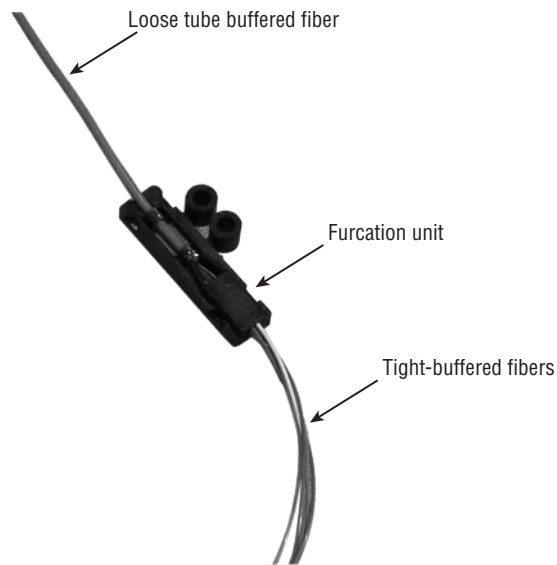
These kits, known as *fanout kits* and *breakout kits*, are designed to adapt groups of coated fibers for connectors by separating them and adding a tight buffer to each one. The buffer protects the fiber and gives it a thickness of 900 μ so that a standard connector can be attached. The exact handling of the fibers depends on the type of kit used.

Fanout Kit

The fanout kit, shown in Figure 7.21, converts loose tube buffered fibers into tight-buffered fibers ready for connectors.

A typical fanout kit contains an enclosure sometimes called a *furcation unit*. The furcation unit attaches to the jacket of the loose tube cable while the unbuffered fibers pass through the unit and out the other end. Hollow tight buffer tubes 900 μ in diameter are applied to the fibers and passed into the furcation unit, which is then closed, locking the buffers in place on the fibers. After the fibers have the buffers applied, connectors can be attached for use in a patch panel or other protected enclosure.

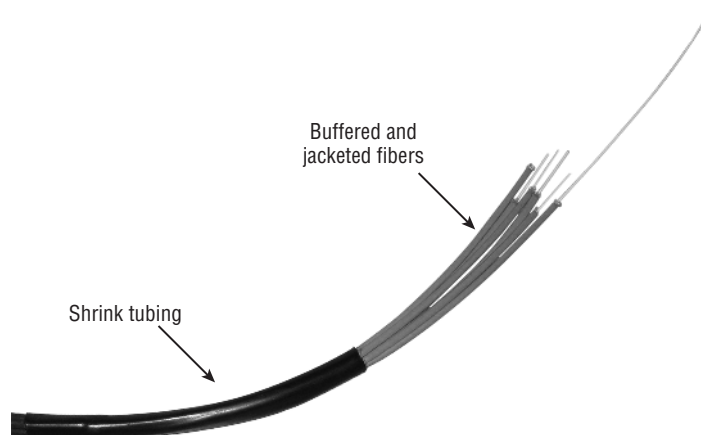
FIGURE 7.21 A fanout kit adds a tight buffer to individual fibers.



Breakout Kit

The breakout kit, shown in Figure 7.22, is similar to the fanout kit in that it spreads the fibers from the loose tube buffer through a furcation kit and provides 900 μ tight buffers to be applied to the fiber. The breakout kit, however, is designed to allow the fiber to be connected directly to equipment with standard connectors.

FIGURE 7.22 A breakout kit adds a tight buffer and a jacket to individual fibers.



In addition to materials provided in the fanout kit, the breakout kit provides a 3 mm diameter jacket with an aramid strength member that slips over the buffer tube and is also locked in the furcation unit. Heat-shrink tubing may also be used to provide strain relief and limit bending where the fibers spread out.

Breakout kits and fanout kits are available to match the number of fibers in the cable being used, and some companies offer kits that can be used with ribbon cable. Different length buffer tubes and jackets are available depending on the amount of fiber that will be spread out from the end of the cable.

Note that different vendors may use a variety of terms or trade names to refer to breakout kits and fanout kits. Some will offer jackets as an add-on to fanout kits to make the breakout kits. You can identify the product that you need by determining whether it is made for fiber that will end up at a patch panel or be connected directly to equipment.

Blown-in Fiber

What if you knew that you wanted to run fiber in a building, but you didn't know exactly what equipment it would be serving, or what the requirements for the fiber were? One solution to this dilemma is *blown-in fiber*.

Although still a relatively recent addition to the installer's arsenal, blown-in fiber can solve a number of problems created by the wide variety of optical fiber systems available.

Blown-in fiber installation starts with a hollow tube about 5 mm in diameter. This tube is installed during construction or renovation and acts as a loose tube buffer for fibers that will run through it later. The tube may also be part of a tube cable, which may carry up to 19 tubes and is available in configurations similar to other cables, including armored, all-dielectric, and waterproof.

To run the fibers through the tube, simply lead the fibers into one end of the tube and blow pressurized air through the tube. The air carries the fibers with it through the tube, similar to the way that pneumatic carriers at drive-through tellers shoot your deposits into the bank building.

Blown-in fiber can be used, as already described, when the fiber needs in a building are not yet known, or when fiber needs to be repaired or upgraded. Blown-in fiber is also useful when a company cannot afford to install all of the fiber that it might potentially use. As a business grows and the new fiber can be economically justified, it can be blown in as needed.

One advantage of blown-in fiber, even when it is installed all at once, is that cuts or damaged sections can be repaired more quickly. In case of an accidental cut from power equipment, for example, the damaged section of tube can be cut out and a new section spliced in, and new fiber can be blown in. This could be a great time savings over splicing multiple broken fibers in a cable. Blown-in fiber can be run 1000 feet or more through a building, around curves and uphill, in a very short time.

Considerations for blown-in fiber include the fact that while the equipment and materials for blowing in fiber cost more than standard installation, it requires fewer installers, so an up-front investment might be offset if there is a lot of fiber to be installed.

Blown-in fiber is also a potential benefit if changes are expected in equipment, network size, or fiber technology over the short term.

NEC Standards for Optical Fiber

The National Electrical Code (NEC) is published by the National Fire Protection Association (NFPA) in conjunction with insurance agencies, architects, and manufacturers to establish safety and fire protection standards for anyone working with electrical equipment. Even though optical fibers (with the exception of composite cables) do not carry electricity, many of them have conductive components or are installed in areas where electrical cables or equipment are present.

The NEC has provided specific guidance for running optical fiber cable within buildings. The code requires cables to be tested for fire resistance and smoke characteristics, determines which types of cables may be run in different areas of a building, and specifies the types of housings, or raceways, that can be used with each type of cable.

Be sure you stay current with the latest NEC guidelines. The information in this chapter can be found in NEC Article 770, “Optical Fiber Cables and Raceways.”

NEC-Listed Cable Types

The NEC recognizes three types of cables in Article 770:

- Nonconductive: Cables containing no metallic members or other electrically conductive materials.
- Conductive: Cables containing non-current-carrying conductive members, including strength members, armor, or sheath.
- Composite: Cables containing optical fiber and current-carrying electrical conductors. These cables may also contain non-current-carrying conductive members.

The NEC classifies fiber optic cables according to their electrical and fire safety characteristics and contains rules for the use of each cable type to minimize hazards. Any cabling run indoors must be listed with the NEC. Any outdoor cabling that is brought into a structure must be terminated after no more than 50 feet and connectorized or spliced to an interior, NEC-listed cable. Table 7.1 shows the cable types along with the markings each cable will contain. Markings should be clearly visible on the cable, as shown in Figure 7.23, and no interior cable should be used unless it contains the NEC cable type on it.

FIGURE 7.23 NEC cable type marking



TABLE 7.1 NEC Cable Types and Description

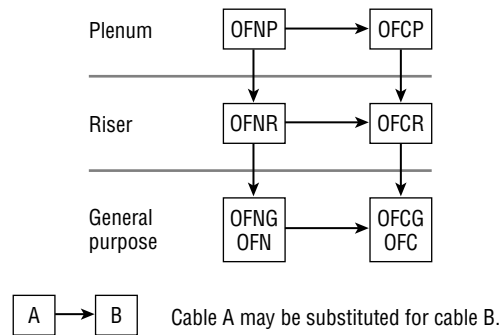
Marking	Type	Location	Permitted Substitutions
OFNP	Nonconductive optical fiber plenum cable	Ducts, plenums, other air spaces	None
OFCP	Conductive optical fiber plenum cable	Ducts, plenums, other air spaces	OFNP
OFNR	Nonconductive optical fiber riser cable	Risers, vertical runs	OFNP
OFCR	Conductive optical fiber riser cable	Risers, vertical runs	OFNP, OFCP, OFNR
OFNG	Nonconductive optical fiber general purpose cable	General purpose use except for risers and plenums	OFNP, OFNR
OF CG	Conductive optical fiber general purpose cable	General purpose use except for risers and plenums	OFNP, OFCP, OFNR, OFCR, OFNG, OFN
OFN	Nonconductive optical fiber general purpose cable	General purpose use except for risers, plenums, and spaces used for environmental air	OFNP, OFNR
OFC	Conductive optical fiber general purpose cable	General purpose use except for risers, plenums, and spaces used for environmental air	OFNP, OFCP, OFNR, OFCR, OFNG, OFN

All cables listed by the NEC must be resistant to the spread of fire, and those listed as suitable for environmental air spaces such as plenums must also have low-smoke characteristics. Note in the substitutions list that nonconductive cables may always be substituted for conductive cables of an equal or lower rating, but conductive cables may never be substituted for nonconductive cables. Figure 7.24 shows the order in which cables may be substituted for one another.



The NEC describes low-smoke-producing cable as having a maximum peak optical density of 0.5 or less and an average optical density of 0.15 or less. This means that there may still be enough smoke to be a health hazard, but building occupants can see through it well enough to escape, and emergency crews can find the fire through the smoke. Fire-resistant cable means that a fire will spread 1.52 m (5 feet) or less if it does ignite, so it will not carry fire between rooms or floors that are otherwise isolated from one another.

FIGURE 7.24 Cable substitution guide



The NEC also describes the *raceways* and *cable trays* that can be used with fiber optic cables and the conditions under which each type of cable may be used. The code prohibits, for example, conductive fibers being run in the same cable tray or raceway with wiring for power, lights, and other electrical power systems. They can, however be run in the same cable tray or raceway as lower-power communication circuits and control systems.

NEC-Listed Raceways

The NEC defines a raceway as an “enclosed channel of metal or nonmetallic materials designed expressly for holding wires, cables, or busbars.”

This definition may include metal or nonmetallic conduit and tubing along with raceways running through concrete or under floors. Where optical fiber cables are concerned, however, the NEC lists three types of raceways:

Plenum optical fiber raceways These run through spaces that carry environmental air, such as above-ceiling voids. The NEC requires these raceways to be fire-resistant and have low smoke-producing characteristics.

Riser optical fiber raceways These run through vertical spaces, often between floors, where flammable materials would give fire and smoke a chance to spread quickly. The NEC lists riser raceways as requiring fire-resistant characteristics capable of preventing the carrying of fire from floor to floor.

General-purpose optical fiber raceways These are used where fire-resistance and low smoke-producing characteristics of the other raceway types are not as critical, including non-environmental air spaces. The NEC calls for these raceways to be resistant to the spread of fire as defined in UL 2024, “Standards for Optical Fiber Cable Raceways.”

Cable Markings and Codes

In addition to the NEC listing code, optical fiber cables have a number of other markings and codes to help identify them and make them easier to use. Many of these markings are optional, so not every cable will have them. It is useful to know what they are, though, so that you can make proper use of them when they do appear.

Markings that appear on the outer cable help identify what is inside the cable and where it originated. They may also include aids for measuring the cable.

Inside the cable, the fiber buffers are usually color-coded with standard colors to make connections and splices easier.

External Markings

A cable’s external markings consist of manufacturer’s information, including the manufacturer’s name and phone number, cable part number or catalog number, and the date the cable was manufactured. Information about the cable itself includes the NEC listing code, which is only applied after UL testing, the fiber type (single-mode or core/cladding size), and cable length markings in meters or feet.

Color Codes

As with copper wiring, optical fibers running in cables must have some way of being distinguished from one another so that they can be connected properly at each end. According to standards used by the Telecommunications Industries Association (TIA) and the Electronic Industries Alliance (EIA), color-coded tight buffers can help identify up to 24 different fibers within a cable.

The standard containing the color sequence for the buffers is detailed in TIA/EIA-598-B, which also describes standards for cable jacket colors to identify the type of fiber contained within. Table 7.2 shows the color-coding scheme for individual fibers bundled in a cable. The Abbreviation column shows the lettering that would appear on fiber that is not color-coded but instead contains abbreviations for the colors that would be used. The *D/* indicates a color that is accompanied by a tracer.

TABLE 7.2 Fiber Color Coding

Position Number	Base Color/Tracer	Abbreviation
1	Blue	BL
2	Orange	OR
3	Green	GR

TABLE 7.2 Fiber Color Coding *(continued)*

Position Number	Base Color/Tracer	Abbreviation
4	Brown	BR
5	Slate	SL
6	White	WH
7	Red	RD
8	Black	BK
9	Yellow	YL
10	Violet	VI
11	Rose	RS
12	Aqua	AQ
13	Blue/Black	D/BL
14	Orange/Black	D/OR
15	Green/Black	D/GR
16	Brown/Black	D/BR
17	Slate/Black	D/SL
18	White/Black	D/WH
19	Red/Black	D/RD
20	Black/Yellow	D/BK
21	Yellow/Black	D/YL
22	Violet/Black	D/VI
23	Rose/Black	D/RS
24	Aqua/Black	D/AQ

Table 7.3 shows the color coding used on cable jackets to indicate the type of fiber they contain, if that is the only type of fiber they contain. Note that jackets may be left in their “raw” state with a colored tracer instead.

TABLE 7.3 Cable Jacket Colors

Fiber Type	Jacket Color Non-military Applications	Jacket Color Military Applications
Multimode (50/125)	Orange	Orange
Multimode (62.5/125)	Orange	Slate
Multimode (100/140)	Orange	Green
Single-mode	Yellow	Yellow

EXERCISE 7.1

Sizing It Up

A cable’s external markings may include *sequential markings*, or numbers that indicate the total length of the cable from a fixed point on the cable. Because fiber optic cable can come in lengths as large as 20 km, sequential markings are useful in determining how much cable is left on a reel, measuring off large runs of cable, or simply determining the length of a piece of cable without pulling out a tape measure.

Sequential markings are accompanied by tick marks occurring every two feet or every meter, depending on the manufacturer. The numbers themselves indicate length, not the number of tickmarks. To measure the length of the cable using the sequential markings, first determine the measurement standard that is being used. Next, subtract the number at the low end from the number at the high end. The difference between the two is the length.

Because most measurements dealing with fiber optics use meters rather than feet, you will want to convert any measurements in feet to the metric system. The formula for converting length is:

$$1 \text{ foot} = 0.3048 \text{ meter}$$

Let’s practice measuring out a length of cable using sequential markings. What is the length in meters of a cable that has sequential markings of 9,846 at one end and 12,218 at the other end, and the markings are measuring the cable in feet?

EXERCISE 7.1 (continued)

First, determine the distance between the markings:

$$12,218 - 9,846 = 2,372 \text{ feet}$$

Now, convert feet to meters:

$$2,372 \times 0.3048 = 722.99 \text{ meters}$$

Remember, though, that the length of the cable is not necessarily the length of the fiber inside of it. In loose tube buffer cable, the fiber is actually slightly longer than the cable to account for some stretch in the outer layers. This fact becomes important if fault location procedures tell you that there is a fault at a specific distance from the end of the fiber. That distance could be short of the measured distance on the outside of the cable since the fiber wanders around on the inside of the tube.

Bend Radius Specifications

Throughout the installation process, optical fiber's light-carrying abilities are threatened by poor handling, damage from tools or accidents, and improper installation procedures.

One of the problems that can severely increase attenuation within a fiber is extreme bending. When fiber is bent too far, the light inside no longer reflects off of the boundary between the core and the cladding, but passes through it to be absorbed in the cladding and coating. To reduce the risk of excessive bending during installation, TIA/EIA-568-B.3 sets out standards for optical fiber cable bending.

The standard states that the bend radius for optical fiber cable running inside a building will not be less than what is recommended by the manufacturer. If there are no manufacturer limits on bend radius, the standard set by TIA/EIA is a radius of no less than 10 times the outside diameter of the cable under no-load conditions, and not less than 15 times the diameter of the cable when it is under a tensile load, or stressed, condition.

For cables running outside a building, the TIA/EIA minimum bend radius is not less than 10 times the cable outside diameter under no-load conditions, and not less than 20 times the cable outside diameter under a tensile load condition.

EXERCISE 7.2

Find the TIA/EIA minimum bend radius of a 5 mm diameter cable.

If a cable 5 mm in diameter is running outside of a building under a tensile load, what is the minimum bend radius according to TIA/EIA-568-B.3?

EXERCISE 7.2 (continued)

Remember that the minimum bend radius for cables running outside a building is 20 times the cable's outside diameter if the cable is under a tensile load.

$$5 \text{ mm} \times 20 = 100 \text{ mm}$$

The minimum bend radius unless specified by the manufacturer is 100 mm.

Summary

This chapter covered the types of cables used to carry optical fibers while protecting them and keeping large numbers of fibers bundled together. It described the materials used in common cables and the different types of cables used to meet different needs.

The chapter described the ways in which fiber can be arranged within cables, along with the methods used to protect the fibers from stretching, damage, and excessive handling. It also described some of the locations in which cables commonly run.

Finally, the chapter discussed some of the standards used in producing and handling optical fiber cables. It mentioned some of the organizations that have produced standards for optical fiber cable safety and efficiency, and described the standards required while handling and installing cables.

Exam Essentials

Understand optical fiber cable construction. Make sure that you understand the basic components of a fiber optic cable. Be able to describe the materials commonly used in fiber optic cables. Be able to explain the differences between some types of cable components.

Describe the types of fiber optic cables. Be able to describe the different types of fiber optic cables. Be able to explain how different types of fiber optic cables are used. Be familiar with the advantages and drawbacks of different types of cable.

Understand how to work with cable. Make sure that you understand the safety requirements for working with fiber optic cable. Be able to describe the locations in which optical fiber cable runs and understand how cable is selected for those locations. Be able to explain industry standards used when selecting and installing cable.

Review Questions

1. Which of the following describes the four components of a fiber optic cable?
 - A. Core, cladding, coating, buffer
 - B. Fiber, buffer, strength member, jacket
 - C. Core, buffer, jacket, coating
 - D. Fiber, buffer, strength member, coating
2. A bundle of fibers running through a buffer with room to move around inside is called:
 - A. Loose tube buffer
 - B. Spacer buffer
 - C. Random buffer
 - D. Tensile buffer
3. One benefit of tight-buffered fibers is that they can:
 - A. Stretch more without breaking
 - B. Take up less space in a cable
 - C. Run for short distances outside the cable
 - D. Carry a signal farther
4. Aramid is used as a strength member because it is:
 - A. Flexible, fire resistant, and strong
 - B. Rigid, puncture resistant, and self-supporting
 - C. Conductive and animal-proof
 - D. Inexpensive
5. An important consideration for jackets used in indoor cables is:
 - A. Temperature resistance
 - B. Animal resistance
 - C. Water resistance
 - D. Fire resistance
6. Duplex cordage is most often used in:
 - A. Applications requiring one-way transmission
 - B. Applications requiring two-way transmission
 - C. Applications requiring an electrical conductor
 - D. Applications requiring two transmission speeds

7. Breakout cables contain:
 - A. Tightly bundled tight-buffered cables
 - B. Fibers in a loose tube buffer
 - C. Simplex cables grouped around a central strength member
 - D. A mix of loose tube and tight-buffered fibers
8. Messenger cable is used to:
 - A. Send information back and forth before the fiber is hooked up.
 - B. Transmit low data rates for basic messaging.
 - C. Pull fiber through buildings during installation.
 - D. Run fiber in a self-supporting cable between two elevated structures.
9. Ribbon cable's primary advantage is its:
 - A. Strength
 - B. Small size
 - C. Low cost
 - D. Fire resistance
10. Optical fiber cable designed to run underwater is called:
 - A. Submersible cable
 - B. Waterproof cable
 - C. Submarine cable
 - D. Transoceanic cable
11. Hybrid cable is designed to:
 - A. Combine multimode and single-mode fibers in one cable.
 - B. Combine plastic and silica fiber in one cable.
 - C. Combine loose tube buffer and tight-buffered fiber in one cable.
 - D. Combine different sizes of multimode fiber in one cable.
12. The National Electrical Code defines composite cable as:
 - A. Carrying optical fiber and conductive strength members in the same run
 - B. Carrying optical fiber in a cable using a variety of materials
 - C. Carrying optical fiber and current-carrying-conductor in the same run
 - D. Carrying tight-buffered and loose tube buffer optical fiber in the same run

13. Simplex cord is an example of:
- A. Heavy-duty cable
 - B. Plenum cable
 - C. Riser cable
 - D. Light-duty cable
14. One of the most important characteristics of plenum cable is:
- A. Low-smoke, no halogen jacket materials
 - B. High strength for pulling
 - C. Rigid jacketing for vertical runs
 - D. High fiber volume
15. What does a fanout kit do?
- A. Converts loose tube buffered fibers into simplex cable
 - B. Converts loose tube buffered fibers into tight-buffered fibers
 - C. Converts tight-buffered fibers into loose tube buffered fibers
 - D. Converts ribbon cable to duplex cordage
16. If you see a cable with “OFNP” printed on it, you know that:
- A. It is single-mode fiber.
 - B. It is loose tube buffer cable.
 - C. It is plenum cable.
 - D. It is armored cable.
17. Which of the following can be substituted for a cable marked “OFN”?
- A. OFC
 - B. OFCP
 - C. OFNP
 - D. OFCR
18. How many fiber color codes are recognized by TIA/EIA-598-B?
- A. 6
 - B. 12
 - C. 18
 - D. 24

19. The first place to look for a fiber's bend radius specification is:
- A. NEC
 - B. Manufacturer's specifications
 - C. TIA/EIA-568-B.3
 - D. TIA/EIA-598-B
20. If the manufacturer does not specify the minimum bend radius for a length of indoor cable under tension, the minimum bend radius should be:
- A. 30 mm
 - B. 5 times the cable diameter
 - C. 15 times the cable diameter
 - D. 20 times the cable diameter

Answers to Review Questions

1. B. The core, cladding, and coating of the fiber itself are considered one component of the cable, with the rest being the buffer, strength member, and jacket, in various configurations.
2. A. A loose tube buffer allows many fibers to run through a buffer without being attached to it. This allows the buffer to take tension and bending without affecting the fiber inside, and allows a large number of fibers to be contained in a cable.
3. C. Tight-buffered cables have protective buffers attached directly to the fiber so that they are protected from damage that would harm bare fibers. This allows the fibers to run outside of a cable and be attached to connectors.
4. A. Aramid yarns, sold under the trade name Kevlar, are flexible, fire resistant, and strong, and are often used in bulletproof vests and fire protection gear.
5. D. The materials used in the jackets of indoor cables must meet certain standards for fire safety, including low smoke output and resistance to fire so they will not carry fire and smoke throughout the building.
6. B. Duplex cordage consists of two tight-buffered cables in jackets attached parallel to each other. This arrangement makes them easy to handle when running paired optical fibers, such as those used in two-way transmissions.
7. C. Breakout cable is designed to carry a number of fibers in a single jacket, but because the fibers are in simplex cables, they can be handled individually once they are broken out of the cable.
8. D. Messenger cable consists of a fiber optic cable attached to a steel or dielectric messenger designed to support the weight of the cable between two poles or other structures.
9. B. Ribbon cable is made of a number of tight-buffered fibers attached to one another side by side and placed in a single jacket. Because they are bonded so closely, more fibers can be run through a cable, saving time during installation.
10. C. Submarine cable configurations range from heavy-duty cable designed to keep water out to heavily armored cable that also carries power for optical amplifiers.
11. A. Hybrid cable is used when users want to run single-mode and multimode fibers at the same time. Running them in a single cable saves time and space, and allows multimode users to upgrade to single mode when budgets and requirements allow.
12. C. Composite cable typically carries twisted-pair wiring for telephones along with optical fiber. The wiring can be used for communications during installation, or for providing power to remote equipment along the cable's route.
13. D. Light-duty cable is designed for basic protection of the fiber during handling. It is not intended for pulling or for heavy use.
14. A. Plenum cable occupies spaces that carry environmental air, so the jacketing material must not give off noxious fumes when it gets hot. These cables are required for environmental air spaces.

15. B. A fanout kit provides an enclosure and tubes for placing tight buffers on fiber from a loose tube buffer cable. The kits are designed to prepare fiber for installation in a patch panel.
16. C. The letters OFNP stand for optical fiber, nonconductive, plenum, meaning that there is no conductive material in the cable and it is rated for use in a plenum or air space.
17. C. Whenever a cable is listed as nonconductive, the only cable that can be substituted for it is another nonconductive cable, such as the OFNP.
18. D. The TIA/EIA-598-B standard recognizes 24 color codes for fiber running within a cable, including colors with tracers running through them. Color names may also be abbreviated for use on noncolored fibers.
19. B. The cable manufacturer will often provide specifications on the bend radius for the cable. If those are not provided, follow the specifications laid out in TIA/EIA-568-B.3.
20. C. The TIA/EIA-568-B.3 specification for minimum bend radius of an indoor cable under tension is 15 times the cable diameter.



Chapter

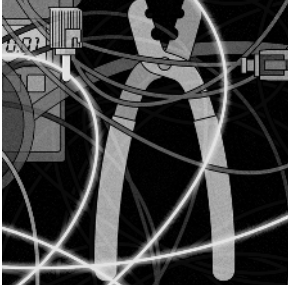
8

Splicing

OBJECTIVES COVERED IN THIS CHAPTER:

✓ Splicing

- Explain the intrinsic factors that affect splice performance.
- Explain the extrinsic factors that affect splice performance.
- Describe the basic parts of a mechanical splicer.
- Describe how to perform a mechanical splice.
- Describe the operation of a fusion splicer.
- Describe how to perform a fusion splice.
- List TIA/EIA-568-B.3 inside plant splice performance requirements.
- List TIA/EIA-758 outside plant splice performance requirements.



You’ve probably heard the saying, “A chain is only as strong as its weakest link.” In fiber optics, any splice is potentially the weakest link. Performed properly, the splice will attenuate the signal only slightly—from about 0.05 dB to 0.15 dB, typically. Performed poorly, a splice can leak light, reflect a signal back down the transmission path, or separate completely, requiring expensive troubleshooting and repair.

In this chapter, we will discuss fiber optic splices. We will describe the factors that affect the way a splice carries optical signals and the effects on the signals themselves. We will also describe different tools and methods used to splice optical fibers as well as standards used to gauge splice performance.

Putting It Together

A splice is a direct, permanent connection between two optical fiber ends. As with electrical wire splices, optical fiber splices may be used to add length to an existing fiber or to repair a broken or damaged fiber. However, while electrical splices only require firm, clean contact between two pieces of wire and permanent fastening using solder or a crimp connector, optical fiber splices require a great deal of precision and careful preparation of the fiber ends if they are to be spliced properly. The fiber cores must align precisely to prevent any loss of light across the splice.

Considering the size of most fibers in use and the tolerances required to align the fiber cores with each other, each successful splice is a triumph of training and technology. There are a number of factors that can work against a good splice, even if you have done everything properly. These can be divided into *intrinsic factors*, or factors related to the actual structure of the fiber, and *extrinsic factors*, which concern the relationship of one fiber to another in the splice.

Intrinsic Factors

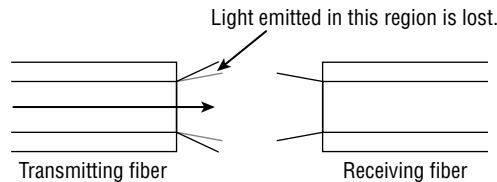
Even when fibers are manufactured within specified tolerances, there are still slight variations from one fiber to another. These variations can cause mismatches between two fiber ends although the ends might be aligned completely within specifications for the type of splice being performed.

Let’s look at the most common types of variations.

NA Mismatch Loss

Numerical aperture (NA) mismatch loss occurs when the numerical aperture of the transmitting fiber is larger than that of the receiving fiber, as shown in Figure 8.1. Because NA is determined by the refractive indices of the core and cladding, it is possible for the two fiber cores to be the same size and still result in an NA mismatch loss. As a result of NA mismatch loss, the transmitting fiber core emits light in a wider area than the cone of acceptance of the receiving fiber covers. Some of the light emitted by the transmitting fiber is therefore lost in the cladding of the receiving fiber because it is below the critical angle.

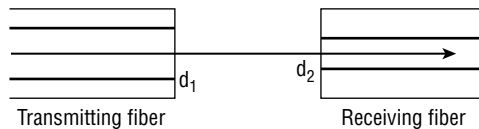
FIGURE 8.1 When NA mismatch loss occurs, the cone of acceptance in the receiving fiber cannot gather all of the light emitted by the transmitting fiber.



Core Diameter Mismatch Loss

Core diameter mismatch loss results when the diameter of the transmitting core is greater than that of the receiving core, as shown in Figure 8.2. Unlike NA mismatch loss, core diameter mismatch loss depends only on the difference in core diameters. Light loss occurs when light at the outer edge of the transmitting core falls outside the diameter of the receiving core and is instead absorbed by the cladding in the receiving fiber.

FIGURE 8.2 Core diameter mismatch loss is caused by a difference in core diameters.



The amount of loss in percentage of light due to core diameter mismatch can be calculated with the formula

$$\text{Loss} = [(d_1)^2 - (d_2)^2] / (d_1)^2$$

where d_1 is the diameter of the transmitting core and d_2 is the diameter of the receiving core.

Core diameter mismatch loss can be more significant if fibers of different types are spliced. For example, if a fiber with a 62.5 μ core is spliced to a fiber with a 50 μ core, the loss reaches 36%, or 1.9 dB. If a 50 μ graded-index multimode fiber is spliced to a 9 μ single-mode fiber, the signal suffers a loss of nearly 97%, or 15 dB.

EXERCISE 8.1**Calculate the percentage of light lost in a splice with core diameter mismatch.**

For the sake of this exercise, the transmitting core (d_1) is 50.1 μ and the receiving core (d_2) is 49.9 μ .

Apply the formula $\text{Loss} = [(d_1)^2 - (d_2)^2] / (d_1)^2$

$$50.1^2 - 49.9^2 = 20$$

$$20/50.1^2 = 0.008, \text{ or } 0.8\%$$

Remember, you can then calculate the decibel loss with the formula

$$\text{dB} = 10\text{Log}_{10}(P_{\text{out}} / P_{\text{in}})$$

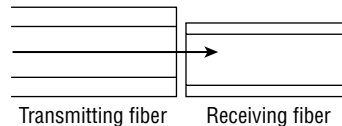
Note that the decibel formula is based on the amount of power remaining, which is calculated by dividing the power output by the power input. We can arrive at the same figure by subtracting the loss of 0.8% from 100% for a value of 99.2%, or 0.992.

Now, $10\text{Log}_{10}(0.992) = 0.03 \text{ dB}$.

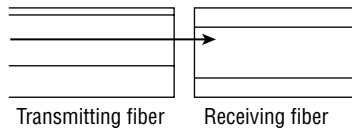
Cladding Diameter Mismatch Loss

Cladding diameter mismatch loss occurs when the cladding diameters of the joined fibers are not the same, causing misalignment of the cores, as shown in Figure 8.3.

FIGURE 8.3 Cladding diameter mismatch loss results from differing cladding diameters.

**Concentricity Loss**

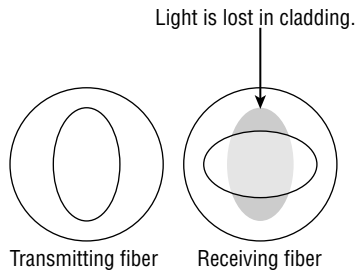
Ideally, fiber cores and their cladding are *concentric*, meaning that they share a common geometric center. This is not always the case, however. In fact, fiber cores are more likely to be off-set from the cladding center by a slight amount. If the transmitting and receiving fiber cores are non-concentric in different directions, they will not mate exactly, and light from the transmitting fiber will be lost, as shown in Figure 8.4.

FIGURE 8.4 Concentricity loss is caused by off-center fiber cores.

The concentricity tolerance is the distance between the cladding core and the fiber core.

Ellipticity Loss

Just as fiber cores and cladding may not be perfectly concentric, fiber cores may also not be perfectly circular. The core may fall within acceptable measurements and still be slightly elliptical, or oval. If the ellipticity of the transmitting and receiving cores do not match exactly, as shown in Figure 8.5, the transmitting fiber's core will overlap the receiving fiber's cladding, causing a light loss.

FIGURE 8.5 Ellipticity loss takes place when the ellipticities of two fibers do not match exactly.

Note that the amount of loss is dependent upon the alignment of the ellipticities of the two cores. Maximum loss occurs when the long or major axes of the cores are at right angles to one another, and minimum loss occurs when the axes of the cores are aligned. This means that it is possible for two fiber ends to have a high loss the first time they are spliced together and a low loss the next time.

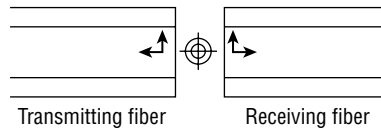
All of the above factors are possible in any manufactured fiber, and in the worst-case scenarios they could cause a loss of 0.1 dB to 0.6 dB. Their effects are kept to a minimum through tightly controlled manufacturing standards in the optical fiber industry. As measurement and manufacturing methods continue to improve, the chance of attenuation due to intrinsic factors should decrease.

Extrinsic Factors

Extrinsic factors causing attenuation in a splice are factors related to the condition of the splice itself. In short, as long as the fiber falls within its nominal specs, a well-made splice will cause less attenuation than a poorly-made splice.

In an ideal splice, the fiber cores are perfectly centered on each other and the core axes are perpendicular to the faces being joined, as shown in Figure 8.6. In addition, the fiber ends should be in firm contact. As we will see, any variation from these conditions can cause attenuation or complete loss of the signal.

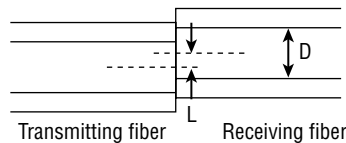
FIGURE 8.6 Conditions for an ideal splice



Lateral Displacement

Lateral displacement results from an offset of the fiber cores' center axes, as shown in Figure 8.7. As the lateral displacement increases, less light from the transmitting fiber makes its way into the receiving fiber. The effect is similar to that of core diameter mismatch loss, but lateral displacement can occur even when the core diameters are the same.

FIGURE 8.7 Lateral displacement is a side-to-side misalignment or centerline offset of the fiber cores.



In larger fibers, a slight amount of lateral displacement is acceptable, since the majority of the cores' surface area is still in contact. In smaller fibers, however, even a slight offset can place the center of the transmitting core entirely outside the area of the receiver core.

The formula for expressing displacement is the ratio of the amount of offset to the core diameter, or

$$\text{Displacement ratio} = L/D$$

where L is the distance of lateral offset and D is the core diameter. The offset is usually expressed as a percentage.

EXERCISE 8.2

Determine the offset of two fibers with core diameters of 50 μ and a lateral offset of 2.5 μ .

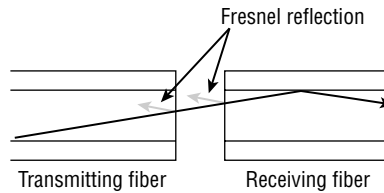
$$\text{Displacement} = 2.5/50 = 0.05, \text{ or } 5\%$$

In multimode fiber, a displacement of 10% can cause a loss of about 0.5 dB. Note that the loss calculations are based on an even distribution of light through the core, and therefore represent the maximum loss for a given displacement. In reality, the amount of light energy in the core increases toward the center, so losses due to lateral displacement may be less than the calculated amounts.

End Separation

Even if the fibers are perfectly aligned, they may still experience loss through *end separation*. End separation, or a gap between the two fibers, as shown in Figure 8.8, may be common or even desirable in some mechanical splices to minimize the risk of fibers being joined with too much pressure, which could fracture the ends.

FIGURE 8.8 End separation is a gap between fiber ends in a mechanical splice.



End separation can cause *gap loss* in two different ways. The first is through *Fresnel reflection*, which takes place when light passes from the refractive index of the fiber into that of the air, and then back into the fiber. Each change in the refractive index causes a certain amount of light to be reflected and therefore lost.

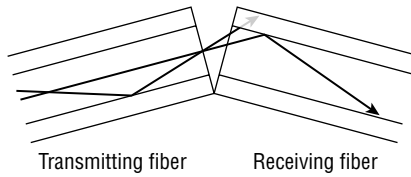
One way to overcome the effects of Fresnel reflection in separated fibers is by using an *index matching gel*, which is a transparent gel having a refractive index close to that of the fibers being spliced. The gel fills the gap and reduces the amount of Fresnel reflection to an acceptable level.

End separation also causes light loss in multimode fibers when the high-order modes leave the transmitting fiber at an angle that makes them miss the cone of acceptance of the receiving fiber. The higher the NA of the fiber, the less end separation the higher-order modes can survive, since the light will travel outside the cone of acceptance more quickly at a lower refractive angle.

Angular Misalignment

As stated previously, an ideal splice requires fiber ends that are perpendicular to the axis of the fiber cores and that mate with one another exactly, so the cores line up with one another.

If the fibers meet each other at an angle, the signal will suffer losses from *angular misalignment*, as shown in Figure 8.9. In this condition, attenuation takes place in two ways. First, the higher-order modes may be outside the cone of acceptance for the receiving fiber. Contrary to the conditions for end separation, this problem diminishes as the fiber's numerical aperture increases, due to the fiber's ability to accept light at a lower refractive angle.

FIGURE 8.9 Angular misalignment results when fiber ends are not perpendicular to fibers.

The only solution for angular misalignment is to prepare the fiber ends properly, making sure that both ends are perpendicular to the fibers, and to ensure that they are in line with one another during splicing.

Now let's look at splicing equipment and methods.

Splicing Equipment

The goal of any optical fiber splice is to join two fiber ends permanently with as little loss in optical quality as possible. Factors influencing the type of splice chosen include:

- Initial cost of equipment
- Cost per splice
- Frequency of splicing operations
- Optical quality of splice

Of course, you always want the highest optical quality for a splice, but in some cases there may be a trade-off if equipment costs outweigh the benefits of a higher quality splice.

Optical fibers may be spliced using two methods: mechanical splicing and fusion splicing. Each method has advantages, so we cannot say that one method is always better than another.

Mechanical Splicers

Mechanical splicers use a plastic tube with a locking mechanism that holds two fibers against each other to make a splice, as shown in Figure 8.10.

Mechanical splicers themselves are relatively inexpensive, but each splicing operation requires a permanent fixture to be applied to the splice to hold the fiber ends together. The need for consumables can drive up the cost per splice and may make other methods more desirable if a large number of splices is anticipated.

Because single-mode fiber requires more precision in splices than multimode and plastic fibers, mechanical splices usually are not performed on single-mode fiber without special equipment, which can drive up the cost per splice.

One of the most commonly used mechanical splicers is manufactured by 3M (www.3m.com).

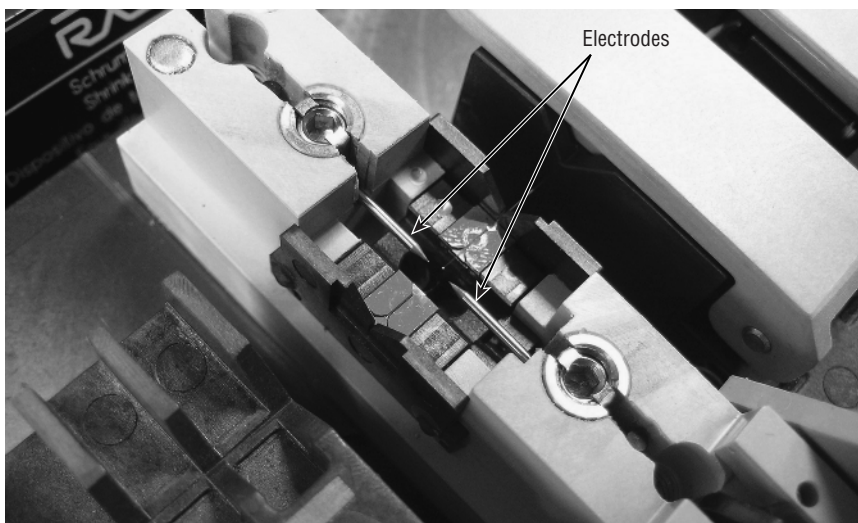
FIGURE 8.10 A mechanical splicer with fiber ends in place



Fusion Splicers

Fusion splicers create a permanent splice by welding the fiber ends to one another with an electrical arc. The splice is then enclosed in heat-shrink tubing with an oven built into the splicer. A fusion splicer is shown in Figure 8.11.

FIGURE 8.11 A fusion splicer with fibers ready to be spliced placed between the electrodes.



Fusion splicers are more expensive than mechanical splicers, usually costing in the thousands of dollars, but they provide a number of features that improve the precision of splices.

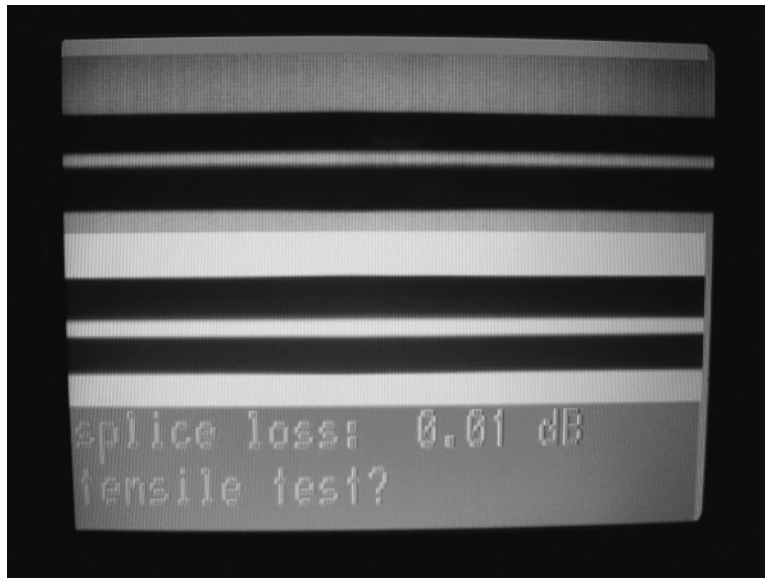
Features built into fusion splicers may include microscopes or high-resolution video imagers, GPS for recalling splice coordinates, computer-controlled functions, networking capability, and USB connectivity.

In addition to a microscope or video imager for aligning and checking the fiber ends, fusion splicers generally have two other standard features: an alignment mechanism and a device for estimating optical power loss.

The alignment mechanism draws the fiber ends into a position that optimizes their exposure to the welding arc and ensures that they are in proper contact for a good splice.

The power loss estimation function, shown in Figure 8.12, tests the splice by imposing macrobends on the transmitting side and the receiving side of the fiber and injecting a light signal through the bend on the transmitting side. The light coming from the macrobend on the receiving side is measured and the loss through the splice is calculated. This function serves two purposes. First, it allows the operator to position the fiber ends for the least loss from intrinsic and extrinsic factors. Second, it allows the operator to review the splice to ensure that it is within specifications.

FIGURE 8.12 Power loss estimation in a fusion splicer



You can find out more about specific features of fusion splicers through their manufacturers' websites. Some of the major manufacturers of fusion splicers are:

Corning www.corningcablesystems.com

Alcoa/AFL Telecommunications www.alcoa.com

Fitel www.fitel.com

Let's look at the procedures used with each type of splicer.

Splicing Procedures

This section will familiarize you with the basic procedures used with each type of splicer. While manufacturers' models will vary, many of the same principles and requirements apply no matter what type of splicer you use. Be sure to read and follow the directions for your particular splicer carefully. The components are designed for precision alignment and pressure, and any variation from the manufacturers' specifications can damage the fibers or produce a poor splice.



Whenever handling optical fiber, remember to follow the safety precautions described in Chapter 6, "Safety."

Mechanical Splicing Procedure

To prepare for mechanical splicing, make sure that the work area is clean, dry, and well-lit. Assemble the following tools before you begin:

- Mechanical splicing tool and splice
- Plastic coating (buffer) stripper
- Reagent grade isopropyl alcohol
- Lint-free wipes
- Cleaver

Once your materials are assembled, proceed with the following steps.

Set-up

1. Open the fiber buffer tubes and expose and clean the fibers.
2. Remove the splice from its protective packaging and load the splice into the assembly tool by pressing firmly at the ends of the splice.

Fiber Preparation

3. Remove from the storage reel or coil the minimum length of fiber required to prepare and splice the fibers—less than one loop if possible.
4. Strip approximately 1 to 2 inches (25 mm to 51 mm) of plastic coating from the fiber using a mechanical stripper.
5. Clean the bare glass by pulling the fiber through an alcohol soaked lint-free wipe. This will remove any fragments or dirt remaining on the fiber.
6. Cleave the fiber to the length specified by the brand of splicer you are using, typically 12–14 mm $\pm$ 0.5 mm. Note: The cleaved ends should be within 2° of perpendicular to the fiber axis and should be free of defects.
7. If a gauge is provided, check the cleave length in the gauge, but do not allow the cleaved end to contact the tool. Do not clean the fibers after they have been cleaved.

Splice Assembly

8. Push the fiber into the clamping mechanism on one side of the splice.
9. Hold the fiber about 6 mm (about 0.25 inch) from the bare fiber and insert the end straight into the splice.
10. Push the fiber into the splice until you feel resistance. When fully inserted, the fiber should be straight or have a slight bow of up to 3 mm (0.1 inch).
11. Prepare the second fiber as described above.
12. Place the second fiber in the clamping mechanism on the opposite side of the splicing tool.
13. Feed the fiber end into the splice until you meet resistance. At this point, the first fiber should have more of a bow in it from contact with the second fiber.
14. Push the first fiber back into the splice until the bows are equal.
15. Pivot the splicing tool handle down until it contacts the top of the splice, then squeeze the tool handle to complete the assembly and lock the spliced ends in place.
16. Remove the fibers from the clamping mechanisms and lift the splice from the tool.

Adjustment

17. If you observe high loss in the splice, use the cap-lifter provided with the splicing tool to remove the splice cap.
18. Repeat the fiber centering and splicing procedure. If acceptable loss is not achieved after two attempts, remove the fiber, then strip, clean, and re-cleave. Re-splice using a new splice.

Fusion Splicing Procedure

Many fusion splicers contain a feature that automatically positions the fiber ends in proper relationship with each other and with the electrodes for the best possible splice. All that is required of the operator is to prepare the fibers properly and place them in the device.

To prepare for fusion splicing, as with mechanical splicing, make sure that the work area is clean, dry, and well-lit. Assemble the following tools before you begin:

- Fusion splicing tool
- Plastic coating (buffer) stripper
- Reagent grade isopropyl alcohol
- Lint-free wipes

- Cleaver
- Heat-shrink tubing

Once your materials are assembled, proceed with the following steps:

Set-up

1. Open the fiber buffer tubes and expose and clean the fibers.
2. Enable power to the fusion splicer.

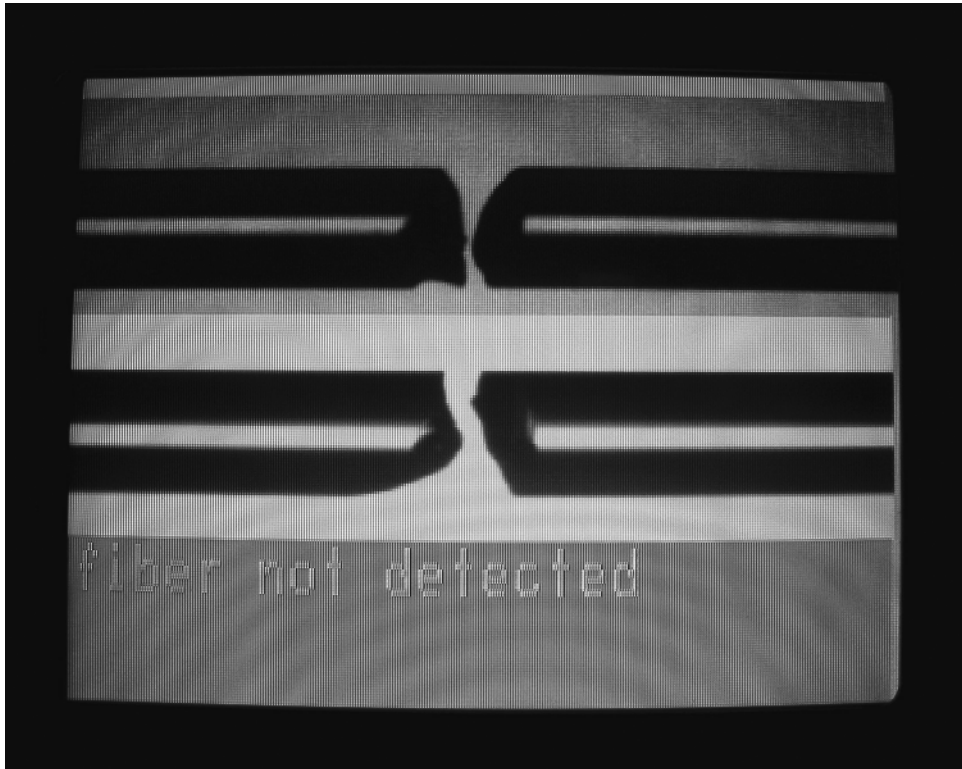
Fiber Preparation

3. Remove from the storage reel or coil the minimum length of fiber required to prepare and splice the fibers—less than one loop if possible.
4. Slide the heat-shrink tubing over one fiber end and move it far enough up the fiber to place it out of the way.
5. Strip approximately 1 to 2 inches (25 mm to 51 mm) of plastic coating from the fiber using a mechanical stripper.
6. Clean the bare glass by pulling the fiber through an alcohol soaked lint-free wipe. This removes any fragments or dirt remaining on the fiber.
7. Cleave the fiber to the length specified by the brand of splicer you are using, typically 12–14 mm $\pm$ 0.5 mm. Note: The cleaved ends should be within 2° of perpendicular to the fiber axis and should be free of defects.

Splice Assembly

8. Position one fiber end in the unit near the electrode and close the fiber clamp next to the electrode.
9. Close the outer clamp on the fiber.
10. Repeat steps 7 and 8 for the other fiber end to be spliced.
11. Close the electrode cover.
12. Select the appropriate splicing program in the splicer's computer control and activate it. The splicer performs the necessary calibrations and positioning, performs the splice, and measures the loss across the splice. The operation may be viewed through the video monitor, if available, as shown in Figure 8.13.
13. Remove the splice and position the heat-shrink tubing over it. Place the splice and tubing in the heat-shrink oven to seal and protect the splice.

FIGURE 8.13 The video monitor of a fusion splicer showing two views of the fiber ends ready to be spliced.



Splice Requirements

As with connectors, TIA/EIA standards specify a maximum permissible loss for splices, depending on their location. For inside plant splices, TIA/EIA-568-B.3 states that optical fiber splices, whether fusion or mechanical, will have a maximum attenuation of 0.3 dB.

For outside plant splices, TIA/EIA-758 says that the splice insertion loss shall not exceed 0.1 dB mean, with a maximum of 0.3 dB, as measured with an optical time domain reflectometer (described in Chapter 15).



Real World Scenario

Fusion or Mechanical?

It may seem like an easy choice between mechanical and fusion splicing: Fusion splices are superior to mechanical splices because they essentially make one fiber out of two. Mechanical splices, no matter how good they are, always have some back reflection and typically have higher loss than fusion splices. The cost per splice is virtually nil with fusion splices, while mechanical splices cost about \$15 to \$25 per splice. In short, if you have a fusion splicer, there is really no need to even look at mechanical splicing.

So why bother with mechanical splices at all?

The only thing favoring mechanical splices is the initial setup cost. Mechanical splicers cost about a tenth of what fusion splicers cost, and additional training is required if workers are to be able to tune a fusion splicer properly. After that, most parameters are the same, including the time it takes to make a splice, the permanence of the splice itself, and the number of people required to make a splice.

The answer is not a simple one. It depends on broader economic factors such as the size of your operation, the number of splices you expect to perform, the number of crews you have working, and the nature of the work you're performing.

For example, if you work in a small operation with a single crew, but you're looking at a big job over the next month involving a few hundred splices, it may be worthwhile to look into renting a fusion splicer. The cost of the rental may be more than offset by what you'll save in consumables for a mechanical splicer.

If, on the other hand, your work is based on sending several crews out to do a few splices each, mechanical splicing may be the way to go, because several splicers will be required.

If you want the joy of fusion splicing without the pain of the purchase price, you can always rent or lease a fusion splicer. Companies such as Fiber Instrument Sales, Inc. (www.fiberinstrumentsales.com) provide rental arrangements for fusion splicers and other high-end fiber equipment if you know you won't need it on a long-term or recurring basis.

Rental rates typically run $\frac{1}{10}$ to $\frac{1}{12}$ of the purchase price of a new splicer per month. Bear in mind, though, that the rental period starts the day that the splicer is shipped to you and ends the day that it is received back at the rental office. In other words, you're going to pay for the days that it is in transit.

If you still want to buy a fusion splicer, but don't want to pay full price, consider a pre-owned or reconditioned unit. These are occasionally available through distributors and still have years of life left in them.

Summary

This chapter covered optical fiber splices and the factors that can affect their quality. We covered the influence of the fiber's structure on losses across the splice, as well as the influence of the construction of the splice itself.

We also discussed the tools and procedures necessary for the most common types of optical fiber splices. We described different types of splicing devices and the benefits of using each type. In addition, we described general procedures used to make splices with each type of device.

Exam Essentials

Understand the intrinsic factors affecting splices. Make sure you understand how fiber structure can affect light loss through a splice. Be able to describe the different intrinsic factors that affect a splice and the types of effects they have.

Understand the extrinsic factors affecting splices. Make sure you understand how the physical relationship between the fibers can affect light loss through a splice. Be able to describe the different extrinsic factors that affect a splice and the types of effects they have.

Be familiar with different types of splices. Make sure you can describe the different types of optical fiber splices and the equipment used to produce them. Be able to describe the benefits and drawbacks of each type of splice.

Be familiar with splicing procedures. Make sure you understand the general steps for producing each type of splice. Be able to describe the tools and procedures required for each type of splice, as well as procedures that apply to any work with fibers.

Understand splice specifications. Make sure you know the specifications for splice loss performance.

Review Questions

1. NA mismatch loss occurs when:
 - A. The numerical aperture of the transmitting fiber is smaller than that of the receiving fiber.
 - B. The numerical aperture of the transmitting fiber is larger than that of the receiving fiber.
 - C. The numerical apertures of both fibers are larger than normal.
 - D. The numerical apertures of both fibers are smaller than normal.
2. Core diameter mismatch loss occurs when:
 - A. The diameter of the transmitting core is greater than that of the receiving core.
 - B. The diameter of the transmitting core is less than that of the receiving core.
 - C. The diameter of the transmitting core is not precisely aligned with the diameter of the receiving core.
 - D. The diameter of the receiving core is at the low end of the acceptable size range.
3. Cladding diameter mismatch loss is caused when:
 - A. The cladding diameter of the transmitting fiber is larger than the cladding of the receiving fiber.
 - B. The cladding diameter of the transmitting fiber is smaller than the cladding of the receiving fiber.
 - C. The cladding diameters of the fibers do not match.
 - D. The cladding diameters of both fibers are slightly larger than normal.
4. If the transmitting and receiving cores are offset from their cladding in different directions, the result may be:
 - A. Concentricity loss
 - B. Centrality loss
 - C. Lateral offset loss
 - D. Slip loss
5. If the transmitting and receiving cores are slightly oval, the splice may experience:
 - A. Symmetrical loss
 - B. Asymmetrical loss
 - C. Oval loss
 - D. Ellipticity loss

6. Lateral displacement of the fibers causes:
 - A. Core diameter mismatch
 - B. Slip loss
 - C. Offset of the cores' axes
 - D. No problems
7. One of the effects of end separation is:
 - A. Improved light transmission
 - B. Loss due to Fresnel reflection
 - C. A weaker splice
 - D. A reduction in light speed across the gap
8. Loss due to angular separation diminishes as:
 - A. Core diameter increases
 - B. Cladding diameter decreases
 - C. Numerical aperture increases
 - D. Fiber diameter increases
9. Which of the following is an advantage of using a mechanical splicer?
 - A. Lower loss across the splice
 - B. Greater precision in alignment
 - C. Lower cost for splicing equipment
 - D. Lower cost per splice
10. The TIA/EIA specifications call for an inside plant splice to have attenuation of no more than:
 - A. 0.1 dB
 - B. 0.15 dB
 - C. 0.2 dB
 - D. 0.3 dB

Answers to Review Questions

1. B. Loss from a numerical aperture (NA) mismatch occurs when light exiting the core of the transmitting fiber is outside of the cone of acceptance of the receiving fiber.
2. A. Core diameter mismatch loss occurs when light from the transmitting core passes into the cladding of the receiving fiber, which is a result of the receiving core being smaller than the transmitting core.
3. C. Cladding diameter mismatch loss occurs when the cladding diameters are different. This causes the cores to be slightly offset from one another since the cladding is used to align the fibers.
4. A. Fiber cores are likely to be offset from the cladding by a slight amount, and if they are offset, or not concentric, in different directions, some light from the transmitting core can be lost in the cladding of the receiving fiber.
5. D. Ellipticity loss results when the ellipticity of the transmitting and receiving cores do not match exactly, causing the light from the transmitting core to overlap the receiving cladding.
6. C. Lateral displacement, a physical offset of the cores' center axes, causes problems similar to core diameter mismatch, but it can occur even when the cores' diameters match exactly.
7. B. Any amount of separation between fiber ends can cause Fresnel reflection, which causes loss as the light enters the gap and again as it enters the receiving fiber.
8. C. As the numerical aperture of the core increases, the fiber cores can accept light at lower refractive angles, such as those created by angular separation between two spliced fibers.
9. C. The primary advantage of a mechanical splicer is its low initial cost. Consumables used in the splicing process, however, create a higher cost per splice.
10. D. TIA/EIA-568-B.3 states that inside plant optical splices, whether fusion or mechanical, will have a maximum attenuation of 0.3 dB.



Chapter

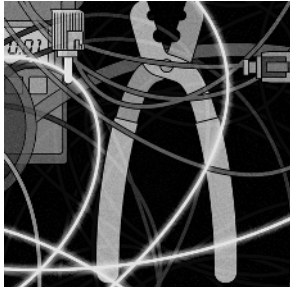
9

Connectors

OBJECTIVES COVERED IN THIS CHAPTER:

✓ Connectors

- Describe the basic parts of a fiber optic connector.
- Describe common connector ferrule materials.
- Explain the intrinsic factors that affect connector performance.
- Explain the extrinsic factors that affect connector performance.
- Describe return reflections in an interconnection.
- Describe flat finish connectors.
- Describe PC finish connectors.
- Describe APC finish connectors.
- Describe the TIA/EIA-568-B.3 recognized connectors.
- Describe small form factor connectors.
- Describe multifiber connectors.
- Describe a pigtail.
- Describe the steps involved in an anaerobic epoxy connector termination and polish.
- Describe the steps involved in a UV epoxy connector termination and polish.
- Describe the steps involved in an oven cured epoxy connector termination and polish.
- Describe the steps involved in a pre-load epoxy connector termination and polish.
- Describe epoxyless connector termination.
- Explain how to properly clean a connector.
- Explain how to examine the endface of a connector per TIA/EIA-455-57B.
- List the TIA/EIA-568-B.3 connector performance requirements.



The ideal traveling environment for a pulse of light is an unbroken optical fiber. At some point, however, that fiber must connect to a piece of equipment or join another fiber in order to extend its length or change the type of fiber being used.

One of the most common methods for *terminating* a fiber, or making its end useful, is to use a connector. A connector is a device that protects the end of the fiber while allowing it to be quickly and reliably joined to equipment, patch panels, or other fibers. Connectors are often used to join two fibers together instead of *splices* because they allow the fibers to be disconnected and reconnected easily. Splices, on the other hand, are permanent connections between two fibers. Connectors can be useful when network assignments must be changed, when equipment must be removed from the link and replaced, or when expansion is anticipated.

This chapter describes several common connectors used in optical fiber termination. It describes the factors that affect connector performance and methods used to improve performance. The chapter also discusses methods used to install connectors so they meet performance standards.

The Fiber Optic Connector

The job of a fiber optic connector is to couple a fiber end mechanically to a piece of hardware or to another fiber so that the cores line up accurately and produce the smallest amount of loss. Inherent in this requirement is the need for the connector to protect the fiber from repeated handling during connection and disconnection, align the fiber end precisely with its counterpart in the connection, and prevent strain on the fiber itself. Strain relief is normally provided by the strength members in the fiber cable, but once the fiber itself is attached to the connector, the connector must take up that job.

Although there are several different types of connectors recognized by industry standards, they all contain common components, shown in Figure 9.1.

Beginning at the working end of the connector, the *ferrule* holds the fiber in place inside a standard-sized housing. The ferrule must hold the fiber exactly centered in its endface for the best possible connection, so its construction is critical. Not only must the hole for the fiber be accurately placed; it must also be sized precisely to receive the exact diameter of the cladding. The fiber's plastic coating is stripped away so the fiber can be inserted in the ferrule.

Ferrules typically may be made of metal, ceramic, or plastic with a selection of hole diameters ranging from slightly larger than the fiber diameter to slightly smaller to allow for minute

variations in the manufactured fiber diameters. For example, for a 125 μ fiber, ferrules are available with hole sizes ranging from 124 μ to 127 μ . Because they must align the fiber end precisely, ferrules must meet several important criteria: they must be strong enough to withstand many cycles of connection and disconnection without bending, cracking, or breaking; they must maintain dimensional stability to ensure proper alignment of the fiber; and they must be of the right shape and consistency to allow polishing, which ensures a good connection.

Ceramic materials such as aluminum oxide and zirconium oxide are among the best materials for ferrules, offering the best combination of characteristics. They are hard enough to protect the fiber end, and their *coefficient of thermal expansion*, the measure of how much a material expands and contracts with temperature changes, is about the same as the fiber itself.

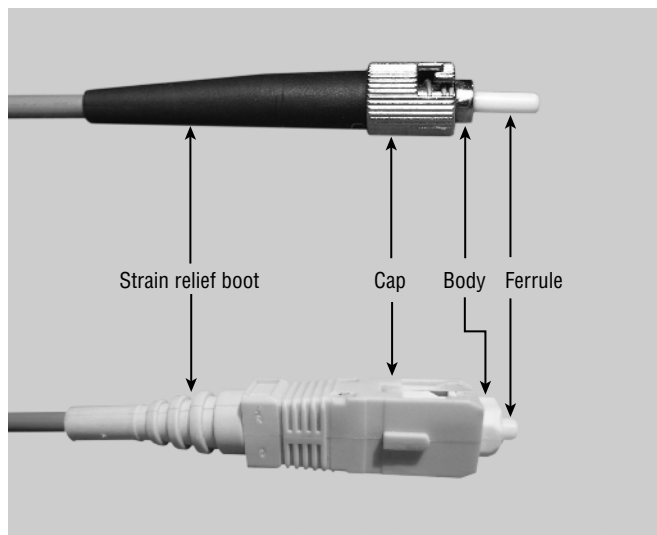
Metal ferrules made of stainless steel are stronger than ceramic, but they are less dimensionally stable. Plastic ferrules are less expensive than metal or ceramic, but they are neither as strong nor as stable as the other materials.

When a connector is built, which we'll describe in detail later in this chapter, the fiber is epoxied into the ferrule with the end protruding slightly beyond the endface of the ferrule. The fiber end is later trimmed and polished with the ferrule endface for a precise fit.

The ferrule fits inside the next component, the body of the connector. The body, which can be either metal or plastic, holds the fiber and the cable in place and transfers any strain placed on the connector to the cable rather than the fiber.

The cap, sometimes called the coupling nut, fits over the body of the connector and provides a means to secure the connector, as shown in Figure 9.2. This can be a locking mechanism, a threaded ring, or a snap attachment, depending on the type of connector.

FIGURE 9.1 Fiber optic connector components



The cable is secured to the connector inside a strain relief boot, which keeps the cable from being pulled at too great an angle against the connector and protects the joint between the cable and the connector.

Unlike electrical connectors that usually have a plug and a receptacle that mate with each other, fiber optic connectors are all plugs. Any connection requires an adapter to hold the connectors together in the proper alignment, so the usual arrangement consists of ferrule endfaces meeting each other inside the adapter, as shown in Figure 9.3.

FIGURE 9.2 Connector cap types

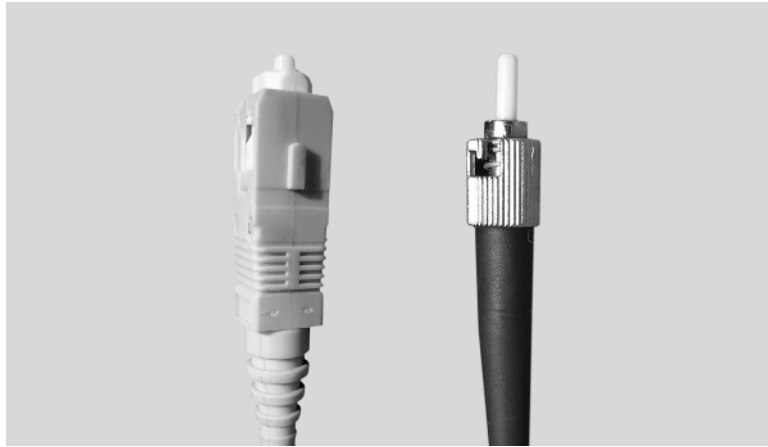


FIGURE 9.3 Typical connector attachment

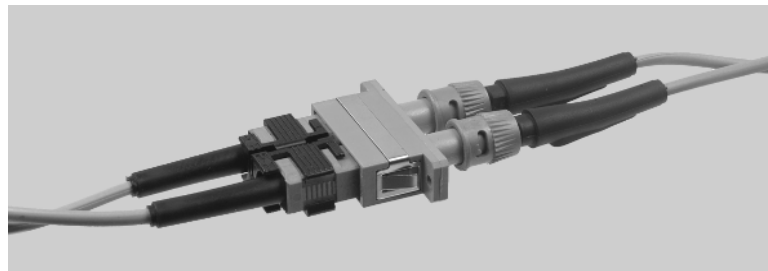


Photo courtesy of Norfolk Wire & Electronics

Connector Performance

Connector performance is dependent on three types of factors: *intrinsic* factors, which are determined by the construction of the connector and the fiber itself; *extrinsic* factors, which are determined by the relationship of the connector to the hardware and the optical fiber to the connector; and *endface finish* factors, which are determined by the precision and geometry of the polished end of the ferrule.

Intrinsic and extrinsic factors also apply to the fiber itself and to fiber splices, which we discussed in Chapter 8, “Splicing.” Endface finish factors have both intended and unintended results when applying the final polish to the ferrule endface and can be divided into two categories: roughness and geometry.

Roughness

Roughness is the measure of how well the fiber endface is polished. If you tried to look through glass or clear plastic with a rough surface, you would see light and some color, but the images would not be clear. Because the fiber core diameter and light wavelengths are measured in microns or nanometers, a surface finish that might seem acceptable in most other situations could still have enough irregularities to deflect light rays as they leave or enter the fiber end, attenuating the signal.

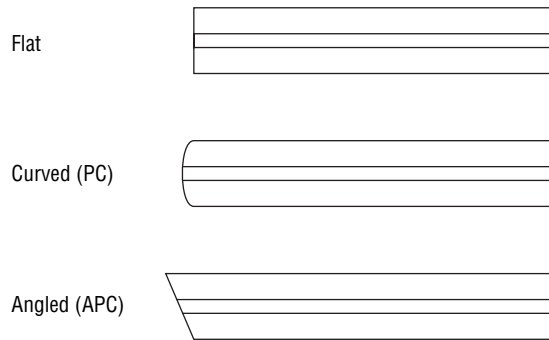
Geometry

Geometry refers to the shape of the ferrule endface. It seems at first that the ideal shape would be a flat surface that would mate with the flat surface of another ferrule, but this geometry actually presents the most potential problems.

As you can see in Figure 9.4, flat endfaces will always have some polishing irregularities. The slightest amount of roughness can keep the fiber ends far enough apart to cause end separation loss due to Fresnel reflection. In addition, some of the reflected light returns down the fiber to the transmitter as *return reflection*, which can interfere with certain types of light sources.

To ensure physical contact (PC) between fiber ends, the best endface geometry is a convex curve. This curve, or *PC finish*, ensures that the highest feature on the endface is the fiber end. When the fiber ends are in direct physical contact, the light behaves as if the connected ends are a continuous piece of fiber and passes through with no back reflection from interconnections.

Another method of reducing return reflection is with an *angled PC (APC) finish*. This type of finish puts an angle of about 8° on the endface, with the intent that it will mate with a similarly angled endface when properly aligned. With the endface and the fiber end angled, any light that is reflected is sent into the cladding, rather than back along the fiber.

FIGURE 9.4 Endface geometry configurations

Connector Types

The purpose of all optical fiber connectors is the same—to align a terminated fiber end with another terminated fiber end or a piece of equipment and hold it firmly in place. The manner in which this has been accomplished, however, has continued to evolve as technology improves and as requirements change. As a result, manufacturers have created many different types of connectors. Most of the connectors are still in use, though some have become more common while others exist only in older systems.

Connectors can be divided into two main groups: single-fiber connectors and multifiber connectors. Single-fiber connectors are designed for only one fiber, but they may be used to break out multiple fibers from a cable for individual termination in patch panels or hardware. Multifiber connectors may be used to connect paired fibers in duplex systems, or to connect entire ribbons of 12 fibers with one connector. They may also be used as a way to break out ribbon cables into individual fibers by using a *pigtail*.

Because there are so many different types of connectors, the Telecommunications Industry Association and Electronics Industry Alliance have not attempted to recognize or create standards for all of the available types. In their TIA/EIA-568-B.3 standard, however, the associations have jointly recognized performance standards for connectors, which will be described at the end of this chapter.

Let's look at the different types of connectors.

Single-Fiber Connectors

Single-fiber connectors have a wide variety of connection methods. Some, including the earliest types, are engaged by pushing and twisting, or by using a threaded sleeve to draw the connector tight. Newer forms, however, are square or rectangular snap-in connectors. These are engaged by a simple push that engages a locking mechanism.

SC Connectors

The SC (subscriber connector), shown in Figure 9.5, is among the most widely used connectors. It is the only single-fiber connector formally recognized by TIA/EIA-568-B.3. Originally developed by Nippon Telephone and Telegraph (NTT), the SC has a standard-sized 2.5 mm ferrule and a snap-in connector that was created as an alternative to connectors that required turning or twisting to keep them in place. In addition, SC connectors can be installed in only one orientation, making them suitable for APC finish connectors or fiber connections in which the cores are required to be aligned to reduce *polarization mode dispersion*.

ST Connectors

The ST (straight tip) connector, shown in Figure 9.6, was developed by AT&T as a variation on a design used with copper coaxial cables. This connector has a metal connector cap that must be twisted to lock into place. The ST is considered a legacy connector, as it has been around for quite some time and can still be found in many installations.

FIGURE 9.5 The SC connector is one of the most common types in use.

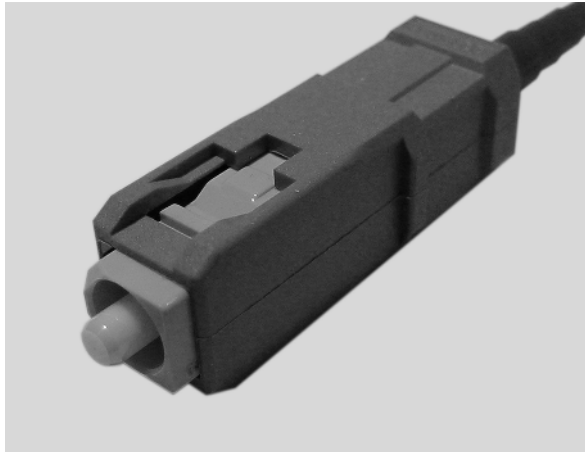


FIGURE 9.6 An ST connector must have its connector cap twisted to lock in place.



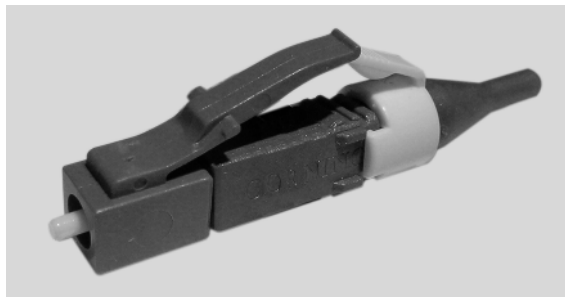
FC Connectors

An FC (face contact) connector is a rugged metal connector with a screw-on connector cap and a 2.5 mm ferrule. Like the SC connector, the FC is used in connections where proper polarization must be maintained. Because the connector is cylindrical, it must be aligned with a built-in key. Note, however, that there are several different standards for the size of the key, meaning that the connector must be properly matched with its adapter.

LC Connectors

The LC connector, shown in Figure 9.7, is a *small form factor* plastic connector. Developed by Lucent Technologies, this snap-in connector is considered to be a smaller version of the SC connector and is sometimes referred to as a mini SC. The small form factor connector has a 1.25 mm ferrule—half the size of a standard connector—that allows it to be used in smaller components and in areas where connectors must be closer together.

FIGURE 9.7 The LC connector is a small form factor version of the SC connector.



D4 Connectors

The D4 connector, also known as a DIN connector, is an older style heavy-duty metal connector with a 2.5 mm ferrule and a threaded connector cap that must be screwed on to secure the connector. It was developed by Siemens and is mostly used by Deutsche Telecom, the parent company of T-Mobile Wireless. It is similar in function to the FC connector, but its profile is slightly smaller, allowing it to be used in smaller spaces than the FC connector.

SMA Connectors

The SMA connector was developed by Amphenol from its line of microwave connectors known as SubMiniature A. The connector has a 3 mm stainless steel ferrule and a connector cap that is threaded on the inside. Because it was originally developed before the invention of single-mode fiber, the SMA connector does not provide as precise a connection as more recent designs. It is still used in military applications and in the delivery of high-power laser light.

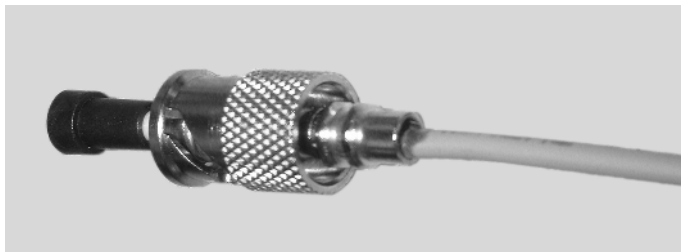
Biconic Connectors

The biconic connector features a stainless steel or ceramic cone-shaped ferrule that helps in aligning fiber ends during connection. Biconic connectors are threaded on the outside for screw-in placement, and although they are considered legacy connectors, they are still used in military applications because of their robustness.

Mini BNC Connectors

The mini BNC, shown in Figure 9.8, is similar in appearance to its counterpart made for copper coaxial cables. The mini BNC (short for either “Bayonet Nut Connector” or “Bayonet-Neill Concelman,” the inventor of the original BNC) is a metal twist-lock connector with a 2.5 mm ferrule. The mini BNC is considered a legacy connector, so while it is rarely installed in new systems, you may find it on older installations.

FIGURE 9.8 The mini BNC connector resembles the BNC connector used in RF applications.



Multiple-Fiber Connectors

Multiple-fiber connectors are mostly designed to support duplex operations and ribbon cables, allowing a greater number of connections in a smaller space. Some duplex connectors resemble larger or ganged versions of single-fiber connectors.

FDDI Connectors

The FDDI connector was developed by the American National Standards Institute (ANSI) for use in Fiber Distributed Data Interface (FDDI) networks. This plastic duplex connector is also called FDDI MIC for *medium interface connector*, a connector that is used to link electronics and fiber transmission systems. It features protective shields around the two 2.5 mm ferrules to prevent damage. The connector locks into place with latches on the sides and can be keyed to ensure proper orientation of the two fiber ends.

ESCON Connectors

The ESCON connector was developed for IBM's ESCON (Enterprise Systems Connection) architecture in the early 1990s. It is similar to the FDDI, but it has a shroud that retracts from the ferrules when the connector is engaged.

SC Duplex Connectors

The SC duplex connector, shown in Figure 9.9, is actually two single SC connectors joined with a plastic clip. This arrangement can be extended into even more plugs, if necessary, but is most commonly applied in the duplex configuration.

FIGURE 9.9 The SC duplex is made of two single SC connectors joined together.

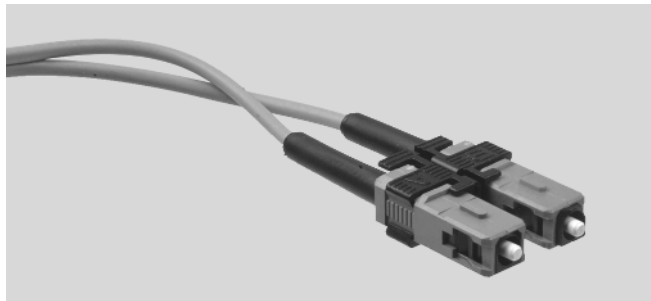


Photo courtesy of Norfolk Wire & Electronics

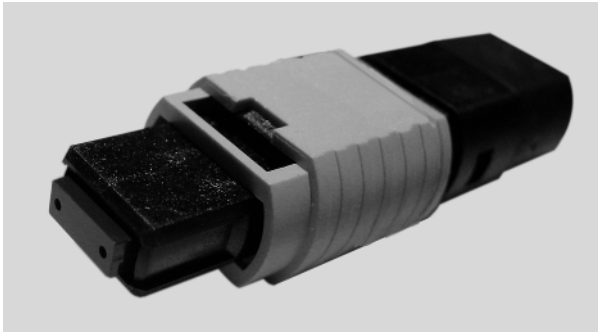
MPO Connectors

The MPO (Multifiber Push On) connector, shown in Figure 9.10, is built on the MT-style ferrule, designed by NTT. The MT (mechanical transfer) ferrule is designed to hold up to twelve fibers in a ferrule 7 mm wide and is ideally suited for ribbon cable connections. In addition,

precision-machined guide pins maintain the close alignment necessary for connecting twelve fibers at once. These guide pins can be arranged as necessary between the mating connectors depending on the way in which they will be used. Connectors designed for multiple fibers are also known as *array connectors*.

The MPO connector has a plastic body that is spring-loaded to keep the connectors together. While not recognized by TIA/EIA standards, the MPO is defined by TIA 604-5, also known as FOCIS 5.

FIGURE 9.10 The MPO connector packs twelve fibers into a single ferrule.



MTP Connectors

The MTP connector is designed by USConec and built around the MT ferrule. It is designed as a high-performance version of the MPO and will interconnect with MPO connectors.

MT-RJ Connectors

The MT-RJ connector, shown in Figure 9.11, was developed to emulate the functionality of the RJ-45 modular connector, which is used in most office networking. It provides a snap-in, duplex connection in a housing that resembles the familiar modular network plug. In fact, the connector is designed to fit in the same physical opening as the RJ-45, so many networking fixtures and wall plates can be retained after the hardware inside them has been upgraded for fiber.

The MT-RJ's single ferrule holds two fibers in a housing smaller than most single-fiber connectors, so it is attractive in many applications where size matters. The connector is built like a mechanical splice, so the fibers are slid into the ferrule and the plug is closed to hold the fibers in place without the need for epoxy.

MT-RJ connectors may be used as most connectors are, with an adapter used to join two connectors together, or in a true plug-and-jack arrangement with bare fibers or a transceiver installed in the jack.

There are many other connector types in use, and more are being added as the technology to align and secure them improves. Table 9.1 gives you a quick reference to the connectors we've discussed and their key features.

FIGURE 9.11 The MT-RJ connector is designed to emulate the RJ-45 modular plug in fit and ease of use.

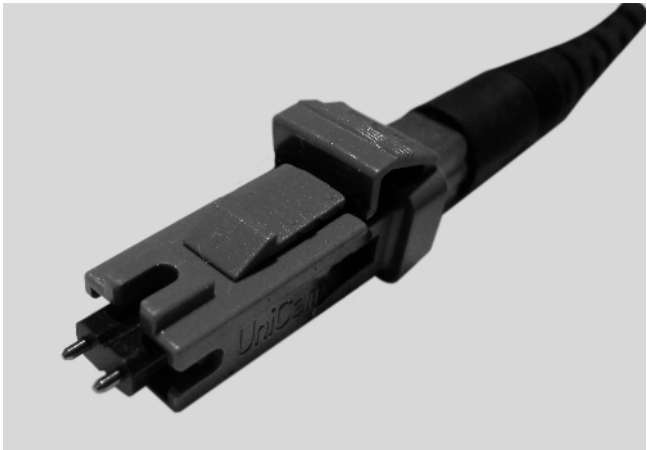


TABLE 9.1 Quick Reference for Fiber Optic Connectors

Connector	No. of Fibers	Connection	Shape	Ferrule Size
Single-Fiber				
SC	1	Push-in	Square	2.5 mm
ST	1	Twist-lock	Round	2.5 mm
FC	1	Screw-on	Round	2.5 mm
LC	1	Push-in	Square	1.25 mm
D4 (DIN)	1	Screw-on	Round	2.5 mm
SMA	1	Screw-on	Round	3 mm
Biconic	1	Screw-on	Round	Conical
Mini BNC	1	Twist-lock	Round	2.5 mm

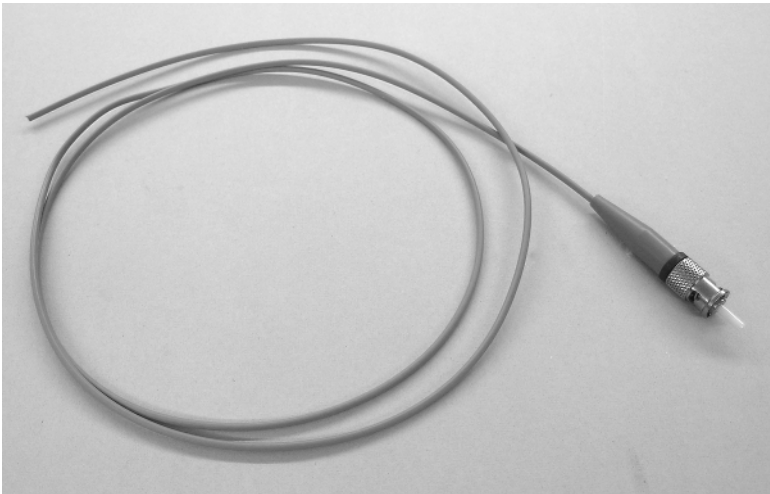
TABLE 9.1 Quick Reference for Fiber Optic Connectors *(continued)*

Connector	No. of Fibers	Connection	Shape	Ferrule Size
Multifiber				
FDDI	2	Push-in	Rectangular	(2) 2.5 mm
ESCON	2	Push-in	Rectangular	(2) 2.5 mm
SC Duplex	2	Push-in	Rectangular	(2) 2.5 mm
MPO	12	Push-in	Rectangular	7 mm
USConec MTP	12	Push-in	Rectangular	7 mm
MT-RJ	2	Push-in	Rectangular	5 mm

Pigtails

Pigtails are fiber ends with connectors factory-attached for future splicing into a system, as shown in Figure 9.12. Typically, a pigtail starts as a manufactured patch cord or jumper with a connector at each end. You can then cut the jumper in half and have two pigtails ready to splice.

FIGURE 9.12 Use a pigtail to splice a connector to an existing fiber.



Pigtails are available with a variety of connectors, depending on your needs. These products have advantages over field terminations because the connectors are factory-installed to exacting standards. This can save time over attaching a connector to the end of a fiber and produces a better connection. They are especially useful in situations where many connectors have to be added to cables in a relatively short time, or in a location where it is easier to make a splice than it is to add a connector.

On the downside, pigtails require hardware to protect the splice and investment in a fusion or mechanical splicer, in addition to the cost of the connector and hardware itself.

Specialized Connectors

While standard commercial connectors are used in most networks and communications links, fiber optic connectors are also available in specialized configurations that feature durable cases, heavy-duty connector bodies, and multiple-fiber connections, such as the TFOCA connector shown in Figure 9.13.

These connectors are often found in military applications where heavy use and destructive environments are common.

FIGURE 9.13 The TFOCA connector is a specialized connector for military applications.



Photo courtesy of STRAN Technologies

Connector Termination

One of the most exacting jobs you will encounter in working with fiber optics is connector termination. If you do the job properly, a fiber the size of a human hair will mate with another fiber or a piece of hardware and transfer 93% or more of the light passing through it. If you don't, the light could stop at the connector and go no further, or so little of the light could transfer that it would essentially be useless. With those encouraging words urging us forward, let's look at the ways in which a fiber can be terminated with a connector.

While some of the procedures involved may vary, the main task is the same: You must insert a bare fiber into a hole in the ferrule so that it sticks out slightly beyond the ferrule endface and secure it there, then you must cut, or *cleave*, the excess fiber and polish the ferrule endface and the fiber together to achieve the proper profile and finish.

While this task sounds simple, it requires some careful planning and execution. Remember that the hole in the ferrule is almost microscopic, since it must be no larger than the fiber itself. Also, the method used to hold the fiber in place must be able to withstand not only the polishing process but also the constant use that the connector will have to endure.

Epoxy

The most common material used to secure the fiber is a two-part epoxy. There are different types of epoxy in common use, and the type selected determines most of the steps in the termination process. Each has its advantages and disadvantages, and the location and circumstances of the work you are performing will determine which type you'll use.

Let's look at each type of epoxy in detail.

Oven-Cured Epoxy

Oven-cured epoxy is probably the best type you can use, but it is also the most cumbersome. The epoxy itself consists of a resin and a hardener, which must be mixed, in the right proportion. Once it is mixed, it has a pot life of only a few minutes, and any epoxy that is unused must be discarded. On the other hand, once a batch is mixed, several connectors can be assembled at the same time, so it is useful in making a large number of terminations at one location.

Once the connector has been assembled, you have to insert it into a specially built oven, shown in Figure 9.14, to cure the epoxy for anywhere from 30 minutes to one hour and then let the assembly cool before you can begin polishing it. The oven will hold a number of connectors, so as each one is built it can be inserted for curing.

While oven-cured epoxy is time-consuming and equipment-intensive, it produces a hard, fully cured bead around the base of the fiber, as shown in Figure 9.15. This bead reduces the risk that the fiber will break inside the ferrule during cleaving, which could ruin the connector.

FIGURE 9.14 An epoxy-curing oven cures epoxy in 30 minutes to one hour.



Photo courtesy of W.R. Systems

FIGURE 9.15 The bead of oven-cured epoxy protects the fiber during cleaving.

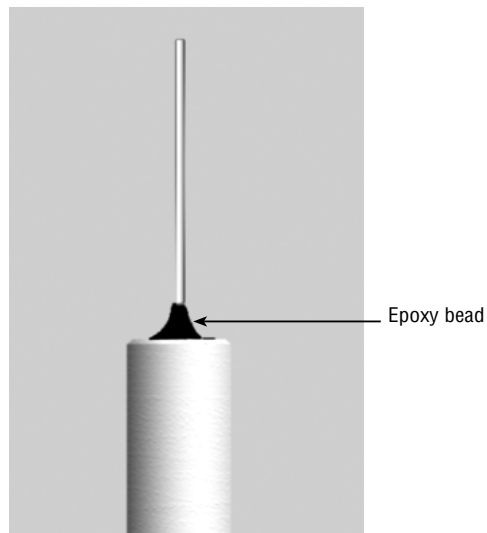


Photo courtesy of W.R. Systems

Oven-cured epoxy is also more stable than other epoxies, meaning that it will stand up to harsher temperature and stress conditions.

UV Epoxy

UV, or ultraviolet-cured, epoxy still requires an energy source for curing, but this time the source is an ultraviolet lamp. Once the epoxy is mixed and applied, the lamp will cure the mix in 60 seconds or less. The procedure requires no warm-up time for an oven and no cool-down time for the connector.

Anaerobic Epoxy

Anaerobic epoxy is usually used when there is no power available or when time is at a premium. While the adhesive still comes in two parts, each part is applied separately to the fiber and the ferrule. When the two are joined, the epoxy hardens and cures in about ten seconds.

While this may seem to be an ideal solution for all situations, there are some drawbacks to anaerobic epoxy.

For one or two connectors, there is a significant time savings. However, anaerobic epoxy cannot be used in batches as oven-cured epoxy can, so you can prepare and work on only one connector at a time. Also, the adhesive does not form a bead the way oven-cured epoxy does, so greater skill is required for cleaving and polishing the fiber.

Tools

Because fiber dimensions are so small, you'll need some specialized tools to help prepare the fiber properly. These tools are designed for handling fiber specifically, and any substitutions will only waste your time and materials.

Shears Even though cabling materials are fairly easy to cut, the aramid fiber strength member is fibrous and loose fitting, and your blade will not have anything firm to cut into. You'll need a good pair of sharp shears to cut through the strength member quickly and cleanly.

Stripper A good stripper has graded openings for removing the outer jacket, tight buffer, and coating so that only the fiber itself is exposed. Some strippers are made for specific fiber sizes, while others will work with a variety of sizes. Several different types of strippers are shown in Figure 9.16.

Scribe The scribe, shown in Figure 9.17, is used for precision work in removing the fiber end once it has been adhered inside the ferrule. To use it, you only need to nick one side of the fiber lightly. This breaks the surface of the cladding, which provides the fiber's tensile strength. You'll need some practice with a scribe to get the right results, since it requires just the right touch to keep from damaging the fiber.

Abrasives Abrasives are used to put the final polish on the fiber end and ensure that it has the correct profile. Depending on the method of polishing you prefer, different types of abrasives are available. Essentially, they are like a very fine sandpaper with a Mylar film or cloth backing and abrasive material adhered to the backing. The abrasive material is either aluminum oxide, available in particle sizes from 5 μ to 0.3 μ , or diamond, available in 0.1 μ . The abrasives shown in Figure 9.18 are abrasive cloths with padded backings.

FIGURE 9.16 Strippers are used to expose the bare fiber.



FIGURE 9.17 A scribe is used for removing the fiber end.



Photo courtesy of Norfolk Wire & Electronics

Puck The puck, shown in Figure 9.19, is used to ensure that the ferrule stays perpendicular to the polishing cloth or film during the polishing process. Even the slightest variation in the polishing angle can produce back reflections that would render the connector unusable.

Alcohol Alcohol is needed to clean the fiber end before it's adhered inside the ferrule and for cleaning the abrasive if it gets contaminated. For safety, keep the alcohol in an approved dispenser as described in Chapter 6, "Safety."

Wipes Use lintless wipes to apply the alcohol while leaving as little residue as possible behind.

Now let's look at the process for terminating with a connector.

FIGURE 9.18 Polishing cloth is used to finish the connector end.

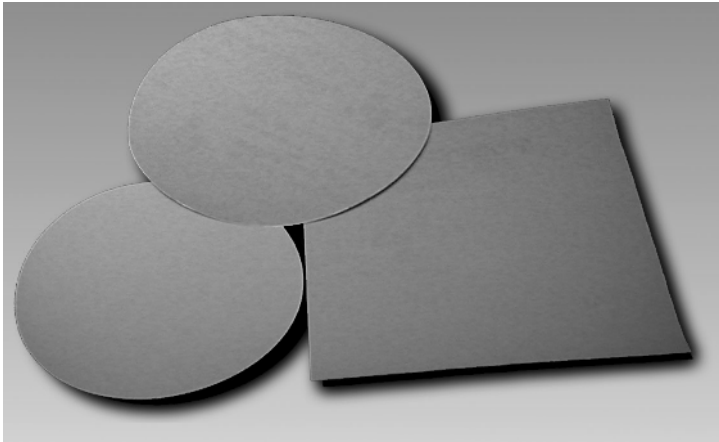


Photo courtesy of W.R. Systems

FIGURE 9.19 A puck helps maintain the right angle for polishing.

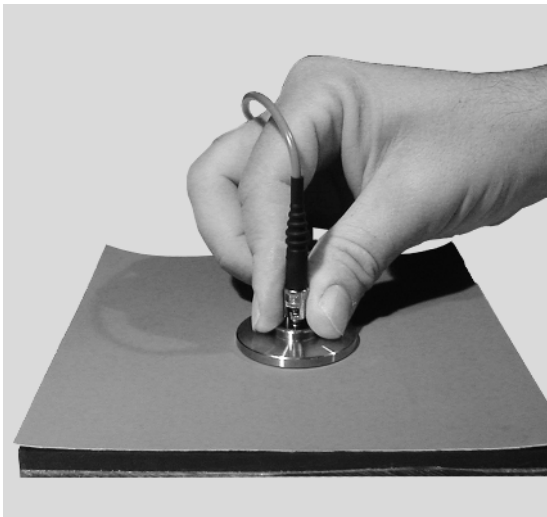


Photo courtesy of W.R. Systems

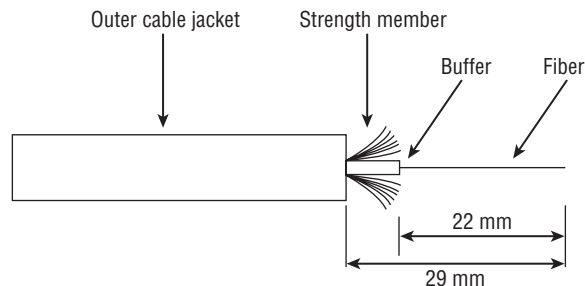
Assembling the Connector

Assemble your tools before you start so you don't have to look for them in the middle of the process. It may seem like a fussy detail, but your work will proceed more smoothly and efficiently if you have prepared all of your tools and materials and laid them out where you can reach them easily when you need them.

Fiber and Connector Preparation

1. Cut the cable two inches longer than you'll need.
2. Install the strain relief boot on the cable end, small diameter first. Important: The strain relief boot will not fit over the connector, so it must be placed on the cable first.
3. Using the appropriate strip chart from Figure 9.20, prepare the cable end with the stripper.
4. Mark the cable at 22 mm ($7/8$ ") and 29 mm ($1\text{-}1/8$ ") back from each end.
5. Remove the cable jacketing at the 22 mm mark and cut the strength member flush with the remaining cable jacketing.
6. Remove the cable jacketing at the 29 mm mark and fan the strength member back with compressed air.
7. Strip away the buffer in $1/4$ " increments to reduce the chance of breaking the fiber.
8. Clean the fiber with an alcohol-moistened wipe and blow dry.
9. Clean and inspect the connector. Ensure that the fiber hole is not obstructed and the ferrule tip is clean. If the hole is obstructed, clean it out with a piece of piano wire. Piano wire can also be used to remove a broken fiber from the connector hole.
10. Dry-fit the connector on the prepared end to ensure a proper fit.

FIGURE 9.20 Strip chart for ceramic and steel ferrules



Heat-Cured Epoxy Application

The following procedures apply to heat-cured epoxies. We'll look at two different types of heat-cured epoxy: Hysol FI-7070 and Tra Con BAF113. Be sure to follow the specific directions for the epoxy you are using.

Tra Con Epoxy: Ceramic Ferrules

This epoxy comes in premeasured packets that mix when you break an internal seal in the package. Once you have ensured that the two parts are mixed according to instructions, butter the bare fiber with epoxy using a toothpick. Apply a small amount of epoxy evenly around the edge of the cable jacketing. Insert the fiber into the rear back shell of the connector. Apply a small bead of epoxy to the ferrule tip, covering only about a third of the ceramic diameter end.

Hysol Epoxy: Ceramic Ferrules

Mix the epoxy according to the manufacturer's instructions. Load the syringe with epoxy and depress the plunger until the syringe is free of air pockets. Insert the syringe tip into the backbone of the connector until it bottoms out against the ceramic ferrule. Maintain pressure and slowly inject the epoxy until a bead appears on the end of the ferrule tip. Continue to inject epoxy until the bead covers about a third of the ferrule diameter end. Release the pressure on the plunger, wait five seconds, and then remove the syringe. Apply a thin coat of epoxy evenly around the inside wall of the rear back shell of the connector with the syringe. Insert the fiber into the rear back shell of the connector.

Anaerobic Epoxy Application

Anaerobic epoxy comes in two parts, like heat-cured epoxy, but instead of resin and hardener the two parts are a primer and an adhesive. When they are mixed and oxygen is removed, the epoxy hardens and cures in about 10 seconds, so it's important to work quickly and efficiently. Preparation of the fiber and ferrule is the same as for heat-cured epoxy.

Once the fiber and ferrule are prepared, place the adhesive portion of the epoxy in the syringe and depress the plunger until the syringe is free of air pockets. Insert the syringe tip into the backbone of the connector until it bottoms out against the ceramic ferrule. Maintain pressure and slowly inject the epoxy until a bead appears on the end of the ferrule tip. Continue to inject epoxy until the bead covers about a third of the ferrule diameter end. Release the pressure on the plunger, wait five seconds, and then remove the syringe. Apply a thin coat of epoxy evenly around the inside wall of the rear back shell of the connector with the syringe.

Using a toothpick, butter the fiber with the primer portion of the epoxy, but do not cover the end of the fiber. This will give you a little bit of extra time as you slide the fiber into the ferrule. Work the primer over the back of the fiber.

Insert the fiber into the rear back shell of the connector. Once you start pushing the fiber into the ferrule, the primer and adhesive will be in an oxygen-free environment and begin curing, so you must keep the fiber in motion until it is all the way into the ferrule. Do not try to back the fiber out or you will break it.

UV Epoxy Application

Assembling a connector with UV epoxy is similar to working with anaerobic epoxy with two major exceptions. First, the ferrule must be specifically made for UV epoxy with a clear glass area surrounding the fiber so that UV light can enter to cure the epoxy. Second, a UV lamp must be used to cure the epoxy.

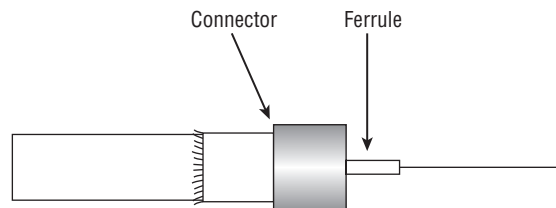
Apply the adhesive and primer just as you would for anaerobic epoxy and insert the fiber into the connector and ferrule. Once the fiber is in place, set the connector under the UV lamp and turn the lamp on. The UV lamp has safety features to keep you from looking directly at the light, but it is important that you do not try to work around these safeguards or you could damage your eyes. The light will cure the epoxy in about 30 seconds.

Assembling the Connector

Once the epoxy has been applied, you must complete the connector assembly as follows:

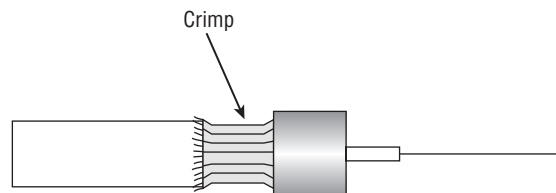
1. Gently feed the fiber through the connector, as shown in Figure 9.21, making sure the strength member fans back as you push the fiber forward.

FIGURE 9.21 Feed the fiber through the connector.



2. Crimp the rear back shell with a .128 hex crimp die, as shown in Figure 9.22.

FIGURE 9.22 Crimp the rear back shell of the connector.



3. If you are using heat-cured epoxy, place the connector in the curing oven and cure the epoxy according to the cure schedules in Table 9.2.

TABLE 9.2 Epoxy Cure Schedules

Epoxy	Room Temperature Cure Time	Heat Cure Time	Heat Cure Temperature	Mix Ratio
BAF113	24 hours	1 hour	100° C	Premeasured
FI-7070	24 hours	30 minutes	100° C	2:1

Pre-Load Epoxy Connector Termination

Some connectors have adhesive already loaded into the ferrule so you can avoid mixing and injecting the material into the connector. These *pre-load*, or *hot melt*, connectors require heat as oven-cured epoxy does, but the process is slightly different. You'll still prepare the fiber in the same way as with other methods, but the connector requires some extra preparation.

To terminate a fiber using a pre-load epoxy connector:

1. Preheat the oven for at least five minutes and assemble the cooling stand.
2. Remove the connector from its packaging and load the connector into the connector holder, then place the connector and holder into the oven for the prescribed time to melt the adhesive.
3. After the fiber has been prepared (don't forget to put the boot onto the cable), remove the connector from the oven.
4. Insert the fiber into the connector.
5. Place the connector into the cooling rack for 2 to 3 minutes.

Polishing the Connector

The final step in the connector termination process is polishing the fiber end. Depending on the type of ferrule and epoxy used and the type of abrasive employed, the polishing process will differ.

The procedures used with ceramic ferrules are designed to produce a PC finish. The ferrule endface is preformed to the correct dimension, so one of the goals of polishing is to ensure that the curve in the fiber end matches the curve of the ferrule endface. It is also important that the apex of the curve be at the center of the fiber, so the core will make proper contact.

To ensure the proper curve, a polishing tool, or *puck*, is used to keep the connector perpendicular to the polishing surface at all times. When you insert the connector into the puck, the ferrule projects from the face of the puck just enough to make contact with the polishing film. The soft pad under the polishing film allows the film to conform to the ferrule endface shape.

Mylar Polishing Film

Mylar polishing film consists of a Mylar film with crystal abrasives fixed to it with adhesive. The film is quite thin, which allows it to conform to the shape of the ferrule when it is placed on a soft backing pad.

The steps for polishing fiber with Mylar film are as follows:

1. Be sure the epoxy is completely cured and allow the connector to cool.
2. Scribe the fiber at the base of the epoxy bead, as shown in Figure 9.23.
3. To wear down the rough nub of the fiber end, hold the connector so that it is facing up and arch a piece of 5 μ polishing film over it. Lightly rub the film in a circle over the connector until you no longer feel the fiber end “grab” the film.
4. Prepare a glass plate and a soft pad, and make sure they are clean.
5. Place a clean piece of 5 μ polishing film over the soft pad, using the glass plate as a clean, smooth backing surface for the pad.
6. Insert the connector into the puck and place the puck on the polishing film.
7. Slowly move the puck in a figure-8 pattern over the polishing film to begin grinding off the epoxy bead. After about 20 or 25 figure-8 strokes, you’ll start to feel less resistance. Stop polishing before you have removed all of the epoxy.
8. Remove the connector from the puck and clean the connector, the puck, and the polishing surface thoroughly with alcohol and compressed air.
9. Place a sheet of 0.3 μ film over the soft pad, insert the connector into the clean puck, and complete the final polish by performing figure-8 strokes. Continue until all of the epoxy is removed.

FIGURE 9.23 Scribing location for oven-cured epoxy bead

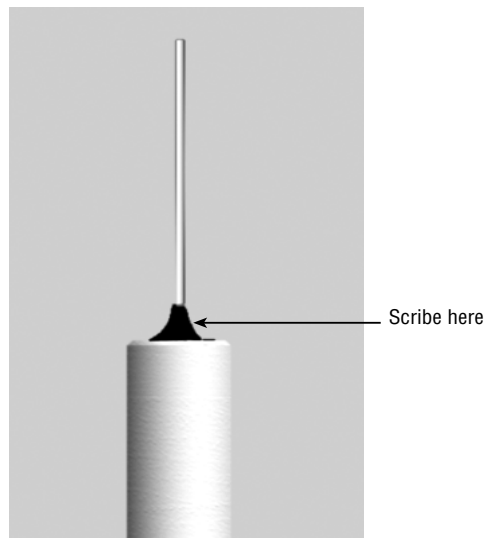


Photo courtesy of W.R. Systems

If you have time, and if the highest quality polish is important, you may want to use an intermediate film of $1\text{ }\mu$ before going to the $0.3\text{ }\mu$ film. A final polish with the $0.1\text{ }\mu$ film will ensure the best optical quality possible for the fiber end, but be careful: the diamond abrasive on the $0.1\text{ }\mu$ film will also wear down the ceramic ferrule and could damage it.

Abrasive Polishing Cloth

Abrasive polishing cloth is a recent addition to the technician's toolkit. It uses abrasive crystals bonded to a cloth backing with flexible latex, allowing the abrasive to move and reducing breakage of the glass fiber. Because of its flexibility, the cloth does not require the "step-down" in abrasive sizes the way Mylar polishing film does. This can cut the polishing time by several minutes. Abrasive polishing cloth is also more durable than polishing film because it has a cloth backing, allowing it to be reused several times before it must be thrown away. The backing also serves as a layer of cushioning, so the soft pad is not required as it is with the film. The backing itself conforms to the ferrule dimensions, as shown in Figure 9.24.

To polish fiber with abrasive polishing cloth:

1. If the connector was heat cured, make sure the epoxy is cured and the connector is cool.
2. Scribe the epoxy bead as shown earlier in Figure 9.23 and pull the fiber end straight up to remove it.
3. To wear down the rough nub of the fiber end, hold the connector so that it is facing up and arch a piece of $1\text{ }\mu$ polishing cloth over it. Lightly rub the cloth in a circle over the connector until you no longer feel the fiber end "grab" the cloth.
4. Prepare a clean, hard rubber pad as a backing for the polishing cloth.
5. Set a piece of $1\text{ }\mu$ polishing cloth on the glass plate.
6. Set the connector in a clean puck and place the puck face down on the polishing cloth.

FIGURE 9.24 The backing of abrasive polishing cloth conforms to the ferrule's curve.

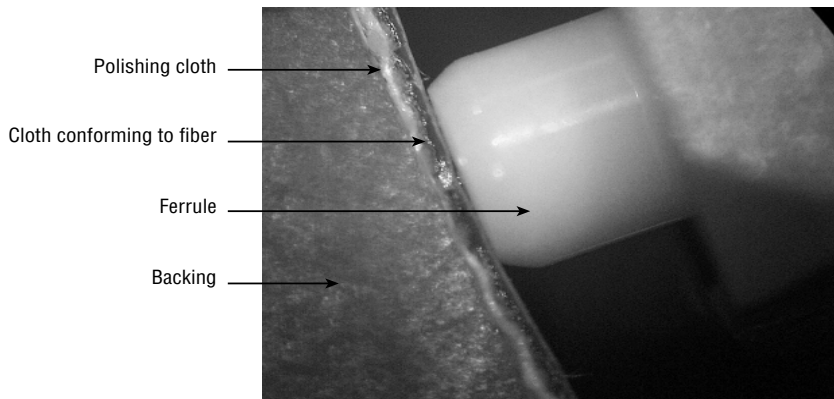


Photo courtesy of W.R. Systems

7. Slowly move the puck in a figure-8 pattern over the polishing cloth to begin grinding off the epoxy bead. After about 20 or 25 figure-8 strokes, you'll start to feel less resistance. Stop polishing before you have removed all of the epoxy.
8. Remove the connector from the puck and clean the connector, the puck, and the polishing surface thoroughly with alcohol and compressed air.
9. Place the polishing cloth on a clean glass plate. Replace the connector in the puck and perform the final polishing by moving the puck in a figure-8 pattern until all of the epoxy is removed.

Endface Examination

Once the connector has been polished, you'll need to inspect it to make sure it will carry signals properly. Remember that the fiber core can be as small as 9 μ , so even the smallest scratch, grain of dirt, drop of oil, or imperfection has the potential to cause unacceptable losses.

There are actually two kinds of losses with which we are concerned: *insertion loss* and *return loss*. Insertion loss affects the light that leaves the fiber end through the connector and is inserted into the next connector or into the hardware to which the fiber is connected. The lower the insertion loss, the better, because you want as much of the light passing through as possible.

Return loss, on the other hand, affects the light that reflects off of the fiber end due to Fresnel reflection and bounces back into the fiber in the opposite direction. You'll want return losses to be as high as possible to minimize the amount of light reflected back into the fiber.

Fortunately, a properly polished fiber end takes both losses in the direction they're supposed to go, decreasing insertion loss and increasing return loss. Scratches, dirt, and other imperfections drive insertion losses up and return losses down.

To inspect the fiber, you'll need a microscope capable of seeing not only the ferrule end but also the fiber itself, including the core. Two such microscopes are shown in Figure 9.25.

Before you inspect the fiber, clean it thoroughly to remove dust, residue, oils, and other contaminants. Do not use alcohol to clean the ferrule endface, as alcohol can leave a thin layer behind on the endface, creating enough of a residue to attenuate the light from the core.

The best cleaning method is to use a dry method such as swabs or commercial dry-cleaning products. Rather than trying to float the contaminants off the way alcohol and other liquid cleaners do, dry methods trap particles in a fine matrix and carry them away from the ferrule completely.

When you inspect the fiber, it should be free of epoxy and scratches. The following are some examples of results you might see upon inspection:

Ideal The fiber end is clear of any dark or light areas and the core illumination is even, as shown in Figure 9.26.

Broken If the fiber has broken during polishing, you'll see part of a smooth surface that suddenly drops off to a black area where the fiber has broken off below the top of the ferrule, as shown in Figure 9.27.

FIGURE 9.25 Microscopes are used for examining polished connectors.



FIGURE 9.26 A good polish yields a clean, even surface.

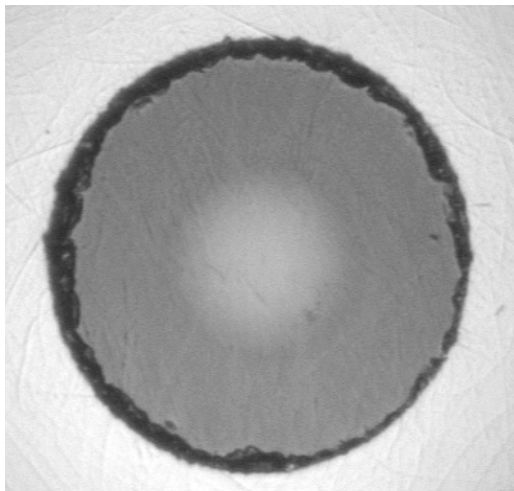


Photo courtesy of W.R. Systems

Chipped If the fiber is chipped, the middle area may look good, but you will see a drop-off to a mottled patch at the edge where the corner of the fiber has chipped off, as shown in Figure 9.28.

Skin oil If you've touched the connector end, you've probably gotten oil from your skin on the connector and the fiber. It will show up as dark streaks or droplets that can interfere with the light from the fiber, as shown in Figure 9.29.

FIGURE 9.27 The fiber has broken during polishing.

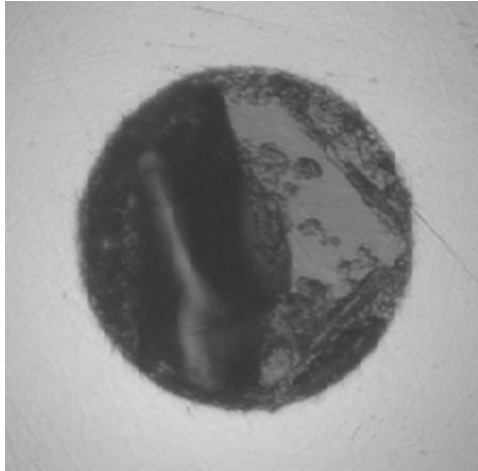


Photo courtesy of W.R. Systems

FIGURE 9.28 The chipped fiber drops off to a lower surface level.

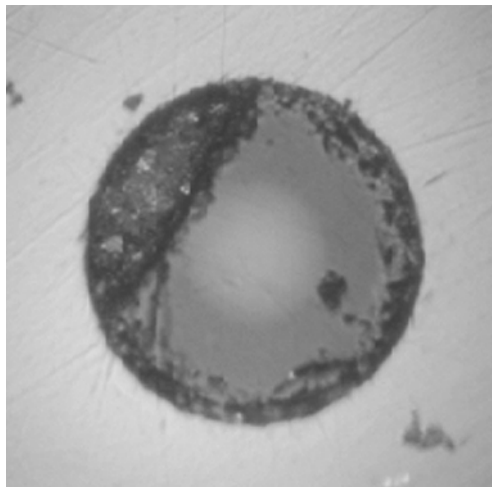


Photo courtesy of W.R. Systems

Bad polish Quite simply, you have a bad polish if you can still see epoxy around the edge of the fiber, as shown in Figure 9.30. It means that you did not polish long enough or press hard enough to wear away the epoxy.

No fiber It happens. Sometimes the fiber slips out of the ferrule during polishing, or it never makes it all the way through. If this happens, you'll simply see a black hole in the center of the ferrule, as shown in Figure 9.31.

FIGURE 9.29 Skin oil comes from careless handling.

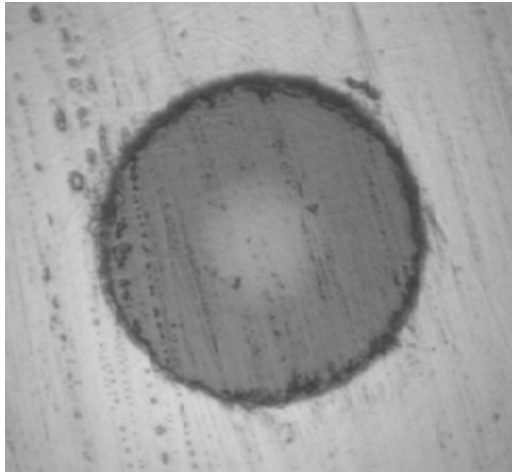


Photo courtesy of W.R. Systems

FIGURE 9.30 Epoxy around the edge of the fiber indicates a bad polish.

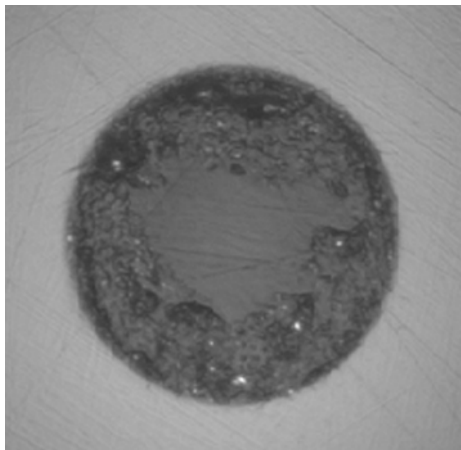


Photo courtesy of W.R. Systems

FIGURE 9.31 If you see a black hole, it means the fiber either slipped out or was never there.

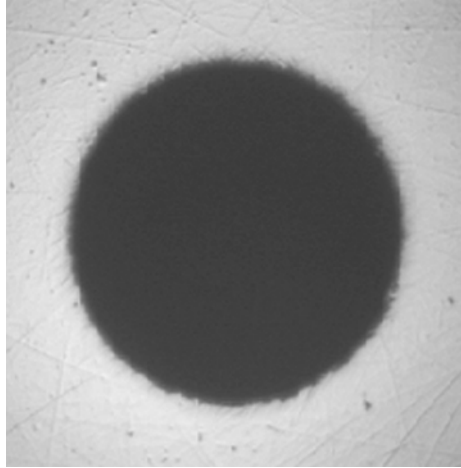


Photo courtesy of W.R. Systems

Connector Performance

If you have assembled and polished the connector properly and cleaned it well, chances are you'll get a good connection with it. It would be nice, however, to be sure that it is right before you put it into service. The best way to test a connector is with a fiber optic power meter, which tests the amount of loss that a connector causes in a light signal.

Power meters use a light source with a known power to send light through a fiber, and a receiver to convert the light into an electrical signal. The receiver is then plugged into a digital multimeter and tested to provide a reference voltage that can be used to determine dBm. The reference fiber is then replaced with the fiber and connector being tested and the voltage recorded again. The difference between the reference measurement dBm and the test measurement dBm is the actual loss in dB.

In order for a connector to meet minimum performance standards in TIA/EIA-568-B.3, it must have a mated pair insertion loss no greater than 0.75 dB.

Summary

This chapter covered the use of connectors as a means of terminating optical fibers. We described the typical construction of connectors and the materials used in connector ferrules.

In this chapter, we discussed factors that affect connector performance and the methods used to improve connector performance through ferrule shaping. We also described the different types

of finishes used on connector ends and the effects of each. We described different types of connectors and their typical uses, along with the use of pigtails and multifiber connectors.

We described the methods for preparing and assembling connectors using different types of epoxies and polishing methods. We also described the results of endface examination through a microscope. Finally, we described the requirements that a connector must meet to conform to TIA/EIA standards.

Exam Essentials

Be familiar with fiber optic connectors. Make sure that you understand the function and components of a fiber optic connector. Be able to describe their construction and materials. Be able to describe factors affecting their performance.

Understand the differences between connectors. Be able to describe different endface finishes and how they affect performance. Be able to describe different types of connectors and their uses. Make sure that you understand the use of pigtails as a means of terminating a fiber.

Be familiar with connector termination. Make sure that you can describe the steps in preparing a fiber for termination. Be able to describe the preparation and assembly of a connector. Be able to describe the proper polishing technique for a completed connector.

Be familiar with cleaning and inspecting connectors. Make sure that you understand how to clean a connector and inspect it for damage or improper construction. Make sure you understand the standards that a connector must meet in terms of light loss.

Review Questions

1. Which part of the connector holds the fiber in place?
 - A. The ferrule
 - B. The cap
 - C. The boot
 - D. The body
2. The best material for a ferrule is _____.
 - A. Metal
 - B. Plastic
 - C. Ceramic
 - D. Glass
3. Intrinsic factors in connector performance are determined by:
 - A. Precision and geometry of the ferrule end
 - B. Construction of the connector and the fiber itself
 - C. Relationship of the fiber to the connector
 - D. The type of connector being used
4. Which geometry of the connector and fiber end ensures a physical contact (PC) finish?
 - A. Flat
 - B. Rough
 - C. Curved
 - D. None
5. Because the SC connector can only be installed in one orientation, it is useful for _____.
 - A. Underwater connections
 - B. Angled physical contact connections
 - C. Tight spaces
 - D. Older connections
6. The LC, which has a ferrule half the size of the SC ferrule, is what kind of connector?
 - A. Small form factor
 - B. Multifiber
 - C. Screw-on
 - D. Twist-lock

7. A connector that combines two to twelve fibers in a single ferrule such as an MT is called a _____.
- A. Multimode connector
 - B. Small form factor connector
 - C. Multiple-fiber connector
 - D. Gang
8. A length of fiber with a factory-installed connector is called a _____.
- A. Quick-connect
 - B. Rat tail
 - C. Pigtail
 - D. Splice-on
9. The best type of epoxy for use in connectors is _____.
- A. Anaerobic
 - B. UV
 - C. One-part
 - D. Heat cured
10. The main drawback of using heat-cured epoxy is:
- A. It causes the most mess.
 - B. It takes the longest to cure.
 - C. It is hard to polish.
 - D. It requires a special connector.
11. In addition to forming the best bond, heat-cured epoxy is preferred because:
- A. It is easier to mix.
 - B. It causes less loss in the fiber.
 - C. It allows connectors to be made in batches.
 - D. It requires less skill to apply.
12. Anaerobic epoxy has as its main advantage:
- A. A quick cure time
 - B. A superior bond
 - C. A long pot life
 - D. Reusable components
13. What is a scribe used to do?
- A. Mark the buffer for stripping.
 - B. Mark the ferrule for polishing.
 - C. Nick the fiber for removing the end.
 - D. Mark the fiber to measure the proper length.

14. Except for the 0.1 μ size, the material used in polishing abrasives is usually _____.
 - A. Diamond
 - B. Aluminum oxide
 - C. Sand
 - D. Glass
15. The puck is used to:
 - A. Keep a firm grip on the connector for polishing.
 - B. Hold down the abrasive material during polishing.
 - C. Keep the abrasive surface clean.
 - D. Keep the ferrule perpendicular to the polishing surface.
16. Of the items below, which is the first step you need to perform?
 - A. Strip the fiber.
 - B. Cut the strength member.
 - C. Place the boot on the cable end.
 - D. Clean the fiber.
17. In a pre-load epoxy connector, how is the epoxy activated?
 - A. With water
 - B. With air
 - C. With lack of oxygen
 - D. With heat
18. One advantage of polishing cloth over polishing film is:
 - A. Polishing cloth does not need cleaning.
 - B. Polishing cloth has a flexible adhesive for the abrasive.
 - C. Polishing cloth stays rigid longer.
 - D. Polishing cloth can be used on any surface.
19. The best magnification for inspecting the endface of a fiber is _____.
 - A. 20X
 - B. 80X
 - C. 400X
 - D. 1000X
20. According to TIA/EIA-568-B.3, what is the maximum loss allowed between two mated connectors?
 - A. 0.1 dB
 - B. 0.25 dB
 - C. 0.75 dB
 - D. 1.0 dB

Answers to Review Questions

1. A. The ferrule holds the fiber in the right position for mating with hardware or another connector.
2. C. Ceramic provides the best combination of strength and stability for the ferrule to withstand heavy use and temperature variations.
3. B. Intrinsic factors are determined by the construction of the fiber and the connector and any flaws they may have as a result of the manufacturing process.
4. C. A curved finish on the ferrule endface and fiber end ensures that the highest point of the connector is the fiber core, which will make contact with a similarly constructed mating connector.
5. B. Because the SC can only be installed one way, it can use the APC connection, which is another way of ensuring physical contact between connected fibers.
6. A. The LC, often called a mini-SC, is a small form factor connector, which means that it can fit into a smaller space than standard-sized connectors.
7. C. Multiple fiber connectors are often used for duplex connections or for ribbon cable connections. They may be smaller than single-fiber connectors, yet contain up to twelve fibers in a single ferrule.
8. C. A pigtail is a factory-attached connector on a length of fiber that can be spliced to a fiber end for quick terminations. Pigtails are useful if a connector with factory polish is required to terminate a fiber.
9. D. Heat-cured epoxy forms the toughest, most durable bond and produces a hard bead that reduces the risk of the fiber breaking inside the ferrule.
10. B. Heat-cured epoxy requires 30 minutes to 1 hour to cure in a 100° C oven. This can be inconvenient when there is poor access to power or when connectors must be added quickly.
11. C. If a number of connectors must be made at the same time, heat-cured epoxy has a pot life that allows several connectors to be made at the same time.
12. A. Anaerobic epoxy cures in about 10 to 15 seconds, making it useful when a connector must be made quickly.
13. C. The scribe is used to break the surface of the cladding so that the end of the fiber can be removed cleanly once the fiber is secure inside the ferrule.
14. B. Aluminum oxide is used as the abrasive material for most grades of polishing media.
15. D. To ensure that the core of the fiber is at the highest point of the curve formed by polishing, the ferrule must be perpendicular to the polishing surface at all times. The puck maintains the correct angle during polishing.
16. C. Before exposing the fiber, place the boot on the cable end. This prevents damage to the bare fiber from the boot passing over it after the fiber has been stripped.

- 17. D. A pre-load epoxy connector must be placed in an oven for at least five minutes to activate the adhesive before the fiber is inserted.
- 18. B. By using a flexible adhesive, the abrasive crystals aren't as rigid on the backing, reducing the chance of damaging the fiber during polishing.
- 19. C. A 400X microscope provides the best view of the ferrule endface and the fiber for inspection after polishing and cleaning.
- 20. C. TIA/EIA standards allow a maximum 0.75 dB loss between two mated connectors.



Chapter

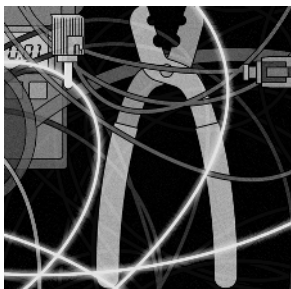
10

Fiber Optic Light Sources

OBJECTIVES COVERED IN THIS CHAPTER:

✓ Fiber Optic Light Sources

- Describe the basic operation and types of LED light sources used in fiber optic communications.
- Describe the basic operation and types of laser light sources used in fiber optic communications.
- Describe LED performance characteristics.
- Describe laser performance characteristics.
- Describe the performance characteristics of a LED transmitter.
- Describe the performance characteristics of a laser transmitter.
- Explain the safety classifications of the light sources used in fiber optic communication.



As discussed in Chapter 1, “History of Fiber Optics,” the idea of transmitting information with light is not new—just the technology that makes it easily possible is. Like optical fiber, light source technology has improved rapidly over the decades. These technological advances have greatly increased data transmission rates and reduced costs. Fiber optic transmitters are available to support every standardized network with a variety of connector choices.

This chapter discusses current fiber optic light source and transmitter technology as it applies to common telecommunication network standards, industrial control systems, and general-purpose systems. The performance standards that we discuss in this chapter do not represent the highest levels achievable in the lab or on the test bench. They represent commonly available parts for standardized networks, the types of networks that a typical fiber optic installer or fiber optic technician are exposed to. All performance values were obtained from manufacturers’ data sheets or networking standards.

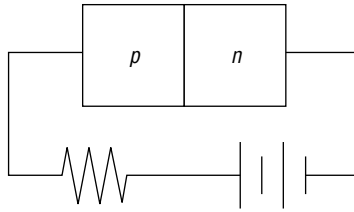
Semiconductor Light Sources

The light sources used in fiber optic communication systems are far different from the light sources used to illuminate your home or office. Fiber optic light sources must be able to turn on and off millions to billions of times per second while projecting a near-microscopic beam of light into an optical fiber. On top of this performance, they must be reasonably priced, highly reliable, easy to use, and available in a small package.

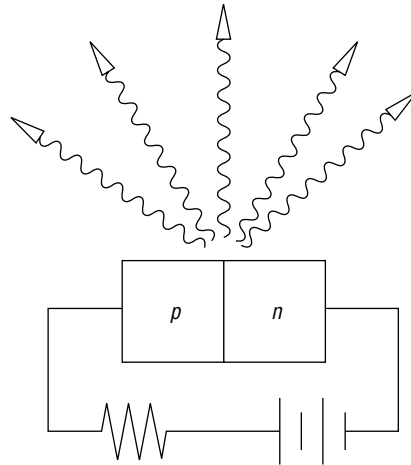
Semiconductor light source technology has made all this possible. Today’s fiber optic communication systems use *light-emitting diodes (LEDs)* and *laser diodes* (from this point forward, the laser diode will be referred to as the *laser*) exclusively. These semiconductor light sources are packaged to support virtually every fiber optic communication system imaginable.

LED Sources

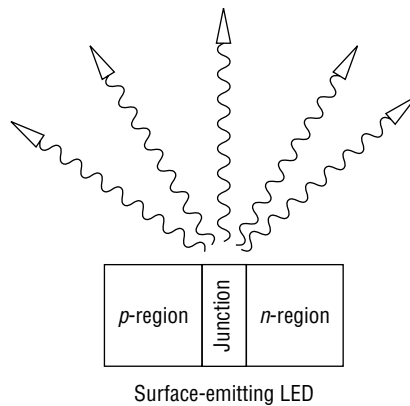
A basic LED light source is a semiconductor diode with a p region and an n region. When the LED is forward biased (a positive voltage is applied to the p region and a negative voltage to the n region), current flows through the LED. As current flows through the LED, the junction where the p and n regions meet emits random photons. This process is referred to as *spontaneous emission*. Figure 10.1 shows a forward-biased LED in a basic electric circuit.

FIGURE 10.1 Forward-biased LED

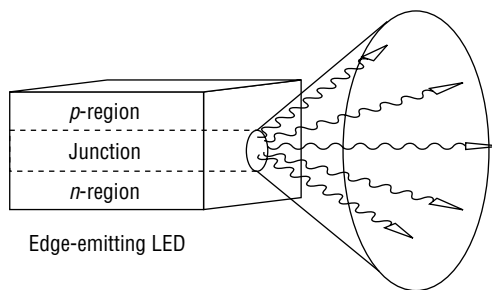
Photons emitted from the junction where the p and n regions meet are not in phase nor launched in the same direction. These out-of-phase photons are called *incoherent light*. This incoherent light cannot be focused so that each photon traverses down the optical fiber. Because of this, only a small percentage of the photons emitted will be coupled into the optical fiber. Figure 10.2 shows the out-of-phase photons being spontaneously emitted from the LED.

FIGURE 10.2 Radiating forward-biased LED

Two types of LEDs, the *surface-emitting LED* and the *edge-emitting LED*, are commonly used in fiber optic communication systems. Surface-emitting LEDs are a *homojunction structure*, which means that a single semiconductor material is used to form the pn junction. Incoherent photons radiate from all points along the pn junction, as shown in Figure 10.3.

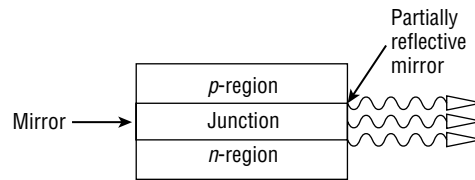
FIGURE 10.3 Radiating surface-emitting LED

Edge-emitting LEDs are a *heterojunction structure*, which means that the pn junction is formed from similar materials with different refractive indices. The different refractive indices are used to guide the light and create a directional output. Light is emitted through an etched opening in the edge of the LED, as shown in Figure 10.4.

FIGURE 10.4 Radiating edge-emitting LED

Laser Sources

The term *laser* is actually an acronym that stands for light amplification by stimulated emission of radiation. Like the LED, the laser is a semiconductor diode with a p region and an n region. Unlike the LED, the laser has an optical cavity that contains the emitted photons with reflecting mirrors on each end of the diode. One of the reflecting mirrors is only partially reflective. This mirror allows some of the photons to escape the optical cavity, as shown in Figure 10.5.

FIGURE 10.5 Radiating laser

Every photon that escapes the optical cavity is a duplicate of the first photon to escape. These photons have the same wavelength, phase relationship, and direction as the first photon. This process of generating light energy is called *stimulated emission*. The photons radiated from the laser have a fixed relationship that is referred to as *coherent light* or coherent radiation. Figure 10.5 shows the in-phase light waves emitted from the laser.

There are three families of lasers used in fiber optic communication systems: *Fabry-Perot*, *distributed feedback (DFB)*, and the *vertical-cavity surface-emitting laser (VCSEL)*. Each laser family has unique performance characteristics that will be discussed in this chapter and is designed to support a specific telecommunication application.

Light Source Performance Characteristics

This section will compare the performance characteristics of the LED and laser light sources. The performance of a light source can be judged in several areas: output pattern, wavelength, spectral width, output power, and modulation speed. These performance areas determine the type of optical fiber that the source can be coupled to, transmission distance, and data rate. Without a doubt, the laser is the hands-down winner when it comes to ultra high-speed long-distance data transmission. However, many applications require only a fraction of the performance that a laser offers; for these applications, an LED is used.

Output Pattern

The LED and laser semiconductors used in fiber optic light sources are packaged to couple as much light as possible into the core of the optical fiber. The output pattern or NA of the light source directly relates to the energy coupled into the core of the optical fiber. The LED has a wide output pattern compared to a laser and does not couple all its light energy into the core of a multimode optical fiber. The output pattern of a laser light source is very narrow, allowing a majority of the light energy to be coupled into the core of a single-mode optical fiber or in some applications a multimode optical fiber.

LED Output Pattern

Unpackaged surface-emitting and edge-emitting LEDs have wide output patterns, as shown earlier in Figures 10.3 and 10.4. An LED light source must be assembled in a package to mate with a specific connector type and optical fiber. However, occasionally there may be a requirement for an LED to be packaged as a *pigtail*. A pigtail is a short length of tight-buffered optical fiber permanently bonded to the light source package, as shown in the photograph in Figure 10.6.

FIGURE 10.6 Photograph of a pigtailed light source



To couple as much light as possible into the core of the optical fiber, the manufacturer typically mounts a micro lens in the shape of a sphere directly on top of the LED. The manufacturer may also use an additional lens to further control the output pattern. Figure 10.7 shows a surface-emitting LED with a series of lenses directing light into the optical fiber.

Even the best lens can't direct all the light energy into the core of the optical fiber. The typical LED light source overfills the optical fiber, allowing light energy to enter the cladding and the core at angles exceeding the NA. These high-order or marginal modes are attenuated over a short distance. Only the light energy that enters the core within the cone of acceptance will propagate to the end of the optical fiber. Figure 10.8 shows a typical LED source overfilling an optical fiber, which is illustrated by the light spot that the source projects.

FIGURE 10.7 Packaged surface-emitting LED

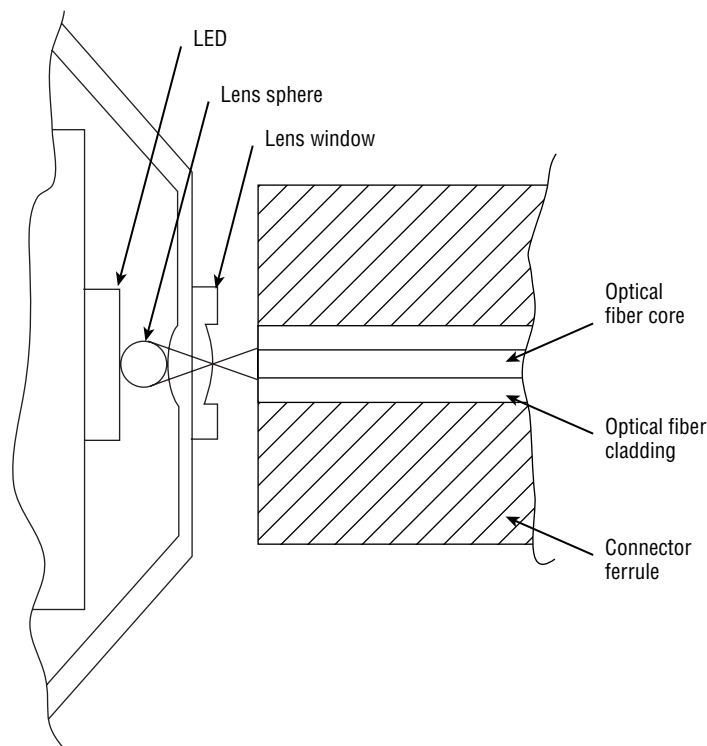
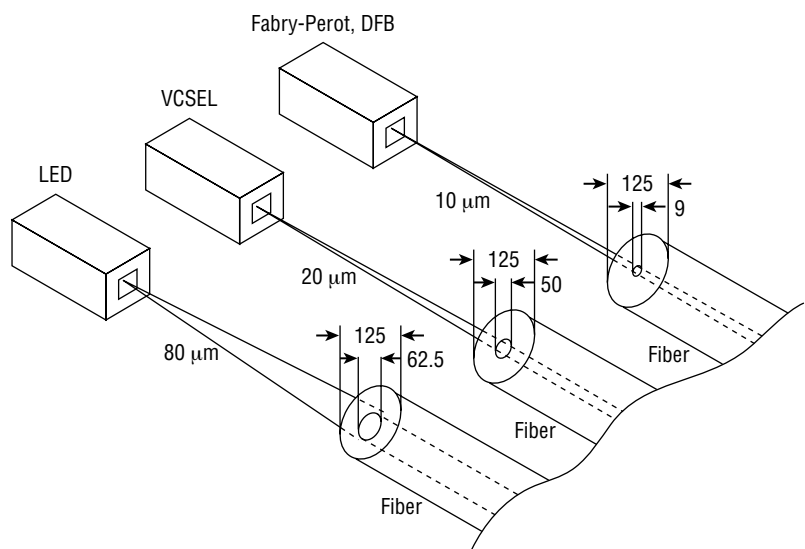


FIGURE 10.8 Laser and LED spot sizes



Laser Output Pattern

Unlike the LED, the laser light source has a narrow output pattern or NA. Like the LED, the laser must be packaged to align the source with the optical fiber and couple as much energy as possible into the core. Lasers can be packaged for a specific connector or bonded to a pigtail.

Laser light sources are designed for either multimode or single-mode applications. VCSELs are currently designed only for multimode optical fiber applications. The VCSEL emits a wider output pattern and larger spot size than the Fabry-Perot or DFB lasers. The output of the VCSEL fills only the center of the core of the multimode optical fiber as shown in Figure 10.8.

Fabry-Perot and DFB lasers are typically designed for single-mode optical fiber. The output pattern and spot size of these lasers are narrower and smaller than the VCSEL. Figure 10.8 shows the typical output pattern of the Fabry-Perot, DFB, and VCSEL laser.

Source Wavelengths

The performance of a fiber optic communication system is dependent on many factors. One key factor is the wavelength of the light source. Short wavelengths (650 nm through 850 nm) have more modes and greater attenuation than longer wavelengths (1300 nm through 1600 nm). The wavelength of the light source can determine system bandwidth and transmission distance.

LED Wavelengths

LED light sources are manufactured from various semiconductor materials. The output wavelength of the LED depends on the semiconductor material that it's manufactured from. Table 10.1 breaks down the semiconductor materials and their associated wavelengths. LED wavelengths are chosen for fiber optic communication systems based on their application. Visible wavelengths (650 nm) are typically used in short-distance low data rate systems with large core diameter optical fiber. These systems are typically found in industrial control systems. Infrared wavelengths (820 nm, 850 nm, and 1300 nm) are typically used for longer distance, higher data rate systems with smaller core optical fiber.

TABLE 10.1 LED Semiconductor Materials

Wavelength	Semiconductor Materials
650 nm	Aluminum (Al), Gallium (Ga), Arsenic (As)
820 nm	Aluminum (Al), Gallium (Ga), Arsenic (As)
850 nm	Aluminum (Al), Gallium (Ga), Arsenic (As)
1300 nm	Indium (In), Gallium (Ga), Arsenic (As), Phosphorus (P)

Laser Wavelengths

Just like the LED, a laser light source output wavelength is dependent on the semiconductor material that it's manufactured from, as shown in Table 10.2. Unlike an LED, the laser can be used for multimode or single-mode applications. VCSELs are designed for gigabit or greater transmission over multimode optical fiber.

TABLE 10.2 Laser Semiconductor Materials

Wavelength	Semiconductor Materials
850 nm	Aluminum (Al), Gallium (Ga), Arsenic (As)
1310 nm	Indium (In), Gallium (Ga), Arsenic (As), Phosphorus (P)
1550 nm	Indium (In), Gallium (Ga), Arsenic (As), Phosphorus (P)

A laser is the only source that can be used with single-mode optical fiber. The wavelength chosen depends on the application. Short wavelengths limit transmission distance due to attenuation. However, many applications do not require long transmission distances. The 850 nm VCSEL is typically used with 50/125 μm or 62.5/125 μm multimode optical fiber in LAN applications.

The Fabry-Perot and DFB lasers can be manufactured to operate at 1310 nm or 1550 nm. Like the VCSEL, the Fabry-Perot laser can be used with multimode optical fiber. At 1310 nm, the Fabry-Perot laser is capable of a 1 Gbps data rate over 500 meters of multimode optical fiber. The 1550 nm Fabry-Perot laser will support Synchronous Optical Network (SONET), OC-48 over 10 km of single-mode optical fiber.



As VCSEL technology improves, it will begin to replace the Fabry-Perot for 1310 nm and 1550 nm applications.

The DFB laser is unique because it can be tuned to a specific wavelength. This allows many lasers to operate around a center wavelength in a multichannel system. Multichannel systems will be discussed in greater detail in Chapter 12, “Passive Components and Multiplexers.”

The DFB laser is virtually a single-wavelength laser and has the least dispersion of laser fiber optic sources. Because of this, the DFB is the choice for long-distance high-speed applications. The 1550 nm DFB laser can support a 2.5 Gbps data rate at a transmission distance of 80 km.

Source Spectral Output

The spectral output of a fiber optic light source can have a significant impact on the bandwidth of a fiber optic communication system. Different wavelengths of light travel at different speeds through an optical fiber. Because of this, a light pulse made up of more than one wavelength will

disperse as each wavelength travels at different velocities down the optical fiber limiting the bandwidth of the system. To achieve the highest bandwidth possible, a fiber optic light source must output a single wavelength. However, the bandwidth requirement for many fiber optic communication systems does not always warrant this narrow spectral output. For those applications, an LED light source is used.

LED Spectral Width

The spectral width of an LED light source is much wider than the spectral width of a laser. LED spectral width is typically described by the term Full Width, Half Maximum (FWHM). When the spectral width of an LED is displayed graphically, FWHM is measured at one-half the maximum intensity across the full width of the curve, as shown in Figure 10.9.

LED spectral widths vary tremendously. A short wavelength 650 nm visible LED may have a spectral width as narrow as 20 nm. The spectral width of a long wavelength 1300 nm infrared LED may be as wide as 170 nm. Because an LED is used only for multimode applications, the wide spectral width has no effect on the bandwidth of the communication system. Figure 10.9 compares the spectral width of various LED wavelengths.

Laser Spectral Width

Laser light sources offer the highest level of performance for a fiber optic communication system. Unlike the LED, lasers output a very narrow spectrum of light. The manufacturer may describe the spectral output width of a laser several ways. These include FWHM, Root Mean Squared (RMS), and 20 dB below peak spectral density.

FIGURE 10.9 Typical FWHM spectral width of 650 nm, 850 nm, and 1300 nm LEDs

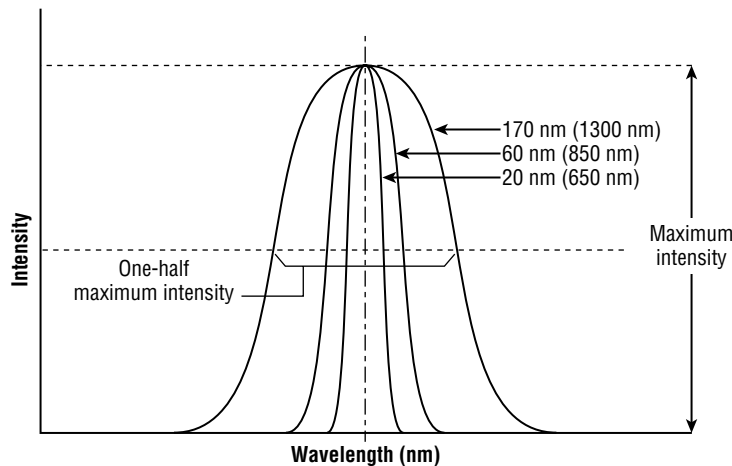
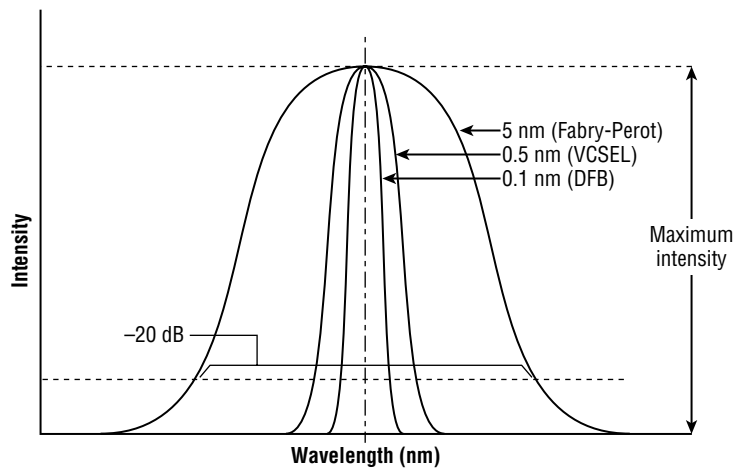


FIGURE 10.10 Typical -20 dB spectral width of Fabry-Perot, VCSEL, and DFB laser

All laser light sources output a narrow spectrum of light. However, there are significant differences in the spectral width of each laser family. The spectral width of the Fabry-Perot laser may span as much as 5 nm, causing dispersion problems in high-speed or long transmission distance systems. VCSELs have shorter cavities than Fabry-Perot lasers, which reduce their spectral output width. The DFB laser is virtually a one-frequency laser with spectral widths less than 0.1 nm.

The DFB laser achieves this narrow spectral output width by using a series of corrugated ridges on the semiconductor substrate to diffract unwanted wavelengths back into the laser cavity. These corrugated ridges form an internal diffraction grating, which is discussed in detail in Chapter 12. The laser manufacturer can set the desired output wavelength with the spacing of the corrugated ridges on the internal diffraction grating. The DFB laser is used in the highest performance systems. Figure 10.10 compares the spectral width of each laser family.

Source Output Power

The output power of the light sources used in fiber optic communication systems varies dramatically depending on the application. LED light sources are typically designed to support transmission distances up to 2 km while laser light sources may support transmission distances in excess of 80 km. Laser optical power output levels can exceed LED optical output power levels by more than 20 dB.

LED Output Power

Today's LED technology seems to be changing daily. High power LED light sources are beginning to replace many incandescent light sources. At this writing, there are single-lamp 1-watt LED white light flashlights available with beams so powerful that viewing them directly is like

viewing a flashbulb. However, the LED light sources used in fiber optic communication output considerably less power than that flashlight.

The optical output power of a typical LED light source used in fiber optic communications is 10 dB or more below the lowest power laser light source. Output power is typically expressed by the manufacturer at the Beginning of Life (BOL) and the End of Life (EOL). An LED light source will lose some output power over its useable lifetime. The industry convention is a degradation of 1.5 dB from BOL to EOL.

LED light sources couple only a small portion of their light energy into the core of the optical fiber. The amount of energy coupled into the optical fiber depends on the optical fiber core size and NA. The larger the core size and greater the NA, the more energy coupled into the core.

Off-the-shelf LED light sources are available to support a wide range of glass, plastic, and PCS optical fiber. Most LED light sources used in telecommunication applications are designed to work with either 50/125 μm or 62.5/125 μm optical fiber. The light energy coupled into the core of a 50/125 μm optical fiber with an NA of 0.20 is approximately 3.5 dB less than the energy coupled into the core of a 62.5/125 μm optical fiber with an NA of 0.275. Because the same LED light source may be used with different size optical fibers, the manufacturer typically measures the output power at the end of one meter of optical fiber with the cladding modes removed.

Laser Output Power

Laser light sources offer the highest output power available for fiber optic communication systems. The higher output power of the laser allows greater transmission distances than with an LED. Laser output power varies depending on the application. VCSELs used in multimode applications have the lowest output power. Fabry-Perot and DFB lasers used in single-mode applications have the greatest output power.

Like the LED, the output power of the laser will diminish over its useable lifetime. Laser manufacturers typically provide BOL and EOL minimum optical power levels. The industry convention for lasers allows for a 1.5 dB reduction in power from BOL to EOL. Optical output power levels are normally expressed as the amount of light coupled into a 1-meter optical fiber.

Source Modulation Speed

There are many advantages to using optical fiber as a communications medium. High bandwidth over long distances is one of them. What limits the bandwidth of today's fiber optic communication systems is not the optical fiber but the light source. The modulation speed of a light source is just one factor that can limit the performance of a fiber optic communication system.

LED Modulation Speed

Today's LED light sources can be modulated to support data rates greater than 400 Mbps. However, most LED light sources are designed to support network standards that do not require a data rate that high. An example of this is the IEEE 802.3 Ethernet standard, which establishes network data rates at 10 Mbps, 100 Mbps, 1 Gbps, and 10 Gbps. The LED can easily support the 10 Mbps and 100 Mbps data. However, current LED technology can't begin to support the 1 Gbps data rate.⁷

Laser Modulation Speed

Laser light sources are constantly evolving. Laser manufacturers have been able to modulate all three-laser families at data rates up to 10 Gbps. However, many applications for laser light sources do not require data rates in the 10 Gbps range. Laser light sources are used in network applications with data rates as low as 10 Mbps when transmission distances greater than 2000 meters are required.

Transmitter Performance Characteristics

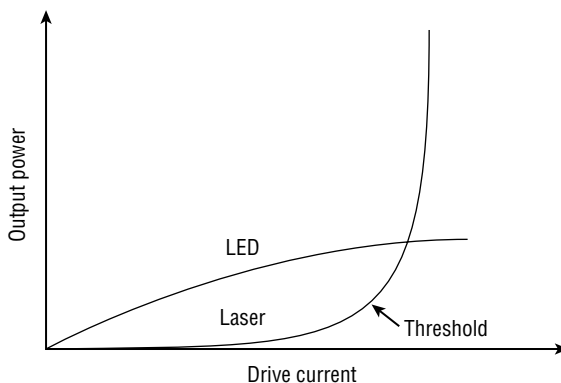
Up to this point, we have discussed only the characteristics of the unpacked and packaged LED and laser light source with no mention of the electronics required to drive these light sources. The incredible data rates discussed in the previous text would not be possible without integrated electronics packaged into the transmitter.

To the fiber optic technician, the LED or laser transmitter is a black box. The manufacturer has neatly integrated everything required to convert the electrical input signal into light energy into the smallest package possible. However, the fiber optic technician must be able to interpret the manufacturer data sheet and verify the optical output of the transmitter.

LED Transmitter Performance Characteristics

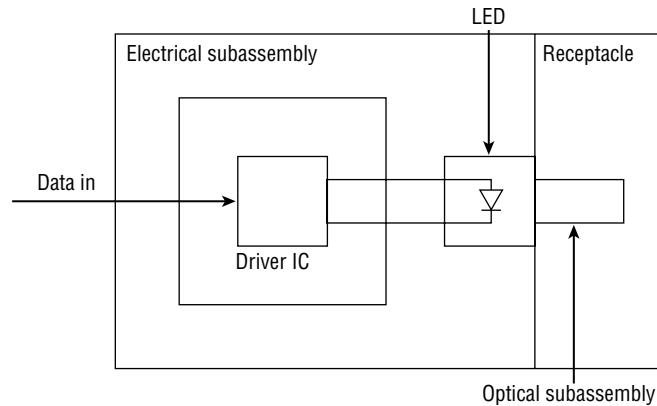
LED transmitters are designed to support multimode optical fibers with core sizes ranging from as small as 50 μm to as large as 1 mm. LED transmitters are directly modulated. (Direct modulation is when the drive current through the LED is varied.) The output power of the LED is directly proportional to the current flow through the LED, as shown in Figure 10.11. In a digital application, the drive current is switched on and off. In an analog application, the drive current is varied.

FIGURE 10.11 LED and laser optical output power versus drive current



A typical LED transmitter contains an electrical subassembly, optical subassembly, and receptacle, as shown in Figure 10.12. The electrical subassembly amplifies the electrical input signal with the driver integrated circuit (IC). The driver IC provides the current to drive the LED in the optical subassembly. The optical subassembly mates with the receptacle to direct light into the optical fiber.

FIGURE 10.12 Block diagram of a typical LED transmitter



Because most fiber optic links are simplex, they require a transmit optical fiber and a receive optical fiber. Typically, the LED transmitter is packaged with the receiver section. Packaging the transmitter and receiver together reduces the overall space required, simplifies circuit board design, and reduces cost. Figure 10.13 is a photograph of a 1300 nm transceiver with a ST receptacle.

FIGURE 10.13 Photograph of a 100 Mbps 1300 nm LED transceiver with ST receptacle



The performance characteristics of the LED transmitter are typically broken up into four groups: recommended operating conditions, electrical characteristics, optical characteristics, and data rate. The recommended operating conditions describe maximum and minimum temperature and voltage ranges that the device can operate in without damage. Table 10.3 shows the typical recommended operating conditions for a 1300 nm 100 Mbps LED transmitter.

TABLE 10.3 LED Transmitter Recommended Operating Conditions

Parameter	Symbol	Min.	Typ.	Max.	Unit
Ambient Operating Temperature	T_A	0		70	°C
Supply Voltage	V_{CC}	4.75		5.25	V
Data Input Voltage—Low	$V_{IL} - V_{CC}$	-1.810		-1.475	V
Data Input Voltage—High	$V_{IH} - V_{CC}$	-1.165		-0.880	V
Data and Signal Detect Output Load	R_L		50		Ω

The electrical characteristics of the LED transmitter describe the supply current requirements, the data input requirements, and the power dissipated by the device. Table 10.4 shows the typical electrical characteristics for a 1300nm 100 Mbps LED transmitter.

The optical characteristics of the LED transmitter at a minimum include output power, center wavelength, and spectral width. Table 10.5 shows the typical optical characteristics for a 1300 nm 100 Mbps LED transmitter.

TABLE 10.4 LED Transmitter Electrical Characteristics

Parameter	Symbol	Min.	Typ.	Max.	Unit
Supply Current	I_{CC}		145	185	mA
Power Dissipation	P_{DISS}		0.76	0.97	W
Data Input Current—Low	I_{IL}	-350	0	-1.475	μA
Data Input Current—High	I_{IH}		14	350	μA

TABLE 10.5 LED Transmitter Optical Characteristics

Parameter	Symbol	Min.	Typ.	Max.	Unit
Optical Output Power: BOL 62.5/125 μm NA = 0.275 Fiber: EOL	P_0	-19 -20	-16.8	-14	dBm avg.
Optical Output Power: BOL 50/125 μm , NA = 0.20 Fiber: EOL	P_0	-22.5 -23.5	-20.3	-14	dBm avg.
Optical Extinction Ratio			0.001 -50	0.03 -35	%dB
Optical Output Power at Logic "0" State	P_0 ("0")			-45	dBm avg.
Center Wavelength	λ_c	1270	1308	1380	nm
Spectral Width—FWHM	$\Delta\lambda$		137	170	nm
Optical Rise Time	t_r	0.6	1.0	3.0	ns
Optical Fall Time	t_f	0.6	2.1	3.0	ns
Duty Cycle Distortion Contrib- uted by the Transmitter	DCD		0.02	0.6	ns p-p
Data-Dependent Jitter Contrib- uted by the Transmitter	DDJ		0.02	0.6	ns p-p

EXERCISE 10.1

Determine the minimum output power for an EOL LED transmitter connected to a 50 μm multimode optical fiber.

- BOL and EOL output power information can only be obtained from the manufacturer's data sheet.
- As mentioned in the text, the performance characteristics of the LED transmitter are typically broken up into four groups: recommended operating conditions, electrical characteristics, optical characteristics, and data rate. BOL and EOL output power information are found under transmitter optical characteristics.
- Table 10.5 contains transmitter optical characteristic information typically found in a manufacturer's data sheet.

EXERCISE 10.1 (continued)

- In the Parameter column of Table 10.5, locate the row that describes EOL optical output power for 50/125 μm optical fiber.
- In the Min column, locate the number that is in line with the EOL optical output power.
- In the Unit column, locate the unit that the optical output power is measured in.
- The minimum output power for an EOL LED transmitter connected to a 50 μm optical fiber is -23.5 dBm.
- This information is required when determining a power budget. Chapter 14, “Fiber Optic System Design Considerations,” will explain how to apply this information to a power budget.

LED transmitters are often designed to support one or more network standards. A 100 Mbps 1300 nm transmitter could support 100 BaseFX or ATM 100 Mbps. Table 10.6 lists the data rates, wavelengths, and transmission distances for various LED transmitters used in IEEE 802.3 Ethernet communication systems.

TABLE 10.6 LED Transmitters for Ethernet Applications

Network	Data Rate	Media	Length
10BaseFL	10 Mb/s	850 nm multimode fiber	6562 feet (2000 m) 62.5/125 μm or 50/125 μm
100BaseFX	100 Mb/s	1300 nm multimode fiber	6562 feet (2000 m) 62.5/125 μm or 50/125 μm
100BaseSX	10/100 Mb/s	850 nm multimode fiber	984 feet (300 m) 62.5/125 μm or 50/125 μm

Laser Transmitter Performance Characteristics

Laser transmitters are designed to support either single-mode optical fiber systems or multi-mode optical fiber systems. The single-mode transmitter is designed to interface with 9/125 μm optical fiber. Until recently, all single-mode transmitters were from the Fabry-Perot or DFB families. However, manufacturers keep improving VCSEL technology and the VCSEL is beginning to emerge as an alternative 1300 nm single-mode optical fiber transmitter.

Like the LED transmitter, the laser transmitter can be broken into three sections: the electrical subassembly, optical subassembly, and receptacle or pigtail, as shown in Figure 10.14. The electrical subassembly amplifies the input signal and provides the drive current for the laser. Unlike the LED, the laser emits very few photons until the drive current passes a threshold level, as shown earlier in Figure 10.11. A feedback circuit that monitors laser output intensity typically controls the drive current provided to the laser to insure a constant output level.



Real World Scenario

Problems Associated with Core Diameter Mismatch

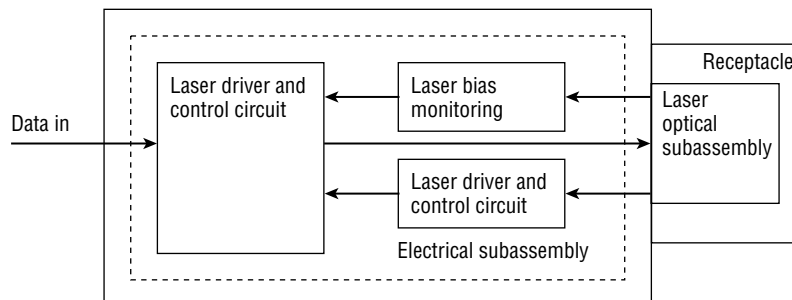
LED transmitters are typically designed to support different core size multimode optical fiber. For example, a 100BaseSX transmitter is designed to operate with 50/125 μm or 62.5/125 μm optical fiber, as shown in Table 10.6. If the LED transmitter receptacle is designed to accept an SC connector, any SC connector regardless of optical fiber size or type can be plugged into the transmitter.

So what happens when a 62.5/125 μm patch cord from the LED transmitter is mated with a 50 μm horizontal cable? Back reflections and attenuation. As you may remember from the “LED Output Power” section, I mentioned that an LED light source couples roughly 3.5 dB more light energy into a 62.5 μm core than a 50 μm core. So when you take the 62.5 μm core from the transmitter and mate it with the 50 μm core horizontal cable, there is roughly a 3.5 dB loss at that connection. As mentioned in Chapter 2, “Principles of Fiber Optic Transmission,” a loss of 3 dB means that half of the power is lost. So if less than half the photons from a light pulse get coupled into the horizontal cable, where do the rest of the photons go?

Well, some of the photons will be reflected back into the cladding and absorbed. However, some of the photons will be reflected back into the core. Many of those photons will travel back to the transmitter where they will be reflected off the transmitter window or lens assembly back into the core. These photons will travel the length of the optical fiber to the receiver. There is a good possibility that the receiver will detect these photons and convert their light energy into electrical energy. Now the circuit decoding the electrical energy is being bombarded with good and bad data that prevents the system from operating.

The fix for this problem is simple: use a 50 μm patch cord. The bottom line is that you should pay careful attention to core diameter to prevent core diameter mismatch. You can’t look at only the color of the patch cord to determine the core diameter. There are orange patch cords with a 62.5 μm core and there are orange patch cords with a 50 μm core. You must read the cable markings from the manufacturer to find out the actual core diameter.

FIGURE 10.14 Block diagram of a typical laser transmitter



The optical subassembly contains a photodiode that receives part of the energy output by the laser. The electrical output of the photodiode is monitored in the electrical subassembly to help control drive current. The optical subassembly may mate with an external modulator, receptacle, or pigtail.

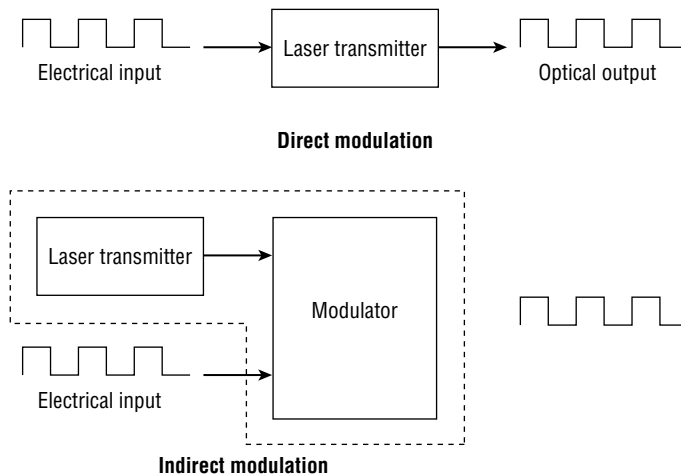
Laser transmitters may or may not be packaged with the receiver section. How the transmitter is packaged depends on the application. Laser transmitters operating up to 2.5 Gbps typically do not require coolers and are often packaged with a receiver section. Laser transmitters operating at 10 Gbps or greater produce enough heat to require an integrated cooler to help the laser maintain a constant temperature. Temperature variations will affect the laser's threshold current, output power, and frequency.

The VCSEL is used in multimode transmitters. Currently VCSEL multimode transmitters support only 850 nm operation with 50/125 μm or 62.5/125 μm optical fiber. The VCSEL can support data rates as high as 2.5 Gbps through multimode optical fiber.

The Fabry-Perot or DFB laser is capable of direct modulation up to 10 Gbps. To achieve data rates greater than 10 Gbps, the laser must be indirectly modulated. Indirect modulation does not vary the drive current as direct modulation does. Maintaining a constant drive current prevents the output wavelength from changing as the electron density in the semiconductor material changes. This is commonly referred to as laser chirp, which causes dispersion and limits bandwidth over long transmission distances.

Indirect modulation requires a constant drive current to the laser. The laser light source outputs a continuous wave (CW) that is modulated by an external device. The external modulation device is typically integrated into the laser transmitter package. Indirect modulation of a laser allows higher modulation rates (up to 40 Gbps) and less dispersion than direct modulation. Figure 10.15 shows a functional block diagram of direct and indirect modulation.

FIGURE 10.15 Block diagram of direct and indirect modulation



The performance characteristics of the laser transmitter are typically broken up into four groups: recommended operating conditions, electrical characteristics, optical characteristics, and data rate. The recommended operating conditions describe the maximum and minimum temperatures and voltage ranges that the device can operate in without damage. Table 10.7 shows the typical recommended operating conditions for a 1300 nm 2.5 Gbps laser transmitter.

TABLE 10.7 Laser Transmitter Recommended Operating Conditions

Parameter	Symbol	Min.	Typ.	Max.	Unit
Ambient Operating Temperature	T_A	0		+70	°C
Supply Voltage	V_{CC}	3.1		3.5	V
Power Supply Rejection	PSR		100		mV _{pp}
Transmitter Differential Input Voltage	V_D	0.3		2.4	V
Data Output Load	R_{DL}		50		Ω
TTL Signal Detect Output Current—Low	I_{OL}			1.0	mA
TTL Signal Detect Output Current—High	I_{OH}	−400			μ mA
Transmit Disable Input Voltage—Low	T_{DIS}			0.6	V
Transmit Disable Input Voltage—High	T_{DIS}	2.2			V
Transmit Disable Assert Time	T_{ASSERT}			10	μ S
Transmit Disable Deassert Time	$T_{DEASSERT}$			50	μ S

The electrical characteristics of the laser transmitter describe the supply current requirements, the data input requirements, and the power dissipated by the device. Table 10.8 shows the typical electrical characteristics for a 1300 nm 2.5 Gbps laser transmitter.

The optical characteristics of the laser transmitter at a minimum include output power, center wavelength, spectral width, and back-reflection sensitivity. Excessive back reflections can interfere with the operation of the laser. Table 10.9 shows the typical optical characteristics for a 1300 nm 2.5 Gbps laser transmitter.

TABLE 10.8 Laser Transmitter Electrical Characteristics

Parameter	Symbol	Min.	Typ.	Max.	Unit
Supply Current	I_{CCR}		115	140	mA
Power Dissipation	P_{DISS}		0.38	0.49	W
Data Output Voltage Swing (single-ended)	$V_{OH} - V_{OL}$	575		930	mV
Data Output Rise Time	t_r		125	150	ps
Data Output Fall Time	t_f		125	150	ps
Signal Detect Output Voltage—Low	V_{OL}			0.8	V
Signal Detect Output Voltage—High	V_{OH}	2.0			V
Signal Detect Assert Time (OFF to ON)	AS_{MAX}			100	μs
Signal Detect Deassert Time (ON to OFF)	ANS_{MAX}			100	μs

TABLE 10.9 Laser Transmitter Optical Characteristics

Parameter	Symbol	Min.	Typ.	Max.	Unit
Optical Output Power 9 μm SMF	P_{OUT}	−10	−6	−3	dBm
Center Wavelength	λ_c	1260		1360	nm
Spectral Width—rms	σ		1.8	4	nm rms
Optical Rise Time	t_r		30	70	ps
Optical Fall Time	t_f		150	225	ps
Extinction Ratio	E_R	8.2	12		dB
Optical Output Eye	Compliant with eye mask Telecordia GR-253-CORE				
Back-Reflection Sensitivity				−8.5	dB
Jitter Generation	pk to pk			70	mUI
	RMS			7	mUI

EXERCISE 10.2

Determine the back-reflection sensitivity for a laser transmitter.

- Back-reflection sensitivity information can be obtained only from the manufacturer’s data sheet.
- As mentioned in the text, the performance characteristics of the laser transmitter are typically broken up into four groups: recommended operating conditions, electrical characteristics, optical characteristics, and data rate. Back-reflection sensitivity information is found under transmitter optical characteristics.
- Table 10.9 contains transmitter optical characteristic information typically found on a manufacturer’s data sheet.
- In the Parameter column of Table 10.9, locate the row that describes back-reflection sensitivity.
- In the Max column, locate the number that is in line with back-reflection sensitivity.
- In the Unit column, locate the unit that back-reflection sensitivity is measured in.
- The maximum back-reflection sensitivity is –8.5 dB.

Laser transmitters can typically support multiple data rate operation. The same transmitter may be used in a 155 Mbps ATM switch or a 2.5 Gbps SONET OC-48 switch. Table 10.10 lists the data rates, wavelengths, and transmission distances for various laser transmitters used in IEEE 802.3 Ethernet communication systems.

TABLE 10.10 Laser Transmitters for Ethernet Applications

Network	Data Rate	Media	Length
100BaseLX	10/100 Mb/s	1300 nm single-mode fiber	9.3 miles (15 km) 9/125 μ m
100BaseLH	10/100 Mb/s	1300 nm single-mode fiber	24.9 miles (40 km) 9/125 μ m
100BaseLH	10/100 Mb/s	1550 nm single-mode fiber	49.7 miles (80 km) 9/125 μ m

Light Source Safety

The light sources used in fiber optic communication systems operate at very low power levels. Unlike an incandescent lamp, their light energy is distributed over a narrow spectrum, typically the infrared spectrum. This often not visible narrow spectrum light energy can be a danger to

the fiber optic installer or fiber optic technician. The level of danger depends on the classification of the light source and the amount of energy coupled into the optical fiber. Laser light sources pose the greatest risk because of their narrow spectral width coherent light.

Classifications

We have learned that fiber optic light sources are very different from the light bulbs used to illuminate our home or office. We shop for light bulbs by their wattage. The wattage shown on the package of the light bulb tells us how much energy the light bulb requires to operate. A 100-watt incandescent light bulb does not output 100 watts of light energy; it requires 100 watts of power to operate. Much of the energy required to operate the bulb is actually emitted as heat, not visible light. We all know how hot a 100-watt incandescent light bulb can get.

That same 100-watt light bulb emits a very broad spectrum of visible light. In other words, the bulb emits many different wavelengths of visible light that combine together to create white light. Each wavelength of light radiating from the bulb represents only a small fraction of the overall light energy output by the bulb. Unlike the laser, the visible light that radiates from the bulb is incoherent. Because the broad visible spectrum of light radiating from the bulb is incoherent and the light energy is spread out over many wavelengths, the bulb will not damage your eye as a laser can. However, this doesn't mean that you should stare at 100-watt light bulbs.

Fiber optic light sources are classified by their ability to damage your eye. LED and laser fiber optic light sources should be classified by the manufacturer. The classification may appear on the device in the form of a label as described in Chapter 6, "Safety," or it may be noted only in the literature about the device.

Most LED fiber optic light sources are Class 1, while laser fiber optic light sources can range from Class 1 to Class 4. Remember that laser fiber optic light sources are infrared and invisible to the naked eye. A laser could damage your eye and you would never know it was happening. The following list shows the light source classifications:

- Class 1: Under normal operating conditions, these lasers do not produce damaging radiation levels.
- Class 2: These lasers present some potential for hazard if viewed directly for long periods of time. However, because of the normal blink response, these lasers do not typically present a hazard.
- Class 3: These medium-power lasers should not be viewed directly.
 - Class 3a: Under normal conditions, these lasers do not produce a hazard when viewed momentarily with the unaided eye. However, these lasers may present a hazard when viewed with collecting optics.
 - Class 3b: Direct viewing or viewing of specular reflections can produce a hazard.
- Class 4: Direct, specular, or diffuse reflections can produce a hazard.

Safety Handling Precautions

There are some fundamental precautions that should always be observed when working with a fiber optic communication system.

- Assume that the fiber optic cable assembly you are handling is energized.
- Never directly view the end of an optical fiber or the end face of a fiber optic connector without verifying that the optical fiber is dark.
- View the end face of a fiber optic connector from at least 6 inches when testing continuity or using a visible fault locator.

Summary

In this chapter, you gained an understanding of LED and laser light sources and transmitters. We looked at how the different performance characteristics of the laser and LED are used in telecommunication networks, industrial control systems, and general-purpose systems. We discussed performance standards that represented commonly available parts for standardized networks.

Exam Essentials

Describe the types of LED light sources used in fiber optic communication and their basic operation. You should know the two types of LEDs used for fiber optic light sources. Make sure you understand the basic theory behind the operation of the LED light sources. Be able to describe how the light is emitted from each LED type and what the phase relationship between the photons is.

Know the types of laser light sources used in fiber optic communication and their basic operation. You should know the three laser families used for fiber optic light sources. Make sure you understand the basic theory of operation of the laser. Be able to describe how light energy is emitted from the laser and the phase relationship between the photons.

List the performance characteristics of an LED light source. Make sure you understand that the unpackaged LED has a wide output pattern. The LED is packaged to direct as much light as possible into the optical fiber. However, the packaged LED still overfills the optical fiber. You need to know the visible and infrared LED wavelengths used in fiber optic light sources. Make sure you can describe the spectral widths of the visible and infrared LED light sources. You should know that the output power of an LED light source decays over time, and know how the manufacturer expresses this. Be able to describe how optical fiber core diameter and NA determine the amount of light energy coupled into the core of the optical fiber. You should know the maximum data rate that an LED light source can transmit.

Know the performance characteristics of a laser light source. Make sure you understand that the laser has a narrow output pattern. You need to know the infrared wavelengths used in each family of laser fiber optic light sources. Make sure you can describe the spectral widths of each family of laser light source. You should know that the output power of a laser light source decays over time, and know how the manufacturer expresses this. You should know the maximum data rate that a laser light source can transmit.

Describe the performance characteristics of an LED transmitter. You should know that the performance characteristics of the LED transmitter are typically broken up into four groups: recommended operating conditions, electrical characteristics, optical characteristics, and data rate. Make sure you understand that recommended operating conditions describe the maximum and minimum temperatures and voltage ranges that the device can operate in without damage. The electrical characteristics of the LED transmitter describe the supply current requirements, the data input requirements, and the power dissipated by the device. The optical characteristics of the LED transmitter at a minimum include output power, center wavelength, and spectral width. Make sure you understand that LED transmitters are typically designed to support network standards and rarely support data rates above 155 Mbps.

Know the performance characteristics of a laser transmitter. Remember that only laser transmitters are designed to support either single-mode optical fiber systems or multimode optical fiber systems. Make sure you understand that the VCSEL is currently only used in multimode transmitters supporting 850 nm operation with 50/125 μm or 62.5/125 μm optical fiber. You should know that the direct modulation of a laser can support data rates up to 10 Gbps; data rates greater than 10 Gbps are achieved with indirect modulation. You should understand that indirect modulation does not vary the drive current as direct modulation does, which prevents the output wavelength from changing. You should remember that indirect modulation of a laser allows higher modulation rates (up to 40 Gbps) and less dispersion than direct modulation. You need to know that the performance characteristics of the laser transmitter are typically broken up into four groups: recommended operating conditions, electrical characteristics, optical characteristics, and data rate. You need to know that excessive back reflections can interfere with the operation of the laser. You should know that laser transmitters have data transmission rates from 100 Mbps to as high as 40 Gbps.

Explain the safety classifications of the light sources used in fiber optic communication.

You should know the safety classifications for fiber optic light sources, which classification is eye safe, and how to view light sources that are not eye safe. You must also know the fundamental safety precautions that should be observed when working around a fiber optic communication system.

Review Questions

1. LEDs emit photons that are out of phase; this is referred to as _____ light.
 - A. Coherent
 - B. Incoherent
 - C. Stimulated
 - D. Spontaneous
2. LEDs emit random photos through a process called _____ emission.
 - A. Coherent
 - B. Incoherent
 - C. Stimulated
 - D. Spontaneous
3. Lasers emit photons that are in phase through a process called _____ emission.
 - A. Coherent
 - B. Incoherent
 - C. Stimulated
 - D. Spontaneous
4. The photons emitted from a laser have the same wavelength and are in phase; this is referred to as _____ light.
 - A. Coherent
 - B. Incoherent
 - C. Stimulated
 - D. Spontaneous
5. LEDs have a _____ output pattern when compared to the output pattern of a laser.
 - A. Narrow
 - B. Wide
 - C. Coherent
 - D. Stimulated
6. LEDs with visible wavelengths are typically used for _____-data rate _____-distance fiber optic communication systems.
 - A. High, long
 - B. Low, long
 - C. High, short
 - D. Low, short

7. Long wavelength 1300 nm LEDs offer _____ bandwidth over _____ transmission distances than short wavelength 850 nm LEDs.
 - A. Higher, longer
 - B. Lower, longer
 - C. Higher, shorter
 - D. Lower, shorter
8. VCSELs are currently available at a _____ wavelength.
 - A. 1550 nm
 - B. 1310 nm
 - C. 850 nm
 - D. 820 nm
9. The spectral width of a laser is _____ than the spectral width of an LED.
 - A. Wider
 - B. Narrower
 - C. Shorter
 - D. Longer
10. The optical output power of a laser is _____ than the optical output power of an LED.
 - A. Wider
 - B. Greater
 - C. Narrower
 - D. Less
11. The amount of light energy coupled into the core of an optical fiber depends on the _____ size and _____ of the optical fiber.
 - A. Cladding, NA
 - B. Cable, NA
 - C. Core, NA
 - D. Core, cable
12. LED radiation should not degrade more than _____ over its lifetime.
 - A. 0.5 dB
 - B. 1.0 dB
 - C. 1.5 dB
 - D. 3.0 dB

13. Laser light sources have a _____ modulation speed than LEDs.
- A. Wider
 - B. Slower
 - C. Narrower
 - D. Higher
14. LED transmitters are designed to support only _____ modulation.
- A. Indirect
 - B. Phase
 - C. Wavelength
 - D. Direct
15. The center wavelength of an LED transmitter would be found under the _____ characteristics of the LED transmitter data sheet.
- A. Operating
 - B. Electrical
 - C. Optical
 - D. Temperature
16. _____ modulation of a laser light source allows the highest modulation rates.
- A. Indirect
 - B. Wavelength
 - C. Direct
 - D. Phase
17. The _____ is the laser primarily used in multimode transmitters.
- A. DFB
 - B. VCSEL
 - C. Fabry-Perot
 - D. LED
18. The back-reflection sensitivity of a laser transmitter would be found under the _____ characteristics of the data sheet.
- A. Optical
 - B. Electrical
 - C. Mechanical
 - D. Physical

19. Only a _____ laser is considered eye safe.
- A. Class 1
 - B. Class 2
 - C. Class 3
 - D. Class 4
20. The end of an optical fiber _____ be viewed directly without verifying that the optical fiber is dark.
- A. May always
 - B. Should never
 - C. May sometimes
 - D. Must always

Answers to Review Questions

1. B. The photons emitted from the junction where the p and n regions meet do not have a fixed relationship; they are out of phase. These out-of-phase light waves are called incoherent light.
2. D. As current flows through the LED, the junction where the p and n regions meet emits random photons. This process is referred to as spontaneous emission.
3. C. Every photon that escapes the optical cavity is a duplicate of the first photon to escape. These photons have the same wavelength, phase relationship, and direction as the first photon. This process of generating light energy is called stimulated emission.
4. A. Every photon that escapes the optical cavity is a duplicate of the first photon to escape. These photons have the same wavelength, phase relationship, and direction as the first photon. This fixed relationship is referred to as coherent light.
5. B. LEDs have wide output patterns and typically use a micro lens within a package to direct as much light as possible into the optical fiber.
6. D. Visible wavelengths attenuate more in the optical fiber, reducing transmission distances. Short wavelengths have more modes that increase modal dispersion and reduce bandwidth.
7. A. Infrared 1300 nm wavelengths attenuate less than 850 nm wavelengths in the optical fiber, increasing transmission distances. Longer wavelengths have fewer modes and less modal dispersion increasing bandwidth.
8. C. At this printing, VCSEL technology supports only a wavelength of 850 nm. However, 1300 nm wavelength VCSELs will be available in the near future.
9. B. LED spectral widths range from 20 nm to 170 nm. Laser spectral widths range from 5 nm to less than 0.1 nm.
10. B. LEDs output less optical power than lasers. Optical output power is the amount of light energy coupled into the core of the fiber. LED optical power output levels vary depending on the core size and NA of the optical fiber.
11. C. The amount of energy coupled into the optical fiber depends on the optical fiber core size and NA. The larger the core size and greater the NA, the more energy coupled.
12. C. Output power is typically expressed by the manufacturer at the Beginning of Life (BOL) and the End of Life (EOL). An LED light source will lose some output power over its useable lifetime. The industry convention is a degradation of 1.5 dB from BOL to EOL.
13. D. Laser light sources may have modulation speeds as great as 10 Gbps in a network application. LED modulation speeds in a network application typically do not exceed 155 Mbps.
14. D. LED transmitters are directly modulated. Direct modulation is when the drive current through the LED is varied. The output power of the LED is directly proportional to the current flow through the LED.

- 15. C. The optical characteristics of the LED transmitter at a minimum include output power, center wavelength, and spectral width.
- 16. A. Indirect modulation of a laser allows higher modulation rates (up to 40 Gbps) and less dispersion than direct modulation.
- 17. B. Current VCSEL multimode transmitters support only 850 nm operation with 50/125 μm or 62.5 μm optical fiber. The VCSEL can support data rates through multimode optical fiber up to 2.5 Gbps.
- 18. A. The optical characteristics of the laser transmitter at a minimum include output power, center wavelength, spectral width, and back-reflection sensitivity. Excessive back reflections can interfere with the operation of the laser.
- 19. A. Under normal operating conditions, Class 1 lasers do not producing damaging radiation levels.
- 20. B. Never directly view the end of an optical fiber or the end face of a fiber optic connector without verifying that the optical fiber is dark.



Chapter

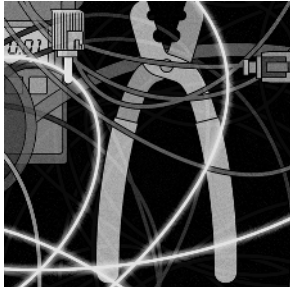
11

Fiber Optic Detectors and Receivers

OBJECTIVES COVERED IN THIS CHAPTER:

✓ Fiber Optic Detectors and Receivers

- Describe the basic operation of a photodiode.
- Describe the operation of a PIN photodiode.
- Describe the operation of an Avalanche Photodiode (APD).
- Describe the performance characteristics of commonly used photodiodes.
- Describe the basic components in a fiber optic receiver.
- Describe receiver performance characteristics.



In Chapter 1, “History of Fiber Optics,” we introduced you to the fiber optic receiver. The job of the receiver is to take light energy from the optical fiber and convert it to electrical energy. In this chapter, we will explain the basic components that make up the fiber optic receiver, starting with the photodiode. You will learn about the effects of optical input power on the performance of the receiver and the performance characteristics of LED and laser receivers.

Photodiode Fundamentals

A *photodiode* in a fiber optic receiver is like the tire on your car. The photodiode is where the rubber meets the road. Light energy from the optical fiber stops at the photodiode. It’s the job of the photodiode to convert the light energy received from the optical fiber into electrical energy. There are different performance-level tires that you can put on your car and there are different performance-level photodiodes that can be incorporated into a receiver. This section of the chapter discusses the fundamentals of basic photodiode operation and the different types of photodiodes that may be used in a receiver.

The best way to imagine a photodiode is to think about a *solar cell*. We have all seen the exhibits at museums where a solar cell or a group of solar cells powers a small boat or car with the light energy from a light bulb. Maybe you own a solar charger for your boat battery or have decorative outdoor lighting that uses solar cells to recharge the batteries.

The solar cell takes the light energy it receives and converts it into electrical energy. In other words, the photons absorbed by the solar cell cause electrons to flow within the solar cell. This electron flow is called a *current*. The current from the solar cell flowing through the motor of the small boat or car causes the motor to rotate. The more current, the faster the motor rotates and the faster the boat or car travels.

Like the LED we learned about in Chapter 10, “Fiber Optic Light Sources,” the basic photodiode is a semiconductor diode with a *p* region and an *n* region, as shown in Figure 11.1. Photons absorbed by the photodiode excite electrons within the photodiode in a process called *intrinsic absorption*. When stimulated with an outside bias voltage, these electrons produce a current flow through the photodiode and the external circuit providing the bias voltage.

The PN photodiode is *reverse biased* when used in an electrical circuit. This is the opposite of how the LED is used in an electrical circuit. Reverse bias means that the *n* region of the photodiode is connected to a positive electrical potential and the *p* region is connected to a negative electrical potential.

In an electrical circuit, as shown in Figure 11.2, light that is absorbed by the photodiode produces current flow through the entire external circuit. As current flows through the *resistor*, it produces a voltage drop across the resistor. This voltage drop is input to an amplifier for amplification.

You may be wondering why the output of the photodiode needs to be amplified, since there is no amplification used with a solar cell. The solar cell is typically supplied photons by a powerful light source such as the sun or a very bright lamp. The photodiode used in a fiber optic receiver gets all of its light energy from the optical fiber connected to it. As we have learned in this book, the core of an optical fiber is extremely small and carries very little light energy in comparison to the energy that a solar cell receives. A photodiode that is stimulated by the light energy in an optical fiber does not produce a great amount of electrical current flow. This is why amplification is required. It's also the reason why different photodiodes have been developed for various fiber optic receiver applications.

FIGURE 11.1 PN photodiode

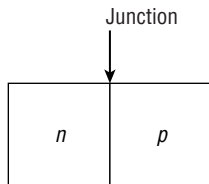
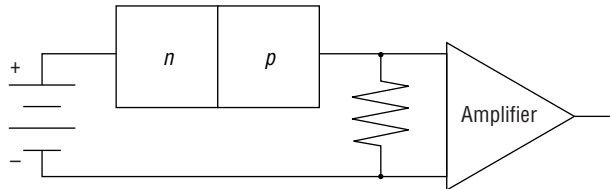


FIGURE 11.2 PN photodiode in an electrical circuit



PIN Photodiode

The PIN photodiode works like a PN photodiode; however, it is manufactured to offer better performance. The better performance comes in the form of improved efficiency and greater speed. Improved efficiency means that it has a better photon-to-carrier conversion ratio. If the same amount of light energy hit a PN photodiode and a PIN photodiode, the PIN photodiode would generate more current flow through an external circuit.

Greater speed means that the diode can turn on and off faster. Remember that in fiber optics the light pulses being sent by the transmitter happen at a very fast rate. The photodiode needs to be able to stop and start electron flow fast enough to keep up with the incoming light pulses.

The PIN photodiode shown in Figure 11.3 is constructed a little differently than the PN photodiode. An intrinsic layer is used to separate the p region and the n region. This creates a large depletion region that absorbs the photons with improved efficiency when compared to the PN photodiode.

FIGURE 11.3 PIN photodiode

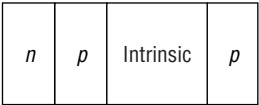


Avalanche Photodiode

The *avalanche photodiode (APD)* works just as its name suggests. On a snow-covered mountain, a small vibration can trigger an avalanche of snow. With the APD, a small bundle of photons can trigger an avalanche of electrons. The APD accomplishes this through a process called *photomultiplication*.

The APD is constructed with one more p region than the PIN photodiode, as shown in Figure 11.4. When the APD is biased very close to its breakdown voltage, it acts like an amplifier with a multiplication factor, or *gain*. An APD with a multiplication factor of 50 sets free on the average 50 electrons for each photon absorbed. The free electrons produce current flow through the electrical circuit connected to the APD.

FIGURE 11.4 Avalanche photodiode



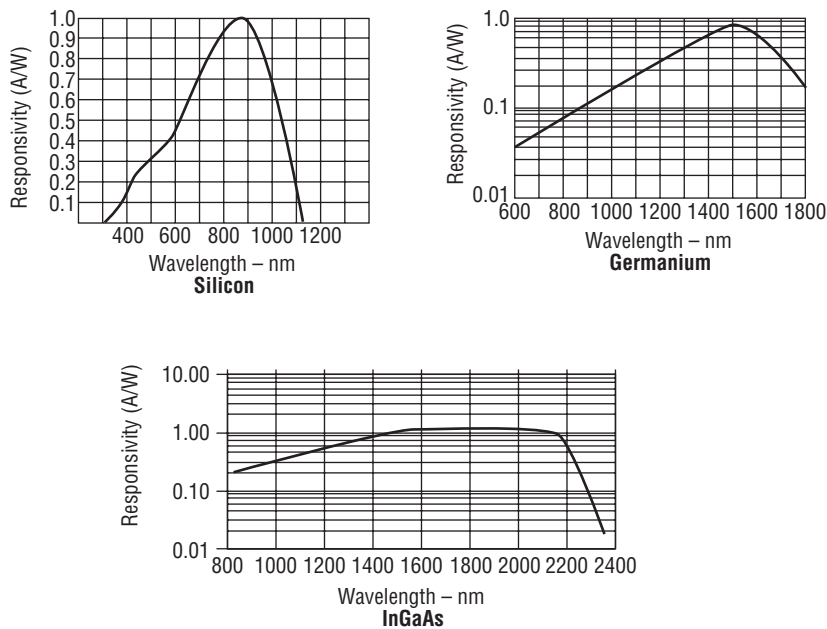
Responsivity

The *responsivity* of the photodiode describes how well the photodiode converts a wavelength or a range of wavelengths of optical energy into electrical current. It's the ratio of the photodiode electrical output current to its optical input power. The greater the responsivity, the greater the electrical current output for a given amount of optical input power.

Responsivity is described in amperes/watt (A/W). A photodiode in a fiber optic receiver will never generate an ampere of electrical current. That's not to say that a photodiode couldn't be built to generate an ampere of electrical current. Remember that a photodiode in a fiber optic receiver receives a very small amount of light energy. A typical receiver works well with an optical input power as low as one microwatt. One microwatt is one millionth of a watt.

The overall responsivity of a photodiode depends on three factors: semiconductor material makeup, wavelength, and diode construction. Photodiodes are constructed for specific wavelengths. Some semiconductor materials perform better at longer wavelengths and some perform better at shorter wavelengths. Silicon photodiodes perform best in the visible and short infrared wavelengths. Germanium and indium gallium arsenic (InGaAs) photodiodes perform best at long infrared wavelengths, as shown in Figure 11.5. Diode construction also plays a large role in responsivity. The responsivity of an APD photodiode may be 100 or more times greater than a PIN photodiode.

FIGURE 11.5 Photodiode semiconductor responsivity



Quantum Efficiency

The responsivity of a photodiode depends on its quantum efficiency. Quantum efficiency describes how efficiently a photodiode converts light energy into electrical energy—that is, photons into free electrons. It is typically expressed as a percentage. A quantum efficiency of 47 percent means that for every 100 photons absorbed, 47 current-generating electrons will be created.

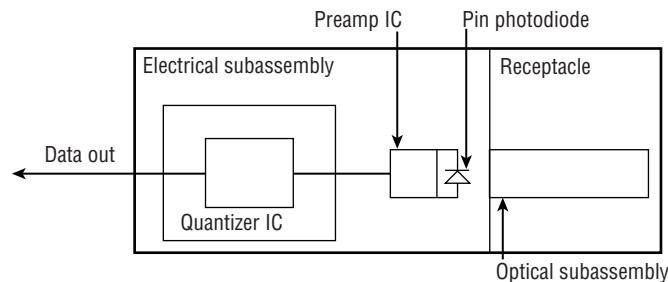
Switching Speed

It has been stressed many times in this book that fiber optic systems transmit and receive millions or billions of light pulses per second. When data is being moved at this rate, there is very little time for the photodiode to switch on and off. Switching speed depends on physical size, construction, and electrical biasing. Photodiodes in fiber optic receivers are very small in size and biased to produce the fastest possible switching time.

Fiber Optic Receiver

Many different fiber optic receiver designs are in use today. The complexity of these receivers varies by application. The job of the receiver is to take the light energy from the optical fiber and convert it into electrical energy. The output of the receiver is designed to interface with electronics that handle the information after the light-to-electricity conversion. A typical receiver can be broken into three subassemblies: electrical subassembly, optical subassembly, and receptacle, as shown in Figure 11.6.

FIGURE 11.6 Block diagram of a typical LED receiver

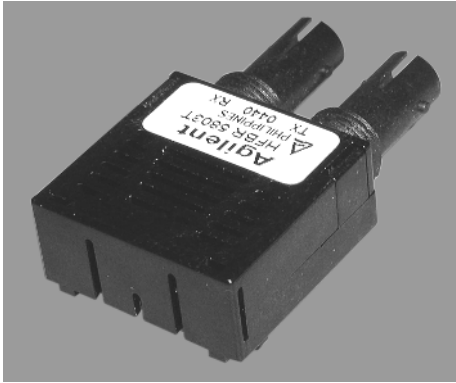
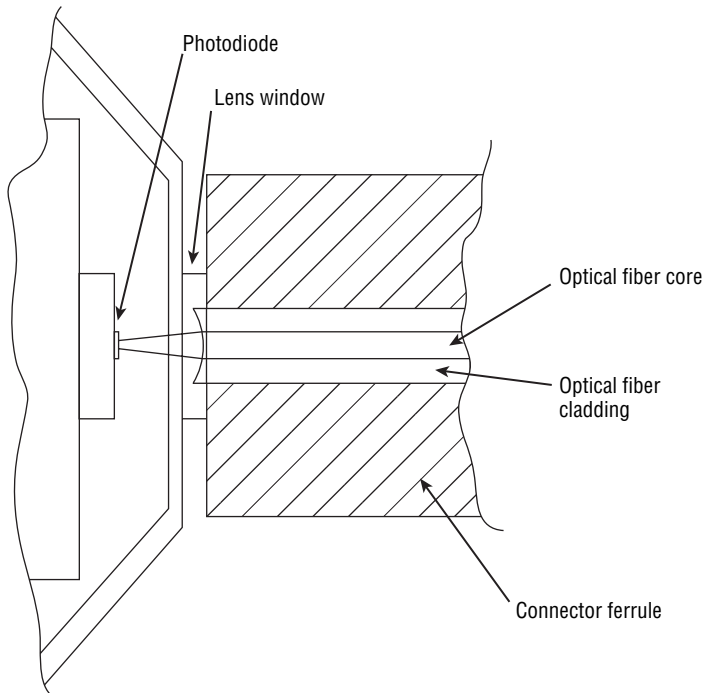


Receptacle

The *receptacle* is the part of the fiber optic receiver that accepts the connector. It also aligns the ferrule so that the optical fiber within the ferrule is perpendicular with the window edge in the optical assembly. Receiver or transceiver modules are manufactured to support a variety of connector types. The transceiver module in Figure 11.7 is designed to accept an ST connector.

Optical Subassembly

The *optical subassembly* guides the light energy from the optical fiber to the photodiode. A window in the optical subassembly makes contact with the optical fiber endface. This window may be shaped like a lens to focus the light energy onto the photodiode, as shown in Figure 11.8, or a lens may be placed between the window and the photodiode.

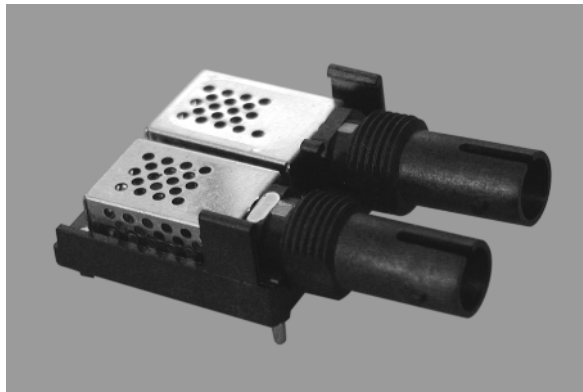
FIGURE 11.7 Transceiver with an ST receptacle**FIGURE 11.8** Photodiode and window

In many receivers, the photodiode and preamplifier are housed in the optical subassembly. Sometimes the optical subassembly is referred to as the photodiode preamplifier subassembly. The outer housing of the receiver or transceiver typically houses the optical subassembly. Typically a metal shield surrounds the optical subassembly to reduce interference from external EMI fields.

Electrical Subassembly

The electrical subassembly is typically built on a multilayer printed circuit board. This printed circuit contains the *quantizer IC* and other passive components required to complete the electrical circuit. The photodiode and preamplifier contained in the optical subassembly are electrically connected to the printed circuit board. A metal shield is typically placed over the electrical subassembly to reduce interference that may be caused by external EMI fields, as shown in Figure 11.9.

FIGURE 11.9 Transceiver module with EMI shield exposed



The photodiode converts the light pulses from the optical fiber to electrical current pulses, as discussed earlier in this chapter. The *transimpedance amplifier*, or preamplifier, amplifies the electrical current pulses from the photodiode and outputs voltage pulses to the quantizer IC. The *limiting amplifier* in the quantizer IC amplifies the voltage pulses and provides a binary decision. It determines whether the electrical pulses received represent a binary 1 or a binary 0.

The quantizer IC also measures the received optical energy. It sets the signal detect line when there is adequate signal strength to convert the light energy into electrical energy. This prevents the electronics from trying to decode a weak signal or noise.

Receiver Performance Characteristics

This section examines several key performance characteristics of a fiber optic receiver. It generalizes *dynamic range* and *operating wavelength*. The specific performance characteristics of LED and laser receivers are covered in detail. The data used in this section was extracted from manufacturer data sheets and represents typical performance characteristics of readily available fiber optic receivers.

Dynamic Range

Fiber optic receivers are limited in the amount of optical input power they can receive. Too much optical input power will saturate the photodiode. Saturating the photodiode prevents the photodiode from turning off after the light pulse has been absorbed. This can cause electrical output pulses of the receiver to overlap, creating a *bit error*.

The dynamic range of the receiver is measured in decibels. It is the difference between the maximum and minimum optical input power that the receiver can accept. If the maximum optical input power was -14 dBm and the minimum optical input power was -32 dBm, the dynamic range would be 18 dB.

The receiver will generate a minimum number of errors when the optical input power is kept within the minimum and maximum values. All receivers generate errors. Error generation is typically described by the receiver's *bit error rate*, or BER. A typical receiver may have a BER of one error in a billion to one error in a trillion. Typically this would be written as 10^{-9} or 10^{-12} , respectively.

Operating Wavelength

Fiber optic receivers are designed to operate within a range of wavelengths. A typically 1300 nm receiver may have an operating wavelength range from 1270 nm to 1380 nm. This is because fiber optic transmitters output optical energy within a wavelength range. The receiver must be able to accept a 1300 nm transmitter that outputs 1275 nm or 1375 nm. A receiver designed for 1300 nm may not perform well or not perform at all when connected to an 850 nm or 1550 nm transmitter.

LED Receiver Performance Characteristics

The performance characteristics of the LED receiver are typically broken up into four groups: recommended operating conditions, electrical characteristics, optical characteristics, and data rate. The recommended operating conditions describe maximum and minimum temperature and voltage ranges that the device can operate in without damage. Table 11.1 shows the typical recommended operating conditions for a 1300 nm 100 Mbps LED receiver.

TABLE 11.1 LED Receiver Recommended Operating Conditions

Parameter	Symbol	Min.	Typ.	Max.	Unit
Ambient operating temperature	T_A	0		70	$^{\circ}\text{C}$
Supply voltage	V_{CC}	4.75		5.25	V
Data input voltage—low	$V_{IL} - V_{CC}$	-1.810		-1.475	V

TABLE 11.1 LED Receiver Recommended Operating Conditions *(continued)*

Parameter	Symbol	Min.	Typ.	Max.	Unit
Data input voltage—high	$V_{IH} - V_{CC}$	-1.165		-0.880	V
Data and signal detect output load	R_L		50		Ω

The electrical characteristics of the LED receiver describe the supply current requirements, data output voltages, signal detect output voltages, rise/fall times, and power dissipated by the device. Table 11.2 shows the typical electrical characteristics for a 1300 nm 100 Mbps LED receiver.

TABLE 11.2 LED Receiver Electrical Characteristics

Parameter	Symbol	Min.	Typ.	Max.	Unit
Supply current	I_{CC}		82	145	mA
Power dissipation	P_{DISS}		0.3	0.5	W
Data output voltage—low	$V_{OL} - V_{CC}$	-1.840		-1.620	V
Data output voltage—high	$V_{OH} - V_{CC}$	-1.045		-0.880	V
Data output rise time	t_r	0.35		2.2	ns
Data output fall time	t_f	0.35		2.2	ns
Signal detect output voltage—low	$V_{OL} - V_{CC}$	-1.840		-1.620	V
Signal detect output voltage—high	$V_{OH} - V_{CC}$	-1.045		-0.880	V
Signal detect output rise time	t_r	0.35		2.2	ns
Signal detect output fall time	t_f	0.35		2.2	ns

The optical characteristics of the LED receiver at a minimum include minimum optical input power, maximum optical input power, and operating wavelength. Table 11.3 shows the typical optical characteristics for a 1300 nm 100 Mbps LED receiver.

TABLE 11.3 Receiver Optical Characteristics

Parameter	Symbol	Min.	Typ.	Max.	Unit
Optical input power minimum at window edge	$P_{IN \text{ Min.}}$ (W)		-33.5	-31	dBm avg.
Optical input power minimum at eye center	$P_{IN \text{ Min.}}$ (C)		-34.5	-31.8	dBm avg.
Optical input power maximum	$P_{IN \text{ Max.}}$	-14	-11.8		dBm avg.
Operating wavelength	λ	1270		1380	nm

EXERCISE 11.1**Determine optical input power dynamic range for an LED receiver.**

- The optical input power dynamic range for the LED receiver described above is the difference between the minimum value for the maximum optical input power and the maximum value for the minimum optical input power. For this exercise, use Table 11.3. Remember that this information should be obtained from the manufacturer data sheet.
- The minimum value for the maximum optical input power is -14 dBm and the maximum value for the minimum optical input is -31.8 dBm.
- The difference between -14 dBm and -31.8 dBm is 17.8 dB.
- The optical input power dynamic range for the LED receiver is 17.8 dB.

LED receivers are typically designed to support one or more network standards. A 100 Mbps 1300 nm receiver could support 100 BaseFX or ATM 100 Mbps. Table 11.4 lists the data rates, wavelengths, and transmission distances for various LED receivers used in IEEE 802.3 Ethernet communication systems.

TABLE 11.4 LED Receivers for Ethernet Applications

Network	Data Rate	Media	Length
10BaseFL	10 Mb/s	850 nm multimode fiber	6562 feet (2000 m) 62.5/125 μm or 50/125 μm
100BaseFX	100 Mb/s	1300 nm multimode fiber	6562 feet (2000 m) 62.5/125 μm or 50/125 μm

TABLE 11.4 LED Receivers for Ethernet Applications *(continued)*

Network	Data Rate	Media	Length
100BaseSX	10/100 Mb/s	850 nm multimode fiber	984 feet (300 m) 62.5/125 μm or 50/125 μm

Laser Receiver Performance Characteristics

The performance characteristics of the laser receiver, like the LED receiver's, are typically broken up into four groups: recommended operating conditions, electrical characteristics, optical characteristics, and data rate. The recommended operating conditions describe maximum and minimum temperature and voltage ranges that the device can operate in without damage. Table 11.5 shows the typical recommended operating conditions for a 1300 nm 2.5 Gbps laser transmitter.

TABLE 11.5 Laser Receiver Recommended Operating Conditions

Parameter	Symbol	Min.	Typ.	Max.	Unit
Ambient operating temperature	T_A	0		+70	$^{\circ}\text{C}$
Supply voltage	V_{CC}	3.1		3.5	V
Power supply rejection	PSR		100		mV_{PP}
Transmitter differential input voltage	V_D	0.3		2.4	V
Data output load	R_{OL}		50		Ω
TTL signal detect output current—low	I_{OL}			1.0	mA
TTL signal detect output current—high	I_{OH}	−400			μa
Transmit disable input voltage—low	T_{DIS}			0.6	V
Transmit disable input voltage—high	T_{DIS}	2.2			V

The electrical characteristics of the laser receiver describe the supply current requirements, data output characteristics, and power dissipated by the device. Table 11.6 shows the typical electrical characteristics for a 1300 nm 2.5 Gbps laser receiver.

TABLE 11.6 Laser Receiver Electrical Characteristics

Parameter	Symbol	Min.	Typ.	Max.	Unit
Supply current	I_{CC}		115	140	mA
Power dissipation	P_{DISS}		0.38	0.49	W
Data output voltage swing (single-ended)	$V_{OH} - V_{OL}$	575		930	mV
Data output rise time	t_r		125	150	ps
Data output fall time	t_f		125	150	ps
Signal detect output voltage—low	V_{OL}			0.8	V
Signal detect output voltage—high	V_{OH}	2.0			V
Signal detect assert time (OFF to ON)	AS_{MAX}			100	μ S
Signal detect deassert time (ON to OFF)	ANS_{MAX}			100	μ S

The optical characteristics of the laser receiver at a minimum include output power, center wavelength, spectral width, and back reflection sensitivity. Excessive back reflections can interfere with the operation of the laser. Table 11.7 shows the typical optical characteristics for a 1300 nm 2.5 Gbps laser receiver.

TABLE 11.7 Laser Receiver Optical Characteristics

Parameter	Symbol	Min.	Typ.	Max.	Unit
Receiver sensitivity	$P_{IN\ MIN}$		−23	−19	dBm avg.
Receiver overload	$P_{IN\ MAX}$	−3	+1		dBm avg.
Input operating wavelength	λ	1260		1570	nm
Signal detect—asserted	P_A		−27.3	−19.5	dBm avg.
Signal detect—deasserted	P_D	−35	−28.7		dBm avg.

TABLE 11.7 Laser Receiver Optical Characteristics (*continued*)

Parameter	Symbol	Min.	Typ.	Max.	Unit
Signal detect—hysteresis	P_H	0.5	1.4	4	dB
Reflectance			-35	-27	dB

EXERCISE 11.2**Determine the optical input power dynamic range for a laser receiver.**

- The optical input power dynamic range for the laser receiver described above is the difference between the minimum value for receiver overload and the maximum value for receiver sensitivity. For this exercise, use Table 11.7. Remember that this information should be obtained from the manufacturer data sheet.
- The minimum value for receiver overload is -3 dBm and the maximum value for receiver sensitivity is -19 dBm.
- The difference between -3 dBm and -19 dBm is 16 dB.
- The optical input power dynamic range for the laser receiver is 14 dB.

Laser transmitters can typically support multiple data rate operation. The same transmitter may be used in a 155 Mbps ATM switch or a 2.5 Gbps SONET OC-48 switch. Table 11.8 lists the data rates, wavelengths, and transmission distances for various laser transmitters used in IEEE 802.3 Ethernet communication systems.

TABLE 11.8 Laser Receivers for Ethernet Applications

Network	Data Rate	Media	Length
100BaseLX	10/100 Mb/s	1300 nm single-mode fiber	9.3 miles (15 km) 9/125 μm
100BaseLH	10/100 Mb/s	1300 nm single-mode fiber	24.9 miles (40 km) 9/125 μm
100BaseLH	10/100 Mb/s	1550 nm single-mode fiber	49.7 miles (80 km) 9/125 μm



Real World Scenario

Is it the Computer Hardware or Software?

Not too long ago, a coworker and I were troubleshooting a communication problem between a piece rack-mounted computing equipment and a router. The equipment and the router were communicating over 50/125 μm multimode optical fiber. Everything would work for a while, and then the equipment and router would stop communicating.

Typically, when things like this happen, the programmers blame the failure on the hardware and we hardware engineers blame the failure on the software. Because my coworker and I had worked together on the design of the network switch in the computing equipment, we immediately became involved when the communication failure occurred. The next phase was troubleshooting to prove that the hardware was not the problem.

During the troubleshooting process, someone questioned whether the receiver on the router was receiving any light energy from the switch's transmitter. Because the two pieces of equipment communicate at a wavelength of 1300 nm, the light is not visible. To quickly answer that question, a power meter was used to measure the optical output power from the transmitters on both pieces of equipment. A mode filter was used on the 1-meter jumper from the transmitter to the power meter.

The measurements obtained were compared to the manufacturer's optical characteristics for both the transmitter and the receiver. The optical output power for each transmitter was within specifications. About the same time that we determined the hardware was functioning properly, the programmers made some minor changes and the problem was resolved.

The fiber optic technician needs to be able to measure the transmitter's optical output power and determine that it's within the acceptable range for the receiver. This is a fundamental skill that will be needed in various troubleshooting scenarios.

Summary

This chapter covered the different photodiodes and basic optical and electronic components found in a typical fiber optic receiver. In it, we examined the general performance characteristics of the different photodiodes and looked at the performance characteristics of LED and laser receivers.

Exam Essentials

Describe the basic operation of a photodiode. Be able to describe the basic operation of a photodiode. Remember that the job of the photodiode is to convert optical energy into electrical energy.

Describe the operation of a PIN photodiode. Be able to describe the basic operation of the PIN photodiode. Remember that the PIN photodiode works like a PN photodiode but offers better performance, with improved efficiency and greater speed.

Describe the operation of an avalanche photodiode (APD). Be able to describe the basic operation of an avalanche photodiode. Remember that a small bundle of photons can trigger an avalanche of electrons. The APD accomplishes this through a process called photomultiplication.

Describe the performance characteristics of commonly used photodiodes. Be able to describe the three performance characteristics of commonly used photodiodes. Remember that the three common characteristics are responsivity, quantum efficiency, and switching speed.

Describe the basic components in a fiber optic receiver. Be able to describe the three basic components of a fiber optic receiver. Remember that a receiver is typically made up of an electrical subassembly, optical subassembly, and receptacle.

Describe receiver performance characteristics. You should know that the performance characteristics of the receiver are typically broken up into four groups: recommended operating conditions, electrical characteristics, optical characteristics, and data rate. Make sure you understand that recommended operating conditions describe the maximum and minimum temperature and voltage ranges that the device can operate in without damage. The electrical characteristics of the receiver describe the supply current requirements, the data output requirements, and the power dissipated by the device. The optical characteristics of the receiver at a minimum include minimum optical input power, maximum optical input power, and operating wavelength.

Review Questions

1. Fiber optic receivers use a(n) _____ to convert the light energy from the optical fiber into electrical energy.
 - A. Photoresistor
 - B. Photodiode
 - C. Phototransistor
 - D. LED
2. When photons are absorbed by a photodiode, they excite electrons within the photodiode in a process called _____ absorption.
 - A. Intrinsic
 - B. Extrinsic
 - C. Internal
 - D. External
3. A small number of photons absorbed by the _____ photodiode can cause a large flow of electrons.
 - A. PN
 - B. PIN
 - C. Avalanche
 - D. Unbiased
4. _____ describes how well a photodiode converts optical energy into electrical current.
 - A. Gain
 - B. Quantum efficiency
 - C. Absorption
 - D. Responsivity
5. _____ range is the difference between the maximum and minimum optical input power that the receiver can accept.
 - A. Maximum
 - B. Minimum
 - C. Dynamic
 - D. Quantum

6. Error generation in a fiber optic receiver is typically described by the receiver's _____.
- A. BRE
 - B. BER
 - C. RER
 - D. FORER
7. The _____ is the part of the fiber optic receiver that accepts the connector and aligns the ferrule.
- A. Receptacle
 - B. Outlet
 - C. Sleeve
 - D. Cap
8. In a fiber optic receiver, the _____ subassembly makes contact with the endface of the optical fiber.
- A. Electrical
 - B. Mechanical
 - C. Receptacle
 - D. Optical
9. The _____ amplifier in the quantizer IC provides a binary decision and determines whether the electrical pulses received represent a binary 1 or a binary 0.
- A. Quantum
 - B. Signal
 - C. Optical
 - D. Limiting
10. The performance characteristics of the fiber optic receiver are typically broken up into _____ groups.
- A. Two
 - B. Three
 - C. Four
 - D. Five

Answers to Review Questions

1. B. In a fiber optic receiver, it's the job of the photodiode to convert the light energy received from the optical fiber into electrical energy.
2. A. Photons absorbed by a photodiode excite electrons within the photodiode in a process called intrinsic absorption.
3. C. With the APD, a small bundle of photons can trigger an avalanche of electrons. The APD accomplishes this through a process called photomultiplication.
4. D. The responsivity of the photodiode describes how well the photodiode converts a wavelength or a range of wavelengths of optical energy into electrical current. It's the ratio of the photodiode electrical output current to its optical input power. The greater the responsivity, the greater the electrical current output for a given amount of optical input power.
5. C. The dynamic range of the receiver is measured in decibels. It is the difference between the maximum and minimum optical input power that the receiver can accept.
6. B. All receivers generate errors. Error generation is typically described by the receiver's bit error rate, or BER.
7. A. The receptacle is the part of the fiber optic receiver that accepts the connector. It also aligns the ferrule so that the optical fiber within the ferrule is perpendicular with the window edge in the optical assembly.
8. D. The optical subassembly guides the light energy from the optical fiber to the photodiode. A window in the optical subassembly makes contact with the endface of the optical fiber.
9. D. The limiting amplifier in the quantizer IC amplifies the voltage pulses and provides a binary decision. It determines whether the electrical pulses received represent a binary 1 or a binary 0.
10. C. The performance characteristics of LED and laser receivers are typically broken up into four groups: recommended operating conditions, electrical characteristics, optical characteristics, and data rate.



Chapter

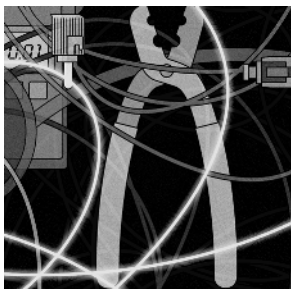
12

Passive Components and Multiplexers

OBJECTIVES COVERED IN THIS CHAPTER:

✓ Passive Components and Multiplexers

- Describe the operation of a tee coupler and application.
- Describe the operation of a star coupler and application.
- Describe the operation of an optical switch.
- Explain when optical attenuators are used.
- Explain the operation of an optical isolator.
- Describe wavelength division multiplexing (WDM).
- Describe dense wavelength division multiplexing (DWDM).
- Explain the basic operation of an optical amplifier.
- Explain when an optical filter is used.



The objective of this chapter is to allow the reader to gain an understanding of fiber optic passive components and multiplexers. This chapter covers not only particular devices and their applications, but also the reasons why the components were chosen and when they should be used.

Fiber optic passive components and multiplexers are elementary items, but necessary in all applications that require the transmission, combining, or distribution of optical signals. Passive components are components that do not require an external energy source. Multiplexers are devices that are used to combine two or more signals into a single output. The term *multiplexing* is used to refer to the process by which the signals are combined.

Some of the optical devices we cover in this chapter are *couplers*, *switches*, *attenuators*, *isolators*, *amplifiers*, and *filters*. We will also examine multiplexers and their associated processes, in particular *wavelength division multiplexing* and *dense wavelength division multiplexing*.

Couplers

In many applications, it may not be possible to have a design of many point-to-point connections. In these cases, optical couplers are used. A fiber optic coupler is a device that combines or splits optical signals. A coupling device may combine two or more optical signals into a single output, or the coupler may be used to take a single optical input and distribute it to two or more separate outputs. Figure 12.1 is an example of a basic four-port coupler.

FIGURE 12.1 Four-port coupler

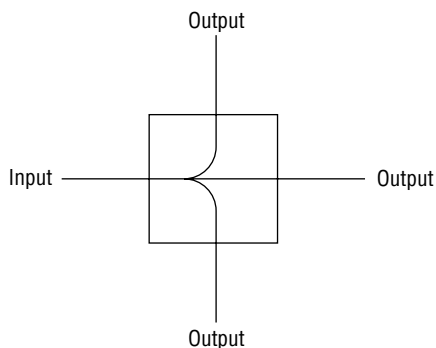
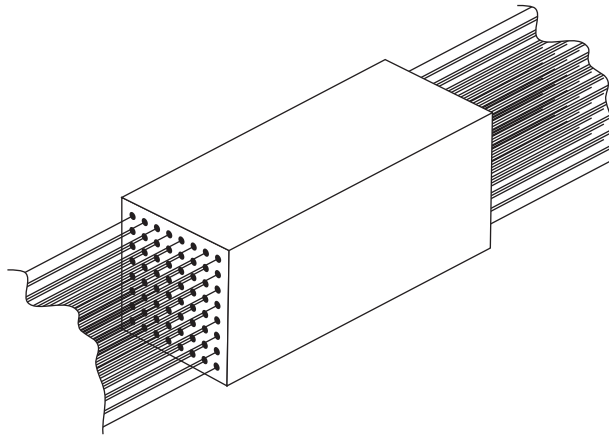


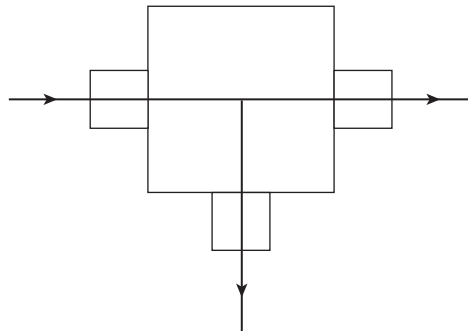
FIGURE 12.2 128-port coupler

Couplers are available with a wide range of input and output ports. A basic coupler may have only one input port and two output ports. Today's technology supports couplers with up to 64 input and 64 output ports, as shown in Figure 12.2.

There are many different types of couplers, and the number of input and output ports is dependent upon the intended usage. Some of the types of *optical couplers* are *optical combiners*, Y couplers, *star couplers*, *tee couplers*, *tree couplers*, and *optical splitters*; in this chapter, we will only focus on the tee coupler and the star coupler.

The Tee Coupler

A *tee coupler* is a three-port optical coupling device that has one input port and two output ports, as shown in Figure 12.3.

FIGURE 12.3 Tee coupler

The tee coupler is a passive device that splits the optical power from the input port into two output ports. The tee coupler is in essence an optical splitter. The uniqueness of the tee coupler is that this type of coupler typically distributes most of the optical input power to one output and only a small amount of power to the secondary output. It should be noted that when the outputs are evenly distributed, the coupler is called a Y coupler. The tee coupler is also referred to as an optical tap, due to the nature of the device. A majority of the power continues forward, but a portion of the signal (determined by the splitting ratio) is tapped to be used for an output port.

The tee coupler is a 1×2 (one by two) coupler, meaning that it has one input port (or connection) and two output ports. As previously stated, the optical output power of the two output ports is typically not evenly distributed. The most common splitting ratios are 90:10 and 50:50 (a Y coupler).

As with any passive device, there are losses. Losses are important considerations in fiber optic design. Keep in mind that there are many types of losses. There are losses from distribution, backscattering, crosstalk, tapping, and coupling. Therefore, it is very important to take into account all losses for a design using tee couplers or any passive device.

A typical use for a tee coupler would be to supply optical signals to a bus type network of in-line or serially connected terminals. Assuming ideal conditions and a 90:10 split on the tee coupler, the first terminal would receive 10% of the optical signal and 90% of the optical signal would go forward to the next tee coupler, and so on, as shown in Figure 12.4.

It's easy to see that it will not take long for the optical signal power levels to decrease to an unacceptable level. Keep in mind that we did not account for any losses; this is an ideal system. If we were to take losses into account, the results (shown in Figure 12.5) would be much different.

FIGURE 12.4 Ideal tee couplers in a bus type network

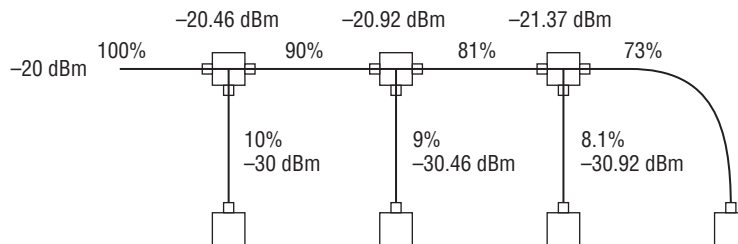


FIGURE 12.5 Real tee couplers in a bus type network

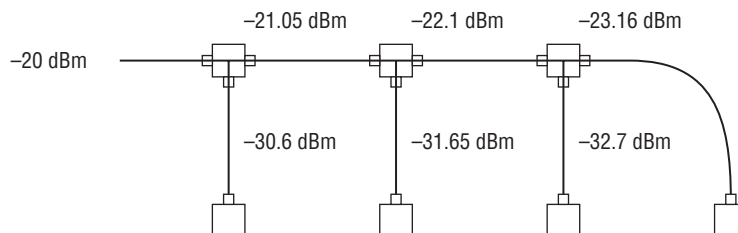
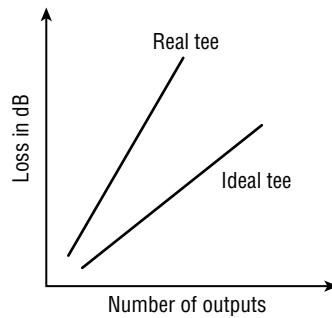


FIGURE 12.6 Comparison of ideal and real tee couplers

Now let's look at some of the common losses and their effect on the network of interconnections. To be realistic, we have to take into account the connector and its losses; we have assumed that there is a 0.3 dB loss for each connector (insertion loss) and another 0.3 dB for the loss due to the coupler (excess loss). We have also assumed that all of the couplers and connectors have the same insertion and excess losses. Looking at Figure 12.5, you can see how these losses affect the optical output power of each port. These losses at each port greatly reduce the number of taps a real tee coupler can support. Figure 12.6 compares the performance differences between ideal tee couplers and real tee couplers.

As shown by the previous examples, taking into account the losses of the device can make a large difference in anticipated optical output power. Hence, by incorporating losses such as crosstalk or directional, you could realize an even greater power loss from input to output. From this example, it should be clearly seen that the losses realized from using the tee coupler devices are real and must be accounted for by the fiber optic technician. Actual coupler losses can always be obtained from the manufacturer data sheet.

EXERCISE 12.1

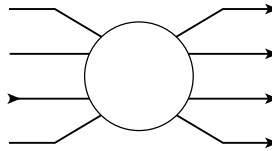
Determine the output power at both output ports of a tee coupler with a 50:50 splitting ratio using the dB rules of thumb.

- Assume that all losses for the coupler and interconnections combined equal 2.5 dB.
- Assume that the input power to the coupler is -20 dBm.
- The first step is to account for the tee coupler losses. This is done by subtracting 2.5 dB from the input power of -20 dBm. The remaining power available to the ports is -22.5 dBm.
- The second step is to split the remaining power using the dB rules of thumb covered in Chapter 2.
- Each output port will receive 50% of the energy. A loss of 50% is a change of 3 dB. The output power at each port will equal -22.5 dBm minus 3 dB. Each output port will have a power output of -25.5 dBm.

The Star Coupler

The star coupler is used in applications that require multiple ports—input and output. The star coupler will distribute optical power equally from two or more input ports to two or more output ports. Figure 12.7 shows a basic star coupler with four input ports and four output ports. Star couplers are available in 2×64 up to 64×64 dimensions.

FIGURE 12.7 Eight-port star coupler



A special version of the star coupler, called a tree coupler, is used when there is one input port and multiple output ports or when there are multiple input ports and one output port.

Star couplers are frequently used in network applications when there are a large number of output terminals. In our tee coupler example, we noted that in the serial tee connections at every coupler, we had to account for device and connection losses. In the star coupler, we are using only one device; therefore, the device and losses are decreased in a large port network. So the larger the network is, the more efficient the star coupler becomes.

Two types of star couplers are commonly used: the reflective star, shown in Figure 12.8, and the transmissive star, shown in Figure 12.9. Couplers are typically considered to be a black box—that is, only the manufacturer knows what's inside. However, many star couplers are made of fused optical fibers. The reflective star is defined as a coupler that realizes a signal at all input and output ports when a signal is applied to only one port. The transmissive star is defined as a coupler that realizes a signal at all output ports when a signal is applied to any of the input ports.

FIGURE 12.8 Fused reflective star coupler

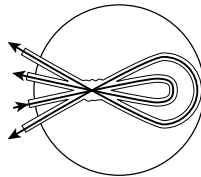
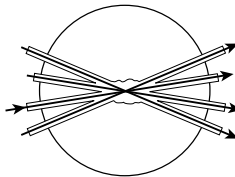
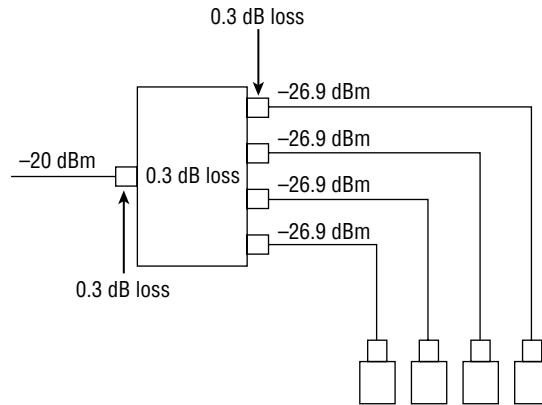


FIGURE 12.9 Fused transmissive star coupler



In our previous example, we looked at the losses in a network of four terminals with tee couplers. Let's compare the losses for the same number of terminals using a star coupler. Figure 12.10 shows the power delivered to each terminal using the same values, 0.3 dB insertion losses and 0.3 dB excess loss (due to only one coupler). This example shows a tree coupler with four output ports and one input port, similar to the sequential tee-coupled workstation network.

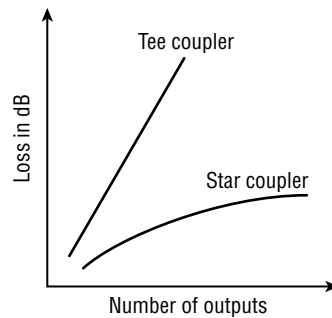
FIGURE 12.10 Four-output port tree coupler



EXERCISE 12.2

Determine the output power at each output port of an 11-port star coupler using the dB rules of thumb.

- Assume that all losses for the coupler and interconnections combined equal 6.5 dB.
- Assume that the input power to the coupler is -7 dBm.
- The first step is to account for the star coupler losses. This is done by subtracting 6.5 dB from the input power of -7 dBm. The remaining power available to the ports is -13.5 dBm.
- The second step is to split the remaining power using the dB rules of thumb covered in Chapter 2, "Principles of Fiber Optic Transmission."
- Each output port will receive 10% of the energy because the energy is distributed evenly between the output ports. A loss of 90% is a change of 10 dB. The output power at each port will equal -13.5 dBm minus 10 dB. Each output port will have a power output of -23.5 dBm.

FIGURE 12.11 Real tee coupler vs. real star coupler comparison chart

The star coupler has, of course, other losses that are not accounted for in this example. We are considering this network fairly ideal. However, this example does clearly show the advantage of a star coupler over a tee coupler. The key advantage here is that there is only one excess loss or loss caused by the coupler. The only remaining losses are from the interconnections. The advantage of the star coupler becomes very apparent as the number of ports increases. A simple loss-comparison chart as shown in Figure 12.11 can help shown the significance in the number of terminals versus loss for the tee and star couplers.

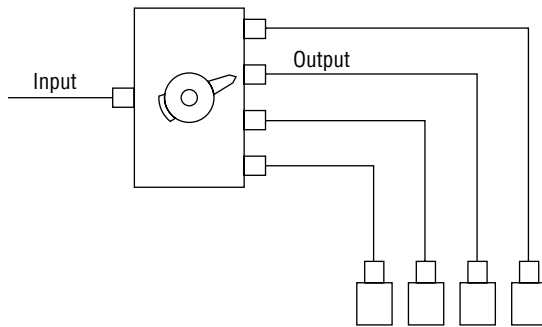
In summary, the larger the number of required outputs, the more likely it will be that a star coupler is used instead of a tee coupler. Both of these devices are very useful and have many applications. As a fiber optic technician, you need to know and remember the designs that each coupler is best suited for and the losses associated with each.

Optical Switches

The next device we will be looking at is an optical switch. The fiber optic switch can be a mechanical, optomechanical, or electronic device that opens or closes an optical circuit. The switch can be used to complete or break an optical path. Passive fiber optic switches will route an optical signal without electro-optical or optoelectrical conversion. However, a passive optical switch may use an electromechanical device to physically position the switch. An optical switch may have one or more input ports and two or more output ports. Figure 12.12 shows a basic optical switch with one input port and four output ports.

As with any other type of switch, the optical switch has many uses dependent upon the complexity of the design. In essence, the switch is the control for making, breaking, or changing the connections within an optical circuit. This definition can be expanded to incorporate the concept of being the control that interconnects or transfers connections from one optical circuit to another.

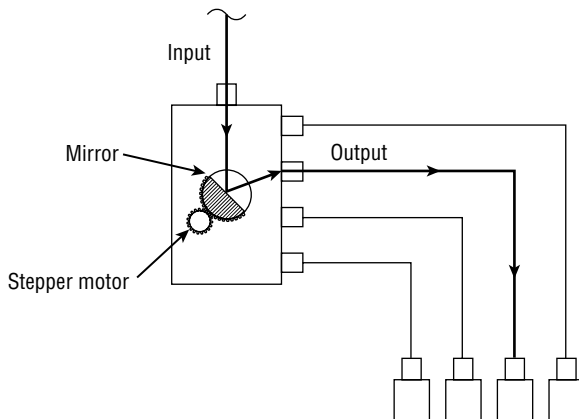
It is important for the fiber optic technician to be aware of the basic switch parameters for an optical switch. Some of the performance parameters to consider are the number of input and output ports (required size of the switch), wavelength range, losses (including insertion and polarization), crosstalk, and data rates.

FIGURE 12.12 Basic optical switch

Optomechanical

An optomechanical switch redirects an optical signal by moving fiber or bulk optic elements by means of mechanical devices. These types of switches are typically stepper motor driven. The stepper will move a mirror that directs the light from the input to the desired output, as shown in Figure 12.13. Although optomechanical switches are inherently slow due to the actual physical movement of the optical elements, their reliability, low insertion losses, and minimal crosstalk make them the most widely deployed type of switch.

The optomechanical switch works on the premise that the input and output fiber optical beams are collimated within the fiber and “matched” within the switching device—the beams are moved within the device to ensure the switched connection from the inputs to the outputs. The optomechanical switch can be physically larger than alternative switches, but there are many micromechanical fiber optic switches becoming available.

FIGURE 12.13 Optomechanical switch

Thermo-Optic

The thermo-optic switch is based on waveguide theory and utilizes waveguides made in polymers or silica. In other words, this switch utilizes the thermal/refractive index properties of the devices material. The principle of this switch relies upon the changing of the waveguide's refractive index due to a temperature change.

The temperature change can be accomplished in many ways, but generally the device is heated by using a resistive heater, which has the effect of slowing down light in one of the paths. The device then combines the light in the two paths in a constructive or destructive effect, making it possible to attenuate or switch the signal. This type of switch is inherently slow due to the time it takes to heat the waveguide. It's like a burner on an electric stove: It takes a while to heat up and a while to cool down.

This type of device has less optical loss than the optomechanical switch. Thermo-optic switches are attractive for several reasons: they work well in low optical power applications, are small in size, and have the potential to be integrated with a number of devices based on silicon wafer theory.

Electro-Optic

Electro-optic refers to a variety of phenomena that occur when an electromagnetic wave in the optical spectrum travels through a material under the stress of an electric field. An electro-optic switch is based upon the changing of the refractive index of a waveguide by using an electric field. This device is semiconductor based and therefore, boasts high speed and low optical power loss similar to that of the thermo-optic devices. This device is still in the research stage; however, the technology is rapidly advancing.

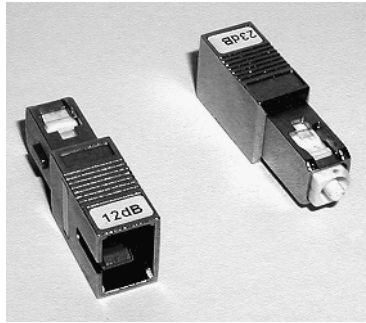
In summary, optical switches can be used in a variety of applications, large and small. The use of a fiber optic switch allows data to be routed where it's needed.

Optical Attenuators

An optical attenuator is a passive device that is used to reduce the power level of an optical signal. The attenuator circuit will allow a known source of power to be reduced by a predetermined factor, which is usually expressed as decibels. Optical attenuators are generally used in single-mode long-haul applications to prevent optical overload.

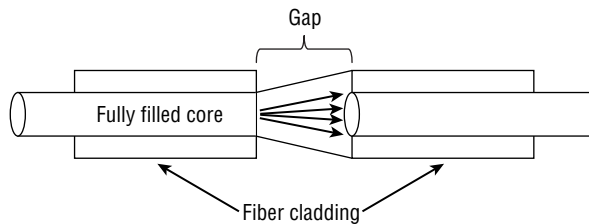
Optical attenuators typically come in two forms of packaging. The bulkhead optical attenuator shown in Figure 12.14 can be plugged into the receiver receptacle. The inline attenuator resembles a patch cord and is used between the patch panel and the receiver.

Optical attenuators use several different principles in order to accomplish the desired power reduction. Attenuators may use the gap-loss, absorptive, or reflective technique to achieve the desired signal loss. The types of attenuators generally used are fixed, stepwise variable, and continuously variable.

FIGURE 12.14 Bulkhead optical attenuator

Gap-Loss Principle

The principle of gap-loss is used in optical attenuators to reduce the optical power level by inserting the device in the fiber path using an in-line configuration. Gap-loss attenuators are used to prevent the saturation of the receiver and are placed close to the transmitter. Gap-loss attenuators use a longitudinal gap between two optical fibers so that the optical signal passed from one optical fiber to another is attenuated. This principle allows the light from the transmitting optical fiber to spread out as it leaves the optical fiber. When the light gets to the receiving optical fiber, some of the light will be lost in the cladding because of gap and the spreading that has occurred. The gap-loss principle is shown in Figure 12.15.

FIGURE 12.15 Gap-loss principle attenuator

Remember that this type of attenuator is used very close to the optical transmitter. The gap-loss attenuator will only induce an accurate reduction of power when placed directly after the transmitter. These attenuators are very sensitive to modal distribution ahead of the transmitter, which is another reason for keeping the device close to the transmitter to keep the loss at the desired level. The farther the gap-loss attenuator is placed away from the transmitter, the less effective the attenuator is, and the desired loss will not be obtained. To attenuate a signal farther down the fiber path, an optical attenuator using absorptive or reflective techniques should be used.



Real World Scenario

Using a Bulkhead Attenuator to Test Receiver Sensitivity

In Chapter 10, "Fiber Optic Light Sources," you learned about the output parameters of a fiber optic transmitter and in Chapter 11, "Fiber Optic Detectors and Receivers," you learned about the input parameters of a fiber optic receiver. In Chapter 16, "Link/Cable Troubleshooting," you will learn how to apply this information to analyze the performance of a fiber optic link. The fiber optic technician needs to be able to test a fiber optic receiver or offer technical assistance to the technician or engineer who is testing the fiber optic receiver.

A fiber optic receiver provides optimal performance when the optical input power is within a certain range. But how do you test the receiver to see if it will provide optimal performance at the lowest optical input powers? One way is to use optical attenuators, such as bulkhead attenuators. Typically only a couple of values are required to complete your testing.

The first step is to measure the optical output power of the fiber optic transmitter with the power meter. Remember from Chapters 10 and 11 that industry standards define transmitter optical output power and receiver optical input power for a particular network standard. If you are testing a 100BaseFX receiver, you should be using a 100BaseFX transmitter. The optical output power of the transmitter should be within the range defined by the manufacturer's data sheet.

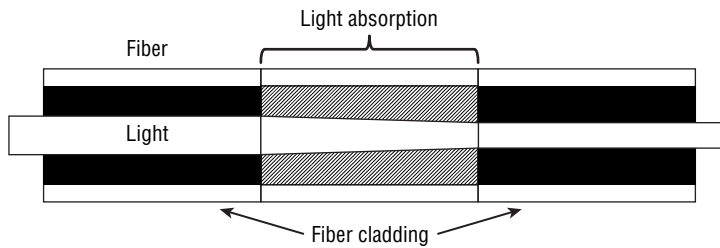
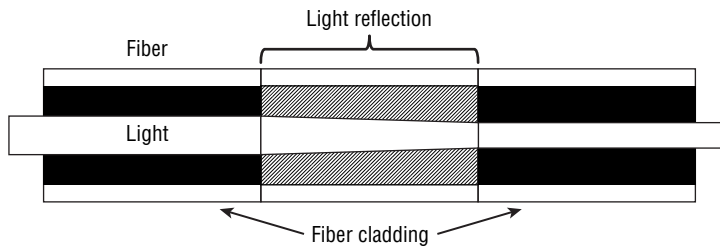
The next step is to connect the transmitter to the receiver and verify proper operation at the maximum optical output power that the transmitter can provide. You need to test the receiver at the minimum optical input power that the receiver can accept while still providing optimal performance. To do this, you need to obtain the lowest optical input power level value from the manufacturer's data sheet.

The next step is to calculate the attenuation level required for the test. Let's say that the transmitter's optical output power was -17 dBm and the minimum optical power level for the receiver is -33 dBm. The difference between -17 dBm and -33 dBm is 16 dB. You would use a 16 dB bulkhead attenuator at the input of the receiver and retest the receiver. If the receiver still operates properly, it's within specifications.

Absorptive Principle

The absorptive principle, or absorption, accounts for a percentage of power loss in optical fiber. This loss is realized because of imperfections in the optical fiber that absorb optical energy and convert it to heat. (See Chapter 5, "Optical Fiber Characteristics," for a detailed discussion of the subject.) This principle can be employed in the design of an optical attenuator to insert a known reduction of power.

The absorptive principle uses the material in the optical path to absorb optical energy. The principle is simple, but can be an effective way to reduce the power being transmitted and/or received. Figure 12.16 shows the principle of the absorption of light.

FIGURE 12.16 Absorptive principle attenuator**FIGURE 12.17** Reflective principle attenuator

Reflective Principle

The reflective principle, or scattering, accounts for the majority of power loss in optical fiber and again is due to imperfections in the optical fiber, which in this case cause the signal to scatter. This topic is also discussed in detail in Chapter 5. The scattered light causes interference in the fiber, thereby reducing the amount of transmitted and/or received light. This principle can be employed in the planned attenuation of a signal. The material used in the attenuator is manufactured to reflect a known quantity of the signal, thus allowing only the desired portion of the signal to be propagated. This principle is shown in Figure 12.17.

Now that we have looked at the principles behind the attenuator theories, we will show some of the types of attenuators. We will examine fixed, stepwise variable, and continuously variable attenuators and when they should be used.

Fixed Attenuators

Fixed attenuators are designed to have a fixed attenuation level. They can theoretically be designed to provide any amount of attenuation that is desired. The output signal is attenuated relative to the input signal. Fixed attenuators are typically used for single-mode applications.

Stepwise Variable Attenuators

A stepwise variable attenuator is a device that changes the attenuation of the signal in known steps such as 0.1 dB, 0.5 dB, or 1 dB. The stepwise attenuator may be used in several applications when dealing with multiple power sources—for example, if there are three inputs available, there may be a need to attenuate the signal at a different level for each of the inputs. Conversely, the stepwise attenuator may also be used in situations where the input signal is steady, yet the output requirements change depending upon the device that the signal is outputted to. The stepwise attenuator should be used in applications where the inputs, outputs, and operational configurations are known.

Continuously Variable Attenuators

A continuously variable attenuator is an attenuator that can be changed on demand. These attenuators generally have a device in place that allows the attenuation of the signal to change on demand. A continuously variable attenuator is used in uncontrolled environments where the input characteristics and/or output needs continually change. This allows the operator to adjust the attenuator to accommodate the changes required quickly and precisely without any interruption to the circuit.

In summary, there are many types of attenuators and many principles upon which they work. The key to choosing the appropriate one is to understand the theory on which each operates and the application that the attenuator will be applied to. Attenuators are readily available commercially for all types and sizes of applications.

Optical Isolator

Many laser-based transmitters and optical amplifiers use an optical isolator because the components that make up the optical circuit are not perfect. Connectors and other types of optical devices that are frequently used will have flaws in the end faces, thus mutating the signal that is being transmitted. As previously discussed, when any device is added to the circuit, it is highly probable that there will be some reflection, absorption, or scattering of the optical signal. These effects on the light beam may cause light energy to be reflected back at the source and interfere with source operation. To reduce the effects of the interference, an optical isolator is used.

The optical isolator comprises elements that will permit only forward transmission of the light; it does not allow for any return beams in the fiber transmission routes or in the amplifiers. There are a variety of optical isolator types, such as polarized (dependent and independent), composite, and magnetic.

Polarized

As mentioned, the polarized optical isolator transmits light in one direction only. This is accomplished by using the polarization axis of the linearly polarized light. The incident light is transformed to linearly polarized light by traveling through the first polarizer. The light then goes through a Faraday rotator; this takes the linearly polarized light and rotates the polarization 45 degrees, then the

light passes through the exit polarizer. The exit polarizer is oriented at the same 45 degrees relative to the first polarizer as the Faraday rotator is. With this technique, the light is passed through the second polarizer without any attenuation. This technique allows the light to propagate forward with no changes, but any light that may travel backward is extinguished entirely.

The loss of backward-traveling light occurs because when the backward light passes through the second polarizer, it is shifted again by 45 degrees, thus being linearly polarized light. The light then passes through the rotator and again is rotated by 45 degrees in the same direction as the initial tilt. So when the light reaches the first polarizer, it is polarized at 90 degrees. And when light is polarized by 90 degrees, it will be “shut out.” Figure 12.18 shows the forward-transmitted light in a dependent polarized optical isolator.

Figure 12.19 shows the backward-traveling light in a dependent polarized optical isolator.

FIGURE 12.18 Forward-transmitted light through a polarized optical isolator

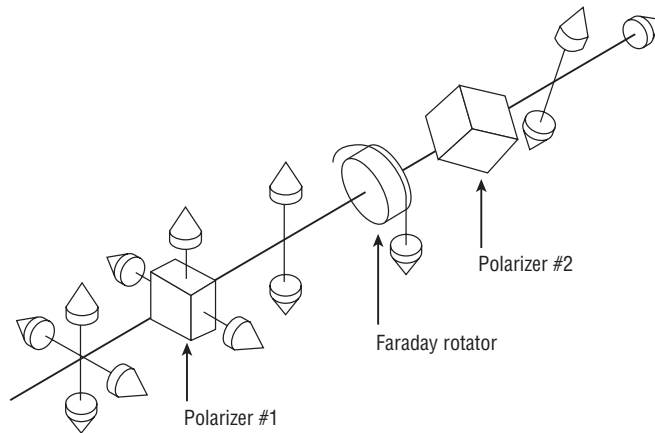
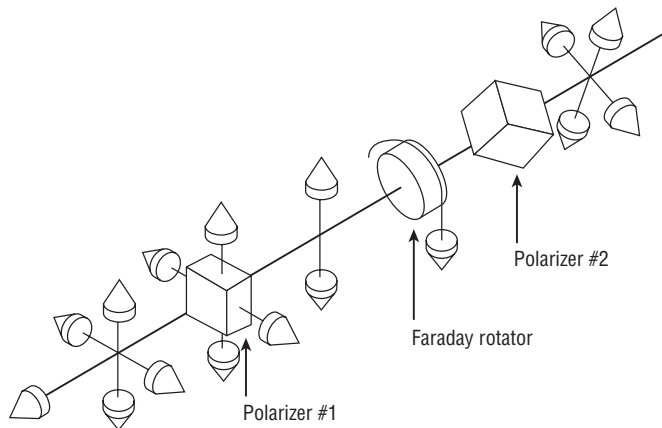


FIGURE 12.19 Reverse-transmitted light through a polarized optical isolator

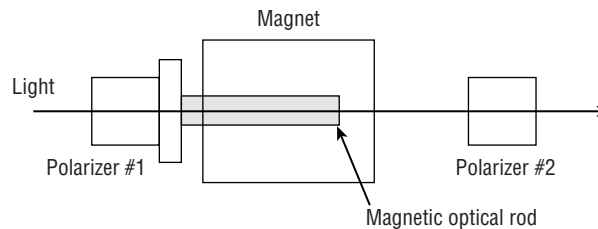


It should be noted that these figures depict the dependent type of polarized optical isolator. There is also an independent polarized optical isolator. The independent device allows all polarized light to pass through, not just the light polarized in a specific direction. The principle of operation is roughly the same as the dependent type, just slightly more complicated. The independent optical isolators are frequently used in optical fiber amplifiers.

Magnetic

Magnetic optical isolators are another name for polarized isolators. The magnetic portion of any isolator is of great importance. As mentioned, there is a Faraday rotator in the optical isolators. The Faraday rotator is a rod composed of a magnetic crystal having a Faraday effect and operated in a very strong magnetic field. The Faraday rotator ensures that the polarized light is in the correctly polarized plane, thus ensuring that there will be no power loss. Figure 12.20 shows a basic magnetic optical isolator.

FIGURE 12.20 Magnetic optical isolator



In summary, optical isolators are used to ensure stabilization of laser transmitters and amplifiers as well as to maintain good transmission performance.

Wavelength Division Multiplexing

Wavelength division multiplexing (WDM) is the combining of different optical wavelengths from two or more optical fibers into just one optical fiber. This combining or coupling of the wavelengths can be very useful in increasing the bandwidth of a fiber optic system. WDM multiplexers are used in pairs: one at the beginning of the fiber to couple the inputs and one at the end of the fiber to decouple and then route the separated wavelengths into separate fibers. An WDM multiplexer can be thought of as an optical fiber highway; the highway can support a very large bandwidth, thus increasing the system's capacity.

Each channel in a WDM multiplexer is designed to transmit a specific optical wavelength. The multiplexer operates very much like a coupler at the beginning of the optical fiber and as a filter at the end of the optical fiber. For example, an eight-channel multiplexer would have the ability to combine eight different channels or wavelengths from separate optical fibers onto one optical fiber. Again, to take advantage of the enormous bandwidth at the end of the optical fiber, another multiplexer will recover the separate wavelengths.

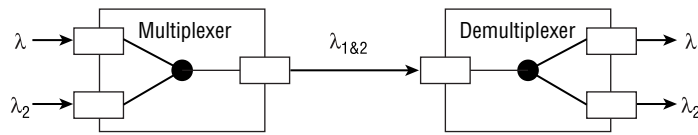
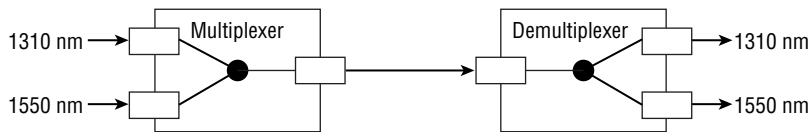
FIGURE 12.21 Simple WDM system

Figure 12.21 shows a simple WDM system composed of multiple light sources, a multiplexer or combiner that combines the wavelengths into one optical fiber, and a *demultiplexer* or splitter that separates the wavelengths to their respective receivers.

WDM multiplexers are available in a variety of sizes, but will most commonly be found with 2-, 4-, 8-, 16-, 32-, and 64-channel configurations. The types of multiplexers are wideband (or crossband), narrowband, and dense.

Wideband or crossband multiplexers are devices that combine a broad range of wavelengths, such as 1310 nm and 1550 nm. A narrowband multiplexer will combine multiple wavelengths with 1000 GHz channel spacing. A dense multiplexer combines wavelengths with 100 GHz channel spacing. Figure 12.22 shows a basic wideband or crossband WDM system.

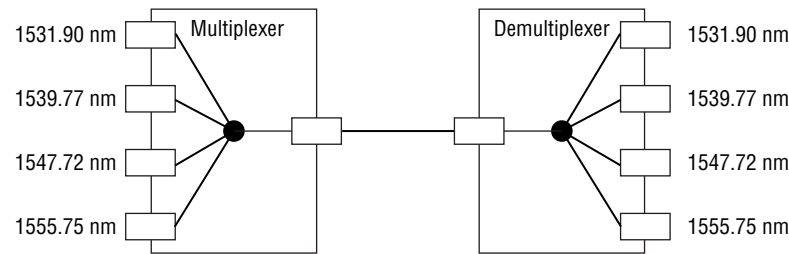
FIGURE 12.22 Basic wideband or crossband WDM system

Narrowband WDM systems have channels spaced 1000 GHz, or approximately 8 nm, apart. Table 12.1 shows the wavelength and frequency data for narrowband WDM systems. Figure 12.23 shows a basic narrowband WDM system.

TABLE 12.1 Narrowband WDM Channel Spacing

λ (nm)	F (THz)
1531.90	195.7
1539.77	194.7
1547.72	193.7
1555.75	192.7

FIGURE 12.23 Basic narrowband WDM system



The industry standard on the dense wavelength division multiplexing (DWDM) multiplexers, as recommended by the International Telecommunications Union (ITU), is 100 GHz, or approximately 0.8 nm, channel spacing. Table 12.2 displays values in the C band, which is the 1550 nm band. The C band uses wavelengths from 1530 to 1565 nm. Other commonly referred-to bands are the S band, with wavelengths from 1525 to 1538 nm, and the L band, with wavelengths from 1570 to 1610 nm.

The C band, which is currently the most popular band in use, is further split into short (blue) and long (red) bands, as shown in Table 12.2.

TABLE 12.2 DWDM 100 GHz Channel Spacing

Short/Blue Band			Long/Red Band		
Channel	λ (nm)	F (THz)	Channel	λ (nm)	F (THz)
65	1525.66	196.5	40	1545.32	194.0
64	1526.44	196.4	39	1546.12	193.9
63	1527.21	196.3	38	1546.92	193.8
62	1527.99	196.2	37	1547.72	193.7
61	1528.77	196.1	36	1548.51	193.6
60	1529.55	196.0	35	1549.32	193.5
59	1530.33	195.9	34	1550.12	193.4
58	1531.12	195.8	33	1550.92	193.3
57	1531.90	195.7	32	1551.72	193.2
56	1532.68	195.6	31	1552.52	193.1

TABLE 12.2 DWDM 100 GHz Channel Spacing (*continued*)

Short/Blue Band			Long/Red Band		
Channel	λ (nm)	F (THz)	Channel	λ (nm)	F (THz)
55	1533.47	195.5	30	1553.33	193.0
54	1534.25	195.4	29	1554.13	192.9
53	1535.04	195.3	28	1554.94	192.8
52	1535.82	195.2	27	1555.75	192.7
51	1536.61	195.1	26	1556.55	192.6
50	1537.40	195.0	25	1557.36	192.5
49	1538.19	194.9	24	1558.17	192.4
48	1538.98	194.8	23	1558.98	192.3
47	1539.77	194.7	22	1559.79	192.2
46	1540.56	194.6	21	1560.61	192.1
45	1541.35	194.5	20	1561.42	192.0
44	1542.14	194.4	19	1562.23	191.9
43	1542.94	194.3	18	1563.05	191.8
42	1543.73	194.2	17	1563.86	191.7
41	1544.53	194.1	16	1564.68	191.6

As shown in Table 12.2, the closer the channels are spaced together, the higher the number of channels that can be inserted into a band. Currently a spacing of 50 GHz is available. It is important to note that as the spacing or the width of each channel decreases, the smaller the spectral width becomes. This is relevant because the wavelength must be stable or sustainable long enough not to drift into an adjacent channel.

Now let's look at a different view of channel spacing. Figure 12.24 shows a four-channel narrowband WDM spectrum using DFB laser transmitters with a spectral width of 1.0 nm measured at -20 dB. Figure 12.25 shows a 32-channel dense WDM spectrum using DFB laser transmitters with a spectral width of 0.3 nm measured at -20 dB.

FIGURE 12.24 Four-channel narrowband WDM spectrum

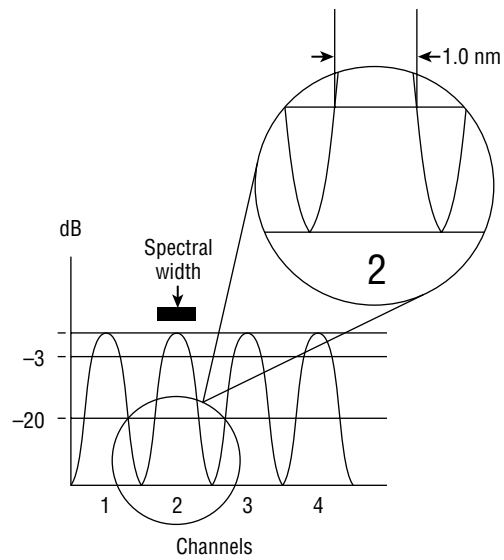
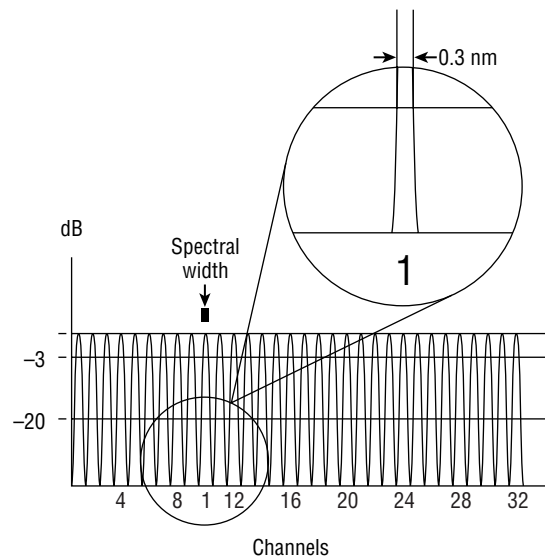


FIGURE 12.25 32-channel dense WDM spectrum



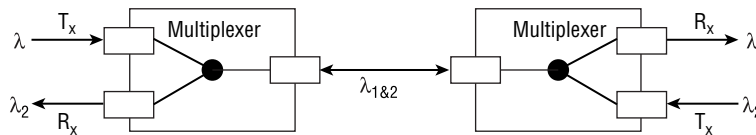
You can quickly see that as the channel spacing decreases, the laser transmitter spectral width must also decrease. To achieve 50 GHz channel spacing, the laser transmitter spectral width needs to be very narrow—or about as narrow as today’s laser technology permits.

Besides having a very narrow spectral width, the laser transmitter cannot drift—it must output the same wavelength at all times. If the laser transmitter’s output wavelength changes even a few tenths of a nanometer, it could drift into the next channel and cause interference problems.

There are different configurations of WDM multiplexers. Everything we have covered up to this point describes a unidirectional WDM system. The unidirectional WDM multiplexers are configured so that the multiplexer only connects to optical transmitters or receivers. In other words, they allow the light to travel in only one direction and they provide only simplex communication over a single optical fiber. Therefore, full-duplex communications require two optical fibers.

A WDM multiplexer that is designed to connect with both transmitters and receivers is called bidirectional; in essence, the multiplexer is designed for optical transmission in both directions using only one optical fiber. Two channels will support one full-duplex communication link. Figure 12.26 shows two bidirectional WDM multiplexers communicating over a single optical fiber.

FIGURE 12.26 Two-channel bidirectional WDM system



As with any other device that is added to a fiber optic network, there are factors that must be considered. As mentioned earlier in the chapter, losses are a factor that every fiber optic technician must take into account. When using WDM multiplexers, remember that the greater the number of channels, the greater the insertion loss. Other specifications to keep in mind when using WDM multiplexers are isolation, PMD, and the spectral bandwidth.

In summary, WDM multiplexers are widely used devices. They provide a way to utilize the enormous bandwidth capacity of optical fiber without the expense of using the fastest laser transmitters and receivers. Just think about it: an eight-channel WDM system using directly modulated 2.5 Gbps laser transmitters carries twice as much data as a single indirectly modulated 10 Gbps laser transmitter. WDM systems allow designers to combine modest performance parts and create an ultra performance system. WDM systems deliver the most bang for the buck!

Optical Amplifier

As optical signals travel through an optical fiber, they are attenuated. In long-haul applications, the signal is attenuated to the point where re-amplification is required. Traditionally, a device commonly referred to as a *repeater* accomplished this re-amplification.

A repeater is basically a receiver and transmitter combined in one package. The receiver converts the incoming optical energy into electrical energy. The electrical output of the receiver drives the electrical input of the transmitter. The optical output of the transmitter represents an amplified version of the optical input signal plus noise.

The technology available today eliminates the need for repeaters. Passive optical amplifiers are now used instead of repeaters. A passive optical amplifier amplifies the signal directly without the need for optical-to-electric and electric-to-optical conversion. There are several different techniques in which to passively amplify an optical signal: erbium doped fiber amplifiers, semiconductor optical amplifiers, and Raman amplification, all of which use a technique called laser pumping.

Erbium doped fiber amplifiers (EDFAs) are generally used for very long fiber links such as under-sea cabling. The EDFAs use a fiber that has been treated or “doped” with erbium and this is used as the amplification medium. The pump lasers operate at wavelengths below the wavelengths that are to be amplified. The doped fiber is energized with the laser pump. As the optical signal is passed through this doped fiber, the erbium atoms transfer their energy to the signal, thereby increasing the energy or the strength of the signal as it passes. With this technique, it is common for the signal to be up to 50 times or 17 dB stronger leaving the EDFA than when it entered.

An example of an EDFA is shown in Figure 12.27. EDFAs may also be used in series to further increase the gain of the signal. Two EDFAs used in series may increase the input signal as much as 34 dB.

Semiconductor optical amplifiers (SOAs) use a similar technique without doping the optical fiber. Unlike the EDFA, which is energized with a laser pump, the SOA is energized with electrical current. The SOAs use an optical waveguide and a direct bandgap semiconductor that is basically a Fabry-Perot laser to inject light energy into the signal, as shown in Figure 12.28. This technique, however, does not offer the high amplification that the EDFAs do. SOAs are typically used in shorter fiber links such as *metropolitan area networks (MANs)*.

FIGURE 12.27 Erbium doped fiber amplifier

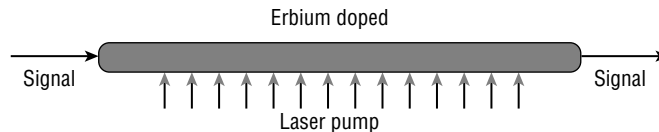
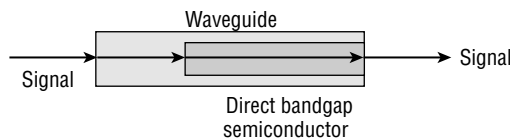


FIGURE 12.28 Semiconductor optical amplifier



One problem with SOAs is that the gain is very hard to control. By using the semiconductor technique and a waveguide, the signal may deplete the gain of a signal at another wavelength. This can introduce crosstalk among channels by allowing the signal at one wavelength to modulate another.

Lastly, *Raman amplification* is a method that uses pump lasers to donate energy to the signal for amplification. However, unlike using EDFAs, this technique does not use doped fiber, just a high-powered pumping laser as shown in Figure 12.29. The laser is operated at wavelengths 60 nm to 100 nm below the desired wavelength of the signal. The laser signal energy and the photons of the transmitted signal are coupled, thereby increasing the signal strength.

FIGURE 12.29 Raman amplification



Raman amplification does not amplify as much as the EDFAs but it does have an advantage in that it generates much less noise.

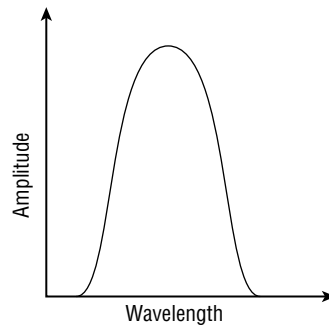
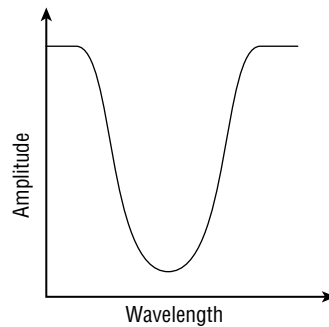
These techniques can be combined together to take advantage of their amplification characteristics. In some cases, Raman and EDF amplifiers are combined in long-haul fiber links to ensure high amplification and decreased noise levels.

In summary, each amplification technique has advantages and disadvantages. The fiber optic technician needs to keep in mind the amplification that the amplifier is being used in. For example, if a signal needed amplification but noise was an issue, a Raman amplifier would most likely be the best choice. If the signal needed to be amplified by just a small amount, an SOA might be best. However, all of these amplification methods have one big advantage: optical amplifiers will amplify all signals on a fiber at the same time. Thus, it is possible to simultaneously amplify multiple wavelengths. But it is important to keep in mind that the power levels must be monitored carefully because the amplifiers can become saturated, thereby causing incorrect operation.

Optical Filter

An optical filter is a device that selectively permits transmission or blocks a range of wavelengths. Optical filters are typically bandpass or band-reject. A bandpass optical filter allows a certain range of optical wavelengths to pass and attenuates the rest, as shown in Figure 12.30. A band-reject optical filter attenuates a band of optical wavelengths and allows the others to pass, as shown in Figure 12.31.

An example of a basic optical filter would be the optical filter used on a traffic light. A typical traffic light contains three optical filters, one red, one yellow, and one green. The bulb behind each optical filter is the same and emits a wide range of visible wavelengths. The optical filters only allow a certain range of wavelengths to pass, creating the red, yellow, or green light.

FIGURE 12.30 Optical bandpass filter response**FIGURE 12.31** Optical band-reject filter response

Bandpass optical filters are designed to transmit a specific waveband or wavelength range. A wideband optical filter may allow wavelengths plus and minus 20 nm off the center frequency to pass. This type of optical filter would be used when signals are separated by several hundred nanometers. You would use this optical filter with a 1310 nm and 1550 nm source.

A narrowband optical filter allows only a very narrow range of optical energy to pass, as shown in Figure 12.32. The bandwidth of a narrowband optical filter may be less than one nanometer. The narrowband optical filter would be used in a DWDM application to reject adjacent optical channels.

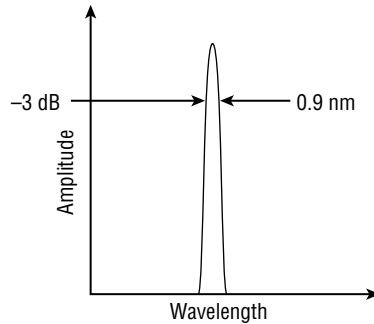
Each of these optical filter types is simple in theory yet is a vital part of some fiber optic systems. As stated, an optical filter is a device that selects the wavelengths it will allow to pass and will reject the others.

Summary

This chapter explored some of the basic passive components used in fiber optic systems and networks. Prior to this chapter, we had looked only at employing optical fiber point to point. However,

in many network applications, different physical configurations are required and one transmitter may send data to multiple receivers. The passive components discussed in this chapter make these types of networks possible.

FIGURE 12.32 Narrowband optical filter response



Exam Essentials

Describe the operation of a tee coupler and application. Be able to describe how the tee coupler distributes optical power. Be able to describe how the tee coupler is used in a fiber optic system.

Describe the operation of a star coupler and application. Be able to describe how the star coupler distributes optical power. Be able to describe how the star coupler is used in a fiber optic system.

Describe the operation of an optical switch. Be able to describe the basic operation of an optical switch and how it's used in a fiber optic system.

Explain when optical attenuators are used. Be able to describe what an optical attenuator is and explain when it is used in a fiber optic system.

Explain the operation of an optical isolator. Be able to explain the basic operation of an optical isolator.

Describe wavelength division multiplexing. Be able to describe wavelength division multiplexing and how it combines different optical wavelengths from two or more optical fibers into just one optical fiber.

Describe dense wavelength division multiplexing. Be able to describe the difference between WDM and DWDM.

Explain the basic operation of an optical amplifier. Be able to describe the basic operation of an optical amplifier and when it is used in a fiber optic system.

Explain when an optical filter is used. Be able to describe an optical filter and when it is used in a fiber optic system.

Review Questions

1. A tee coupler is a _____-port device.
 - A. Two
 - B. Three
 - C. Four
 - D. Five
2. A tee coupler with a 50:50 splitting ratio is also referred to as a(n) _____ coupler.
 - A. C
 - B. L
 - C. Y
 - D. X
3. The tee coupler is used with what type of network architecture?
 - A. Bus
 - B. Star
 - C. Ring
 - D. Parallel
4. A coupler with only one input port and multiple output ports is called a _____ coupler.
 - A. Tee
 - B. Tree
 - C. Star
 - D. Branch
5. With a(n) _____ star coupler, a signal supplied to any input fiber (input port) will produce signals at each output fiber (output port).
 - A. Transmissive
 - B. Reflective
 - C. Conductive
 - D. Inverting
6. With a(n) _____ star coupler, a signal applied to any fiber (port) will produce an output at every fiber (port).
 - A. Transmissive
 - B. Reflective
 - C. Conductive
 - D. Inverting

7. An optical _____ is used to complete or break an optical path.
- A. Coupler
 - B. Filter
 - C. Switch
 - D. Attenuator
8. An optical _____ is used to attenuate the power level of an optical signal.
- A. Coupler
 - B. Filter
 - C. Switch
 - D. Attenuator
9. An optical attenuator should typically be just before a _____.
- A. Transmitter
 - B. Splice
 - C. Receiver
 - D. Coupler
10. An optical _____ permits the transmission of light in only one direction.
- A. Coupler
 - B. Isolator
 - C. Attenuator
 - D. Splice

Answers to Review Questions

1. B. A tee coupler is a three-port optical coupling device that has one input port and two output ports.
2. C. The most common splitting ratios for tee couplers are 90:10 and 50:50. A tee coupler with a 50:50 splitting ratio is also known as a Y coupler.
3. A. A typical use for a tee coupler is in a bus type network.
4. B. A special version of the star coupler called a tree coupler is used when there is only one input port and multiple output ports or when there are multiple input ports and one output port.
5. A. The transmissive star is defined as a coupler that realizes a signal at all output fibers (output ports) when a signal is applied to any input fiber (input port).
6. B. The reflective star is defined as a coupler that realizes a signal at all fibers when a signal is applied to any fiber.
7. C. The optical switch can be a mechanical or an electronic device that opens and closes optical circuits. The optical switch can be used to make or break an optical path.
8. D. The optical attenuator is a passive device that is used to reduce the power level of an optical signal.
9. C. Optical attenuators are typically used just before the receiver to attenuate the input signal into the receiver.
10. B. The optical isolator is composed of elements that will permit only the forward transmission of light.



Chapter

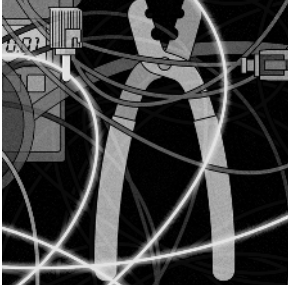
13

Cable Installation and Hardware

OBJECTIVES COVERED IN THIS CHAPTER:

✓ Cable Hardware and Installation

- Explain manufacturer installation cable specifications.
- Explain the static and dynamic loading on a fiber optic cable during installation.
- Describe commonly used installation hardware.
- Describe tray and duct installation.
- Describe conduit installation.
- Describe direct burial installation.
- Describe aerial installation.
- Describe blown fiber installation.
- Explain cable grounding and bonding per NEC article 250.
- Describe patch panel preparation.
- Describe racks and cable preparation.
- Describe splice enclosure preparation.
- Describe wall outlet installation.
- Explain how to label a cable in accordance with TIA/EIA-606A.



Under the right conditions, an optical fiber will carry light thousands of kilometers almost instantly. If the cable is not installed properly, though, the light it is supposed to carry may not even travel from one part of an office building to another.

Proper installation depends on a thorough knowledge of the strengths and limits of a fiber optic cable, as well as the methods for protecting the cable both during the installation and over its lifetime. The cable must endure the pulling that is necessary for it to be put in place; environmental conditions that threaten to freeze, soak, or otherwise damage it; and the daily stresses that result from its location or position.

The rules governing fiber optic cable installation are designed to minimize the short-term and long-term stresses on the cable as well as ensure that the installation conforms to codes governing fire safety. Some of these requirements were discussed in Chapter 7, “Fiber Optic Cables,” with regard to the structure of the cable itself. The rules governing cable construction can only be effective, though, if the cable is installed properly.

This chapter describes the requirements for a successful fiber optic cable installation. It describes the conditions affecting fiber optic cable and the methods for installing the cable so that the effects of those conditions are minimized. The chapter describes regulations concerning electrical safety for cable installations as well as methods for routing cables in different situations. The chapter also discusses the ways in which cables are enclosed and terminated, including proper labeling methods.

Installation Specifications

Many fiber optic cables are created for specific types of service and must be installed in accordance with their manufacturer’s specifications. While some specifications are concerned with the particular duty or job a cable will perform, others apply to traits shared by all cables. Two of these specifications are *minimum bend radius* and *maximum tensile rating*. Both specifications apply to conditions faced by fiber optic cable while it is being installed and, once it has been installed, to its normal working conditions.

Minimum Bend Radius

As we have seen in previous chapters, optical fiber depends on the maintenance of total internal reflection to carry an optical signal. Macrobends, or very tight bends in the fiber, can decrease the light’s angle of incidence enough to cause some or all of the light to pass into the cladding, severely attenuating the signal or cutting it off completely.

To reduce the risk of macrobends during and after installation, manufacturers often specify a minimum bend radius for their cables. This does not mean that bends approaching the minimum radius will not also cause some light to pass into the cladding; that will occur whenever there is a bend in the fiber. These amounts are negligible, though, and occur only in the highest-order modes.

Another, more basic reason for the minimum bend radius is that the silica fibers, while flexible, are not indestructible. Bending beyond the minimum radius can cause the fiber to break inside the cable, which would require splicing at the very least and replacement of the entire cable as a worst-case scenario.

The minimum bend radius varies depending on whether or not the cable is bent during the installation process or remains bent during its service life. It also varies according to the amount of tension being placed on the cable.

Maximum Tensile Rating

Recall that the strength member of a fiber optic cable is bonded directly to the connector, isolating any forces pulling on the cable or the connector from the fiber itself.

Even though the strength member in an optical fiber cable absorbs a great deal of the tension placed on the cable, it cannot protect the fiber inside completely. Tensile loading, or stretching force, creates an increasing risk to fiber as the force increases.

The first effect of tensile loading is a temporary attenuation, which goes away once the load is removed. As the load increases, though, the fiber is damaged enough to cause permanent attenuation. Finally, under enough load, the fiber can crack and break.

While tensile loading on a cable is damaging enough if the cable is straight, the effects are increased if the fiber is bent while under tensile loading. For this reason, a cable's maximum tensile rating is set according to the minimum bend radius for the cable on the assumption that some part of the cable will be subjected to the minimum bend radius.

Because fibers must be pulled into place during installation, sometimes over long distances, greater tensile loads are allowed for short periods to accommodate the installation process. The loads encountered during installation are called *dynamic loads* because they can increase or decrease in a very short time and they are rarely constant. These loads must not be allowed to persist for the life of the installation, however.

The long-term loads on a fiber optic cable are called *static loads*. These are the consistent loads that a cable will sustain throughout its installed life. Because a cable must endure static loads indefinitely, the maximum static load set by the manufacturer is much smaller than the dynamic load.

Let's look at a specification chart for tensile load and bend radius for some typical optical fiber cables.

Diameter	inch	0.114	0.187
	mm	2.9	4.8
Weight	lb/kft	6	12
	kg/km	9	18

Minimum Bend Radius	Install	inch	1.8	2.8
		cm	4.5	7.1
	Long Term	inch	1.2	1.9
		cm	3.0	4.7
Maximum Load	Install	lb	50	150
		n	220	660
	Long Term	lb.	15	66
		n	66	198

The cables shown in the chart are a single fiber and a duplex cable designed for interior use. They consist of tight-buffered fiber, a strength member, and an outer jacket.

Note that the weight of the cable over distance is included in the chart. The reason for including the weight is that it can become a factor in adding to the tensile load if the fiber is to be suspended or hanging vertically for long distances. In the first fiber, for example, the long-term load limit is 15 pounds, while its weight is 6 pounds per thousand feet. If the cable must support itself for a length of 500 feet, it already has 3 pounds of tensile loading on it.

Note also that the manufacturer’s specified bend radius for installation is about 15 times the cable diameter, while the long-term bend radius is about 10 times the cable diameter. This greater minimum bend radius for installation allows a greater tensile load—about three times as much as the long-term tensile load—to be used for pulling or wrestling the cable into position. Remember that the figures used in this instance are for a cable that is installed inside a building. Cables that will be installed outside of a building will typically have a larger minimum bend radius.

While the minimum bend radius and maximum tensile load are the manufacturers’ recommendations for installation, it is still a good idea to avoid approaching these numbers if at all possible. As we stated earlier, any bend can cause some light to pass into the cladding, increasing attenuation, if even slightly.

Installation Hardware

Fiber optic cables and the fibers within them have a number of specialized requirements for their installation and connection. From cable-pulling tools to cable-protecting enclosures, installation hardware has been specially designed and built to meet the needs of fiber optic cables in almost any environment and situation.

Let’s take a look at some of the hardware commonly used in fiber optic installation.

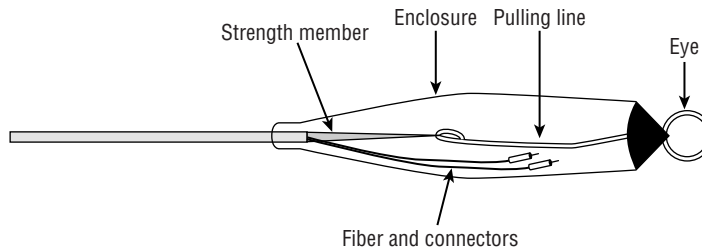
Pulling Eye

Sooner or later, you will need to run a cable through a wall, conduit, or other inaccessible space. An indispensable tool for this job is the pulling eye, shown in Figure 13.1. This device is specially designed to attach to the cable's strength member at one end and a pulling line at the other. The pulling line, sometimes called a messenger line, has been fed through the space to be occupied by the cable. The line is then used to pull the eye through the space with the cable attached.

The pulling eye also uses a sheath that encloses the fiber ends to protect them from damage while the cable is being pulled.

Some optical fiber cables are sold with the pulling eye already attached.

FIGURE 13.1 The pulling eye is used to pull cable through conduit.



Pullbox

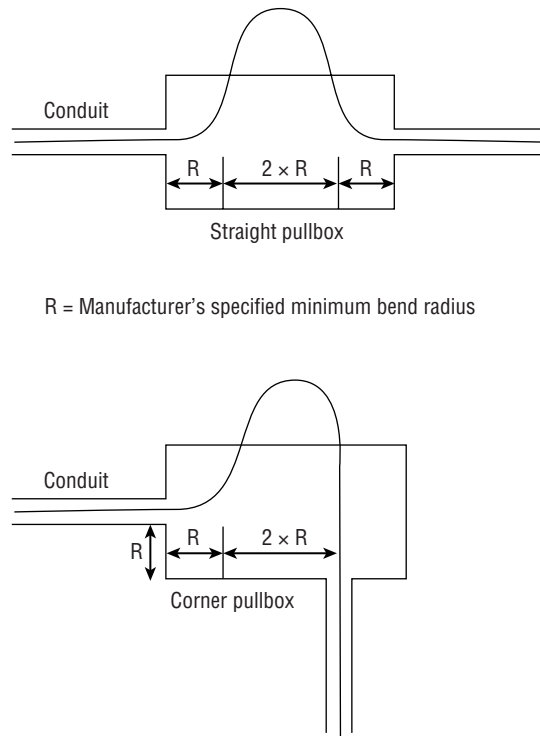
Optical fiber is small enough and light enough to be relatively easy to pull through conduits, compared to electrical cable. Even optical fiber cable, however, will exert friction and increase the tensile loading if it goes through enough turns or if there is a large amount of it already in the conduit.

To make the cable easier to pull and to ease the tensile load on it, pullboxes are installed at intervals in the conduit. Typically, pullboxes are installed after straight runs of 250 to 300 feet and every time a set of turns totals 180° or more. The purpose of a pullbox is to create an intermediate opening for pulling the cable to reduce the length that is being pulled through the conduit and to reduce the number of turns through which the cable must be pulled at any one time.

To use a pullbox, pull the fiber through the box and out the large opening. Once the fiber has been pulled as far as it will go, feed it into the other side of the box and down the conduit. All of the conduit length and turns before the pullbox have just been eliminated from any further pulling on the cable, reducing the load on it.

There are two types of pullboxes, as shown in Figure 13.2: straight and corner. Straight pullboxes are installed in-line in the conduit and have an opening at least four times the minimum bend radius of the cable being pulled. This length prevents the cable from exceeding its minimum bend radius as the last of it is pulled through the box.

FIGURE 13.2 Straight and corner pullboxes ease tensile loading on fiber during installation.



R = Manufacturer's specified minimum bend radius

Corner pullboxes are installed at angles in the conduit and typically require a length of three times the cable's bending radius and a depth that equals the bending radius. This requirement prevents the cable from dragging against a sharp turn when it is pulled through.

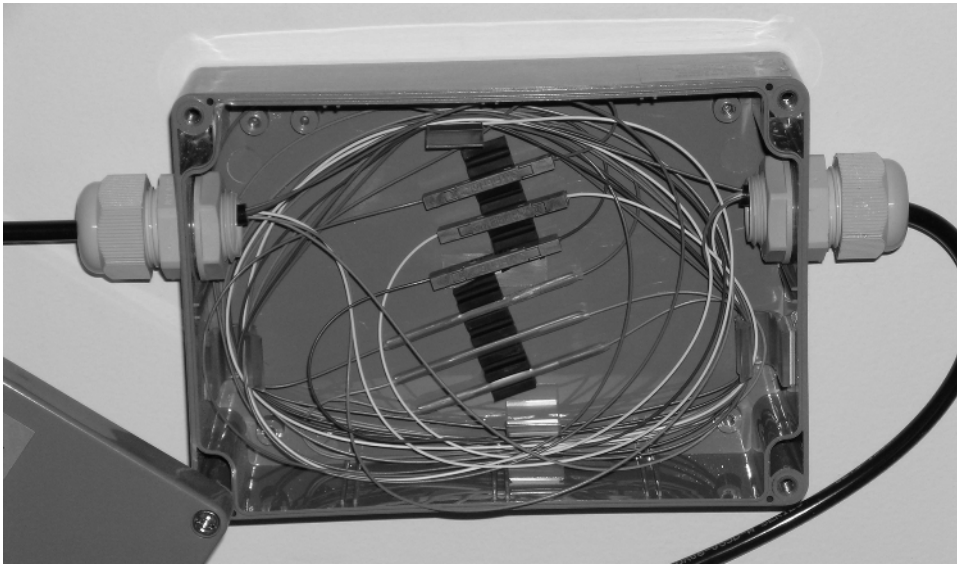
Splice Enclosures

Any time you have a splice in an optical fiber, whether it is a mechanical or a fusion splice, you must protect it from exposure and strain. Splice enclosures such as the one shown in Figure 13.3 take many forms, depending on their location and specific application. Some have been adapted from electrical splice enclosures used in the telecommunications industry for aerial and underground cable, while others are designed specifically for optical fibers and are used in internal installations.

Typically, splice enclosures will incorporate the following features:

- Strain relief that ensures that the strength member will carry all of the tensile loading
- An organizing panel that holds the actual splices in an orderly fashion
- Space for looping extra cable in case another splice is required

FIGURE 13.3 A splice enclosure protects splices from damage and tensile loads.



Splice enclosures may be radial, meaning that a cable enters from one direction and the cable to which it is spliced exits in another direction, or axial, in which the cables enter and leave in the same direction.

Patch Panels

A patch panel, as seen in Figure 13.4, allows signals to be routed from fiber cables to different destinations by plugging in, or patching, connectors to fixed points on the panel.

FIGURE 13.4 A patch panel allows fiber connections to be changed.

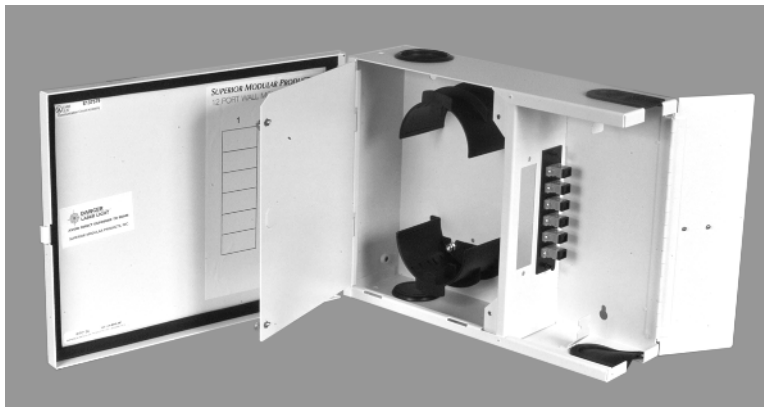


Photo courtesy of Norfolk Wire & Electronics

Patch panels are often used as termination points for cables carrying many fibers when the signals from those fibers must be distributed throughout an office or other work area. Typically, the fibers from the cable have a breakout or fanout kit attached and the connectors are hooked into one side of the patch panel. These attachments may be permanent or movable, depending on the amount of flexibility desired for the panel.

On the other side of the panel are spaces for patch cords to carry signals to equipment or further distribution, as necessary.

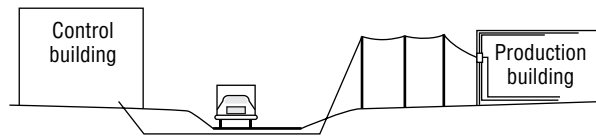
Installation Methods

Optical fiber has already reached into most of the places that once only knew copper cable. As technology, regulations, and pricing permit, fiber will ultimately replace copper for most signal-carrying applications, even into the home.

Many fiber installations resemble those used for copper wiring and have built on the lessons learned from it. There are some installation requirements and methods unique to fiber, however, that are required to protect the cable and ensure the highest quality transmission.

Let's look at a typical application for optical fiber that uses a variety of installation methods. In our example, shown in Figure 13.5, a manufacturing plant is using fiber to carry instrumentation and control signals between the production building and another building several hundred meters away (called the control building in the diagram). The cable carrying the signals must be collected in a central area and routed into a cable that runs through trays and ductwork to the outside. The cable then runs from the building and is strung across several poles until it reaches a road. There, the cable runs underground until it enters the next building and is distributed to data collection and control systems.

FIGURE 13.5 A sample fiber installation scenario



Let's look at installation for each of these situations.

Tray and Duct

Tray and duct installation is used inside structures and is similar to installation methods used for electrical wiring. Because most optical fiber is nonconductive, however, and is typically lighter than copper wiring, some of the requirements and restrictions for copper do not apply to fiber.

When the fiber cable rests in trays or horizontal ductwork, as shown in Figure 13.6, the weight of the cable is usually not a factor as long as the runs remain on the same level. If cabling is run vertically, however, the cable will have to support itself for short runs or be secured using either cable clamps or hangers. Be sure to follow the cable manufacturer's specifications for cable rise.

If possible, install clamps for vertical runs every 1 to 2 m to keep tensile loading to a minimum. If long unsupported vertical runs are necessary, you must clamp simplex cable every 210 m, and duplex cable every 300 m at the absolute maximum.

There are two methods for clamping cables for vertical runs. The first, which can also be used to secure cables horizontally, are clamps that secure the cable directly to the surface by placing pressure against it. These should be installed carefully to prevent crushing the cable with excessive force.

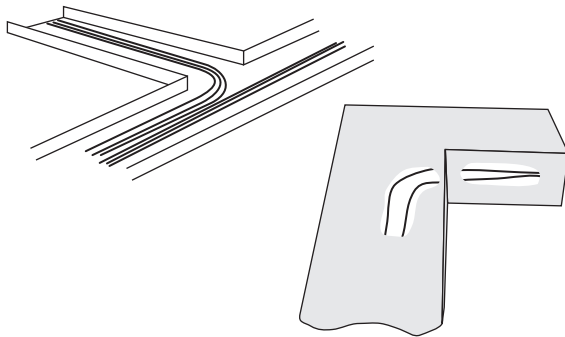
The second type of vertical clamp is a tight wire mesh that wraps around the cable and is secured to a hanger. This method has the advantage of reducing pressure against the cable itself while still supporting it. It also allows the cable to be removed more easily, since the mesh wrap is slipped over the hanger, not permanently attached.

When installing cable in a tray, be aware of whether the tray will also be occupied by other cables, especially electrical cables. Aside from safety considerations surrounding the use of conductive or hybrid optical fiber cables, the weight of the much larger copper cables can cause problems if they are piled on top of the fiber. If there is a risk of this happening, you may want to use armored cable to protect the fiber inside from the crushing weight of other cables.

If you are laying the cable into the tray, rather than pulling it, you can meet the manufacturer's minimum bend radius specification. If you will be pulling it, however, be sure to leave some extra radius to account for the tensile loading.

Whether you are installing in a tray or in a duct, do not allow the fiber to contact any sharp edges or bends. Ideally, you should be working with components that do not have these hazards built in, but in most cases you will have to account for them by keeping cable tension as low as possible during installation.

FIGURE 13.6 Tray and duct installation



Conduit

Conduit installation uses dedicated conduits for the cable runs, which are installed by feeding pulling lines through the conduit, attaching them to a pulling eye attached to the cable's strength member, and then pulling the cable through the conduit.

Conduit may be run inside structures or underground, and in many cases conduits may already be in place for other applications, such as power or telephone lines. When you are installing cable in conduits within a structure, be sure that you have allowed enough room in the conduit for the amount of cable you are installing, as shown in Figure 13.7. If possible, account for future expansion by installing extra fiber at the same time as the initial installation. This will prevent the need for trying to run cable through conduits that are already populated.

The National Electrical Code (NEC) specifies the following fill ratios by cross-sectional area for conduit:

1 cable – 53%

2 cables – 31%

3 or more cables – 40%

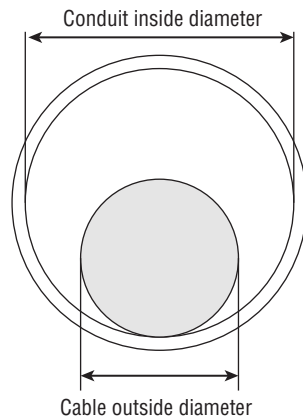
To determine the fill ratio, use the following formula:

$$\text{Fill Ratio} = (OD^2\text{cable}_1 + OD^2\text{cable}_2 \dots / ID^2\text{conduit})$$

where *OD* is the outside diameter of the cable and *ID* is the inside diameter of the conduit.

If a large conduit is already in place and contains other cables, or if it is likely that other cables will be run through the conduit in the future, you may want to install an *innerduct*, a conduit sized for the optical fiber cable that will protect it from other activities in the larger conduit.

FIGURE 13.7 The conduit must leave room for fiber to be pulled.



EXERCISE 13.1**Determine the minimum conduit size for a cable with an OD of 0.4 in.**

Use the formula for determining fill ratio:

$$\text{Fill Ratio} = (OD^2\text{cable}_1 + OD^2\text{cable}_2 \dots / ID^2\text{conduit})$$

Apply the NEC standard of 53% for a single cable.

$$0.53 = (0.16 / ID^2\text{conduit})$$

$$ID^2\text{conduit} = 0.302$$

$$ID\text{conduit} = 0.55"$$

The inside diameter of the conduit must be at least 0.55", but in practice it is always a good idea to allow more room for pulling to keep tension to a minimum and allow room for turns.

Direct Burial

To run our cable under the roadway, we can install it by direct burial. As the name implies, this method can be as simple as placing a suitable cable directly in the ground. Direct burial methods also include placing a cable within a protective pipe and burying it.

For short runs, a backhoe can be used to dig a trench in which the cable is laid and then covered up. This may be more effective where a narrow road has to be torn up as well. For longer runs that require more efficient methods, a cable-laying plow is available. This device is designed to open a trench, lay the cable, and cover it up again while on the move. It is a more complicated machine, but it is useful when long distances must be covered.

When using direct burial methods, be sure to dig the trench deep enough to be below the frost line. In some areas, this can be as much as 30 inches deep. Remember that underground cable runs must also be protected from rain and other water seepage, digging and trenching operations, gnawing animals, and earth movement.

Aerial

As our cable leaves the production building, it's going to be spliced onto a fiber that is part of a messenger cable, which will be strung along a series of poles in an aerial installation. These poles already carry power lines, but they will not affect the fiber itself because it is not subject to interference from electrical induction.

If a messenger cable is not used, the cable will have to be lashed to steel wire running between the poles to get the necessary support. Either way, cables in aerial installations must be able to withstand loading from high winds, ice, birds and climbing animals, and even windblown debris such as branches.

Blown Fiber

In new construction or renovation, it's a good idea to consider blown fiber as an alternative to conventional methods. Blown fiber, shown in Figure 13.8, can be installed in long runs of conduit without the hazards and strains associated with pulling.

In a blown fiber installation, a special conduit about 5 mm in diameter is installed to carry the fiber during construction. Once construction is complete and it is time to install the fiber, the blown fiber cable is fed into the conduit and air is blown through the system, carrying the cable with it. Using this method, fiber can be blown up to 1000 feet or more through a conduit system.

FIGURE 13.8 Blown fiber uses special components to make installation easier.



Electrical Safety

Even though optical fiber cable itself does not carry electrical power, there may be some circumstances in which you will have to contend with electricity. If a cable contains conductive components such as armor or metal strength members, and the cable is likely to come in contact with electrical circuits, there is a chance that the metal could become a path for an electrical current in an accident, potentially leading to fire or personal injury.



Real World Scenario

Leaving Slack for Repairs

No matter where you install cable, you'll want to leave some slack for the inevitable repairs that will have to be performed. How much slack should you leave? That depends on the situation. Look at the location, the number of turns that the fiber takes as it passes through conduits, trays or ductwork, and the likelihood that the fiber will be disturbed after you have installed it. Consider the following cautionary tale:

During the construction of a strip mall in Virginia, cable installers ran cable underground in an area where construction was ongoing. After the cable was laid, backhoe operators accidentally pulled up the conduit carrying the cable seven times, requiring the cable to be spliced and relaid in the ground. After the sixth event, all of the slack in the cable had been used up in repairs and the cable had to be replaced.

For the record, the location of the cable was clearly marked.

So how much slack should you leave in such an environment? You can't justify planning for an unlimited number of backhoe accidents, but you can prepare for the first few and reduce the chance that the cable will have to be replaced.

Consider that a backhoe operator or other observer may not notice that a conduit has been pulled up until it has cleared the ground by a foot or more. Add to that the depth of the burial—let's use three feet for our example—and the width of the backhoe bucket, which may be another two to three feet. Remember that for every foot that the cable is lifted, two feet of cable are actually being pulled up, because it's happening on both sides of the bucket.

Now you have, as a worst-case scenario, eight feet of cable plus the amount lying across the bucket, giving you a potential of 11 feet of cable that you'll want to add to the length of a run, just to protect it from one chance encounter with a backhoe.

When you leave the slack, don't forget to leave it in such a way that it will pay out easily if it's pulled. If you don't, it can get tangled and you'll lose any advantage you might have had by providing all of that extra length.

In the event that a conductive component in a cable comes in contact with an electrical current, that current is going to seek the most direct path to the ground. If you happen to touch the cable or anything to which the cable is connected, that path could go through you. Depending on the voltage and current involved, you could face severe injury or even death.

Plant electrical systems provide a safeguard against such accidents by joining all of the non-current-carrying components together and tying them to ground. If a current-carrying wire accidentally comes in contact with any non-current-carrying component, the circuit is

completed and an overcurrent protection device such as a circuit breaker shuts the system down completely.

For this reason, NEC article 250 requires that any electrically conductive materials that are likely to become energized be grounded and bonded. Grounding provides a direct path to the ground for the current to follow in case conductive components are energized. Because the human body is not as efficient a conductor as copper, which is used in grounding, the current will follow the grounding path rather than pass through you.

Bonding provides a permanent electrical path between conductive materials so that in the event any of the materials are accidentally energized, the current is immediately routed back to the power supply ground circuit, tripping the circuit breaker and shutting off power to the system.

Hardware Management

Eventually, fiber optic cables end up in some kind of cabinet, panel, or enclosure. Whether it's a patch panel, rack, splice enclosure, or wall outlet, the preparation of these items is crucial to the performance of the entire network. In these locations, the fiber or cable is terminated and connected in some way, either to another fiber, a connector, or a piece of hardware, and this process provides the greatest likelihood for mistakes and mismanagement.

Poor hardware preparation can lead to confusion, poor connections, excess strain on cables and fibers, and inefficient troubleshooting. In addition, a poorly organized hardware space reflects badly on the installer.

Here are some guidelines for good hardware management:

Cleanliness

A clean working environment is essential to fiber work. Whether you are splicing fibers, terminating cables, or mounting cables in cabinets, it's important to keep dust, trash, water, and other contaminants away from your work area.

If possible, block off the space in which you are working to prevent exposure to contaminants. Use air filters to draw dust and grit out of the air, and try to limit the amount of nonessential work traffic in the area where you are working. Crews have been known to park their work trucks over manholes to keep dirt from the street from falling inside while they work. In other areas, it may be enough to use good housekeeping practices to ensure that the work area won't be contaminated.

Organization

Once a cable enters any kind of enclosure, it is likely to be joined by anywhere from one to dozens of other cables. If these cables are not organized from the very beginning, they can quickly become a rat's nest of tangled, interwoven lines.

Complicating things further, it is important to leave extra cable available at the end of a run in case splices are necessary, so there must be some way to contain the extra cable.

Most manufactured enclosures have built-in cable management devices, including clips, attachment points or studs for ties, and moldings for looping cable. These features will help you organize cable for a cleaner, more efficient layout.

When you are running cable into an enclosure, don't just take the shortest route from the entry point to the cable's intended port. Run the cable so that it is bundled with others going to the same area, and never run cables so that they are blocking access to other ports in the cabinet.

Secure cables in place wherever possible. This helps relieve strain on the cable and keeps it from assuming new and interesting configurations after the cabinet is closed.

Organization and neatness are also essential for efficient troubleshooting. It's much easier to trace connections and links if the cables are organized and neatly bundled than if they are spread all over the cabinet and tangled together.

Labeling

Labeling, as we will see later in this chapter, is key to a well-run network. Labeling helps you find the right port or cable quickly and easily, and it makes troubleshooting more efficient.

All cables must be labeled within 300 mm (about 12") of their ends. The labeling must be done in a consistent and easily readable format, which will be described shortly.

It's also important to label ports and outlets with labels that will not come off easily. In large systems, you can lose a great deal of time just trying to remember which outlet is connected to which panel in a room at the other end of the building.

Documentation

Even after the cable is properly organized and labeled, you'll need to keep good records of the network, down to the designations and locations of each port and cable. Good documentation can speed up troubleshooting and system repair and modification, and even improve safety, because it reduces the amount of time spent climbing around in spaces just to find out where a cable leads.

Documentation, like labeling, needs to follow a consistent and easily readable format, which is described in TIA/EIA-606-A. It should identify not only the cables and hardware but the spaces occupied by the network and the locations of outlets and type of installation used.

Labeling Requirements

Even the simplest fiber optic network can instantly become confusing if it is improperly or insufficiently labeled to show where cables originate, how they are connected, and where they go.

It may be tempting to come up with a handy system of your own for labeling, but unless you can predict all of the changes that will take place as the system grows, you will soon find yourself with a cumbersome mess on your hands. In addition, if you don't keep track of your labeling system, it will be difficult if not impossible for others to decipher it.

The TIA/EIA standard for labeling is contained in TIA/EIA-606-A. This document contains detailed information on conventions for labeling optical fiber connections within buildings and between buildings that share a communication network. The standard specifies the methods used to determine how fiber links are identified and a numbering/lettering system that is standardized according to a top-down classification.

As an example of labeling according to TIA/EIA-606-A, imagine working with a wall outlet that has a port labeled:

1A-B002

This particular number, known as a horizontal link identifier, immediately communicates the following information:

1 – The fiber originates in a telecommunications space (TS) on the first floor of the building.

A – The TS is designated as A.

B – The fiber originates in patch panel B in TS A.

002 – The fiber originates in port 002 of the patch panel.

Using this labeling system, we can very quickly trace the fiber to a specific connection point, even in a very large building.

While the above example is fairly simplistic, it shows that by using a consistent labeling system, it is possible to communicate fiber routings accurately. Other numbers may be used to show *backbones*, or links between telecommunication spaces, interbuilding cabling, and other major connections.

Summary

This chapter covered equipment and methods used in installing fiber optic cable. We described specifications used by manufacturers to ensure that their cables continue to operate properly and the ways to meet those specifications during installation.

The chapter discussed hardware commonly used in cable installation to make installation easier and reduce the potential for damage to the cable. We also described installation methods used in some common scenarios.

The chapter described safety measures to be employed when installing cable with conductive components and the reasons for these measures.

We described techniques for ensuring proper installation of cable in various enclosures, and we discussed the importance of organization and preparation of these enclosures for proper operation of the network.

Finally, we described the standard for labeling cables and ports to ensure a consistent and easily readable system.

Exam Essentials

Be familiar with installation specifications. Make sure that you understand the manufacturer's specifications for minimum bend radius and maximum tensile load. Be able to describe methods for maintaining these standards during installation.

Be familiar with installation hardware. Be able to describe hardware used during the installation process and as part of the permanent cable installation. Make sure that you understand how the hardware is used and what role it serves in installation.

Understand installation methods. Be able to describe different installation methods in common use. Be able to describe requirements for each type of installation.

Understand the importance of grounding and bonding. Be able to describe grounding and bonding. Understand how these procedures are used when installing fiber optic cables. Be able to describe the conditions that require grounding and bonding.

Be familiar with hardware preparation. Understand the importance of preparing hardware for cable installation. Understand methods used to keep the work environment clean and the cable organized. Be able to describe organization techniques. Understand the importance of labeling. Be able to describe the labeling standard for cables.

Review Questions

1. A cable's minimum bend radius specification is greater:
 - A. During installation
 - B. After installation
 - C. When the temperature is higher
 - D. When it is in a cable tray
2. A cable's maximum tensile load rating:
 - A. Is lower for installation than for long-term use
 - B. Is higher for installation than for long-term use
 - C. Is the same for all conditions
 - D. Does not apply during installation
3. Tensile loads encountered during installation are also called _____.
 - A. Static loads
 - B. Duct loads
 - C. Horizontal loads
 - D. Dynamic loads
4. A pulling eye is attached to:
 - A. The cable's outer jacket
 - B. The cable's strength member
 - C. The fiber
 - D. The armor
5. A pullbox is used to:
 - A. Pull multiple cables through a conduit.
 - B. Pull conduit into place.
 - C. Relieve tensile loads on cable during pulling.
 - D. Hold cable taut until pulling can resume.
6. When is a splice enclosure used?
 - A. Whenever a fiber has been spliced
 - B. When a splice must be placed underground
 - C. When a splice must be placed underwater
 - D. Splice enclosures are optional.

7. What is the purpose of a patch panel?
 - A. It takes the place of a splice enclosure.
 - B. It fills a hole where fiber has been installed.
 - C. It is used to route signals between cables and other fiber hardware.
 - D. It provides a permanent link between two pieces of hardware.
8. A fiber cable in a tray may have to be armored if:
 - A. The tray is not horizontal.
 - B. The fiber has to be pulled through the tray.
 - C. The tray is metal.
 - D. It is sharing the tray with electrical cables.
9. The most common method for installing cable in a conduit is _____.
 - A. Laying
 - B. Pulling
 - C. Pushing
 - D. Pre-installing
10. Installing a cable by opening a trench and laying it in the ground is also called _____.
 - A. Trenching
 - B. Grounding
 - C. Layering
 - D. Direct burial
11. The type of cable commonly used in aerial installations is called _____.
 - A. Messenger cable
 - B. Aerial cable
 - C. Strung cable
 - D. Flying cable
12. Blown fiber installation is an alternative to _____.
 - A. Direct burial
 - B. Aerial installation
 - C. Pulling
 - D. Cable trays

- 13.** Bonding is required for:
- A.** Any cable that is run vertically
 - B.** Any cable that is run near power lines
 - C.** Any cable with conductive, non-current-carrying components
 - D.** Any cable that carries electricity
- 14.** When working with splices, connectors, and enclosures, it's important to have:
- A.** A clean environment
 - B.** Complete silence
 - C.** Plenty of help
 - D.** An independent power source
- 15.** When installing a fiber optic network, it is important to label the components in accordance with:
- A.** Your personal cable identification system
 - B.** The company's internal labeling system
 - C.** NEC article 250
 - D.** TIA/EIA-606-A

Answers to Review Questions

1. A. A cable's minimum bend radius is greater when there is higher tension on the cable, which is likely to occur during installation.
2. B. A cable's maximum tensile load rating is higher for installation because the loads are not expected to last for the life of the cable.
3. D. Because the loads encountered during installation are not constant and occur as a result of short-term forces being applied to the cable, they are referred to as dynamic loads.
4. B. The pulling eye is attached to the strength member, which is designed to take the tensile loads imposed on the cable while it is being pulled.
5. C. A pullbox is used in long conduit runs or after multiple turns in a conduit to provide an intermediate pulling location for the cable, which relieves the tensile loads that would be imposed if the cable had to be pulled through the entire system at once.
6. A. A splice enclosure is used whenever two fibers have been spliced to protect the splices and to take tension off the cables.
7. C. A patch panel allows signals to be routed from fiber cables to different destinations through the use of patch cords.
8. D. If the fiber cable is running through a tray with electrical cables, the weight of the electrical cables may be enough to crush the fiber if it is not armored.
9. B. Most cable is installed in conduit by pulling it through with a line attached to a pulling eye, which is attached to the cable's strength member.
10. D. Direct burial is a method of laying a fiber optic cable directly in the ground. The cable must be suitably protected from damage and from the elements.
11. A. Messenger cable is used in aerial installations. It has a messenger line built into the cable to take the loads imposed on cable that is strung between poles.
12. C. Blown fiber uses pressurized air to blow a fiber cable through conduits without the need for pulling.
13. C. Bonding ties all conductive, non-current-carrying components together so that if they are accidentally energized, current will return from them to the power source and trip a circuit breaker. It is required by NEC article 250.
14. A. A clean environment is essential to working with splices, connectors, and enclosures. Contaminants, moisture, and dirt can make the job more difficult and harm sensitive components.
15. D. TIA/EIA-606-A describes the standard labeling system used to identify fiber networks for almost any size system. The standard makes it easy to identify the exact origin or destination of a cable anywhere in a building or group of buildings.



Chapter

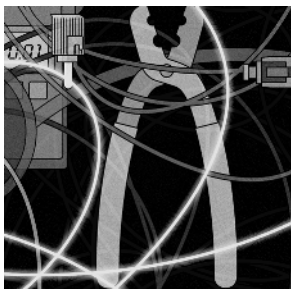
14

Fiber Optic System Design Considerations

OBJECTIVES COVERED IN THIS CHAPTER:

✓ Fiber Optic System Design Considerations

- List the considerations for a basic fiber optic system design.
- Compare the bandwidth advantages of optical fiber over twisted pair and coaxial copper cables.
- Compare the attenuation advantages of optical fiber over twisted pair and coaxial copper cables.
- Explain the electromagnetic immunity advantages of fiber optic cable over copper cable.
- Describe the weight-saving advantages of fiber optic cable over copper cable.
- Describe the size advantages of fiber optic cable over copper cable.
- Describe the security advantages of fiber optic cable over copper cable.
- Compare the safety advantages of fiber optic cables over copper cables.
- Explain how to apply TIA/EIA-568-B.3 to analyze multimode fiber optic link performance.
- Explain how to apply TIA/EIA-568-B.3 to analyze single-mode fiber optic link performance.
- Prepare a basic optical link power budget and explain its importance.



Up to this point, you have learned about transmitters, receivers, couplers, attenuators, connectors, splices, and fiber optic cable. These pieces are part of the building block of a fiber optic system.

A basic fiber optic system contains a transmitter, receiver, fiber optic cable, and connectors.

There are many ways to approach fiber optic system design and there are many different fiber optic systems. Fortunately there are industry standards that simplify fiber optic system design. This chapter focuses on the basic design considerations for a fiber optic system, compares optical fiber to copper, explains how to break down a fiber optic link to analyze performance, and shows how to prepare a power budget.

Basic Fiber Optic System Design Considerations

Before beginning a basic fiber optic system design, two questions need to be answered. How much data needs to be moved and what is the transmission distance? Throughout this book, you have learned how the physical properties of light and of the optical fiber determine bandwidth and transmission distance. Now let's take some of these lessons learned and apply them in the design of a basic fiber optic system.

We mentioned earlier that before you begin a design, the data rate and transmission distance must be known. Let's look at two rules of thumb that will help you design a link to meet your data rate and transmission distance expectations. Rule of thumb number one: For data rates up to 155 Mbps with a transmission distance no greater than 2000 m, choose an LED transmitter and multimode optical fiber. Rule of thumb number two: For data rates greater than 155 Mbps and transmission distances greater than 2000 m, choose a laser transmitter and single-mode optical fiber.

You may remember from Chapter 10, "Fiber Optic Light Sources," that LED transmitters are typically not designed for data rates exceeding 155 Mbps, whereas a laser transmitter that supports data rates from 155 Mbps through 2.5 Gbps is readily available. Laser transmitters are also available to support data rates greater than 2.5 Gbps.

The two rules of thumb just discussed are not all-encompassing. They are intended to be a general guideline. There are many applications where these rules of thumb do not apply; however, those applications are not basic systems.

The Advantages of Optical Fiber over Copper

So far in this chapter, you have learned the rules of thumb for a basic system design without hearing about any of the factors that affect system performance. The next section will look at seven different performance areas within a system and compare the performance of optical fiber to Category 5e and coaxial copper cable.

Performance in this comparison will be evaluated in the areas of bandwidth, attenuation, *electromagnetic immunity*, size, weight, safety, and security. The fiber optic system will operate at a 1300 nm wavelength with an LED source. System performance data for the optical fiber and Category 5e copper cable will be derived from Commercial Building Telecommunication Cabling Standard, ANSI/TIA/EIA-568-B-2001.

Bandwidth

“Bandwidth” is very popular buzzword these days. We are bombarded with commercials advertising high-speed downloads from a *cable modem* or *digital subscriber line (DSL)* or satellite. The competition to sell us bandwidth is fierce. With the *Federal Communications Commission (FCC)* recently approving fiber to the home, soon we will have more bandwidth available to us than ever before.

In earlier chapters, we looked at the physical properties of the optical fiber and fiber optic light source that limit bandwidth. You learned that single-mode systems with laser transmitters offer the greatest bandwidth and that multimode systems with LED transmitters offer the least. You also learned that the bandwidth of the optical fiber is inversely proportional to its length. In other words, as the length of the optical fiber increases, the bandwidth of the optical fiber decreases.

So how does length affect the bandwidth of a copper cable? Well, when it comes to cable length, copper suffers just like optical fiber does. As the length of a copper cable is increased, the bandwidth for that cable decreases. So if both copper and optical fiber lose bandwidth over distance, why is optical fiber superior to copper? To explain that, we will look at the minimum bandwidth requirements defined in TIA/EIA-568-B.3 for multimode optical fiber and TIA/EIA-568-B.1 for *unshielded twisted-pair (UTP) Category 5e* cable. Remember that the values defined in TIA/EIA-568-B.1 and B3 are the minimum values that a manufacturer needs to achieve. There are many manufacturers offering optical fiber and copper cables that greatly exceed these minimum requirements. However, for this comparison, only values defined in these two standards will be used.

One of the problems we run into doing this comparison is cable length. TIA/EIA-568-B.1 defines Category 5e performance at a maximum physical length of 100 m. TIA/EIA-568-B.3 does not define a maximum bandwidth at a maximum optical fiber length. TIA/EIA-568-B.3 defines multimode optical fiber bandwidth in MHz•km. To do this comparison, then, we will need to calculate the bandwidth limitations for 100 m of multimode optical fiber using the formula bandwidth multiplied by length is less than or equal to 500 MHz:

$$(BW \times L \leq 500 \text{ MHz})$$



TIA/EIA-568-B.3 section 5.5.1 states that Category 5e multipair balanced 100 Ω backbone cabling that supports data applications up to 100 MHz should be limited to a total distance of 90 m. The 90 m distance allows for an additional 5 m at each end for equipment cables or patch cords. The total combined length cannot exceed 100 m.

Table 14.1 contains the optical fiber transmission performance parameters defined in section 4.2 of TIA/EIA-568-B.3. For this comparison, we will look at the performance of a 62.5/125 μm optical fiber at 1300 nm. The minimum information transmission capacity for this optical fiber is 500 MHz•km. You learned earlier that modal dispersion is primarily responsible for the bandwidth limitations of multimode optical fiber. The bandwidth-limiting effects of modal dispersion decrease as optical fiber length decreases. This 100 m optical fiber would have a bandwidth 10 times greater than 500 MHz, which is 50 times greater than the Category 5e UTP cable.

TABLE 14.1 Optical Fiber Cable Transmission Performance Parameters

Optical Fiber Cable Type	Wavelength (nm)	Maximum Attenuation (dB/km)	Minimum Information Transmission Capacity for Overfilled Launch (MHz•km)
50/125 μm multimode	850	3.5	500
	1300	1.5	500
62.5/125 μm multimode	850	3.5	160
	1300	1.5	500
Single-mode inside plant cable	1310	1.0	N/A
	1550	1.0	N/A
Single-mode outside plant cable	1310	0.5	N/A
	1550	0.5	N/A

You will notice in Table 14.1 that TIA/EIA-568-B.3 does not define information transmission capacity for single-mode optical fibers. So how would a single-mode optical fiber stack up to a Category 5e UTP cable at 100 m? Without pushing the bandwidth performance envelope, there are single-mode systems available that provide Gbps performance at 80 km transmission distances. An 80 km optical fiber is 800 times longer than a 100 m Category 5e UTP cable. A Gbps represents a 500 MHz bandwidth, or information transmission capacity. The single-mode system is thus capable of moving five times the data over a distance 800 times greater than the Category 5e UTP cable—and we are not even looking at a high-performance system. Optical fiber offers incredible bandwidth advantages over Category 5e UTP cable.



How do we get from Gbps to MHz? Remember that in Chapter 2, “Principles of Fiber Optic Transmission,” we learned about bits and symbols. Baud rate is the number of symbols per second. For most applications, the baud rate equals the bit rate. In other words, a 1 Gbps transmission would transmit one billion symbols per second. As you learned in Chapter 2, one complete cycle of a square wave contains two symbols, a high symbol and a low symbol. So a 500 MHz square wave transmits one billion symbols per second, or 1 Gbps.

Now that we have seen how the bandwidth of multimode optical fiber with a 1300 nm source greatly exceeds that of Category 5e cable, let’s do a comparison with RG6 *coaxial cable*. In this comparison, the performance data for the RG6 coaxial cable is taken from the average of several manufacturers’ data sheets. TIA/EIA-568-B does not define performance parameters for RG6 coaxial cable.

An RG6 coaxial cable with a transmission frequency of 1 GHz has roughly the same 100 m transmission distance characteristics as a Category 5e cable. We know from the previous comparison that a 100 m length of 62.5/125 μm multimode optical fiber will support transmission frequencies up to 5 GHz over that same distance with a 1300 nm light source. In this comparison, the multimode optical fiber offers a bandwidth advantage five times greater than the RG6 coaxial cable. However, the problem with this comparison is that there are no LED transmitters available that support 1 GHz or 5 GHz transmission frequencies. These transmission frequencies would have to be accomplished with a laser transmitter.

These comparisons have demonstrated the bandwidth advantages of optical fiber over copper cable. The comparisons were done at very short distances because at this point we have not addressed the effects of attenuation in a copper cable or optical fiber. The effects of attenuation limit the bandwidth of copper cables, though not of optical fiber. The next comparison will describe the effects of attenuation on copper cable and optical fiber in great detail. Once you understand the effects of attenuation, you will see just how much more bandwidth over distance an optical fiber offers over a copper cable.

Attenuation

All transmission mediums lose signal strength over distance. As you know, this loss of signal strength is called attenuation and is typically measured in decibels. Optical fiber systems measure attenuation using optical power. Copper cable systems typically use voltage drop across a defined load at various transmission frequencies to measure attenuation. The key difference here is not that optical fiber uses power and copper uses voltage. The key difference is that attenuation in copper cables is measured at different frequencies. This is not the case with optical fiber, where attenuation is measured with a continuous wave light source that is not modulated.

The attenuation in a copper cable increases as the transmission frequency increases. Table 14.2 shows the maximum worst pair insertion loss for a 100 m horizontal Category 5e and Category 3 cable as defined in TIA/EIA-568-B.2 section 4.3.4.7. This table clearly shows the effects that transmission frequency has on a copper cable.

TABLE 14.2 Horizontal Cable Insertion Loss, Worst Pair*

Frequency (MHz)	Category 3 (dB)	Category 5e (dB)
0.772	2.2	1.8
1.0	2.6	2.0
4.0	5.6	4.1
8.0	8.5	5.8
10.0	9.7	6.5
16.0	13.1	8.2
20.0		9.3
25.0		10.4
31.25		11.7
62.5		17.0
100.0		22.0

* For a length of 100 m (328 ft)

The maximum allowable attenuation in an optical fiber is defined in section 4.2 of TIA/EIA-568-B.3. Table 14.3 is the attenuation portion of the optical fiber cable transmission performance parameters. This table defines attenuation for both multimode and single-mode optical fibers. You will notice that there is no column for transmission frequency in this table. That is because optical fiber does not attenuate as transmission frequency increases.

Now that we know how optical fiber and copper cable attenuate, let's do a comparison. The first comparison will put optical fiber up against Category 5e cable. The distance will be 100 m and the transmission frequency will be 100 MHz.

Looking at Table 14.2, you'll see that the worst-case attenuation for the Category 5e cable at 100 MHz is 22 dB. What this means is that the Category 5e cable loses 99.37% of its energy over a distance of 100 m at that transmission frequency. If you look at Table 14.3 and do the math, you'll see that multimode optical fiber being operated at 1300 nm has a loss of only 0.15 dB. The optical fiber loses only 3% of its light energy over the 100 m distance. This means that the Category 5e cable loses 150 times more energy than the optical fiber.

TABLE 14.3 Optical Fiber Cable Attenuation Performance Parameters

Optical Fiber Cable Type	Wavelength (nm)	Maximum Attenuation (dB/km)
50/125 μm multimode	850	3.5
	1300	1.5
62.5/125 μm multimode	850	3.5
	1300	1.5
Single-mode inside plant cable	1310	1.0
	1550	1.0
Single-mode outside plant cable	1310	0.5
	1550	0.5

That comparison clearly shows the attenuation advantages of optical fiber over Category 5e cable. Now let’s compare the same optical fiber at the same wavelength to an RG6 coaxial cable. For this comparison, the RG6 attenuation characteristics are taken from the average of several different manufacturers’ published data sheets, as shown in Table 14.4.

TABLE 14.4 RG6 Cable Insertion Loss*

Frequency (MHz)	RG6 (dB)
1.0	1.0
5.0	1.8
10.0	2.3
20.0	3.3
50.0	5.0
100.0	6.2
200.0	9.2
300.0	11.0
400.0	12.5
500.0	14.0

TABLE 14.4 RG6 Cable Insertion Loss* (*continued*)

Frequency (MHz)	RG6 (dB)
1000.0	19.4
3000.0	35.1

* For a length of 100 m (328 ft)

As you look at Table 14.4, you can see that RG6 coaxial cable easily outperforms Category 5e cable. However, it does not even begin to approach the multimode optical fiber operating at 1300 nm. The RG6 coaxial cable has 14 dB of attenuation at a transmission frequency of 500 MHz over a distance of 100 m. Again, the multimode optical fiber operating at 1300 nm has 0.15 dB of attenuation.

Whereas the Category 5e cable lost 99.37% of its signal strength, in the above comparison the RG6 coaxial cable lost only 74.88% of its energy over the same distance. Remember that the RG6 coaxial cable is being compared to the optical fiber at five times the transmission frequency used for the Category 5e comparison. As you know, the multimode optical fiber operating at 1300 nm loses only 3% of its light energy over the 100 m distance. This means that the RG6 coaxial cable lost roughly 25 times more energy than the multimode optical fiber.

These comparisons should make it clear that optical fiber has an enormous attenuation advantage over copper cable. The comparisons were done only with multimode optical fiber. Outside plant single-mode optical fiber greatly outperforms multimode optical fiber.

Single-mode optical fiber links are capable of transmission distances greater than 80 km without re-amplification. You can see from the above comparisons that Category 5e cable would require re-amplification every 100 m at a transmission frequency of 100 MHz. RG6 coaxial cable would only require re-amplification every 300 m at a transmission frequency of 100 MHz.

An 80 km RG6 coaxial cable link would require roughly 264 amplifiers for a 100 MHz transmission frequency; the single-mode optical fiber link would only require one at transmission frequencies greatly exceeding the RG6 coaxial cable. This means that the RG6 coaxial cable link is 266 times more likely to fail than the optical fiber link. Optical fiber links are unsurpassed in transmission distance and reliability.

Electromagnetic Immunity

Electromagnetic interference (EMI) is electromagnetic energy, sometimes referred to as *noise*, that causes undesirable responses, degradation, or complete system failure. Systems using copper cable are vulnerable to the effects of EMI because a changing electromagnetic field will induce current flow in a copper conductor. Optical fiber is a dielectric or an insulator, and current does not flow through insulators. What this means that EMI has no effect on the operation of an optical fiber.

Let's take a look at some examples of EMI-induced problems in a copper system. A Category 5e cable has four pairs of twisted conductors. The conductors are twisted to keep the impedance uniform along the length of the cable and to decrease the effects of EMI by canceling out opposing fields. Two of the conductor pairs are used to transmit and two of the conductor pairs are used to receive.

For this example, only one pair is transmitting and one pair is receiving. The other two pairs in the cable are not used. You can look at each pair of conductors as an antenna. The transmitting pair is the broadcasting antenna and the receiving pair is picking up that transmission just like your car radio antenna.

The data on the transmitting pair is broadcast and picked up by the receiving pair. This is called crosstalk. If enough current is induced into the receiving pair, the operation of the system can be affected. This is one reason why it is so critical to maintain the twists in a Category 5e cable.

Would we have this problem with an optical fiber? Does crosstalk exist in optical fiber? Regardless of the number of transmit and receive optical fibers in a cable assembly, crosstalk does not exist. To have crosstalk in an optical fiber cable assembly, light would have to leave one optical fiber and enter one of the other optical fibers in the cable assembly. Because of total internal reflection, under normal operating conditions light never leaves the optical fiber. Therefore, crosstalk does not exist in optical fiber cable assemblies.

Now let's take a look at another EMI scenario. Copper cables are being routed through a manufacturing plant. This manufacturing plant houses large-scale electromechanical equipment that generates a considerable amount of EMI, which creates an EMI-rich environment.

Routing cables through an EMI-rich environment can be difficult. Placing the cables too close to the EMI-generating source can induce unwanted electrical signals strong enough to cause systems to malfunction or stop operating. Copper cables used in EMI-rich environments typically require electrical shielding to help reduce the unwanted electrical signals. In addition to the electrical shielding, the installer must be aware of the EMI-generating sources and ensure that the copper cables are routed as far as possible from these sources.

Routing copper cables through an EMI-rich environment can be challenging, time-consuming, and expensive. However, optical fiber cables can be routed through an EMI-rich environment with no impact on system performance. The fiber optic installer is free to route the optical fiber as efficiently as possible. Optical fiber is very attractive to every industry because it is totally immune to EMI.

Size and Weight

The size of any cable must always be taken into consideration when preparing for an installation. Many times fiber optic cables will be run through existing conduits or raceways that are partially or almost completely filled with copper cable. This is one area where small fiber optic cable has an advantage over copper cable.

Let's do a comparison and try to determine the reduced-size advantage that fiber optic cable has over copper cable. As you learned in Chapter 4, "Optical Fiber Construction and Theory," a coated optical fiber is typically 250 μm in diameter. You learned that ribbon fiber optic cables sandwich up to 12 coated optical fibers between two layers of Mylar tape. Twelve of these ribbons stacked on top of each other form a cube roughly 3 mm by 3 mm. This cube can be placed

inside a buffer and surrounded by a strength member and jacket to form a cable. The overall diameter of this cable would be only slightly larger than an RG6 coaxial cable or a bundle of four Category 5e cables.

So how large would a copper cable have to be to offer the same performance as the 144 optical fiber ribbon cable? That would depend on transmission distance and the optical fiber data rate. Because we have already discussed Category 5e performance, let's place a bundle of Category 5e cables up against the 144 optical fiber ribbon cable operating at a modest 2.5 Gbps data rate over a distance of just 100 m.

A Category 5e cable contains four conductor pairs and as defined in TIA/EIA-568-B.2 is 0.25 inches in diameter. Each pair is capable of a 100 MHz transmission over 100 m. As you learned earlier in this chapter, a 100 MHz transmission carries 200 million symbols per second. If each symbol is a bit, the 100 MHz Category 5e cable is capable of a 200 Mbps transmission rate. When the performance of each pair is combined, a single Category 5e cable is capable of an 800 Mbps transmission rate over a distance of 100 m.

Now let's see how many Category 5e cables will be required to provide the same performance as the 144 optical fiber ribbon cable. The 144 optical fiber ribbon cable has a combined data transmission rate of 360 Gbps. When we divide 360 Gbps by 800 Mbps, we see that 450 Category 5e cables are required to equal the performance of this modest fiber optic system.

When 450 Category 5e cables are bundled together, they are roughly 5.3 inches in diameter. As noted earlier in this chapter, the 144 optical fiber ribbon cable is approximately the size of four Category 5e cables bundled together. The Category 5e bundle has a volume roughly 112.5 times greater than the 144 optical fiber ribbon cable. In other words, Category 5e bundles need 112.5 times more space in the conduit than the 144 optical fiber ribbon cable.

The comparison we just performed is very conservative. The distance we used was kept very short and the transmission rate for the optical fiber was kept low. We can get even a better appreciation for the cable size reduction fiber optic cable offers if we increase the transmission distance and the data rate.

In this comparison, let's increase the transmission distance to 1000 m and the data transmission rate to 10 Gbps. The bandwidth of a copper cable decreases as distance increases, just as with fiber optic cables. Because we have increased the transmission distance by a factor of ten, it's fair to say that the Category 5e cable bandwidth will decrease by a factor of ten over 1000 m.

With a reduction in bandwidth by a factor of ten, we will need ten times more Category 5e cables to equal the old 2.5 Gbps performance. In other words, we need 4500 Category 5e cables bundled together. In this comparison, however, the bandwidth has been increased from 2.5 Gbps to 10 Gbps. This means we have to quadruple the number of Category 5e cables to meet the bandwidth requirement. We now need 18,000 Category 5e cables bundled together. Imagine how many cables we would need if the transmission distance increased to 80,000 m. We would need a whopping 1,440,000 Category 5e cables bundled together.

The above comparisons vividly illustrate the size advantage that fiber optic cables have over copper cables. The advantage becomes even more apparent as distances increase. The enormous capacity of such a small cable is exactly what is needed to install high bandwidth systems in buildings where the conduits and raceways are almost fully populated with copper cables.

Now that we have calculated the size advantages of optical fiber over Category 5e cable, let's look at the weight advantages. It is pretty easy to see that thousands, tens of thousands, or millions

of Category 5e cable bundled together will easily outweigh a ribbon fiber optic cable roughly one half of an inch in diameter. It's difficult if not impossible to state that a fiber optic cable would weigh so many times less than a copper cable performing the same job—there are just too many variables in transmission distance and data rate. However, it's not difficult to imagine the weight savings that fiber optic cables offer over copper cables. These weight savings are being employed in commercial aircraft, military aircraft, and the automotive industries, just to mention a few.

Security

We know that optical fiber is a dielectric and because of that it is immune to EMI. So why is an optical fiber secure and virtually impossible to tap? Because of total internal reflection, optical fiber does not radiate.

In Chapter 5, “Optical Fiber Characteristics,” you learned about macrobends. Excessive bending on an optical fiber will cause some of the light energy to escape the core and cladding, and possibly penetrate the coating, buffer, strength member, and jacket. This energy is detectable by means of a *fiber identifier*.

Fiber identifiers detect light traveling through an optical fiber by inserting a macrobend. Photodiodes are placed against the jacket or buffer of the fiber optic cable on opposite sides of the macrobend. The photodiodes detect the light that escapes from the fiber optic cable. The light energy detected by the photodiodes is analyzed by the electronics in the fiber identifier. The fiber identifier can typically determine the presence and direction of travel of the light.

If the fiber identifier can insert a simple macrobend and detect the presence and direction of light, why is fiber secure? Detecting the presence of light and determining the source of the light does not require much optical energy. However, as you learned earlier in this chapter, a fiber optic receiver typically has a relatively small window of operation. In other words, the fiber optic receiver typically needs at least 10% of the energy from the transmitter to accurately decode the signal on the optical fiber. Inserting a macrobend in a fiber optic cable and directing 10% of the light energy into a receiver is virtually impossible. A macrobend this severe would also be very easy to detect with an *optical time domain reflectometer (OTDR)*. There is no transmission medium more secure than optical fiber.



The fiber identifier and OTDR are covered in detail in Chapter 15, “Test Equipment and Link/Cable Testing.”

Safety

Electrical safety is always a concern when working with copper cables. Electrical current flowing through copper cable poses shock, spark, and fire hazards. Optical fiber is a dielectric that cannot carry electrical current, hence it presents no shock, spark, or fire hazard.

Because optical fiber is a dielectric, it also provides electrical isolation between electrical equipment. Electrical isolation eliminates ground loops, eliminates the potential shock hazard

when two pieces of equipment at different potentials are connected together, and eliminates the shock hazard when one piece of equipment is connected to another with a ground fault.

Ground loops are typically not a safety problem. Ground loops are typically an equipment operational problem. They create unwanted noise that can interfere with equipment operation. A common example of a ground loop is the hum or buzz you hear when an electric guitar is plugged into an amplifier with a defective copper cable or electrical connection. Connecting two pieces of equipment together with optical fiber removes any path for current flow, which eliminates the ground loop.

Using copper cable to connect two pieces of equipment that are at different electrical potentials poses a shock hazard. It's not uncommon for two grounded pieces of electrical equipment separated by distance to be at different electrical potentials. Connecting these same two pieces of equipment together with optical fiber poses no electrical shock hazard.

If two pieces of electrical equipment are connected together with copper cable and one develops a ground fault, there is now a potential shock hazard at both pieces of equipment. Everyone is likely to experience, or hear of someone experiencing, a ground fault at least once in their life. A common example is when you touch an appliance such as an electric range or washing machine and experience a substantial electrical shock. If the piece of equipment that shocked you was connected to another piece of equipment with a copper cable, there is a possibility that someone touching the other piece of equipment would also be shocked. If the two pieces of equipment were connected with optical fiber, the shock hazard would exist only at the faulty piece of equipment.

Nonconductive fiber optic cables offer some other advantages, too. They do not attract lighting. They can be run through areas where faulty copper cables could pose a fire or explosion hazard. The only safety requirement that Article 770 of the NEC places on nonconductive fiber optic cables is the type of jacket material. When electrical safety, spark, or explosion hazards are a concern, there is no better solution than optical fiber.



Article 770 of the NEC is discussed in detail in Chapter 7.

Link Performance Analysis

Now that we have discussed fiber optic link design considerations and the advantages of optical fiber over copper cable, let's look at how to analyze the performance of a fiber optic link. This section focuses on link performance analysis using Optical Fiber Cabling Components Standard TIA/EIA-568-B.3.

Industry standard TIA/EIA-568-B.3 defines optical fiber components' performance. We will use three parts of this standard to analyze the performance of a link: section 4.2, cable transmission performance; section 5.6, optical fiber splice; and annex A, optical fiber connector performance specifications.

When analyzing link performance using TIA/EIA-568-B.3, you are performing a worst-case analysis. In other words, your link should perform no worse than the standard. Typically a link will outperform the standard or greatly exceed the standard. The standard sets the minimum requirements for cable transmission performance, splice performance, and connector performance.

Cable Transmission Performance

TIA/EIA-568-B.3 section 4.2 addresses the performance of 50/125 μm multimode optical fiber, 62.5/125 μm multimode optical fiber, and single-mode inside and outside plant optical fiber. Maximum attenuation and minimum information transmission capacity is defined for each fiber type by wavelength, as shown in Table 14.5.

TABLE 14.5 Optical Fiber Cable Transmission Performance Parameters

Optical Fiber Cable Type	Wavelength (nm)	Maximum Attenuation (dB/km)	Minimum Information Transmission Capacity for Over-filled Launch (MHz•km)
50/125 μm multimode	850	3.5	500
	1300	1.5	500
62.5/125 μm multimode	850	3.5	160
	1300	1.5	500
Single-mode inside plant cable	1310	1.0	N/A
	1550	1.0	N/A
Single-mode outside plant cable	1310	0.5	N/A
	1550	0.5	N/A

Splice and Connector Performance

TIA/EIA-568-B.3 section 5.6 addresses the performance of fusion or mechanical optical fiber splices. The standard states that a fusion or mechanical splice shall not exceed a maximum optical attenuation of 0.3 dB when measured in accordance with ANSI/EIA/TIA-455-34 or ANSI/EIA/TIA-455-59.

This section also defines the minimum return loss for mechanical or fusion optical fiber splices. Multimode mechanical or fusion splices shall have a minimum return loss of 20 dB while single-mode mechanical or fusion splices shall have a minimum return loss of 26 dB when measured in accordance with ANSI/EIA/TIA-455-107. The minimum single-mode return loss for broadband analog video CATV applications is 55 dB.

TIA/EIA-568-B.3, annex A describes the optical fiber connector performance requirements. Section A.3 states that all multimode connectors, adapters, and cable assemblies shall meet the

requirements at both 850 nm and 1300 nm wavelengths. All single-mode connectors, adapters, and cable assemblies shall meet the requirement at both 1310 nm and 1550 nm wavelengths.

Section A.3.2 states the maximum insertion loss for a mated connector pair. The maximum insertion loss of a mated multimode or single-mode connector pair is 0.75 dB. Multimode-mated pairs shall be tested in accordance with FOTP-171 methods A1 or D1, or FOTP-34 method A2. Single-mode mated pairs shall be tested in accordance with FOTP-171 methods A3 or D3, or FOTP-34 method B.



Real World Scenario

Using a Power Budget to Troubleshoot a Fiber Optic System

This chapter explains how to calculate a power budget for a multimode fiber optic link and a single-mode fiber optic link. A power budget tells the fiber optic installer or technician the maximum allowable loss for every component in the fiber optic link. The sum of these losses equals the maximum acceptable loss for the link per the TIA/EIA-568-B.3 standard.

The maximum acceptable loss for the link is the worst-case scenario. Every component in the fiber optic link should outperform the TIA/EIA-568-B.3 standard. Links with a total loss just under the maximum acceptable loss may not function properly. This is especially true with short links.

To illustrate, we'll tell a story about a short fiber optic link on a submarine. Of course, all fiber optic links on a submarine are short. This link, when tested with the light source and power meter, had a total loss under the maximum allowable value. However, when the link was connected to the receiver and transmitter, the fiber optic system did not function properly.

Because the link was short, the loss for the optical fiber was negligible. The calculated loss came from the multiple interconnections. The maximum allowable loss for the interconnections was 3.0 dB. The measured loss for the link was 2.1 dB. This is 0.9 dB below the maximum allowable. The typical performance for a link with this number of interconnections would have been less than 1.0 dB.

One of the interconnections in the link had a loss much greater than the maximum allowable. This bad interconnection was reflecting 20+% of the light energy from the transmitter back at the transmitter. This back-reflected energy was reflected toward the receiver at the transmitter interconnection. The receiver detected the back-reflected energy from the transmitter interconnection, causing the system to malfunction.

Just because the total loss for a fiber optic link is below the maximum allowable doesn't mean your system will perform without problems. If your system is having problems, start by evaluating each component in the fiber optic link.

Power Budget

Now that the link performance requirements have been identified, let's put them to use in a power budget. A power budget as defined in IEEE Standard 802.3 is the minimum optical power available to overcome the sum of attenuation plus power penalties of the optical path between the transmitter and receiver. These losses are calculated as the difference between the minimum transmitter launch power and the minimum receive power.

Multimode Link Analysis

Table 14.6 lists some of the typical optical characteristics for a 1300 nm LED transmitter. In this table, we see that manufacturers typically list three values for optical output power. There is a typical value and there are the extreme maximum and minimum values. The minimum value represents the least amount of power that the transmitter should ever output. This is the output power level we will use in calculating our power budget.

TABLE 14.6 LED Transmitter Optical Characteristics

Parameter	Symbol	Min.	Typ.	Max.	Unit
Optical output power	BOL P_0	-19	-16.8	-14	dBm avg.
62.5/125 μm , NA = 0.275 fiber	EOL	-20			
Optical output power	BOL P_0	-22.5	-20.3	-14	dBm avg.
50/125 μm , NA = 0.20 fiber	EOL	-23.5			
Optical extinction ratio			0.001 -50	0.03 -35	% dB
Optical output power at logic "0" state	P_0 ("0")			-45	dBm avg.
Center wavelength	λ_c	1270	1300	1380	nm
Spectral width—FWHM	Δ		137	170	nm
Optical rise time	t_r	0.6	1.0	3.0	ns
Optical fall time	t_f	0.6	2.1	3.0	ns
Duty cycle distortion contributed by the transmitter	DCD		0.02	0.6	ns p-p
Duty cycle jitter contributed by the transmitter	DDJ		0.02	0.6	ns p-p

Table 14.7 lists some of the typical optical characteristics for a 1300 nm receiver. In this table, the manufacturers list the maximum and minimum optical input power, and under each of those there is a typical and a minimum value. The receiver will perform best when the input power is between the minimum value for maximum optical input power and maximum value for minimum optical input power.



Remember from Chapter 10 that the manufacturer-stated optical output power of a typical laser transmitter is measured after 1 m of optical fiber. Therefore we do not have to account for the connector loss at the transmitter. Remember from Chapter 11, “Fiber Optic Detectors and Receivers,” that the manufacturer typically states minimum optical input power for the receiver at the window edge. Therefore we do not have to account for the connector loss at the receiver.

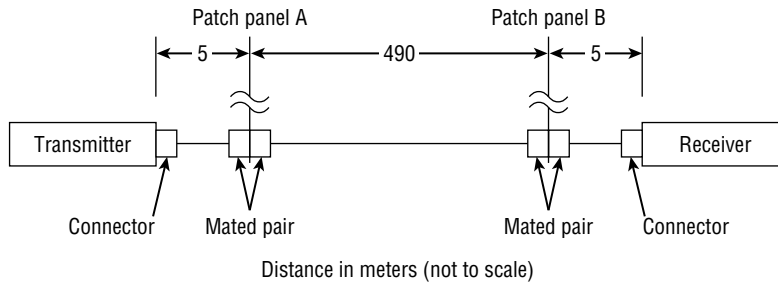
TABLE 14.7 LED Receiver Optical Characteristics

Parameter	Symbol	Min.	Typ.	Max.	Unit
Optical input power minimum at window edge	$P_{IN\ Min.}\ (W)$		-33.5	-31	dBm avg.
Optical input power minimum at eye center	$P_{IN\ Min.}\ (C)$		-34.5	-31.8	dBm avg.
Optical input power maximum	$P_{IN\ Max.}$	-14	-11.8		dBm avg.
Operating wavelength	λ	1270		1380	nm

Now let’s do the math and determine the received power window of operation for the receiver. If power to the receiver falls within this window, the receiver will have a low bit error rate. The minimum value for the optical output power is -20 dBm at EOL. The minimum optical output power at EOL should always be used to account for the aging of the LED light source. The maximum value for the minimum optical input power is -31 dBm. The difference is 11 dB, which is the window of operation where the receiver will provide the best performance. This is also our power budget.

To better understand a power budget, let’s take a look a basic fiber optic link as shown in Figure 14.1 and calculate the maximum loss allowable per EIA/TIA-568-B.3. After we have calculated the maximum allowable loss, we can compare maximum loss to the power budget and determine the minimum power to the receiver. If the minimum power to the receiver falls within the window of operation, the link should support low bit error rate data transmission.

The first link we are going to look at is a 490 m span of 62.5/125 μm multimode optical fiber. A diagram of this link is shown in Figure 14.1. Each end of the optical fiber is connectorized and plugged into a patch panel. On the other side of each patch panel, there is a 5 m patch cord. One patch cord connects to the transmitter and the other to the receiver. The total link length is 500 m.

FIGURE 14.1 Basic fiber optic link

You learned earlier in Table 14.3, “Optical Fiber Cable Attenuation Performance Parameters,” that TIA/EIA-568-B.3 defines the maximum attenuation for a 62.5/125 μm optical fiber at both 850 nm and 1300 nm. Because the transmitter selected for this example has a 1300 nm output, we will only evaluate link performance at 1300 nm. The maximum allowable attenuation per TIA/EIA-568-B.3 is 1.5 dB per km of 62.5/125 μm optical fiber. The total link length was 500 m with a maximum allowable attenuation of 0.75 dB.

The link shown in Figure 14.1 has six connectors. One connector is plugged into the transmitter, one is plugged into the receiver, and the remaining four are mated together in pairs at each patch panel. When evaluating connector loss, only the mated pairs are accounted for. The loss at the transmitter and receiver is ignored. You may remember from Chapter 10 that the manufacturer-stated optical output power of a typical LED transmitter is measured after 1 m of optical fiber with all cladding modes removed. Therefore we do not have to account for the connector loss at the transmitter. As you learned in Chapter 11, the manufacturer typically states minimum optical input power for the receiver at the window edge. Therefore we do not have to account for the connector loss at the receiver.

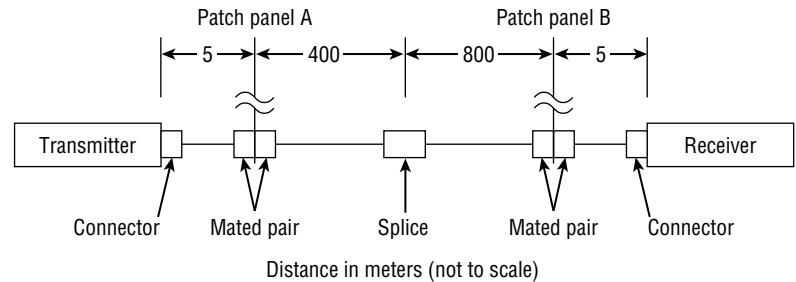
Because the transmitter and receiver manufacturers have accounted for connector loss, we only have to account for mated pair loss. In the link shown in Figure 14.1, there are two mated pairs, one at each patch panel. As you learned earlier in this chapter in the “Splice and Connector Performance” section, TIA/EIA 568-B.3 states that the maximum allowable insertion loss for a mated connector pair is 0.75 dB. Because there are two mated pairs on our link, the maximum interconnection loss for our link is 1.5 dB.

At this point, we have identified the maximum allowable loss for the optical fiber and the interconnections. The maximum allowable loss for the link per TIA/EIA-568-B.3 is the sum of these two losses. The maximum allowed loss for the link in Figure 14.1 is 2.25 dB.

Now that all the performance parameters of the link have been identified, let’s determine the minimum power that should be available to the receiver. The minimum power available to the receiver is equal to the minimum optical output power of the LED transmitter minus the loss of the link. The minimum optical output power for the LED transmitter is -20 dBm, the link loss is 2.25 dB, and the minimum power available at the receiver should be -22.25 dBm. The power budget for the link in Figure 14.1 was 11 dB. The maximum loss allowable for this link per TIA/EIA-568-B.3 is 2.25 dB. The loss for this link is small and provides 8.75 dB of headroom for the life of the LED transmitter.

The link we are going to examine in Figure 14.2 is a multimode link with a splice. The transmitter and receiver used with this link operate at 850 nm. The minimum EOL optical output power for the 850 nm LED transmitter is -22.5 dBm. The minimum optical input power for the receiver is -32.5 dBm.

FIGURE 14.2 Multimode fiber optic link with splice



Looking at Figure 14.2, we see two patch panels with 1200 m of 62.5/125 μm multimode optical fiber between them. There is a mechanical splice 400 m away from patch panel A. Each end of the optical fiber is connectorized and plugged into the patch panel. Patch cords 5 m in length each are used to connect the LED transmitter to one patch panel and the receiver to the other patch panel.

Table 14.8 is a blank power budget calculation table. This table can be used to calculate the power budget for a link, sum all the losses in the link, and calculate headroom.

TABLE 14.8 Blank Multimode TIA/EIA-568-B.3 Power Budget Calculation Table

#	Description	Value
1	Minimum EOL optical output power	dBm
2	Minimum optical input power	dBm
3	Subtract line 2 from line 1 to calculate the power budget.	dB
4	Km of optical fiber	
5	Number of interconnections	
6	Number of splices	
7	Multiply line 4 $\times$ 3.5.	
8	Multiply line 4 $\times$ 1.5.	
9	Multiply line 5 $\times$ 0.75.	

TABLE 14.8 Blank Multimode TIA/EIA-568-B.3 Power Budget Calculation Table (*continued*)

#	Description	Value
10	Multiply line 6 $\times$ 0.3.	
11	Add lines 7, 9, and 10 for total link loss at 1300 nm.	dB
12	Add lines 8, 9, and 10 for total link loss at 1300 nm.	dB
13	Subtract line 11 from line 3 for headroom at 850 nm.	dB
14	Subtract line 12 from line 13 for headroom at 1300 nm.	dB

Table 14.9 has all the values for the link filled in. The power budget for the link is 10 dB. The loss for the link is 6 dB and the headroom for the link is 4 dB.

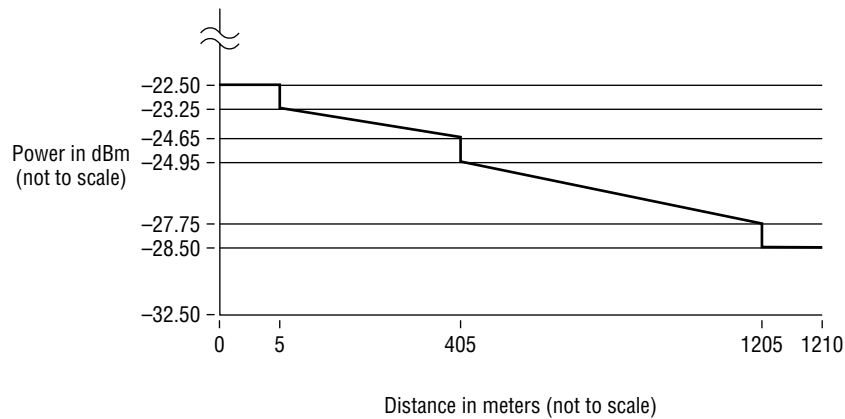
TABLE 14.9 Completed Multimode TIA/EIA-568-B.3 Calculation Table for an 850 nm Transmitter

#	Description	Value
1	Minimum EOL optical output power	-22.5 dBm
2	Minimum optical input power	-32.5 dBm
3	Subtract line 2 from line 1 to calculate the power budget.	10.0 dB
4	Km of optical fiber	1.210
5	Number of interconnections	2.0
6	Number of splices	1.0
7	Multiply line 4 $\times$ 3.5.	4.2
8	Multiply line 4 $\times$ 1.5.	1.8
9	Multiply line 5 $\times$ 0.75.	1.5
10	Multiply line 6 $\times$ 0.3.	0.3
11	Add lines 7, 9, and 10 for total link loss at 1300 nm.	6.0 dB
12	Add lines 8, 9, and 10 for total link loss at 1300 nm.	3.6 dB
13	Subtract line 11 from line 3 for headroom at 850 nm.	4.0 dB
14	Subtract line 12 from line 3 for headroom at 1300 nm.	

Up to this point, we have only looked at calculating the power budget for a link using a table. However, the power budget can be plotted graphically. The graphical representation of the losses in the link very much resembles the trace of an optical time domain reflectometer (OTDR). The OTDR will be covered in great detail in Chapters 15 and 16.

Figure 14.3 is the graphical representation of the link we analyzed in Figure 14.2. When analyzing a link graphically, power or loss is noted on the vertical scale and distance is noted on the horizontal scale. Power gain and loss may be drawn to scale; however, there is not usually sufficient room on your paper to draw distance to scale.

FIGURE 14.3 Graphical representation of a multimode fiber optic link



The first step in setting up the graphical representation of the link is to assign values for the vertical axis. As shown in Table 14.6, the maximum value assigned is the minimum EOL optical output power, -22.5 dB. The minimum value assigned is the minimum EOL optical output power minus the loss for the link. The loss for this link is 6 dB, so -28.5 dB is the minimum value assigned.

The next step is to assign values to the vertical axis that represent the before and after power levels for the interconnections. Working from the transmitter to the receiver, the first interconnection is a mated connector pair at patch panel A. There is a 5 m patch cord between the transmitter and patch panel A. At a wavelength of 850 nm, the maximum loss for the 5 m patch cord is only 0.0175 dB. This very small value represents only a fraction of a percent of the loss for this link and can be ignored in the graphical representation. Typically you can ignore all patch cord loss when generating a graphical representation of a link. This simplifies the drawing and reduces the time required to generate the drawing.

The amount of light energy entering the first mated pair is -22.5 dBm. The maximum loss allowed by TIA/EIA-568.3 for a mated pair is 0.75 dB. Using this value, the power exiting the mated pair should be no less than -23.25 dBm. In a linear fashion working from top to bottom, the next value recorded on the vertical axis is -23.25 dBm.

The next value on the vertical axis represents the light energy entering the mechanical splice. There is 400 m of optical fiber from the first mated pair to the mechanical splice. The loss for the 400 m of optical fiber is 1.4 dB. The light energy entering the mechanical splice is -24.65 dBm and

is recorded on the vertical axis. The maximum loss allowed by TIA/EIA-568-B.3 for a mechanical splice is 0.3 dB. Using this value, the power exiting the mechanical splice should be no less than -24.95 dBm, which is recorded on the vertical axis.

Between the mechanical splice and patch panel B is 800 m of optical fiber. The loss of the 800 m of optical fiber is 2.8 dB. The light energy entering the mated pair at patch panel B is -27.75 dBm, which should be recorded on the vertical axis. As we already know, TIA/EIA-568-B.3 allows a maximum loss of only 0.75 dB for a mated pair. The minimum amount of light energy exiting the mated pair should be -28.5 dB. If we ignore the loss for the 5 m patch cord between patch panel B and the receiver, the power available to the receiver is -28.5 dBm. This is the last value recorded on the vertical axis.

The next step in this process is to record the horizontal axis values. Working from left to right starting at the transmitter, the first value recorded on the horizontal axis is zero. It is from this point that we will record the successive segments of optical fiber.

The first segment of optical fiber is the 5 m patch cord. With an overall length of 1200 m, it's not practical to draw the 5 m patch cord to scale unless you have a very large sheet of paper. Looking at Figure 14.3, you can see that we made the first segment on the horizontal axis long enough to clarify that the 5 m patch cord is a separate segment of optical fiber. The second value recorded on the horizontal axis is 5 m.

The second segment is a 400 m span of optical fiber from patch panel A to the mechanical splice. The next value recorded on the horizontal axis is 405 m.

The third segment of optical fiber is an 800 m span from the mechanical splice to patch panel B. The next value recorded on the horizontal axis is 1205 m.

The last segment of optical fiber is the 5 m patch cord that connects the receiver to patch panel B. The last value recorded on the horizontal axis is 1210 m.

With all the values for the vertical and horizontal axes recorded, the last step is to draw a series of lines that graphically represent the recorded values. (You may want to use a straight edge for this.) The first line segment will be a horizontal line from -22.5 dBm on the vertical axis to the 5 m point on the horizontal axis. This line represents the patch cord that connects the transmitter to patch panel A. The line is horizontal because the patch cord loss is insignificant. At the end of this, a vertical line will be drawn that runs from -22.5 dBm to -23.25 dBm; it represents the loss for the mated pair at patch panel A.

The third line will extend from the 5 m point on the horizontal axis to the 405 m point. It will be a sloping line from left to right that goes from -23.25 dBm on the vertical axis to -24.65 dBm. This line represents the loss for the 400 m section of optical fiber from patch panel A to the mechanical splice. At the end of this line a vertical line is drawn from -24.65 dBm to -24.95 dBm to represent the loss for the mechanical splice.

The fifth line will extend horizontally from the 405 m point to the 1205 m point. This line will slope from -24.95 dBm on the vertical axis to -27.75 dBm; it represents the loss for the 800 m section of optical fiber from the mechanical splice to patch panel B. At the end of this line, a vertical line will be drawn from -27.75 dBm to -28.5 dBm, representing the loss for the mated pair at patch panel B.

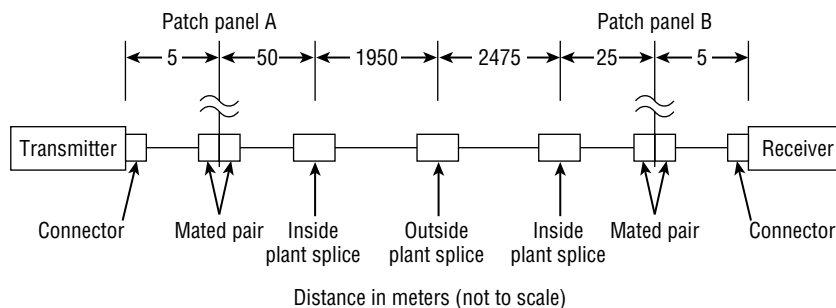
The last line drawn is a horizontal line from the 1205 m point on the horizontal axis to the 1210 m point. This line represents the patch cord from patch panel B to the receiver. There is no slope to this line because the loss for the patch cord is insignificant. If everything has been drawn correctly, your drawing should look like Figure 14.3.

Single-Mode Link Analysis

As with multimode optical fiber, TIA/EIA-568-B.3 defines performance requirements for single-mode optical fiber, connectors, and splices. However, TIA/EIA-568-B.3 only addresses *inside plant* splice performance. *Outside plant* splice performance is addressed in TIA/EIA-758. TIA/EIA-758 states that splice insertion loss shall not exceed 0.1 dB mean (0.3 dB maximum) when measured with OTDR testing.

Figure 14.4 is a basic single-mode optical fiber link with 4510 m of optical fiber spliced in three locations between two patch panels located in separate buildings. The transmitter is connected to patch panel A with a 5 m patch cord and the receiver is connected to patch panel B with a 5 m patch cord. There is an inside plant splice 50 m from patch panel A, another 25 m from patch panel B, and an outside plant splice 2000 m from patch panel A.

FIGURE 14.4 Single-mode fiber optic link



Before we can analyze the link and do a power budget, we need to look at how manufacturers typically describe the performance characteristics for laser transmitters and receivers. Table 14.10 lists some of the typical optical power characteristics for a 1310 nm laser transmitter. In this table, we see that manufacturers typically list three values for optical output power. There is a typical value and there are the extreme maximum and minimum values. The minimum value represents the least amount of power the transmitter should ever output. This is the output power level we will use in calculating our power budget.

Table 14.11 lists some of the typical optical power characteristics for a 1310 nm receiver. In this table, the manufacturers list the maximum and minimum optical input power, and under each of those there is a typical and a minimum value. The receiver will perform best when the input power is between the minimum value for maximum optical input power and maximum value for minimum optical input power.

With all the transmitter and receiver values defined, we can calculate the received power window of operation for the receiver using the power budget calculation table shown in Table 14.12. The minimum value for the optical input power is -10 dBm. The minimum value for the optical input power is -19 dBm. The difference is 9 dB, which is the window of operation where the receiver will provide the best performance. This is also our power budget for this transmitter-receiver combination.

TABLE 14.10 Laser Transmitter Optical Characteristics

Parameter	Symbol	Min.	Typ.	Max.	Unit
Optical Output Power 9 μm SMF	P_{OUT}	−10	−6	−3	dBm
Center Wavelength	λ_c	1260		1360	nm
Spectral Width—rms	σ		1.8	4	nm rms
Optical Rise Time	t_r		30	70	ps
Optical Fall Time	t_f		150	225	ps
Extinction Ratio	E_R	8.2	12		dB
Optical Output Eye	Compliant with eye mask Telecordia GR-253-CORE				
Back-Reflection Sensitivity				−8.5	dB
Jitter Generation	pk to pk			70	mUI
	RMS			7	mUI

TABLE 14.11 Laser Receiver Optical Characteristics

Parameter	Symbol	Min.	Typ.	Max.	Unit
Receiver sensitivity	P_{IN} MIN		−23	−19	dBm avg.
Receiver overload	P_{IN} MAX	−3	+1		dBm avg.
Input operating wavelength	λ	1260		1550	nm
Signal detect — asserted	P_A		−27.3	−19.5	dBm avg.
Signal detect — deasserted	P_D	−35	−28.7		dBm avg.
Signal detect — hysteresis	P_H	0.5	1.4	4	dB
Reflectance			−35	−27	dB

TABLE 14.12 Blank Single-Mode TIA/EIA-568-B.3, TIA/EIA-758 Power Budget Calculation Table

#	Description	Value
1	Minimum EOL optical output power	dBm
2	Minimum optical input power	dBm
3	Subtract line 2 from line 1 to calculate the power budget.	dB
4	Km of inside plant optical fiber	
5	Km of outside plant optical fiber	
6	Number of interconnections	
7	Number of inside plant splices	
8	Number of outside plant splices	
9	Multiply line 4 $\times$ 1.0.	
10	Multiply line 5 $\times$ 0.5.	
11	Multiply line 6 $\times$ 0.75.	
12	Multiply line 7 $\times$ 0.3.	
13	Multiply line 8 $\times$ 0.1.	
14	Add lines 9, 10, 11, 12, and 13 for total link loss.	dB
15	Subtract line 14 from line 3 for outside plant headroom.	dB

Now that the power budget has been calculated, the power budget calculation table can be completed. The link shown in Figure 14.4 contains two mated pairs, two inside plant splices, one outside plant splice, and 4510 m of optical fiber. The maximum allowable loss for the two mated pairs is 1.5 dB. The maximum allowable loss for the two inside plant splices is 0.6 dB, and for the outside plant splice 0.1 dB.

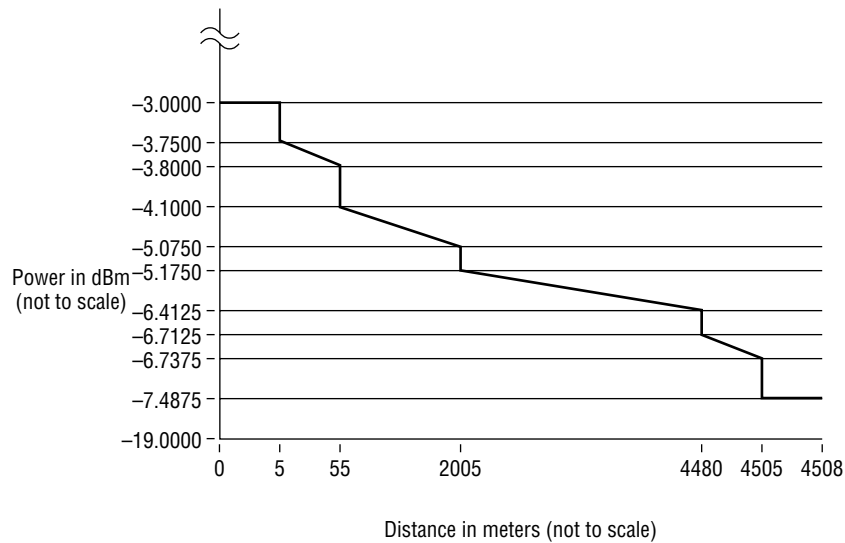
The 4510 m of optical fiber is broken up into inside plant and outside plant. The 4425 m of optical fiber between the two inside plant splices will be evaluated as outside plant and the remaining 85 m will be evaluated as inside plant. The maximum allowable loss for the outside

plant optical fiber is rounded down to 2.2 dB. The maximum allowable loss of the inside plant optical fiber is 0.085, which is rounded up to 0.1 dB. The maximum allowable loss for this link is rounded up to 4.5 dB. Our power budget was 9 dB; that leaves us 4.5 dB of headroom. Table 14.13 is the completed power budget calculation table with the rounded values.

With the power budget calculation table completed, you should be able to quickly sketch out the graphical representation of the link. Figure 14.5 is the completed graphical representation for the link we just discussed without rounding.

TABLE 14.13 Completed Single-Mode TIA/EIA-568-B.3, TIA/EIA-758 Power Budget Calculation Table

#	Description	Value
1	Minimum EOL optical output power	-10.0 dBm
2	Minimum optical input power	-19.0 dBm
3	Subtract line 2 from line 1 to calculate the power budget.	9.0 dB
4	Km of inside plant optical fiber	0.085
5	Km of outside plant optical fiber	4.425
6	Number of interconnections	2
7	Number of inside plant splices	2
8	Number of outside plant splices	1
9	Multiply line 4 $\times$ 1.0.	0.1
10	Multiply line 5 $\times$ 0.5.	2.2
11	Multiply line 6 $\times$ 0.75.	1.5
12	Multiply line 7 $\times$ 0.3.	0.6
13	Multiply line 8 $\times$ 0.1.	0.1
14	Add lines 9, 10, 11, 12, and 13 for total link loss.	4.5 dB
15	Subtract line 14 from line 3 for outside plant headroom.	4.5 dB

FIGURE 14.5 Graphical representation of a single-mode fiber optic link

Summary

This chapter outlined the design considerations for a fiber optic link, explained the performance advantages that optical fiber has over copper cable, and explained in detail how to analyze the performance of a fiber optic link. Optical fiber is the highest performance transmission medium available today. Optical fiber boasts incredible bandwidth with very little attenuation over great distances. Fiber optic cables the size of a garden hose can hold hundreds of optical fibers each capable of carrying billions of bits of data per second. Fiber optic systems are emerging everywhere, and in the very near future fiber to your home will be a reality.

Exam Essentials

List the main considerations when designing a basic fiber optic system. Remember that there are many ways to approach fiber optic system design and there are many different fiber optic systems. Before beginning a fiber optic system design, two questions need to be answered: How much data needs to be moved and what is the transmission distance? Also remember that there are two rules of thumb to help you design a link to meet your data rate and transmission distance expectations.

Describe the bandwidth advantages of optical fiber over twisted-pair and coaxial copper cables.

Remember that single-mode systems with laser transmitters offer the greatest bandwidth and that multimode systems with LED transmitters offer the least bandwidth. The bandwidth of optical fiber and copper cable is inversely proportional to its length. In other words, as the length of the optical fiber or copper cable increases, the bandwidth of the optical fiber or copper cable decreases.

Describe the attenuation advantages of optical fiber over twisted-pair and coaxial copper cables.

Remember that all transmission mediums lose signal strength over distance and the loss of signal strength, or attenuation, is typically measured in decibels. Optical fiber systems measure attenuation using optical power. Copper cable systems typically use voltage drop across a defined load at various transmission frequencies to measure attenuation.

Remember that attenuation in copper cables is measured at different frequencies. This is not the case with optical fiber, where attenuation is measured with a continuous wave light source that is not modulated. The attenuation in a copper cables increases as the transmission frequency increases. This is not the case with optical fiber, where transmission frequency has no impact on attenuation.

You need to know that the maximum allowable attenuation in an optical fiber is defined in section 4.2 of TIA/EIA-568-B.3 and listed in Table 14.3.

Describe the electromagnetic advantages of optical fiber over copper cable. You need to know that electromagnetic interference (EMI) is electromagnetic energy that causes undesirable responses, degradation, or complete system failure. Systems using copper cable are vulnerable to the effects of EMI because a changing electromagnetic field will induce current flow in a copper conductor. Optical fiber is a dielectric, or an insulator, and current does not flow through insulators. Thus EMI has no effect on the operation of an optical fiber and fiber optic cables do not suffer from crosstalk like copper cables do.

Describe the size advantages of fiber optic cable over copper cable. You need to know that a copper cable and fiber optic cable with similar performance characteristics will differ greatly in size. There is no rule of thumb for size difference; however, the fiber optic cable may be up to hundreds of times smaller than the copper cable.

Describe the weight-saving advantages of fiber optic cable over copper cable. You need to know that a copper cable and fiber optic cable with similar performance characteristics will greatly differ in weight. There is no rule of thumb for weight difference; however, the fiber optic cable may be up to hundreds of times lighter than the copper cable.

Describe the security advantages of fiber optic cable over copper cable. You need to know that because of total internal reflection, optical fiber does not radiate, making it virtually impossible to tap. Optical fiber is the most secure transmission medium available.

Describe the safety advantages of fiber optic cable over copper cable. You need to know that optical fiber is a dielectric that cannot carry electrical current and presents no shock, spark, or fire hazard. Optical fiber also provides electrical isolation between electrical equipment. Electrical isolation eliminates ground loops, eliminates the potential shock hazard when two pieces

of equipment at different potentials are connected together, and eliminates the shock hazard when one piece of equipment is connected to another with a ground fault.

Describe how to analyze the performance of a multimode fiber optic link using TIA/EIA-568-B.3.

You need to know that TIA/EIA-568-B.3 section 4.2 addresses the performance of 50/125 μm multimode optical fiber, 62.5/125 μm multimode optical fiber, and single-mode inside and outside plant optical fiber.

You should also know that TIA/EIA-568-B.3 section 5.6 addresses the performance of fusion or mechanical optical fiber splices. Multimode-mated pairs should be tested in accordance with FOTP-171 methods A1 or D1, or FOTP-34 method A2.

Describe how to analyze the performance of a single-mode fiber optic link using TIA/EIA-568-B.3 and TIA/EIA-758. You should know that TIA/EIA-568-B.3 section 5.6 addresses the performance of fusion or mechanical optical fiber splices. Single-mode mated pairs shall be tested in accordance with FOTP-171 methods A3 or D3, or FOTP-34 method B. Remember that TIA/EIA-758 defines outside plant performance.

Describe how to prepare a power budget for a basic multimode or single-mode optical fiber link.

You should know that a power budget, as defined in IEEE Standard 802.3, is the minimum optical power available to overcome the sum of attenuation plus power penalties of the optical path between the transmitter and receiver. You should also be able to prepare a multimode or single-mode power budget using the power budget calculation tables. From the power budget calculation table, you need to be able to draw a graphical representation of the fiber optic link.

Review Questions

1. When beginning a basic fiber optic system design, what should be the first things considered?
 - A. Data rate, transmission distance
 - B. Cable diameter, cable length
 - C. Inside plant, outside plant
 - D. Buffer type, jacket material
2. Optical fiber offers _____ bandwidth and _____ attenuation than twisted-pair or coaxial cable.
 - A. Less, more
 - B. Equal, more
 - C. Greater, more
 - D. Greater, less
3. Per TIA/EIA-568-B.3, the minimum bandwidth for one km of 62.5/125 μm optical fiber operating with a 850 nm light source is _____.
 - A. 160 MHz
 - B. 200 MHz
 - C. 500 MHz
 - D. 1000 MHz
4. Per TIA/EIA-568-B.3, the minimum bandwidth for one km of 50/125 μm optical fiber operating at 850 nm is _____.
 - A. 160 MHz
 - B. 200 MHz
 - C. 500 MHz
 - D. 1000 MHz
5. Per TIA/EIA-568-B.3, the minimum bandwidth for 2000 m of 62.5/125 μm optical fiber operating with a 1300 nm light source is _____.
 - A. 160 MHz
 - B. 200 MHz
 - C. 250 MHz
 - D. 500 MHz

6. Per TIA/EIA-568-B.3, the minimum bandwidth for 1250 m of 50/125 μm optical fiber operating with an 850 nm light source is _____.
- A. 160 MHz
 - B. 200 MHz
 - C. 400 MHz
 - D. 500 MHz
7. TIA/EIA-568-B.3 states that the maximum allowable attenuation for 1000 m of 50/125 μm optical fiber operating with an 850 nm light source is _____.
- A. 1.0 dB
 - B. 1.5 dB
 - C. 3.0 dB
 - D. 3.5 dB
8. TIA/EIA-568-B.3 states that the maximum allowable attenuation for 1000 m of 62.5/125 μm optical fiber operating with a 1300 nm light source is _____.
- A. 1.0 dB
 - B. 1.5 dB
 - C. 3.0 dB
 - D. 3.5 dB
9. TIA/EIA-568-B.3 states that the maximum allowable attenuation for 1000 m of inside plant single-mode optical fiber operating with a 1310 nm light source is _____.
- A. 0.5 dB
 - B. 1.0 dB
 - C. 1.5 dB
 - D. 3.5 dB
10. TIA/EIA-568-B.3 states that the maximum allowable attenuation for 1000 m of outside plant single-mode optical fiber operating with a 1550 nm light source is _____.
- A. 0.5 dB
 - B. 1.0 dB
 - C. 1.5 dB
 - D. 3.5 dB
11. TIA/EIA-568-B.3 states that the maximum allowable attenuation for 5200 m of inside plant single-mode-mode optical fiber operating with a 1550 nm light source is _____.
- A. 2.6 dB
 - B. 5.2 dB
 - C. 5.6 dB
 - D. 7.8 dB

12. TIA/EIA-568-B.3 states that the maximum allowable attenuation for 12,500 m of outside plant single-mode optical fiber operating with a 1310 nm light source is _____.
- A. 6.25 dB
 - B. 6.5 dB
 - C. 12.5 dB
 - D. 18.75 dB
13. Optical fiber is a _____, therefore it is immune to the effects of EMI.
- A. Conductor
 - B. Composite
 - C. Dielectric
 - D. Radiator
14. Fiber optic cables are _____ and _____ than copper cables with comparable transmission frequency performance.
- A. Larger, heavier
 - B. Smaller, lighter
 - C. Smaller, heavier
 - D. Larger, lighter
15. Fiber optic cables do not _____ light energy, therefore they are virtually immune to being tapped, which makes them very secure.
- A. Rotate
 - B. Receive
 - C. Detect
 - D. Radiate
16. Optical fiber does not _____ electricity and offers safety advantages over copper cables.
- A. Rotate
 - B. Conduct
 - C. Detect
 - D. Radiate
17. The maximum interconnection loss as defined by TIA/EIA-568-B.3 for a multimode connector pair is _____.
- A. 0.1 dB
 - B. 0.3 dB
 - C. 0.5 dB
 - D. 0.75 dB

18. The maximum interconnection loss as defined by TIA/EIA-568-B.3 for a single-mode connector pair is _____.
- A. 0.1 dB
 - B. 0.3 dB
 - C. 0.5 dB
 - D. 0.75 dB
19. The maximum insertion loss as defined by TIA/EIA-568-B.3 for a mechanical or fusion splice is _____.
- A. 0.1 dB
 - B. 0.3 dB
 - C. 0.5 dB
 - D. 0.75 dB
20. The maximum insertion loss as defined by TIA/EIA-758 for an outside plant splice is _____.
- A. 0.1 dB
 - B. 0.3 dB
 - C. 0.5 dB
 - D. 0.75 dB

Answers to Review Questions

1. A. Before beginning a fiber optic system design, two questions need to be answered: How much data needs to be moved and what is the transmission distance?
2. D. Optical fiber offers the highest bandwidth and the least attenuation of any medium currently available.
3. A. TIA/EIA-568-B.3 defines multimode optical fiber bandwidth in MHz•km. Optical fiber transmission performance parameters are defined in section 4.2 of TIA/EIA-568-B.3. The minimum information transmission capacity for 1000 m of 62.5/125 μ m optical fiber operating at 850 nm is 160 MHz.
4. C. TIA/EIA-568-B.3 defines multimode optical fiber bandwidth in MHz•km. Optical fiber transmission performance parameters are defined in section 4.2 of TIA/EIA-568-B.3. The minimum information transmission capacity for 1000 m of 50 /125 μ m optical fiber operating at 850 nm is 500 MHz.
5. C. TIA/EIA-568-B.3 defines multimode optical fiber bandwidth in MHz•km. Optical fiber transmission performance parameters are defined in section 4.2 of TIA/EIA-568-B.3. The minimum information transmission capacity for 2000 m of 62.5/125 μ m optical fiber operating at 1300 nm is 250 MHz.
6. C. TIA/EIA-568-B.3 defines multimode optical fiber bandwidth in MHz•km. Optical fiber transmission performance parameters are defined in section 4.2 of TIA/EIA-568-B.3. The minimum information transmission capacity for 1250 m of 50/125 μ m optical fiber operating at 850 nm is 400 MHz.
7. C. TIA/EIA-568-B.3 defines multimode optical fiber attenuation in dB/km. Optical fiber transmission performance parameters are defined in section 4.2 of TIA/EIA-568-B.3. The maximum allowable attenuation for 1000 m of 50/125 μ m optical fiber operating at 850 nm is 3.5 dB.
8. B. TIA/EIA-568-B.3 defines multimode optical fiber attenuation in dB/km. Optical fiber transmission performance parameters are defined in section 4.2 of TIA/EIA-568-B.3. The maximum allowable attenuation for 1000 m of 62.5/125 μ m optical fiber operating at 1300 nm is 1.5 dB.
9. B. TIA/EIA-568-B.3 defines single-mode optical fiber attenuation in dB/km. Optical fiber transmission performance parameters are defined in section 4.2 of TIA/EIA-568-B.3. The maximum allowable attenuation for 1000 m of inside plant single-mode optical fiber operating at 1310 nm is 1.0 dB.
10. A. TIA/EIA-568-B.3 defines single-mode optical fiber attenuation in dB/km. Optical fiber transmission performance parameters are defined in section 4.2 of TIA/EIA-568-B.3. The maximum allowable attenuation for 1000 m of outside plant single-mode optical fiber operating at 1550 nm is 0.5 dB.

11. B. TIA/EIA-568-B.3 defines single-mode optical fiber attenuation in dB/km. Optical fiber transmission performance parameters are defined in section 4.2 of TIA/EIA-568-B.3. The maximum allowable attenuation for 5200 m of inside plant single-mode optical fiber operating at 1550 nm is 5.2 dB.
12. A. TIA/EIA-568-B.3 defines single-mode optical fiber attenuation in dB/km. Optical fiber transmission performance parameters are defined in section 4.2 of TIA/EIA-568-B.3. The maximum allowable attenuation for 1000 m of outside plant single-mode optical fiber operating at 1550 nm is 0.5 dB.
13. C. Optical fiber is a dielectric or an insulator. Current does not flow through insulators, so EMI has no effect on the operation of an optical fiber.
14. B. Compared to a copper cable with similar transmission frequency performance, a fiber optic cable will always be smaller and lighter. The difference in size and weight widens considerably as the transmission distance increases.
15. D. Because of total internal reflection, optical fiber does not radiate light energy. Optical fiber is virtually immune to being tapped, which makes it the medium of choice for secure communications.
16. B. Electrical safety is always a concern when working with copper cables. Electrical current flowing through copper cable poses shock, spark, and fire hazards. Optical fiber is a dielectric that cannot carry electrical current and presents no shock, spark, or fire hazard.
17. D. Section A.3.2 of TIA/EIA-568-B.3 states the maximum insertion loss for a mated connector pair. The maximum insertion loss of a mated multimode connector pair is 0.75 dB. Multimode mated pairs shall be tested in accordance with FOTP-171 methods A1 or D1, or FOTP-34 method A2.
18. D. Section A.3.2 of TIA/EIA-568-B.3 states the maximum insertion loss for a mated connector pair. The maximum insertion loss of a mated single-mode connector pair is 0.75 dB. Single-mode mated pairs shall be tested in accordance with FOTP-171 methods A3 or D3, or FOTP-34 method B.
19. B. TIA/EIA-568-B.3 section 5.6 addresses the performance of fusion or mechanical optical fiber splices. The standard states that a fusion or mechanical splice shall not exceed a maximum optical attenuation of 0.3 dB when measured in accordance with ANSI/EIA/TIA-455-34 or ANSI/EIA/TIA-455-59.
20. A. TIA/EIA-758 states that splice insertion loss shall not exceed 0.1 dB mean (0.3 dB maximum) when measured with OTDR testing.



Chapter

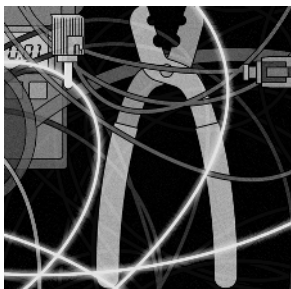
15

Test Equipment and Link/Cable Testing

OBJECTIVES COVERED IN THIS CHAPTER:

✓ Test Equipment and Link/Cable Testing

- Describe the basic operation of a continuity tester.
- Explain how to test the continuity of an optical fiber with a continuity tester.
- Describe the basic operation of a visible fault locator.
- Explain how to locate faults with a visible fault locator.
- Describe the basic operation of a fiber identifier.
- Explain how to test a cable with a fiber identifier.
- Describe the basic operation of an optical return loss test set.
- Explain how to test optical return loss with an optical return loss test set.
- Describe the basic operation of a single-mode and multimode light source and optical power meter.
- Explain the difference between a test jumper and a patch cord.
- Explain the purpose of a mode filter.
- Explain how to perform optical loss measurement testing of a cable plant using TIA/EIA-526-14A methods A, B, and C.
- Explain how to measure the optical loss in a patch cord with a light source and optical power meter.
- Describe the basic operation of a optical time domain reflectometer (OTDR).
- Explain how to test a cable plant with an OTDR.



This chapter explains the basic operation and application of the essential fiber optic test equipment available to the fiber optic installer and technician. The basic theory, operation, and application of each piece of test equipment will be explained. Many of the test methods described in this chapter are based on current industry standards. How to test to these standards is described in detail.

Continuity Tester

The continuity tester is a basic and essential tool for every fiber optic toolkit. It is also one of the least expensive tools in your toolkit. This low-cost tool will allow you to quickly verify the continuity of an optical fiber.

The continuity tester is basically a modified flashlight. Many of the continuity testers available today are just that, modified flashlights. Some flashlights, such as the one pictured in Figure 15.1, have been modified to use an LED instead of an incandescent lamp. Others may just receive a brighter incandescent lamp, like the tester pictured in Figure 15.2.

FIGURE 15.1 LED continuity tester



FIGURE 15.2 Incandescent continuity tester

The job of the continuity tester is to project light from the LED or incandescent lamp into the core of the optical fiber. This is typically accomplished by attaching a receptacle to the lamp end of the flashlight. The receptacle is designed to center and hold the connector ferrule directly above the LED or incandescent lamp. When the connector ferrule is inserted into the receptacle, the endface is typically just above the light source. This approach eliminates the need for a lens to direct light into the core of the optical fiber. However, it directs only a fraction of the light emitted by the lamp or LED into the core of the optical fiber.

Because there is no lens used to focus light energy into the optical fiber, the continuity tester works best with multimode optical fiber. The measured optical output power of an LED continuity tester is typically less than -36 dBm when measured at the end of one meter of $62.5/125\text{ }\mu\text{m}$ optical fiber. The optical output power is reduced another 3 to 4 dB when used with $50/125\text{ }\mu\text{m}$ optical fiber and 16 to 20 dB when used with single-mode optical fiber. The continuity tester works best with multimode optical fiber; however, it can be used with single-mode optical fiber. For best results with single-mode optical fiber, dim the lights in the test area if possible.

The low output power of the continuity tester combined with the high attenuation of visible light by an optical fiber limit the length of optical fiber that can be tested. A multimode optical fiber will attenuate a 650 nm light source roughly 7 dB per kilometer. This high attenuation limits the use of the continuity tester to multimode optical fibers no greater than 2 km in length.

LED continuity testers have a couple of advantages over incandescent lamp testers. They typically feature a red LED that is easy to see. They require far less power from the batteries than an incandescent lamp. An LED continuity tester may provide 10 or more times longer battery life than an incandescent lamp. The LED lamp will never need replacing, unlike an incandescent lamp that may last only 10 hours. LED lamps are also shock resistant.

Whether you are using an LED or an incandescent lamp continuity tester, operation and testing are identical. The continuity tester can test only for breaks in the optical fiber. It does not have sufficient power to identify the location of a break.

The first step when using the continuity tester is to clean and visually inspect the endface of the connector before inserting it into the continuity tester. You need to visually inspect the connector to verify that there is no endface damage. A shattered endface will not allow light to be coupled into the core of the optical fiber under test.

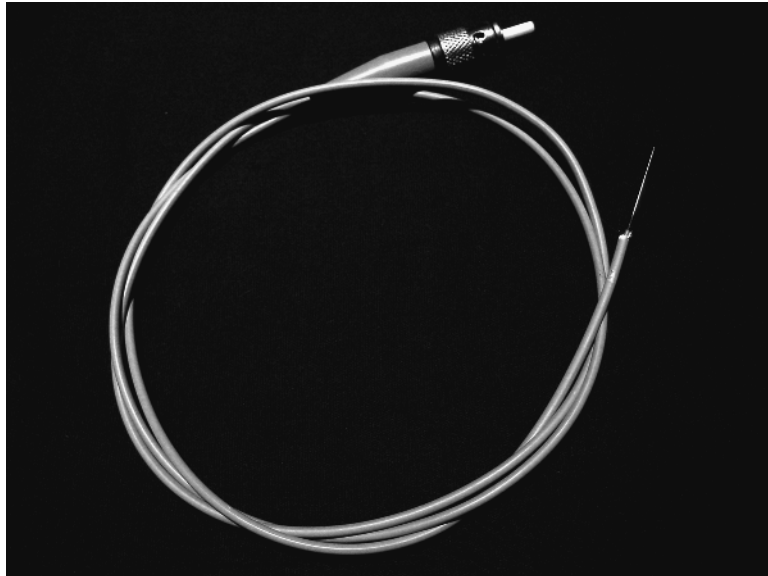
After the connector has been cleaned and inspected, you need to verify that the continuity tester is operating properly. Turn the continuity tester on and verify that it is emitting light. This could save you from embarrassment. You don't want to tell the customer that there is a break in their fiber optic cable when the real problem is dead batteries in your continuity tester.

Depending on where the other end of the fiber optic cable to be tested is located, you may need an assistant to help you. With the continuity tester turned on, insert the ferrule of the connector under test into the receptacle. If light is being emitted from the other end of the optical fiber, there is good continuity. This means only that there are no breaks in the optical fiber. This does not mean that there are no macrobends or high-loss interconnections in the fiber optic cable or link under test.

The continuity tester is often used to verify that there are no breaks in a reel of fiber optic cable before it is installed. There are a couple of ways you could approach testing the reel. One way would be to install a connector on either end of the cable. The other end of the cable should have the jacket and strength member stripped back so that the buffer is exposed. You should remove a small amount of buffer to expose the optical fiber under test. This will allow you to clearly see the light from the continuity tester, ensuring accurate results.

Another approach is to use a pigtail with a mechanical splice or alignment sleeve. The pigtail would have a connector on one end that will mate with the continuity tester receptacle. The other end should have the jacket, strength member, and buffer stripped back so that 10 to 15 mm of optical fiber is exposed, as shown in the photograph in Figure 15.3. The optical fiber should have a perpendicular cleave.

FIGURE 15.3 Pigtail prepared for temporary mechanical splice



One end of the cable under test should be prepared just like the nonconnectorized end of the pigtail. The other end should have the jacket and strength member stripped back so that the buffer is exposed. A small amount of buffer should be removed to expose the optical fiber under test.

To test for continuity, insert the pigtail connector into the continuity tester receptacle. Turn on the continuity tester and verify that light is being emitted from the exposed optical fiber on the opposite end of the pigtail. Insert the exposed optical fiber from the pigtail into one side of a mechanical splice or alignment sleeve. Insert the optical fiber from the cleaved end of the cable into the other side of the mechanical splice or alignment sleeve and check for continuity.

Visible Fault Locator

Like the continuity tester, the *visible fault locator (VFL)* is an essential tool for every fiber optic toolkit. Unlike the continuity tester, it is not one of the least expensive tools in your toolkit. The VFL will allow you to quickly verify the continuity of an optical fiber, identify breaks or macrobends in the optical fiber, and identify a poor fusion splice in multimode or single-mode optical fiber.

The big difference between the continuity tester and the VFL is the light source and optical output power of the light source. The VFL typically uses a red laser light source. The optical output power of the laser is typically 1 mW or greater. Because of the high optical output power, you should never view the output of the VFL directly.

The VFL is available in different shapes and sizes. Some may look like a pen, others may be built into an optical time domain reflectometer (OTDR), and some may look like a small test equipment box. Figure 15.4 is a photograph of a VFL with an ST receptacle.

FIGURE 15.4 VFL with ST receptacle



The VFL fills the core of the optical fiber with light from the laser. The light from the laser escapes the optical fiber at a break or macrobend. The light escaping from the optical fiber will typically illuminate the buffer surrounding the optical fiber. Macrobends are not always visible through the jacket but are typically visible through the buffer. Breaks may be visible through the jacket of the fiber optic cable depending on jacket color, thickness, number of optical fibers in the cable, and amount of strength member.

Unlike the continuity tester, the VFL is not limited to testing multimode optical fibers no greater than 2 km in length. The VFL can be used to verify continuity of multimode or single-mode optical fiber beyond 2 km in length. Due to attenuation of the 650 nm laser light source by the optical fiber, macrobends may not be detectable beyond 1 km in multimode optical fiber and 500 meters in single-mode optical fiber. The same holds true for finding breaks in the optical fiber through the jacket of the fiber optic cable.

The first step before using the VFL for testing an optical fiber is to measure the output power of the VFL. This can be done with a multimode test jumper and an optical power meter. Connect the output of the VFL to the test jumper and the other end of the test jumper to the power meter input. Set the power meter at the shortest wavelength possible. This is typically 850 nm. The measured optical output power of the VFL should be no less than -7 dBm with a 62.5/125 μm optical fiber. If the measured optical output power is less than -7 dBm, replace the battery and retest. Most VFLs will have a measured optical output power greater than -3 dBm with a fresh battery. Optical output power below -7 dBm will limit the effectiveness of the VFL to identify breaks and macrobends.

The second step before using the VFL is to clean and visually inspect the endface of the connector on the optical fiber to be tested before inserting it into the VFL. You need to visually inspect the connector to verify that there is no endface damage. A shattered endface will not allow light to be coupled into the optical fiber under test. A shattered endface may also damage the VFL. Many VFLs use a short pigtail from the laser to the receptacle. The other end of the pigtail has a connector that will mate with the receptacle. A shattered endface can damage the optical fiber in the connector on the end of the pigtail.

To identify a macrobend in the optical fiber, visually inspect the length of the cable under test. If you locate a bend in the cable that is glowing, this is a macrobend. Figure 15.5 is a photograph of simplex cordage with a macrobend.

To identify a break in the optical fiber, visually inspect the length of the cable under test. If you locate a small spot on the cable that is emitting red light, this is a break. Figure 15.6 is a photograph of multimode cordage with a break.

The VFL can be used to test the continuity of an optical fiber in the same manner described in the “Continuity Tester” section earlier in this chapter. The VFL, however, will couple 1,000 times or more light energy into the optical fiber than the LED continuity tester. You should never view the output of the VFL or the endface of a connector being tested by the VFL directly.

The VFL will test continuity and identify breaks or macrobends in the optical fiber. The VFL is often used in conjunction with an OTDR to help identify the actual location of the fault. Chapter 16, “Link/Cable Troubleshooting,” describes how to use the VFL to troubleshoot macrobends or breaks in an optical fiber.

FIGURE 15.5 Photograph of the light from a VFL identifying a macrobend in simplex cordage

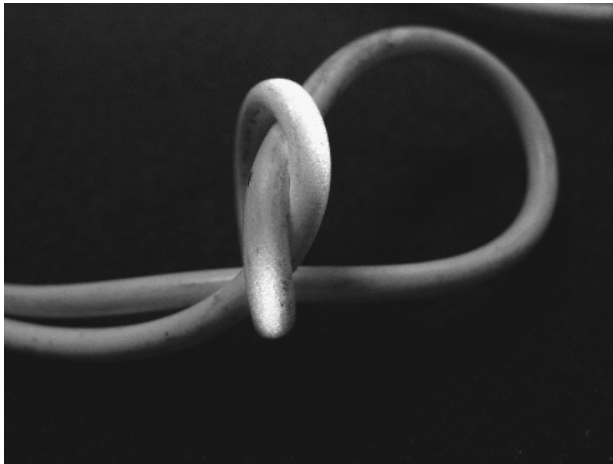


FIGURE 15.6 Photograph of the light from a VFL identifying a break in simplex cordage



Fiber Identifier

The fiber identifier is the fiber optic installer or technician's infrared eyes. By placing a slight macrobend in an optical fiber or fiber optic cable, it can detect infrared light traveling through the optical fiber and determine the direction of light travel. Some fiber identifiers can also detect test pulses from an infrared light source.

The fiber identifier typically contains two photodiodes that are used to detect the infrared light. The photodiodes are mounted so that they will be on opposite ends of the macrobend of the optical fiber or fiber optic cable being tested. Figure 15.7 is a photograph showing the location of the photodiode assemblies in the fiber identifier. The photodiode assemblies look like two small glass lenses.

The fiber identifier can typically be used with coated optical fiber, tight-buffered optical fiber, a single optical fiber cable, or a ribbon cable. Each of these must be placed in the center of the photodiodes during testing. Selecting the correct attachment for the optical fiber or optical fiber cable type under test typically does this. The photograph in Figure 15.8 shows three attachments.

To test an optical fiber or fiber optic cable, select the correct attachment and install it on the fiber identifier. The photograph in Figure 15.9 shows the fiber identifier ready to test a 900 μm tight-buffered optical fiber. Center the tight-buffered optical fiber over the photodiode assemblies and insert a macrobend into the tight-buffered optical fiber by pushing the slide into the tight-buffer optical fiber and compressing it as shown in the photograph in Figure 15.10.

FIGURE 15.7 Photodiode assemblies in the fiber identifier



FIGURE 15.8 Fiber identifier optical fiber and fiber optic cable attachments



FIGURE 15.9 Fiber identifier ready to test a 900 μm tight-buffered optical fiber



FIGURE 15.10 Fiber identifier compressing a tight-buffered optical fiber



If there is sufficient infrared light energy traveling through the optical fiber, one of the directional arrows on the fiber identifier should illuminate. The directional arrow is pointing in the direction the light is traveling. If a test pulse is being transmitted through the optical fiber, the directional arrow and test pulse indicator should illuminate.

The fiber identifier works best with coated optical fiber or tight-buffered optical fiber. Like the VFL, the fiber identifier does not always work with single optical fiber cables. How well it works with a cable depends on the amount of strength member within the cable, jacket thickness, and amount of light energy available from the optical fiber.

The fiber identifier can be used by itself or in conjunction with an OTDR. It can be used to identify traffic in a working fiber optic link or can be used to help identify the location of a fault. Fault location techniques are discussed in detail in Chapter 16.

Optical Return Loss Test Set

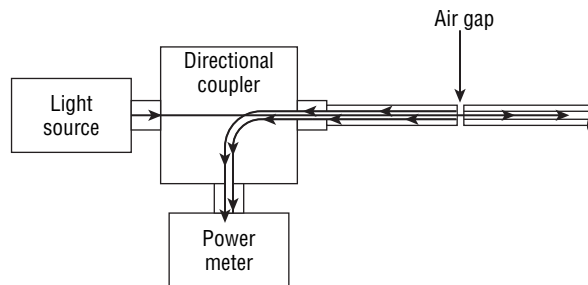
Optical return loss (ORL) testing is performed with an optical return loss test set. ORL testing measures the amount of optical light energy that is reflected back to the transmit end of the fiber optic cable. The energy being reflected back, or back reflections, comes from mechanical interconnections, passive devices, fiber ends, and Rayleigh scattering caused by impurities within the optical fiber.

Besides reducing the amount of light transmitted, back reflections can cause various laser light source problems. They can cause the laser's output wavelength to vary. They can also cause fluctuations in the laser's optical output power and possibly permanently damage the laser light source.

The impact that back reflections have on the laser light source can cause problems in analog and digital systems. In digital systems, they can increase the BER. In analog systems, they reduce the *signal to noise ratio (SNR)*.

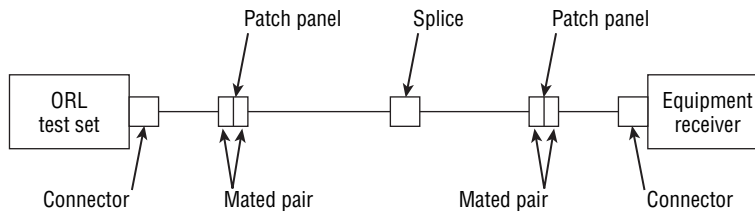
The ORL test set measures return loss using an *optical continuous wave reflectometer (OCWR)*. A light source within the ORL test set continuously transmits light through a directional coupler, as shown in Figure 15.11. Light energy returned from the optical fiber is channeled to the photodiode of a power meter. The light energy measured by the power meter is the return loss.

FIGURE 15.11 Directional coupler in an ORL test set



ORL measurements of a fiber optic link should be taken with all patch cords and equipment cords in place. All system equipment should be turned off. The receive connector should remain plugged into the equipment receiver. The transmit connector should be unplugged from the equipment transmitter and plugged into the ORL test set after the test set has been calibrated, as shown in Figure 15.12. The test set should be calibrated as described in the manufacturer's operation manual. Prior to performing ORL testing, clean and inspect all connectors. Allow sufficient time for the ORL test set to warm up and stabilize.

FIGURE 15.12 ORL test set connected to fiber optic link and system equipment



ORL test sets are available for both multimode and single-mode optical fiber. The multimode ORL test uses an LED light source with a typical output power of -20 dBm. The return loss measurement range is typically between 10 and 45 dB. Accuracy is normally within 0.5 dB.

The single-mode ORL test set uses a laser source with a typical output power of -10 dBm. The return loss measurement range is typically between 0 to 50 dB or 0 to 60 dB. Accuracy in the 0 to 50 dB range is normally 0.5 dB with a decrease in the 0 to 60 dB range of 1 dB.

Light Source and Optical Power Meter

The light source and optical power meter work hand in hand with each other. They are typically referred to as an *optical loss test set*, or *OLTS*. There are many different OLTSs available to the fiber optic installer or technician. Some support only multimode testing, some single-mode, and a few can be used for both. The multimode OLTS is typically the lowest priced, followed by the single-mode and then the combination multimode and single-mode.

This section of the chapter explains the basic operation of the multimode and single-mode OLTS. The OLTS is required by TIA/EIA-568-B.1 for optical fiber transmission performance testing. Multimode testing should be performed in accordance with TIA/EIA-526-14A, method B. Single-mode testing should be performed in accordance with TIA/EIA-526-7, method A.1.

Multimode

The multimode OLTS consist of an LED-based light source and an optical power meter. The typical light source consists of an 850 nm LED and a 1300 nm LED, like the one shown in the photograph in Figure 15.13. The optical power meter is typically selectable for 850 nm, 1300 nm, and 1550 nm, like the one shown in the photograph in Figure 15.14.

FIGURE 15.13 Multimode 850 nm and 1300 nm light source**FIGURE 15.14** Multimode 850 nm, 1300 nm, and 1550 nm optical power meter

There are two types of LED light sources available, filtered and unfiltered. A filtered light source does not overfill the optical fiber. An unfiltered light source does overfill the optical fiber with high-order modes. TIA/EIA-568-B.3 requires that multimode insertion loss measurements be performed with a light source that meets the launch requirements of TIA/EIA-455-50B. This launch requirement can be achieved within the test equipment, which is the case with the filtered light source or with an external mandrel wrap on the transmit test jumper. Most LED light sources in use today are unfiltered and require the mandrel wrap. The mandrel wrap is discussed in great detail later in this chapter.

The optical output power of an unfiltered LED light source is typically -20 dBm for both the 850 nm and 1300 nm LEDs when measured with a $62.5/125$ μm optical fiber. The optical output power increases roughly 4 dB when used with 100/140 μm optical fiber and decreases roughly 3 dB when used with a 50/125 μm optical fiber.



Remember that TIA/EIA-568-B.3 recognizes only 50/125 μm or 62.5/125 μm multimode optical fiber.

Most multimode light sources are designed to operate between the temperatures of 0 to +40° C or 0 to +50° C. Within these temperature ranges, they should provide a stable optical output power that typically varies less than ± 0.2 dB over an eight-hour period. To achieve this performance, the light source should be allowed to warm up before beginning testing. The manufacturer specifies warm-up times.

The multimode power meter typically uses a single photodiode with a switch to select the proper wavelength. The selector switch compensates for the responsivity of the photodiode at the different wavelengths. The output of the optical power meter is in dBm, where 0 dBm is equal to 1 mW of optical power.

Many power meters will support both multimode and single-mode testing. Acceptable optical fiber types may range from 9/125 μm through 100/140 μm . The optical input power range is typically +3 dBm to -50 dBm.

As with the light sources, most optical power meters are designed to operate between the temperatures of 0 and +40° C or 0 and +50° C. When operated within this temperature range, absolute accuracy is typically within ± 0.25 dB, relative accuracy within ± 0.15 dB, and repeatability within ± 0.04 dB. The values stated here are typical. Actual values can be obtained from the manufacturer's data sheet.

Single-Mode

The single-mode OLTS consists of a laser-based light source and an optical power meter. The typical light source consists of a 1310 nm laser and a 1550 nm laser. Some models have a separate output port for each laser and others combine both wavelength lasers into a single output port. Another popular laser combination is 1550 nm and 1625 nm.

Unlike the LED multimode light source, the single-mode light source does not require a mandrel wrap. There is no mandrel wrap required by TIA/EIA-568-B because the laser does not overfill the core of the optical fiber with high-order modes. Single-mode link attenuation measurements should be performed in accordance with TIA/EIA-526-7, method A.1.

The optical output power of the laser is typically around -5 dBm into a 9/125 μm single-mode optical fiber. Some single-mode light sources have the ability to attenuate the output of the laser. Most single-mode light sources are designed to operate between the temperatures of 0 to +40° C or 0 to +50° C. Within these temperature ranges, they should provide a stable optical output power that typically varies no more than ± 0.1 dB over an eight-hour period. To achieve this performance, the light source should be allowed to warm up before beginning testing. The manufacturer specifies warm-up times.

The single-mode optical power meter is typically designed to handle a range of wavelengths. It may measure common multimode optical fiber wavelengths and single-mode optical fiber wavelengths. A typical power meter will be selectable for 850 nm, 1300 nm, 1310 nm, 1480 nm, 1550 nm, and 1625 nm.

Single-mode optical power meters are typically designed to operate between the temperatures of 0 to +40° C or 0 to +50° C. When operated within this temperature range, they provide the best accuracy. Absolute accuracy is typically within ± 0.25 dB when measured at +25° C with an optical input power of -10 dBm, relative accuracy within ± 0.15 dB, and repeatability within ± 0.04 dB. The values stated here are typical. Actual values can be obtained from the manufacturer's data sheet.

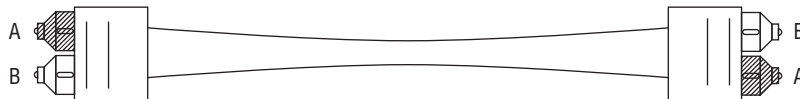
Patch Cord

TIA/EIA-568-B.3 section 6 sets the performance specifications for optical fiber patch cords recognized in premises cabling standards. The optical fiber patch cord is used at cross-connects to connect optical fiber links. They are also used as equipment or work area cords to connect telecommunications equipment to horizontal or backbone cabling.

An optical fiber patch cord is a two-fiber cable. It uses the same connector type and optical fiber type as the optical fiber cabling that it is connected to. The patch cord must meet the cable transmission performance requirements and physical cable specifications of sections 4.2 and 4.3 of TIA/EIA-568-B.3. The patch cord must also meet the connector and adapter requirements of section 5.2 of TIA/EIA-568-B.3.

Optical fiber patch cords used for either cross-connection or interconnection to equipment shall be orientated as defined in section 6.4 of TIA/EIA-568-B.3 and shown in Figure 15.15. Position A goes to position B on each optical fiber. The connector that plugs into the receiver is position A and the connector that plugs into the transmitter is position B.

FIGURE 15.15 Optical fiber patch cord terminal configuration



Test Jumper

The terms “patch cord” and “jumper” are often interchanged. A *patch cord* as defined by TIA/EIA-568-B.3, section 6 is a two-fiber cable. However, the term “patch cord” is often used to describe a single-fiber cable. IEEE standard 802.3 defines a jumper cable assembly as an optical assembly used for bidirectional transmission and reception of information. A test jumper can be a single-fiber cable or a multi-fiber cable. This section of the chapter focuses on multimode test jumpers as described in TIA/EIA-526-14A.

In the fiber optic industry, the *test jumper* has several names. The test jumper connected to the light source is typically called the transmit jumper. The test jumper connected to the optical

power meter is typically called the receive jumper. The U.S. Navy calls the test jumper a measurement quality jumper, or MQJ. You may also see the test jumper referred to as a reference jumper. Regardless of the name, the test jumper is a critical part of your optical power measurement equipment.

Test jumpers must have a core diameter and numerical aperture nominally equal to the optical fiber being tested. You cannot test a 50/125 μm link with a 62.5/125 μm test jumper or vice versa. Per TIA/EIA-526-14A, jumpers shall be no less than one meter in length and no greater than 5 meters in length. The termination of the test jumper shall be compatible with the cable being tested.

Test jumpers should be cleaned and inspected prior to making measurements. The endface of each connector should be evaluated under a microscope. They should always meet or exceed the good criteria of TIA/EIA-455-57B. This means that there can be no scratches, notches, or chips in the endface of the optical fiber. You should also clean and inspect each connector of the cable under test. Mating a damaged connector with a test jumper connector can destroy the test jumper connector.

Test jumpers should be tested for insertion loss prior to performing any of the TIA/EIA-526-14A methods. They should be tested the same way a patch cord is tested. This test method is described later in this chapter. The maximum acceptable loss for a test jumper is 0.4 dB. This value is not defined by the standard but serves as a good rule of thumb. Many test jumpers will have a measured loss of less than 0.2 dB.

Mode Filter

The *mandrel wrap* or *mode filter* is required by TIA/EIA-568-B.1 and TIA/EIA-568-B.3 for light sources that do not meet the launch requirements of TIA/EIA-455-50B, method A. It is required because the attenuation in a short link of multimode optical fiber may be higher than calculated when making insertion loss measurements due to the power lost in the high-order modes. The high-order modes are caused by an LED light source that overfills the optical fiber.

When measuring the insertion loss of a multimode link with an LED source that overfills the optical fiber, a mandrel wrap must be used on the transmit reference jumper. The reference jumper should have five non-overlapping turns around a smooth mandrel. The diameter of the mandrel depends on the optical fiber core diameter, as shown in Table 15.1. Figure 15.16 is a photograph of a reference jumper with a mandrel wrap attached to an LED light source.

An unfiltered LED light source overfills an optical fiber by launching high- and low-order modes into the core and cladding, as shown in Figure 15.17. The high-order modes experience more attenuation than the low-order modes at interconnections, splices, and bends. This causes higher than expected loss in short multimode links. The mandrel wrap takes advantage of the high loss of high-order modes caused by excessive bending of the optical fiber. The mandrel wrap removes the high-order modes by inserting a series of macrobends in the transmit jumper, as shown in Figure 15.18.

FIGURE 15.16 Transmit test jumper with mandrel wrap

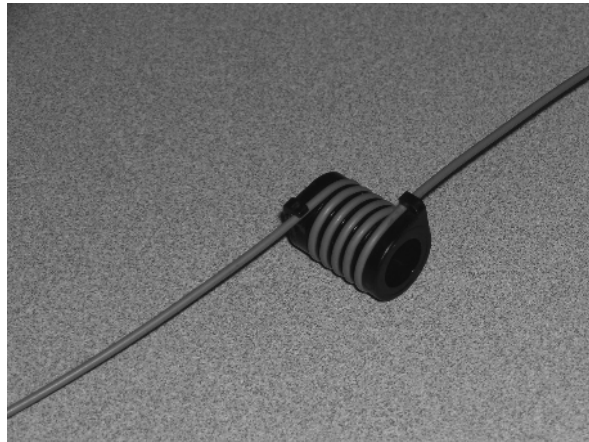
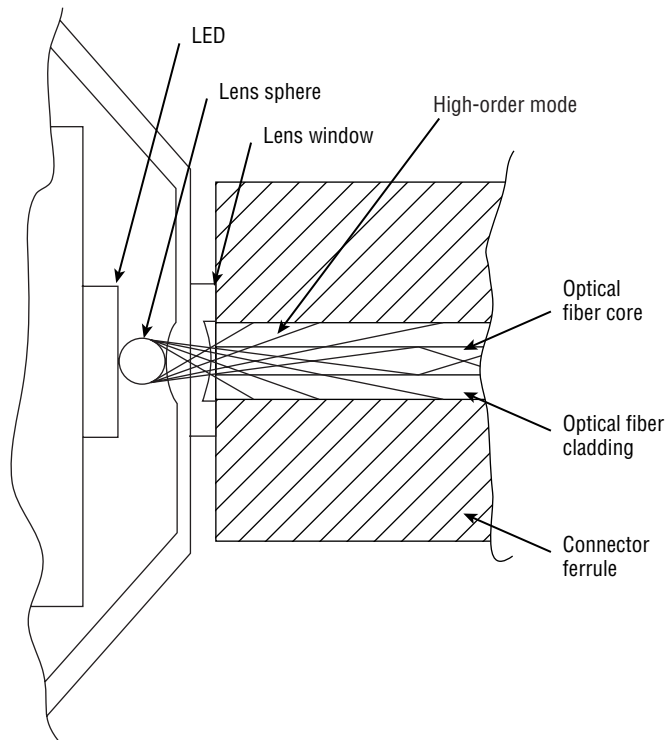


FIGURE 15.17 LED source overfilling a multimode optical fiber



The five small-radius non-overlapping loops around the mandrel wrap significantly attenuate the high-order modes with minimal attenuation of the low-order modes. The mandrel wrap is actually a low-order or low-pass mode filter. It allows the low-order modes to pass with very little attenuation while greatly attenuating the high-order modes. The mandrel wrap should always be on the reference jumper at every stage of testing when making insertion loss measurements.

FIGURE 15.18 Macrobend attenuating loosely coupled modes in a multimode optical fiber

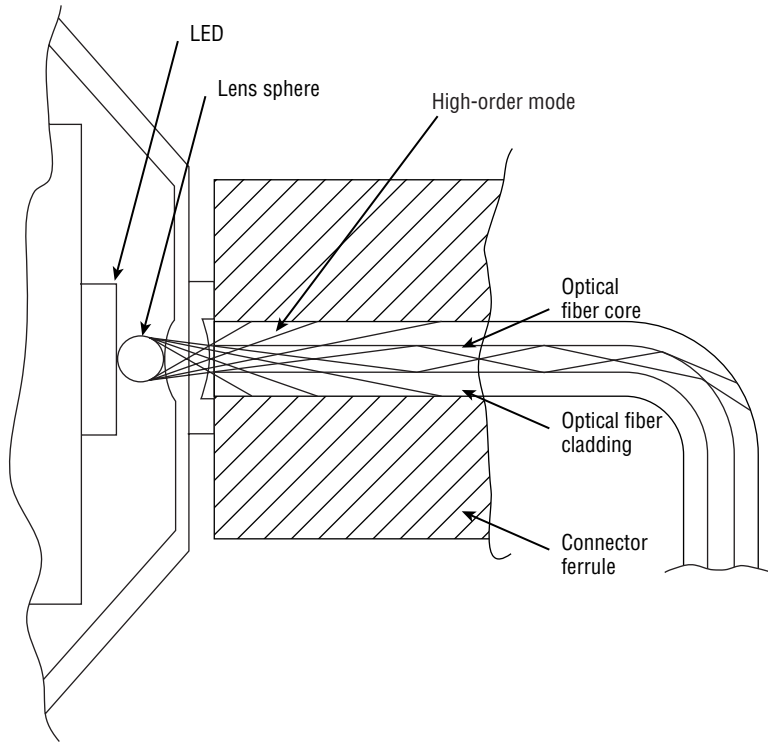


TABLE 15.1 Mandrel Diameters for Multimode Optical Fiber

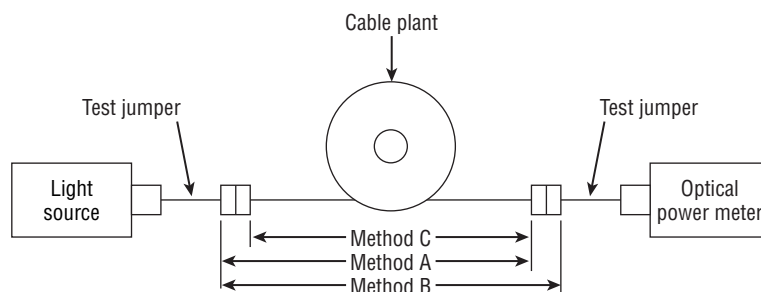
Fiber Core Size (μm)	Mandrel Diameter for 900 μm Buffered Optical Fiber (mm)	Mandrel Diameter for 3 mm Jacketed Cable
50	25	22
62.5	20	17

TIA/EIA-526-14A Optical Loss Measurement

TIA/EIA-526-14A provides three methods for testing the cable plant. The difference between the three methods is how the reference power measurement is taken. Each method uses a different number of test jumpers. Method A uses two test jumpers to obtain the reference power measurement, method B uses one test jumper, and method C uses three test jumpers. Each of these methods is described in detail in this section of the chapter.

Test methods A, B, and C each measure the optical loss of the cable plant at different points, as shown in Figure 15.19. Test method A measures the loss of the cable plant plus one connection loss. Test method B measures the loss of the cable plant plus two connection losses. Test method C measures the loss of the cable plant only. Test method B is the only method that can be used when testing to TIA/EIA-568-B.

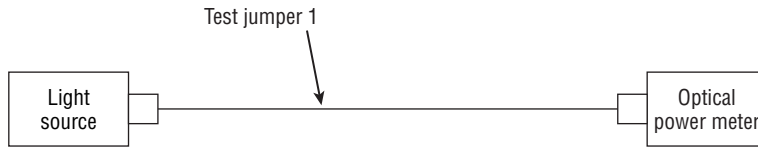
FIGURE 15.19 Cable plant measured values for methods A, B, and C



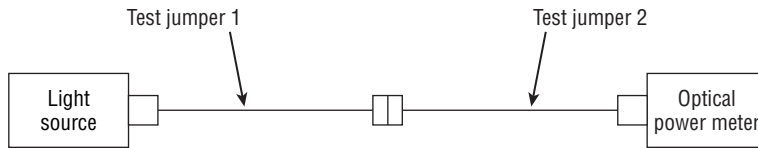
The first step before performing any of the optical power loss measurements is to clean and inspect the test jumpers and connectors at the ends of the cable plant to be tested. Verify that your test jumpers have the same optical fiber type as the cable plant you are going to test. Install the mandrel wrap on test jumper 1 as defined in TIA/EIA-568-B.1. Test jumper 1 will be connected to the light source.

While you are cleaning and inspecting the test jumpers and cable plant connectors, turn on the light source and optical power meter. Set both of them to the test wavelength. Allow sufficient time for the light source and optical power meter to warm up and stabilize. Warm-up information can be found in the manufacturer's data sheet.

After sufficient time has passed for the light source and optical power meter to stabilize, you need to measure the loss of the test jumpers. Connect test jumper 1 as shown in Figure 15.20. Record the optical power displayed on the optical power meter. This number is the reference power measurement in dBm. This number is typically around -20 dBm with a $62.5/125$ μm multimode optical fiber and -23 dBm with a $50/125$ μm multimode optical fiber. These numbers can vary from OLTS to OLTS.

FIGURE 15.20 Test jumper 1 optical reference power measurement

Connect test jumper 2 as shown in Figure 15.21. Record the optical power displayed on the optical power meter. The difference between the reference power measurement and this value is the connection loss for test jumpers 1 and 2. The difference between this value and the reference power measurement should be less than or equal to 0.4 dB. If this number is greater than 0.4 dB, re-clean the connectors and re-test. If the loss still exceeds 0.4 dB, replace test jumper 2. Repeat this test for test jumper 3 if method C is being used.

FIGURE 15.21 Test jumper 2 optical reference power measurement

Method A

Method A uses two jumpers to establish the optical power reference. Connect test jumpers 1 and 2 as shown in Figure 15.21. Record the optical power displayed by the optical power meter. This number is the reference power measurement in dBm. This number is typically around -20 dBm with a $62.5/125\ \mu\text{m}$ multimode optical fiber and -23 dBm with a $50/125\ \mu\text{m}$ multimode optical fiber. These numbers can vary from OLTS to OLTS.

After recording the reference power measurement, separate the test jumpers at their point of interconnection as shown in Figure 15.22. Do this without disturbing their attachment to the light source and optical power meter. Attach the test jumpers to the cable plant as shown in Figure 15.23.

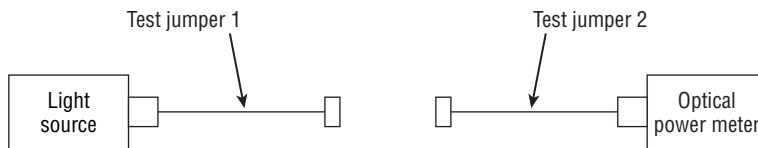
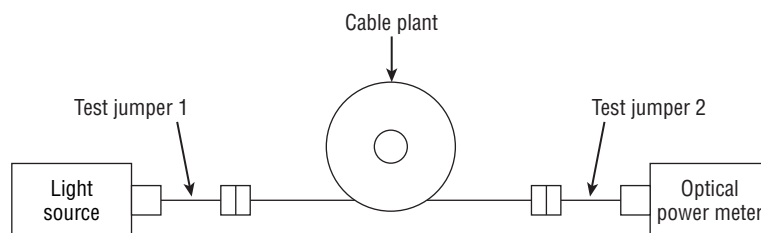
FIGURE 15.22 Test jumpers 1 and 2 separated

FIGURE 15.23 Test jumpers 1 and 2 connected to the cable plant

With the test jumpers attached, record the optical power displayed on the optical power meter. This is your test power measurement. The loss for the cable plant is the difference between the reference power measurement and the test power measurement.

To obtain the optical power loss for the cable plant, subtract the test power measurement from the reference power measurement. If the reference power measurement was -20.4 dBm and the test power measurement was -21.6 dBm, the optical power loss for the cable plant would be 1.2 dB.

Method B

Method B uses one jumper to establish the optical power reference. However, two test jumpers are required to perform the test. This method is required by TIA/EIA-568-B.3 section 11.3 to measure link attenuation. Per TIA/EIA-568-B.1 section 11.3.3.1, horizontal links shall be tested at 850 nm and 1300 nm in one direction. Per section 11.3.3.2, multimode backbone links shall be tested in one direction at both 850 nm and 1300 nm. Per section 11.3.3.3, centralized links shall be tested at 850 nm or 1300 nm in one direction; 850 nm is recommended. Link attenuation using this method is equal to cable attenuation plus connector insertion loss plus splice insertion loss.



Single-mode backbone links should be tested at 1310 nm and 1550 nm in accordance with ANSI/TIA/EIA-526-7 method A.1.

Connect test jumper 1 as shown earlier in Figure 15.20. Record the optical power displayed by the optical power meter. This number is the reference power measurement in dBm. This number is typically around -20 dBm with a $62.5/125$ μm multimode optical fiber and -23.5 dBm with a $50/125$ μm multimode optical fiber. These numbers can vary from OLTs to OLTs.

After recording the reference power measurement, disconnect the test jumper from the optical power meter. Do not disturb the jumper's attachment to the light source. Attach test jumper 2 to the optical power meter as shown in Figure 15.22. Attach the test jumpers to the cable plant as shown in Figure 15.23.

With the test jumpers attached, record the optical power displayed on the optical power meter. This is your test power measurement. The loss for the cable plant is the difference between the reference power measurement and the test power measurement.

To obtain the optical power loss for the cable plant, subtract the test power measurement from the reference power measurement. If the reference power measurement was -20.7 dBm and the test power measurement was -23.6 dBm, the optical power loss for the cable plant would be 2.9 dB.

Method C

Method C uses three jumpers to establish the optical power reference. Connect test jumpers 1, 2, and 3 as shown in Figure 15.24. Record the optical power displayed by the optical power meter. This number is the reference power measurement in dBm. This number is typically around -20 dBm with a $62.5/125$ μm multimode optical fiber and -23.5 dBm with a $50/125$ μm multimode optical fiber. These numbers can vary from OLTS to OLTS.

After recording the reference power measurement, separate the test jumpers at their point of interconnection as shown in Figure 15.25. Do this without disturbing their attachment to the light source and optical power meter. Attach the test jumpers to the cable plant as shown earlier in Figure 15.23.



Test jumper 3 is not used during testing. It is used only to establish reference power.

With the test jumpers attached, record the optical power displayed on the optical power meter. This is your test power measurement. The loss for the cable plant is the difference between the reference power measurement and the test power measurement.

To obtain the optical power loss for the cable plant, subtract the test power measurement from the reference power measurement. If the reference power measurement was -20.2 dBm and the test power measurement was -21.6 dBm, the optical power loss for the cable plant would be 1.4 dB.

FIGURE 15.24 Method C reference power measurement

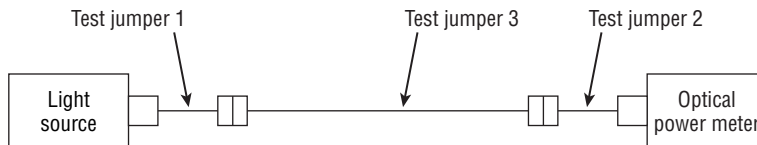
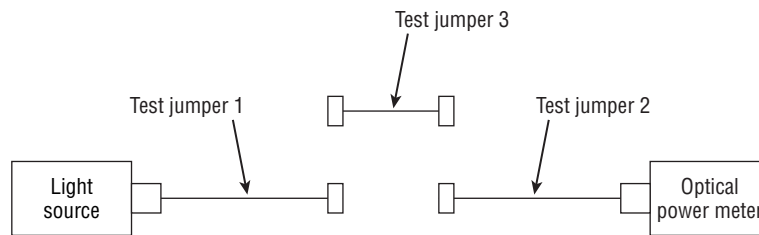


FIGURE 15.25 Removal of test jumper 3 from method C reference power measurement

Patch Cord Optical Power Loss Measurement

Multimode patch cord optical loss power measurement is performed using the steps described in TIA/EIA-526-14A method A. The patch cord is substituted for the cable plant. Because patch cords are typically no longer than 5 m, the loss for the optical fiber is negligible and testing can be performed at 850 nm or 1300 nm. The loss measured in this test is the loss for the patch cord connector pair. TIA/EIA-568-B.3 states that the maximum loss for a connector pair is 0.75 dB.

After setting up the test equipment as described in TIA/EIA-526-14A method A, clean and inspect the connectors at the ends of the patch cords to be tested. Verify that your test jumpers have the same optical fiber type and connectors as the patch cords you are going to test. Ensure that there are no sharp bends in the test jumpers or patch cord during testing.

Because both patch cord connectors are easily accessible, optical power loss should be measured in both directions. The loss for the patch cord is the average of the two measurements. If the loss for the patch cord exceeds 0.75 dB in either direction, the patch cord needs to be repaired or replaced.

OTDR

So far in this chapter, you have learned about tools and test equipment that can be used to test a fiber optic link or cable. Of all the tools and test equipment discussed, none provide the fiber optic technician with more information about the fiber optic cable or link than the *optical time domain reflectometer* (OTDR). The OTDR allows the fiber optic technician to evaluate the loss and reflectance of interconnections and splices. It will measure the attenuation rate of an optical fiber and locate faults.

This section of the chapter focuses on basic OTDR theory, setup, and testing. The OTDR is a complex piece of test equipment with many variables that must be programmed correctly before testing can be performed. Proper OTDR setup and cable preparation will ensure accurate test results.

OTDR Theory

The OTDR is nothing more than a device that launches a pulse or pulses of light into one end of an optical fiber and records the amount of light energy that is reflected back. Unlike all the test equipment discussed up to this point, the OTDR allows the fiber optic technician to see with great resolution what is happening in the fiber optic link or cable under test. With the OTDR, the fiber optic link or cable is no longer a black box. The fiber optic technician can see how light passes through every segment of the fiber optic link.

Light reflecting back in an optical fiber is the result of reflection or backscatter. Reflections are when the light traveling through the optical fiber encounters changes in the refractive index. These reflections are called Fresnel reflections. Backscatter, or Rayleigh scattering, results from evenly distributed compositional and density variations in the optical fiber. Photons are scattered along the length of the optical fiber. The photons that travel back toward the OTDR as shown in Figure 15.26 are considered backscatter.

OTDRs come in many different shapes and sizes, as shown in Figure 15.27. The newer models are almost pocket-sized while others require a shoulder strap. The small OTDR has size advantages; however, the small screen permits only a limited viewing area.

The OTDR is typically a battery-powered device. It's battery powered because many places where the OTDR is used have no electrical power available. It is a good idea to bring a charged spare battery with you when you are performing testing.

Many OTDRs can be configured to test both multimode and single-mode optical fibers. A typical OTDR may hold up to three light source modules, like the OTDR shown in Figure 15.28. Modules can be added or removed as testing requirements change. Some OTDRs even contain a visible fault locator like the one shown in Figure 15.29.

FIGURE 15.26 Backscattered photons

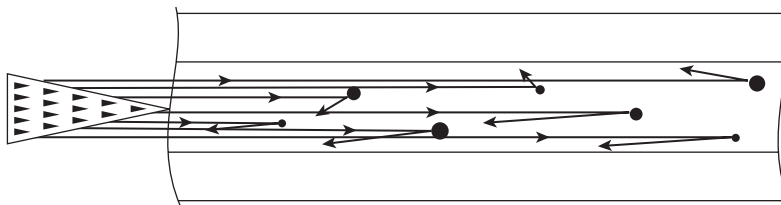


FIGURE 15.27 Large and small OTDRs



FIGURE 15.28 Light source module locations on an OTDR

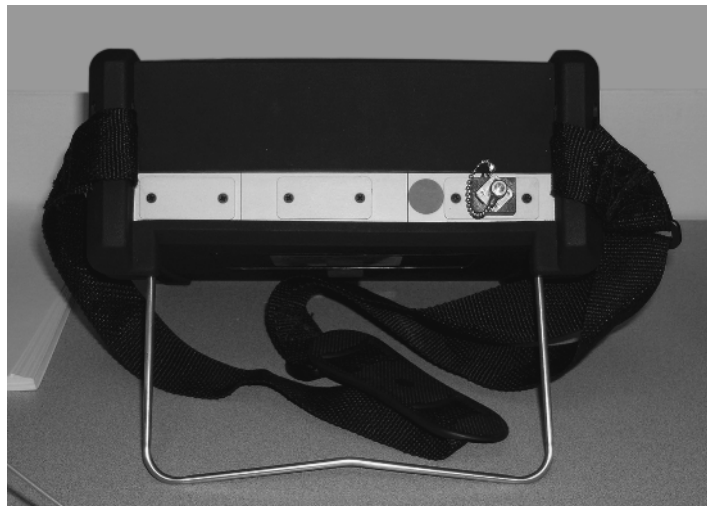
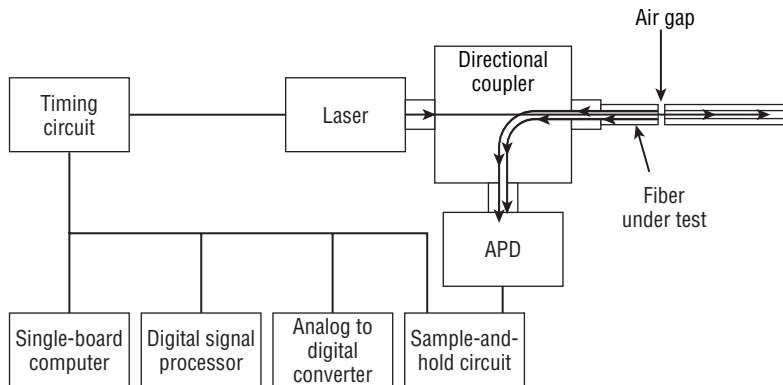


FIGURE 15.29 The visible fault locator built into an OTDR**FIGURE 15.30** OTDR block diagram

A typical OTDR can be broken up into eight basic components: the *directional coupler*, *laser*, *timing circuit*, *single-board computer*, *digital signal processor (DSP)*, *analog to digital converter*, *sample-and-hold circuit*, and *avalanche photodiode*. Figure 15.30 is a block diagram of the OTDR showing how light is launched from the laser through the directional coupler into

the optical fiber. The directional coupler channels light returned by the optical fiber to the avalanche photodiode.

The avalanche photodiode converts the light energy into electrical energy. The electrical energy is sampled at a very high rate by the sample-and-hold circuit. The sample-and-hold circuit maintains the instantaneous voltage level of each sample long enough for the analog to digital converter to convert the electrical value to a numerical value. The numerical value from the analog to digital converter is processed by the DSP and the result is sent to the single-board computer to be stored in memory and displayed on the screen. This entire process is typically repeated many times during a single test of an optical fiber and coordinated by the timing circuit.

For the OTDR to produce accurate results, the refractive index of the optical fiber under test must be known. The refractive index of an optical fiber is different for each wavelength tested. The fiber optic technician must enter the correct refractive index into the OTDR for each wavelength. The correct refractive index for a fiber optic cable can be obtained from the manufacturer.

The OTDR samples light energy from the optical fiber at precise intervals. Each sample taken by the OTDR represents the round-trip time for light energy in the optical fiber. Let's assume that the OTDR is taking 500 million samples per second or one sample every two nanoseconds. If the refractive index for the optical fiber under test were equal to 1.5, every five samples would represent the distance of 1 meter, as shown in Figure 15.31. The following formula can be used to find distance based on time and refractive index. In this formula, the speed of light is rounded up to 3×10^8 m/s:

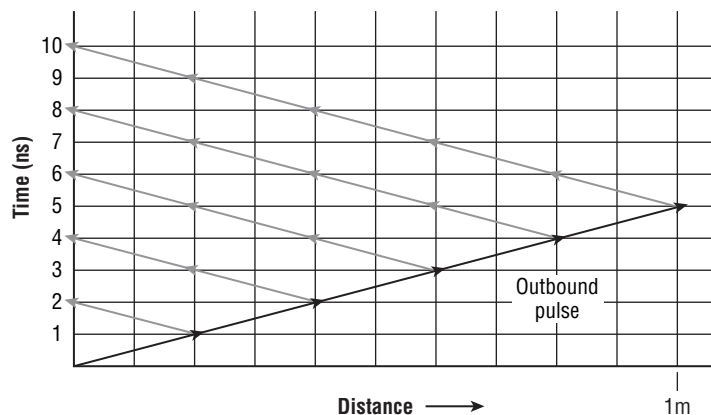
$$\text{Distance} = ((\text{time in ns})/2) \times (\text{speed of light in free space})/\text{refractive index}$$

For the above example:

$$\text{Distance} = (10 \text{ ns}/2) \times (3 \times 10^8)/1.5$$

$$\text{Distance} = 1 \text{ m}$$

FIGURE 15.31 OTDR sampling at 2 ns rate



EXERCISE 15.1

Calculate the distance to the end of an optical fiber.

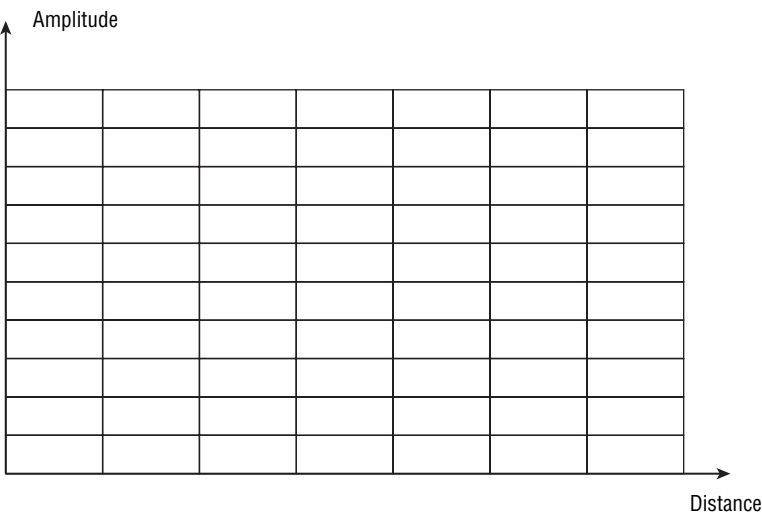
After 970 ns, a large back reflection is received by the avalanche photodiode in the OTDR. This large back reflection represents the end of the optical fiber. The refractive index for the optical fiber under test is 1.475. With the information provided, calculate the length of the optical fiber.

- To solve this, use the formula
$$\text{Distance} = (\text{time in ns})/2 \times (\text{speed of light in free space})/\text{refractive index}$$
- Plug in the values provided above and the formula looks like this:
$$(970 \text{ ns})/2 \times (3 \times 10^8)/1.475$$
- When the formula is solved, the distance equals 98.64 m.

OTDR Display

The OTDR displays time or distance on the horizontal axis and amplitude on the vertical axis, as shown in Figure 15.32. The horizontal axis can typically be programmed to display distance in feet, meters, or kilometers. The vertical axis is not programmable. The vertical axis displays relative power in dB.

FIGURE 15.32 OTDR display



The OTDR creates a trace like the one shown in Figure 15.33. The trace shows event loss, event reflectance, and optical fiber attenuation rate. The OTDR can horizontally and vertically zoom in on any section of the trace. This permits a more detailed inspection of the optical fiber or event.

The light pulses from OTDR produce a blind spot or dead zone like the one shown in Figure 15.34. The dead zone is an area where the avalanche photodiode has been saturated by the reflectance of a mechanical interconnection. The size of the dead zone depends on the length of each light pulse and the amount of light reflected back toward the avalanche photodiode.

FIGURE 15.33 Event-filled OTDR trace

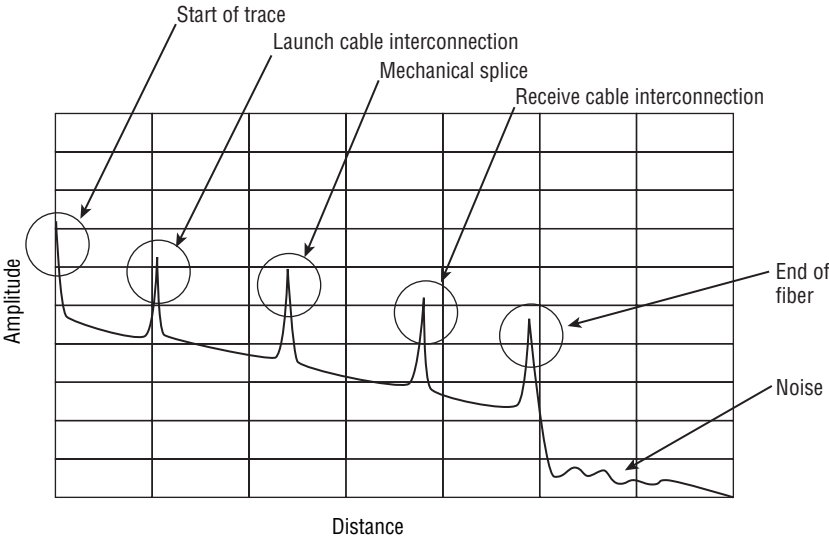
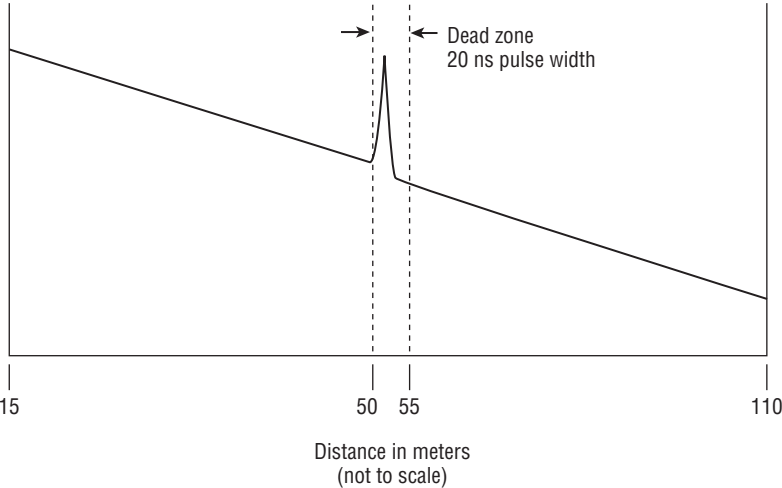


FIGURE 15.34 OTDR trace dead zone



An interconnection that has very little reflectance will not saturate the avalanche photodiode. This type of interconnection will produce the smallest dead zone. The dead zone will typically be no longer than three times the length of the light pulse as seen by the OTDR. Dividing the pulse width in nanoseconds by 10 can approximate the length of the light pulse as seen by the OTDR in meters. A pulse width of 20 ns will yield an ideal dead zone of 2 m.

The length of the dead zone depends on the reflectance. High reflectance interconnections saturate the avalanche photodiode and produce a greater than ideal dead zone, as much as ten times greater than ideal.

OTDR Setup

Prior to testing, the OTDR needs to be set up correctly to provide the most accurate readings. When setting up the OTDR, you need to select the correct fiber type, wavelength or wavelengths, range and resolution, pulse width, averages, refractive index, thresholds, and backscatter coefficient. This process takes only a couple of minutes and ensures the most accurate results possible.

There are many different OTDRs on the market and it is impossible to describe the setup for each. This section of the chapter focuses on general setup parameters. Some OTDRs may have additional parameters and some may not include all of the parameters discussed.

Fiber Type

Each light source or light source module in an OTDR is designed for one or several specific optical fiber types. A multimode module should not be used to test a single-mode optical fiber, and vice versa. Before heading for the test site, ensure that your OTDR has the correct module for the optical fiber to be tested.

Wavelength

The wavelength that your OTDR can test with depends on the light source module or modules in your OTDR. Some light source modules contain a single laser while others contain two different wavelength lasers. A light source with two lasers allows testing of the optical fiber at two wavelengths without disconnecting the cable under test. This simplifies testing and reduces testing time.

Range and Resolution

The distance range of an unzoomed trace displayed on the OTDR and the distance between data points is determined by range and resolution. As a general rule of thumb, the OTDR range should be set to 1.5 times the length of the fiber optic link. If the range is set too short, the results will be unpredictable and the entire link may not be displayed. If the range is set too long, the trace will fill only a small portion of the display area.

Increasing the range automatically increases the distance between the data points. When the distance between the data points is increased, resolution is reduced. When setting range, choose the first range value that exceeds your fiber optic link length. Selecting a 2 km range for a 1.3 km link will yield more accurate results than selecting a 20 km range.

Pulse Width

The pulse width determines the size of the dead zone and the maximum length optical fiber that can be tested. A short pulse width produces a small dead zone. However, a short pulse width reduces the length of optical fiber that can be tested.

The pulse width should be selected so that the trace never disappears into the noise floor. If the pulse width is set properly, the trace will stay smooth until the end of the fiber optic link. If the pulse is set too wide, events may be lost in the dead zone.

Averages

When setting the averages parameter on an OTDR, select the number of averages that produce a smooth trace in the least amount of time. If too few averages are taken, the trace will appear noisy because the noise floor is too high. If too many samples are taken, the trace will be smooth; however, valuable testing time will be wasted.

Refractive Index

As mentioned earlier in the chapter, the manufacturer should specify the refractive index or the group index of refraction for an optical fiber. The refractive index of similar optical fibers does not vary much from manufacturer to manufacturer. Most values are typically within 1% of each other. If the value for the optical fiber you are testing is not known, use the values shown in Table 15.2. Entering a low refractive index will produce measurements that are too long, and entering a high refractive index will produce measurements that are too short.

TABLE 15.2 Refractive Index Default Values

Wavelength	Refractive Index
850 nm	1.4960
1300 nm	1.4870
1310 nm	1.4675
1550 nm	1.4681

Thresholds

Thresholds can typically be set for end of fiber, event loss, and reflectance. Many OTDRs have a default value preset for each of these. Depending on the testing you are performing, the default values may be too sensitive or not sensitive enough.

Most OTDRs will generate event tables automatically based on the threshold settings. To capture a majority of the events in the event table, set the thresholds on the sensitive side. A good starting place would be to set the values as shown in Table 15.3. If these values are too sensitive, you can always increase them.

TABLE 15.3 Threshold Default Values

Event	Default Value
End of fiber	6.0 dB
Event loss	0.05 dB
Reflectance	−65 dB

Backscatter Coefficient

The OTDR uses the backscatter coefficient to calculate reflectance. As with the refractive index, the optical fiber manufacturer specifies backscatter coefficient. Backscatter coefficient does vary from manufacturer to manufacturer. However, the variation is typically not that great and default values shown in Table 15.4 can be used with good results when the manufacturer’s specified value is not known.

TABLE 15.4 Backscatter Coefficient Default Values

Wavelength	Backscatter Coefficient
850 nm	68.00
1300 nm	76.00
1310 nm	80.00
1550 nm	83.00

Cable Plant Test Setup

Now that the OTDR has been configured with the correct light source modules and the setup parameters have been entered, the OTDR can be connected to the fiber optic link. There are many different ways that the OTDR can be used to test a single optical fiber cable or fiber optic link. In this chapter, the test setup will include a launch cable and receive cable. Launch and receive cables may also be referred to as *pulse suppression cables*.

Launch and receive cables allow the OTDR to measure the insertion loss at the near and far ends of the fiber optic link. Like test jumpers, launch and receive cables should be constructed of optical fiber similar to the optical fiber under test. Launch and receive cables are available from many different manufacturers and come in all shapes and size. Some launch and receive cables are designed for rugged environments like the one in Figure 15.35. Others are designed to be compact, almost pocket-sized, like the one in Figure 15.36.

You should select the length of the launch and receive cables based on the pulse width you are using to test the optical fiber. Launch and receive cables are typically the same length. A good rule of thumb is to have at least 1 meter of launch or receive cable for each nanosecond of pulse width, with the minimum length cable being 100 m. A 2 km link being tested with a 20 ns pulse width would require at least a 100 m launch and receive cable. A 10 km link being tested with a 200 ns pulse would require at least a 200 m launch and receive cable.

After the launch and receive cables have been selected, they can be attached to the fiber optic link as shown in Figure 15.37. Remember to clean and inspect the connectors on the cable under test and on the launch and receive cables. Dirty or damaged connectors on the launch and receive cables interconnection can cause high near-end or far-end connection insertion loss.



Real World Scenario

If Your Pulse Suppression Cables Don't Match, You May See a Ghost

One day I was watching some students test a reel of multi-fiber cable with the OTDR. The Army had recently used this cable in the desert. Each fiber in the cable had been terminated on both ends.

After cleaning and visually inspecting the connectors, the pulse suppression cables were connected to both ends of the cable. The OTDR was connected to the launch cable and each optical fiber was tested. Looking at the OTDR trace, the students concluded that each optical fiber had multiple breaks because the OTDR trace had multiple back reflections every 100 meters.

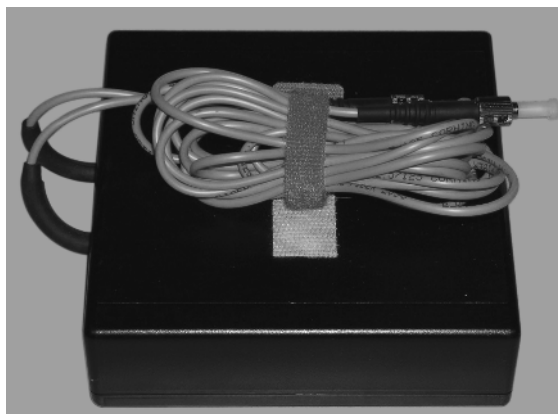
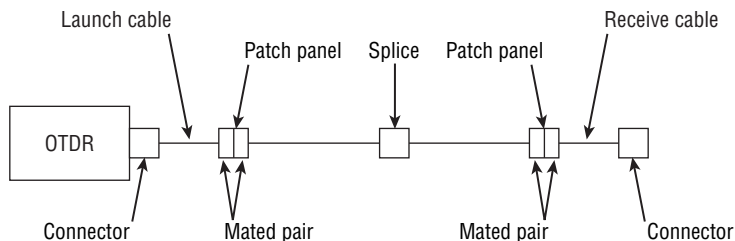
I asked the students to test the continuity of each optical fiber with the continuity tester. Each optical fiber had good continuity. I then asked the students the core diameter of the optical fiber they were testing and the pulse suppression cables.

It turned out that the repetitive pulses or ghosts on the OTDR trace were caused by the core diameter mismatch between the pulse suppression cables and the optical fiber under testing. The students were testing a 50/125 μ m optical fiber with 62.5/125 μ m pulse suppression cables.

FIGURE 15.35 Rugged launch/receive cable



Photo courtesy of OptiConcepts

FIGURE 15.36 Compact launch/receive cable**FIGURE 15.37** OTDR connected to fiber optic link under test with launch and receive cables

Testing and Trace Analysis

Testing the cable plant or a fiber optic link with the OTDR should be done in both directions. Multimode optical fiber should be tested at both 850 nm and 1300 nm. Single-mode optical fiber should be tested at both 1310 nm and 1550 nm. Event loss and optical fiber attenuation is typically the average of the bidirectional values.

There are many standards on how to test both multimode and single-mode optical fiber with an OTDR. This section of the chapter describes several techniques that will allow you to measure the distances and losses in a typical cable plant or fiber optic link. The chapter focuses on short links—the type you would typically find in a commercial building cabled to the TIA/EIA-568-B Commercial Building Telecommunications Standard.

All measurement techniques discussed in this chapter utilize the 2-point method. All OTDRs can measure loss using the 2-point method, but not all OTDRs can automatically perform 2-point subtraction. This section of the chapter assumes that your OTDR can perform 2-point subtraction. If your OTDR can't, you will have to perform the calculations manually.

Some OTDRs are capable of *least-squares averaging (LSA)*. LSA can be used to measure attenuation. This section of the chapter does not address LSA measurements. If your OTDR can perform LSA measurements, consult the operator's manual for how to perform them.

When testing with the OTDR, always observe the manufacturer's precautions. Never directly view the end of an optical fiber being tested with an OTDR. Viewing the end of an optical fiber being tested with an OTDR directly or with a microscope may cause eye damage.

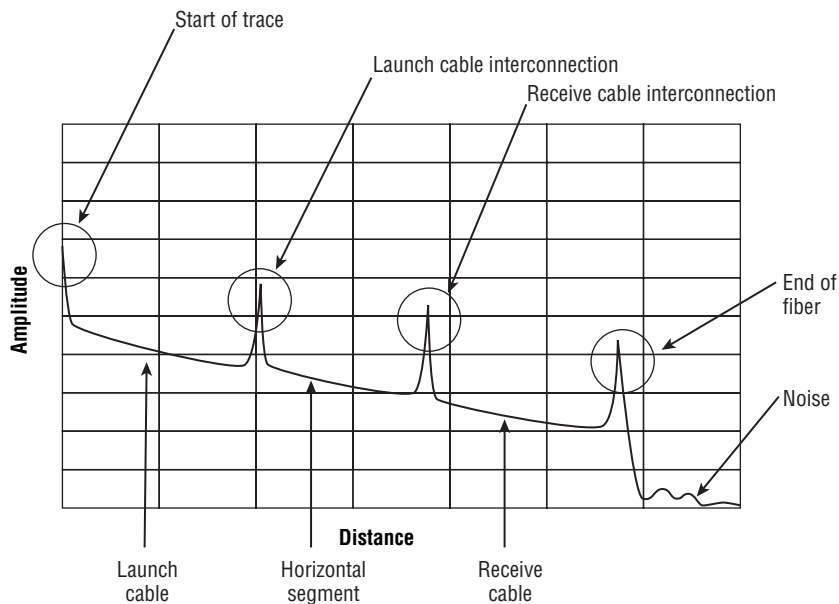
Baseline Trace

The first step in trace analysis is to generate the baseline trace. You should not press the test button on the OTDR to generate a baseline trace until you ensure the following:

- All connectors have been cleaned and inspected and are undamaged.
- Launch and receive cables have optical fiber similar to the optical fiber under test.
- Launch and receive cables are the correct length.
- Launch and receive cables are properly connected to each end of the fiber optic link under test.
- The correct fiber type, wavelength, range and resolution, pulse width, averages, refractive index, and backscatter coefficient have been entered into the OTDR.

Figure 15.38 is a drawing of a baseline trace as presented on the OTDR screen. This baseline trace contains a launch cable, horizontal segment, and receive cable. The launch and receive cables are 100 m in length. The horizontal segment is 85 m in length.

FIGURE 15.38 Baseline trace of horizontal segment



Looking at the trace from left to right there is a large back reflection at the input to the launch cable. Because a 20 ns pulse width was selected, the trace is smooth within 10 m. The smooth trace slopes gradually to the back reflection caused by the connector pair where the launch cable and horizontal segment are connected together.

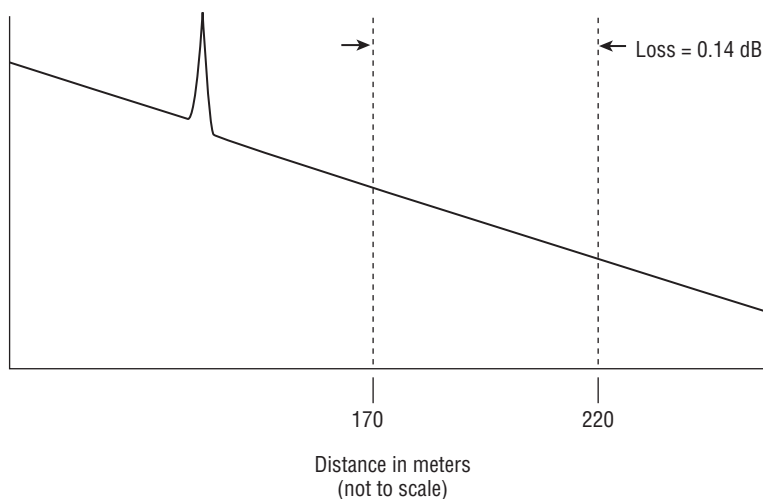
The trace becomes smooth again 10 m after the interconnection back reflection. The trace remains smooth up to the back reflection caused by the connector pair where the receive cable and horizontal segment interconnect. The trace again becomes smooth 10 m after the interconnection back reflection until a large back reflection is generated by the end of the receive cable. The receive cable back reflection is followed by a large reduction in amplitude, then the trace disappears into the noise floor.

Measuring the Attenuation of a Partial Length of Optical Fiber

When using the 2-point method, the first thing you should do is measure the attenuation for a partial segment of the cable under test. This should be done for each wavelength that you are testing the optical fiber with. It has to be done only one time for each cable type being tested. The data should be recorded—this information will be needed to accurately measure interconnection loss.

After taking the baseline trace, position the two cursors on a smooth section of the optical fiber. The longer the section, the more accurate your results will be, because noise will have less impact on the overall measurement. The trace in Figure 15.39 has the A and B cursors 50 m apart on the smooth section of the horizontal segment of the trace. The loss for this 50 m segment at a wavelength of 850 nm is approximately 0.14 dB.

FIGURE 15.39 Measuring the attenuation of a cable segment using the 2-point method





Because of ripple on the smooth trace, you will notice that the A-B loss changes slightly every time a cursor moves. Ripple caused by noise will cause variations from measurement to measurement. This variation may be as much as 0.05 dB.

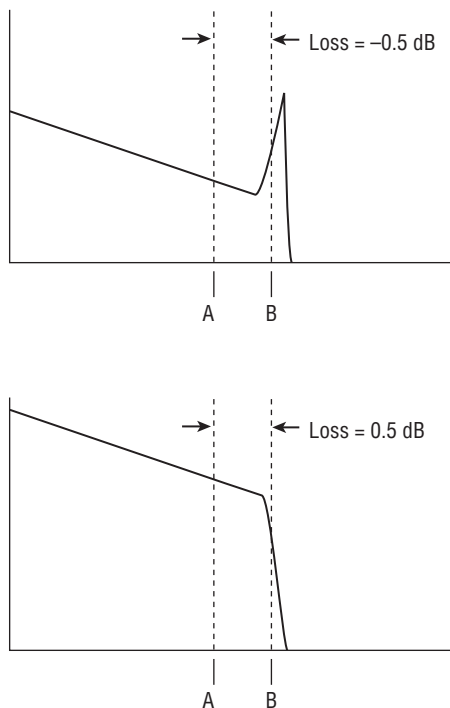
Measuring the Distance to the End of the Optical Fiber

A break in an optical fiber in a fiber optic cable is not the end of the optical fiber. However, a break in an optical fiber looks like the end of the optical fiber on the OTDR display. The same method is employed to measure the distance to a break in an optical fiber or measure the distance to the end of an optical fiber.

Light from the OTDR exiting the optical fiber under test typically produces a strong back reflection. The strong back reflection is caused by the Fresnel reflection when the light from the optical fiber hits the air. However, a break in an optical fiber is not always exposed to the air. The break may be exposed to index matching gel from a mechanical splice, gel from a loose buffer, or water.

The same method can be used to find the distance to the end of an optical fiber whether a Fresnel reflection happens or not. Figure 15.40 has two OTDR traces. The top trace is the end of an optical fiber with a Fresnel reflection. The bottom trace is the end of an optical fiber without a Fresnel reflection.

FIGURE 15.40 Measuring the distance to the end of an optical fiber using the 2-point method



To measure the distance to the end of the optical fiber after zooming in on the back reflection, place the A cursor on a smooth section of the trace just in front of the back reflection or the drop in the trace. Move the B cursor toward the A cursor until it is in the leading edge of the back reflection. Keeping moving the B cursor toward the A cursor until the A-B loss is ± 0.5 dB. It should be 0.5 dB of loss for the nonreflective trace and 0.5 dB of gain for the reflective trace. The length for the entire span is the distance for the B cursor.

Measuring the Length of a Cable Segment

The first step in measuring the length of a cable segment is to horizontally zoom in on the cable segment as shown in Figure 15.41. Place the cursors in the leading edge of the reflective events for that segment. The cursors should intersect the leading edge of the reflective event at the same vertical height above the smooth part of the trace as shown in Figure 15.41. The distance between cursors is the length of the cable segment.

Measuring Interconnection Loss

The first step in measuring interconnection loss is to horizontally zoom in on the interconnection. Position the A cursor just in front of the back reflection, then position the B cursor on a smooth area on the trace after the interconnection back reflection. The loss for the interconnection and the optical fiber between the cursors is displayed on the OTDR screen. The loss for the link shown in Figure 15.42 is 0.4 dB. The distance between the A and B cursors is 50 m.

FIGURE 15.41 Measuring the length of a cable segment

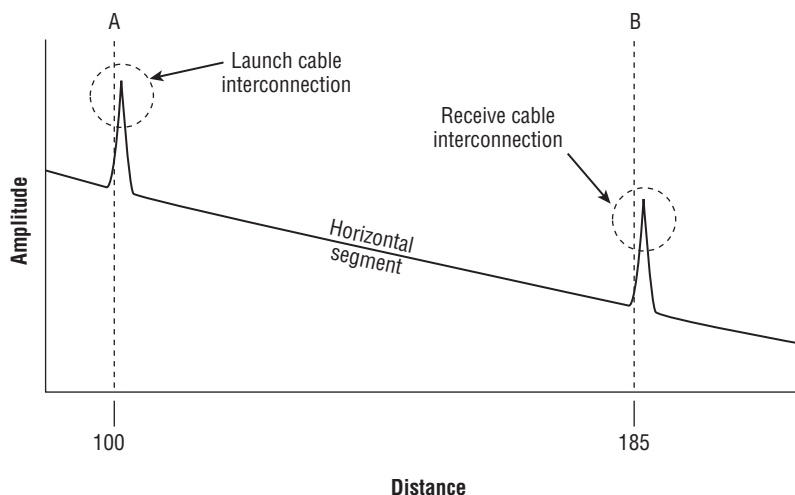
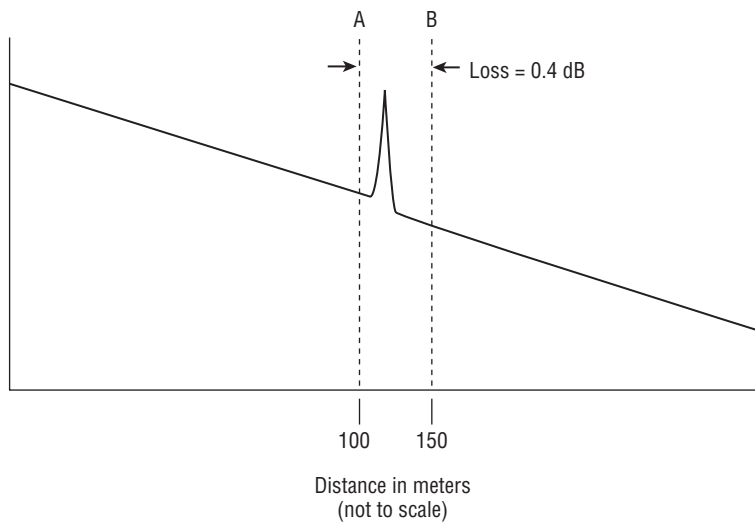


FIGURE 15.42 Measuring interconnection loss with the OTDR

The B cursor should always be in a smooth section of the trace. When a long pulse width is used, the dead zone may be very large. The distance between the A and B cursors may need to be several hundred meters. In that case, you measure the attenuation of a partial length of optical fiber no less than several hundred meters.

To find the loss for only the interconnection, the loss for the optical fiber between cursors A and B needs to be subtracted from the A-B loss displayed by the OTDR. This loss was previously at 0.14 dB for 50 m. To find the actual interconnection loss, subtract 0.14 dB from the 0.4 dB A-B loss. The loss for only the interconnection is approximately 0.26 dB.

Measuring the Loss of a Fusion Splice or Macrobend

A fusion splice or macrobend does not produce a back reflection. A macrobend will always appear as a loss in the form of a dip in the smooth trace. A fusion splice may appear as a dip or a bump in the trace. The fusion splice will appear as a dip in the trace when tested in both directions when the optical fibers fusion-spliced together have the same backscatter coefficient.

When optical fibers with different backscatter coefficients are fusion-spliced together, the splice may appear as a loss in one direction and a gain in the form of a bump in the trace when tested in the other direction. This is referred to as a gainer. To find the loss for the fusion splice, the splice must be tested in both directions and the results averaged together. The losses or loss and gain should be added together and the sum divided by 2. This is the loss for the fusion splice.

To find the loss of a fusion splice or macrobend, horizontally zoom in on the event. The loss from a fusion splice or macrobend is typically very small and will require vertical zoom in addition to horizontal zoom. Place the A cursor on the smooth part of the trace before the dip in the

trace. Place the B cursor on the smooth part of the trace after the dip, as shown in Figure 15.43. The loss for this event is 0.25 dB. The loss for the event includes the loss for the fusion splice or macrobend plus the 50 m of optical fiber between the cursors. Subtract the loss for the 50 m of optical fiber that was previously measured at 0.14 dB from the event loss. The loss for this fusion splice or macrobend is 0.11 dB.

FIGURE 15.43 Measuring the loss of a fusion splice or macrobend

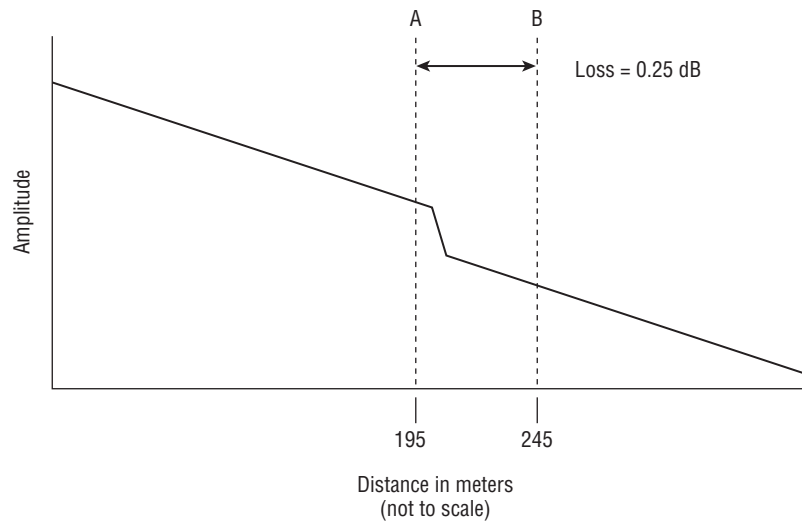
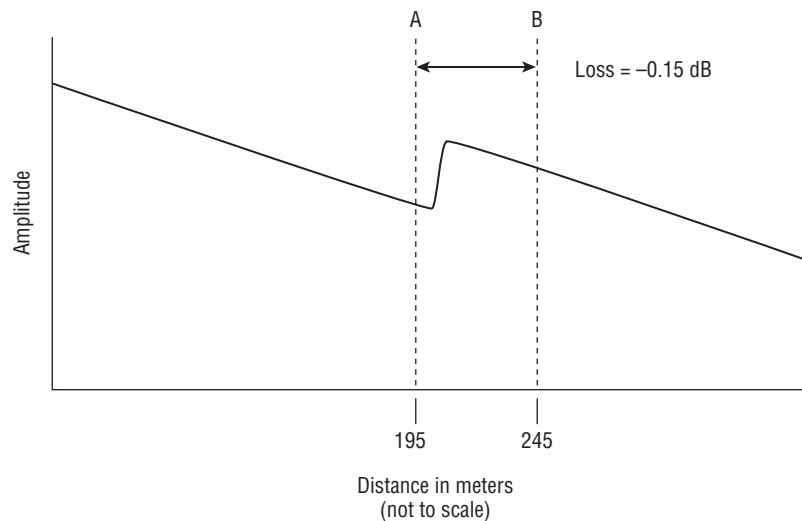


FIGURE 15.44 Measuring the gain of a fusion splice



To find the gain of a fusion splice, horizontally zoom in on the event. The gain from a fusion splice is typically very small and will require vertical zoom in addition to horizontal zoom. Place the A cursor on the smooth part of the trace before the bump in the trace. Place the B cursor on the smooth part of the trace after the bump, as shown in Figure 15.44. The gain for this event is 0.15 dB. The gain for the event includes the gain for the fusion splice plus the 50 m of optical fiber between the cursors. Add the value for the loss for the 50 m of optical fiber that was previously measured at 0.14 dB to the event gain. The gain for this fusion splice is 0.29 dB.

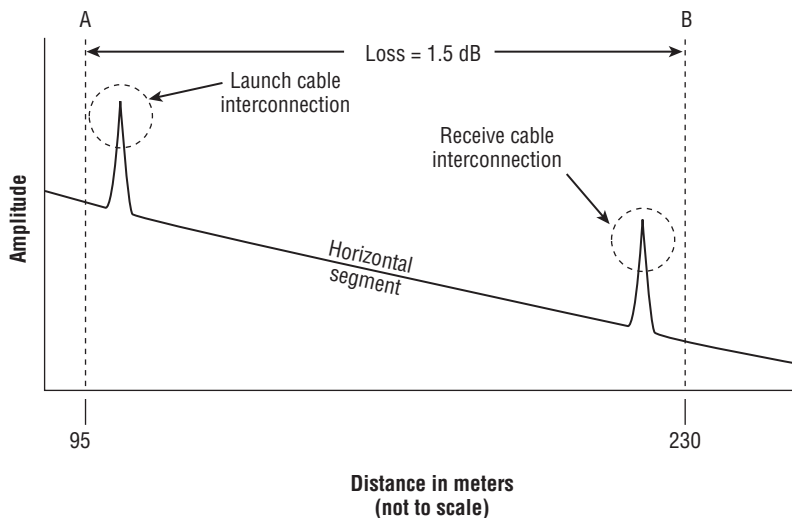
Measuring the Loss of a Cable Segment and Interconnections

To find the loss for a cable segment plus the interconnections, the exact length of the segment must be known. This can be done by using the method described to measure the length of a cable segment. The length of the cable segment in this example, as shown in Figure 15.45, is 85 m.

The first step is to horizontally zoom in on the cable segment. After zooming in on the cable segment, place the A cursor on a smooth section of the trace just in front of the leftmost cable segment interconnection back reflection. Place the B cursor on the smooth part of the trace after the rightmost cable segment interconnection back reflection. Looking at the A-B distance on the OTDR, move the B cursor to the right until the distance equals the length of the cable segment plus 50 m, as shown in Figure 15.45. The A-B loss is 1.5 dB.

To find the loss for the cable segment plus the interconnections, subtract the loss for the 50 m of optical fiber from the 1.5 dB loss for the cable segment and interconnections. The previously measured loss for 50 m of optical fiber at the 850 nm is 0.14 dB. The loss for the cable segment and the interconnections is 1.36 dB.

FIGURE 15.45 Measuring the loss of a cable segment and interconnections



Documentation

After you test the cable plant with the OTDR, you need to properly document the results of the test. The documentation should include at a minimum the following:

- The date of the testing
- The number or identification of fiber or cable tested
- The test procedure number
- OTDR make, model, serial number, and calibration dates
- Pulse width
- Range and resolution
- Number of averages
- Wavelength(s)
- Launch conditions

Summary

This chapter has taken an in-depth look at the key pieces of fiber optic test equipment that the fiber optic technician should be familiar with. How well the fiber optic technician can test the performance of a cable plant or fiber optic link depends on their overall knowledge of the test equipment. Test equipment technology will continue to evolve; however, the basic theory behind the equipment will remain the same. The fiber optic technician needs to understand the basic theory behind the test equipment and its application. Once the basic theory is mastered, the fiber optic technician should be able to quickly learn how to operate new pieces of test equipment as they become available.

Exam Essentials

Describe the basic operation of a continuity tester. Be able to explain basic operation of the incandescent lamp and LED continuity tester.

Explain how to test the continuity of an optical fiber with a continuity tester. Be able to explain how to test the continuity of an optical fiber with the continuity tester. Know the maximum optical fiber length that can typically be tested with a continuity tester.

Describe the basic operation of a visible fault locator. Be able to describe the difference between the VFL and the continuity tester. Know the performance characteristics of the VFL with multimode and single-mode optical fiber.

Explain how to locate faults with a visible fault locator. Be able to explain how the VFL is used to identify breaks and macrobends in an optical fiber.

Describe the basic operation of a fiber identifier. Be able to describe how the fiber identifier detects infrared light by placing a slight macrobend in the optical fiber or fiber optic cable.

Explain how to test a cable with a fiber identifier. Be able to explain how to select the proper attachment for the fiber identifier and test a fiber optic cable.

Describe the basic operation of an ORL loss test set. Remember that the ORL test set uses an optical continuous wave reflectometer to measure return loss. Be able to explain how the light source continuously transmits light through a directional coupler into the optical fiber under test. The energy reflected back by the optical fiber is the return loss.

Explain how to test optical return loss with an optical return loss test set. Be able to describe how the ORL test set is connected to the link under test. Remember to allow sufficient time for the test set to warm up and stabilize. Don't forget that the test set must be calibrated before use.

Describe the basic operation of a single-mode and multimode light source and optical power meter. Be able to describe the difference between a multimode and single-mode light source and optical power meter. Know that the optical power meter displays optical power in dBm.

Explain the difference between a test jumper and a patch cord. Be able to explain the qualities required for a patch cord per TIA/EIA-568-B.3 and the qualities required for a test jumper per TIA/EIA-526-14A.

Explain the purpose of a mode filter. Be able to describe the basic operation of a mode filter. Know that a mode filter removes high-order modes from an optical fiber and that the mode filter is always placed on the light source test jumper.

Explain how to perform optical loss measurement testing of a cable plant using TIA/EIA-526-14A methods A, B, and C. Be able to describe how to test a cable plant using one, two, or three test jumpers with a light source and power meter.

Explain how to measure the optical loss in a patch cord with a light source and optical power meter. Be able to describe how to test a patch cord with a light source and power meter using two test jumpers.

Describe the basic operation of an OTDR. You should remember that the eight basic components of a typical OTDR are the directional coupler, laser, timing circuit, single-board computer, DSP, analog to digital converter, sample-and-hold circuit, and avalanche photodiode. Be able to describe how the OTDR launches pulses of light into an optical fiber and measures the energy returned from the optical fiber over a predetermined period of time.

Explain how to test a cable plant segment with an OTDR. Be able to describe how to properly set up the OTDR and connect it to the cable plant or fiber optic link under test. Know how to measure fiber attenuation, interconnection loss, splice loss, the distance to the end of the optical fiber, and the loss for a cable segment.

Review Questions

1. The continuity tester should only be used to verify that there are no _____ in the optical fiber.
 - A. Breaks
 - B. Shorts
 - C. Macrobends
 - D. Microbends
2. The VFL can be used to test the continuity of an optical fiber and locate _____ in a fiber optic cable.
 - A. Shorts
 - B. Macrobends
 - C. Dispersion
 - D. Reflectance
3. The fiber identifier typically contains _____ photodiode(s) to measure the light energy escaping from the macrobend in the fiber optic cable.
 - A. One
 - B. Two
 - C. Three
 - D. Four
4. The ORL test set is used to measure the light energy being reflected back toward the transmit end of the _____.
 - A. Interconnection
 - B. Splice
 - C. Macrobend
 - D. Optical fiber
5. An unfiltered LED light source _____ the optical fiber with high-order modes.
 - A. Underfills
 - B. Overfills
 - C. Fills
 - D. Moderately fills

6. Method B uses _____ jumper(s) to establish the optical power reference.
 - A. One
 - B. Two
 - C. Three
 - D. Four
7. The OTDR will only provide accurate length measurements when the correct _____ has been programmed in for the optical fiber under test.
 - A. Backscatter coefficient
 - B. Attenuation rate
 - C. Connector count
 - D. Refractive index
8. The OTDR displays amplitude in _____ on the _____ axis.
 - A. dB, vertical
 - B. dBm, horizontal
 - C. dBm, vertical
 - D. dB, horizontal
9. Launch and receive cables should be selected based on the _____ being used to test the optical fiber.
 - A. Pulse width
 - B. Pulse amplitude
 - C. Wavelength
 - D. Backscatter coefficient
10. A fusion splice or _____ does not produce a _____ on the OTDR trace.
 - A. Break, dip
 - B. Macrobend, dip
 - C. Macrobend, back reflection
 - D. Break, back reflection

Answers to Review Questions

1. A. The continuity tester can only verify the continuity of the optical fiber. No continuity means that there is a break in the optical fiber.
2. B. The VFL can be used to locate macrobends in a fiber optic cable. The area in the fiber optic cable where the macrobend exists will glow when illuminated by the VFL.
3. B. The fiber identifier typically contains two photodiodes that are used to detect the infrared light. The photodiodes are mounted so that they will be on opposite ends of the macrobend of the optical fiber or fiber optic cable being tested.
4. D. ORL testing measures the amount of optical light energy that is reflected back to the transmit end of the fiber optic cable. The energy being reflected back, or back reflections, comes from mechanical interconnections, passive devices, fiber ends, and Rayleigh scattering caused by impurities within the optical fiber.
5. B. An unfiltered light source does overfill the optical fiber with high-order modes. TIA/EIA-568-B.3 requires that multimode insertion loss measurements be performed with a light source that meets the launch requirements of TIA/EIA-455-50B.
6. A. Method B uses one jumper to establish the optical power reference. However, two test jumpers are required to perform the test. This method is required by TIA/EIA-568-B.3 section 11.3 to measure link attenuation.
7. D. For the OTDR to produce accurate results, the refractive index of the optical fiber under test must be known. The refractive index of an optical fiber is different for each wavelength tested. The fiber optic technician must enter the correct refractive index into the OTDR for each wavelength. The correct refractive index for a fiber optic cable can be obtained from the manufacturer.
8. A. The OTDR displays time or distance on the horizontal axis and amplitude in dB on the vertical axis. The horizontal axis can typically be programmed to display distance in feet, meters, or kilometers. The vertical axis is not programmable. The vertical axis displays relative power in dB.
9. A. Launch and receive cables are typically the same length. A good rule of thumb is to have at least 1 meter of launch or receive cable for each nanosecond of pulse width, with the minimum length cable being 100 m in length. A 2 km link being tested with a 20 ns pulse width would require at least a 100 m launch and receive cable. A 10 km link being tested with a 200 ns pulse would require at least a 200 m launch and receive cable.
10. C. A fusion splice or macrobend does not produce a back reflection. A macrobend will always appear as a loss in the form of a dip in the smooth trace. A fusion splice may appear as a dip or a bump in the trace. The fusion splice will appear as a dip in the trace when tested in both directions when the optical fibers fusion-spliced together have the same backscatter coefficient.



Chapter

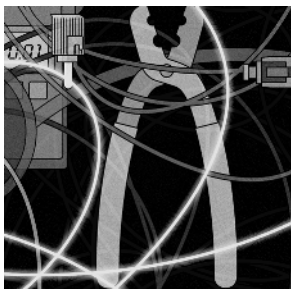
16

Link/Cable Troubleshooting

OBJECTIVES COVERED IN THIS CHAPTER:

✓ Link/Cable Troubleshooting

- Explain how to properly perform a visual inspection of a connector.
- Explain how to properly evaluate a connector per TIA/EIA-455-57B and locate faults.
- Describe continuity tester fault location techniques.
- Describe visible fault locator fault location techniques.
- Describe cable identifier fault location techniques.
- Describe OTDR fault location techniques.
- Describe common restoration practices.



So far you have learned about fiber optic theory, cables, connectors, splices, transmitters, receivers, and many different passive components. We have shown you how to predict the performance of a link and test that link to industry standards using various pieces of test equipment. However, we have not discussed what to do when the link doesn't work.

This chapter takes the knowledge and skills you have learned up to this point and applies them to troubleshooting a fiber optic link or cable. You will learn basic techniques that will allow you to quickly analyze and determine the fault. You will learn fault location techniques with devices as simple as a flashlight and as complicated as an OTDR.

Connector Inspection

The weakest link in any fiber optic installation is likely to be the connector. When you look at a fiber optic connector, you don't see anything exciting. Fiber optic connectors look simple. They have very few moving parts and don't require electricity. How hard can it be to properly install a fiber optic connector?

As you learned in Chapter 9, "Connectors," the fiber optic connector provides a way to connect an optical fiber to a transmitter, receiver, or other fiber optic device. The optical fiber is physically much smaller than the connector and virtually invisible to the naked eye. The naked eye can't tell the difference between a perfectly polished fiber optic connector and a fiber optic connector with the optical fiber shattered.

Fiber optic connectors can have poor performance for the many reasons that are described in detail in Chapter 9. However, in our experience, poor surface finishes and dirt are the primary causes of poor connector performance. Many times the poor surface finish is actually caused by dirt or contamination. We can't stress enough how important it is for the fiber optic installer or fiber optic technician to make every effort possible to ensure that the connector is kept clean. Remember that a dirt particle invisible to the naked eye can damage a fiber optic connector beyond repair.

Dirt particles on the connector endface are not the only cause of poor connector performance. Dirt anywhere on the connector ferrule or inside the connector receptacle can cause poor connector performance. Many times epoxy that runs onto the side of the connector ferrule during oven curing can go unnoticed during the polishing process. This epoxy will prevent the connector ferrule from aligning properly in the connector receptacle and produce a high loss interconnection.

The first step in troubleshooting any fiber optic link problem is to visually inspect and clean the connectors. Close attention should be paid to the sides of the connector ferrule to ensure that there is no epoxy on the surface. Excess epoxy needs to be removed from the connector ferrule before the connector is inserted into a receptacle. The endface of the connector should be cleaned following the procedures in Chapter 9. After the connector has been properly cleaned, it should be evaluated with an *inspection microscope*.

Connector Endface Evaluation

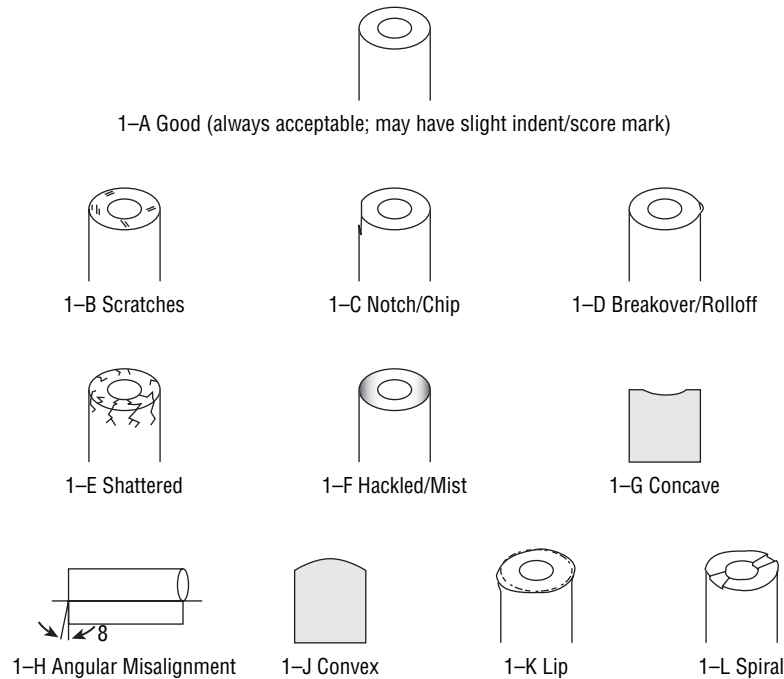
The endface of a fiber optic connector can only be properly evaluated with an inspection microscope. As you know from Chapter 15, “Test Equipment and Link/Cable Testing,” there are many different inspection microscopes available to the fiber optic installer and fiber optic technician. Typically, a multimode connector can be evaluated with a 100X microscope while a single-mode connector requires a minimum of 200X magnification. A 400X microscope works great for both multimode and single-mode. Figure 16.1 is a photograph of two hand-held microscopes. The smaller microscope is a 100X and the larger is a 400X.

TIA/EIA-455-57B provides a guideline for examination of an optical fiber endface. Figure 16.2 contains possible core and cladding fiber endface results. Each endface in Figure 16.2 is labeled with a number and a letter. The number and letter combinations are listed in Table 16.1. This table is used to identify always acceptable, usually acceptable, and often acceptable endface results.

FIGURE 16.1 100X and 400X microscopes



FIGURE 16.2 TIA/EIA-455-57B core and cladding endface results



Typical fiber endface results with core and cladding

TABLE 16.1 TIA/EIA-455-57B Core and Cladding Endface Results

Figure	Appearance Acceptability		
	Always Acceptable	Usually Acceptable ⁽¹⁾	Often Acceptable ⁽¹⁾
1	1-A	1-B, 1-C, 1-D, 1-G, 1-H ⁽²⁾	1-E, 1-F, 1-J, 1-K, 1-L
2	2-A	2-B, 2-C, 2-D, 2-H ⁽³⁾	2-E, 2-F, 2-J, 2-K

⁽¹⁾ Tighter or looser limits may be specified by the fiber optic test procedure (FOTP)
⁽²⁾ Unless defect extends into the core
⁽³⁾ Unless defect extends too far, as defined by the detail specification

Looking at Table 16.1, you will see that one endface result is always acceptable. This is endface finish 1-A. Note that 1-A is drawn to have no imperfections, or to be “cosmetically perfect.” However, the note next to the figure states that a slight indent or score mark is acceptable.

When you are evaluating the endface of a connector with a microscope, the magnification level of the scope has everything to do with the detection of imperfections. Earlier in the text, we mentioned that most multimode fiber optic connectors could be properly evaluated with a 100X inspection microscope. Viewing a multimode endface with this magnification level will reveal only gross defects—small scratches from the polishing abrasive are typically undetectable.

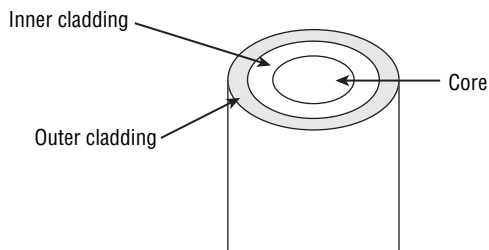
The same connector viewed with a 400X inspection microscope may show many small scratches that were caused by the polishing abrasive. The endface that looked cosmetically perfect with 100X magnification became really ugly with 400x magnification. Imagine how that endface would look with 600X or 800X magnification.

The fiber optic installer and fiber optic technician need to remember that typically a multimode fiber optic connector endface does not have to be cosmetically perfect to provide a low-loss interconnection. In our classes, we demonstrate this to the students. We take a patch cord with cosmetically perfect connectors on each end and measure the loss. Then we take another patch cord with connector endfaces that have been polished with only a 1 μm abrasive and measure the loss. The students are surprised when the cosmetically perfect patch cord has more loss than the less-than-cosmetically-perfect patch cord. For most applications, a multimode connector endface does not need to be polished with an abrasive finer than 1 μm . A 1 μm abrasive will leave visible scratches in the endface when viewed with a 400X microscope. However, this same endface may appear scratch free when viewed with a 100X microscope.

To evaluate the endface of a multimode connector, you need to divide the endface into three parts: the core, the inner cladding, and the outer cladding. As shown in Figure 16.3, the outer cladding is the area from the center of the cladding to the connector ferrule. The inner cladding is the area from the center of the cladding to the core.

Field-terminated multimode connectors are typically polished with an aluminum oxide abrasive. As you learned in Chapter 9, aluminum oxide is harder than the optical fiber but softer than the ceramic ferrule. Because the aluminum oxide does not polish down the ceramic ferrule during the polishing process, chipping around the epoxy ring extends into the outer cladding. Sometimes this chipping may extend in excess of 10 μm from the epoxy ring. This is typical, which means that this is acceptable.

FIGURE 16.3 Multimode connector endface



A good rule of thumb when evaluating a multimode connector is to accept all endface finishes that have no defects in the core and inner cladding. The connector endface should be defect free when viewed with a 100X microscope. Minor scratches and pitting are acceptable when viewed with a 200X microscope or a 400X microscope.

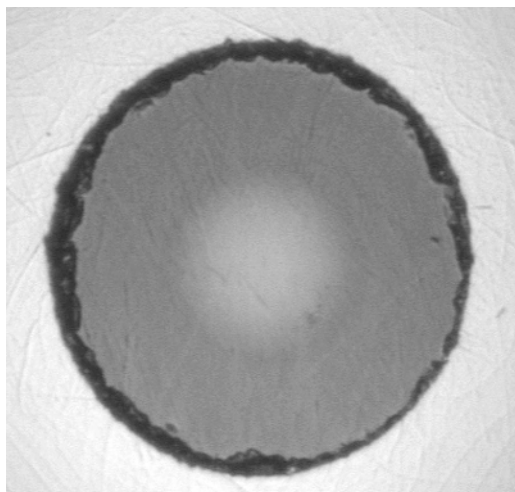
Figure 16.4 is a photograph of a multimode endface polished with a 1 μm aluminum oxide abrasive viewed with a 400X video microscope. Note the minor chipping along the epoxy ring and the minor scratches on the inner cladding and core. This is an acceptable multimode endface. This connector endface viewed with a 100X microscope would look cosmetically perfect.

Single-mode connector evaluation requires 400X magnification. The endface of a single-mode connector should look cosmetically perfect. Remember from Chapter 9 that a single-mode connector endface is typically finished with a 0.1 μm abrasive. A single-mode endface should have a very thin epoxy ring in comparison to a multimode endface. There should be no pitting or chipping along the epoxy ring. The inner cladding, outer cladding, and core should be defect free.

In some applications, single-mode terminations are performed in the field and not finished with a 0.1 μm abrasive. These terminations will have endface finishes similar to a field-terminated multimode endface. The endface should be evaluated as you would evaluate a multimode endface.

Up to this point, we have focused only on the cosmetics of the endface. We have not covered the geometry of the endface. When a connector is viewed with a microscope, you see only a two-dimensional view—you can't see depth or height. As you learned in Chapter 9, virtually all connector ferrules have a radius. This means that a properly polished optical fiber should have the same radius as the ferrule.

FIGURE 16.4 Multimode connector endface viewed with a 400X microscope



TIA/EIA-455-57B does not provide guidance when it comes to measuring or evaluating ferrule or endface radius. Ferrule or endface radius cannot be measured with a microscope; it can only be measured with an interferometer. An *interferometer* is an expensive piece of test equipment that is not typically carried out into the field. However, some portable video microscopes are now incorporating interferometers.

In the early stages of troubleshooting, a visual inspection of the connector endface is all that is necessary. If the connector endface meets the requirements established by TIA/EIA-455-57B, you can go to the next step, which would be to check optical fiber continuity. You don't need to be concerned about connector endface radius, unless testing with the OTDR reveals a high interconnection loss or high interconnection back reflection.

High interconnection loss or back reflection from a mated connector pair with acceptable endface finishes is usually caused by an air gap. The air gap may be caused by dirt that is not allowing the ferrule to properly seat in the receptacle. The air gap may be present because the connector was not inserted properly into the receptacle. The air gap may also be the result of one or both connectors having a concave or flat as opposed to convex endface finish.

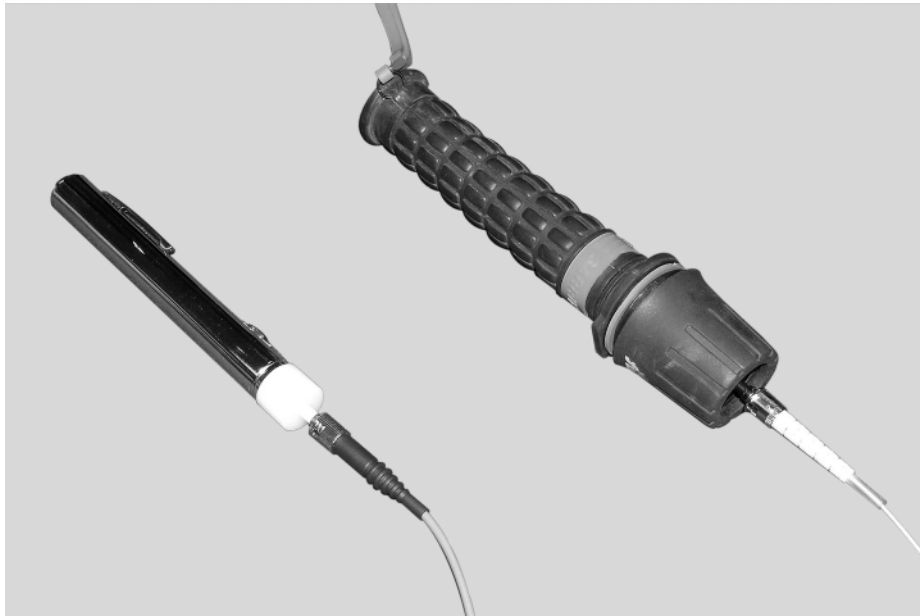
Continuity Tester Fault Location Techniques

As you learned in Chapter 15, the continuity tester is really no more than a flashlight. There are many different continuity testers on the market. Some use red LED light sources; others use incandescent lights. If you don't have a continuity tester, you can just use a flashlight. Figure 16.5 shows an LED and incandescent continuity tester. As you can see, these are modified flashlights.

The key to using a flashlight is alignment. Remember, only light that enters an optical fiber through the cone of acceptance will propagate the length of the optical fiber. You may have to practice aligning the flashlight on a known good patch cord before testing a longer fiber optic cable. It is much easier to use a flashlight to test multimode optical fiber than single-mode optical fiber.

The only drawback to using a flashlight besides manual alignment is the fact that you have to hold the connector and flashlight together. This could be a real challenge when you are working by yourself and the other end of the fiber optic cable is in another building. If that's the case, you may have to tell the boss to break down and spend the \$30.00 on a continuity tester.

As you learned in Chapter 15, the optical output power of an LED or incandescent continuity tester is relatively low when compared to an LED or laser transmitter. Most of the LED continuity testers we have tested output somewhere between -30 dBm and -40 dBm. The output of an incandescent continuity tester can actually exceed -30 dBm, depending on the lamp used and how well the lamp is focused. LED and incandescent continuity testers should be made eye safe so that you can directly view the connector endface emitting the light energy.

FIGURE 16.5 Incandescent and LED continuity testers

So what can you do with a continuity tester? Basically, you can check the continuity of an optical fiber—that is, check to see if an optical fiber allows light to pass from one end to the other. A continuity tester tests only for breaks in the optical fiber. Macrobends or microbends can't be detected with a continuity tester.

Let's work through an example and take a step-by-step approach to troubleshooting a fiber optic link with a continuity tester. As you already know and as has been stated in this book many times, the first step in troubleshooting a fiber optic link is to clean the connector endface. If you have forgotten how to clean a connector endface, refer to Chapter 9.

With a clean connector in hand, the next step is to examine the connector endface with an inspection microscope. As you learned in Chapter 9, a 100X microscope is adequate for multi-mode applications. Single-mode applications require a 200X or 400X microscope. If your examination of the endface of each connector on opposite ends of the fiber optic cable shows no defects, you can go to the next step. However, if your examination reveals a defect, there is no sense in testing continuity until the connector is repaired or replaced.

At this point, the connectors at each end of the fiber optic cable are clean and you are ready to test cable continuity. Remember: Before attaching the connector to the continuity tester, verify that the continuity tester works. To verify continuity tester operation, turn it on and see if it emits light. This seems like beginner-level stuff, but how many times have you heard the story about the repairman who shows up to fix a computer only to find that it's simply not plugged in?



Real World Scenario

Can a Flashlight Keep Your Company from Going out of Business?

Over the years, many students have shared stories about their experiences in the field. Several of these stories I typically share with each class. The story that really applies to this chapter is about a small company that made its livelihood doing aerial copper cable installations.

One day, this company took a job doing an aerial installation of a multifiber cable. Because this company had never installed a fiber optic cable, they handled it like a copper cable. After several days of hard work, the cable was in place and the installation company called a fiber optic contractor to complete the installation.

The first thing the fiber optic contractor did was test the cable with the OTDR. The first optical fiber tested was broken several hundred meters from the OTDR. The next optical fiber tested was also broken. Testing from the opposite end of the cable revealed at least two breaks in each optical fiber tested.

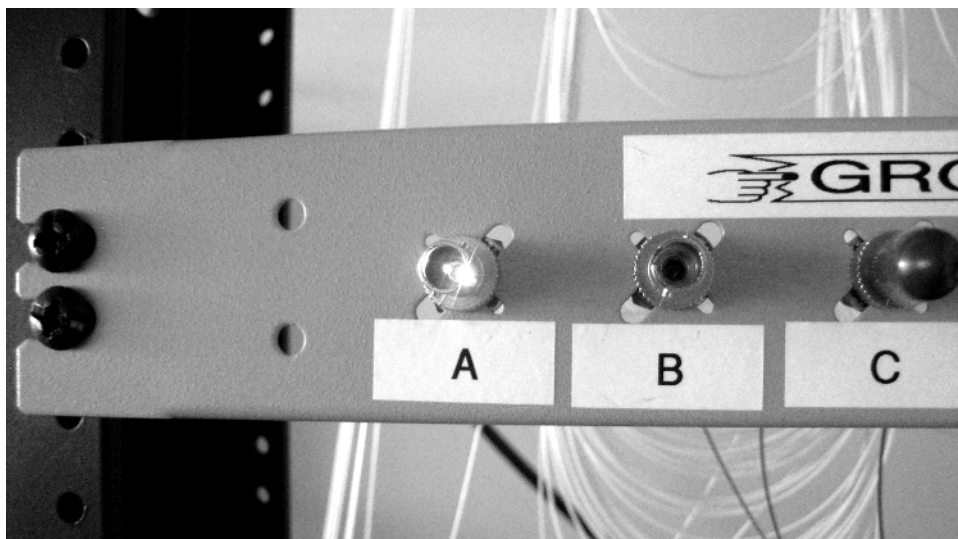
This small company had to install a new fiber optic cable and absorb the cost of the cable and the labor. Needless to say, this small company almost went out of business because of that. Eventually they determined that the fiber optic cable had minor damage from what could have been a forklift.

The company never wanted to make this mistake again, so they sent their sharpest employee to my fiber optic installer course. You can imagine the look on this person's face when I demonstrated how to use a continuity tester to look for a break in an optical fiber.

The moral of the story is: Always test your fiber optic cable before installation. If your company doesn't have an OTDR, use a continuity tester. If you don't have a continuity tester, use a flashlight.

Now that you know the continuity tester works, plug in or attach a connector at one end of the fiber optic cable. With the continuity tester on, observe the connector at the other end of the fiber optic cable. If the connector at the other end of the fiber optic cable is not emitting light, the optical fiber in the fiber optic cable is broken. If the connector at the other end of the fiber optic cable is emitting light, you have continuity and know that the optical fiber is not broken.

Figure 16.6 is a photograph of a portion of a patch panel with two ST type connector receptacles. Each connector receptacle is populated on the back side. In other words, a fiber optic cable is connectorized and plugged into the back side of each receptacle. On the other end of each fiber optic cable, a continuity tester is attached and turned on. Receptacle A shows good continuity. No light is emitted from receptacle B, indicating a broken optical fiber.

FIGURE 16.6 Continuity test of two fiber optic cables

So far you have learned how to test the continuity of the optical fiber in a fiber optic cable. Now let's look at the limitations of the continuity tester. Receptacle A is emitting light, so you know that there is no break in the optical fiber. Unfortunately, that is all you can tell about the optical fiber in this cable.

Our eyes respond to light in a *logarithmic* manner, which means that we can't detect small changes in optical power. A change of 0.2dB or 0.5 dB, for example, is not really detectable. This means that we can't look at the light being emitted and tell if there is a macrobend or microbend in the optical fiber. The continuity tester is just that, a continuity tester. It can only tell you that the optical fiber is not broken.

Another drawback to the continuity tester is the fact that it uses visible light. You may remember from Chapter 5, "Optical Fiber Characteristics," that visible light is a poor choice for fiber optic transmission because of attenuation. The high attenuation of visible light in an optical fiber limits the use of the continuity tester to cables no longer than 2 km.

Remember that the continuity tester is basically a flashlight with no lens system to direct the light energy into the core of the optical fiber. Back in Chapter 10, "Fiber Optic Light Sources," we explained that a 62.5 μm core accepts more light energy than a 50 μm core when plugged into the same transmitter. This means that the continuity tester will not direct much light energy into the core of a single-mode optical fiber.

The continuity tester can be used with single-mode optical fiber. However, the light emitted from the connector at the opposite end of the cable will be dim in comparison to the multimode optical fiber. This is one application where the performance of the multimode optical fiber literally outshines the performance of the single-mode optical fiber.

Visible Fault Locator

The continuity tester is a valuable piece of test equipment. However, due to its low output power, it is limited to just testing the continuity of the optical fiber. The visible fault locator (VFL) is a continuity tester with a focused visible laser source instead of an unfocused LED or incandescent lamp. The powerful laser in the VFL allows the fiber optic technician to test continuity and visibly identify breaks or macrobends in the optical fiber.

VFLs come in all shapes and sizes. Some are the size and shape of a large pen while others may have the size and shape of a digital multimeter. Regardless of the design, all of them perform the same function. VFLs are also incorporated into some OTDRs. Incorporating the VFL into the OTDR reduces the number of pieces of test equipment that you need to carry to the job site.

The laser light sources used in most VFLs typically output 1 mW of optical energy somewhere in the 650 nm range. The output of the VFL is roughly 1,000 to 10,000 times greater than the output of an LED continuity tester. This means that you should never look directly at the endface of a connector emitting light from the VFL. The VFL is an eye hazard, and proper safety precautions need to be taken when operating the VFL. You can refer to Chapter 6, “Safety,” to find out detailed information on light source classification and safety.

Now let's take a look at how to employ the VFL to locate faults in a fiber optic cable. As with the continuity tester, the first thing you'll need to do is clean the connector endface and inspect it with a microscope. If the endface finish is acceptable, the VFL can be connected to a connector at one end of the fiber optic cable. The connector at the other end of the fiber optic cable should not be viewed directly during this testing.

The VFL is designed to fill the core of an optical fiber with visible light. Depending on the cable type, the VFL may illuminate faults in an optical fiber through the buffer, strength member, and even jacket. However, the VFL performs best with tight-buffered optical fiber.

Figure 16.7 is a photograph of a broken tight-buffered optical fiber. You can see in the photograph how the VFL illuminates the break. The light energy from the VFL penetrates the buffer, allowing the fiber optic technician to quickly identify exactly where the fault is located.

The VFL and the OTDR work hand in hand with each other when it comes to locating breaks in an optical fiber. The OTDR can provide the fiber optic technician the distance to the break. The VFL allows the fiber optic technician to see the break in the optical fiber.

Fiber optic cables are not the only place where the optical fiber may break. The optical fiber may break inside the connector or connector ferrule. Unless the optical fiber is broken at the endface of the connector, it is not visible with a microscope.

Often, students connect cables that look great when viewed with the microscope but fail continuity testing. When this happens, the hardest part is determining which connector contains the break in the optical fiber. Without a VFL in the classroom, students would have to cut the cable in half and use the continuity tester to identify the bad connection.

The VFL will often identify the bad termination or connector. Figure 16.8 is a photograph of an ST connector with a broken optical fiber in the ferrule of the connector. Looking at the photograph, you can see VFL illuminating the break in the optical fiber. The output of the VFL is so powerful that it penetrates the ceramic ferrule. Figure 16.9 is a photograph of an ST connector without a break in the optical fiber.

FIGURE 16.7 Broken tight-buffered optical fiber

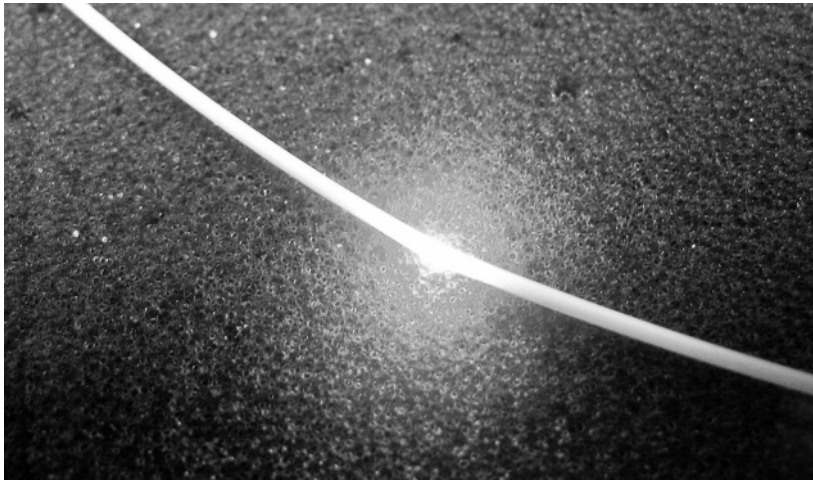
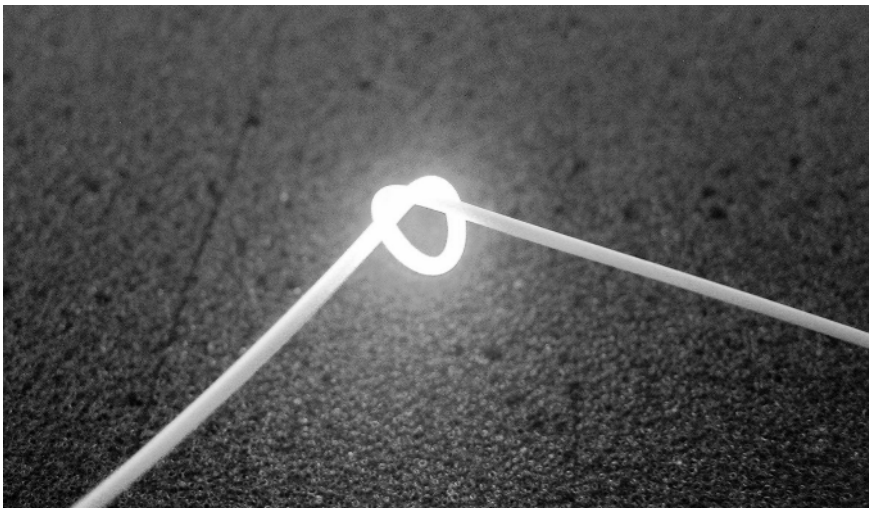


FIGURE 16.8 Broken optical fiber in ST connector ferrule being tested with VFL



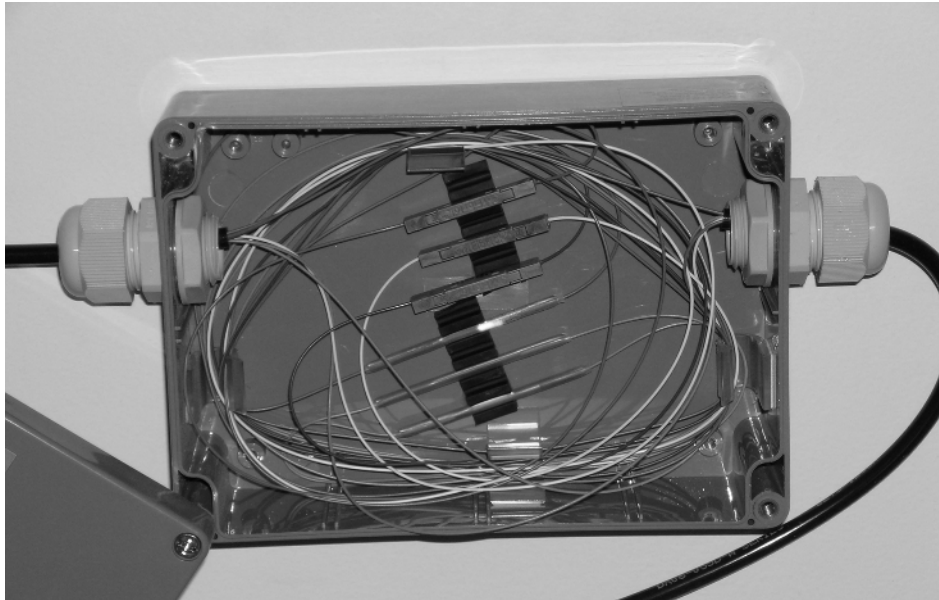
FIGURE 16.9 Good ST connector being tested with VFL

The VFL can also be used to locate a macrobend in an optical fiber. However, macrobends do not allow nearly as much light to penetrate the buffer as does a break in the optical fiber. Locating a macrobend with the VFL may require darkening the room. Figure 16.10 is a photograph of a severe macrobend in a tight-buffered optical fiber. You can see from the photograph how the VFL illuminates the macrobend.

FIGURE 16.10 Macrobend in a tight-buffered optical fiber

Macro bends and high-loss fusion splices appear the same on an OTDR trace. The VFL allows the fiber optic technician to identify a high-loss fusion splice. Figure 16.11 is a photograph of three fusion splices and three mechanical splices inside a splice enclosure. The splice illuminated by the VFL is a high-loss fusion splice. Believe it or not, the loss from this splice was not great enough to impact system performance. The loss for this splice was only 0.6 dB.

FIGURE 16.11 High-loss fusion splice being tested with VFL



Fiber Identifier

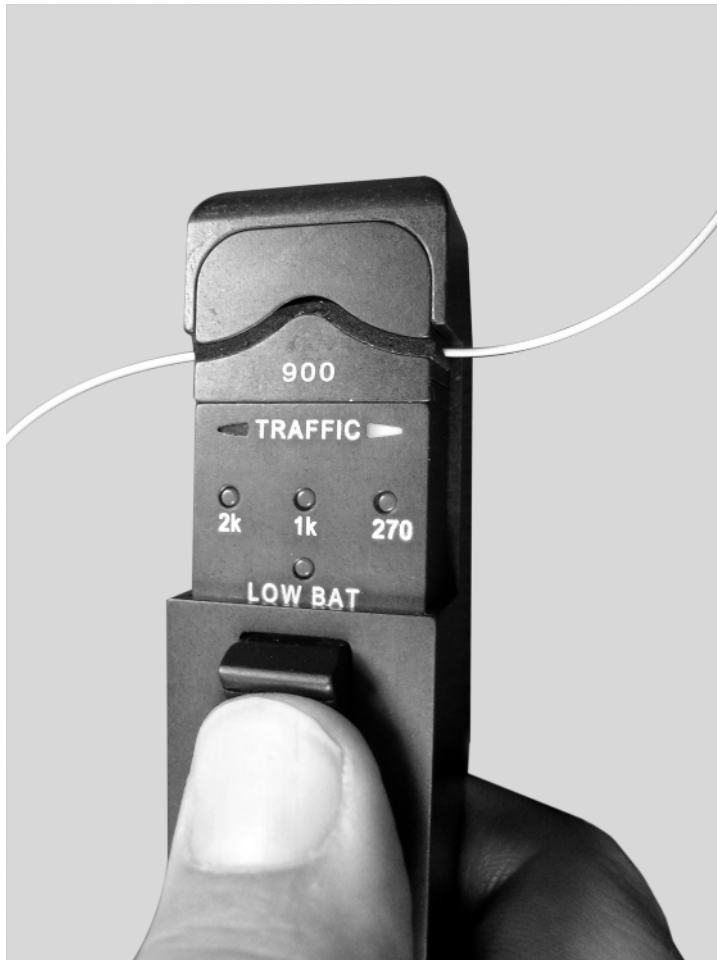
The fiber identifier is a piece of test equipment that allows the fiber optic technician to see through the jacket, strength member, and buffer of the fiber optic cable. As you learned in Chapter 15, the fiber identifier is designed to place a macrobend in the fiber optic cable under test. Photodiodes in the fiber identifier detect light penetrating through the fiber optic cable. The electronics in the fiber identifier measure the detected light energy and display the direction of light travel through the optical fiber.

The fiber identifier is used very much like the VFL when it comes to troubleshooting. One key difference is that the fiber identifier replaces your eyes. Another difference is that fiber optic cable under test typically does not have to be disconnected from an active circuit—it can remain plugged into the transmitter and receiver. The infrared light traveling through the optical fiber during normal operation is often enough to perform most tests. However, sometimes an additional infrared light source is required to adequately troubleshoot.

Up to this point, we have only discussed troubleshooting with test equipment that emits visible light. You may have noticed that we keep mentioning that the fiber identifier works with infrared light. We have not mentioned using visible light with the fiber identifier. Visible light can be used with the fiber identifier; however, most fiber identifiers perform best with longer wavelength infrared light sources.

The photograph in Figure 16.12 shows a fiber identifier clamped around a tight-buffered optical fiber. This tight-buffered optical fiber is connected between a 1310 nm laser transmitter and receiver. You can see in the photograph that the arrow is pointing to the direction the light is traveling through the optical fiber. This is an example of a fiber identifier being used to detect traffic on an optical fiber.

FIGURE 16.12 Fiber identifier



The fiber identifier can also be used with an external light source. Often the external light source is an OTDR. Many OTDR manufacturers build or program in a pulsed output function. When set for a pulsed output, the OTDR emits a continuous pulse train at a predetermined frequency. The electronics in the fiber identifier can detect preset frequencies and illuminate the corresponding LED. This feature can be very helpful when you are trying to identify an unmarked tight-buffered optical fiber within a bundle of tight-buffered optical fibers. This feature can also be helpful when you are trying to approximate the location of a break in the optical fiber.

The fiber identifier can be used with the OTDR to narrow down the location of a break in an optical fiber when a VFL is not available or when the light from the VFL is not visible through the jacket of the fiber optic cable. If the index of refraction is correct, the OTDR should provide an accurate distance to the fault. Remember from Chapter 15 that the OTDR measures the length of optical fiber to the fault, not the length of fiber optic cable. The cable length may be shorter than the optical fiber length. This is especially true if loose buffer cable exists in the fiber optic link.

Once you have found the approximate location of the fault with the OTDR, set the OTDR or infrared light source to pulse at a predetermined frequency. Clamp the fiber identifier on the faulted fiber optic cable several meters before the approximate location of the fault. Check the fiber identifier for the predetermined frequency. If the fiber identifier does not detect the predetermined frequency, move the fiber identifier several meters closer to the OTDR or infrared light source and recheck for the predetermined pulse. If you have selected the correct fiber optic cable to test and you're confident about the distance to the fault, you should detect the predetermined frequency. If you still do not detect the frequency, double-check everything and retest. If you still do not detect the predetermined frequency, there may not be enough optical energy for the fiber identifier to function properly.

If you are able to detect the predetermined frequency, move the fiber identifier down the fiber optic cable away from the OTDR or infrared light source in one-meter increments. Continue to do this until the fiber identifier no longer detects the predetermined pulse. You now know within one meter where the break in the optical fiber is located. At this point, you may want to disconnect the OTDR or infrared light source and connect the visible fault locator. The visible fault locator may illuminate the exact location of the fault. If the visible fault locator does not illuminate and conditions permit, darken the area around the fault. This may allow you to see the illuminated fault.

OTDR Fault Location Techniques

This section of the chapter is not designed to teach you how to operate the OTDR or use an OTDR to test a fiber optic link. OTDR operation and OTDR fiber optic link testing are covered in Chapter 15. This section of the chapter is designed to help you use the OTDR to troubleshoot or locate faults. Fiber optic links exist in virtually every place imaginable. We will not attempt to address every possible scenario; rather, we will discuss how to look for and find typical faults in a fiber optic link.

Many troubleshooting scenarios never require an OTDR. The OTDR may be used exclusively to troubleshoot or the OTDR may be brought in after it is determined that there is a fault in a fiber optic link. Remember that the OTDR is used to locate the fault in a fiber optic cable or to evaluate connector or splice performance. A faulty fiber optic cable can be identified with a simple continuity test.

Let's say you just finished testing a fiber optic link with a fiber optic source and power meter. You know the length of the link and you have calculated the loss for the link, connectors, and splice as outlined in Chapter 14, "Fiber Optic System Design Considerations." Your calculated loss for the link is 2.5 dB @ 1300 nm, as shown in Table 16.2. Your measured loss for the link is 3.75 dB.

TABLE 16.2 Completed Multimode TIA/EIA-568-B.3 Power Budget Calculation

#	Description	Value
1	Minimum EOL optical output power	-20.0 dBm
2	Minimum optical input power	-31.0 dBm
3	Subtract line 2 from line 1 to calculate the power budget.	11.0 dB
4	Kilometers of optical fiber	0.467
5	Number of interconnections	2.0
6	Number of splices	1.0
7	Multiply line 4 x 3.5.	1.64
8	Multiply line 4 x 1.5.	0.7
9	Multiply line 5 x 0.75.	1.5
10	Multiply line 6 x 0.3.	0.3
11	Add lines 7, 9, and 10 for total link loss at 1300 nm.	3.44 dB
12	Add lines 8, 9, and 10 for total link loss at 1300 nm.	2.5 dB
13	Subtract line 11 from line 3 for headroom at 850 nm.	7.56 dB
14	Subtract line 12 from line 3 for headroom at 1300 nm.	8.5 dB

EXERCISE 16.1

Estimate the minimum performance improvement for a fiber optic link after a faulty inside plant mechanical splice with a measured loss of 1.7 dB has been repaired.

- You have just tested a fiber optic link with the OTDR and discover that a mechanical splice has a loss of 1.7 dB.
 - In Chapter 8, “Splicing,” you learned that per TIA/EIA-568-B.3 the maximum loss for a mechanical splice is 0.3 dB.
 - The splice you just tested exceeds TIA/EIA-568-B.3 by 1.4 dB.
 - The performance for the fiber optic link after the splice is repaired should improve by no less than 1.4 dB.
 - Therefore the minimum performance improvement for the fiber optic link after the splice is repaired is 1.4 dB.
 - The maximum performance would be 1.7 dB. To obtain that performance improvement, the repaired splice would have no measurable loss.
-

As mentioned several times in this chapter, the first thing you should do is clean and perform a visual inspection of the connectors. You should also clean and inspect the receptacle that each connector is plugged into. After cleaning, you should retest with the fiber optic source and the power meter. If the cleaning does not improve link performance to the point where it meets or exceeds the calculated loss, it's time to start troubleshooting.

You learned earlier in this chapter that a cosmetically perfect connector might not necessarily be a low-loss connector. A connector can have a great endface finish; however, the geometry of the connector can be very distorted. Remember that you can't see height or depth with a two-dimensional microscope. But with the OTDR, you can quickly measure the loss for a mated connector pair and look at the light energy reflected back toward the OTDR.

Chapter 15 taught you how to test a fiber optic link with the OTDR. You probably remember that pulse suppression jumpers or cables must be attached to both ends of the fiber optic link under test. Ensure that the pulse suppression jumpers are long enough for your application, as described in Chapter 15.

With the *pulse suppression jumpers* attached, test the fiber optic link as described in Chapter 15. After the link has been tested, the high-loss component or components need to be identified. The measured loss for this link with the fiber optic source and power meter was 1.25 dB greater than the calculated value. Remember that the calculated value is the worst-case scenario. A well-constructed fiber optic link should test well under the calculated value.

The next step is to evaluate the OTDR trace. There are many possible causes for the loss. The only thing that testing has revealed so far is that there are no breaks in the optical fiber in the link. If there was a break, the loss for the link would typically be in excess of 30 dB.

You typically approach any troubleshooting scenario looking for a single fault. However, at times you will discover more than one fault. This troubleshooting scenario is going to be approached as if there were a single fault.

Based on testing, this link suffers from higher than acceptable loss. This loss could be caused by an interconnection, splice, or macrobend. It's the fiber optic technician's job to be able to identify which of these three possible causes is the problem.

Let's say that a bad interconnection is the problem. A bad interconnection would look like Figure 16.13 on the OTDR. Look at the amount of energy entering the interconnection and exiting the interconnection. This interconnection has a loss of 2.25 dB. Remember that the maximum allowable loss for an interconnection, per TIA/EIA-568-B.3, is 0.75 dB. After this interconnection is repaired, the measured loss for the link should improve by no less than 1.5 dB ($2.25 \text{ dB} - 0.75 \text{ dB} = 1.5 \text{ dB}$). The link loss will now be below the calculated value by no less than 0.25 dB ($-1.25 \text{ dB} + 1.5 \text{ dB} = 0.25 \text{ dB}$).

The next possible problem is a bad splice. This link contains one mechanical splice and does not contain a fusion splice. A poor fusion splice looks very similar to a macrobend and will be covered in the next paragraph. A bad mechanical splice looks just like a bad interconnection. The bad mechanical splice trace is shown in Figure 16.14. You should immediately notice that there is virtually no difference between the bad interconnection trace and this trace. The only difference is the amount of loss. When you look at the energy entering the splice and the energy leaving the splice, you'll see that a loss of 1.75 dB is realized. This loss is 1.45 dB greater than the maximum amount allowable by TIA/EIA-568-B.3.

The last possible problem with this fiber optic link is a macrobend. Although 1.25 dB or greater is a significant loss and not a very typical macrobend, this amount of loss is possible from a macrobend. Macrobends and bad fusion splices look the same on the OTDR. Neither the fusion splice nor the macrobend produces a back reflection. Remember that back reflections are produced only from mechanical interconnections. As with the previous two examples, look at the energy entering the macrobend and the energy exiting the macrobend, as shown in Figure 16.15. The loss for this macrobend is 1.8 dB. When this macrobend is repaired the measured loss for the fiber optic link should decrease by 1.8 dB.

FIGURE 16.13 Bad interconnection

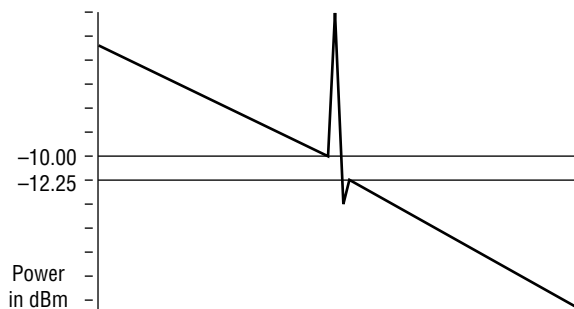
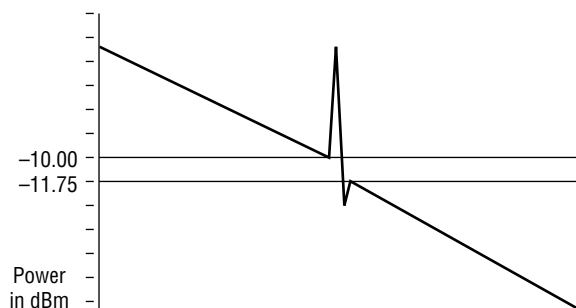
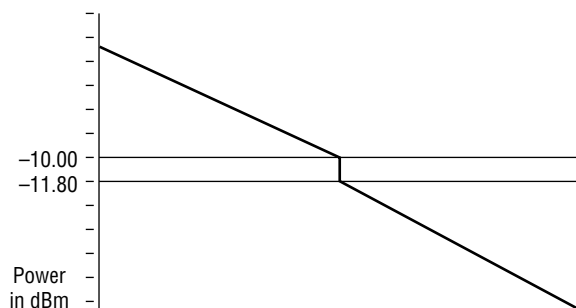


FIGURE 16.14 Bad mechanical splice**FIGURE 16.15** Macrobend

We have not yet addressed using the OTDR to find a break in the optical fiber. A break in an optical fiber can be detected with a continuity tester or visible fault locator. However, the OTDR can locate the exact location of the break. You find the break in an optical fiber with the OTDR just as you find the length of an optical fiber. Chapter 15 describes in detail how to measure the length of an optical fiber.

Restoration Practices

So far in this chapter, you have learned about some of the tools available to the fiber optic technician to troubleshoot a fiber optic link. In Chapter 10 and in Chapter 11, "Fiber Optic Detectors and Receivers," you learned how to understand the operating specifications of a fiber optic transmitter and receiver. This section of the chapter will outline a logical approach to troubleshooting a single fault to restore a fiber optic link that has stopped working. We will proceed under the assumption that you have access to all of the tools described in this chapter.

The first step in any restoration is to ask the customer three things: what they believe the problem is, when they believe it occurred, and what the last thing done to the system was. The best technicians know that many problems are operator error or customer induced. For example, the only problem may be that the customer changed the connections at the patch panel and made an improper connection, or the customer just rearranged the furniture in their office and damaged a fiber optic cable in the process.

After speaking with the customer, you should begin to assess the situation and attempt to identify the fault. Fiber optic cables do not just break, splices do not just go bad, and interconnections do not just fail unless someone has handled them. However, electronics do go bad and electronics do fail. Once you have identified the fiber optic link that has the problem, the next step is to make sure that all cables are connected properly.

If all the cables are connected properly, the next step is to check and see if the electronics are functioning properly. Most fiber optic transmitters output light energy as soon as they are powered up. In other words, the transmitter will output a series of pulses typically at a constant frequency when power is applied. The first step is to look for this light energy at the receiver end of the fiber optic link and measure it. The measured energy should be within the input specification range for the receiver, which is covered in detail in Chapter 11. If the measured energy is within the specification for the receiver, then the problem is most likely the receiver. If the measured energy is not within the range of the receiver, then the problem may be in the fiber optic link or the transmitter.

The next step is to measure the optical output power of the transmitter and verify that it is within the specification, as discussed in detail in Chapter 10. If the measured output power of the transmitter is within specification, the problem is somewhere in the fiber optic link. If the measured output power of the transmitter is not within specification, the problem is with the transmitter.

If the transmitter and receiver are functioning to specification, the next place to look is the fiber optic link. The first things to test in the fiber optic link are the patch cords. You should clean and examine the connectors on each patch cord as described earlier in this chapter, and test the patch cord for loss as described in Chapter 15. You should also clean and inspect the connectors on the back side of the patch panel. Faulty patch cords should be replaced and bad connections on the main cable span should be repaired. If any repairs are made, the system should be retested.

After you inspect the patch cords and connectors, if no faults have been located, the next step is to test the entire cable span. Fiber optic links should be well documented, though that is not always the case. The best scenario would be having that documentation for a comparison to your testing. However, if the customer can't provide documentation, you should still be able to analyze the link and locate faults.

The next step is to connect the OTDR as outlined in Chapter 15 and test the cable plant. If documentation exists, compare the OTDR traces and repair the faulty cable or splice. If documentation does not exist, evaluate the OTDR trace as outlined in Chapter 15 and earlier in this chapter and repair the faulty cable or splice. You may need to employ the VFL or fiber identifier to help you locate the exact physical location of the fault.

Summary

This chapter has provided the basic troubleshooting guidelines and walked you through a step-by-step approach to troubleshoot and restore service to a fiber optic link. The fiber optic technician needs to be able to employ all of the tools at his/her disposal to restore service as quickly as possible. Remember that time is money for the customer; the quicker the system is brought back online, the happier your customer will be and the better chance you have at being called back for the next problem.

Exam Essentials

Explain how to properly perform a visual inspection of a connector. Remember that the weakest link in any fiber optic installation is the connector. Know that you should first visually inspect and clean the connectors. Know to look for and remove any excess epoxy from the connector ferrule. Know that the endface of the connector should be cleaned and the connector evaluated with an inspection microscope.

Explain how to properly evaluate a connector per TIA/EIA-455-57B and locate faults. Remember that the endface of a fiber optic connector can be properly evaluated only with an inspection microscope. Know that a multimode connector can be evaluated with a 100X microscope and a single-mode connector requires a minimum of 200X magnification.

Remember that TIA/EIA-455-57B provides a guideline for examination of an optical fiber endface. Be able to examine a connector endface and determine whether it has an acceptable finish.

Remember that you can accept all endface finishes on a multimode connector that have no defects in the core and inner cladding. Know that the connector endface should be defect free when viewed with a 100X microscope. Know that minor scratches and pitting are acceptable when viewed with a 200X or 400X microscope.

Describe continuity tester fault location techniques. Remember that you can only check the continuity of an optical fiber with the continuity tester. Know that a continuity tester only tests for breaks in the optical fiber and cannot detect macrobends or microbends.

Describe visible fault locator fault location techniques. Remember that the VFL is a continuity tester with a focused visible laser source instead of an unfocused LED or incandescent lamp. The powerful laser in the VFL allows the fiber optic technician to test continuity and visibly identify breaks or macrobends in the optical fiber.

Know that the VFL is an eye hazard and proper safety precautions need to be taken when operating the VFL.

Remember that when the VFL is connected to one end of an optical fiber in a fiber optic cable, it fills the core with visible light. Depending on the cable type, the VFL can illuminate faults in an optical fiber through the buffer, strength member, and even jacket. However, the VFL performs best

with tight-buffered optical fiber. The VFL can also be used to illuminate bad terminations or splices.

Remember that the VFL is typically used in conjunction with the OTDR to localize the physical location of the fault.

Describe fiber identifier fault location techniques. Remember that the fiber identifier allows the fiber optic technician to see through the jacket, strength member, and buffer of the fiber optic cable. The photodiodes in the fiber identifier detect light penetrating through the fiber optic cable. The electronics in the fiber identifier measure the detected light energy and display the direction of light travel through the optical fiber.

Remember that clamping the fiber identifier around a fiber optic cable in various locations allows you to localize faults.

Describe OTDR fault location techniques. Remember that many troubleshooting scenarios never require an OTDR. The OTDR may be used exclusively to troubleshoot or the OTDR may be brought in after it is determined there is a fault in a fiber optic link. Remember that the OTDR is used to locate the fault in a fiber optic cable or evaluate connector or splice performance.

Remember to evaluate the OTDR trace for loss caused by macrobends. Interconnection and splice loss should not exceed the TIA/EIA-568-B.3 maximum values.

Describe common restoration practices. Remember that the first step in any restoration is to ask the customer what they believe the problem is, when they believed it occurred, and what the last thing done to the system was. Remember that many problems are operator error or customer induced.

Know that the next step is to assess the situation and attempt to identify the fault. Know that once you have identified the malfunctioning fiber optic link, you should first make sure that all cables are connected properly.

Know that the next step is to see whether the electronics are functioning to specifications, and the final step is to check the fiber optic link. Know that you first test the patch cords of the fiber optic link and then use the OTDR to test the cable plant.

Review Questions

1. The continuity tester can only be used to locate _____ in an optical fiber.
 - A. Breaks
 - B. Macrobends
 - C. Microbends
 - D. Shorts
2. The _____ can be used to illuminate breaks and macrobends in an optical fiber.
 - A. OTDR
 - B. Continuity tester
 - C. Fiber identifier
 - D. VFL
3. The _____ can be used to detect infrared light traveling through an optical fiber in a fiber optic cable.
 - A. OTDR
 - B. Continuity tester
 - C. Fiber identifier
 - D. VFL
4. A general rule of thumb for evaluating a multimode connector is that defects in the _____ are acceptable for most applications.
 - A. Inner core
 - B. Inner cladding
 - C. Outer core
 - D. Outer cladding
5. The _____ step in any restoration is to ask the customer what they believe the problem is, when they believe it occurred, and what the last thing done to the system was.
 - A. First
 - B. Second
 - C. Third
 - D. Last

6. A high-loss _____ and a macrobend have similar OTDR traces.
 - A. Interconnection
 - B. Mechanical splice
 - C. Fusion splice
 - D. Air gap
7. The _____ of a connector endface is not visible with a microscope.
 - A. Radius
 - B. Loss
 - C. Density
 - D. Scratches
8. High interconnection loss or high interconnection back reflection from a mated connector pair with cosmetically perfect endface finishes may be caused by a(n) _____.
 - A. Scratches
 - B. Air gap
 - C. Nicks
 - D. Index matching gel
9. Mechanical splices and mated connector pair interconnections almost always produce _____ on the OTDR trace.
 - A. Sags
 - B. High loss
 - C. Low loss
 - D. Back reflections
10. _____ provides a guideline for examination of an optical fiber endface.
 - A. TIA/EIA-568-B.3
 - B. TIA/EIA-758-B
 - C. TIA/EIA-455-57B
 - D. TIA/EIA-598-B

Answers to Review Questions

1. A. With a continuity tester, you can check the continuity of an optical fiber—in other words, check to see if an optical fiber allows light to pass from one end to the other. A continuity tester tests only for breaks in the optical fiber and cannot detect macrobends or microbends.
2. D. The visible fault locator is designed to fill the core of an optical fiber with visible light. Depending on the cable type, the VFL may illuminate faults in an optical fiber through the buffer, strength member, and even the jacket. The VFL performs best with tight-buffered optical fiber.
3. C. The fiber identifier allows the fiber optic technician to see through the jacket, strength member, and buffer of the fiber optic cable. The fiber identifier is used very much like the VFL when it comes to troubleshooting. The key differences are that the fiber identifier replaces your eyes and that the fiber optic cable under test typically does not have to be disconnected from an active circuit. The infrared light traveling through the optical fiber during normal operation is often enough to perform most tests. However, sometimes an additional infrared light source is required to adequately troubleshoot.
4. D. A good rule of thumb when evaluating a multimode connector is to accept all endface finishes that have no defects in the core and inner cladding. This means that the connector endface should be defect free when viewed with a 100X microscope. Minor scratches and pitting are acceptable when viewed with a 200X or 400X microscope.
5. A. The first step in any restoration is to ask the customer what they believe the problem is, when they believe it occurred, and what the last thing done to the system was. The best technicians know that many problems are operator error or customer induced. In other words, the problem may stem from the customer having changed the connections at the patch panel and made an improper connection or having rearranged the furniture and damaged a fiber optic cable in the process.
6. C. Macrobends and bad fusion splices look the same on the OTDR. Neither the fusion splice nor the macrobend produces a back reflection. Remember that back reflections are produced only from mechanical interconnections.
7. A. When a connector is viewed with a microscope, you see only a two-dimensional view; you can't see depth or height. As you learned in Chapter 8, virtually all connector ferrules have a radius. This means that a properly polished optical fiber should have the same radius as the ferrule. The radius of the connector is not visible with a microscope.
8. B. High interconnection loss or high interconnection back reflection from a mated connector pair with acceptable endface finishes is usually caused by an air gap. The air gap may be caused by dirt that is not allowing the ferrule to properly seat in the receptacle. The air gap may be present because the connector was not inserted properly into the receptacle. The air gap may also be the result of one or both connectors having a concave instead of convex endface finish.
9. D. Neither the fusion splice nor the macrobend produces a back reflection. Remember that back reflections are produced only from mechanical interconnections such as mechanical splices and mated pair interconnections.
10. C. TIA/EIA-455-57B provides a guideline for examination of an optical fiber endface.



Glossary

A

absorption When referring to light, the conversion of specific wavelengths of light energy into heat through contact with impurities such as water or ions of copper or chromium. Absorption is one of the ways in which light can be lost during transmission through optical fiber.

acceptance angle The angle within which light can enter a fiber core of a given numerical aperture and still reflect off of the boundary layer between the core and the cladding. The acceptance angle is also known as the *cone of acceptance*.

amplifier A device used to increase the power of an optical signal.

amplitude modulation The creation of a signal by changing the strength of a carrier such as a beam of light.

analog A signal that varies continuously through time in response to an input. An analog signal is infinitely variable within a specified range.

analog to digital (A/D) converter A device that converts analog signals into digital signals for storage or transmission.

angle of incidence The angle of a ray of light striking a surface or boundary as measured from a line drawn perpendicular to the surface. Also called *incident angle*.

angle of refraction The angle of a ray of light that is refracted as it passes through an interface, as measured from a line drawn perpendicular to the interface.

angled PC (APC) A connector in which the ferrule is polished at an angle to ensure physical contact with the ferrule of another APC connector.

angular misalignment The offset of fiber cores caused by the fiber ends meeting at an angle.

array connector A connector designed for use with multiple fibers.

attenuation Loss of power. In fiber optics, light energy is attenuated as it travels through fiber and hardware, due to impurities and manufacturing defects.

attenuator A passive device used to reduce the power of an optical signal.

avalanche photodiode A photodiode that multiplies the effect of the photons it absorbs, acting as an amplifier.

B

backbone In networking, the part of a local communication system that carries data between branching points.

bait The mold or form on which silicon dioxide soot is deposited to create the optical fiber preform.

bandwidth The amount of information in the form of light pulses per second that an optical fiber can carry before the information is distorted or lost due to dispersion. In fiber optics, bandwidth is usually measured in megahertz (MHz), or millions of cycles per second, per kilometer of fiber.

binary A value that can only be expressed as one of two states. Binary values may be “1” or “0”; “high” or “low” voltage or energy; or “on” or “off” actions of a switch or light signal. Binary signals are used in digital data transmission.

bit A binary digit. The smallest piece of data possible in a digital communication system.

bit error An error in data transmission resulting from a bit being a different value when it is received from the value at which it was transmitted.

bit error rate (BER) Also known as *bit error ratio*, the ratio of bit errors to the total number of bits transmitted.

bit rate The actual number of light pulses per second being transmitted through a fiber optic link. Bit rates are usually measured in megabits per second (mbps) or gigabits per second (gbps).

blown-in fiber Fiber that is fed through pre-installed conduit using air pressure to blow the fiber through the conduit.

breakout kit A collection of components used to add tight buffers and jackets to individual fibers from a loose tube buffer cable. Breakout kits are designed to allow individual fibers to be terminated with standard connectors.

byte A binary “word” consisting of eight bits.

C

cable modem A modem that transmits and receives signals through copper coaxial cable.

cable tray A shallow tray used to support and route cables through building spaces.

Category 5e An enhanced version of Category 5 or “cat 5” cable used for Gigabit Ethernet connections over short distances.

central tube ribbon cable A cable that carries optical fibers in a multiple stacked ribbon arrangement through a loose tube in the center of the cable.

chromatic dispersion The distortion of optical signals in single-mode fiber caused by the combined effects of *waveguide dispersion* and material dispersion.

cladding The component of an optical fiber surrounding the core. The cladding is not designed to carry light, but it has a refractive index only slightly lower than that of the core. The cladding may be made of glass or plastic.

cleave To cut a section of fiber by scoring the outside and pulling off the end.

clock The timing signal used to control digital data transmission.

coating The first true protective layer of an optical fiber. The coating is made of plastic and is added to the fiber immediately after the fiber is drawn. In addition to protecting the fiber from nicks and scratches, the coating adds tensile strength to the fiber.

coaxial cable A copper two-wire cable consisting of a central wire surrounded by an insulator and a braided outer conductor that also serves as electromagnetic shielding for the cable. The entire cable is surrounded with a heavy protective outer insulator.

coefficient of thermal expansion The measure of a material's change in size in response to temperature variation.

coherent light Light in which photons have a fixed or predictable relationship. Coherent light is typically emitted from lasers.

concentric Sharing a common geometric center.

cone of acceptance A cone-shaped region extending outward from an optical fiber core and defined by the core's *acceptance angle*.

cordage Fiber optic cable designed for use as patch cable or short-term connections. Cordage may be either single-fiber (simplex) or double fiber (duplex). Cordage is not meant for permanent installations.

core The light-carrying component of an optical fiber. The core has a higher refractive index than the surrounding cladding and is typically made of glass or plastic.

coupler See *optical coupler*.

critical angle The smallest angle of incidence at which light passing through a material of a higher refractive index will be reflected off the boundary with a material of a lower refractive index. The angle is measured from a line perpendicular to the boundary between the two materials, known as *normal*. The critical angle is necessary for total internal reflection to occur.

crosstalk The interference of two or more signals with one another.

current The flow of electrons through a conductor.

D

decibel A relative measurement of signal strength used to measure gain or loss of optical power in a system. The decibel scale is a logarithmic scale used to measure the ratio of a signal's transmitted strength to its received strength. For example, a loss of 3 decibels (dB) in a system means that about half of the original signal is left. A loss of another 3 dB means that half of the remaining signal is left.

demodulate To retrieve a signal from a carrier and convert it into a usable form.

demultiplexer A device that separates signals of different wavelengths for distribution to their proper receivers.

dense wavelength division multiplexing A form of multiplexing that separates channels by transmitting them on different wavelengths at intervals of about 0.8 nm.

dielectric A material that does not conduct electrical current. Optical fibers are made of dielectric materials.

differential group delay (DGD) The total difference in travel time between the two polarization states of light traveling through an optical fiber. This time is usually measured in picoseconds, and can differ depending on specific conditions within a fiber and the polarization state of the light passing through it.

digital A signal that uses *binary* values to carry information. Digital signals are used to communicate information between computers or computer-controlled hardware.

digital signal processor (DSP) A device that manipulates or processes data that has been converted from analog to digital form.

digital subscriber line (DSL) A method of delivering high-speed Internet connections over standard copper telephone lines.

directional coupler A device that samples or tests data traveling in one direction only.

dispersion The spreading of light rays along the propagation path due to one or more factors within the medium through which the light is traveling. If dispersion becomes too great, individual signal components can overlap one another and degrade the quality of the optical signal. Dispersion is one of the most common factors limiting the amount of data that can be carried in optical fiber and the distance the signal can travel while still being usable.

dispersion-shifted fiber An optical fiber specially designed with a *zero-dispersion point* that occurs in the same wavelength as one of the fiber's points of low attenuation—about 1550 nm.

distributed feedback (DFB) laser A semiconductor laser specially designed to produce a narrow spectral output for long-distance fiber optic communications.

dopants Impurities that are deliberately introduced into the materials used to make optical fibers. Dopants are used to control the refractive index of the material for use in the core or the cladding.

duplex (1) A link that can carry a signal in two directions for transmitting and receiving data. (2) An optical fiber cable or cord carrying two fibers.

dynamic loads Loads such as tension or pressure that change over time, usually within a short period.

dynamic range The difference between the maximum and minimum optical input power that an optical receiver can accept.

E

edge-emitting LED An LED that produces light through an etched opening in the edge of the LED.

electromagnetic immunity Protection from the interfering or damaging effects of electromagnetic radiation such as radio waves or microwaves.

encode To convert analog data into digital data.

end separation The separation of fiber ends, usually taking place in a mechanical splice or between two connectors.

endface finish The condition of the end of a connector ferrule. Endface finish is one of the factors affecting connector performance.

erbium doped fiber amplifier (EDFA) An optical amplifier that uses a length of fiber doped with erbium and energized with a pump laser to inject energy into a signal.

extrinsic factors When describing a fiber connection, factors contributing to attenuation that are determined by conditions of a splice or connector, as opposed to conditions in the fiber itself.

F

Fabry-Perot A type of laser used in single-mode optical fiber transmission. The Fabry-Perot laser typically has a spectral width of about 5 nm.

fanout kit A collection of components used to add tight buffers to optical fibers from a loose tube buffer cable. A fanout kit typically consists of a *furcation unit* and measured lengths of tight buffer material.

Federal Communications Commission (FCC) The federal agency responsible for regulating broadcast and electronic communications in the United States.

ferrule A metal or ceramic cylinder designed to hold the fiber firmly in the connector for accurate positioning.

fiber identifier A testing device that displays the direction of travel of light within a fiber by introducing a *macro bend* and analyzing the light that escapes the fiber.

figure-8 A fiber optic cable with a strong supporting member incorporated for use in aerial installations. See also: *messenger cable*.

filter A device that blocks certain wavelengths to permit selective transmission of optical signals.

finish The condition of the ferrule and the fiber endface.

four-wave mixing The creation of new light wavelengths from the interaction of two or more wavelengths being transmitted at the same time within a few nanometers of each other. Four-wave mixing is named for the fact that two wavelengths interacting with each other will produce two

new wavelengths, causing distortion in the signals being transmitted. As more wavelengths interact, the number of new wavelengths increases exponentially.

frequency The number of times that corresponding parts of successive waves pass the same point in a fixed period, usually one second. Frequency is typically expressed in cycles per second, or Hertz.

Fresnel reflection Reflection of a small amount of light passing from a medium of one refractive index into a medium of another refractive index.

full-duplex A system in which signals may be transmitted in two directions at the same time. A full-duplex system requires at least two separate fibers—one for each direction of transmission. Full-duplex systems are often used for systems such as long-distance telephone connections, in which signals are transmitted and received at the same time.

furcation unit A component used in breakout kits and fanout kits for separating individual optical fibers from a cable and securing tight buffers and/or jackets around the fibers.

G

gain An increase in power.

graded-index A fiber core with a refractive index that gradually gets lower from the center of the core to the outside of the core. Graded-index fiber is most commonly used to correct the problems that can occur in *multimode fiber*.

H

half-duplex A system in which signals may be sent in two directions, but not at the same time. In a half-duplex system, one end of the link must finish transmitting before the other end may begin.

heterojunction structure An LED design in which the *pn* junction is formed from similar materials that have different refractive indices. This design is used to guide the light for directional output.

homojunction structure An LED design in which the *pn* junction is formed from a single semiconductor material.

hot melt A connector with built-in adhesive that must be preheated before the fiber can be installed.

I

incoherent light Light in which the electric and magnetic fields of photons are completely random in orientation. Incoherent light is typically emitted from light bulbs and LEDs.

index matching gel A clear gel used between fibers that are likely to have their ends separated by a small amount of air space. The gel matches the refractive index of the fiber, reducing light loss due to Fresnel reflection.

index of refraction See *refractive index*.

innerduct A separate duct running within a larger duct to carry fiber optic cables.

insertion loss Light or signal energy that is lost as the signal passes through the fiber end in the connector and is inserted into another connector or piece of hardware. A good connector minimizes insertion loss to allow the greatest amount of light energy through.

inside plant A description of fiber or fiber specifications with regard to their installation within a structure.

intelligence A signal that is transmitted by imposing it on a carrier such as a beam of light to change the amplitude of the carrier.

interface The boundary layer between two media of different refractive indices.

intrinsic factors When describing a fiber connection, factors contributing to attenuation that are determined by the condition of the fiber itself.

isolator A device that permits only forward transmission of light and blocks any reflected light.

J

jacketed ribbon cable A cable carrying optical fiber in a ribbon arrangement with an elongated jacket that fits over the ribbon.

L

laser A semiconductor diode that emits coherent light. Laser is an acronym for *light amplification by stimulated emission of radiation*. Lasers are used to provide the high-powered, tightly controlled light wavelengths necessary for high-speed, long-distance optical fiber transmissions.

laser diodes Semiconductor devices designed to produce laser light for fiber optic communications.

least-squares averaging A method of measuring attenuation in a fiber optic signal that reduces the effect of high-frequency noise on the measurement.

light-emitting diode (LED) A semiconductor device that produces incoherent light. LEDs are used in most fiber optic communication systems that do not require long distances or high data rates.

loose-buffered Also known as *loose tube buffered*, optical fiber that is carried loosely in a buffer many times the diameter of the fiber. Loose-buffered fiber is typically terminated with a breakout kit or a fanout kit and connected to a patch panel.

M

macrobend An external bend in an optical fiber with a radius small enough to change the angle of incidence and allow light to pass through the interface between the core and the cladding rather than reflect off of it. Macrobends can cause signal attenuation or loss by allowing light to leave the fiber core.

mandrel wrap A device used to remove high-order modes caused by overfilling in a length of fiber for insertion loss measurements. High-order modes caused by overfilling are normally attenuated soon after they are launched and must not be factored when testing for insertion loss.

maximum coupling angle The maximum angle, measured from the center line of the fiber core, at which light entering the fiber will reflect off of the core/cladding interface. See also: *acceptance angle*.

maximum tensile rating A manufacturer's specified limit on the amount of tension, or pulling force, that may be applied to a fiber optic cable.

media In fiber optics, the material or materials through which light travels.

medium interface connector (MIC) A connector that is used to link electronics and fiber transmission systems.

messenger cable A cable with a strong supporting member attached to it for use in aerial installations. See also: *figure-8*.

metropolitan area network (MAN) An interconnected group of local area networks (LANs) within a metropolitan area.

microbend Deformation of the core/cladding interface that changes the angle of incidence, allowing light to pass through the interface rather than reflect off of it. Microbends are typically caused by crushing or other damage to the fiber.

minimum bend radius A fiber manufacturer's specified limit on the amount of bending that a fiber can take before its signal-carrying capability is diminished.

modal dispersion A type of dispersion caused when parts of an optical signal take different paths through a fiber. Modal dispersion potentially can cause parts of a signal to arrive in a different order from the one in which they were transmitted, rendering the signal unusable.

mode field diameter In a single-mode fiber, the actual diameter of the light beam traveling through the core and part of the cladding. The mode field diameter is usually slightly greater than the core diameter.

mode filter See *mandrel wrap*.

modes In an optical fiber, the possible paths light can take through the fiber core. A high-order mode is a path that results in numerous reflections off the core/cladding interface. A low-order mode results in fewer reflections. A zero-order mode is a path that goes through the fiber without reflecting off the interface at all. The number of modes in an optical fiber is determined by the

diameter of the core, the wavelength of the light passing through it, and the refractive indices of the core and cladding. The number of modes increases as the core diameter increases, the wavelength decreases, or the difference between refractive indices increases.

modulate (1) To convert data into a signal that can be transmitted by a *carrier*. (2) To control.

multimode fiber A fiber with a core diameter large enough to allow light to take more than one possible path through it.

multiplexing Transmitting multiple data channels in the same signal.

N

noise Electromagnetic energy that is not considered part of the signal.

nonzero-dispersion-shifted fiber A type of single-mode optical fiber designed to reduce the effects of chromatic dispersion while minimizing four-wave mixing.

normal A path drawn perpendicular to the *interface*, or boundary layer between two media, that is used to determine the *angle of incidence* of light reaching the interface.

numerical aperture A dimensionless number that expresses the ability of an optical fiber core to collect light. The numerical aperture is determined by the refractive indices of the core and cladding. The numerical aperture is also used to determine the fiber's *acceptance angle*.

Nyquist Minimum The calculated minimum effective sampling rate for a given analog signal based on its highest expected frequency. The Nyquist Minimum requires sampling to take place at a minimum of twice the expected highest frequency of an analog signal. For example, if an analog signal's highest frequency is expected to be 10 kHz, it must be sampled at a rate of at least 20 kHz.

O

operating wavelength The wavelength at which a fiber optic receiver is designed to operate. Typically, an operating wavelength includes a range of wavelengths above and below the stated wavelength.

optical combiner A device used to combine fiber optic signals.

optical continuous wave reflectometer (OCWR) A device that measures *optical return loss*, or the loss of signals due to reflection back toward the transmitter.

optical coupler A device used to combine or split signals in an optical fiber system.

optical loss test set A set of devices consisting of a light source and an optical power meter used for measuring loss through optical fiber.

optical return loss Optical loss in a fiber optic link caused by signals being reflected back toward the transmitter.

optical splitter A device used to split fiber optic signals.

optical subassembly The portion of a fiber optic receiver that guides light from the optical fiber to the photodiode.

optical time domain reflectometer (OTDR) A device used to test a fiber optic link, including fiber and connectors, by launching an optical signal through the link and measuring the amount of energy that is reflected back. The OTDR is a troubleshooting device that can pinpoint faults throughout a fiber optic link.

outside plant A description of fiber or fiber specifications with regard to their installation outside of any structure.

P

patch cord A two-fiber optical cable used for testing and temporary connections.

PC The abbreviation for physical contact, which describes a connector that places the fiber end in direct physical contact with the fiber end of another connector.

photodiode A component that converts light energy into electrical energy. The photodiode is used as the receiving end of a fiber optic link.

photomultiplication The release of multiple electrons for every photon that is absorbed. Photomultiplication is used to amplify the effects of light reaching a photodiode.

photon A basic unit of light when it exhibits qualities of a particle.

pigtail A short length of optical fiber with a connector or hardware device such as a light source package installed by a manufacturer.

polarization mode The specific orientation of the electric and magnetic components of a light wave along its path of propagation.

polarization mode dispersion Dispersion caused by imperfections in a single-mode fiber slowing down a polarization mode of the signal. When one polarization mode lags behind another, the signal spreads out and can become distorted.

preform A short, thick glass rod that forms the basis for an optical fiber during the manufacturing process. The preform is created first, and then melted and drawn under constant tension to form the long, thin optical fiber.

pre-load A connector with built-in adhesive that must be preheated before the fiber can be installed.

puck A metal disc that holds a connector in the proper position against an abrasive medium for polishing the fiber after it has been installed.

pulse code modulation (PCM) The process of converting an analog signal to a digital signal by sampling it at regular intervals and turning each sample into digital data, which is transmitted sequentially and returned to analog format after it is received.

Q

quantum A basic unit, usually used in reference to energy. A quantum of light is called a *photon*.

quantizer IC An integrated circuit (IC) that measures received optical energy and interprets each voltage pulse as a binary 1 or 0.

quantizing error Inaccuracies in analog to digital conversion caused by the inability of a digital value to match an analog value precisely. Quantizing error is reduced as the number of *bits* used in a digital sample increases, since more bits allow greater detail in expressing a value.

R

raceways Structures used within building spaces to support and guide electrical and optical fiber cables.

Raman amplification An amplification method using a pump laser to donate energy to a signal to amplify it without using a doped length of fiber.

Rayleigh scattering The redirection of light caused by atomic structures and particles along the light's path. Rayleigh scattering is responsible for some attenuation in optical fiber, because the scattered light is typically absorbed when it passes into the cladding.

receptacle The part of a fiber optic receiver that accepts a connector and aligns the ferrule for proper optical transmission.

refraction The bending of light as it passes from one medium into another. Refraction occurs as the velocity of the light changes between materials of two different refractive indices.

refractive index The value given to a medium to indicate the velocity of light passing through it relative to the speed of light in a vacuum. By comparing the refractive indices of two materials, you can determine how much light will bend, or *refract*, as it passes from one material to another.

refractive index profile A graphical description of the relationship between the refractive indices of the core and the cladding in an optical fiber.

repeater A device that receives, amplifies, and transmits a signal to extend its travel in a communication link. Repeaters are commonly used to overcome attenuation in long-distance systems.

responsivity The measure of how well a photodiode converts a wavelength or range of wavelengths of optical energy into electrical current.

return loss The amount of loss in an optical signal reflected back from the connector. A good connection provides a high return loss to minimize optical return.

return reflection Light energy that is reflected from the end of a fiber through *Fresnel reflection*.

rip cord A length of string built into optical fiber cables that is pulled to split the outer jacket of the cable without using a blade.

S

sample To measure values of analog data at regular time intervals and convert those measured values into digital data. The more often a signal is sampled, the more accurately the original information can be reproduced in a digital form.

sample-and-hold circuit A circuit that samples an analog signal such as a voltage level and then holds the voltage level long enough for the *analog to digital converter* to change the level to a numerical value.

scattering The redirection of light caused by atomic structures and particles along the light's path. See also: *Rayleigh scattering*.

semiconductor optical amplifier (SOA) A laser diode with fibers at each end instead of mirrors. The light from the fiber at either end is amplified by the diode and transmitted from the opposite end.

sequential markings Markings on the outside of a fiber optic cable to aid in identification and measuring. Sequential markings provide indications of the fiber and cable type, along with regular tick marks or numbers used to measure long cable runs.

serial In data transmission, data that is carried as signals that follow one another.

sheath The outer jacket of a fiber optic cable.

signal to noise ratio The amount of noise in a signal relative to the strength of the signal itself. The signal to noise ratio is commonly measured in decibels (dB). The higher the ratio, the cleaner the signal that is being received at that power level.

simplex (1) A link that can only carry a signal in one direction. (2) A fiber optic cable or cord carrying a single fiber. Simplex cordage is mainly used for patch cords and temporary installations.

single-board computer A circuit board containing the components needed for a computer that performs a prescribed task. A single-board computer is often the basis of a larger piece of equipment that contains it.

single-mode fiber A fiber with a core diameter large enough for light to take only one possible path through it.

sintering A process in optical fiber manufacturing in which the soot created by heating silicon dioxide is compressed into glass to make the fiber preform.

small form factor A connector that is designed to take up less physical space than a standardized connector. Small form factor connectors are used where space is at a premium.

solar cell A device used to convert light energy into electrical current.

spectral width The range of wavelengths within a light source. A laser may have a spectral width of 1 to 3 nanometers, while an LED may have a spectral width of 20 to 170 nanometers.

splice The permanent connection of one fiber end to another through fusing or mechanical connection.

spontaneous emission The emission of random photons at the junction of the p and n regions in a light-emitting diode when current flows through it.

star coupler A device that distributes optical power equally from two or more input ports to two or more output ports. Star couplers may have up to 64 input and output ports.

static loads Loads such as tension or pressure that remain constant over time, such as the weight of a fiber optic cable in a vertical run.

step-index A fiber that has a core with a single refractive index and a cladding with a single refractive index, and only one boundary between the two.

stimulated emission The process in which a photon interacting with an electron triggers the emission of a second photon with the same phase and direction as the first. Stimulated emission is the basis of a laser.

surface-emitting LED An LED in which incoherent light is emitted at all points along the pn junction.

switch A mechanical, optical, or optomechanical device that completes or breaks an optical path or routes an optical signal.

T

tee coupler A device used for splitting optical power from one input port to two output ports.

tensile strength Resistance to pulling or stretching forces.

terminate To add a component such as a connector or a hardware connection to a bare fiber end.

test jumper A single- or multifiber cable used for connections between an optical fiber and test equipment.

tight-buffered An optical fiber with a buffer that matches the outside diameter of the fiber, forming a tight outer protective layer.

timing circuit In an optical time domain reflectometer, the circuit used to coordinate and regulate the test procedures.

total attenuation The loss of light energy due to the combined effects of scattering and absorption.

total internal reflection The reflection of all of the light in a medium of a given refractive index off of the interface with a material of a lower refractive index. Total internal reflection takes place at the interface between the core and the cladding of an optical fiber.

tractor A machine used to pull a preform into an optical fiber using constant tension.

transimpedance amplifier In a fiber optic receiver, a device that receives electrical current from the photodiode and amplifies it before sending it to the *quantizer IC*.

tree coupler A coupler with one input port and three or more output ports, or with three or more input ports and one output port.

U

unshielded twisted-pair (UTP) (1) A pair of wires twisted together with no electromagnetic shielding around them. (2) A cable containing one or more unshielded twisted pairs, typically Category 3, 4, or 5 cable.

V

vertical-cavity surface-emitting laser (VCSEL) A type of laser that emits coherent energy along an axis perpendicular to the *pn* junction.

visible fault locator (VFL) A testing device consisting of a red laser that fills the fiber core with light, allowing a technician to find problems such as breaks and macrobends by observing the light through the buffer, and sometimes the jacket, of the fiber.

W

waveguide A material or component that serves as a guide for electromagnetic waves along its length. In fiber optics, the fiber core is a waveguide.

waveguide dispersion The spreading of a signal in a single-mode fiber as some of the light passes through the cladding and travels at a higher velocity than the signal in the core due to the cladding's lower refractive index. Waveguide dispersion is one component of *chromatic dispersion*.

wavelength The distance between two corresponding points in a series of waves. Wavelength is preferred over the term *frequency* when describing light.

wavelength division multiplexing A method of carrying multiple channels through a fiber at the same time by using different wavelengths of light within a small spectral range.

window In optical transmission, a wavelength at which attenuation is low, allowing light to travel greater distances through the fiber before requiring a *repeater*.

Z

zero-dispersion point In an optical fiber of a given refractive index, the narrow range of wavelengths within which all wavelengths travel at approximately the same speed. The zero-dispersion point is useful in reducing *chromatic dispersion* in single-mode fiber.

zipcord Duplex fiber optic cordage consisting of two tight-buffered fibers with outer jackets bonded together, resembling electrical wiring used in lamps and small appliances.

Index

Note to the reader: Throughout this index **boldfaced** page numbers indicate primary discussions of a topic. *Italicized* page numbers indicate illustrations.

Numbers

3M, mechanical splicers, 170
8-port star coupler, 276

A

A/D (analog-to-digital) conversion, **21–23**
abrasives, 199, 201
 polishing cloth, 207–208
absolute power gains and losses, **32–33**
absorption, **95–96**, 96, 430
absorptive principle attenuators, **282**, 283
acceptance angle (cone of acceptance), 98, 98, 430
aerial cable, 309
 messenger cable for, 140, 141
air gap, 407
air, refractive index (RI), 50
Alcoa/AFL Telecommunications, fusion splicers, 172
alcohol, 200
 isopropyl, 119
 residue from, 208
aluminum oxide, 185
American National Standards Institute (ANSI), 192
 laser service groups, **114–115**
Amphenol, 191
amplifiers, 430
 optical, **291–293**
amplitude, 17
amplitude modulation, **17–18**, 18, 430
anaerobic epoxy, 199
 application, 203
 safety, 120
analog signal, 430
 conversion to digital, **21–23**
 digital signal conversion to, **23–24**, 24
 sampling, 21
 transmission, **18–19**
analog to digital (A/D) converter, 430
angle of incidence, 54, 430
 in refraction calculation model, 51

angle of refraction, 430
 in refraction calculation model, 51
angled PC (APC) finish, 187, 188, 430
angular misalignment in splicing, **169–170**, 170, 430
Aramid yarns, 135, 135
armoured cable, 140, 140
array connectors, 193, 430
assessment test, xxxi–xxxiv
AT&T, fiber optic telephone systems, 7
attenuation, 18, **94–97**, 430
 absorption, **95–96**, 96
 of AM signal, 19
 exam essentials, 103
 measuring for partial length of optical fiber, 390, 390
 optical fiber vs. copper, **325–328**
 scattering, 96, **96–97**
attenuators, 430. *See also* optical attenuators
avalanche photodiode, 380, 430
averages, and OTDR setup, 384

B

Babinet, Jacques, 3
backbones, 430
 labeling, 314
backscatter, 377, 377
 coefficient and OTDR setup, 385
bait, 69, 430
band-reject optical filter, 293
 response, 294
bandpass optical filter, 293
 response, 294
bandwidth, 87, 431
 dispersion and, **94**
 optical fiber vs. copper, **323–325**
baseline trace, 389, 389–390
bel, 27
Bell, Alexander Graham, 4
Bell Laboratories, 45
bend radius
 minimum, 67, **300–301**
 specifications, 155

bending losses, 100
 biconic connectors, 191, 194
 bidirectional WDM multiplexers, 291, 291
 binary numbering system, 19–20, 431
 bit, 20, 431
 bit error, 259, 431
 bit error rate (BER), 259, 431
 bit rate, 431
 and signal loss from dispersion, 87, 88
 bit rate error, for receiver, 336
 blind spot, in light pulses from OTDR, 382, 382–383
 blown-in fiber, 148, 310, 310, 431
 body of connector, 185
 Boys, Charles Vernon, 4–5
 breaking strength, for optical fiber, 67
 breakout cable, 139, 139–140
 breakout kit, 147, 147–148, 431
 breaks in fiber, VFL for identifying, 360, 361
 broadband analog video CATV applications, minimum signal-mode return loss for, 333
 Brouwer, William, 5–6
 buffered cable, 131–134
 bulkhead optical attenuators, 281
 to test receiver sensitivity, 282
 burial, for cable installation, 309
 bytes, 20, 431

C

cable, 131
 basics, 130–131
 bend radius specifications, 155
 components, 131–137, 132
 buffer, 131–134
 jacket, 136–137
 strength members, 134–136
 duty specifications, 145–146
 exam essentials, 156
 markings and codes, 152–154
 sequential markings, 154–155
 substitution guide, 151
 termination methods, 146–148
 transmission performance, 333
 types, 137–145
 armoured cable, 140, 140
 breakout cable, 139, 139–140
 composite cable, 145, 145
 cordage, 138–139
 distribution cable, 139
 hybrid cable, 145
 messenger cable, 140–141, 141
 ribbon cable, 141, 141–142, 142
 submarine cable, 143, 143–144
 cable installation
 electrical safety, 310–312
 exam essentials, 315
 hardware, 302–306
 patch panels, 305, 305–306
 pullbox, 303, 304
 pulling eye, 303, 303
 splice enclosures, 304–305, 305
 hardware management, 312–313
 methods, 306–310
 aerial cable, 309
 blown-in fiber, 148, 310, 310, 310, 310
 conduit, 308, 308–309
 direct burial, 309
 slack for repairs, 311
 tray and duct, 306–307, 307
 specifications, 300–302
 maximum tensile rating, 301–302
 minimum bend radius, 300–301
 cable modem, 431
 cable rise, weight considerations, 307
 cable trays, 151, 431
 cap of connector, 185, 185
 Category 5e, 431
 centi-, 44
 central tube ribbon cable, 142, 431
 ceramic materials, for ferrules, 185
 chemicals
 exposure, 122–123
 safety, 118–120
 anaerobic epoxy, 120
 isopropyl alcohol, 119
 solvents, 119–120
 chipped fiber end, 210, 210
 chromatic dispersion, 76, 90–93, 431
 cladding of optical fiber, 6, 16, 62, 63, 431
 diameter mismatch loss in splicing, 166, 166
 refractive index (RI), 50
 clamping cable for vertical runs, 307
 cleaning fiber, 208
 cleanliness in work environment, 312, 402
 cleaving fiber, 197, 431
 clock, 432
 clock signal, 25
 coating of optical fiber, 62, 63, 64, 432
 coaxial cable, 325, 432
 coefficient of thermal expansion, 185, 432
 coherent light, 432
 Colladon, Daniel, 3
 collapse of trench, 122

- color coding
 - for cable, 152–154
 - optical fiber coating, 63, 64
 - communication failure, troubleshooting, 265
 - composite cable, 145, 145, 149
 - computers, digital data for, 19
 - concentric, 432
 - concentricity loss, in splicing, 166, 167
 - conductive cable, 149
 - conduit, for cable installation, 308, 308–309
 - cone of acceptance, 98, 98, 432
 - connections, and noticeable loss, 54
 - connectors, 16, 17, 184–186, 185
 - assembly process, 202–208
 - anaerobic epoxy application, 203
 - completing assembly, 204, 204
 - fiber and connector preparation, 202
 - heat-cured epoxy application, 203
 - polishing connector, 205–208
 - pre-load epoxy connector termination, 205
 - UV epoxy application, 204
 - cap types, 186
 - endface examination, 208–212, 209, 403–407
 - absence of fiber, 212
 - defects, 210, 211, 404
 - microscopes, 403
 - exam essentials, 213
 - for glass vs. plastic fibers, 67
 - index matching gel, 55
 - inspection, 402–407
 - performance, 187, 212, 333–334
 - termination, 197–208
 - epoxy, 197–199
 - tools, 199–200, 200, 201
 - types, 188–196
 - biconic connectors, 191
 - D4 connectors, 191
 - ESCON connectors, 192
 - FC connectors, 190
 - FDDI connectors, 192
 - LC connectors, 190, 190
 - mini BNC connectors, 191, 191
 - MPO connectors, 192–193, 193
 - MT-RJ connectors, 193, 193, 194
 - MTP connectors, 193
 - pigtail, 195, 195–196
 - quick reference, 194–195
 - SC connectors, 189, 189
 - SC duplex connectors, 192, 192
 - SMA connectors, 191
 - specialized connectors, 196
 - ST connectors, 189, 190
 - TFOCA connector, 196
 - typical attachment, 186
 - continuity tester, 356–359
 - fault location techniques, 407–410, 408, 410
 - incandescent tester, 357
 - LED continuity tester, 356
 - continuously variable attenuators, 284
 - copper cable
 - advantages of optical fiber over, 323–332
 - attenuation, 325–328
 - bandwidth, 323–325
 - electromagnetic immunity, 328–329
 - safety, 331–332
 - security, 331
 - size and weight, 329–331
 - attenuation, 95
 - vs. fiber, 16
 - cordage, 138–139, 432
 - core of optical fiber, 16, 62, 63, 432
 - diameter mismatch
 - loss in splicing, 165, 165–166
 - problems from, 236
 - numerical aperture, 72
 - Corning, 7
 - fusion splicers, 172
 - couplers, 272–278
 - 128-port, 273
 - exam essentials, 295
 - four-port, 272
 - star coupler, 276, 276–278
 - tee coupler, 273, 273–275
 - in bus type network, 274
 - coupling nut of connector, 185
 - critical angle, 52, 98, 432
 - formula, 53
 - reflection of light exceeding, 53
 - crossband multiplexers, 287
 - crosstalk, 5, 329, 432
 - current, 252, 432
 - Curtiss, Lawrence, 6
 - cutoff wavelength, 76
-
- ## D
- D4 connectors, 191, 194
 - data transmission, 3–4
 - analog, 18–19
 - analog vs. digital, 20–21
 - digital, 19–20
 - dispersion and usability, 74
 - exam essentials, 34–35
 - dB (decibels), 26–31
 - gains in percentages, 29–30

- losses in percentages, 28–29
 - rules of thumb, 31
 - dead zone on trace, in light pulses from OTDR, 382
 - decibels (dB), 26–31, 95, 432
 - formula for Fresnel reflection light loss in, 54
 - gains in percentages, 29–30
 - losses in percentages, 28–29
 - rules of thumb, 31
 - demodulate, 432
 - demodulating light, 17
 - demultiplexer, 26, 287, 433
 - dense multiplexer, 287
 - channel spacing, 288–289
 - dense wavelength division multiplexing, 433
 - design issues
 - advantages of optical fiber over copper, 323–332
 - attenuation, 325–328
 - bandwidth, 323–325
 - electromagnetic immunity, 328–329
 - safety, 331–332
 - security, 331
 - size and weight, 329–331
 - basics, 322
 - exam essentials, 346–348
 - link performance analysis, 332–345
 - cable transmission performance, 333
 - power budget, 335–341
 - single-mode link analysis, 342–345
 - splice and connector performance, 333–334
 - Deutsche Telecom, 191
 - DFB (distributed feedback) laser family, 223, 226
 - output power, 230
 - spectral width, 229
 - spot sizes, 225
 - wavelengths, 227
 - dielectric, 433
 - dielectric cable, 136
 - differential group delay, 93, 433
 - digital data transmission, 19–20
 - analog conversion to, 21–23
 - conversion to analog, 23–24, 24
 - digital signal, 433
 - digital signal processor (DSP), 433
 - digital subscriber line (DSL), 433
 - digital waveform, 20
 - direct burial, for cable installation, 309
 - directional coupler, 433
 - dirt particles, 402
 - dispersion, 87–94, 433
 - and bandwidth, 94
 - chromatic, 76, 90, 90–93
 - effects on signal, 87
 - exam essentials, 103
 - material, 89, 89
 - modal, 74, 74–75, 88
 - polarization-mode, 93
 - profile for optical fiber, 91
 - waveguide, 89–90, 90
 - of wavelengths, 47
 - dispersion-shifted fiber (DSF), 76–77, 91, 433
 - refractive index profile, 92
 - displacement ratio, 168
 - distance, measuring to end of fiber, 391, 391–392
 - distributed feedback (DFB) laser family, 223, 226, 433
 - spot sizes, 225
 - distribution cable, 139
 - documentation, 312
 - of OTDR results, 396
 - dopants, 69, 433
 - draw rate, speed, and fiber thickness, 69
 - DSF. *See* dispersion-shifted fiber (DSF)
 - DSL (digital subscriber line), 433
 - DSP (digital signal processor), 433
 - duplex, 433
 - duplex cordage, 138–139
 - duplex link, 14
 - DuPont Company, 5
 - dynamic loads, 301, 433
 - dynamic range, 433
-
- ## E
- EDFAs (erbium doped fiber amplifiers), 292, 292, 434
 - edge-emitting LED, 221, 222, 434
 - EIA (Electronic Industries Alliance), 64
 - electrical safety, 120–121
 - in cable installation, 310–312
 - shock hazard, 332
 - electrical signal, conversion to light, 17–18
 - electro-optic switches, 280
 - electromagnetic energy
 - light as, 42–45
 - as sine wave, 43
 - three-dimensional nature, 42
 - electromagnetic immunity, 434
 - optical fiber vs. copper, 328–329
 - electromagnetic spectrum, 45–47, 46
 - Electronic Industries Alliance (EIA), 64
 - color coding standards, 152
 - ellipticity loss, in splicing, 167, 167

EMD (equilibrium mode distribution), 101
 emergencies, 122–123
 encode, 434
 end separation in splicing, 169, 169, 434
 endface of connector
 examination, 208–212, 209, 403–407
 absence of fiber, 212
 defects, 210, 211, 404
 microscopes, 403
 finish, 434
 endoscope, 6
 energy loss, 26
 engineering controls, 112
 Enterprise Systems Connection (ESCON), 192
 epoxy, 197–199
 application, 203–204
 equation. *See* formula
 equilibrium mode distribution (EMD), 101
 equipment. *See* test equipment
 erbium doped fiber amplifiers (EDFAs), 292, 292, 434
 error in signal capture, quantizing, 22–23
 ESCON connectors, 192, 195
 Ethernet applications
 laser receiver for, 264
 laser transmitters for, 240
 LED receiver for, 261–262
 LED transmitters for, 235
 ethyl alcohol, refractive index (RI), 50
 European standard for optical fiber, 64
 extrinsic factors, 434
 in splicing, and attenuation, 164, 167–170
 eyes
 laser damage to, 67, 241
 protective equipment, 113

F

Fabry-Perot laser family, 223, 226, 434
 modulation, 237
 output power, 230
 spectral width, 229
 spot sizes, 225
 wavelengths, 227
 face contact (FC) connector, 190
 fanout kit, 146, 147, 434
 Faraday rotator, 284–285
 FC connectors, 190, 194
 FDDI connectors, 192, 195
 Federal Communications Commission (FCC), 323, 434

feet measurements, converting metric to, 154–155
 ferrule, 184–185, 185, 434
 radius evaluation, 407
 fiber. *See* optical fiber
 Fiber Distributed Data Interface (FDDI), 192
 fiber identifier, 331, 361–364, 362, 363, 434
 for troubleshooting, 414–416, 415
 fiber optic link, 14–17, 15
 components
 connectors, 16, 17
 optical fibers, 15–16, 16
 receiver, 15, 15
 transmitter, 15, 15
 fiber optic receiver, 256–258
 block diagram, 256
 electrical subassembly, 258
 exam essentials, 266
 optical subassembly, 256–257
 performance characteristics, 258–264
 dynamic range, 259
 LED receiver, 259–262
 operating wavelength, 259
 receptacle, 256
 fiberglass in cable, 136, 136
 figure 8 cable, 140, 434
 filter, 434
 optical, 293–294
 narrowband response, 295
 response, 294
 finish, 434
 fire, 123
 from isopropyl alcohol, 119
 fire extinguisher, 123
 fire-resistant cable, 150
 Fitel, fusion splicers, 172
 fixed attenuators, 283
 flashlight, for continuity testing, 407
 formula
 acceptance angle, 99
 to calculate refraction with Snell's law, 52
 critical angle, 53
 displacement ratio, 168
 distance to end of fiber, 380–381
 for energy in photon, 45
 fill ratios for conduit, 308, 309
 Fresnel reflection light loss in decibels, 54
 light loss in splice with core diameter mismatch, 166
 light loss through Fresnel reflection, 54–55
 modes for step-index multimode fiber, 99
 number of modes, 72
 number of new waves from transmitting multiple, 92

- numerical aperture of core, 72
- refraction principle, 3
- refractive index (RI), 50, 89
 - for relationship of wavelength and frequency, 43
- forward-biased LED, 220, 221
- forward-transmitted light, through polarized
 - optical isolator, 285
- four-wave mixing, 77, 91, 434–435
 - real world scenario, 92
- frequency, 435
 - for electromagnetic energy, 43
 - and wavelengths, 44
- Fresnel, Augustin, 54
- Fresnel reflection, 54–55, 377, 435
 - from end separation, 169
- full-duplex system, 14, 435
- furcation unit, 146, 147, 435
- fusion splicers, 171, 171–172
 - measuring loss, 393–395, 394
 - vs. mechanical, real world scenario, 177
 - procedures, 174–175
 - video monitor, 176

G

- gain, 435
 - absolute power, 32–33
 - calculating decibel value for signal power, 27
 - rules of thumb, 31
- gap loss, from end separation, 169
- gap-loss principle attenuators, 281, 281
- gel, in loose tube buffered cable, 133
- general-purpose optical fiber raceways, 151
- geometry, of ferrule endface, 187, 188
- ghosts on OTDR trace, 386
- giga-, 44
- gigahertz, 7
- glass fibers, 65
 - vs. plastic, 66–67
- glass, refractive index (RI), 50
- gloves, 113
- graded-index, 435
- graded-index fiber
 - and modal dispersion, 88
 - multimode, 75
- ground loops, 332
- grounding for non-current-carrying components, 311–312
- GTE, fiber optic telephone systems, 7

H

- half-duplex system, 14, 435
- Hansell, Clarence Weston, 5
- hard-clad silica (HCS), 66
- hardhats, 113
- hardware. *See also* test equipment
 - for cable installation, 302–306
 - patch panels, 305, 305–306
 - pullbox, 303, 304
 - pulling eye, 303, 303
 - splice enclosures, 304–305, 305
 - management, 312–313
 - troubleshooting communication failure, 265
- HCS (hard-clad silica), 66
- heat-cured epoxy, application, 203
- heavy-duty cable, 146
- heterojunction structure, 222, 435
- high-order mode, 73, 73
- Hirschowitz, Basil, 6
- home entertainment systems, plastic fiber for, 66
- homes, fiber optic installation, 8
- homojunction structure, 221, 435
- Hopkins, Harold, 6
- hot melt, 435
- hybrid cable, 145
- Hysol epoxy, 203

I

- IEEE standard 802.3, 368
- incandescent continuity tester, 357
 - vs. LED continuity tester, 357
- incoherent light, 221, 435
- independent optical isolators, 286
- index matching gel, 55, 169, 435
- index of refraction, 3
- infrared wavelengths, 46, 96
 - eye damage from, 114
- injuries, 122
- innerduct, 308, 436
- insertion loss, 208, 436
- inside plant, 436
 - maximum allowable loss for fiber, 345
 - splice performance, 342
- installation. *See* cable installation
- intelligence, 17, 436
- interconnection loss, measuring, 392–393, 393
- intercontinental cable, 144
- interface, 436
 - in refraction calculation model, 51

interferometer, 407
 interleaving, 26
 intrinsic absorption, 252
 intrinsic factors, 436
 in splicing, and attenuation, 164–167
 isolator, 436
 isopropyl alcohol, safety, 119

J

jacket, 136–137
 color coding, 154
 jacketed ribbon cable, 142, 436
 joule, 45
 Joule, James Prescott, 45
 jumper cable assembly, 368

K

Kao, Charles K., 7
 Kapany, Narinder, 6
 Kevlar yarns, 135, 135
 kilo-, 44

L

labeling, 313–314
 ladders, 121–122
 Lamm, Heinrich, 5
 laser diodes, 436
 laser light sources, 7, 222–223, 436
 as acronym, 222
 damage to eyes, 67
 modulation speed, 231
 output power, 230
 safety, 115
 service groups, 114–115
 transmitter
 block diagram, 236
 electrical characteristics, 239
 for Ethernet applications, 240
 optical characteristics, 239
 performance characteristics, 235–240
 recommended operating conditions, 238
 warning placards, 116
 wavelengths, 227
 laser pumping, 292

laser receiver
 for Ethernet applications, 264
 performance characteristics, 262–264
 electrical characteristics, 263
 optical characteristics, 263–264
 recommended operating conditions, 262
 lateral displacement in splicing, 168, 168–169
 launch cable, for OTDR, 386
 LC connectors, 190, 190, 194
 leasing fusion splicers, 177
 least-squares averaging, 389, 436
 LED continuity tester, 356
 vs. incandescent tester, 357
 LED light sources, 220–222, 366
 modulation speed, 230
 output pattern, 224
 output power, 229–230
 packaged surface-emitting, 224
 for plastic fiber, 67
 transmitter
 block diagram, 232
 electrical characteristics, 233
 for Ethernet applications, 235
 optical characteristics, 234
 performance characteristics, 231–233
 recommended operating conditions, 233
 wavelengths, 226
 LED receiver
 for Ethernet applications, 261–262
 optical characteristics, 336
 performance characteristics, 259–262
 electrical characteristics, 260
 optical characteristics, 261
 recommended operating conditions,
 259–260
 length of cable segment, measuring, 392, 392
 light
 controlling course, 4–6
 early communication forms, 2–3
 as electromagnetic energy, 42–45
 exam essentials, 56
 refraction, 47–51
 light-duty cable, 145
 light-emitting diodes (LEDs), 436. *See also* LED
 light sources
 light signal, conversion of electrical signal to,
 17–18
 light sources
 exam essentials, 242–243
 performance characteristics, 223–231
 output pattern, 223–226
 source modulation speed, 230–231

- source output power, 229–230
- source spectral output, 227–229
- source wavelengths, 226–227
- safety, 114–115, 240–242
 - classifications, 241
- semiconductor, 220–223
 - laser sources, 222–223
 - LED sources, 220–222
- transmitter performance characteristics, 231–240
 - laser transmitter, 235–240
 - LED transmitter, 231–233
- limiting amplifier, 258
- link, 14. *See also* fiber optic link
 - transmitting many channels over single, 26
- link performance analysis, 332–345
 - cable transmission performance, 333
 - power budget, 335–341
 - single-mode link analysis, 342–345
 - splice and connector performance, 333–334
- Little Rock (USS), fiber optic telephone link
 - installation, 7
- long runs for submarine cable, 144
- loose-buffered cable, 131, 132, 132–133, 133, 436
- loss
 - absolute power, 32–33
 - calculating decibel value for signal power, 27
 - rules of thumb, 31
 - of energy, 26
 - real world scenario, 33
 - from tee coupler, 274–275
- low-order mode, 73, 73
- low-smoke, no halogen (LSNH) for cable jacket, 137, 146
- low-smoke-producing cable, NEC definition for, 150
- LTGF (loose tube, gel-filled) cable, 133
- Lucent Technologies, 190
- Lucite, 5
- manufacturing optical fiber, 68, 68–71
- markings, external, on cable, 152
- mated pair loss, account for, 337
- material dispersion, 89, 89
- Material Safety Data Sheet (MSDS), 118
- maximum allowable loss
 - for inside plant optical fiber, 345
 - for outside plant optical fiber, 344–345
- maximum coupling angle, 98, 437
- maximum tensile rating, 301–302, 437
- MCI, 8
- MCVD (modified chemical vapor deposition), 69, 69–70
- measurement quality jumper (MQJ), 369
- mechanical splicers, 170, 171
 - vs. fusion, real world scenario, 177
 - procedures, 173–174
- media, 437
 - velocity of light through, 48
- medium interface connector, 192, 437
- mega-, 44
- messenger cable, 140–141, 141, 303, 437
- metal ferrules, 185
- metric measurements, conversion to feet, 154–155
- metropolitan area network (MAN), 437
- micro-, 44
- microbends, 93, 100, 100, 437
 - continuity tester and, 408
- microscopes for fiber inspection, 208, 403, 405
- microwave range, of electromagnetic spectrum, 45
- microwave towers, 8
- mili-, 44
- mini BNC connectors, 191, 191, 194
- mini SC connector, 190
- minimum bend radius, 67, 300–301, 437
- minimum return loss, 333
- modal dispersion, 74, 74–75, 88, 437
- mode field diameter, 437
 - of step-index fiber, 76
- mode filter (mandral wrap), 366, 369–370, 370, 437
- modes for optical fiber, 71–77, 437–438
 - formula for number of, 72
 - real world scenario, 78
- modified chemical vapor deposition (MCVD), 69, 69–70
- modulate, 438
- modulated light signal, 4
- MPO connectors, 192–193, 193, 195
- MQJ (measurement quality jumper), 369
- MSDS (Material Safety Data Sheet), 118
- MT-RJ connectors, 193, 193, 194, 195
- MTP connectors, 193, 195

M

- macrobends, 100, 100, 437
 - attenuating loosely coupled modes, 370
 - continuity tester and, 408
 - measuring loss, 393–395, 394
 - reducing risk, 301
 - troubleshooting problems on OTDR, 419, 420
 - VFL for identifying, 360, 361
- magnetic optical isolators, 286, 286
- mandrel wrap, 366, 369–370, 370, 437

multi-fiber connectors, 188
 ESCON connectors, 192
 FDDI connectors, 192
 MPO connectors, 192–193, 193
 MT-RJ connectors, 193, 193, 194
 MTP connectors, 193
 pigtail, 195, 195–196
 quick reference, 194–195
 SC duplex connectors, 192, 192
 specialized connectors, 196
 TFOCA connector, 196

multi-mode connector, evaluating endface, 405–406, 406

multi-mode fiber, 73
 core and cladding, 65
 end separation in splicing, 169
 graded-index, 75
 practical uses, 78
 step-index, 73–75
 practical uses, 78

Multifiber Push On (MPO) connector, 192–193, 193

multimode fiber, 438
 graphic representation for optic link, 340, 340–341
 LED source overfilling, 370
 mandrel diameters for, 371
 optic link with splice, 338, 338

multimode OLTS, 365–367, 366

multiplexers, 272

multiplexing, 21, 26, 26, 272, 438
 exam essentials, 295
 wavelength division (WDM), 286–291, 287
 channel spacing, 289, 290, 291

Mylar polishing film, 205–207

N

nano-, 44

narrowband multiplexers, 287, 288

National Electrical Code (NEC), 145, 146
 cable type marking, 149
 cable types and description, 150
 fill ratios for conduit, 308
 standards for optical fiber, 149–151

National Fire Protection Association (NFPA), 149

Nature, 6

Nippon Telephone and Telegraph (NTT), 189

noise, 18, 328, 438

nonconductive cable, 149
 substitution for, 150

nonzero-dispersion-shifted fiber (NZ-DSF), 77, 92, 438

normal, 438
 in refraction calculation model, 51

North American standard for optical fiber, 64

numerical aperture of core, 72, 98–99, 438
 mismatch in splicing, 165, 165

Nyquist Minimum, 25, 438

O

O'Brien, Brian, 6

Occupational Safety and Health Administration (OSHA), 112

ocean, cable under, 144

OCWR (optical continuous wave reflectometer), 364, 438

OFC cable marking, 150

OFCE cable marking, 150

OFCEP cable marking, 150

OFCE cable marking, 150

OFN cable marking, 150

OFNG cable marking, 150

OFNP cable marking, 150

OFNR cable marking, 150

OLTS. *See* optical loss test set (OLTS)

operating wavelength, 438

optical amplifiers, 291–293

optical attenuators, 280–284
 absorptive principle, 282, 283
 bulkhead, 281
 to test receiver sensitivity, 282
 continuously variable, 284
 exam essentials, 295
 fixed attenuators, 283
 gap-loss principle, 281, 281
 reflective principle, 283
 stepwise variable, 284

optical combiner, 438

optical continuous wave reflectometer (OCWR), 364, 438

optical coupler, 438

optical densities, 48

optical fiber, 15–16, 16
 clean-up of stray ends, 118
 components, 62, 62–66
 materials, 64–66
 standards, 64, 65
 dispersion profile, 91
 exam essentials, 78
 formula for distance to end, 380–381

- history
 - exam essentials, 8–9
 - integration and application, 7–8
 - manufacturing technology development, 4–7
 - research on extending, 6–7
 - manufacturing, 68, 68–71
 - modes, 71–77
 - real world scenario, 78
 - narrowband response, 295
 - National Electrical Code (NEC) standards, 149–151
 - specifications, 102
 - tensile strength, 67
 - optical filter, 293–294
 - narrowband response, 295
 - response, 294
 - optical isolators, 284–286, 285
 - optical loss measurement (TIA/EIA-526-14A), 372, 372–375
 - method A, 373–374
 - method B, 374–375
 - method C, 375
 - optical loss test set (OLTS), 365, 438
 - multimode, 365–367, 366
 - single-mode, 367–368
 - optical power meter, 365–368
 - optical return loss, 438
 - optical splitter, 439
 - optical subassembly, 439
 - optical switches, 278–280, 279
 - optical time domain reflectometer (OTDR), 331, 376–396, 378, 439
 - block diagram, 379
 - cable plant test setup, 386
 - components, 379
 - dead zone on trace, 382
 - display, 381–383
 - documentation, 396
 - event-filled trace, 382
 - fault location techniques, 416–420
 - sampling at 2 ns rate, 380
 - setup, 383–385
 - testing and trace analysis, 388–395
 - attenuation of partial length of fiber, 390, 390
 - baseline trace, 389–390
 - distance measurement to end of fiber, 391, 391–392
 - interconnection loss, 392–393, 393
 - length of cable segment, 392, 392
 - loss measurement of cable segment and interconnections, 395, 395
 - loss measurement of fusion splice or macrobend, 393–395
 - theory, 377–380
 - use with fiber identifier, 416
 - use with visible fault locator, 411
 - visible fault locator (VFL), 379
 - optomechanical switches, 279, 279
 - organization, in hardware management, 312–313
 - OSHA (Occupational Safety and Health Administration), 112
 - OTDR. *See* optical time domain reflectometer (OTDR)
 - outside plant, 439
 - maximum allowable loss for fiber, 344–345
 - splice performance, 342
 - outside vapor deposition (OVD), 70, 70
 - oven-cured epoxy, 197, 198
-
- P**
- passive optical amplifiers, 292
 - patch cord, 368, 439
 - optical power loss measurement, 376
 - patch panels, 305, 305–306
 - PC finish, 187, 188
 - PC (physical contact), 187, 439
 - PCM (pulse code modulation), 25, 25, 439
 - PCS (plastic-clad silica), 65
 - percentages, decibel losses expressed in, 28–29
 - personal protective equipment, 113
 - Peters, C. Wilbur, 6
 - photodiode, 252–253, 439
 - avalanche photodiode, 254–256
 - exam essentials, 266
 - PIN photodiode, 253–254
 - PN photodiode, 253
 - quantum efficiency, 255
 - responsivity, 254–255, 255
 - switching speed, 256
 - photomultiplication, 254, 439
 - photons, 45, 439
 - photophone, 4
 - physical contact (PC), 187
 - pico-, 44
 - pigtail, 188, 195, 195–196, 224, 224, 358, 358–359, 439
 - Planck, Max, 45
 - Planck's constant, 45

plasma chemical vapor deposition (PCVD), 70–71
 plastic-clad silica (PCS), 65
 plastic ferrules, 185
 plastic fiber, 66
 real world scenario, 66–67
 plenum cable, 146
 plenum optical fiber raceways, 151
 polarity of light, 93
 in optical fiber cross-section, 93
 polarization mode, 439
 polarization-mode dispersion (PMD), 93, 439
 polarized optical isolators, 284–286, 285
 polished fiber endface, 187
 polishing cloth, 201
 polishing connector, 205–208
 polyethylene for cable jacket, 136
 polyvinyl chloride (PVC) for cable jacket, 136
 polyvinyl difluoride (PVDF) for cable jacket, 136
 power budget, 335–341
 graphic plot, 340
 for transmitter-receiver combination, 342, 344
 for troubleshooting fiber optic system, 334
 power loss estimation, by fusion splicer, 172, 172
 power measurements, converting dBm to, 32–33
 pre-load, 439
 pre-load epoxy connector termination, 205
 preform, 68, 439
 proportional voltage, from digital data conversion,
 23–24
 puck, 200, 201, 205, 206, 439
 pullbox, 303, 304
 pulling eye, 303, 303
 pulse code modulation (PCM), 25, 25, 439
 pulse suppression cables, for OTDR, 386
 pulse suppression jumpers, 418
 pulse width, and OTDR setup, 384

Q

quantizer IC, 258, 439
 quantizing error, 439
 and digital signal quality, 22–23
 quantum, 45, 439
 quantum efficiency of photodiode, 255

R

raceways, 151, 439
 radio stations, digital signal broadcasts, 20

Raman amplification, 293, 293, 439
 range, and OTDR setup, 383
 Rayleigh scattering, 96, 377, 439
 receive cable, for OTDR, 386
 receive jumper, 369
 receiver, 15, 15
 digital data for display, 21
 receptacle, 439
 reel of fiber optic cable, testing for breaks, 358
 reflections, 377
 Fresnel, 54–55, 377, 435
 from end separation, 169
 reflective principle attenuators, 283
 reflective star coupler, 276, 276
 refraction, 47–51, 439
 causes of, 48–51, 49
 model to calculate, 51
 Snell's law to calculate amount, 52
 refraction principle, formula, 3
 refractive index profiles, 73–76, 74, 439
 dispersion-shifted fiber, 76–77
 multi-mode step-index fiber, 73–75
 multimode graded-index fiber, 75
 single-mode step-index fiber, 75–76
 refractive index (RI), 50, 439
 core vs. cladding, 63, 72
 equation, 50
 and OTDR setup, 384
 relay systems with light, 2
 renting fusion splicers, 177
 repeaters, 95, 291–292, 439
 repetitive pulses on OTDR trace, 386
 resistor, 253
 resolution, and OTDR setup, 383
 respirator, 113
 responsivity, 439
 of photodiode, 254–255, 255
 return loss, 208, 439
 return reflection, 187, 439
 reverse biased photodiode, 252
 reverse-transmitted light, through polarized optical
 isolator, 285
 ribbon cable, 141, 141–142, 142
 size and weight, vs. copper, 329–330
 ripcord, 137, 439
 riser cable, 146
 riser optical fiber raceways, 151
 RJ-45 connector, MT-RJ connectors vs., 193
 Roth (Viennese researcher), 5
 roughness, and connector performance, 187
 Ruess (Viennese researcher), 5

S

safety

- basics, 112–114
 - engineering controls, 112
 - good work habits, 113–114
 - personal protective equipment, 113
- chemicals, 118–120
 - anaerobic epoxy, 120
 - isopropyl alcohol, 119
 - solvents, 119–120
- emergencies, 122–123
- exam essentials, 124
- handling fiber, 117
- light sources, 114–115, 240–242
 - classifications, 241
- on-site, 120–122
 - electrical, 120–121, 310–312
 - ladders, 121–122
 - trenches, 122
- optical fiber vs. copper, 331–332
- of plastic fiber, 66
- sample, 441
- sample-and-hold circuit, 441
- sampling rate, for digitizing analog signal, 21, 22
- SC connectors, 189, 189, 194
- SC duplex connectors, 192, 192, 195
- scattering, 96, 96–97, 441
- scribe, 199, 200
- scribing location, for oven-cured epoxy bead, 206
- security, optical fiber vs. copper, 331
- segment of cable
 - measuring length, 392, 392
 - measuring loss, 395, 395
- semiconductor light sources, 220–223
 - laser sources, 222–223
 - LED sources, 220–222
- semiconductor optical amplifiers (SOAs), 292, 292–293, 441
- sequential markings, 441
 - on cable, 154–155
- serial, 441
- serial data transmission, 25
- SG π lasers, 114–115
- sharks, 144
- shattered endface, and VFL, 360
- shears, 199
- sheath, 137, 137, 441
- short runs for submarine cable, 143
- Siemens, 191
- signal, link for sending, 14
- signal power
 - calculating decibel value of gain or loss, 27
 - rules of thumb, 31
- signal to noise ratio, 364, 441
- silicon dioxide, 63
- simplex, 441
- simplex cordage, 138, 138
- simplex link, 14
- sine wave, electromagnetic energy as, 43
- single-board computer, 441
- single-fiber connectors, 188, 189–191
 - biconic connectors, 191
 - D4 connectors, 191
 - FC connectors, 190
 - LC connectors, 190, 190
 - mini BNC connectors, 191, 191
 - quick reference, 194
 - SC connectors, 189, 189
 - SMA connectors, 191
 - ST connectors, 189, 190
- single-mode fiber, 441
 - graphic representation for optic link, 346
 - and modal dispersion, 88
 - practical uses, 78
- single-mode glass fiber, core and cladding, 65
- single-mode link analysis, 342–345
- single-mode OLTS, 367–368
- single-mode step-index fiber, 75–76
 - light propagation, 76
- sintering, 70, 441
- size of cable, optical fiber vs. copper, 329–331
- skin oil on fiber, 210, 211
- SMA connectors, 191, 194
- small form factor, 441
- small form factor plastic connector, 190
- Smith, David, 5
- Snell, Willebrord, 3
- Snell's law, 52
- SOAs. *See also* semiconductor optical amplifiers (SOAs)
- software, troubleshooting communication failure, 265
- solar cell, 441
- solvents, safety, 119–120
- specialized connectors, 196
- spectral output of light source, 227–229
- spectral width, 76, 89, 441
 - laser, 228–229
 - LED light sources, 228, 228
 - reduced, 93
- splice enclosures, 304–305, 305
- splicing, 441
 - basics, 164
 - extrinsic factors, 164, 167–170
 - intrinsic factors, 164–167
 - vs. connectors, 184

- equipment, 170–172
 - fusion splicers, 171, 171–172
 - mechanical splicers, 170, 171
 - exam essentials, 178
 - fusion vs. mechanical, real world scenario, 177
 - measuring loss, 393–395, 394
 - performance, 333–334
 - procedures
 - fusion splicers, 174–175
 - mechanical splicers, 173–174
 - requirements, 176
 - troubleshooting problems on OTDR, 419, 420
 - spontaneous emission, 220, 442
 - Sprint, 8
 - ST connectors, 189, 190, 194
 - stairstep voltage, conversion to smooth waveform, 24
 - Standard Telecommunications Laboratories, 7
 - star coupler, 276, 276–278, 442
 - static loads, 301, 442
 - steel, for cable jacket, 137
 - step-index, 442
 - step-index fiber, 73
 - multi-mode, 73–75
 - calculating number of modes, 99
 - single-mode, 75–76
 - light propagation, 76
 - stepwise variable attenuators, 284
 - stimulated emission, 223, 442
 - straight tip (ST) connector, 189, 190
 - strain relief boot, 185, 186
 - in connector assembly, 202
 - strain relief, from connector, 184
 - stretching, 67
 - stripper, 199, 200
 - stripping fiber, 202, 202
 - submarine cable, 143, 143–144
 - submarine, fiber optic link on, 334
 - subscriber (SC) connector, 189, 189
 - surface-emitting LED, 221, 222, 442
 - switches, 442
 - optical, 278–280, 279
 - switching speed, of photodiode, 256
 - telecommunications switching systems, plastic
 - fiber for, 66
 - tensile rating, maximum, 301–302
 - tensile strength, 67, 442
 - tera-, 44
 - terminate, 442
 - termination methods. *See also* connectors
 - for cable, 146–148
 - test equipment
 - continuity tester, 356–359
 - incandescent tester, 357
 - LED continuity tester, 356
 - exam essentials, 396–397
 - fiber identifier, 361–364, 362, 363
 - light source and optical power meter, 365–368
 - mode filter, 369–370
 - optical return loss test set, 364–365
 - optical time domain reflectometer (OTDR), 376–396, 378
 - block diagram, 379
 - cable plant test setup, 386
 - dead zone on trace, 382
 - display, 381–383
 - event-filled trace, 382
 - sampling at 2 ns rate, 380
 - setup, 383–385
 - testing and trace analysis, 388–395
 - theory, 377–380
 - visible fault locator (VFL), 379
 - patch cord, 368
 - optical power loss measurement, 376
 - test jumper, 368–369
 - TIA/EIA-526-14A optical loss measurement, 372, 372–375
 - visible fault locator (VFL), 359, 359–360
 - test jumper, 368–369, 442
 - testing, glass vs. plastic fiber, 67
 - TFOCA connector, 196
 - thermal expansion, coefficient of, 185
 - thermo-optic switches, 280
 - 3M, mechanical splicers, 170
 - thresholds, and OTDR setup, 384–385
 - TIA. *See also* Telecommunications Industry Association (TIA)
 - TIA/EIA-455-50B standard, 366
 - TIA/EIA-455-57B standard, 403
 - TIA/EIA-526-14A optical loss measurement, 372, 372–375
 - method A, 373–374
 - method B, 374–375
 - method C, 375
 - TIA/EIA-568-B standard, 152, 323, 324
-
- T**
- T-Mobile Wireless, 191
 - tee coupler, 273, 273–275, 442
 - in bus type network, 274
 - Telecommunications Industry Association (TIA), 64. *See also* TIA/EIA
 - color coding standards, 152

TIA/EIA-568-B.1 standard, 369
 TIA/EIA-568-B.3 standard, 188, 189, 332, 366, 369
 on inside plant splice performance, 342
 section 4.2 on 50/125 μm multimode optical fiber, 333
 TIA/EIA-606-A standard, 314
 TIA/EIA-758 standard, 342
 TIA/EIA-recognized optical fibers, 102
 TIA/EIA standards, maximum permissible loss for splices, 176
 tight-buffered cable, 131, 134, 135, 442
 timing circuit, 442
 TIR. *See* total internal reflection (TIR)
 total attenuation, 97, 97, 442
 total internal reflection (TIR), 3–4, 51–54, 52, 53, 442
 and modes, 71–72
 Tra Con epoxy, 203
 trace analysis, 388–395
 attenuation of partial length of fiber, 390, 390
 baseline trace, 389, 389–390
 distance measurement to end of fiber, 391, 391–392
 interconnection loss, 392–393, 393
 length of cable segment, 392, 392
 loss measurement of cable segment and interconnections, 395, 395
 loss measurement of fusion splice or macrobend, 393–395
 trace, by OTDR, 382
 tractor, 69, 442
 transimpedance amplifier, 258, 442
 transmission. *See* data transmission
 transmissive star coupler, 276, 276
 transmit jumper, 368, 370
 transmitter, 15, 15
 transmitter performance characteristics, 231–240
 laser transmitter, 235–240
 LED transmitter, 231–233
 tray and duct cable installation, 306–307, 307
 tree coupler, 276, 443
 trenches, 122
 troubleshooting
 communication failure, 265
 connector endface evaluation when, 402–407
 continuity tester fault location techniques, 407–410, 408, 410
 exam essentials, 422–423
 fiber identifier for, 414–416, 415
 OTDR fault location techniques, 416–420
 with power budget, 334
 restoration practices, 420–421
 visual fault locator (VFL), 411–414, 412, 413
 Tyndall, John, 3

U

ultraviolet, 46
 Underwriters Laboratories (UL) 1666, 146
 unidirectional WDM multiplexers, 291
 unshielded twisted-pair (UTP), 443
 U.S. Navy, fiber optic telephone link installation, 7
 USConec, 193
 UV epoxy, 199
 application, 204

V

vacuum, refractive index (RI), 50
 van Heel, Abraham, 5–6
 vapor axial deposition (VAD), 70, 71
 vertical cable runs, weight considerations, 307
 vertical-cavity surface-emitting laser (VCSEL), 223, 225, 443
 in multimode transmitters, 237
 output power, 230
 spectral width, 229
 visible fault locator (VFL), 359, 359–360, 411–414, 412, 413, 443

W

warm-up times, for multimode light sources, 367
 water
 refraction of light through, 47
 refractive index (RI), 50
 wave nature of light, and refraction, 48
 waveform
 amplitude, 17
 digital, 20
 waveguide, 7, 443
 waveguide dispersion, 89–90, 90, 443
 wavelength division multiplexing (WDM), 77, 286–291, 287, 443
 channel spacing, 289, 290, 291
 wavelengths, 7, 443
 and attenuation, 95
 dispersion of, 47
 of electromagnetic energy, 43
 electromagnetic energy characteristics dependent on, 45
 and frequency, 44
 light sources, 226–227
 and OTDR setup, 383
 weight, optical fiber vs. copper, 329–331
 Wheeler, William, 4

white light, refraction, 49–50, 50
wideband multiplexers, 287
window, 443
wipes, 200
work environment, cleanliness in, 312
work habits, and safety, 113–114

Y

Y coupler, 274

Z

zero-dispersion point, 77, 91, 443
zero-dispersion-shifted fiber, 77, 77
zero-order mode, 73, 73
zipcord, 138, 443
zirconium oxide, 185